

Biblia Profesional Completa

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Guia de lectura de la version completa

0.1 Guia de lectura de la version completa

Esta version completa existe para revisar todo el corpus sin perder listas, tablas, empresas, datos de mercado ni materiales operativos. No intenta condensar la investigacion como la version editorial.

0.1.1 Que incluye

Seccion	Cantidad
Modulos Markdown de raiz incluidos completos	47
Markdown de Research incluidos completos	18
PDFs de Research inventariados y enlazados	12
Historic indexado como obsoleto	9
Lineas Markdown fuente incluidas	47228
Bytes Markdown fuente incluidos	1615581
Bloques duplicados largos reemplazados por nota	0
Caracteres duplicados no reimpresos	0

0.1.2 Regla de deduplicacion

Solo se omiten bloques largos cuyo contenido normalizado ya aparecio antes. La primera aparicion permanece completa. La segunda aparicion se reemplaza por una nota con la fuente original. Si el texto contiene informacion nueva, distinta o parcialmente diferente, permanece.

0.1.3 Regla de verdad

Tema	Regla
Historic	Es historico/desactualizado y no gobierna decisiones actuales
Research	Es evidencia fuente; si contradice modulos actuales, se subordina a la regla canonica
Pasaporte portugues	Esperado alrededor de 2028; no es autorizacion laboral UE actual
Perfil publico	Real-Time 3D Developer / Unity Technical Artist
TwinSight	Technical visualization / assembly inspection prototype; no full production digital twin
Tracker/hojas	Mandan para estado operativo vivo
PDF completo	Snapshot de revision; no sustituye Obsidian ni archivos fuente

0.1.4 Auditoria principal

ID	Severidad	Categoria	Hallazgo	Resolucion
AUD-001	High	Work authorization	El pasaporte portugues no debe presentarse como autorizacion laboral UE actual.	Regla canonica: pasaporte/citizenship esperado alrededor de 2028; hasta emision legal, aplicar como Colombia-based remote contractor/B2B.
AUD-002	Low	Historic data	Existen datos historicos como 'July 2026' que contradicen la regla actual.	Historic queda fuera del cuerpo del PDF y se indexa como obsoleto; no corrige ni domina los modulos actuales.
AUD-003	Medium	Seniority	Aparece 'Senior Technical Artist' en benchmarks, riesgos o textos negativos; no debe convertirse en titulo personal.	Usar 'Real-Time 3D Developer / Unity Technical Artist'. Senior solo como benchmark aspiracional o rol a largo plazo.
AUD-004	Medium	TwinSight framing	TwinSight puede relacionarse con digital twin, pero no debe presentarse como digital twin industrial completo sin IoT/live state.	Framing canonico: Unity WebGL technical visualization, assembly inspection prototype, digital-twin-adjacent.
AUD-005	Medium	AI tooling	ARA/AI tooling es ruta secundaria; no debe implicar seniority de AI engineer productivo.	ARA se presenta como Python automation / AI tooling prototype, subordinado a TwinSight salvo que se vuelva portfolio-ready.
AUD-006	High	Public proof	GitHub, demo live, CV publico y portfolio son condiciones de credibilidad pendientes hasta que existan URLs reales.	Mantener placeholders explicitos y no afirmar disponibilidad publica hasta verificar links.
AUD-007	Medium	Metrics	Metricas de TwinSight deben coincidir con tesis final antes de usarse publicamente.	Mantener metricas como condicionales si no han sido verificadas contra documento final de tesis.
AUD-008	Medium	Market freshness	Empresas, vacantes, salarios y remote policies son datos temporales.	El PDF es snapshot; para aplicar se usa tracker vivo y verificacion semanal/top targets.
AUD-009	Low	Duplicates	Hay variantes/duplicados, especialmente 28D y agent_A4.	Se incluyen por trazabilidad, marcados como duplicate/reference; la version complete/canonica manda.
AUD-010	Medium	Legal/tax	Impuestos, contractor setup, residencia y permisos requieren validacion profesional antes de decisiones reales.	Tratar los modulos como estrategia informativa; validar con contador/abogado antes de mudanza, facturacion internacional o aceptacion de oferta.

Mapa de fuentes completo

Fuente	Seccion	Estado	Lineas	Bloques omitidos	Caracteres omitidos
00_master_audit_blueprint.md	root	canonical	1145	0	0
01_source_of_truth_profile.md	root	canonical	929	0	0
02_market_role_fit_and_positioning.md	root	canonical	838	0	0
03_salary_benchmark_and_recruitment_colombia.md	root	canonical	808	0	0
04_eu_portugal_germany_mobility_strategy.md	root	canonical	694	0	0
05_language_strategy_and_recruitment.md	root	canonical	962	0	0
06_education_courses_rebelway_masters_phd.md	root	canonical	740	0	0
07_portfolio_strategy_and_project_architecture.md	root	canonical	776	0	0
08_twinsight_x500_case_study.md	root	canonical	893	0	0
08B_twinsight_case_study_final_structure.md	root	canonical	1111	0	0
09_ara_ai_tools_and_secondary_projects.md	root	canonical	840	0	0
10_github_linkedin_cv_strategy.md	root	canonical	1210	0	0
11_company_targets_job_boards_recruiters.md	root	canonical	879	0	0
12_application_system_and_recruiter.md	root	canonical	1327	0	0
13_interview_negotiation_and_offer_risk.md	root	canonical	816	0	0
14_30_60_90_execution_plan.md	root	canonical	927	0	0
15_final_integrated_strategy.md	root	canonical	981	0	0
16_final_consistency_audit_and_gap_register.md	root	canonical	740	0	0
17_cv_base_and_role_variations.md	root	canonical	1152	0	0
18_linkedin_final_rewrite.md	root	canonical	1285	0	0
19_github_readme_templates.md	root	canonical	1413	0	0
19B_github_readme_twinsight_final.md	root	canonical	787	0	0
20_portfolio_copy_and_site_structure.md	root	canonical	1246	0	0
21_twinsight_demo_video_script_and_shotlist.md	root	canonical	912	0	0
21B_twinsight_demo_video_final_plan.md	root	canonical	896	0	0
22_outreach_templates_english_spanish.md	root	canonical	1164	0	0
23_interview_answer_bank.md	root	canonical	902	0	0
24_offer_evaluation_scorecard.md	root	canonical	1050	0	0
26_weekly_execution_dashboard.md	root	canonical	1095	0	0
27_strategy_scenarios_conservative_balanced_aggressive.md	root	canonical	1089	0	0
28_competitor_profile_benchmark.md	root	canonical	889	0	0
28B_public_cv_resume_benchmark.md	root	canonical	916	0	0
28C_github_profiles_benchmark.md	root	reference	737	0	0
28D_artstation_portfolio_benchmark(1).md	root	duplicate	271	0	0
28D_artstation_portfolio_benchmark.md	root	duplicate	439	0	0
28D_artstation_portfolio_benchmark_complete.md	root	canonical	855	0	0
29_portfolio_case_study_benchmark.md	root	canonical	1011	0	0
29B_demo_reels_benchmark.md	root	canonical	882	0	0
29C_artstation_breakdown_benchmark.md	root	reference	838	0	0
30_live_job_postings_market_snapshot.md	root	canonical	221	0	0
31_application_targeting_matrix_and_search_protocol.md	root	canonical	1073	0	0
32_application_materials_gap_closure_plan.md	root	canonical	1007	0	0
33_portfolio_asset_production_blueprint.md	root	canonical	1025	0	0
34_weekly_application_execution_system.md	root	canonical	1128	0	0
35_interview_defense_and_technical_screen_system.md	root	canonical	952	0	0
36_public_profile_launch_sequence.md	root	canonical	1167	0	0
README.md	root	canonical	22	0	0
Research/agent_A4_artstation_portfolio_benchmark(1).md	research	research_duplicate	139	0	0
Research/agent_A4_artstation_portfolio_benchmark.md	research	research_duplicate	48	0	0
Research/agent_B2_digital_simulation_live_jobs_research_snapshot	research	research_snapshot	104	0	0
Research/cerrando_sistema_research	research	research_snapshot	340	0	0
Research/D2_conferences_events_demo_venues.md	research	research_snapshot	272	0	0
Research/deep-research-report_00.md	research	research_snapshot	461	0	0
Research/deep-research-report_01_structure.md	research	research_snapshot	469	0	0
Research/deep-research-report_02.md	research	research_snapshot	108	0	0
Research/deep-research-report_03.md	research	research_snapshot	106	0	0

Fuente	Seccion	Estado	Lineas	Bloques omitidos	Caracteres omitidos
Research/deep-research-report_04.md	research	research_snapshot	89	0	0
Research/deep-research-report_05.md	research	research_snapshot	108	0	0
Research/deep-research-report_06.md	research	research_snapshot	114	0	0
Research/deep-research-report_11.md	research	research_snapshot	148	0	0
Research/deep-research-report_13.md	research	research_snapshot	172	0	0
Research/deep-research-report_C3.md	research	research_snapshot	85	0	0
Research/deep-research-report_C4.md	research	research_snapshot	22	0	0
Research/E1_technical_volume_research_opportunities.md	research	research_snapshot	471	0	0
Research/paquete_continuidad_research.md	research	research_snapshot	932	0	0

Part I

Modulos raiz completos

Chapter 1

00 — Master Audit Blueprint

Fuente: 00_master_audit_blueprint.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1145

Ultima modificacion: 2026-06-11 11:12

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Este archivo define la arquitectura de auditoría que se usará para reconstruir, sección por sección, la investigación y estrategia laboral internacional de Alexander Woodcock Salomón.

Contenido completo depurado

1.0.1 Auditoría estructural para reconstrucción de estrategia laboral internacional

Perfil: Alexander Woodcock Salomón

Fecha de corte operativa: 2026-06-11

Estado del documento: Blueprint de auditoría. No es estrategia final. No aplica correcciones aún.

Modo usado para esta sección: análisis interno de archivos + razonamiento estratégico. **No requiere Deep Research ni modo agente.**

1.0.2 0. Propósito del documento

Este archivo define la arquitectura de auditoría que se usará para reconstruir, sección por sección, la investigación y estrategia laboral internacional de Alexander Woodcock Salomón.

No corrige todavía el plan. Su función es:

1. congelar los hechos base que no deben volver a confundirse;
 2. identificar contradicciones entre documentos;
 3. separar datos vigentes, datos desactualizados, supuestos, aspiraciones y afirmaciones no defendibles;
 4. definir qué secciones deben reconstruirse;
 5. especificar qué falta investigar;
 6. decidir qué secciones requieren Deep Research, modo agente, ambos o ninguno;
 7. fijar criterios de calidad para aprobar cada archivo posterior.
-

1.0.3 1. Fuentes consideradas

1.1 Fuentes principales cargadas

Archivo	Uso en esta auditoría	Estado
deep-research-report (8).md	Informe anterior que será auditado y reconstruido. Contiene estrategia general, CV, LinkedIn, GitHub, portfolio, salarios, movilidad, scripts, plan 30/60/90.	Material base, pero insuficiente y con debilidades metodológicas.
Pegado text(12).txt	Prompt original condensado con perfil, proyectos, objetivos, restricciones y entregables requeridos.	Fuente primaria de intención estratégica.
alexander_woodcock_context.md	Contexto maestro previo: identidad, restricciones, hardware, runway, herramientas, perfil psicológico, planes antiguos.	Útil, pero algunas fechas/estados están desactualizados.
plan_audit.md	Auditoría antigua de tesis/portfolio con énfasis en retopología, documentación de optimización y riesgos de cronograma.	Parcialmente obsoleta porque la tesis ya está completa.
master_schedule.md	Plan anterior con fases tesis → portfolio → job hunt.	Obsoleto como cronograma. Útil como evidencia de prioridades y proyectos planeados.
ticktick_schedule.md	Agenda granular antigua.	Obsoleta como calendario. Útil para entender cursos, tareas pensadas, intención de portfolio y secuenciación.
job_search_strategy.md	Estrategia anterior tipo “Trojan Horse”: Junior Technical Artist / QA Automation / Tools.	Útil como hipótesis, pero debe revalidarse con mercado actual.
portfolio_strategy.md	Estrategia previa de portfolio: Optimization Doctor, Toolsmith, Shader studies, Mobile Hero, Human Portrait.	Útil como arquitectura de portfolio, pero debe revalidarse contra estado real de proyectos.
estrategia_idiomas_mercado_laboral_tecnica_investigacion.md	Investigación de idiomas y mercados con fuentes.	Útil, pero debe auditarse y actualizarse en secciones legales/diomas/salarios.
Courses 2025.xlsx	Inventario de cursos adquiridos.	Debe convertirse en matriz de ROI, no en lista decorativa.

1.2 Observaciones sobre las fuentes

- Los archivos antiguos hablan de una tesis en desarrollo. **Eso ya no es válido.**
- La tesis debe tratarse como: **completa, pendiente de sustentación.**
- Los archivos antiguos no deben usarse como calendario vigente.
- Las metas, proyectos y riesgos sí aportan contexto estratégico.
- Las fuentes salariales y legales del informe anterior no son suficientemente robustas.
- La lista de empresas del informe anterior es superficial: da ejemplos, pero no una base operativa de 100+ targets verificados.
- El Excel de cursos no fue integrado con suficiente rigor en el informe anterior.
- La investigación anterior mezcló español e inglés, afirmaciones fuertes y datos débiles.

1.0.4 2. Fuente de verdad congelada

Esta sección define los hechos base que todo documento posterior debe respetar.

2.1 Identidad y estado académico

Campo	Valor congelado
Nombre	Alexander Woodcock Salomón
Ubicación	Colombia
Ciudad actual relevante	Pasto, Nariño, Colombia
Idioma nativo	Español
Inglés	C1 autoevaluado / funcional para entrevistas técnicas
Alemán	Ninguno o A1 incipiente, no profesional
Portugués	Relevante estratégicamente, no competencia profesional actual confirmada
Carrera actual	Ingeniería Multimedia, UNAD
Tesis	Completada, pendiente de sustentación
Proyecto de tesis	TwinSight X500 / WebGL-Thesis-Proposal

Campo	Valor congelado
Carrera previa	Ingeniería Electrónica, UNAL, aproximadamente 8 semestres / cerca de 80% cursado
Forma internacional sugerida para UNAL	"Completed advanced coursework in Electronic Engineering" o "Electronic Engineering coursework — Universidad Nacional de Colombia"
Experiencia formal target-field	Baja / no industrial fuerte
Experiencia relevante	Proyectos académicos, personales, freelance web simple, proyectos 3D/Unity/Python

2.2 Restricciones vigentes

Restricción	Implicación
No hay autorización laboral UE actual	No afirmar "authorized to work in EU".
Pasaporte/citizenship portugués es expectativa futura	No usarlo como derecho laboral actual.
Alemania depende de escenario futuro de pareja/matrimonio/residencia	No usarlo como plan operativo de corto plazo.
Prioridad: remoto desde Colombia	El plan debe optimizar contractor/B2B, remote-first, LATAM nearshore y global remote.
No hay CV final	CV se construye después de cerrar posicionamiento.
No hay portfolio robusto publicado	Portfolio MVP y TwinSight case study son prioridad.
Cursos sin certificado	Usarlos como aprendizaje/skill proof, no como certificaciones formales.
Riesgo de dispersión	La estrategia debe limitar proyectos activos y evitar sobrecarga de cursos.

2.3 Posicionamiento base provisional

Technical Artist & Real-Time 3D Developer especializado en Unity, WebGL, C#, Blender, CAD-to-realtime optimization, interactive technical visualization y Python/AI-assisted production tooling.

Este posicionamiento aún debe validarse contra mercado. No es definitivo.

1.0.5 3. Matriz de datos: hecho, pendiente, supuesto, aspiración y no afirmable

3.1 Hechos defendibles

Afirmación	Estado	Uso permitido
TwinSight X500 es el proyecto flagship	Defendible	CV, LinkedIn, portfolio, entrevistas
Tesis completa y pendiente de sustentación	Defendible según contexto actualizado del usuario	CV con wording: "pending defense"
Unity, C#, WebGL, Blender, CAD-to-realtime son el núcleo técnico	Defendible	Posicionamiento principal
ARA Framework es prototipo de AI research automation	Defendible si se presenta como prototype / proof-of-concept	Portfolio secundario
Inglés C1 autoevaluado	Defendible con matiz	"Professional working proficiency" si se sostiene en entrevista
Cursos adquiridos no certificados	Defendible	"Self-study / professional development"

3.2 Hechos pendientes de verificación

Afirmación	Qué falta verificar
Estado público real del repo TwinSight	Revisar GitHub público, README, estructura, licencia, assets, demo, branches.
Demo funcional WebGL	Confirmar URL, carga, performance, compatibilidad móvil.

Afirmación	Qué falta verificar
Métricas finales de tesis	Confirmar versión final: triángulos, SUS, NASA-TLX, participantes, task-condition records.
Estado funcional de ARA	Confirmar qué agentes corren, outputs, estabilidad, seguridad de API keys.
Estado real de ai-news-aggregator	Confirmar si corre, si hay demo, qué falta para publicarlo.
Estado real de FitApp-Free	Confirmar si aporta al posicionamiento o distrae.
ArtStation actual	Revisar si hay piezas publicadas, calidad y consistencia.
LinkedIn actual	Revisar texto real, no solo capturas/contexto.
GitHub actual	Revisar repos públicos, pinned, ruido y actividad.
Excel de cursos	Clasificar por ROI real, no por volumen.

3.3 Supuestos estratégicos válidos pero no cerrados

Supuesto	Riesgo
Remote contractor desde Colombia es la vía más realista inicial	Requiere validación legal/tributaria y targets reales.
Technical visualization / digital twin puede pagar mejor que game tech art junior	Requiere investigación de mercado y salarios.
USD 3k/mes es viable en 3–6 meses	Requiere benchmark realista por rol y seniority.
USD 6k/mes es viable a 12–24 meses	Requiere experiencia demostrable, portfolio fuerte y/o nicho mejor pagado.
Alemán es mejor idioma estratégico que francés/japonés	Debe cruzarse con pasaporte portugués, mercados objetivo y ROI temporal.
Rebelway puede aportar a portfolio	Depende de qué curso y qué demo output se produzca.

3.4 Aspiraciones válidas

Aspiración	Uso correcto
Trabajar remoto para empresa extranjera	Objetivo estratégico
Ingresos USD 1.5k → 3k → 6k	Metas por etapa, no promesas
Entrar a technical art / real-time 3D internacional	Narrativa base
Usar futuro pasaporte portugués para movilidad UE	Plan mediano/largo plazo
Aumentar perfil con cursos y portfolio	Plan de skill-building

3.5 Afirmaciones no permitidas

Afirmación	Motivo
"EU work authorized"	Falso actualmente.
"Portuguese citizen"	No confirmado actualmente.
"German work authorization through marriage"	No existe sin matrimonio/residencia/proceso legal.
"Certified in CG Cookie / Alive / Rebelway"	No hay certificados formales.
"Production-ready AI research platform" para ARA	Riesgo de sobreventa.
"Senior Technical Artist"	No corresponde al nivel formal actual.
"Full-stack developer" como identidad principal	Diluye el perfil si no hay evidencia robusta.
"Industry experience in Unity/Tech Art"	No hay empleo formal fuerte en ese campo.

1.0.6 4. Diagnóstico del informe anterior deep-research-report (8).md

4.1 Valor útil del informe anterior

El informe anterior sí aporta:

- una primera estructura general;
- identificación de TwinSight como activo principal;
- reconocimiento de ARA como activo secundario;
- enfoque en remoto desde Colombia;
- necesidad de CV, LinkedIn, GitHub y portfolio;
- preocupación por AI-assisted development;
- primeras ideas de scripts y plan 30/60/90;
- primera lista de roles y mercados;
- recordatorio de no sobreprometer EU work authorization.

4.2 Debilidades críticas

Área	Debilidad
Salarios	Rangos demasiado amplios, mezclan seniority, ubicación, employee, contractor y remoto.
Mercado	Usa etiquetas “Alta/Media” sin suficiente evidencia cuantitativa.
Empresas	Da ejemplos, no una lista usable de 100+ targets con links, roles, remote likelihood y approach.
Legal Colombia	Da una guía general, pero falta verificación con DIAN, seguridad social, IVA/exportación de servicios y misclassification.
UE/Portugal/Alemania	Algunas afirmaciones pueden estar incompletas o desactualizadas; requiere fuentes oficiales.
LinkedIn	No entrega todavía rewrite completo con 15 titulares y About versions.
CV	Da estructura, pero no una versión lista ni diferenciación robusta de 4 variantes.
GitHub	Recomienda limpieza, pero no audita repos reales.
Portfolio	Recomienda case studies, pero no define assets mínimos ni orden de producción con criterios de decisión.
Cursos	No integra el Excel de cursos.
Rebelway	No analiza ROI concreto.
Educación formal	No analiza especializaciones, maestrías o doctorados.
Riesgo operativo	No modela suficientemente runway, energía, dispersión, ni tradeoff portfolio vs aplicaciones.
Idiomas	Usa estrategia previa, pero no la cruza con corto plazo laboral.
Datos	Varias afirmaciones requieren fuentes actuales 2025–2026.

4.3 Problemas de tono y formato

- Mezcla español e inglés de manera irregular.
- Tiene frases de estrategia sin operacionalización.
- En algunos puntos suena más a recomendación genérica que a sistema de ejecución.
- No separa claramente:
 - hechos;
 - inferencias;
 - hipótesis;
 - acciones;
- investigación pendiente.
- No incluye criterios de validación para aprobar cada sección.

1.0.7 5. Contradicciones y conflictos entre documentos

5.1 Estado de tesis

Fuente	Dice	Estado actual
master_schedule.md	Tesis en fase de retopo/desarrollo, entrega Feb 6	Obsoleto
ticktick_schedule.md	Tareas de retopo, Unity import, testing, submission	Obsoleto

Fuente	Dice	Estado actual
plan_audit.md	Retopo no iniciado, baseline pending	Obsoleto parcialmente
Contexto actual del usuario	Tesis completa, pendiente sustentación	Fuente vigente
Pegado text(12).txt	Thesis finished, pending defense	Vigente

Regla: todo lo que suponga tesis en desarrollo debe reinterpretarse como contexto histórico, no como plan vigente.

5.2 Cronograma y runway

Fuente	Dice	Problema
master_schedule.md	Deadline May 2026, fases Dec–May	Cronograma viejo.
job_search_strategy.md	Job hunt empieza Mar 9, panic protocol May 1	Obsoleto en fechas, útil como lógica de contingencia.
alexander_woodcock_context.md	Runway May 2026, capital líquido bajo	Debe actualizarse si la situación financiera cambió.
Usuario actual	Quiere plan futuro real y prometedor	Requiere nuevo calendario desde estado actual.

Regla: el nuevo plan 30/60/90 debe construirse desde la fecha actual y estado tesis completa.

5.3 Posicionamiento

Fuente	Enfoque	Riesgo
job_search_strategy.md	Junior Technical Artist / QA Automation “Trojan Horse”	Puede ser táctico, pero quizá reduce upside si se vuelve eje principal.
portfolio_strategy.md	Optimization Doctor / Toolsmith / Math Wizard / Mobile Hero	Bueno para portfolio, pero puede sobrecargar.
deep-research-report Prompt original	Technical Artist / Real-Time 3D / XR / AI tooling Evitar diseñador multimedia genérico	Amplio, pero falta jerarquía. Correcto.

Regla: hay que definir roles primarios, secundarios y de contingencia. No todos son equivalentes.

5.4 Cursos y portfolio

Fuente	Dice	Riesgo
master_schedule.md	Hacer VFX reel, Human Mobile Hero, Toolsmith, shaders	Puede dispersar si se ejecuta todo antes de aplicar.
portfolio_strategy.md	4 pilares de portfolio	Buena estructura, pero debe priorizarse.
Courses 2025.xlsx	165 cursos no certificados; fuerte concentración en 3D/VFX/Houdini/Blender/UE	Alto potencial, pero riesgo de acumulación sin output.

Regla: curso solo cuenta si produce artefacto: demo, case study, shader, reel, tool, breakdown.

5.5 EU / idiomas

Fuente	Dice	Riesgo
estrategia_idiomas...	Inglés > alemán > portugués funcional > francés condicionado	Coherente, pero debe actualizarse y cruzarse con corto plazo.

Fuente	Dice	Riesgo
Prompt original	Portugués estratégicamente relevante por ciudadanía futura	Correcto, pero no debe convertirse en foco laboral inmediato.
Deep report	Menciona Portugal/Alemania con afirmaciones legales	Debe verificarse con fuentes oficiales.

Regla: el corto plazo es inglés + portfolio + remoto desde Colombia. Alemán/portugués son mediano plazo.

1.0.8 6. Inventario del Excel de cursos

6.1 Extracción preliminar

El archivo Courses_2025.xlsx contiene aproximadamente **165 entradas no vacías** de cursos o recursos.

Distribución preliminar por áreas:

Área	Conteo aproximado
3D Art	67
VFX	18
Game Development	13
Compositing	9
3D Art + VFX	7
Game Development + VFX	6
AI	6
Storytelling	6
Video Editing	4
Web Development	4
2D Art / Painting / Filmmaking / otros	menor

Herramientas más frecuentes:

Software / stack	Conteo aproximado
Blender	41
Houdini	30
Unreal Engine	27
Maya	21
ZBrush	13
Nuke	13
Photoshop	13
After Effects	9
Substance	8
Mari / Ziva / C4D / Marmoset / otros	menor

Academias más frecuentes:

Academia	Conteo aproximado
Rebelway	20
Coloso	11
The Gnomon Workshop	10
CG Circuit	8
Wingfox	6
Compositing Academy	5
Blender Bros	4
Flipped Normals	4
CG Cookie	3

6.2 Diagnóstico inicial

El Excel demuestra acceso a formación amplia, pero también riesgo de dispersión. El problema no es falta de cursos; el problema es **convertir cursos en evidencia laboral**.

6.3 Regla de uso de cursos

Cada curso debe clasificarse así:

Pregunta	Decisión
¿Produce evidencia visible en portfolio?	Priorizar
¿Fortalece TwinSight o el perfil Technical Artist?	Priorizar
¿Tiene certificado formal verificable?	Si no, no presentarlo como certificación
¿Es solo aspiracional o "nice to have"?	Posponer
¿Diluye hacia cine/VFX puro sin conexión con real-time?	Posponer o usar solo a largo plazo
¿Puede convertirse en demo corto en 2–4 semanas?	Potencial alto

6.4 Cursos que deben evaluarse en archivo posterior

El archivo 06_education_courses_rebelway_masters_phd.md debe clasificar:

- cursos para corto plazo laboral;
- cursos para portfolio Technical Artist;
- cursos para VFX/Houdini/Rebelway;
- cursos para Blender hard-surface / CAD-to-realtime;
- cursos para WebGL / interactive web;
- cursos para Unreal, si conviene o no;
- cursos de AI;
- cursos de compositing/video para demo reel;
- cursos que no deben mencionarse.

1.0.9 7. Arquitectura modular final de archivos

La investigación corregida se reconstruirá en estos archivos:

Orden	Archivo	Función
00	00_master_audit_blueprint.md	Auditoría maestra y mapa de reconstrucción.
01	01_source_of_truth_profile.md	Perfil real, hechos defendibles, límites de afirmación.
02	02_market_role_fit_and_positioning.md	Roles, posicionamiento, mercado y priorización.
03	03_salary_benchmark_and_remote_colombia.md	Salarios, remoto, contractor Colombia, modelos de contratación.
04	04_eu_portugal_germany_mobility_strategy.md	Movilidad UE, Portugal, Alemania, legalidad y wording.
05	05_language_strategy_and_roi.md	Idiomas, ROI, horizonte temporal, relación con mercados.
06	06_education_courses_rebelway_masters_phd.md	Cursos, Rebelway, certificaciones, especializaciones, maestrías y doctorados.
07	07_portfolio_strategy_and_project_architecture.md	Arquitectura de portfolio, proyectos, orden, assets mínimos.
08	08_twinsight_x500_case_study.md	Case study completo de TwinSight.
09	09_ara_ai_tools_and_secondary_projects.md	ARA, AI News, FitApp, Blender portrait, proyectos secundarios.
10	10_github_linkedin_cv_strategy.md	GitHub, LinkedIn, CV, keywords, perfiles públicos.
11	11_company_targets_job_boards_recruiters.md	Empresas, job boards, recruiters, comunidades.
12	12_application_system_and_tracker.md	Sistema de aplicaciones, tracker, métricas, cadencia.
13	13_interview_negotiation_and_offer_risk.md	Entrevistas, negociación, red flags, tests técnicos.

Orden	Archivo	Función
14	14_30_60_90_execution_plan.md	Plan operativo desde estado actual.
15	15_final_integrated_strategy.md	Síntesis final integrada. Solo se hace al final.

1.0.10 8. Reglas de reconstrucción

Cada archivo posterior debe incluir obligatoriamente:

1. objetivo de la sección;
2. fuentes usadas;
3. premisas;
4. auditoría de lo anterior;
5. qué estaba correcto;
6. qué estaba desactualizado;
7. qué estaba débil;
8. qué faltaba;
9. investigación adicional necesaria;
10. decisiones estratégicas;
11. versión corregida o estructura final;
12. checklist de aprobación;
13. recomendación de herramienta para la siguiente sección.

1.0.11 9. Criterios de calidad por archivo

Un archivo solo se considera aprobado si cumple:

Criterio	Requisito
Vigencia	No usa cronogramas obsoletos como si fueran actuales.
Verificabilidad	Distingue hechos de supuestos.
Coherencia	No contradice el source of truth.
Especificidad	Está conectado a TwinSight, ARA, cursos, GitHub, LinkedIn y metas.
Acción	Produce decisiones, no solo análisis.
Evidencia	Usa fuentes externas cuando corresponde.
No sobreventa	No exagera experiencia, legalidad, ciudadanía, certificados ni producción.
Modularidad	Puede unirse al documento final sin reescritura completa.
Priorización	Distingue corto, mediano y largo plazo.
Riesgo	Incluye riesgos y mitigaciones.

1.0.12 10. Secciones que requieren Deep Research, agente o ninguno

10.1 Tabla de herramientas por archivo

Archivo	Deep Research	Modo agente	Motivo
00_master_audit_blueprint.md	No	No	Auditoría interna estructural.
01_source_of_truth_profile.md	No	No	Depende de datos propios y archivos.
02_market_role_fit_and_positioning.md	Si	Opcional / útil	Requiere demanda real de roles, títulos, keywords, vacantes.

Archivo	Deep Research	Modo agente	Motivo
03_salary_benchmark_and_remote_c6Iombia.md	Sí	Sí	Requiere salarios actualizados, modelos contractor/EOR, Colombia.
04_eu_portugal_germany_mobility_5Strategy.md	Sí	Sí, con control	Requiere fuentes oficiales y extracción de requisitos.
05_language_strategy_and_roi.md	Sí parcial	Opcional	Requiere actualizar demanda por país/idioma, pero ya hay base.
06_education_courses_rebelway_ma5sters_phd.md	Sí	Sí	Requiere comparar cursos, certificaciones, programas, costos, ROI.
07_portfolio_strategy_and_project4No/partial.md	No/partial	Opcional bajo	Requiere más criterio interno que web. Agente puede revisar ejemplos.
08_twinsight_x500_case_study.md	No	No	Debe basarse en tesis y evidencia propia.
09_ara_ai_tools_and_secondary_p4No/partial.md	No/partial	No / opcional	Depende del estado real de repos.
10_github_linkedin_cv_strategy.m4No/partial.md	No/partial	Opcional	Redacción interna; agente solo para keywords o auditoría pública.
11_company_targets_job_boards_re6ruiters.md	Sí	Sí, muy recomendado	Alto volumen de navegación y extracción de links.
12_application_system_and_tracke4No.md	No	No	Diseño operativo.
13_interview_negotiation_and_off6Si/partial.md	Sí/partial	Sí	Tests, negociación y red flags pueden apoyarse en datos actuales.
14_30_60_90_execution_plan.md	No	No	Plan interno según restricciones.
15_final_integrated_strategy.md	No	No	Síntesis de todo lo validado.

10.2 Uso óptimo de los 40 usos de modo agente

No gastar agente en redacción. Usarlo en extracción repetitiva y navegación.

Bloque	Usos estimados	Valor
Empresas 100+ targets	8–12	Muy alto
Salarios por rol/mercado	5–7	Alto
Legal Colombia/EOR/contractor	3–5	Alto
UE/Portugal/Alemania	3–5	Alto
Maestrías/especializaciones/certificaciones	5–7	Alto
Job boards/comunidades/recruiters	3–5	Alto
Validación de LinkedIn/GitHub público	2–3	Medio
Portfolios de referencia	2–3	Medio
Reserva	5–8	Para verificación/fallos

Regla: cada uso de agente debe producir tabla con fuentes, no conclusiones finales.

1.0.13 11. Auditoría por futuras secciones

11.1 01_source_of_truth_profile.md

Objetivo: cerrar el perfil real y defendible.

Debe incluir:

- identidad profesional;
- tesis completa pendiente de sustentación;
- educación UNAD / UNAL;
- habilidades fuertes;
- habilidades secundarias;
- proyectos reales;
- cursos;
- restricciones legales;
- límites de afirmación;

- claims prohibidos;
- matriz de evidencia.

Faltantes actuales:

- estado real final de TwinSight;
- estado real de portfolio/GitHub/LinkedIn;
- disponibilidad laboral exacta post-sustentación;
- situación administrativa actual: RUT, facturación, cuenta internacional;
- situación financiera actual.

Herramientas recomendadas:

Deep Research: **No**.

Modo agente: **No**.

11.2 02_market_role_fit_and_positioning.md

Objetivo: validar roles y posicionamiento contra mercado real.

Debe analizar:

- Unity Technical Artist;
- Real-Time 3D Developer;
- Unity WebGL Developer;
- Interactive 3D Developer;
- Technical Visualization Developer;
- CAD-to-Realtime Technical Artist;
- XR / Simulation Developer;
- Digital Twin Visualization Developer;
- Game Technical Artist;
- Tools Developer;
- Python AI Tooling Developer;
- Full-stack como contingencia.

Faltantes actuales:

- conteo de vacantes reales por título;
- requisitos frecuentes;
- remoto vs local;
- seniority requerido;
- portfolio expectations;
- ATS keywords;
- probabilidad real de entrada;
- roles con mayor ROI para TwinSight.

Herramientas recomendadas:

Deep Research: **Sí**.

Modo agente: **Opcional / útil** para extraer vacantes reales.

11.3 03_salary_benchmark_and_remote_colombia.md

Objetivo: rehacer salarios con segmentación realista y analizar contractor desde Colombia.

Debe separar:

- Colombia local;
- Colombia contractor internacional;

- LATAM nearshore;
- US remote contractor;
- US employee;
- Canada;
- UK;
- Germany;
- Netherlands;
- Spain;
- Portugal;
- global remote;
- freelance;
- EOR employee.

Faltantes actuales:

- separar junior real vs junior fuerte vs mid;
- brutos vs netos;
- employee vs contractor;
- mercado game vs enterprise;
- tarifas horarias vs mensual;
- deducciones y seguridad social en Colombia;
- efecto de impuestos/cambio COP/USD;
- evaluación de USD 1.5k / 3k / 6k por horizonte.

Herramientas recomendadas:

Deep Research: **Sí.**

Modo agente: **Sí.**

11.4 04_eu_portugal_germany_mobility_strategy.md

Objetivo: separar claramente movilidad futura de derecho laboral actual.

Debe analizar:

- ciudadanía portuguesa esperada;
- qué habilita ser ciudadano UE;
- diferencia entre ciudadanía, residencia fiscal y contratación remota;
- Alemania por matrimonio / reunificación;
- EU Blue Card;
- visas digitales;
- España / Portugal / Alemania / Netherlands / Irlanda;
- wording permitido en CV/LinkedIn/recruiter conversations.

Faltantes actuales:

- fuentes oficiales actualizadas;
- estado legal específico de ruta sefardí portuguesa;
- implicaciones fiscales de vivir en Colombia con pasaporte UE;
- riesgo de sobreprometer;
- escenarios de aplicación: remoto desde Colombia vs mudanza UE.

Herramientas recomendadas:

Deep Research: **Sí, obligatorio.**

Modo agente: **Sí, con control estricto.**

11.5 05_language_strategy_and_roi.md

Objetivo: actualizar estrategia de idiomas con ROI laboral.

Debe integrar:

- inglés C1 → C2 funcional;
- alemán B1/B2 para DACH;
- portugués A2/B1 funcional;
- francés condicionado;
- holandés solo si Netherlands;
- idiomas asiáticos no prioritarios salvo objetivo personal.

Faltantes actuales:

- actualización 2026 de mercados por idioma;
- relación idioma ↔ roles concretos;
- prioridad en próximos 3–6 meses;
- costo de oportunidad;
- plan semanal realista;
- certificados posibles de inglés/alemán/portugués.

Herramientas recomendadas:

Deep Research: **Sí parcial.**

Modo agente: **Opcional bajo.**

11.6 06_education_courses_rebelway_masters_phd.md

Objetivo: convertir el Excel de cursos en roadmap educativo con ROI.

Debe incluir:

- matriz de cursos adquiridos;
- Rebelway a corto/mediano/largo plazo;
- cursos que conviene terminar;
- cursos que conviene ignorar;
- cursos que deben generar demo;
- certificaciones con valor real;
- Unity certification;
- Coursera/Google Data Analytics si aplica;
- especializaciones;
- maestrías;
- doctorados;
- programas UE/LatAm/online;
- costo, duración, idioma, becas, ROI.

Faltantes actuales:

- no existe matriz por ROI;
- no hay priorización por output;
- no se distingue curso certificado vs no certificado;
- no se evalúa formal education;
- no se cruza con mercado laboral.

Herramientas recomendadas:

Deep Research: **Sí.**

Modo agente: **Sí.**

11.7 07_portfolio_strategy_and_project_architecture.md

Objetivo: definir portfolio como sistema de evidencia laboral.

Debe decidir:

- cuántos proyectos mostrar;
- orden de proyectos;
- TwinSight como flagship;
- ARA secundario;
- Blender portrait como pipeline técnico;
- AI News / FitApp como secundarios o privados;
- GitHub Pages vs sitio propio;
- PDF portfolio;
- demo reel;
- screenshot list;
- evitar overbuilding.

Faltantes actuales:

- criterios de corte;
- qué se necesita antes de aplicar;
- qué puede esperar;
- assets mínimos;
- timeline real;
- matriz esfuerzo/impacto;
- definición de MVP.

Herramientas recomendadas:

Deep Research: **No**.

Modo agente: **Opcional bajo** para revisar ejemplos, no necesario.

11.8 08_twinsight_x500_case_study.md

Objetivo: convertir TwinSight en caso de estudio laboral robusto.

Debe incluir:

- problema;
- contexto académico;
- rol personal;
- pipeline CAD → Blender → Unity WebGL;
- features;
- métricas;
- validación SUS/NASA-TLX;
- limitaciones;
- AI-assisted development;
- assets manufacturer/CAD;
- screenshots;
- demo script;
- CV bullets;
- interview pitch.

Faltantes actuales:

- versión final del proyecto;
- capturas reales;
- demo real;

- README final;
- visual evidence;
- limitaciones honestas;
- texto para portfolio;
- uso de IA explicado sin sobreventa.

Herramientas recomendadas:

Deep Research: **No**.

Modo agente: **No**.

11.9 09_ara_ai_tools_and_secondary_projects.md

Objetivo: auditar proyectos secundarios y decidir uso estratégico.

Debe cubrir:

- ARA Framework;
- ai-news-aggregator;
- FitApp-Free;
- Blender portrait;
- deus-ex-machina;
- skills-introduction-to-github;
- otros repos privados.

Faltantes actuales:

- estado funcional real;
- costo de publicarlos;
- riesgo de dilución;
- fit por rol;
- repo/public/private/archive decision matrix;
- claims permitidos;
- README / demo requirements.

Herramientas recomendadas:

Deep Research: **No / parcial**.

Modo agente: **No**, salvo auditoría pública de GitHub.

11.10 10_github_linkedin_cv_strategy.md

Objetivo: consolidar presencia pública y materiales de aplicación.

Debe incluir:

- auditoría GitHub real;
- LinkedIn full rewrite;
- 15 headline options;
- About 3 versions;
- experience rewrite;
- education rewrite;
- projects;
- featured section;
- skills;
- CV base;
- 4 variantes CV;
- keyword bank;

- AI-assisted wording;
- remote Colombia wording;
- EU future wording.

Faltantes actuales:

- no hay rewrite completo;
- no hay CV final;
- no hay auditoría pública real;
- no hay decisiones de repos;
- no hay checklist de edición.

Herramientas recomendadas:

Deep Research: **No / parcial.**

Modo agente: **Opcional** para revisar perfiles/repos públicos y keywords.

11.11 11_company_targets_job_boards_recruiters.md

Objetivo: crear base operativa real de oportunidades.

Debe incluir:

- 100+ empresas;
- país;
- sector;
- roles objetivo;
- por qué encaja;
- remote likelihood;
- contractor likelihood;
- idioma;
- salario potencial;
- careers link;
- LinkedIn;
- approach;
- priority/probability/compensation/portfolio-fit score;
- job boards;
- comunidades;
- recruiters;
- newsletters;
- search queries.

Faltantes actuales:

- la lista anterior no es operativa;
- faltan links;
- faltan criterios;
- faltan recruiters;
- faltan roles activos;
- faltan filtros;
- falta probabilidad real.

Herramientas recomendadas:

Deep Research: **Sí.**

Modo agente: **Sí, muy recomendado.**

11.12 12_application_system_and_tracker.md

Objetivo: crear sistema ejecutable de aplicaciones.

Debe incluir:

- rutina semanal;
- cadencia de aplicaciones;
- A/B/C priority;
- reglas de customización;
- outreach;
- follow-up;
- tracker columns;
- KPIs;
- revisión semanal;
- criterios para pivotar;
- criterios para rechazar ofertas;
- contingencia BPO/web/freelance si aplica.

Faltantes actuales:

- métricas más concretas;
- relación con energía/tiempo;
- reglas de decisión;
- plantilla lista para usar;
- balance portfolio/applications.

Herramientas recomendadas:

Deep Research: **No**.

Modo agente: **No**.

11.13 13_interview_negotiation_and_offer_risk.md

Objetivo: preparar defensa del perfil, negociación y filtros de riesgo.

Debe incluir:

- preguntas técnicas por rol;
- technical tests;
- pitch TwinSight 2/5/15 min;
- cómo explicar baja experiencia formal;
- cómo explicar IA;
- salary expectation scripts;
- contractor negotiation;
- red flags;
- offer checklist;
- payment terms;
- scope creep;
- IP/NDA;
- trial tasks.

Faltantes actuales:

- listas técnicas específicas por rol;
- scripts de negociación más sólidos;
- red flags internacionales;
- criterios de aceptar/rechazar;
- cálculo neto contractor.

Herramientas recomendadas:

Deep Research: **Sí parcial.**

Modo agente: **Sí** para recopilar patterns y datos.

11.14 14_30_60_90_execution_plan.md

Objetivo: plan realista desde tesis completa, no desde tesis en desarrollo.

Debe incluir:

- fecha de inicio real;
- defensa pendiente;
- portfolio MVP;
- CV/LinkedIn/GitHub;
- aplicación inicial;
- cursos mínimos;
- no sobrecarga;
- milestones;
- weekly metrics;
- contingencias;
- runway financiero actualizado;
- plan de revisión.

Faltantes actuales:

- fechas actuales;
- estado financiero actual;
- disponibilidad real;
- prioridad post-sustentación;
- secuencia exacta;
- definición de “suficiente para aplicar”.

Herramientas recomendadas:

Deep Research: **No.**

Modo agente: **No.**

11.15 15_final_integrated_strategy.md

Objetivo: integrar todo en una estrategia final coherente.

Debe incluir:

- resumen ejecutivo;
- estrategia recomendada;
- rutas alternativas;
- plan completo;
- anexos;
- checklists;
- decisiones finales;
- secuencia de ejecución.

Regla: no debe escribirse hasta que los archivos 01–14 estén aprobados.

Herramientas recomendadas:

Deep Research: **No.**

Modo agente: **No.**

1.0.14 12. Registro de riesgos estructurales

12.1 Riesgo de sobreespecialización incorrecta

Problema: apuntar solo a Game Technical Artist puede reducir probabilidad por competencia alta y menor entrada junior.

Mitigación: posicionar como Technical Artist / Real-Time 3D / Technical Visualization con ruta game como secundaria.

12.2 Riesgo de dispersión

Problema: cursos, VFX, Houdini, Human Portrait, ARA, FitApp, AI News, GitHub, portfolio, idiomas y job hunt compiten por atención.

Mitigación: matriz esfuerzo/impacto. Solo activos que soporten empleo en 3–6 meses.

12.3 Riesgo de sobreventa

Problema: presentar proyectos incompletos como producción o AI-assisted code como expert-level development.

Mitigación: wording honesto: prototype, research tool, AI-assisted, implemented/integrated/debugged by author.

12.4 Riesgo salarial

Problema: USD 6k/mes puede ser irrealista a corto plazo si se aplica como junior sin experiencia formal.

Mitigación: metas escalonadas:

- USD 1.5k: survival / mínimo inicial;
- USD 3k: contractor/nearshore viable;
- USD 6k: mid-level o nicho internacional a 12–24 meses.

12.5 Riesgo legal

Problema: asumir EU work authorization o manejo tributario incorrecto en Colombia.

Mitigación: fuentes oficiales, contador, wording conservador, contrato claro.

12.6 Riesgo de portfolio incompleto

Problema: aplicar sin caso de estudio visual puede reducir respuestas.

Mitigación: MVP mínimo: TwinSight case study + demo + screenshots + CV + LinkedIn.

12.7 Riesgo de análisis infinito

Problema: seguir investigando y no aplicar.

Mitigación: cada archivo debe producir decisiones y límites temporales.

1.0.15 13. Matriz esfuerzo/impacto preliminar

Acción	Impacto laboral	Esfuerzo	Prioridad preliminar
TwinSight case study	Muy alto	Medio	Máxima
Demo video TwinSight	Muy alto	Medio	Máxima
LinkedIn rewrite	Alto	Bajo	Alta
CV base + 2 variantes	Alto	Medio	Alta

Acción	Impacto laboral	Esfuerzo	Prioridad preliminar
GitHub cleanup TwinSight	Alto	Medio	Alta
Portfolio MVP GitHub Pages	Alto	Medio	Alta
ARA README/demo mínimo	Medio/Alto	Medio	Media
ai-news public release	Medio	Alto	Baja/Media
FitApp release	Bajo/Medio	Alto	Baja
Blender portrait breakdown	Medio	Medio	Media
Rebelway VFX reel	Medio	Alto	Media/largo plazo
Alemán intensivo ahora	Bajo corto plazo	Alto	Baja corto plazo
Inglés interview practice	Alto	Bajo/Medio	Alta
Lista 100 empresas	Alto	Alto	Alta, con agente
Legal contractor Colombia	Alto	Medio	Alta
Maestrías/doctorados	Bajo corto plazo	Alto	Largo plazo

1.0.16 14. Definición de “MVP laboral”

Antes de aplicar de forma intensiva, el mínimo aceptable debería ser:

1. CV base en inglés;
2. LinkedIn corregido;
3. GitHub sin ruido visible grave;
4. TwinSight README recruiter-friendly;
5. TwinSight case study básico;
6. demo video 60–90 segundos;
7. 8–12 screenshots de calidad;
8. script de explicación de IA;
9. tracker de aplicaciones;
10. lista inicial de 30 empresas;
11. setup administrativo mínimo definido, aunque no totalmente ejecutado.

No esperar a tener:

- portfolio perfecto;
- 100 empresas completas;
- ARA totalmente pulido;
- FitApp público;
- Rebelway reel;
- alemán B1;
- maestría definida.

1.0.17 15. Orden recomendado de trabajo

Etapla 1 — Sin Deep Research / sin agente

1. 00_master_audit_blueprint.md
2. 01_source_of_truth_profile.md

Etapla 2 — Deep Research + agente selectivo

1. 02_market_role_fit_and_positioning.md
2. 03_salary_benchmark_and_remote_colombia.md
3. 04_eu_portugal_germany_mobility_strategy.md
4. 06_education_courses_rebelway_masters_phd.md
5. 11_company_targets_job_boards_recruiters.md

Etapa 3 — Síntesis interna

1. 05_language_strategy_and_roi.md
2. 07_portfolio_strategy_and_project_architecture.md
3. 08_twinsight_x500_case_study.md
4. 09_ara_ai_tools_and_secondary_projects.md
5. 10_github_linkedin_cv_strategy.md
6. 12_application_system_and_tracker.md
7. 13_interview_negotiation_and_offer_risk.md
8. 14_30_60_90_execution_plan.md

Etapa 4 — Integración final

1. 15_final_integrated_strategy.md

1.0.18 16. Checklist de aprobación del presente archivo

Criterio	Estado
Corrige la premisa de tesis en desarrollo	Sí
No produce estrategia final todavía	Sí
Divide la reconstrucción en archivos modulares	Sí
Indica qué requiere Deep Research/agente	Sí
Incluye cursos y Rebelway como sección futura	Sí
Incluye maestrías/doctorados como sección futura	Sí
Incluye contradicciones entre documentos	Sí
Define claims permitidos y prohibidos	Sí
Define criterios de calidad	Sí
Define siguiente sección	Sí

1.0.19 17. Recomendación para la siguiente sección

Siguiente archivo: 01_source_of_truth_profile.md

Deep Research: No activar.

Modo agente: No activar.

Motivo: la siguiente sección debe consolidar hechos propios y límites de afirmación. No depende de datos externos ni navegación web. Requiere precisión interna: perfil, tesis, proyectos, habilidades defendibles, habilidades aspiracionales, cursos, legalidad no afirmable y narrativa base.

1.0.20 18. Preguntas de confirmación antes de 01_source_of_truth_profile.md

Antes de producir el archivo 01, conviene confirmar o dejar como “pendiente”:

1. ¿Fecha estimada de sustentación?
2. ¿La tesis ya tiene documento final aprobado por asesor o solo está lista para defensa?
3. ¿Existe demo WebGL funcional pública?
4. ¿TwinSight tiene capturas finales disponibles?
5. ¿Las métricas finales son exactamente: 6.5M → 95,617 tris, SUS 91.88, NASA-TLX 8.69 vs 19.89, 12 participantes, 96 registros?
6. ¿El repo principal debe limpiarse o puede publicarse como está?
7. ¿Hay RUT/facturación electrónica ya habilitados?

8. ¿Situación financiera/runway actual?
9. ¿El pasaporte portugués sigue estimado en ~2 años o cambió la previsión?
10. ¿Qué cursos de Rebelway están realmente disponibles para ti ahora?

Si no se responden, el archivo 01 puede marcar esos puntos como “pendiente de confirmación” sin bloquear.

Fin de 00_master_audit_blueprint.md.

Chapter 2

01 — Source of Truth Profile

Fuente: 01_source_of_truth_profile.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 929

Ultima modificacion: 2026-06-11 11:18

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Perfil: Alexander Woodcock Salomón **Fecha de corte operativa:** 2026-06-11 **Estado del documento:** Fuente de verdad operativa. No es CV, LinkedIn, portfolio ni estrategia final. **Modo usado para esta sección:** análisis interno de archivos + contexto actualizado del usuario. **No requiere Deep Research ni modo agente.** **Regla crítica:** todo documento posterior debe respetar esta fuente de verdad. Si una sección posterior contradice este archivo, debe justificarlo c

Contenido completo depurado

2.0.1 Perfil maestro verificable para reconstrucción de estrategia laboral internacional

Perfil: Alexander Woodcock Salomón

Fecha de corte operativa: 2026-06-11

Estado del documento: Fuente de verdad operativa. No es CV, LinkedIn, portfolio ni estrategia final.

Modo usado para esta sección: análisis interno de archivos + contexto actualizado del usuario. **No requiere Deep Research ni modo agente.**

Regla crítica: todo documento posterior debe respetar esta fuente de verdad. Si una sección posterior contradice este archivo, debe justificarlo con nueva evidencia o corregirse.

2.0.2 0. Propósito

Este archivo fija los hechos, límites, matices y claims defendibles sobre Alexander Woodcock Salomón antes de reconstruir el resto de la investigación laboral.

Su función es evitar que los documentos posteriores mezclen:

- información vigente con cronogramas antiguos;
- hechos demostrables con aspiraciones;
- habilidades defendibles con habilidades en desarrollo;
- rutas migratorias futuras con derechos laborales actuales;

- cursos adquiridos con certificaciones formales;
- proyectos funcionales con proyectos incompletos o no auditados;
- estrategia principal con rutas secundarias o de contingencia.

Este documento **no decide todavía la estrategia laboral final**. Define el perímetro de verdad desde el cual se redactarán las secciones posteriores.

2.0.3 1. Fuentes usadas y estatus

Fuente	Tipo	Uso en este archivo	Estatus
Pegado text(12).txt	Prompt original largo	Perfil, objetivos, proyectos, restricciones, entregables esperados	Vigente como fuente primaria de intención
deep-research-report (8).md	Informe anterior	Material a auditar: mercado, CV, LinkedIn, GitHub, portfolio, scripts, movilidad, salarios	Útil pero insuficiente; no debe tratarse como verdad cerrada
alexander_woodcock_context.md	Contexto maestro antiguo	Identidad, restricciones, hardware, herramientas, runway antiguo, stack, perfil psicológico, planes	Parcialmente histórico; debe revalidarse
plan_audit.md	Auditoría antigua de tesis/portfolio	Criterios de optimización, documentación de profiling, riesgo operativo	Parcialmente obsoleto; tesis ya no está en desarrollo
master_schedule.md	Cronograma antiguo	Evidencia de prioridades, proyectos y secuenciación previa	Obsoleto como calendario
ticktick_schedule.md	Agenda granular antigua	Evidencia de tareas planeadas, cursos y prioridades técnicas	Obsoleto como calendario
job_search_strategy.md	Estrategia antigua	Hipótesis "Trojan Horse": Junior Technical Artist / QA Automation / Tools	Hipótesis útil; requiere revalidación
portfolio_strategy.md	Estrategia antigua de portfolio	Optimization Doctor, Toolsmith, Math Wizard, Mobile Hero, Human Portrait	Útil como arquitectura conceptual; requiere priorización
estrategia_idiomas_mercado_labor_investigacion_idiomas/m	Investigación de idiomas/m	Base para estrategia de inglés, alemán, portugués, francés, UE	Útil; requiere actualización en secciones externas
Courses 2025.xlsx	Inventario de cursos adquiridos	Insumo para matriz de formación/ROI	Debe analizarse en la sección 06; no son certificaciones
00_master_audit_blueprint.md	Auditoría maestra ya generada	Marco estructural de reconstrucción	Vigente

2.0.4 2. Estado maestro congelado

2.1 Identidad

Campo	Valor operativo
Nombre completo	Alexander Woodcock Salomón
Ubicación	Colombia
Ciudad/región reportada	Pasto, Nariño, Colombia
Zona horaria	UTC-5 / Colombia
Idioma nativo	Español
Inglés	C1 autoevaluado; funcional para entrevistas técnicas si se sostiene en práctica
Alemán	Ninguno / no profesional
Portugués	Estratégicamente relevante, no competencia profesional confirmada
Perfil actual deseado	Technical Artist / Real-Time 3D Developer / Unity WebGL / CAD-to-realtime / Tools

2.2 Estado académico

Elemento	Valor operativo
Carrera actual	Ingeniería Multimedia — UNAD
Estado de tesis	Tesis completa, pendiente de sustentación
Proyecto de tesis	TwinSight X500 / WebGL-Thesis-Proposal
Carrera previa	Ingeniería Electrónica — Universidad Nacional de Colombia, UNAL
Estado UNAL	Aproximadamente 8 semestres / cerca de 80% del currículo, no título final confirmado
Forma internacional recomendada para UNAL	Completed advanced coursework in Electronic Engineering – Universidad Nacional de Colombia
Forma que debe evitarse	Electronic Engineer, BSc Electronic Engineering, degree completed, si el título no fue obtenido

2.3 Estado laboral

Elemento	Valor operativo
Experiencia formal en Technical Art / Unity / Real-Time 3D	Baja / no fuerte todavía
Experiencia freelance	Si, especialmente web/proyectos simples, pero no debe venderse como experiencia industrial senior
Experiencia no target	Call center / YMCA camps / experiencia en inglés y comunicación
Principal evidencia profesional actual	Proyectos: TwinSight X500, ARA Framework, Blender portrait, proyectos web/AI secundarios
Nivel profesional defendible	Entry-level fuerte / junior fuerte / early-career technical artist-real-time developer, según rol
Nivel que no debe afirmarse	Senior Technical Artist, Lead, Production Engineer, EU-based worker, industry-proven specialist

2.4 Estado legal y movilidad

Elemento	Valor operativo
Nacionalidad actual relevante	Colombiana, salvo evidencia adicional no confirmada aquí
Autorización laboral UE actual	No confirmada. Operativamente: no existe
Ciudadanía/pasaporte portugués	Expectativa futura / proceso estratégico, no derecho actual
Ruta Alemania por pareja/matrimonio	Escenario posible futuro, condicionado a hechos no actuales
Prioridad de corto plazo	Remoto desde Colombia
Modelo laboral más realista inicial	Contractor/B2B, freelance internacional, remote-first, EOR si empresa lo ofrece
Claim prohibido	Authorized to work in the EU hasta que sea legalmente cierto

2.5 Estado financiero y operativo

Elemento	Estado
Runway financiero	Datos antiguos indican presión y bajo capital; debe actualizarse
Deadline financiero antiguo	Archivos antiguos hablan de May 2026; tratar como histórico hasta reconfirmación
OPEX	Vive con padres / bajo costo relativo según contexto anterior; debe reconfirmarse
Urgencia laboral	Alta pero no debe forzar mala estrategia ni sobreventa
Setup contractor	No confirmado: RUT, facturación electrónica, contador, Wise/Payoneer/Deel/Remote, ARL/EPS/pensión

Regla: ninguna sección posterior debe hacer cálculos de supervivencia, runway o cadencia de aplicaciones sin reconfirmar datos financieros actuales.

2.0.5 3. Corrección de premisas desactualizadas

Los archivos antiguos son útiles como contexto, pero no como estado actual. La siguiente tabla fija cómo deben interpretarse.

Tema en archivos antiguos	Estado anterior documentado	Estado operativo actual	Uso permitido
Retopología del dron UVs / baking / Unity import	Pendiente / no iniciada Planeados	Ya no debe asumirse pendiente No asumir pendientes sin nueva confirmación	Solo como historial de proceso Verificar si quedan assets o capturas por generar
Testing SUS / NASA-TLX	Planeado	Métricas ya reportadas en prompt	Confirmar versión final de datos antes de publicar
Entrega tesis Feb 6 / fases Dec-May	Cronograma viejo	Obsoleto	No usar como plan actual
Job hunt Mar 9 / Panic Protocol May 1	Plan viejo	Obsoleto en fechas	Puede inspirar contingencia, no agenda vigente
Portfolio Bridge Feb-Mar	Plan viejo	Obsoleto	Reconvertir en backlog moderno
Cursos paralelos durante tesis	Plan viejo	Ya no aplica como restricción de tesis	Mantener como inventario de aprendizaje
Shaders creados / VFX reel planeado	Contexto útil	Requiere verificar estado real	No afirmar como portfolio publicado

Regla operativa: toda mención a “tesis en desarrollo” debe reemplazarse por “tesis completa, pendiente de sustentación”, salvo cuando se describa retrospectivamente el proceso técnico.

2.0.6 4. Matriz de verdad: hechos, pendientes, supuestos, aspiraciones y no afirmables

4.1 Hechos defendibles actualmente

Hecho	Nivel de confianza	Uso permitido
Alexander estudia Ingeniería Multimedia en UNAD	Alto	CV, LinkedIn, portfolio
La tesis está completa y pendiente de sustentación	Alto según actualización del usuario	CV con wording prudente: Thesis completed; defense pending
TwinSight X500 / WebGL-Thesis-Proposal es el proyecto principal	Alto	Principal caso de estudio
El proyecto usa Unity WebGL / C# / Blender / CAD-to-realtime	Alto según prompt/contexto	Posicionamiento, CV, LinkedIn
El proyecto transforma documentación/CAD de dron en visualización técnica 3D interactiva	Alto	Portfolio, entrevistas
Se reportan métricas: 6.5M+ triángulos a 95,617, SUS 91.88, NASA-TLX 8.69 vs 19.89, 12 participantes, 96 registros	Medio-Alto; requiere confirmar versión final antes de publicación	Bullets cuantificados si son finales y verificables
ARA Framework existe como prototipo de automatización de investigación con LLMs	Medio-Alto	Proyecto secundario, AI tooling
Hay cursos adquiridos sin certificación formal	Alto	Desarrollo profesional, no certificación Professional working proficiency si se sostiene oralmente; evitar sobreprometer si no se ha validado
Inglés C1 es autoevaluado	Medio	
La prioridad laboral es remoto desde Colombia	Alto	Estrategia principal

4.2 Datos pendientes de verificar

Dato	Por qué importa	Dónde se verificará
URL pública funcional de TwinSight	Necesaria para portfolio y aplicaciones	Sección 08 / revisión GitHub-demo
Estado actual del repo WebGL-Thesis-Proposal	Determina limpieza, README y riesgo de assets	Sección 10 / GitHub audit
Estado de ArtStation	Define si sirve para aplicar o debe ocultarse temporalmente	Sección 07 / 10
Estado actual de LinkedIn real	El informe anterior se basa en capturas/contexto parcial	Sección 10
Estado de ARA Framework	Decide si publicarlo o mantenerlo privado	Sección 09

Dato	Por qué importa	Dónde se verificará
Estado de ai-news-aggregator	Decide si vale pulirlo o descartarlo	Sección 09
Estado de FitApp-Free	Probablemente no es prioritario; confirmar	Sección 09
Estado del retrato Blender	Determina si puede publicarse como breakdown técnico	Sección 07 / 09
Cursos completados vs solo adquiridos	Evita presentar acumulación como competencia real	Sección 06
Acceso actual a Rebelway	Determina ROI real de cursos de VFX/Houdini	Sección 06
Setup legal Colombia para contractor	Necesario antes de aceptar pagos internacionales	Sección 03 / 04
Estado real de la ruta portuguesa	No debe inferirse	Sección 04
Runway financiero actual	Define agresividad del plan	Sección 14
Fecha real de sustentación	Define cadencia de portfolio y aplicaciones	Sección 14

4.3 Supuestos estratégicos válidos pero no cerrados

Supuesto	Estado	Riesgo
El mejor eje inicial es Technical Artist / Real-Time 3D Developer	Plausible	Debe validarse con mercado y roles reales
Technical Visualization / Digital Twin / Simulation puede tener mejor ROI que game tech art puro	Plausible	Requiere datos de salarios y demanda
USD 1,500/mes es alcanzable rápido	Plausible	Depende de mercado, portfolio y modalidad contractor
USD 3,000/mes en 3–6 meses es posible pero no garantizado	Hipótesis	Debe validarse con benchmarks LATAM/global remote
USD 6,000/mes en 12–24 meses es alcanzable si hay portafolio fuerte y experiencia acumulada	Hipótesis ambiciosa	Requiere ruta clara a mid-level o nicho mejor pagado
ARA puede abrir ruta secundaria AI tooling	Plausible	Puede diluir el perfil si se sobrepresenta
Rebelway/Houdini puede mejorar tech art/VFX a mediano plazo	Plausible	Riesgo de dispersión si se prioriza antes del portfolio principal
Alemán es inversión estratégica por DACH/UE	Plausible	Bajo ROI en 0–6 meses; requiere horizonte largo

4.4 Aspiraciones legítimas

Aspiración	Cómo debe formularse
Trabajar remoto desde Colombia para empresa extranjera	Objetivo principal de corto plazo
Llegar a USD 3,000/mes	Meta realista condicionada, no promesa
Llegar a USD 6,000/mes	Meta de 12–24 meses, requiere evidencia y mejor mercado
Eventual movilidad UE	Ruta estratégica futura, no derecho actual
Aplicar a roles de videojuegos	Interés personal, pero no necesariamente mejor ROI inicial
Aplicar a simulación, digital twin, technical visualization	Ruta racional si paga mejor y encaja con TwinSight
Aprender alemán/portugués	Inversión mediano-largo plazo, no prioridad por encima de inglés+portfolio en corto plazo

4.5 Afirmaciones prohibidas o de alto riesgo

Afirmación	Motivo
EU work authorized	No existe autorización confirmada actualmente
Portuguese citizen	No está confirmado actualmente
German resident	No está confirmado actualmente
Certified by Rebelway / CG Cookie / Alive	Cursos sin certificado formal no son certificaciones
Senior Technical Artist	No hay experiencia industrial formal suficiente
Production-ready AI research platform para ARA	Riesgo de sobreventa
Full-stack developer como identidad principal	Diluye el posicionamiento 3D/Unity si no hay evidencia fuerte
Industry-proven Unity developer	No hay experiencia formal industrial fuerte aún
Built everything manually without AI	Falso si hubo asistencia IA
AI built the project	También falso si hubo dirección, integración, depuración y validación propia

2.0.7 5. Posicionamiento operativo provisional

5.1 Posicionamiento base

Technical Artist & Real-Time 3D Developer especializado en Unity, WebGL, C#, Blender, CAD-to-realtime optimization, interactive technical visualization y Python/AI-assisted production tooling.

Este posicionamiento es provisional pero fuerte porque conecta:

- Unity/C#;
- WebGL/Web/mobile deployment;
- Blender/asset optimization;
- CAD-to-realtime;
- visualización técnica;
- experiencia de tesis;
- base de ingeniería electrónica;
- tooling/automation con Python e IA.

5.2 Qué NO debe ser el posicionamiento principal

Posicionamiento	Problema
Multimedia Designer	Demasiado genérico y mal pagado
3D Generalist	Compite contra artistas con reels más fuertes; no aprovecha ingeniería
Game Artist	Falta portfolio artístico específico de videojuegos
Full-stack Developer	Ruta saturada, desvía de TwinSight
AI Engineer	ARA no basta para competir contra perfiles ML/software fuertes
VFX Houdini Artist	Cursos disponibles, pero no evidencia profesional actual suficiente
UX/UI Designer	TwinSight tiene UI, pero no es la identidad principal

5.3 Narrativa primaria sugerida

Perfil de ingeniería multimedia con base previa en electrónica, especializado en convertir modelos CAD y sistemas técnicos en experiencias 3D interactivas en tiempo real usando Unity, WebGL, C#, Blender y optimización de assets.

5.4 Narrativa secundaria permitida

Además del eje 3D/Unity, desarrolla herramientas Python e IA para automatizar investigación, documentación y flujos de producción.

5.5 Jerarquía de roles provisional

Nivel	Roles
Primarios	Unity Technical Artist, Real-Time 3D Developer, Interactive 3D Developer, Unity WebGL Developer, Technical Visualization Developer
Secundarios fuertes	CAD-to-Realtime Technical Artist, Digital Twin Visualization Developer, XR/Simulation Developer, Tools Developer for art/production pipelines
Secundarios tácticos	Python AI Tools Developer, Research Automation Developer, QA Automation technical entry route
Contingencia	Full-stack junior, frontend/React, bilingual support/BPO temporal, freelance Unity bug fixing
No prioritarios ahora	Pure 3D Artist, pure VFX Houdini Artist, pure game artist, pure data analyst

Nota: esta jerarquía debe validarse en 02_market_role_fit_and_positioning.md con investigación de mercado actual.

2.0.8 6. Inventario maestro de proyectos

6.1 TwinSight X500 / WebGL-Thesis-Proposal

Campo	Valor operativo
Prioridad	Máxima
Tipo	Tesis / flagship portfolio project
Dominio	Unity WebGL, visualización técnica, CAD-to-realtime, technical UI
Tema	Visualizador web 3D interactivo para inspección/ensamblaje de dron Holybro X500 V2
Claim central	"Seeing parts is not enough. Users need to understand spatial relationships."
Stack reportado	Unity, C#, WebGL, URP, UI Toolkit, Shader Graph, Blender, CAD/STEP pipeline
Funciones reportadas	selección de componentes, exploded view, clipping/cross-section, paneles técnicos, modos visuales, mobile/web performance
Métricas reportadas	6.5M+ triángulos → 95,617; SUS 91.88; NASA-TLX 8.69 vs 19.89; 12 participantes; 96 records
Contribución personal	programación/integración Unity, interacción, UI, optimización, documentación, evaluación, dirección técnica con asistencia IA
Riesgos	assets de fabricante/CAD, estado público del repo, demo funcional, explicación de IA, métricas finales verificables

Uso permitido

- Proyecto principal de CV.
- Primer caso de estudio del portfolio.
- Primer Featured item en LinkedIn.
- Primer pinned repo si está limpio.
- Base para roles Unity WebGL, Interactive 3D, Technical Visualization, CAD-to-realtime, Digital Twin, Simulation.

Claims defendibles si se verifican

- Developed a Unity WebGL interactive 3D technical visualization prototype for drone assembly inspection.
- Optimized CAD-derived assets for real-time web deployment.
- Implemented interactive inspection features including component selection, exploded view and cross-section/clipping.
- Evaluated usability and perceived workload using SUS, NASA-TLX Raw and Think-Aloud.
- Reduced CAD-derived geometry from 6.5M+ triangles to 95,617 optimized triangles.

Claims que requieren cuidado

Claim	Condición
Mobile-first	Solo si la demo fue diseñada/probada realmente para móvil
Thermal visualization	Aclarar si es cualitativa/visual, no simulación física/FEA
Digital twin	Usar solo si se define como visualization/prototype, no gemelo digital industrial completo
AI-assisted implementation	Explicar sin debilitar credibilidad ni ocultar uso de IA
Academic validation	Usar si protocolos/resultados están completos y documentados

Claims prohibidos

- Production-ready industrial digital twin.
- FEA thermal simulation si no hay análisis físico real.
- Commercial product deployed to clients si no es cierto.
- Built all source code manually without AI.
- Senior-level Unity architecture si no se valida con revisión técnica.

6.2 ARA Framework

Campo	Valor operativo
Prioridad	Secundaria
Tipo	AI research automation prototype
Dominio	Python, LangGraph, LLM orchestration, multi-agent workflow, research automation
Estado reportado	Casi funcional / necesita pulido
Stack reportado	Python 3.12+, LangGraph, LangChain, Groq LLaMA 3.3-70B, Redis, Semantic Scholar, Playwright MCP, MarkItDown MCP, PDF processing
Arquitectura	5 agentes: Niche Analyst, Literature Researcher, Technical Architect, Implementation Specialist, Content Synthesizer
Riesgo	Diluir posicionamiento principal; sobreventa como producto enterprise

Uso permitido

- Proyecto secundario para demostrar tooling mindset.
- Ruta alternativa para Python/AI tools.
- Soporte a narrativa de automatización de producción/investigación.
- Puede estar en portfolio secundario si tiene README, demo y límites claros.

Claims defendibles

- Prototype for AI-assisted academic research automation.
- Designed a multi-agent workflow using LangGraph and Python.
- Integrated academic search, browser automation and markdown report generation components.

Claims prohibidos o de alto riesgo

- Production-ready research automation platform.
- Enterprise AI product.
- Autonomous scientific research system.
- Reliable academic citation engine sin validación fuerte.

6.3 ai-news-aggregator

Campo	Valor operativo
Prioridad	Baja-media, según estado real
Tipo	Full-stack / AI news aggregation project
Stack reportado	FastAPI, React, PostgreSQL, Redis, Python backend
Estado reportado	Funcionó antes, necesita ajuste
Uso potencial	Ruta secundaria full-stack/AI
Riesgo	Distrae del posicionamiento Unity/3D si se sobredimensiona

Uso permitido

- Solo incluir si hay demo funcional, README, screenshots y arquitectura limpia.
- Útil para roles Python/AI tooling o full-stack secundario.
- No debe desplazar a TwinSight.

Reglas

- Si no corre: mantener privado.
- Si corre parcialmente: presentarlo como prototype.
- Si no tiene relación con visualización/3D/tools: no poner en CV principal.

6.4 FitApp-Free

Campo	Valor operativo
Prioridad	Baja para estrategia Technical Artist
Tipo	Fitness/product/app logic project
Estado	En desarrollo
Uso potencial	Product systems thinking / full-stack contingency
Riesgo	Alta dispersión; poca relación con Unity/Technical Art

Uso permitido

- No incluir en CV principal para Technical Artist.
- Podría incluirse en portfolio secundario solo si tiene demo y una narrativa de sistemas/product logic.
- No dedicarle tiempo antes del portfolio TwinSight, LinkedIn, CV y aplicaciones.

6.5 Hyperrealistic Blender Portrait / CG Cookie Human

Campo	Valor operativo
Prioridad	Media
Tipo	3D art / character pipeline / technical breakdown
Fuente de aprendizaje	CG Cookie HUMAN — completado según Excel/contexto
Valor	Demuestra Blender, topology, grooming, materials, lighting, end-to-end character workflow
Riesgo	Si se presenta solo como render, compite como character artist y diluye perfil técnico

Uso correcto Presentarlo como:

Technical Art / real-time adaptation breakdown, no como “soy character artist puro”.

Debe mostrar:

- topology;
- UVs;
- bake maps;
- materials;
- grooming/hair strategy;
- lighting;
- optimization considerations;
- posibilidad de mobile/real-time conversion si se realiza.

Uso incorrecto

- Publicar solo beauty render sin breakdown.
- Usarlo como pieza principal del portfolio por encima de TwinSight.
- Afirmar experiencia profesional de character artist si solo fue curso/proyecto personal.

6.6 Proyectos/repositorios antiguos o menores

Proyecto/repositorio	Estado operativo	Uso recomendado provisional
deus-ex-machina	Viejo / GameMaker-style	Auditar. Probablemente no destacar salvo que tenga valor histórico o README contextual.
skills-introduction-to-github	Tutorial	Ocultar/no pinear/no mostrar a reclutadores.
Chocolateria	Irrelevante para estrategia actual	Ignorar salvo razón fuerte.
Bosque de Sombras Manuales	Irrelevante	Ignorar.
alexander-fullstack-portfolio	Test/prototipo	Auditar si sirve para portfolio site.
alexander-3d-portfolio	Test/prototipo	Auditar si sirve para portfolio site.
delarge95.github.io	Potencial portfolio host	Alta utilidad si se convierte en sitio principal.
3DWeb	Potencial web 3D	Auditar.
job-search-strategy	Documentación interna	No mostrar como repo profesional si contiene información personal.
ludex_framework	Desconocido	Auditar antes de decidir.
WoodSleiman	Desconocido	Auditar antes de decidir.
WebGL_tesis	Relacionado con tesis	Auditar duplicación con repo principal.

2.0.9 7. Inventario maestro de habilidades

7.1 Habilidades primarias defendibles

Estas habilidades pueden aparecer en la narrativa principal si están respaldadas por TwinSight, tesis, código o portfolio.

Categoría	Habilidades
Engine / realtime	Unity, C#, Unity WebGL, URP, UI Toolkit
3D / asset pipeline	Blender, retopology, UVs, baking, high-poly to low-poly, CAD-to-realtime optimization
Rendering / visual modes	Shader Graph, basic HLSL, real-time rendering concepts, x-ray/blueprint/solid/wireframe/ghosted visual modes if implemented
Technical visualization	interactive 3D UI, technical inspection, assembly visualization, cross-section/clipping
Performance	triangle reduction, mesh optimization, web/mobile performance considerations, profiling concepts
Documentation / evaluation	technical documentation, SUS, NASA-TLX Raw, Think-Aloud, academic validation
Tooling	Git/GitHub, Python, AI-assisted coding/productivity
Foundations	multimedia engineering, computer graphics fundamentals, math/physics/electronics background from UNAL

7.2 Habilidades secundarias que requieren validación por proyecto

Habilidad	Evidencia requerida antes de afirmarla fuerte
FastAPI	Demo running / repo / endpoint documentation
React	UI functional / deployed app / component structure
PostgreSQL	schema, migrations, queries, deployment proof
Redis	real usage, not only dependency
LangGraph	running graph, agent states, checkpointing evidence
LLM orchestration	demos, logs, architecture diagram
Multi-agent systems	stable agent workflow, reproducible output

Habilidad	Evidencia requerida antes de afirmarla fuerte
Playwright/MCP	scripts or recorded automation
Semantic Scholar integrations	API calls, rate limit handling, output validation
Full-stack apps	deployed or locally reproducible app
GameMaker/GML	old project evidence; likely not central
Houdini/VFX	course outputs or reel, not just access
Unreal/Niagara	course outputs or demonstrable project
Three.js/WebGPU	project proof if added later

7.3 Habilidades aspiracionales o de mediano plazo

Habilidad	Condición para mover a defendible
Advanced HLSL	shader code examples or technical breakdown
WebGPU	prototype/demo
Three.js / React Three Fiber	web 3D project
XR foundations	Unity XR prototype, headset/mobile AR test
Houdini procedural tools	HDA/demo, not course only
Realtime VFX	flipbook, Niagara/Unity VFX demo, reel
Pipeline tools	Unity Editor/Window, Blender Python tool, batch processor
Technical writing	published case studies, README quality
German B1/B2	exam or functional interview capability
Portuguese A2/B1	practical use/learning status

7.4 Habilidades que no deben liderar el perfil ahora

Habilidad / identidad	Motivo
Data Analyst	No conecta directamente con TwinSight ni portfolio 3D actual
Pure Full-stack Developer	Mercado saturado; evidencia no principal
Pure AI Engineer / ML Engineer	Falta base formal/proyectos ML robustos
Pure Houdini FX Artist	Mucho acceso a cursos, poca evidencia profesional actual
Pure Character Artist	Retrato Blender es útil, pero no suficiente como eje
UX/UI Designer	UI Toolkit ayuda, pero no es el valor diferencial principal

2.0.10 8. Formación, cursos y credenciales

8.1 Política general

Los cursos adquiridos son un activo de aprendizaje y producción de portfolio. **No son certificaciones**, salvo que exista certificado verificable, institución formal y credencial emitida.

Uso correcto:

- Professional development.
- Self-directed training.
- Coursework / self-study in [topic].
- Mencionar cursos solo si están conectados a un output visible.

Uso incorrecto:

- Certified Technical Artist.
- Rebelway Certified.
- CG Cookie Certified.
- Enumerar decenas de cursos en CV.
- Presentar acceso a cursos como competencia profesional ya adquirida.

8.2 Cursos y áreas detectadas en Courses 2025.xlsx

El Excel contiene una lista extensa de cursos adquiridos, con columnas como:

- Nombre;
- Área;
- Software;
- Academia;
- Año;
- Link;
- Archivo;
- Clave;
- Enfoque;
- Habilidades;
- Weeks;
- Video Hours;
- Estado;
- Notas.

Áreas detectadas:

Área	Uso potencial
Blender / 3D Art	Soporte a asset pipeline, modeling, topology, shading
Hard Surface	Útil para drones, props, CAD-to-realtime, game-ready assets
Character pipeline	Útil si se convierte en technical breakdown
Houdini / VFX	Mediano-largo plazo para proceduralism, FX, technical art
Unreal / Niagara	Secundario; útil si se abre ruta Unreal/VFX
Compositing / Nuke / DaVinci	Útil para demo reel, presentation polish, VFX; bajo impacto CV principal
Python scripting	Útil para tools developer, Blender/Houdini automation
Web 3D / Spline / Blender to Web	Útil si se conecta a Unity WebGL / interactive 3D
AI / Machine Learning	Útil si se conecta a ARA o tooling, no como ML engineer claim

8.3 Cursos específicos relevantes ya identificados

Curso / grupo	Estado reportado	Valor operativo
CG Cookie HUMAN — Realistic Portrait Creation with Blender	Completado según Excel	Útil para Blender portrait / pipeline breakdown
Alive! — 3D Animation Mechanics & Dynamic Motion	Curso adquirido/no certificado según prompt	Útil para fundamentos de animación, no credential fuerte
Blender Bros Hard Surface / Hard Ops / Box Cutter	Adquiridos según Excel	Útiles para hard-surface asset pipeline
Interactive Content Creation Unleashed: From Blender to Web	Adquirido	Potencial conexión con web 3D / thesis
CG Cookie scripting / Blender Python	Adquirido	Potencial para tools/pipeline
Houdini-Course.com	En progreso según Excel	Mediano-largo plazo; no eje actual sin output
Rebelway Houdini Fundamentals	Acceso detectado	Alto valor de skill, alto riesgo de dispersión
Rebelway Intro to UE	Acceso detectado	Útil si se decide Unreal/VFX; no prioridad sobre Unity actual
Rebelway Advanced Asset Creation	Acceso detectado	Potencial para procedural assets
Rebelway Stylized Realtime FX for Games	Acceso detectado	Potencial si se produce realtime FX reel
Rebelway Python for Houdini Artists	Acceso detectado	Potencial alto para Tools/VFX pipeline
Rebelway VEX for Houdini Artists	Acceso detectado	Mediano-largo plazo
Rebelway Intro to Machine Learning	Pausado según Excel	No priorizar antes de fundamentos/proyecto claro

8.4 Regla de ROI para cursos

Un curso solo pasa a la estrategia principal si produce al menos uno de estos outputs:

1. portfolio case study;

2. demo reel segment;
3. shader/tool breakdown;
4. GitHub repo limpio;
5. ArtStation technical breakdown;
6. improvement visible en TwinSight;
7. evidence for a specific job title.

Sin output, el curso se trata como consumo formativo, no como activo laboral.

2.0.11 9. Idiomas y movilidad: fuente de verdad operacional

9.1 Idiomas actuales

Idioma	Estado operativo	Uso laboral
Español	Nativo	LATAM, España, Colombia, networking local
Inglés	C1 autoevaluado	Principal idioma laboral internacional
Alemán	No profesional	No usar como ventaja laboral actual
Portugués	Relevante estratégicamente	No usar como competencia profesional actual salvo evidencia
Francés / otros	No confirmados	No relevantes actualmente

9.2 Regla de corto plazo

En 0–6 meses, el orden correcto es:

1. defensa/sustentación;
2. portfolio TwinSight;
3. CV/LinkedIn/GitHub;
4. aplicaciones y networking;
5. inglés técnico/interview practice;
6. cursos solo si producen portfolio;
7. idiomas adicionales solo en intensidad baja.

9.3 Movilidad UE

Ruta	Estado	Uso en estrategia
Portugal / pasaporte futuro	No confirmado como derecho actual	Plan mediano/largo plazo
Alemania por pareja/matrimonio	Condicionado a hechos futuros	Plan contingente, no base operativa
EU local jobs	No asumir elegibilidad actual	Aplicar solo si remote/contractor o sponsorship posible
Remote desde Colombia para EU company	Posible según empresa, contractor/EOR	Prioridad investigable

9.4 Claim permitido a reclutadores

Versión prudente:

Based in Colombia and available for remote international contractor roles. Open to future EU mobility if legally viable; no current EU work authorization should be assumed.

Versión que debe evitarse:

I am EU work authorized / I can work in Germany / I will have Portuguese citizenship soon.

2.0.12 10. Restricciones operativas y riesgos personales

10.1 Riesgos reportados en contexto antiguo

Riesgo	Implicación estratégica
Analysis paralysis	Limitar opciones y producir entregables concretos
Shiny Object Syndrome	Evitar muchos cursos/proyectos paralelos
Perfeccionismo	Definir MVP de portfolio y deadlines
Bajo capital / presión financiera antigua	Reconfirmar runway y diseñar plan realista
Hardware con GTX 980 Ti	Evitar depender de UE5/RT/heavy rendering como eje inmediato
Tendencia a construir sistemas grandes	Priorizar empleabilidad antes que arquitectura perfecta

10.2 Reglas de mitigación

Regla	Justificación
TwinSight primero	Es el activo más alineado con el mercado
Portfolio MVP antes que portfolio perfecto	Empleabilidad > perfeccionismo
Máximo 2 proyectos visibles al inicio	Reduce dispersión narrativa
Cursos solo con output	Evita consumo infinito
No pivotar a AI/full-stack salvo decisión estratégica	Evita dilución
No usar EU como excusa para retrasar aplicaciones remotas	EU es mediano plazo
No construir 100 documentos antes de aplicar	Los documentos deben habilitar acción, no reemplazarla

2.0.13 11. Mapa de evidencia requerida por activo

11.1 Evidencia mínima para TwinSight

Evidencia	Prioridad
Demo video 60–120 s	Muy alta
Case study escrito	Muy alta
Screenshots UI principales	Muy alta
Before/after optimization table	Alta
GitHub README recruiter-friendly	Alta
Technical README / architecture diagram	Media-alta
SUS / NASA-TLX results visualization	Alta si los datos son finales
Explanation of AI-assisted development	Alta
Asset/legal note for CAD/manufacture models	Alta
WebGL live demo	Muy alta si funciona estable

11.2 Evidencia mínima para ARA

Evidencia	Prioridad
README con alcance y límites	Alta
Architecture diagram	Alta
Example input/output	Alta
Running demo or screen recording	Media-alta
Security/API key cleanup	Muy alta antes de publicar
Clear “prototype” wording	Muy alta

11.3 Evidencia mínima para Blender portrait

Evidencia	Prioridad
Beauty render	Media
Wireframe/topology	Alta
UV layout	Alta
Material/shader breakdown	Alta
Grooming/lighting breakdown	Media-alta
Optimization/real-time adaptation notes	Alta si se usa para technical art

11.4 Evidencia mínima para LinkedIn/CV/GitHub

Activo	Evidencia mínima
CV	1-page ATS version, 4 variants later
LinkedIn	headline, About, experience, projects, featured, skills
GitHub	pinned repos, README, screenshots, repo hygiene, no tutorial repos destacados
Portfolio	homepage, TwinSight case study, contact CTA, project links

2.0.14 12. Narrative stack: qué historia debe contar el perfil

12.1 Historia principal

Alexander debe presentarse como un perfil técnico-creativo que une ingeniería, 3D y software para construir experiencias interactivas de visualización técnica.

Estructura narrativa:

1. base previa en ingeniería electrónica;
2. transición a ingeniería multimedia;
3. especialización en Unity/WebGL/Blender;
4. tesis como prueba fuerte: TwinSight X500;
5. capacidad de convertir CAD/documentación técnica en visualización 3D interactiva;
6. capacidad secundaria de automatizar con Python/IA;
7. objetivo: roles remotos internacionales en real-time 3D, technical visualization, Unity, XR/simulation/digital twin.

12.2 Historia secundaria

ARA y proyectos AI muestran que Alexander no es solo artista 3D ni solo Unity user, sino alguien con mentalidad de tooling, automation y research systems.

Debe formularse así:

I also build Python/AI-assisted tooling prototypes for research automation and production workflows.

No debe formularse así:

I am an AI engineer specialized in autonomous agents.

12.3 Historia de transición académica

Forma correcta:

I completed advanced coursework in Electronic Engineering at Universidad Nacional de Colombia before moving into Multimedia Engineering, where I focused on computer graphics, real-time 3D, interaction and technical visualization.

Forma incorrecta:

I left engineering and became a designer.

12.4 Historia sobre IA asistida

Forma correcta:

I used AI tools as coding assistants for scaffolding and iteration, while retaining responsibility for architecture, integration, debugging, validation and final implementation decisions.

Forma incorrecta:

AI built the project.

Forma igualmente incorrecta:

I did not use AI.

2.0.15 13. Red flags que deben evitarse en documentos públicos

Red flag	Por qué daña
Demasiados roles en headline	Parece perfil incoherente
Llenar CV con cursos	Sustituye evidencia por consumo de contenido
Presentar proyectos incompletos como productos	Riesgo en entrevista técnica
Usar métricas no verificadas	Riesgo reputacional
Publicar repos con claves, paths personales o código roto	Riesgo técnico/profesional
Decir "senior" por proyectos personales	Desalineación con reclutadores
Enfocar demasiado videojuegos sin portfolio de juegos	Reduce probabilidad
Hablar de EU como si ya hubiera permiso	Riesgo legal/credibilidad
Mezclar FitApp, ARA, TwinSight, VFX, full-stack y character art al mismo nivel	Dilución
No aclarar AI-assisted development si preguntan	Riesgo de percepción de engaño

2.0.16 14. Decisiones que quedan abiertas

Antes de cerrar estrategia final, se deben responder o investigar estas preguntas:

14.1 Perfil y disponibilidad

1. ¿Fecha estimada de sustentación?
2. ¿Disponibilidad real semanal después de sustentación?
3. ¿Runway financiero actualizado?
4. ¿Ingreso mínimo mensual neto necesario?
5. ¿Hay RUT activo?
6. ¿Hay facturación electrónica habilitada?
7. ¿Hay contador o asesor tributario?
8. ¿Hay Payoneer, Wise, Deel, Remote u otro setup?

14.2 TwinSight

1. ¿Cuál es la URL final del repo correcto?
2. ¿Existe demo WebGL pública?
3. ¿Corre en móvil actualmente?
4. ¿Cuáles son las métricas finales exactas?
5. ¿Qué partes de assets son propias, CAD del fabricante o reconstruidas?
6. ¿Qué licencia/nota legal conviene usar?
7. ¿Qué screenshots ya existen?
8. ¿Hay video demo o debe producirse?
9. ¿Qué código fue IA-assisted y cómo explicarlo?

14.3 Proyectos secundarios

1. ¿ARA corre end-to-end?
2. ¿Tiene API keys o datos sensibles que limpiar?
3. ¿ai-news-aggregator corre actualmente?
4. ¿FitApp tiene demo o solo planificación?
5. ¿El Blender portrait está terminado y renderizable?
6. ¿Qué repos están públicos ahora?
7. ¿Qué repos están pineados?

14.4 Cursos

1. ¿Qué cursos están realmente completados?
2. ¿Qué cursos están en progreso?
3. ¿Qué cursos tienen output visible?
4. ¿Qué acceso Rebelway sigue disponible?
5. ¿Qué curso podría producir el portfolio artifact de mayor ROI en 2–4 semanas?
6. ¿Hay certificados emitidos por alguno? Si no, no se tratan como certificaciones.

14.5 Mercado

1. ¿Qué roles tienen mayor frecuencia remota real desde LATAM?
2. ¿Qué títulos usan empresas de technical visualization?
3. ¿Qué roles pagan USD 3k/mes a perfiles early-career contractors?
4. ¿Qué empresas aceptan Colombia/LATAM contractor?
5. ¿Dónde encaja mejor TwinSight: games, XR, simulation, digital twin, product visualization, defense-adjacent, education, drones?

Estas preguntas se resolverán en archivos posteriores, no en esta sección.

2.0.17 15. Implicaciones para archivos posteriores

Archivo futuro	Regla derivada de esta fuente de verdad
02_market_role_fit_and_positioning.md	Validar roles contra mercado; no asumir que Technical Artist puro es el mejor sin datos
03_salary_benchmark_and_remote_colombia.md	Separar employee/contractor, local/global, junior/mid; no usar rangos globales como si fueran remotos desde Colombia
04_eu_portugal_germany_mobility_strategy.md	No afirmar derechos UE actuales; usar fuentes oficiales
05_language_strategy_and_roi.md	Inglés primero; alemán/portugués según horizonte; no sacrificar portfolio por idiomas en corto plazo
06_education_courses_rebelway_masters_phd.md	Clasificar cursos por output/ROI; no por acumulación
07_portfolio_strategy_and_project_architecture.md	TwinSight primero; ARA secundario; limitar dispersión
08_twinsight_x500_case_study.md	Basarse solo en hechos, métricas y assets verificables
09_ara_ai_tools_and_secondary_projects.md	Decidir inclusión por evidencia, no por potencial abstracto

Archivo futuro	Regla derivada de esta fuente de verdad
10_github_linkedin_cv_strategy.md	Redactar desde claims defendibles; corregir tesis/ARA inconsistency
11_company_targets_job_boards_recruiters.md	Buscar empresas que encajen con TwinSight y remoto Colombia, no solo nombres grandes
12_application_system_and_tracker.md	Sistema simple, medible, no sobreplanificado
13_interview_negotiation_and_offer_risk.md	Preparar respuestas para low formal experience, AI-assisted work, salary, remote Colombia
14_30_60_90_execution_plan.md	Plan desde estado actual, no desde cronogramas viejos
15_final_integrated_strategy.md	Sintetizar lo validado; no introducir claims nuevos

2.0.18 16. Criterios de aprobación de esta sección

Esta sección se considera aprobada si:

1. fija que la tesis está completa y pendiente de sustentación;
2. separa hechos, supuestos, aspiraciones y claims prohibidos;
3. identifica TwinSight como activo principal;
4. mantiene ARA como secundario y no como eje principal;
5. trata cursos como aprendizaje, no certificación;
6. evita afirmar autorización laboral UE;
7. conserva rutas Portugal/Alemania como mediano-largo plazo;
8. limita la dispersión de proyectos;
9. define qué debe verificarse antes de CV, LinkedIn, GitHub y portfolio;
10. deja claro que el siguiente paso requiere investigación externa para roles/mercado.

2.0.19 17. Próximo archivo recomendado

El siguiente archivo debe ser:

```
Bloque de código: text
02_market_role_fit_and_positioning.md
```

Recomendación de herramientas para la siguiente sección

Herramienta	Activar	Motivo
Deep Research	Sí	Se necesita validar demanda actual, roles reales, títulos, keywords y fit por mercado.
Modo agente	Sí, si está disponible	Útil para recolectar ofertas reales, títulos frecuentes, requirements, remote restrictions y keywords desde job boards/careers pages.
Modo normal	Sí	Necesario para sintetizar y decidir jerarquía estratégica después de recolectar datos.

Objetivo de la siguiente sección

Determinar, con evidencia actual, cuáles roles debe priorizar Alexander y cuáles deben quedar como secundarios o contingencia:

- Unity Technical Artist;
- Real-Time 3D Developer;
- Unity WebGL Developer;
- Interactive 3D Developer;

- Technical Visualization Developer;
- CAD-to-Realtime Technical Artist;
- XR / Simulation / Digital Twin Developer;
- Game Technical Artist;
- Tools Developer;
- Python AI Tools Developer;
- Full-stack secondary route.

La sección 02 debe producir una matriz de roles basada en:

- probabilidad de entrada;
- salario potencial;
- ajuste con TwinSight;
- ajuste con habilidades reales;
- remoto desde Colombia;
- velocidad a primera entrevista;
- riesgo de sobreventa;
- evidencia de portfolio requerida;
- keywords ATS;
- mercados objetivo.

Chapter 3

02 — Market Role Fit and Positioning

Fuente: 02_market_role_fit_and_positioning.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 838

Ultima modificacion: 2026-06-11 11:41

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Perfil: Alexander Woodcock Salomón **Fecha de corte operativa:** 2026-06-11 **Estado del documento:** sección corregida y terminada para integración posterior. **Modo usado para esta sección:** Deep Research previo + reconstrucción analítica manual + verificación web puntual. **Herramientas adicionales recomendadas para esta sección:** no usar modo agente todavía. Reservarlo para la sección 11_company_targets_job_boards_recruiters.md. **Regla crítica:** esta sección dec

Contenido completo depurado

3.0.1 Auditoría y reconstrucción del encaje entre perfil, roles objetivo y mercado laboral 2025–2026

Perfil: Alexander Woodcock Salomón

Fecha de corte operativa: 2026-06-11

Estado del documento: sección corregida y terminada para integración posterior.

Modo usado para esta sección: Deep Research previo + reconstrucción analítica manual + verificación web puntual.

Herramientas adicionales recomendadas para esta sección: no usar modo agente todavía. Reservarlo para la sección 11_company_targets_job_boards_recruiters.md.

Regla crítica: esta sección decide el eje de posicionamiento profesional. No redacta todavía CV, LinkedIn, GitHub ni portfolio final.

3.0.2 0. Propósito de esta sección

Este archivo define qué familias de roles debe perseguir Alexander en una búsqueda laboral internacional de 3–6 meses, idealmente remota desde Colombia, y qué rutas deben tratarse como secundarias, futuras o de contingencia.

La sección corrige la investigación previa en tres puntos:

1. **Separar familias de rol, job titles, industrias y keywords.** El informe anterior mezclaba Technical Artist, Unity Developer, Digital Twin, CAD-to-realtime, WebGL, XR, AI Tools y Full-stack como si fueran categorías equivalentes. No lo son.
2. **Priorizar por probabilidad, no solo por atractivo.** Un rol con salario alto puede ser poco útil si exige seniority, autorización laboral local, producción AAA, residencia en EE. UU./UE o experiencia formal no demostrada.
3. **Alinear cada rol con evidencia real.** TwinSight X500, ARA Framework, el retrato Blender y los proyectos secundarios no prueban lo mismo. Cada rol requiere un paquete de evidencia distinto.

Este archivo no contiene benchmarking salarial completo. Los salarios se tratan solo como señal de potencial. La validación detallada corresponde a `03_salary_benchmark_and_remote_colombia.md`.

3.0.3 1. Fuentes y calidad de evidencia

1.1 Fuentes internas revisadas

Fuente	Uso en esta sección	Estado
<code>01_source_of_truth_profile.md</code>	Fuente de verdad operativa: tesis completa, perfil, claims permitidos/prohibidos	Vigente
<code>deep-research-report (8).md</code>	Informe anterior: útil como borrador, pero con análisis superficial en roles/salarios	Debe auditarse, no copiarse
Deep Research de sección 02	Insumo actualizado sobre job postings, títulos, salarios y roles	Útil, pero requiere corrección estructural
<code>portfolio_strategy.md</code>	Conceptos útiles: Optimization Doctor, Toolsmith, Math Wizard, Mobile Hero	Útil como arquitectura de portfolio, no como estrategia final
<code>job_search_strategy.md</code>	Hipótesis "Trojan Horse": QA Automation / Tools / Junior Tech Artist	Requiere revalidación; QA Automation no debe ser ruta principal sin evidencia
<code>estrategia_idiomas_mercado_laboral_tecnica_idiomas_rendidos UE y rol de alemán/portugués</code>		Reutilizable en secciones 04 y 05
<code>Courses 2025.xlsx</code>	Insumo de skills/cursos	No se analiza aquí; sección 06

1.2 Fuentes externas verificadas o utilizadas como contexto

Esta sección usa fuentes externas solo para sostener afirmaciones de contexto técnico y patrones de mercado, no para cerrar salarios definitivos.

Fuente	Qué aporta	Tipo de evidencia	Limitación
Unity Manual — Web platform	Unity permite publicar gráficos 3D interactivos en navegador mediante WebGL	Fuente oficial técnica	No es fuente de demanda laboral
Khronos WebGL	WebGL es estándar web abierto, cross-platform, plugin-free, basado en OpenGL ES	Fuente oficial técnica	No es fuente de salarios
LangChain/LangGraph GA announcement	LangGraph Platform fue presentado para despliegue de agentes stateful/long-running	Fuente oficial/vendor	No prueba que ARA sea producto comercial
Job postings de Technical Artist / Unity / XR / Digital Twin	Patrones de requisitos: C#, Python, DCC tools, shader/pipeline, experiencia	Evidencia de mercado puntual	No representa todo el mercado
Salary/job aggregators Papers recientes de digital twins con Unity/Web/3D	Señales de potencial salarial Validan que Unity/Web 3D aparece en simulación, digital twin e industrial visualization	Evidencia débil/media Evidencia técnica/sectorial	Debe consolidarse en sección 03 No es evidencia directa de contratación

1.3 Auditoría de la investigación previa

El Deep Research de esta sección produjo una dirección general correcta, pero no debe usarse como texto final sin reconstrucción.

Elemento	Evaluación
Dirección general hacia Unity / Real-Time 3D / Technical Visualization	Correcta
Tratamiento de TwinSight como evidencia principal	Correcto
Tratamiento de ARA como secundario	Parcialmente correcto, pero demasiado descartado
Clasificación de CAD-to-realtime como rol de baja prioridad	Incorrecta si se interpreta como skill diferenciador; correcta solo si se interpreta como job title literal
Agrupación de roles Technical Artist / Unity TA / Game TA / Shader TA	Demasiado gruesa
Datos salariales dentro de role-fit	Mezclados y parcialmente débiles; mover a sección 03
Remote-from-Colombia	Insuficientemente segmentado
Evidence required before applying	Incompleto
Role rejection/downplay	Correcto en intención, insuficiente en granularidad
Scoring objetivo	Faltante

3.0.4 2. Corrección taxonómica: rol, keyword, skill e industria no son lo mismo

La reconstrucción parte de separar cuatro capas:

2.1 Familias de rol

Una familia de rol agrupa vacantes que podrían aparecer con títulos distintos pero comparten responsabilidades centrales.

Familia	Ejemplos de job titles
Real-Time 3D / Interactive 3D Developer	Real-Time 3D Developer, Interactive 3D Developer, 3D Application Developer, Unity Developer
Unity Technical Artist	Unity Technical Artist, Technical Artist — Unity, Technical Artist — Optimization, Runtime Technical Artist
Technical Visualization / Digital Twin / Simulation	Technical Visualization Developer, Digital Twin Developer, Simulation Developer, Industrial 3D Developer, HMI Unity Developer
Unity WebGL / Web 3D	Unity WebGL Developer, WebGL Developer, 3D Web Developer, Frontend 3D Developer, Three.js Developer
Tools / Pipeline / Automation	Tools Developer, Pipeline Developer, Unity Editor Tools Developer, Technical Artist — Tools, Python Automation Developer
XR / AR / VR	XR Developer, AR/VR Developer, Unity XR Developer, Simulation XR Developer
AI Tools / LLM Applications	LLM Application Developer, AI Tools Developer, Agentic Workflow Developer, Python AI Developer
Game Technical Artist	Game Technical Artist, Technical Artist — Games, Shader Technical Artist, VFX Technical Artist
Full-stack fallback	Full-stack Developer, Python/FastAPI Developer, React Developer
3D Artist fallback/downplay	3D Artist, Blender Artist, 3D Generalist, Prop Artist, Character Artist

2.2 Search keywords

Algunas etiquetas son mejores como términos de búsqueda o diferenciadores, no como título principal.

Keyword / skill	Cómo usarlo
CAD-to-realtime	Diferenciador fuerte para technical visualization, digital twin, Unity industrial, product visualization
WebGL	Keyword fuerte para despliegue web, browsers, interactive 3D y Unity Web
Unity URP	Keyword técnico de rendering/performance
UI Toolkit	Keyword de UI runtime/editor en Unity
Blender optimization	Keyword de pipeline 3D y asset preparation
Retopology / baking / UVs	Evidencia de optimización de assets, no título laboral
SUS / NASA-TLX / Think-Aloud	Diferenciador UX/research para technical visualization, no core de technical art
LangGraph / LLM orchestration	Diferenciador secundario para AI tooling, no eje principal

2.3 Industrias objetivo

Las industrias objetivo no deben confundirse con rol. El mismo rol puede venderse a sectores distintos.

Industria	Relevancia para Alexander
Industrial visualization	Alta: TwinSight encaja directamente
Digital twin / simulation	Alta a mediano plazo; requiere adaptar narrativa técnica
XR training / education	Media-alta; requiere XR proof si se apunta directamente
Product visualization / configurators	Media-alta; WebGL/Unity/Blender encajan
Drones / robotics / hardware visualization	Alta como narrativa de especialización inicial
Game studios	Media; emocionalmente atractiva, pero competitiva y portfolio-demanding
Creative technology studios	Media-alta; buen fit visual/técnico si hay demo fuerte
AI tooling startups	Media; ARA ayuda, pero perfil principal no debe girar completamente
Generic full-stack SaaS	Baja como estrategia principal; útil como fallback

3.0.5 3. Modelo de scoring

Se usa una escala de 1 a 5.

Score	Interpretación
5	Muy favorable
4	Favorable
3	Viable con condiciones
2	Débil / riesgoso
1	No recomendado como foco

3.1 Variables evaluadas

Variable	Definición
Fit with TwinSight	Qué tan directamente TwinSight prueba capacidades exigidas por el rol
Immediate probability	Probabilidad relativa de obtener entrevistas/oferta en 3–6 meses
Remote Colombia feasibility	Probabilidad de que el rol admita remoto desde Colombia, contractor o EOR
Salary upside	Potencial de crecer hacia USD 3k–6k/mes, sin detallar rangos definitivos
Portfolio readiness	Qué tan cerca está Alexander de tener evidencia suficiente para aplicar
Competition risk	Riesgo de competir contra perfiles más establecidos; puntuación alta = menor riesgo
Overclaim risk	Riesgo de que el rol obligue a afirmar capacidades no demostradas; puntuación alta = menor riesgo
12–24m leverage	Capacidad del rol de construir trayectoria internacional de alto valor

3.0.6 4. Matriz principal de priorización por familia de rol

Rank	Familia de rol	Fit TwinSight	Prob. 3–6m	Remote COL	Upside	Portfolio readiness	Menor competencia	Menor overclaim	Leverage 12–24m	Total /40	Bolivia compiles
1	Real-Time 3D / Interactive Technical Visualization Developer	5	4	4	4	4	3	4	5	33	Primaria
2	Unity WebGL / Interactive 3D Developer	5	4	4	3	4	3	4	4	31	Primaria
3	Technical Visualization / Digital Twin / Simulation Developer	5	3	3	5	3	3	4	5	31	Primaria / as controlada
4	Unity Technical Artist — optimization/tools/runtime	4	3	3	4	3	2	3	5	27	Primaria sec
5	Tools / Pipeline / Python Automation Developer	3	3	4	4	3	3	3	4	27	Secundaria
6	XR / AR / VR Developer	3	3	3	4	2	3	3	4	25	Secundaria
7	AI Tools / LLM Application Developer	2	3	4	5	2	2	2	4	24	Secundaria se fortalece
8	Game Technical Artist	3	2	2	4	2	2	2	4	21	Aspiracional inicial
9	Shader Technical Artist	2	2	3	4	2	2	2	4	21	Subskill, no
10	Full-stack Developer	1	3	5	4	2	2	2	3	22	Fallback ecoeje
11	3D Artist / Blender Artist / Generalist	2	2	3	2	2	1	3	2	17	Downplay / secondary
12	AI/ML Engineer	1	1	3	5	1	1	1	3	16	Rechazar co posicionamiento

4.1 Lectura de la matriz

La matriz no dice que el salario más alto esté en la prioridad 1. Dice que la mejor relación entre **evidencia actual + probabilidad + coherencia + upside** está en roles de 3D interactivo, visualización técnica y Unity/WebGL.

El rol más coherente no es “Technical Artist genérico”. Es:

Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization, and WebGL deployment.

Versión en español:

Desarrollador 3D en tiempo real / Technical Artist Unity enfocado en visualización técnica interactiva, optimización CAD-to-realtime y despliegue WebGL.

3.0.7 5. Diagnóstico por familia de rol

5.1 Real-Time 3D / Interactive Technical Visualization Developer

Prioridad: primaria.

Uso recomendado: eje de posicionamiento principal.

Dimensión	Evaluación
Títulos comunes	Real-Time 3D Developer, Interactive 3D Developer, Unity Developer, 3D Application Developer, Technical Visualization Developer
Encaje con TwinSight	Muy alto. TwinSight es exactamente una aplicación interactiva 3D técnica con Unity/WebGL
Encaje con ARA	Bajo-medio. Aporta automatización y tooling, no core 3D
Encaje con Blender portrait	Medio. Prueba pipeline 3D si se muestra técnico
Requisitos típicos	Unity/C#, interacción runtime, UI, performance, assets optimizados, Git, documentación, comunicación con stakeholders técnicos
Portfolio esperado	Demo funcional, case study, explicación del problema, capturas UI, breakdown técnico, video corto
GitHub esperado	README claro, estructura comprensible, si el código es publicable; si no, demo/video suficiente
Remoto desde Colombia	Medio-alto. Más factible que defensa/on-site simulation; menos saturado que full-stack
Riesgo	Que se perciba como solo proyecto académico si no se empaqueta como producto/case study
Proof mínimo	Demo video, case study TwinSight, métricas de optimización/usabilidad, README recruiter-friendly

Conclusión: esta familia debe ser el eje. Convierte el proyecto de tesis en evidencia laboral directa sin forzar el perfil hacia videojuegos puros o AI puro.

5.2 Unity WebGL / Interactive 3D Developer

Prioridad: primaria.

Uso recomendado: ruta de entrada remota y contractor.

Dimensión	Evaluación
Títulos comunes	Unity WebGL Developer, WebGL Developer, 3D Web Developer, Web-based 3D Developer, Frontend 3D Visualization Developer
Encaje con TwinSight	Muy alto por WebGL, browser delivery, Unity, UI e interacción
Encaje con ARA	Bajo. Solo ayuda si se muestra como tooling/automation
Encaje con proyectos web	Medio-alto si hay portfolio web o GitHub Pages
Requisitos típicos	Unity Web builds, performance, asset size, loading, browser constraints, sometimes JavaScript/Three.js/Babylon.js

Dimensión	Evaluación
Portfolio esperado	Live build o video si el build no es público; capturas browser/mobile; métricas de carga/performance
GitHub esperado	README con build instructions, screenshots, known limitations, hosted demo si posible
Remoto desde Colombia	Alto relativo. Web-based roles tienen mejor compatibilidad con contractor remoto
Riesgo	Que Unity WebGL sea visto como nicho si no se acompaña de broader “interactive 3D”
Proof mínimo	Web demo o video, build screenshots, explanation of WebGL constraints, optimization table

Conclusión: esta es la ruta más útil para convertir TwinSight en oportunidad remota. No debe limitarse a “Unity WebGL” como título único; debe acompañarse de Interactive 3D, Web-based 3D, technical visualization y, si aplica, Three.js/WebGPU como skill-gap futuro.

5.3 Technical Visualization / Digital Twin / Simulation Developer

Prioridad: primaria aspiracional controlada.

Uso recomendado: foco industrial de alto potencial, especialmente 12–24 meses.

Dimensión	Evaluación
Títulos comunes	Technical Visualization Developer, Digital Twin Developer, Simulation Developer, HMI Unity Developer, Industrial Visualization Developer
Encaje con TwinSight	Muy alto: hardware real, CAD/manufacturer assets, visualización técnica, piezas, ensamblaje, relaciones espaciales
Encaje con UNAL Electrónica	Alto: base técnica, física, sistemas, lógica de hardware
Encaje con ARA	Medio: automatización/documentación puede sumar en ingeniería
Requisitos típicos	Unity/Unreal, C#/C++, 3D visualization, CAD/BIM/industrial data, simulation logic, sometimes robotics/IoT/cloud
Portfolio esperado	Case study técnico, no solo visual; problema, datos, pipeline CAD, performance, interacción, validación
GitHub esperado	Menos importante que case study/demo si el proyecto usa assets sensibles
Remoto desde Colombia	Medio. Algunos roles son híbridos/on-site por industria, defensa o hardware
Riesgo	Sobrevender como “digital twin” si TwinSight no tiene sincronización con sensores/datos reales
Proof mínimo	Usar “technical visualization” antes que “digital twin” para TwinSight, salvo que haya data linkage real

Corrección terminológica crítica: TwinSight puede venderse como **technical visualization / interactive assembly visualization / CAD-to-realtime prototype**. No debe llamarse “digital twin completo” si no tiene interacción bidireccional con datos reales, sensores, telemetría o estado operacional.

Conclusión: muy fuerte como narrativa de mercado, pero debe controlarse la terminología para evitar sobreclaiming.

5.4 Unity Technical Artist — optimization/tools/runtime

Prioridad: primaria secundaria.

Uso recomendado: aplicar cuando el rol pida Unity, optimization, shaders, tooling, runtime integration o asset pipeline; no cuando sea AAA senior/character/VFX-heavy.

Dimensión	Evaluación
Títulos comunes	Unity Technical Artist, Technical Artist, Runtime Technical Artist, Optimization Technical Artist, Tools Technical Artist
Encaje con TwinSight	Alto en optimización, shaders, visual modes, runtime interaction y pipeline CAD→Unity

Dimensión	Evaluación
Encaje con Blender portrait	Medio si se presenta como mobile/real-time pipeline
Encaje con ARA	Medio si se vende como tooling/automation mindset
Requisitos típicos	Unity, C#, shaders, DCC tools, asset optimization, pipeline, artist-engineer bridge, sometimes Python/Maya/Houdini/Substance
Portfolio esperado	Breakdowns, before/after optimization, shader demos, tools demos, technical explanation
GitHub esperado	Útil para tools/scripts, no siempre indispensable para art roles
Remoto desde Colombia	Medio. Game studios y tech art pueden ser region-limited
Riesgo	Muchos postings piden 3–5+ años y producción profesional; no competir como senior
Proof mínimo	TwinSight case study + shader/visual mode breakdown + optimization process + optional editor tool

Conclusión: debe usarse como familia de aplicación, pero con subtítulo claro: Unity Technical Artist focused on optimization and interactive technical visualization. Evitar Game Technical Artist como etiqueta principal hasta tener portfolio de juego más fuerte.

5.5 Tools / Pipeline / Python Automation Developer

Prioridad: secundaria controlada.

Uso recomendado: complemento de tech art y ruta alternativa si ARA se limpia.

Dimensión	Evaluación
Títulos comunes	Tools Developer, Pipeline Developer, Technical Artist — Tools, Unity Editor Tools Developer, Python Automation Developer
Encaje con TwinSight	Medio: scripts, UI, workflow automation, asset setup si se extraen herramientas
Encaje con ARA	Alto como proof-of-concept, pero debe estar estable y documentado
Encaje con AI News Aggregator	Medio si se audita como app funcional
Requisitos típicos	Python, C#, APIs, automation, internal tools, Git, documentation, debugging
Portfolio esperado	Tool demo, problem solved, before/after workflow time, README, code sample
GitHub esperado	Alto; para tools sí importa ver código/arquitectura
Remoto desde Colombia	Alto relativo
Riesgo	Diluir perfil si se presenta como ruta principal separada de 3D
Proof mínimo	Publicar o documentar una herramienta concreta: Unity Editor Window, asset importer, batch setup, LOD/material tool

Conclusión: esta ruta no debe liderar el perfil, pero sí debe aparecer como diferenciador: Alexander no solo produce assets o escenas; también puede automatizar pipelines.

5.6 XR / AR / VR Developer

Prioridad: secundaria futura.

Uso recomendado: aplicar selectivamente a roles Unity XR junior/mid si no exigen SDKs avanzados o experiencia shipped.

Dimensión	Evaluación
Títulos comunes	XR Developer, AR/VR Developer, Unity XR Developer, Immersive Developer, Simulation Developer XR
Encaje con TwinSight	Medio: 3D interaction and visualization transfer, but not XR-specific
Encaje con hardware/electronics background	Medio-alto
Requisitos típicos	Unity XR, Meta Quest, HoloLens/MRTK, ARCore/ARKit, interaction design, performance, spatial UX
Portfolio esperado	XR demo, headset build, interaction prototype, performance considerations
GitHub esperado	Medium; video demo more important if headset demo not accessible

Dimensión	Evaluación
Remoto desde Colombia	Medium; many XR roles need hardware, labs, or on-site collaboration
Riesgo	Overclaiming XR if there is no deployed headset project
Proof mínimo	Small XR prototype or explicit statement: "Unity 3D background transferable; XR SDK learning in progress"

Conclusión: XR is adjacent, not immediate core. It can become strong if a short XR proof is added later.

5.7 AI Tools / LLM Application Developer

Prioridad: secundaria only if ARA becomes presentable.

Uso recomendado: not part of main LinkedIn headline unless targeting AI tooling separately.

Dimensión	Evaluación
Títulos comunes	LLM Application Developer, AI Tools Developer, Agentic Workflow Developer, Python Automation Developer
Encaje con ARA	High conceptually; medium evidentially until polished
Encaje con TwinSight	Low, except AI-assisted tooling/documentation
Requisitos típicos	Python, API integration, RAG, LLM orchestration, evaluation, deployment, security, observability, vector DBs
Portfolio esperado	Working demo, architecture diagram, README, eval examples, cost control, failure handling
GitHub esperado	High
Remoto desde Colombia	High relative
Riesgo	Competes against software engineers with production SaaS/AI experience; profile dilution
Proof mínimo	Stable ARA demo, documented workflow, sanitized repo, limitations, example outputs, no inflated claims

Critical distinction: ARA supports AI-assisted tooling and automation. It does not currently justify positioning as AI Engineer, ML Engineer, or Production LLM Architect unless additional evidence is built.

5.8 Game Technical Artist

Prioridad: aspirational, not first wave.

Uso recomendado: target selectively for junior/associate Unity tech art, mobile game optimization, tools, or shader internships/contracts.

Dimensión	Evaluación
Títulos comunes	Game Technical Artist, Associate Technical Artist, Technical Artist — Unity, Shader Technical Artist, VFX Technical Artist
Encaje con TwinSight	Medium. It proves Unity/optimization, not game production pipeline
Encaje con Blender portrait	Medium if real-time/game-ready breakdown exists
Requisitos típicos	Shipped games, game engine production, DCC pipeline, shaders, profiling, collaboration with artists/designers
Portfolio esperado	Game-ready assets, shader demos, optimization examples, tools, VFX, breakdowns
GitHub esperado	Useful for tools; less central for art-heavy TA
Remoto desde Colombia	Medium-low depending on studio; region constraints common
Riesgo	High competition, portfolio not yet game-focused
Proof mínimo	Optimization case study + shader studies + tool demo + game-facing asset/pipeline breakdown

Conclusión: emotionally relevant but not the highest-probability route. Use as selected target, not core search axis.

5.9 Shader Technical Artist

Prioridad: subskill, not target role now.

Uso recomendado: strengthen portfolio language, not job title.

Dimensión	Evaluación
Current evidence	Shader Graph, visual modes, basic HLSL
Missing evidence	Advanced HLSL, custom lighting, render features, performance profiling, shader debugging
Risk	Overclaiming if applying to shader-specific roles
Recommended use	Shader Graph / basic HLSL / visual modes as skill cluster

Conclusión: build as supporting skill for Unity Technical Artist, not as role target in first 3–6 months.

5.10 Full-stack Developer

Prioridad: fallback economic route.

Uso recomendado: only if financial pressure forces broader applications.

Dimensión	Evaluación
Evidence	FastAPI/React/PostgreSQL/Redis listed as secondary and needing validation
Fit with strategy	Low; dilutes the 3D/Unity narrative
Remote feasibility	High
Competition	High
Risk	Becoming generic junior full-stack without strong differentiator
Proof minimum	Functional deployed app, tests, README, auth/data model, frontend polish

Conclusión: keep as contingency, not main profile.

5.11 3D Artist / Blender Artist / Generalist

Prioridad: downplay as job target.

Uso recomendado: portfolio support, not professional identity.

Dimensión	Evaluación
Evidence	Blender portrait and asset optimization
Risk	Competing against dedicated artists with stronger ArtStation portfolios
Strategic value	Supports Technical Artist credibility if framed as pipeline/process
Wrong framing	"3D Artist looking for character/prop roles"
Correct framing	"Understands art pipeline and optimization constraints"

Conclusión: do not lead with 3D Artist. Use Blender to prove technical art empathy and asset pipeline competence.

3.0.8 6. Role-by-role mapping of the original 23 requested roles

Original role	Final classification	Priority	Notes
Unity Technical Artist	Family: Unity Technical Artist	High-medium	Target when focused on optimization/tools/runtime, not senior AAA
Technical Artist	Umbrella keyword/family	Medium-high	Use with qualifier: Unity / real-time / optimization / visualization
Real-Time 3D Developer	Primary family	High	Strongest neutral role title
Unity Developer	Search keyword/family	Medium-high	Use if role has 3D/visualization; avoid generic mobile/game roles if weak fit
Unity WebGL Developer	Primary subfamily	High	Strong fit with TwinSight
Interactive 3D Developer	Primary family	High	Strongest remote-friendly framing
WebGL Developer	Secondary/adjacent	Medium-high	Strong if combined with 3D/Unity/Three.js
3D Web Developer	Secondary/adjacent	Medium-high	Useful search term; may require Three.js/WebGPU
Technical Visualization Developer	Primary/industry family	High	Excellent fit
CAD-to-Realtime Technical Artist	Keyword/differentiator	High as skill, low as title	Do not search only this title
Digital Twin Visualization Developer	Primary aspirational	Medium-high	Use carefully; avoid claiming full digital twin if no data sync
Simulation Developer	Primary aspirational/adjacent	Medium	Good with engineering background; remote varies
XR Developer	Secondary future	Medium	Needs XR proof
AR/VR Developer	Secondary future	Medium	Similar XR; hardware proof needed
3D Product Visualization Developer	Adjacent	Medium	Good for configurators/product/web 3D
Game Technical Artist	Aspirational	Medium-low	Not first wave unless junior/tool/Unity optimization
Tools Developer / Pipeline Developer	Secondary controlled	Medium-high	Strong if tool demo exists
Shader Technical Artist	Subskill/future specialization	Low now	Needs stronger shader portfolio
Python Automation Developer	Secondary controlled	Medium	Good if connected to tools/pipeline
AI Tools Developer	Secondary speculative	Medium-low	Requires polished ARA and clear demo
LLM Application Developer	Secondary speculative	Medium-low	High market upside but profile dilution risk
Full-stack Developer	Fallback	Low-medium	Economic contingency only
3D Artist / Blender Artist / 3D Generalist	Downplay	Low	Portfolio support only

3.0.9 7. Project-to-role evidence map

Project	Strongly proves	Partially proves	Does not prove
TwinSight X500	Unity, WebGL, C#, interactive 3D, technical visualization, CAD-to-realtime optimization, UI, visualization modes, academic validation	Tech art, digital twin adjacent, simulation adjacent, product visualization, UX research	Shipped commercial production, full digital twin, XR deployment, AAA game production, senior technical art
ARA Framework	Python automation, LLM orchestration awareness, agent workflow design, research tooling, documentation	Tools/pipeline mindset, AI tooling route, backend/API if implemented	Production ML engineering, enterprise AI product, primary technical art evidence
ai-news-aggregator	Full-stack/AI app if functional	Python/backend, product logic, data pipeline	Real-time 3D, technical art, Unity
FitApp-Free	Product thinking, app logic if functional	Full-stack/backend, data modeling	Technical art, Unity, 3D visualization
Hyperrealistic Blender portrait	Blender skill, topology, materials, art pipeline	Technical art if optimized for realtime/mobile	Character artist seniority, shipped game character pipeline unless properly converted
Older game/project repos	Possible historical interest	Coding progression	Current professional readiness unless cleaned/contextualized

3.0.10 8. Evidence required before applying by role family

8.1 Minimum viable evidence package for primary roles

For the first application wave, the minimum package should be:

Asset	Required?	Purpose
TwinSight case study	Yes	Main proof of role fit
60–120s TwinSight demo video	Yes	Recruiter-accessible evidence
Short technical breakdown	Yes	Show pipeline, architecture, optimization
GitHub README recruiter version	Yes	Explain project without requiring code review
Screenshots: UI, exploded view, clipping, visual modes	Yes	Visual proof
Metrics table	Yes	Triangles, participants, SUS, NASA-TLX; avoid unsupported performance claims
AI assistance disclosure script	Yes	Interview risk control
Live build	Optional but high value	Strongest proof if stable and legally safe
Code sample	Optional/controlled	Useful if repo is clean

8.2 Evidence by role

Role family	Must show before serious applications	Nice-to-have
Real-Time 3D / Interactive 3D	Demo video, UI/interactions, architecture, metrics	Live WebGL build
Unity WebGL / Web 3D	Browser deployment, loading/performance notes, mobile/browser screenshots	Three.js/WebGPU micro-demo
Technical Visualization / Digital Twin	Problem framing, CAD pipeline, relationships between parts, technical information panels	Data-driven mockup or telemetry concept
Unity Technical Artist	Optimization breakdown, visual modes/shaders, pipeline, before/after	Editor tool demo
Tools/Pipeline	Tool demo, GitHub repo, README, before/after workflow time	Unity EditorWindow polished
XR/AR/VR	Transferable Unity proof plus small XR prototype	Quest/HoloLens demo video
AI Tools/LLM	Working ARA demo, architecture diagram, examples, limitations	Deployment or evaluation harness
Game Technical Artist	Shader demos, optimization case, game-ready asset breakdown	Small playable Unity scene
Full-stack fallback	Deployed app, backend/frontend screenshots, API docs	Tests and Docker

3.0.11 9. Recommended positioning architecture

9.1 Primary positioning statement

Use this as the strategic anchor:

Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization, and WebGL deployment.

Spanish operational version:

Desarrollador 3D en tiempo real / Technical Artist Unity enfocado en visualización técnica interactiva, optimización CAD-to-realtime y despliegue WebGL.

9.2 Why this positioning works

Criterion	Evaluation
Uses strongest evidence	TwinSight directly supports it
Avoids generic multimedia label	Yes
Avoids pure art competition	Yes
Avoids overclaiming AI/ML	Yes
Keeps game industry possible	Yes, through technical art / Unity
Keeps industrial/digital twin possible	Yes
Fits remote contractor search	Better than defense/on-site simulation or pure game art
Builds toward 12–24m upside	Yes, via real-time 3D, simulation, visualization and tooling

9.3 Secondary positioning statements

Use only in targeted variants, not all at once.

Route	Statement
Unity WebGL / Interactive 3D	Unity WebGL Developer building browser-based interactive 3D applications for technical visualization.
Technical Visualization / Digital Twin	Technical Visualization Developer translating CAD and engineering assets into interactive real-time 3D systems.
Unity Technical Artist	Unity Technical Artist focused on optimization, runtime visualization systems, shaders, and asset pipelines.
Tools / Python Automation	Python/Unity tools developer building automation prototypes for research and production workflows.
XR / Simulation	Unity developer with real-time 3D visualization background, preparing for XR/simulation applications.

9.4 What not to use as primary positioning

Avoid as main identity	Reason
3D Artist	Competes in saturated art market; underuses engineering/Unity evidence
AI Engineer	ARA does not prove ML engineering or production AI yet
Full-stack Developer	Dilutes differentiation; no strongest evidence
Game Technical Artist alone	Too narrow and high competition; current strongest proof is technical visualization, not game production
Digital Twin Developer alone	Risk of overclaiming if no real sensor/data integration
Shader Technical Artist	Current shader evidence likely insufficient for shader-specialist roles
Senior Technical Artist	Not supported by formal industry experience

3.0.12 10. ATS keyword clusters

10.1 Core universal cluster

Use across most variants:

Bloque de codigo: text

Unity, C#, Real-Time 3D, WebGL, Interactive 3D, Technical Visualization, Blender, CAD-to-Realtime, 3D Optimization, Asset
↪ Optimization, Retopology, UV Mapping, Baking, URP, UI Toolkit, Shader Graph, Git, Technical Documentation, Performance
↪ Optimization

10.2 Unity Technical Artist cluster

Bloque de codigo: text

Technical Artist, Unity Technical Artist, Unity, C#, URP, Shader Graph, HLSL, Real-Time Rendering, Optimization, Asset
↪ Pipeline, Tools, Editor Scripting, Blender, Retopology, Baking, Visual Modes, X-Ray, Wireframe, Blueprint, Technical UI

10.3 Interactive 3D / WebGL cluster

Bloque de codigo: text
Interactive 3D Developer, Real-Time 3D Developer, Unity WebGL, WebGL, WebAssembly, Browser-based 3D, 3D Visualization,
↔ Technical UI, Runtime Interaction, Mobile Optimization, Loading Performance, Frontend 3D, Three.js, WebGPU

10.4 Technical Visualization / Digital Twin / Simulation cluster

Bloque de codigo: text
Technical Visualization, Digital Twin, Simulation, Industrial Visualization, CAD, CAD Optimization, CAD-to-Realtime, Assembly
↔ Visualization, Product Visualization, HMI, Real-Time Monitoring, Unity, C#, 3D UI, Engineering Visualization, Drones,
↔ Robotics, Hardware Visualization

10.5 Tools / Pipeline / Python cluster

Bloque de codigo: text
Tools Developer, Pipeline Developer, Python Automation, Unity Editor Tools, C#, Python, Workflow Automation, LangGraph,
↔ LangChain, LLM Orchestration, Research Automation, API Integration, Redis, Playwright, Semantic Scholar, Technical
↔ Documentation

10.6 XR cluster

Bloque de codigo: text
XR Developer, AR Developer, VR Developer, Unity XR, Meta Quest, HoloLens, MRTK, ARCore, ARKit, Spatial Interaction, Immersive
↔ Training, Simulation, 3D Interaction, Real-Time Performance

10.7 Terms to use carefully

Term	Condition for use
Digital Twin	Use as target market/adjacent field; do not call TwinSight a full digital twin unless data integration exists
AI Tools Developer	Use only when ARA is presentable and limitations are explicit
Shader Technical Artist	Use only as skill or future direction unless shader portfolio is strong
Full-stack	Use only in fallback CV variant
Senior	Avoid unless role is not level-sensitive and phrasing is not self-title

3.0.13 11. Strategic contradictions to avoid

Contradiction	Why it is dangerous	Correction
Claiming thesis is ARA while thesis is TwinSight	Damages credibility and creates inconsistency across LinkedIn/CV	Thesis = TwinSight X500. ARA = secondary project
Presenting as Technical Artist + AI Developer + Full-stack + 3D Artist equally	Dilutes recruiter signal	Primary = Real-Time 3D / Unity Technical Visualization. Others secondary
Calling TwinSight a complete digital twin	Could be technically false if no live data/sensor sync	Use interactive technical visualization or digital twin-adjacent prototype
Leading with game industry while strongest proof is industrial/technical visualization	Lowers probability and increases competition	Target games selectively; lead with visualization/Unity/WebGL
Using CAD-to-realtime as only job title	Few postings use this exact title	Use as differentiating skill under visualization/simulation roles
Presenting ARA as production AI platform	Risk if unstable or not deployed	Present as prototype/tooling proof with limitations
Claiming EU work authorization	Legally false until citizenship/residence/work rights exist	Say remote from Colombia; EU mobility future only if true
Claiming certified training from non-certified courses	Misrepresentation	Completed self-directed training / coursework, not certification
Applying as Shader TA without shader portfolio	Technical interview risk	Use shaders as supporting evidence for Unity TA
Applying as full-stack while portfolio says technical art	Profile incoherence	Use separate fallback CV only if needed

3.0.14 12. Recommended first-wave role families

The first 3–6 month search should target **3–5 role families**, not 23 titles.

12.1 Primary first-wave targets

1. Real-Time 3D / Interactive Technical Visualization Developer **Why:** best overall fit. Converts TwinSight directly into employability evidence.

Search titles: Real-Time 3D Developer, Interactive 3D Developer, Unity Developer, 3D Application Developer, Technical Visualization Developer.

Markets: global remote, LATAM nearshore, creative tech, education tech, industrial visualization, product visualization.

2. Unity WebGL / Interactive 3D Developer **Why:** strongest remote-compatible route from TwinSight.

Search titles: Unity WebGL Developer, WebGL Developer, 3D Web Developer, Frontend 3D Developer, Interactive Web Developer.

Markets: web 3D studios, configurators, marketing/creative tech, technical education, product visualization.

3. Technical Visualization / Digital Twin / Simulation Developer **Why:** strongest long-term upside and best use of engineering + multimedia background.

Search titles: Technical Visualization Developer, Digital Twin Developer, Simulation Developer, Industrial Unity Developer, HMI Unity Developer.

Markets: manufacturing, robotics, drones, aerospace, energy, medical visualization, industrial training.

4. Unity Technical Artist — optimization/tools/runtime **Why:** preserves technical art identity while avoiding pure art/generalist competition.

Search titles: Unity Technical Artist, Technical Artist, Runtime Technical Artist, Technical Artist – Optimization, Technical Artist – Tools.

Markets: game studios selectively, simulation studios, Unity agencies, XR studios, mobile/interactive companies.

12.2 Secondary controlled target

5. Tools / Pipeline / Python Automation Developer **Why:** ARA and future extracted Unity tools can provide differentiation.

Search titles: Tools Developer, Pipeline Developer, Python Automation Developer, Unity Tools Developer, Technical Artist – Tools.

Markets: studios, automation startups, internal tools teams, technical content/research automation.

3.0.15 13. Roles to downplay or reject in first wave

Role	Decision	Reason
3D Artist / Blender Artist	Downplay	Saturated, art-first, weaker salary leverage, underuses Unity/WebGL
Character Artist	Reject as current target	Blender portfolio alone is not enough; not primary career direction
Shader Technical Artist	Downplay to subskill	Current shader evidence likely insufficient for specialist roles
Game Technical Artist AAA	Selective only	High competition, often requires shipped titles and production experience
AI Engineer / ML Engineer	Reject as current target	ARA is not enough for ML/AI engineering claims
Full-stack Developer	Fallback only	Viable for survival but dilutes positioning
QA Manual	Reject unless survival fallback	Does not align with salary/upside; old “Trojan Horse” needs revalidation

Role	Decision	Reason
QA Automation / SDET	Contingency only	Could use engineering background, but not preferred if 3D/Unity route is viable

3.0.16 14. Remote-from-Colombia implications by role

Role family	Remote worldwide probability	Contractor compatibility	Main barrier
Unity WebGL / Interactive 3D	High	High	Need strong demo/live proof
Real-Time 3D / Technical Visualization	Medium-high	Medium-high	Some companies prefer local/hybrid
Tools / Python Automation	High	High	Need code quality and functional tooling proof
Unity Technical Artist	Medium	Medium	Studio region restrictions, portfolio expectations
XR / Simulation	Medium-low	Medium	Hardware/on-site/lab constraints
Digital Twin / Industrial Simulation	Medium-low	Medium	Enterprise/local/industrial access, domain expectations
Game Technical Artist	Medium-low	Medium	Competition, time zone, studio constraints
Full-stack fallback	High	High	Heavy competition, less differentiation
3D Artist	Medium	Medium	Saturated global freelance market

Implication: the strongest remote-first combination is not “game technical artist”. It is **Unity WebGL + interactive 3D + technical visualization**, with technical art as supporting identity.

3.0.17 15. Short-term vs long-term role strategy

15.1 First 3–6 months

Focus on roles that can accept an academically strong but industry-entry profile:

1. Interactive 3D Developer.
2. Unity WebGL Developer.
3. Real-Time 3D Developer.
4. Technical Visualization Developer.
5. Unity Technical Artist — junior/associate/optimization/tools/runtime.

Avoid first-wave applications to senior AAA technical art, shader specialist, pure AI engineer, pure full-stack, and pure 3D artist roles unless there is a specific favorable posting.

15.2 Months 6–12

Expand after portfolio feedback:

- Add XR prototype if XR demand appears.
- Add Unity Editor Tool if tools roles appear.
- Add Three.js/WebGPU micro-project if Web 3D roles appear.
- Publish ARA if AI/tooling roles are producing leads.
- Build stronger shader breakdowns if Unity TA roles demand shaders.

15.3 Months 12–24

Move toward higher-upside specialization:

- Technical Visualization Engineer.
- Digital Twin / Simulation Developer.
- Unity Technical Artist — tools/optimization.
- XR/Simulation Developer.
- AI-assisted pipeline/tools developer for 3D production.

Future EU/Portuguese citizenship, if legally obtained, improves access to EU on-site/hybrid roles, but does not change current remote-from-Colombia positioning.

3.0.18 16. Role-specific risk register

Risk	Affected roles	Severity	Mitigation
Academic-only perception	All primary roles	High	Case study must be industry-facing, not thesis-facing
No formal studio experience	Technical Artist, Game TA, XR	High	Emphasize project evidence, metrics, systems, documentation
No live deployed demo	WebGL/Interactive 3D	Medium-high	Provide stable video if live build is risky
AI-assisted development skepticism	All coding roles	Medium-high	Prepare transparent explanation of architecture/debugging/ownership
Overclaiming digital twin	Digital Twin/Simulation	High	Use precise terms: technical visualization, assembly visualization
Weak shader evidence	Shader TA/Tech Art	Medium	Treat shaders as supporting skill, not specialization
ARA instability	AI Tools/Python	Medium-high	Keep secondary until cleaned and documented
Too many directions	All	High	Use one primary positioning and targeted variants only
Remote restrictions	EU/US employee roles	High	Prioritize contractor/EOR/remote-global filters

3.0.19 17. Decisions locked by this section

17.1 Strategic decisions

1. The main role family is **Real-Time 3D / Interactive Technical Visualization**.
2. The best marketable title is not Multimedia Engineer and not 3D Artist; it is a hybrid of Real-Time 3D Developer and Unity Technical Artist.
3. CAD-to-realtime is a **core differentiator**, not a standalone job title.
4. Digital Twin is a **target market/adjacent role**, not a claim for TwinSight unless the system has real data integration.
5. ARA is a **secondary proof of tooling/automation**, not the professional center.
6. Full-stack is a **fallback**, not a positioning route.
7. Game Technical Artist is a **selective/aspirational route**, not the first filter.
8. Shader Technical Artist is a **future specialization**, not immediate target.

17.2 Required downstream consequences

Future section	Consequence
03_salary_benchmark_and_remote_colombia.md	Benchmark salaries by the role families defined here, not all 23 raw titles equally

Future section	Consequence
07_portfolio_strategy_and_project_architecture.md	Portfolio must lead with TwinSight as technical visualization, not generic thesis
08_twinsight_x500_case_study.md	Case study must explicitly support roles 1–4 above
09_ara_ai_tools_and_secondary_projects.md	ARA must be framed as secondary tools/automation route
10_github_linkedin_cv_strategy.md	CV/LinkedIn must use one core headline and targeted variants
11_company_targets_job_boards_recruiters.md	Company list should prioritize interactive 3D, Unity WebGL, visualization, digital twin, simulation and tools
13_interview_negotiation_and_offer_risk.md	Interview prep must handle AI-assistance, academic project framing and no formal studio experience

3.0.20 18. Open questions for later sections

These are not blockers for role positioning, but must be answered before final strategy.

18.1 Salary / compensation

- What are realistic contractor rates from Colombia for Unity/WebGL/interactive 3D?
- How do salaries differ between game studios, simulation companies, industrial visualization and web 3D agencies?
- What is the probability of reaching USD 1.5k, 3k and 6k by timeline?
- Which role family gives the best short-term salary vs long-term upside tradeoff?

Handle in: 03_salary_benchmark_and_remote_colombia.md.

18.2 Portfolio evidence

- Is there a live TwinSight build suitable for recruiters?
- Are screenshots and demo footage ready?
- Are metrics verified in the final thesis document?
- Can any Unity tool be extracted safely for GitHub?
- Is ARA functional enough to publish?

Handle in: 07, 08, 09, 10.

18.3 Market targeting

- Which companies actually hire remote LATAM for Unity/WebGL/technical visualization?
- Which companies require local authorization?
- Which job boards produce low-noise opportunities?
- Which recruiters specialize in Unity/3D/XR/simulation?

Handle in: 11_company_targets_job_boards_recruiters.md, likely with mode agent.

18.4 Skills gap

- Should Alexander prioritize Three.js/WebGPU, XR SDKs, Shader/HLSL, Unity profiling, or Python tools first?
- Which one produces the fastest portfolio proof?
- Which one most improves the first-wave role families?

Handle in: 06 and later skills roadmap sections.

3.0.21 19. Final role recommendation

The first search wave should target these 5 families, in this order:

1. **Real-Time 3D / Interactive Technical Visualization Developer**
2. **Unity WebGL / Interactive 3D Developer**
3. **Technical Visualization / Digital Twin / Simulation Developer**
4. **Unity Technical Artist — optimization/tools/runtime**
5. **Tools / Pipeline / Python Automation Developer**

The profile should not be marketed as a generic multimedia engineer, generic 3D artist, AI engineer, full-stack developer, or senior technical artist.

The most precise positioning for the rest of the strategy is:

Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization, and WebGL deployment.

This positioning gives enough range to apply across technical visualization, simulation, Unity WebGL, product visualization, creative technology, digital twin-adjacent and selected game technical art roles without making the profile incoherent.

3.0.22 20. Sources

[^unity_web]: Unity Manual — Web platform. Unity states that Unity supports developing games on the Web platform and that WebGL lets developers publish real-time interactive 3D graphics into a browser. <https://docs.unity3d.com/Manual/webgl.html>

[^khronos_webgl]: Khronos Group — WebGL: 3D Graphics for the Web. Khronos defines WebGL as a low-level 3D graphics API based on OpenGL ES that brings plugin-free 3D to the web and is implemented directly in browsers. <https://www.khronos.org/webgl/>

[^langgraph_ga]: LangChain — LangGraph Platform is now Generally Available. LangChain announced LangGraph Platform GA in May 2025 for deploying and managing long-running, stateful agents; relevant to ARA as a tooling/agent workflow context, not as proof of production-grade AI engineering. <https://www.langchain.com/blog/langgraph-platform-ga>

[^unity_dt_robotics]: Xiang et al. — Development of a Simple and Novel Digital Twin Framework for Industrial Robots in Intelligent Robotics Manufacturing, 2024. The paper describes a digital twin framework using Unity 3D and a web-based platform for real-time visualization and control in intelligent manufacturing. <https://arxiv.org/abs/2410.14934>

[^unity_hpc_dt]: Bergeron et al. — Supercomputer 3D Digital Twin for User Focused Real-Time Monitoring, 2024. The paper reports use of Unity 3D for digital twin visualization of HPC systems. <https://arxiv.org/abs/2410.00688>

[^deep_research_note]: The prior Deep Research output for this section used job postings and salary aggregators such as ZipRecruiter, Indeed and Wellfound. Those sources are treated here as market signals, not as final compensation evidence. Detailed salary validation is deferred to 03_salary_benchmark_and_remote_colombia.md.

Chapter 4

03 — Salary Benchmark and Remote-from-Colombia Feasibility

Fuente: 03_salary_benchmark_and_remote_colombia.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 808

Ultima modificacion: 2026-06-11 12:36

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Perfil: Alexander Woodcock Salomón **Fecha de corte operativa:** 2026-06-11 **Estado del documento:** sección corregida y terminada para integración posterior. **Modo usado para esta sección:** Deep Research previo + recolección estructurada con modo agente + verificación web puntual + reconstrucción analítica manual. **Herramientas adicionales recomendadas para esta sección:** no usar más Deep Research ni modo agente para esta sección salvo que se quiera auditar con un c

Contenido completo depurado

4.0.1 Auditoría y reconstrucción del benchmark salarial, contratación remota y viabilidad administrativa 2025–2026

Perfil: Alexander Woodcock Salomón

Fecha de corte operativa: 2026-06-11

Estado del documento: sección corregida y terminada para integración posterior.

Modo usado para esta sección: Deep Research previo + recolección estructurada con modo agente + verificación web puntual + reconstrucción analítica manual.

Herramientas adicionales recomendadas para esta sección: no usar más Deep Research ni modo agente para esta sección salvo que se quiera auditar con un contador colombiano.

Regla crítica: esta sección no es asesoría tributaria, legal, migratoria ni contable. Sirve para estrategia laboral y estimación de escenarios. Las decisiones fiscales deben validarse con contador en Colombia.

4.0.2 0. Propósito de esta sección

Este archivo define qué rangos salariales y modelos de contratación son realistas para Alexander como candidato ubicado en Colombia, con perfil de **Unity / Real-Time 3D / Technical Visualization / Technical Artist**, portafolio fuerte en TwinSight X500 y experiencia formal limitada en industria.

La sección cumple cuatro funciones:

1. **Auditar el benchmark salarial del informe anterior.** El deep research original mezclaba salarios globales, salarios locales, contractor rates y job postings sin suficiente separación.
2. **Separar empleado, contractor, EOR y freelance.** Para Alexander, el valor estratégico de una cifra salarial depende de si puede accederse desde Colombia.
3. **Modelar metas de USD 1.5k, USD 3k y USD 6k mensuales.** No basta con saber que existen salarios altos; hay que estimar probabilidad y condiciones.
4. **Definir implicaciones administrativas básicas en Colombia.** RUT, facturación, IVA, seguridad social, pagos internacionales y riesgo de mala clasificación afectan el ingreso neto.

Esta sección alimenta directamente:

- 04_eu_portugal_germany_mobility_strategy.md
- 10_github_linkedin_cv_strategy.md
- 11_company_targets_job_boards_recruiters.md
- 13_interview_negotiation_and_offer_risk.md
- 14_30_60_90_execution_plan.md

4.0.3 1. Fuentes utilizadas y auditoría de calidad

1.1 Fuentes internas usadas

Fuente	Uso	Estado
01_source_of_truth_profile.md	Perfil real, límites de afirmación	Vigente
02_market_role_fit_and_positioning.md	Familias de rol priorizadas	Vigente
Deep Research sección 03	Benchmark salarial inicial	Útil, pero genérico
Recolección con agente	Datos por fuente y rol	Útil, requiere limpieza
deep-research-report (8).md	Informe base a auditar	Parcialmente débil

1.2 Fuentes externas usadas

Fuente	Tipo	Confianza
Howdy LATAM salary report	Payroll / EOR data	Media-alta
Lemon.io Unity rates	Contractor marketplace	Media
Qubit Labs Unity guide	Recruiting / outsourcing	Media-baja
ZipRecruiter	Salary aggregator	Media-baja
ARVR Jobs salary insights	Job-posting aggregation	Media
HireTalent LATAM Python guide	Recruiting / salary guide	Media
Colombia Move	Guía práctica tributaria	Media-baja
SIA Seguros	Guía práctica seguridad social	Media-baja
Papaya Global	Contractor payment guide	Media-baja
Mismo	Contractor compliance guide	Media-baja

1.3 Lectura crítica de fuentes

No todas las fuentes tienen el mismo peso. Las fuentes de payroll y contratos reales son mejores que los agregadores de salarios. Los agregadores como ZipRecruiter son útiles para señales de mercado, pero mezclan ubicación, nivel, industria, tamaño de empresa, empleador local y trabajo remoto. Las guías de outsourcing suelen mostrar cifras desde la perspectiva de empresas que quieren contratar más barato, por lo que deben tratarse como **datos comerciales**, no como verdad neutral.

El informe anterior falló principalmente en tres puntos:

1. **Confundió salario local con salario accesible.** Un salario de EE. UU. no implica que una empresa pague ese monto a un contractor ubicado en Colombia.
2. **No separó gross de net.** USD 3,000/mes brutos como contractor no son USD 3,000/mes disponibles.
3. **Usó rangos demasiado amplios.** Para decidir estrategia, hace falta separar rol, seniority, modalidad contractual y restricción geográfica.

4.0.4 2. Metodología de benchmark

2.1 Categorías de salario

Categoría	Significado
Local employee	Nómina local en país específico
EOR employee	Empleado legal por intermediario
Direct contractor	Prestación de servicios / B2B
Marketplace contractor	Plataforma tipo Lemon.io
Freelance project	Proyecto puntual por alcance
Job-posting salary	Oferta o agregador de vacantes
Strategic estimate	Inferencia razonada

2.2 Conversión mensual

Para efectos estratégicos:

```
Bloque de código: text
monthly gross = annual gross / 12
monthly contractor full-time = hourly rate × 160 hours
annual contractor full-time = hourly rate × 2,080 hours
```

Estas conversiones son aproximadas. Un contractor rara vez factura 160 horas productivas todos los meses durante 12 meses sin interrupción. Para modelar ingreso conservador, conviene usar 120–140 horas facturables/mes.

2.3 Confidence rating

Rating	Uso
Alta	Payroll, fuente oficial, contrato real
Media	Marketplace, job posting amplio
Baja	Agregador, muestra pequeña, vendor
Muy baja	Snippet, SEO, dato no verificable

4.0.5 3. Conclusión ejecutiva salarial

Para Alexander, las rutas más coherentes con el objetivo económico son:

1. **Unity WebGL / Interactive 3D / Real-Time 3D contractor desde Colombia.** Es la ruta más alineada con TwinSight y con el objetivo de USD 1.5k–3k mensual en 3–6 meses.
2. **Technical Visualization / Digital Twin / Simulation.** Es la ruta con mayor upside a 12–24 meses, pero requiere empaquetar TwinSight como caso técnico serio y buscar empresas fuera de videojuegos.
3. **Unity Technical Artist — optimization/tools/runtime.** Es útil como etiqueta profesional, pero no debe confundirse con Game Technical Artist AAA.

- 4. **Python Automation / AI Tools.** Es ruta secundaria. Puede ayudar a monetizar ARA, pero no debe liderar la búsqueda inicial.
- 5. **3D Artist / Blender Generalist.** Debe quedar como fallback débil o complemento visual. No maximiza salario ni explota suficientemente TwinSight.

La meta de **USD 1.5k/mes** es realista en corto plazo. La meta de **USD 3k/mes** es alcanzable, pero exige CV, LinkedIn, caso TwinSight, demo y outreach bien ejecutados. La meta de **USD 6k/mes** no debe tratarse como objetivo base en 3–6 meses; es una meta de 12–24 meses si se logra entrar en technical visualization, simulation, XR, digital twin o contractor US/EU con suficiente evidencia.

4.0.6 4. Benchmark salarial por familia de rol

4.1 Real-Time 3D / Interactive 3D / Unity Developer

Esta familia es la base salarial más útil porque los datos de mercado para Unity Developer son más abundantes que para Unity WebGL Technical Visualization Developer.

Señales principales

- Howdy reporta que los salarios promedio de desarrolladores en LATAM en 2025 se ubican aproximadamente entre **USD 53k y USD 63k/año**, con Colombia alrededor de **USD 55,894/año** en su dataset de payroll. También reporta bandas LATAM de **USD 18k–28k junior**, **USD 35k–48k mid** y **USD 55k–70k senior**.^[^howdy]
- Lemon.io reporta que un **Middle Unity Developer en Colombia** factura alrededor de **USD 27–35/h**, mientras que senior Unity developers en EE. UU. tienen mediana de **USD 57/h** y strong seniors en San Francisco llegan a **USD 72–89/h**.^[^lemon-core]
- Qubit Labs reporta salarios mid-level Unity por país, con cifras como EE. UU. **USD 116,998**, Canadá **USD 140,000**, Alemania **USD 61,197**, UK **USD 58,709**, España **USD 58,719** y Brasil **USD 27,000**.^[^qubit-country]

Matriz operativa

Mercado	Rango útil	Relevancia
Colombia / LATAM contractor	USD 27–35/h	Alta
LATAM EOR software dev	USD 35k–70k/año	Alta
US employee	USD 100k–150k/año	Baja inmediata
EU employee	USD 58k–95k/año	Media futura
Global freelance	USD 20–50/h	Media

Lectura para Alexander Esta es la ruta con mejor relación entre evidencia disponible y empleabilidad. TwinSight prueba Unity, C#, interacción 3D, UI técnica, optimización de assets y despliegue web. El riesgo es que muchas vacantes Unity están orientadas a juegos, mobile o VR. La búsqueda debe priorizar Interactive 3D, Technical Visualization, WebGL, Simulation, Product Visualization y Unity Developer fuera de game-only pipelines.

4.2 Unity Technical Artist / Technical Artist

El rol de Technical Artist tiene upside alto, pero la evidencia salarial es menos limpia. En Estados Unidos, ZipRecruiter reporta para Technical Artist un promedio de **USD 138,713/año**, con un rango típico entre **USD 127k y USD 152k**.^[^zip-ta]

En UK, ITJobsWatch mostró datos muy limitados para Technical Artist, con pocos anuncios y salario mediano histórico alrededor de **GBP 68,500** en 2025, pero sin suficiente información para mediana 2026.^[^itjobswatch-ta]

Matriz operativa

Mercado	Señal salarial	Relevancia
US Technical Artist	USD 127k–152k	Baja inmediata
UK Technical Artist	~GBP 68.5k	Baja-media
LATAM Technical Artist	Not found	Baja
Unity TA contractor	Proxy Unity rate	Media

Lectura para Alexander Technical Artist debe usarse como parte del posicionamiento, pero con precisión. No conviene apuntar a **Game Technical Artist AAA** como ruta principal inmediata. Sí conviene apuntar a:

- Unity Technical Artist — Optimization
- Runtime Technical Artist
- Technical Artist — Tools
- Technical Artist — Interactive 3D
- Technical Artist — Visualization

El portafolio debe mostrar breakdowns técnicos. Sin shader breakdown, editor tools, profiling, optimization before/after y demo ejecutable, el título puede sonar sobredimensionado.

4.3 Unity WebGL / Web 3D Developer

No hay suficiente data salarial específica para Unity WebGL Developer. La mayoría de fuentes lo absorben dentro de Unity Developer, WebGL Developer, Interactive 3D Developer, Frontend 3D Developer o Three.js Developer.

Proxies razonables

Proxy	Rango útil
Middle Unity Colombia	USD 27–35/h
Mid Unity global	USD 73k–125k/año
WebGL / 3D frontend	Variable
Product visualization	Variable

Lectura para Alexander Esta ruta es estratégica aunque la data salarial sea incompleta, porque TwinSight es una prueba directa de Unity WebGL. La dificultad es que las vacantes explícitas Unity WebGL Developer son menos frecuentes que las de Unity general. La búsqueda debe usar combinaciones:

```
Bloque de código: text
Unity WebGL Developer
Interactive 3D Developer
WebGL 3D Developer
Real-Time 3D Developer
Technical Visualization Developer
3D Web Application Developer
Unity Product Visualization
Unity Simulation WebGL
```

El objetivo no es encontrar solo ofertas con el título exacto. El objetivo es usar WebGL como diferenciador dentro de una ruta de visualización técnica.

4.4 Technical Visualization / Digital Twin / Simulation

Esta es la ruta con mejor upside a 12–24 meses. En EE. UU., ZipRecruiter reporta para Digital Twin Engineer un promedio de **USD 139,368/año** y para Simulation Software Engineer un promedio de **USD 137,846/año**, con rango 25–75 aproximadamente **USD 104k–162k** para Simulation Software Engineer.[^zip-dt][^zip-sim]

Matriz operativa

Rol	US promedio	Relevancia
Digital Twin Engineer	USD 139k/año	Media futura
Simulation Software Engineer	USD 138k/año	Media futura
Technical Visualization	Not found	Alta estratégica
Industrial 3D Developer	Not found	Alta estratégica

Lectura para Alexander TwinSight encaja muy bien en esta ruta porque convierte documentación/CAD/ensamblaje en experiencia 3D interactiva. Sin embargo, muchas ofertas de digital twin y simulation pueden exigir:

- ingeniería mecánica, sistemas, IoT o datos en tiempo real;
- experiencia con sensores, CAD industrial, PLM, BIM o cloud;
- autorización laboral local;
- experiencia formal previa;
- seguridad industrial o defense-adjacent restrictions.

La ruta no debe venderse como Digital Twin Engineer si TwinSight no integra datos vivos, IoT, sensores o sincronización en tiempo real. La formulación más honesta es:

Bloque de código: text
Interactive technical visualization / CAD-to-realtime Unity prototype with digital-twin-adjacent relevance.

4.5 XR / AR / VR Developer

ARVR Jobs reporta datos altos para EE. UU.: junior **USD 84,500–135,000**, mid **USD 145,062–223,459** y senior **USD 163,857–248,827**. La propia fuente advierte que los datos son limitados y que provienen de postings agregados de compañías como Apple, Meta, Google y Snap.[^arvr]

Matriz operativa

Mercado	Señal salarial	Viabilidad
US big tech XR	Muy alta	Baja inmediata
Startup XR remote	Media-alta	Media
LATAM Unity XR contractor	Proxy Unity rate	Media
EU XR	Variable	Futura

Lectura para Alexander XR no debe ser foco inmediato a menos que se cree una demo específica en XR. TwinSight puede adaptarse conceptualmente a XR, pero actualmente prueba mejor visualización web que AR/VR. El rol debe quedar como ruta de expansión, no como target principal inicial.

4.6 Tools / Pipeline Developer

Los datos específicos de Tools Developer son escasos. La fuente de salario debe modelarse por proxies:

- Technical Artist con herramientas.
- Unity Developer con editor scripting.
- Software Engineer general.
- Python Automation Developer.

Lectura para Alexander Esta ruta puede ser valiosa si ARA y scripts de pipeline se presentan correctamente. No obstante, para ganar entrevistas como Tools Developer, se necesita evidencia adicional:

- editor tools en Unity;
- automatización Blender/Python;
- pipeline CAD-to-realtime documentado;
- before/after de optimización;
- scripts limpios y públicos;
- repositorio con README técnico.

Sin esa evidencia, Tools Developer debe ser secundario.

4.7 Python Automation / AI Tools / LLM Application Developer

HireTalent reporta para Python Developer remoto en LATAM una mediana de **USD 44k/año**, entry alrededor de **USD 30k/año**, mid **USD 46k/año** y senior **USD 62k/año**. Para Colombia, reporta rango entry-to-senior de **USD 30k–66k/año**, con mensualidad aproximada de **USD 3k–6k**.^[hiretalent-python]

Matriz operativa

Ruta	Rango útil	Prioridad
Python Automation	USD 30k–62k	Media
LLM App Developer	Variable	Media-baja
AI Engineer	Alto	Baja actual
ML Engineer	Alto	No viable ahora

Lectura para Alexander ARA puede ayudar, pero no debe ser el centro de la búsqueda inicial. La ruta AI Tools puede servir como fallback o complemento si se arma una demo funcional y honesta. No debe venderse como ML/AI engineering. La formulación defendible es:

Bloque de código: text
Python automation and LLM-assisted workflow tooling.

4.8 Full-stack Developer fallback

Full-stack es fallback, no eje. Puede ofrecer acceso a empleos más abundantes, pero diluye la ventaja de Twin-Sight. Además, competiría contra perfiles con experiencia comercial directa en React, backend, bases de datos, deployment y QA.

La única forma razonable de usarlo es:

Bloque de código: text
Unity / 3D web developer with enough full-stack literacy to integrate APIs, dashboards and deployment workflows.

4.9 3D Artist / Blender Generalist fallback

ZipRecruiter reporta un promedio de **USD 82,070/año** para 3D Artist en EE. UU., con rango amplio de **USD 41k–142k**.^[^zip-3dartist]

Lectura para Alexander Esta ruta debe rechazarse como principal. El mercado de artistas 3D puros es muy competitivo, visualmente exigente y no aprovecha lo diferencial del perfil: ingeniería, Unity, WebGL, technical visualization, CAD-to-realtime, programación y evaluación académica.

El retrato Blender debe usarse como soporte visual y técnico, no como reposicionamiento a character artist.

4.0.7 5. Benchmark por región

5.1 Colombia y LATAM

Howdy reporta para LATAM un rango promedio de **USD 53k–63k/año** en software developers, con Colombia en **USD 55,894/año**. También reporta bandas LATAM de **USD 18k–28k junior**, **USD 35k–48k mid**, **USD 55k–70k senior**.^[^howdy]

Lemon.io reporta que un Middle Unity Developer en Colombia factura **USD 27–35/h**.^[^lemon-core]

Lectura Para Alexander, LATAM contractor es el mejor primer mercado si el portafolio queda bien presentado. El target bruto razonable inicial no debe ser el promedio LATAM alto completo, sino una banda escalonada:

Etapas	Bruto mensual razonable
Entrada rápida	USD 1.5k–2.5k
Buen primer contrato	USD 2.5k–4k
Contractor fuerte	USD 4k–6k
Exceptional LATAM	USD 6k+

5.2 Estados Unidos

EE. UU. muestra el mayor upside: Technical Artist, Digital Twin, Simulation y XR pueden superar **USD 130k/año** en datos de agregadores y postings.^{[^zip-ta][^zip-dt][^zip-sim][^arvr]}

Lectura La barrera no es salarial. La barrera es accesibilidad:

- remote US-only;
- autorización laboral;
- contractor compliance;
- competencia senior;
- necesidad de experiencia formal;
- requisitos de seguridad, hardware, defense o residencia.

La estrategia correcta no es aplicar masivamente a empleos US-only, sino buscar:

- remote LATAM;
- worldwide contractor;
- US startups con contractor;
- creative tech studios;
- product visualization firms;
- technical visualization vendors;
- agencies que ya contraten LATAM.

5.3 Canadá

Qubit Labs reporta salarios altos para Unity en Canadá, especialmente Toronto, con mid **USD 105,983** y senior **USD 145,000**.^[^qubit-city]

Lectura Canadá puede pagar bien, pero muchas ofertas son Canada-only. No es mercado prioritario salvo que la oferta especifique contractor o remote worldwide.

5.4 Reino Unido

Qubit Labs reporta Unity London mid **USD 77,512** y senior **USD 81,476**, mientras ITJobsWatch muestra baja cantidad de datos para Technical Artist.[^qubit-city][^itjobswatch-ta]

Lectura El UK puede ser interesante para estudios y creative tech, pero remoto desde Colombia no es común. Prioridad media-baja inicial.

5.5 Alemania

Qubit Labs reporta Unity Berlin mid **USD 83,475** y senior **USD 111,300**.[^qubit-city]

Lectura Alemania tiene sentido a mediano plazo por posible ruta de pareja/UE y sectores industriales. En corto plazo, sin alemán ni autorización laboral UE, debe tratarse como ruta futura. Para remote contractor desde Colombia, conviene buscar empresas alemanas que acepten proveedores externos, no nómina alemana.

5.6 España y Portugal

Qubit Labs reporta España alrededor de **USD 58,719/año** para mid Unity.[^qubit-country] Portugal aparece en Lemon.io como parte de las bancas Unity de Europa occidental, pero sin cifra específica en la tabla visible.[^lemon-location]

Lectura España/Portugal son interesantes por idioma/cultura y futura ciudadanía portuguesa, pero no son los mercados salariales más altos. Sirven mejor como:

- puente UE;
- mercado de estudios/creative tech;
- relocation futura;
- networking hispanohablante;
- contratación europea más flexible.

5.7 Netherlands / Nordics

Qubit Labs reporta Netherlands **USD 76,073**, Norway **USD 73,419** y Denmark **USD 82,225** para mid Unity.[^qubit-country]

Lectura Buen potencial, pero generalmente requieren autorización laboral o presencia europea. Prioridad futura, no inmediata.

5.8 Australia

Qubit Labs reporta Sydney mid Unity **USD 85,668** y senior **USD 104,582**.[^qubit-city]

Lectura No es buen mercado inicial por zona horaria y restricciones. Solo aplicar si la oferta es remote worldwide.

4.0.8 6. Modalidades de contratación

6.1 Comparación operativa

Modelo	Acceso desde Colombia	Recomendación
Empleado Colombia	Alto	Solo si paga bien
Contractor Colombia	Alto	Útil localmente
Foreign contractor	Alto-medio	Prioridad
EOR Colombia	Medio	Bueno si empresa seria
Foreign employee directo	Bajo	Poco realista
Freelance marketplace	Medio	Complementario
Relocation EU/US	Bajo actual	Futuro

6.2 Foreign contractor

Es el modelo más útil en los próximos 3–6 meses. Permite cobrar en USD o EUR sin que la empresa abra entidad local. Requiere:

- contrato de prestación de servicios;
- RUT;
- facturación electrónica si aplica;
- soporte de pagos;
- cumplimiento de seguridad social;
- contador;
- claridad sobre propiedad intelectual;
- evitar subordinación excesiva.

6.3 EOR

Un Employer of Record puede funcionar si una empresa quiere tratarte como empleado legal en Colombia sin abrir entidad. Howdy describe el EOR como entidad que actúa como empleador legal y maneja payroll, impuestos, beneficios y compliance.[^howdy]

Para Alexander, EOR puede ser positivo porque reduce riesgo de informalidad y mejora estabilidad. La desventaja es que la empresa asume costo adicional y puede preferir candidatos en países donde ya tenga payroll.

6.4 Riesgo de misclassification

Mismo resume la diferencia clave entre contractor y empleado: el contractor tiene autonomía, usa sus herramientas, puede trabajar con varios clientes y factura por proyecto/servicio; el empleado trabaja bajo dirección, horario y beneficios. La subordinación es uno de los elementos críticos para determinar relación laboral.[^mismo]

Para reducir riesgo:

- evitar lenguaje de “employee” en contrato contractor;
- definir entregables y milestones;
- conservar autonomía horaria;
- usar equipo propio;
- facturar formalmente;
- evitar exclusividad total si no está bien compensada;
- documentar independencia.

4.0.9 7. Implicaciones administrativas en Colombia

7.1 Checklist básico

Elemento	Estado requerido
RUT	Obligatorio
CIIU correcto	Pendiente contador
Facturación electrónica	Probablemente sí
Cuenta para pagos	Necesaria
Seguridad social	Mensual
Soportes de pago	Obligatorios
Contrato / SOW	Recomendado
Contador	Recomendado

7.2 RUT

Colombia Move describe el RUT como el registro fiscal necesario para operar formalmente como independiente; incluye actividad económica CIIU y datos ante DIAN.[^colombiamove-rut]

Para Alexander, el RUT debe configurarse antes de facturar de forma sostenida a clientes extranjeros. La elección del CIIU debe validarse con contador. Posibles áreas: software development, consultoría, diseño, servicios digitales o actividades creativas/técnicas, según el modelo real de facturación.

7.3 Facturación electrónica

Colombia Move señala que si se factura a clientes colombianos y se superan ciertos ingresos, la factura electrónica DIAN-compliant es necesaria, y que DIAN ofrece una herramienta gratuita.[^colombiamove-invoice]

Para clientes extranjeros, el tratamiento puede variar. La recomendación estratégica es no improvisar: configurar facturación electrónica y tener soporte documental de contratos y pagos desde el inicio.

7.4 Retención en la fuente

Para clientes colombianos, Colombia Move describe una retención típica del **10%** sobre honorarios profesionales y aclara que clientes extranjeros no aplican esa retención local.[^colombiamove-withholding]

Esto importa para comparar local vs internacional. Una oferta local de COP puede tener retenciones y flujo de caja distinto a una oferta extranjera en USD.

7.5 IVA / exportación de servicios

Colombia Move indica que el IVA general es **19%** y que muchos servicios profesionales son IVA-aplicables, pero también señala que ingresos por clientes extranjeros normalmente se tratan como exportación de servicios y suelen estar exentos, si se documenta correctamente.[^colombiamove-iva]

Esta parte debe validarse con contador. No se debe asumir automáticamente IVA 0% para todo servicio remoto. La pregunta práctica no es solo “si el cliente está afuera”, sino si el servicio se utiliza exclusivamente en el exterior, cómo se factura, cómo se documenta y qué responsabilidades figuran en el RUT.

7.6 Seguridad social independiente

SIA Seguros resume que en 2026 todo independiente con ingresos iguales o superiores a 1 SMMLV debe cotizar a salud, pensión y ARL si aplica. También indica que para contratistas por prestación de servicios el IBC es **40% del valor mensual del contrato**, y para independientes por cuenta propia el Decreto 0379 de 2026 permite aplicar el 40% sobre ingreso neto después de costos en ciertos casos.[^sia-ibc]

Los porcentajes base reportados son:

Concepto	Porcentaje
Salud	12.5%
Pensión	16%
ARL nivel I	0.522%
Base habitual	40% ingreso

La regla de costos netos no debe aplicarse sin asesoría. Para contrato de prestación de servicios, SIA indica explícitamente IBC sobre 40% del valor mensual del contrato.[^sia-contratistas]

7.7 Pagos internacionales

Papaya Global destaca que pagos a contractors en Colombia deben considerar documentación tributaria, integración bancaria, DIAN, soporte COP/USD, AML y RUT.[^papaya-payments]

Para Alexander:

- Wise, Payoneer, PayPal, SWIFT o Deel no reemplazan la obligación fiscal.
- El pago recibido debe poder reconciliarse con factura, contrato y declaración.
- Hay que revisar comisiones, tasa de cambio y trazabilidad.

4.0.10 8. Modelo gross-to-net: advertencia operacional

Un ingreso bruto contractor no equivale a ingreso disponible. La planeación debe reservar dinero para:

- seguridad social;
- impuesto de renta;
- contador;
- comisiones de plataforma;
- spread cambiario;
- software/hardware;
- meses sin contrato;
- vacaciones no pagas;
- salud privada o complementaria si aplica.

Modelo conservador inicial

Bruto mensual	Reserva sugerida	Disponible estimado
USD 1,500	25–35%	USD 975–1,125
USD 3,000	25–35%	USD 1,950–2,250
USD 6,000	30–40%	USD 3,600–4,200

Esto no es cálculo tributario. Es una regla de prudencia para no confundir facturación con salario neto.

4.0.11 9. Realismo de metas de ingreso

9.1 Meta USD 1,500/mes

Variable	Evaluación
Probabilidad	Alta
Horizonte	0–3 meses
Ruta	LATAM / freelance / contractor
Condición	CV + LinkedIn + demo

USD 1,500/mes es viable incluso con experiencia formal limitada si TwinSight se presenta bien y se aceptan contratos junior/semimid. No debe ser la meta aspiracional; debe ser el piso mínimo aceptable.

9.2 Meta USD 3,000/mes

Variable	Evaluación
Probabilidad	Media
Horizonte	3–6 meses
Ruta	Unity WebGL / Interactive 3D
Condición	Caso TwinSight fuerte

USD 3,000/mes brutos es razonable para contractor internacional junior-fuerte o mid-junior. El dato de Lemon.io para Middle Unity Developer en Colombia (**USD 27–35/h**) sugiere que el mercado puede soportarlo, pero no garantiza acceso inmediato.[^lemon-core]

9.3 Meta USD 6,000/mes

Variable	Evaluación
Probabilidad 3–6 meses	Baja
Probabilidad 12–24 meses	Media
Ruta	Digital twin / simulation / XR
Condición	Evidencia + entrevistas

USD 6,000/mes requiere facturar alrededor de **USD 37.5/h a 160 h/mes** o más, antes de impuestos y reservas. Es alcanzable en LATAM contractor fuerte o en roles US/EU, pero no debe ser promesa de corto plazo. Para hacerlo realista se requiere:

- demo TwinSight pulida;
- case study técnico;
- portfolio público;
- GitHub limpio;
- narrativa de CAD-to-realtime;
- buen inglés técnico;
- negociación contractor;
- búsqueda directa a empresas de visualización/simulación, no solo job boards.

4.0.12 10. Rutas salariales recomendadas

10.1 Ruta de corto plazo: USD 1.5k–3k

Priorizar:

```
Bloque de código: text
Unity WebGL Developer
Interactive 3D Developer
Real-Time 3D Developer
Technical Visualization Developer
Unity Developer – 3D UI / visualization
```

No priorizar:

```
Bloque de código: text
3D Artist
Game Technical Artist AAA
Pure Full-stack Developer
AI Engineer
Machine Learning Engineer
```

10.2 Ruta de mediano plazo: USD 3k–6k

Priorizar:

Bloque de codigo: text
Unity Technical Artist – optimization/tools/runtime
CAD-to-realtime visualization specialist
Technical Visualization Developer
Digital Twin Visualization Developer
Simulation Developer
XR Simulation Developer

Condiciones mínimas:

- TwinSight live build o video claro.
- Métricas de optimización verificables.
- Before/after CAD-to-web.
- Código o arquitectura explicable.
- Narrativa honesta sobre AI-assisted development.

10.3 Ruta de alto upside: USD 6k+

Priorizar solo después de preparar evidencia:

Bloque de codigo: text
US/EU contractor technical visualization
Industrial Unity simulation
Digital twin implementation partner
XR enterprise simulation
Unity tools/pipeline engineer
LLM workflow automation consultant

4.0.13 11. Red flags de datos salariales

Señal	Riesgo
"Remote" sin país	Puede ser US-only
Salario US alto	Puede requerir work authorization
Promedio Glassdoor/Zip	Mezcla seniority
Vendor report	Sesgo comercial
XR big tech	No representa startups
Digital twin salary	Puede requerir IoT/engineering
Contractor rate alto	Puede asumir seniority real
Full-time contractor	No siempre 160h/mes

4.0.14 12. Datos que faltan o deben validarse después

Esta sección está suficiente para estrategia inicial, pero no para cierre fiscal ni negociación final. Faltan:

1. Validación con contador colombiano sobre RUT, CIU, facturación, IVA y exportación de servicios.
2. Confirmación de obligaciones exactas para pagos en USD/EUR, Wise, Payoneer, PayPal, SWIFT o Deel.
3. Muestra de ofertas reales en sección 11 para confirmar qué roles aceptan Colombia remoto.
4. Datos más finos para Technical Artist en Canadá, Alemania, España, Portugal, Netherlands y Nordics.
5. Datos de Unity WebGL Developer como categoría independiente.
6. Comparación Deel vs Remote vs Oyster vs Plane vs Papaya para contratación EOR/contractor.
7. Benchmarks reales de contratos en Upwork/Toptal/Braintrust/Arc/Lemon.io para Unity WebGL y 3D web.

4.0.15 13. Decisión estratégica final de esta sección

La mejor ruta salarial para Alexander no es perseguir el salario global máximo, sino maximizar la intersección entre:

Bloque de codigo: text

TwinSight evidence
+ Unity/WebGL/C# skills
+ Colombia remote eligibility
+ contractor feasibility
+ technical visualization niche
+ realistic short-term employability

La estrategia salarial debe organizarse en tres niveles:

- 1. **Piso aceptable:** USD 1.5k/mes bruto.
- 2. **Objetivo operativo 3–6 meses:** USD 3k/mes bruto.
- 3. **Objetivo estratégico 12–24 meses:** USD 6k/mes bruto.

El candidato no debe presentarse como senior, AI engineer, digital twin engineer pleno ni technical artist AAA. La formulación más fuerte y defendible para salario es:

Bloque de codigo: text

Unity / Real-Time 3D Developer focused on interactive technical visualization, CAD-to-realtime optimization and WebGL
↔ deployment.

4.0.16 14. Próxima sección recomendada

Archivo siguiente:

Bloque de codigo: text

04_eu_portugal_germany_mobility_strategy.md

Recomendación de herramientas

Herramienta	Activar
Deep Research	Sí
Modo agente	No todavía

Motivo

La sección 04 requiere fuentes oficiales y actualizadas sobre:

- ciudadanía portuguesa por vía sefardí;
- derechos laborales con ciudadanía portuguesa futura;
- Alemania por matrimonio/pareja;
- residencia, Anmeldung, permiso laboral, reunificación familiar;
- EU Blue Card;
- trabajo remoto desde Colombia para empresas europeas;
- qué se puede decir a reclutadores sin sobreafirmar.

Primero conviene usar Deep Research para síntesis legal/migratoria. El modo agente se reserva para más adelante si necesitamos abrir formularios oficiales, recopilar links específicos o comparar rutas por país.

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Chapter 5

04 — EU / Portugal / Germany Mobility Strategy

Fuente: 04_eu_portugal_germany_mobility_strategy.md

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Resumen de orientacion

This file defines the legal and operational mobility framework for Alexander Woodcock Salomón, a Colombia-based Unity / Real-Time 3D / Technical Visualization candidate.

Contenido completo depurado

5.0.1 1. Purpose and scope

This file defines the legal and operational mobility framework for Alexander Woodcock Salomón, a Colombia-based Unity / Real-Time 3D / Technical Visualization candidate.

It is not tax, immigration, or legal advice. It is a structured decision framework based on official EU/German sources and reputable tax references available as of **11 June 2026**.

The goal is to separate:

- current legal reality;
- high-probability future EU mobility through Portuguese citizenship;
- possible German relocation through marriage to a German citizen;
- recruiter-facing wording that is accurate and defensible;
- structural trade-offs between Colombia, Portugal, and Germany.

5.0.2 2. Source-quality audit

The first Deep Research run for this section did not produce a usable EU mobility file. It returned salary and remote-work content that belonged to section 03. Therefore, this section was rebuilt manually using the verified mobility and tax sources listed below.

Primary source categories used

- EU official guidance on work permits and EU citizen residence.
- German federal guidance on EU Blue Card, family reunification, freelancing, and health insurance.
- German residence law references.
- PwC tax summaries for Germany and Portugal.
- Portugal tax and social security references from PwC.

Source limitations

The Portuguese Sephardic citizenship route itself was not re-audited as a legal case file. The user states that the Portuguese passport is highly likely within approximately two years. This document treats that as a **high-probability future condition**, not as a current right.

The exact Germany family-reunification process depends on the German mission, local immigration office, marital status, documents, and residence location. The analysis therefore treats marriage to a German citizen as a real strategic route, but not as automatic permission to move without paperwork.

5.0.3 3. Current source-of-truth status

Legal status now

Alexander is currently based in Colombia and does **not** currently have EU work authorization.

He may apply internationally as:

- Colombia-based remote contractor;
- freelancer / B2B service provider;
- candidate open to EOR if the employer supports Colombia;
- future EU-mobile candidate once Portuguese citizenship is completed.

He should not state that he is currently authorized to work in the EU.

High-probability future status

The user expects a Portuguese passport within approximately two years through the Sephardic Jewish ancestry route. Once a Portuguese passport is actually issued, Alexander becomes an EU citizen with the general right to live and work across EU/EEA member states, subject to local registration, taxation, and social-security rules.

Germany relationship route

The user has a German partner. If they legally marry and relocate to Germany, family reunification can become a practical route to residence and work authorization.

The key distinction is:

- marriage itself is not enough for employment rights abroad;
- residence status must be processed correctly;
- once the relevant German residence title is granted for family reunification, employment can generally be permitted.

5.0.4 4. Mobility-state matrix

State	Work authorization	Recruiter wording
Current Colombia	No EU work right	Remote contractor from Colombia
Portuguese passport pending	No EU work right yet	Future EU citizenship expected
Portuguese passport issued	EU work right	EU citizen, authorized in EU
Married + Germany residence title	Germany work right	Authorized to work in Germany

State	Work authorization	Recruiter wording
Employer-sponsored Blue Card	Germany work right	Visa sponsorship required

5.0.5 5. EU-wide baseline: what changes with Portuguese citizenship

Official EU guidance states that EU nationals generally do not need a work permit to work anywhere in the EU. It also states that self-employed EU nationals never require a work permit in another EU country.

This means that after the Portuguese passport is issued, Alexander can frame himself as an EU citizen for employment purposes in EU/EEA contexts. That materially improves his access to:

- Germany;
- Portugal;
- Spain;
- Netherlands;
- Ireland;
- Nordic countries;
- other EU/EEA labor markets.

However, EU freedom of movement does not remove local compliance obligations. If Alexander lives in another EU country for more than three months, that country may require residence registration. EU guidance states that after three months in another EU country, registration may be required, although EU citizens can start work without waiting for the registration certificate.

Practical consequence

Portuguese citizenship is a major mobility advantage, but it is not equivalent to frictionless global remote employment. It solves the **right-to-work barrier** inside the EU. It does not solve:

- income tax residency;
- social security affiliation;
- payroll compliance;
- employer permanent-establishment risk;
- health-insurance registration;
- local banking and invoicing setup.

5.0.6 6. Portugal route

5.0.7 6.1 What Portugal becomes after citizenship

Once Alexander receives the Portuguese passport, Portugal becomes both:

- an EU citizenship anchor;
- a possible low-friction EU base.

Portugal is strategically relevant because it can provide:

- EU work authorization;
- easier access to EU employers;
- proximity to Spain and wider European markets;
- potentially lower cost of living than Germany or Netherlands;
- Portuguese-language leverage if developed.

5.0.8 6.2 Rights as a Portuguese citizen

As a Portuguese citizen, Alexander would have the general right to:

- live in Portugal;
- work in Portugal;
- work in other EU/EEA countries;
- provide services as self-employed, subject to local rules;
- move across the EU under free-movement rules.

This right begins only when citizenship/passport status is legally completed. Until then, it should not be used as current work authorization.

5.0.9 6.3 Living in Portugal while working remotely

If Alexander becomes resident in Portugal while working for non-Portuguese employers, the issue is no longer mainly immigration. The main issues become:

- tax residency;
- Portuguese social security;
- whether he is an employee or freelancer;
- whether the foreign employer needs payroll, EOR, or local registration;
- whether invoices are issued as independent professional services.

A foreign employer cannot simply treat Portugal residence as irrelevant if Alexander works there as an employee. If he is a contractor, the setup is simpler, but Portuguese tax and social-security obligations still apply.

5.0.10 6.4 Portugal tax residency

PwC summarizes Portuguese tax residency as follows: an individual is generally considered tax resident if they spend more than 183 days in Portugal during a 12-month period that begins or ends in the relevant fiscal year, or if they maintain a habitual residence in Portugal.

Portugal taxes residents on worldwide income. Non-residents are generally taxed only on Portuguese-source income.

Portugal's 2026 personal income tax rates are progressive, ranging from 12.5% to 48%, with additional solidarity rates above high-income thresholds.

5.0.11 6.5 Portugal social security

For employment, PwC reports standard Portuguese social security contributions of:

- 11% employee contribution;
- 23.75% employer contribution.

For self-employed workers, PwC reports a 21.4% contribution rate. Relevant remuneration is generally calculated as 70% of services income or 20% of product-related income.

Portugal also applies additional contribution exposure to contracting companies when a high percentage of the worker's income comes from one company. PwC reports 10% when 80% or more of the worker's income comes from one company, and 7% when 50% to 80% comes from one company.

Practical consequence

Portugal can work well as a European base, but contractor economics must be calculated carefully. A gross remote salary may look attractive but can lose value through:

- income tax;
- social security;
- accountant costs;
- currency conversion;
- invoicing obligations;
- possible VAT obligations.

5.0.12 6.6 Portugal registrations likely needed

For a Portuguese-resident worker or freelancer, likely administrative steps include:

- NIF / Portuguese tax identification;
- social security registration;
- local address or residence record;
- activity registration if freelancing;
- invoicing setup;
- accountant review if billing foreign clients;
- bank account or EU-compatible payment infrastructure.

These steps should be confirmed with a Portuguese accountant once the passport timeline is concrete.

5.0.13 6.7 NHR / IFICI caution

Portugal's previous Non-Habitual Resident regime was substantially changed. Various post-NHR incentive frameworks have existed or been proposed, but eligibility is technical and cannot be assumed.

For this strategy, Portuguese tax incentives should be treated as **optional upside**, not as a planning foundation.

Do not build the career strategy on assuming:

- automatic 20% flat taxation;
- automatic NHR eligibility;
- automatic tax incentives for technical-art or Unity work;
- automatic benefit from moving to Portugal.

Portugal remains valuable primarily because of EU citizenship and market access, not because of guaranteed tax advantages.

5.0.14 7. Germany route

5.0.15 7.1 Germany as target market

Germany is strategically important for Alexander because it combines:

- strong industrial visualization market;
- simulation and digital twin demand;
- automotive and manufacturing ecosystems;
- XR and enterprise 3D use cases;
- possible relocation route through a German partner;
- higher salary ceiling than Portugal or Spain.

Germany is not the easiest immediate route because of language, bureaucracy, health insurance, payroll rules, and work authorization. But it may be one of the strongest 12–24 month routes if legal status is solved.

5.0.16 7.2 Marriage / family reunification route

The user's German-partner context is strategically significant.

If Alexander legally marries his German partner and relocates to Germany through the correct family-reunification process, the route can provide residence and work authorization.

German federal guidance for spouses joining EU citizens states that third-country spouses usually need a family-reunification visa if they are not visa-exempt, and once a corresponding residence title is granted, they are immediately entitled to take up any kind of employment.

Practical consequence

Marriage can materially improve the Germany route, but only after:

- legal marriage is completed;
- documents are prepared;
- the correct visa/residence process is followed;
- German authorities grant the residence title.

Until then, Alexander should not claim German work authorization.

5.0.17 7.3 What this enables in Germany

Once residence and work authorization are granted, Germany becomes more realistic for:

- Unity Developer;
- Technical Artist;
- Simulation Developer;
- Digital Twin Visualization Developer;
- XR Developer;
- Tools / Pipeline Developer;
- technical visualization roles in industrial companies.

This can be stronger than applying from Colombia because German employers often prefer or require:

- local residence;
- EU or German work authorization;
- German payroll eligibility;
- ability to attend hybrid/on-site meetings;
- health insurance and tax registration.

5.0.18 7.4 Blue Card route

The EU Blue Card is an employer-sponsored route for qualified professionals. German federal guidance lists the 2026 general salary threshold as **€50,700 gross annual salary**. For shortage occupations and young professionals, the 2026 threshold is **€45,934.20**.

For IT specialists without a formal degree, Germany allows an EU Blue Card route if the person has a specific job offer, meets the lower salary threshold, and has at least three years of relevant IT work experience within the last seven years at a university-graduate level.

Relevance to Alexander

The Blue Card is not the primary route right now because:

- he does not yet have strong formal industry experience;
- Unity / Technical Artist roles may not always map neatly to shortage occupation categories;
- employers may prefer candidates already in the EU;
- German language can matter for non-startup companies.

However, the Blue Card becomes relevant if Alexander secures:

- a German job offer;
- sufficient salary;
- clear degree or experience equivalence;
- role classification accepted by German authorities.

5.0.19 7.5 Freelance / self-employment route in Germany

Germany has a freelance residence route for certain self-employed professionals. German federal guidance states that non-EU citizens intending to work freelance in Germany generally need a residence permit for freelance employment.

To qualify, a freelancer must generally show:

- ability to finance the undertaking;
- ability to make a living;
- relevant permissions for regulated professions if applicable;
- health insurance;
- business plan and supporting documents;
- pension provisions if older than 45.

The permit is usually initially limited to a maximum of three years. A settlement permit can become possible later if the activity is successful.

Relevance to Alexander

This route is theoretically possible but not recommended as the primary path unless he has:

- paying EU clients;
- a clear freelance business case;
- predictable income;
- German accountant support;
- strong documentation.

For the next 3–6 months, Germany freelance immigration is less practical than remote contracting from Colombia.

5.0.20 7.6 Germany health insurance

Health insurance is compulsory in Germany. German federal guidance distinguishes statutory and private health insurance and notes that proof of health insurance may be required for visa purposes.

For most employees, statutory health insurance applies. Contributions depend on income. Spouses and children may sometimes be included without extra contributions if they are not working.

Practical consequence

Germany's salary ceiling is higher, but net-income modeling must include:

- health insurance;
- pension insurance;
- unemployment insurance;
- long-term care insurance;
- income tax;
- possible church tax;
- cost of living.

5.0.21 7.7 Germany tax and social security

PwC summarizes German tax residence as being triggered by a dwelling available for use or habitual abode. A habitual abode generally occurs when a person is physically present in Germany for more than six months.

German tax residents are generally taxed on worldwide income. Non-residents are generally taxed on German-source income.

PwC summarizes Germany's 2026 progressive income tax rates as:

- 0% up to €12,096 for single taxpayers;
- 14%–42% from €12,097 to €68,429;
- 42% up to €277,825;
- 45% above €277,826.

Germany also has social-security contributions. PwC reports contribution rates for pension, unemployment, health insurance, and long-term care insurance, usually split between employer and employee.

Practical consequence

Germany can provide higher gross salaries than Portugal, Spain, or Colombia. But the gross-to-net difference is significant. A German offer should be evaluated by net salary, not headline gross salary.

5.0.22 8. EU-wide remote work considerations

5.0.23 8.1 Right to work is not the same as payroll compliance

Portuguese citizenship or German residence may solve work authorization, but it does not automatically solve remote work compliance.

A worker living in Portugal or Germany while employed by a foreign company may trigger obligations related to:

- local payroll;
- income tax withholding;
- social-security contributions;
- employer registration;
- EOR use;
- permanent establishment risk;
- local labor law protections.

5.0.24 8.2 Remote employee vs contractor

The difference matters.

As an employee, the employer may need to run payroll or use EOR in the country of residence.

As a contractor, the worker generally invoices independently. But if the contractor works like an employee, misclassification risk may arise.

Indicators of employee-like relationship include:

- fixed schedule;
- exclusivity;
- employer-provided tools;
- daily direct supervision;
- inability to subcontract;
- long-term dependency on one client;
- integration into the company's internal hierarchy.

5.0.25 8.3 Permanent establishment risk

A foreign company may be concerned that a remote employee in Portugal or Germany creates tax exposure for the company. This is especially sensitive if the worker:

- negotiates contracts;
- signs deals;
- manages local business operations;
- represents the company locally;
- operates as a senior decision-maker.

For Alexander's likely early roles, permanent-establishment risk is lower if he works as an individual contributor. But employers may still prefer EOR or contractor arrangements for compliance.

5.0.26 9. Recruiter-facing work authorization statements

5.0.27 9.1 Current wording — Colombia, no EU passport yet

Use:

I am currently based in Colombia and available for remote contractor/B2B work with international companies. I can also work through an EOR if the company supports Colombia.

Avoid:

I am authorized to work in the EU.

Avoid:

I can work in Germany without sponsorship.

Avoid:

I have Portuguese citizenship.

5.0.28 9.2 Portuguese passport pending

Use only if relevant:

I am in the final/advanced pathway toward Portuguese citizenship and expect EU mobility within approximately two years, but I am currently applying as a Colombia-based remote contractor.

Avoid overusing this in short applications. It can distract from immediate employability.

5.0.29 9.3 Once Portuguese passport is issued

Use:

I am an EU citizen through Portuguese citizenship and authorized to work in EU/EEA countries. I am open to remote, hybrid, or relocation-based roles depending on the position.

This becomes a strong ATS and recruiter filter advantage for EU roles.

5.0.30 9.4 If married and Germany residence title is granted

Use:

I am legally resident in Germany and authorized to work there.

If still in process, use:

I am pursuing family-based relocation to Germany and will require the appropriate residence/work authorization process before starting local employment.

5.0.31 9.5 Blue Card wording

Use:

I would require employer sponsorship for a German EU Blue Card unless another residence route is completed first.

Do not claim Blue Card eligibility unless the job offer, salary, degree/experience match, and authority requirements are confirmed.

5.0.32 10. Colombia vs Portugal vs Germany: structural comparison

Base	Main advantage	Main risk
Colombia	Lower cost base	No EU work right
Portugal	EU citizenship base	Tax/social setup
Germany	Higher salary ceiling	Bureaucracy/language

5.0.33 10.1 Colombia contractor

Best for the next 3–6 months.

Strengths:

- no relocation friction;
- immediate availability;
- lower cost of living;
- attractive to foreign employers seeking LATAM contractors;
- time-zone compatibility with US markets;
- can build portfolio while applying.

Weaknesses:

- no EU employee access;
- lower salary ceiling;
- weaker legal protection;
- must manage taxes and social security independently;
- foreign employers may reject Colombia if they do not support contractor or EOR arrangements.

5.0.34 10.2 Portugal resident after citizenship

Best as a 12–24 month EU mobility base.

Strengths:

- full EU citizenship rights after passport;
- simpler EU work authorization;
- better access to European employers;
- Portuguese language may become useful;
- lower cost than Germany, Netherlands, or Nordics in many cities.

Weaknesses:

- salaries can be lower than Germany or Netherlands;
- tax and social-security obligations reduce net income;
- remote employment still requires payroll/EOR/contractor compliance;
- tax incentives cannot be assumed.

5.0.35 10.3 Germany resident through marriage/family route

Best if relocation with German partner becomes real.

Strengths:

- direct access to German labor market after residence/work authorization;
- high relevance to industrial visualization, simulation, digital twins, automotive, manufacturing, and XR;
- higher salary ceiling than Portugal or Spain;
- spouse/family route may reduce dependence on employer sponsorship;
- stronger legal and social protections.

Weaknesses:

- language barrier;
- bureaucracy;
- health insurance and tax complexity;
- higher cost of living;
- family route requires correct documents and legal processing;
- German employers may still prefer German-speaking candidates for many roles.

5.0.36 11. Strategic interpretation by timeline

5.0.37 11.1 Next 0–6 months

Primary route:

- Colombia-based remote contractor.
- International applications to remote-first companies.
- No EU work authorization claims.
- Mention future EU mobility only when strategically useful.

Best role families:

- Real-Time 3D / Interactive Technical Visualization Developer;
- Unity WebGL / Interactive 3D Developer;
- Unity Technical Artist with optimization/runtime angle;
- Tools / Python automation only as secondary.

5.0.38 11.2 Next 6–18 months

Primary route:

- prepare EU-ready portfolio;
- track Portuguese citizenship process;
- start building Portugal/Germany professional networks;
- learn enough German to reduce Germany friction;
- create EU-compatible CV and LinkedIn variants;
- monitor Blue Card salary-fit roles.

5.0.39 11.3 After Portuguese passport

Primary route:

- apply as EU citizen;
- include EU work authorization clearly;
- target Germany, Netherlands, Ireland, Portugal, Spain, Nordics;
- consider relocation only if net income and career signal justify it.

5.0.40 11.4 If marriage/Germany route becomes active

Primary route:

- verify documents with German mission and local immigration authority;
- prepare language/administrative requirements;
- evaluate relocation budget;
- apply to Germany-based roles once work authorization pathway is concrete;
- consider German-resident remote roles after compliance review.

5.0.41 12. Decision thresholds

Do not relocate to Europe only for a marginal salary improvement.

Relocation becomes rational if at least one condition is true:

- net income after tax and rent is materially better than Colombia;
- role is a strong career signal;
- role gives EU industry experience;
- role is in digital twin, simulation, XR, industrial visualization, or technical art;
- partner/family relocation makes Germany strategically necessary;
- EU citizenship is already completed and relocation friction is low.

Relocation is weak if:

- salary is close to Colombia remote contractor income;
- role is low-skill 3D production;
- employer requires expensive relocation without support;
- German bureaucracy is unresolved;
- the role is not aligned with TwinSight / real-time 3D positioning;
- job requires German fluency Alexander does not yet have.

5.0.42 13. Open questions requiring professional verification

Before making binding decisions, verify:

1. Exact status and timing of Portuguese citizenship/passport.
2. Whether Portuguese citizenship is already approved, pending, or only expected.
3. Whether Portuguese tax incentives apply to Alexander's future profile.
4. Whether work from Portugal would be employee, contractor, or freelancer.
5. Whether the German spouse route requires A1 German in the specific case.
6. Which German mission handles the visa if applying from Colombia.
7. Whether the marriage route applies before or after the German partner has German residence established.
8. Whether a Germany-based role can start before residence title approval.
9. Whether a foreign employer can hire Alexander through EOR in Germany or Portugal.
10. Net salary simulation with a local accountant before relocation.

5.0.43 14. Legal-risk rules for the career strategy

Do not claim now

- EU work authorization.
- German work authorization.
- Portuguese citizenship already completed.
- ability to relocate immediately without sponsorship.
- Blue Card eligibility without job offer and salary match.

Safe to claim now

- Colombia-based remote contractor availability.
- openness to EOR if supported.
- future EU mobility expected through Portuguese citizenship.
- willingness to relocate under the correct legal route.
- German partner context only if relevant and disclosed carefully.

Safe to claim after passport

- EU citizen.
- authorized to work in EU/EEA.
- available for EU employment.
- open to relocation/hybrid roles in Europe.

Safe to claim after Germany residence title

- legally resident in Germany.
- authorized to work in Germany.
- available for German payroll or local contracts.

5.0.44 15. Practical action checklist**5.0.45 Now — Colombia phase**

- Keep applying as Colombia-based remote contractor.
- Do not claim EU work authorization.
- Prepare an EU-ready CV variant.
- Track Portugal citizenship timeline.
- Maintain documentation for German route.
- Start German language study only as strategic background, not as the main 3–6 month job-search lever.

5.0.46 6–12 months

- Prepare relocation budget scenarios.
- Research Portugal accountant options.
- Research Germany immigration-law consultation.
- Build EU target-company list.
- Start networking with Germany/Portugal technical visualization companies.
- Create a German-market LinkedIn keyword variant.

5.0.47 Once Portuguese passport is issued

- Update LinkedIn work authorization.
- Update CV header and availability line.
- Apply to EU roles as authorized candidate.
- Consider Portugal as administrative base.
- Consider Germany, Netherlands, Ireland, and Nordics for higher upside.

5.0.48 If marriage/Germany route activates

- Confirm documents with German authorities.
- Verify visa/residence requirements.

- Budget for relocation and health insurance.
- Prepare German-language basics.
- Apply to Germany roles once timeline is credible.
- Avoid signing employment contracts before legal start eligibility is clear.

5.0.49 16. Final mobility conclusion

The strongest mobility logic is staged.

For the next 3–6 months, the practical route is **remote international contracting from Colombia**. This preserves flexibility, avoids immigration delays, and lets Alexander monetize TwinSight while completing thesis defense and portfolio packaging.

The Portuguese passport is a high-value future asset. Once obtained, it changes the strategy materially because Alexander becomes an EU citizen and can compete for EU roles without work-permit friction.

Germany is the highest-upside European route if the relationship/marriage pathway becomes concrete. It is especially relevant for industrial visualization, digital twins, simulation, XR, and Unity-based enterprise applications. But Germany should be treated as a structured relocation pathway, not as an immediate assumption.

The correct recruiter narrative is therefore:

- now: Colombia-based contractor;
- later: EU citizen once Portuguese passport is issued;
- conditional: Germany-resident worker if family reunification through marriage is completed;
- fallback: Blue Card only if employer sponsorship and salary thresholds align.

5.0.50 17. Sources

- Your Europe — Work permits: https://europa.eu/youreurope/citizens/work/work-abroad/work-permits/index_en.htm
- Your Europe — Registering residence abroad: https://europa.eu/youreurope/citizens/residence/documents-formalities/registering-residence/index_en.htm
- Make it in Germany — EU Blue Card: <https://www.make-it-in-germany.com/en/visa-residence/types/eu-blue-card>
- Make it in Germany — Spouses joining EU citizens: <https://www.make-it-in-germany.com/en/living-in-germany/family-reunification/spouses-eu-citizens>
- Make it in Germany — Visa for self-employment / freelancing: <https://www.make-it-in-germany.com/en/visa-residence/types/self-employment>
- Make it in Germany — Health insurance: <https://www.make-it-in-germany.com/en/living-in-germany/money-insurance/health-insurance>
- German Residence Act, section 4: https://www.gesetze-im-internet.de/englisch_aufenthg/englisch_aufenthg.html
- PwC Germany — Individual residence: <https://taxsummaries.pwc.com/germany/individual/residence>
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- PwC Germany — Other taxes: <https://taxsummaries.pwc.com/germany/individual/other-taxes>
- PwC Portugal — Individual residence: <https://taxsummaries.pwc.com/portugal/individual/residence>
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Chapter 6

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Resumen de orientacion

This section defines the language strategy for Alexander Woodcock Salomón's international job-search plan.

Contenido completo depurado

6.1 Language Strategy and ROI Framework

6.1.1 0. Scope

This section defines the language strategy for Alexander Woodcock Salomón's international job-search plan.

The analysis is built around the already defined positioning:

Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization, Unity WebGL deployment, technical UI, and Python-assisted tooling.

The languages evaluated are:

1. **English** — currently self-assessed around C1.
2. **German** — beginner / A1 context at most.
3. **Portuguese** — strategically relevant because a Portuguese passport is expected with high probability in approximately two years.

The target employment context is:

- short term: **remote work from Colombia** within 3–6 months;
- medium term: **higher-compensation international work** within 12–24 months;
- mobility horizon: **Portugal / Germany / EU** once legal pathways mature;
- primary roles: Unity, Real-Time 3D, Technical Visualization, Digital Twin, Simulation, XR, and Tools/Pipeline roles;
- secondary roles: Python automation / AI tooling / LLM application development.

This section does **not** replace immigration, tax, or legal advice. It only defines language ROI as a career-planning variable.

6.1.2 1. Audit of the Deep Research output

The Deep Research output was useful as a directional draft, but it had quality problems that must be corrected before integration.

1.1 Valid findings to keep

The following points are directionally correct:

- **English is the highest short-term ROI language** for international remote roles.
- **German has low immediate ROI for remote-from-Colombia roles**, but high integration and mobility ROI if Germany becomes the relocation target.
- **Portuguese has medium strategic ROI** because of the expected Portuguese passport and possible Portugal residence, but it does not replace English for technical roles.
- **German should not displace English or portfolio work** in the first 3–6 months.
- **Portuguese is easier to acquire from Spanish than German**, so a functional B1/B2 path is realistic within 12–24 months if studied consistently.
- **Language certification is optional for most tech hiring**, but useful for immigration, academia, and credibility when the language claim is uncertain.

1.2 Weak or rejected parts

Several points in the Deep Research output were too broad or weakly sourced:

Issue	Audit result
"Certify English as soon as possible"	Too strong. Certification is useful but not mandatory for most tech roles.
"German spouse route has no language requirement"	Risky. A1 German may be required in many family reunification cases.
"Portuguese B2 in 6–12 months"	Possible but only with structured study. Not automatic.
"German A1–B1 in 6 months"	Possible only under high intensity. Not a baseline.
"Portuguese increases salary directly"	Weak evidence. It mainly improves local integration and Portugal/Brazil access.
"English demand proven by France job-post data"	Weak proxy for global technical hiring. Keep only as partial signal.
"LinkedIn post as source for Berlin English-first jobs"	Weak source. Use as low-confidence signal only.
"CELPE-Bras as primary Portuguese certificate"	Incomplete. For Portugal, CAPLE/Camões exams are more relevant.

1.3 Corrective direction

This file corrects the language strategy by separating:

- **employment ROI;**
- **mobility ROI;**
- **salary ROI;**
- **daily-life integration ROI;**
- **certification ROI;**
- **time cost;**
- **opportunity cost against technical skill development.**

The key correction is:

For the first 3–6 months, language work must not compete heavily against portfolio, TwinSight case study, GitHub cleanup, LinkedIn/CV, applications, and interview preparation. English polishing is high ROI; German and Portuguese are medium-term investments.

6.1.3 2. Source-quality framework

Language-market evidence is harder to quantify than salary data. Therefore, this section uses confidence labels.

Label	Meaning
Verified	Official source, recognized exam body, legal source, or directly observed requirement.
Strong inference	Consistent with job-market structure, role requirements, and regional hiring patterns.
Weak signal	Blog, job-board article, LinkedIn post, Reddit, or small sample.
Strategic assumption	Reasoned planning assumption for Alexander's specific situation.

Sources used conceptually

This section references the following source categories:

- **CEFR framework** for language levels.
- **Cambridge / IELTS / TOEFL** for English certification.
- **Goethe-Institut / telc** for German certification.
- **CAPLE / Camões / University of Lisbon** for European Portuguese certification.
- **CELPE-Bras / INEP** for Brazilian Portuguese certification.
- **EU work-mobility rules** for the future Portuguese passport scenario.
- **German family reunification and residence rules** for the marriage-to-German-citizen scenario.
- **Job-market evidence from previous sections** for the priority roles.

6.1.4 3. Core conclusion

The correct language priority is:

1. **English: maintain and professionalize immediately.**
2. **German: start now at low-to-moderate intensity because of the Germany route.**
3. **Portuguese: start or maintain gradually; accelerate closer to Portuguese passport / Portugal relocation.**

A more precise recommendation is:

Horizon	Priority
0–3 months	English for interviews, portfolio, LinkedIn, CV, demo narration.
3–6 months	English C1 proof through output; German A1/A2 foundation if Germany route remains likely.
6–12 months	German A2/B1; Portuguese A2/B1.
12–24 months	German B1/B2 if moving to Germany; Portuguese B1/B2 if moving to Portugal.
After Portuguese passport	German remains useful but no longer required for EU work rights.

The short-term rule:

If a language hour competes with a portfolio/job-search hour before the first international offer, the portfolio/job-search hour usually has higher ROI.

The exception:

If relocation to Germany through marriage becomes concrete before the Portuguese passport is issued, German A1 becomes an administrative-risk reducer and should be prioritized.

6.1.5 4. Current language state

Language	Current status	Reliability
Spanish	Native	Verified
English	Self-assessed C1	Needs proof
German	Beginner / A1 context	Needs structured study
Portuguese	Not confirmed	Needs diagnostic

4.1 English

Alexander's English is already one of the strongest assets in the profile because it was developed through prior work experience and is sufficient for technical communication. However, "C1 self-assessed" is not the same as verified hiring readiness.

For job-search purposes, English must be evaluated across five outputs:

1. CV and LinkedIn.
2. Portfolio case studies.
3. GitHub README files.
4. Technical interview speech.
5. Demo video narration.

The most important English weakness to avoid is not grammar. It is **unclear technical storytelling**.

4.2 German

German is strategically relevant because:

- the partner has German citizenship;
- legal marriage could make Germany a realistic relocation route;
- German improves daily life, bureaucracy, and access to non-English local roles;
- many German industrial, simulation, digital twin, engineering, and SME contexts still value German.

However, German is **not** the highest-return variable for getting the first remote job from Colombia.

4.3 Portuguese

Portuguese is strategically relevant because:

- a Portuguese passport is expected in approximately two years with high probability;
- Portuguese citizenship would create EU work-mobility rights;
- Portugal may become a practical EU base;
- Portuguese improves local integration in Portugal and access to Brazil/Iusophone markets.

However, Portuguese before the passport is issued has limited direct effect on work authorization.

6.1.6 5. Language ROI by strategic route

5.1 Route A — Remote work from Colombia

Language	ROI	Reason
English	Very high	Required for global remote work.
German	Low	Rarely required for remote-from-Colombia roles.
Portuguese	Low/medium	Useful for Brazil/Portugal clients, but not core.

For remote international roles, English dominates.

Most strong opportunities in Unity, WebGL, technical visualization, XR, AI tooling, and full-stack contracting will operate in English when they are international. German or Portuguese can occasionally differentiate, but they rarely compensate for weak portfolio evidence.

5.2 Route B — Portugal after passport

Language	ROI	Reason
English	High	Tech roles often operate internationally.
Portuguese	Medium/high	Integration, bureaucracy, local jobs, networking.
German	Low	Not relevant unless targeting Germany.

Once the Portuguese passport exists, Portuguese is not needed for EU work authorization itself. It is needed for:

- local administration;
- housing;
- taxes;
- social integration;
- small or medium Portuguese companies;
- long-term local credibility.

For tech roles in Lisbon/Porto startups or remote European companies, English may remain sufficient professionally.

5.3 Route C — Germany through marriage before Portuguese passport

Language	ROI	Reason
English	High	Needed for tech roles.
German A1	High legal/admin risk reduction	May be required for spouse reunification.
German B1	High life ROI	Daily life, bureaucracy, integration.
German B2	Medium/high career ROI	Opens local non-English roles.
Portuguese	Low	Not useful for Germany route.

If the Germany route becomes active before Portuguese citizenship is complete, German has a different ROI profile. It stops being only “nice to have” and becomes a practical relocation variable.

Important correction:

Marriage to a German citizen can support a residence/work pathway, but it does not mean German language can be ignored. A1 German may be required for spouse/family reunification unless an exception applies. This must be verified with the German embassy/consulate at the time of application.

5.4 Route D — Germany after Portuguese passport

Language	ROI	Reason
English	High	Tech work.
German	Medium/high	Integration and local market access.
Portuguese	Low	Passport matters more than language.

If Alexander has a Portuguese passport, he should be an EU citizen and generally able to live/work in Germany under EU free movement rules. In that case, German is not primarily a work-authorization language. It becomes an employability and life-integration language.

6.1.7 6. Language ROI by role family

Role family	English	German	Portuguese
Real-Time 3D / Interactive 3D	Critical	Medium in Germany	Low/medium
Unity WebGL / Web 3D	Critical	Low/medium	Low
Technical Visualization	Critical	High in Germany	Medium in Portugal/Brazil
Digital Twin / Simulation	Critical	High in Germany	Medium
Unity Technical Artist	Critical	Medium	Low
Game Technical Artist	Critical	Medium in German studios	Low
XR / AR / VR	Critical	Medium/high in Germany	Medium
Tools / Pipeline Developer	Critical	Medium	Low
Python Automation / AI Tools	Critical	Low/medium	Low
Full-stack fallback	Critical	Medium by country	Medium by country
3D Artist / Generalist	High	Medium	Medium

6.1 Technical Visualization / Digital Twin / Simulation

German has stronger value here than in game roles. The reason is structural: many German industrial, manufacturing, engineering, automotive, aerospace, logistics, and simulation contexts involve German-speaking stakeholders.

For Alexander, this matters because TwinSight X500 is not a pure game-art project. It is closer to technical visualization, spatial understanding, assembly inspection, and CAD-to-realtime workflows.

Therefore:

- English is enough for international technical roles.
- German expands the German industrial market.
- Portuguese helps in Portugal/Brazil contexts, but salary upside may be weaker than Germany/US/Netherlands/UK.

6.2 Game Technical Artist

English remains the default for international game studios. German helps in German studios, but does not replace a strong art/tech portfolio. Portuguese is low ROI unless targeting Brazilian/Portuguese studios, which may not align with the highest compensation strategy.

6.3 Unity WebGL / Interactive 3D

This route is language-light. Recruiters care more about:

- live WebGL demo;
- performance;
- C# quality;
- front-end integration;
- case study;
- visual proof.

English is necessary. German/Portuguese only matter by client location.

6.4 AI Tools / LLM Application Development

English dominates. German and Portuguese matter mainly if the tool handles local-language corpora, customers, or compliance-heavy use cases.

ARA Framework does not justify a language strategy by itself.

6.1.8 7. Language ROI vs technical skill ROI

7.1 Ranking by short-term employability

For the first 3–6 months, the highest ROI actions are:

Action	Short-term ROI
TwinSight case study	Very high
Demo video in English	Very high
LinkedIn/CV in English	Very high
GitHub README cleanup	High
English interview practice	High
German A1	Medium if Germany route active
Portuguese A1/A2	Low/medium
Formal English certificate	Medium
German B2 push	Low short-term
Portuguese B2 push	Low short-term

The conclusion is direct:

Technical evidence + English presentation beats additional language study for the first offer.

7.2 When language beats technical skill

Language becomes higher ROI than a new technical course when:

- a relocation date is concrete;
- a visa or family reunification process requires language proof;
- a recruiter asks for local language;
- a target company operates in German/Portuguese internally;
- the role involves client-facing technical visualization in that country.

7.3 When technical skill beats language

Technical skill beats language when:

- targeting remote global roles;
- applying to English-first startups;
- building TwinSight proof;
- preparing technical interviews;
- polishing Unity/WebGL performance;
- developing portfolio breakdowns;
- cleaning GitHub repositories.

6.1.9 8. English strategy

8.1 Objective

The objective is not “learn English.” The objective is:

Convert self-assessed C1 English into professionally visible, interview-ready, recruiter-trustworthy English.

This means:

- write CV and LinkedIn in clean international English;
- explain TwinSight fluently in 60, 120, and 300 seconds;
- answer technical interview questions without freezing;
- write GitHub documentation clearly;
- narrate a demo video in polished technical English;
- negotiate rates and contractor conditions.

8.2 Target level

Use case	Minimum	Target
Remote contractor interviews	B2+	C1
Technical presentations	B2+	C1
GitHub documentation	B2	C1
Client-facing work	C1	C1/C2-like clarity
Master's applications	B2/C1	IELTS/TOEFL proof

8.3 Certification decision

English certification is useful but not mandatory for most tech hiring.

Scenario	Certification value
Remote technical job	Medium
Contractor/freelance	Low/medium
EU master's	High
Visa / immigration	Medium/high
Recruiter trust	Medium
Technical interview	Low

Recommendation:

Do not delay applications to obtain IELTS/TOEFL/Cambridge. Build the portfolio and apply. Certify English only if it supports a concrete path: master's, visa, scholarship, or repeated recruiter doubt.

8.4 Best English outputs for portfolio

The highest-ROI English deliverables are:

1. **TwinSight one-liner.**
2. **TwinSight 60-second pitch.**
3. **TwinSight 2-minute demo script.**
4. **README executive summary.**
5. **Case study problem-solution-impact section.**
6. **Interview explanation of AI-assisted development.**
7. **Salary expectation script.**

These should be created before spending time on generic English courses.

6.1.10 9. German strategy

9.1 Strategic role of German

German has three separate values:

1. **Legal/admin value:** possible A1 requirement for spouse/family reunification before Portuguese passport.
2. **Life value:** housing, bureaucracy, healthcare, taxes, daily integration.
3. **Employment value:** access to German-speaking industrial, SME, public-sector, simulation, and engineering roles.

9.2 Correct legal framing

The correct framing is:

If Alexander relocates to Germany before the Portuguese passport, through marriage to a German citizen, he may need to comply with spouse/family reunification requirements, including possible proof of basic German A1 unless an exception applies. Once a residence permit is issued with work authorization, he can work under that permit. This must be checked with the German mission at the time of application.

This corrects the weaker Deep Research phrasing that implied German might not matter for the spouse route.

9.3 German ROI by level

Level	Value
A1	Visa/admin threshold risk reduction
A2	Basic survival
B1	Daily life and integration
B2	Local job market access
C1	Professional fluency for German-first roles

9.4 Target by time horizon

Horizon	German target
0–3 months	A1 foundation
3–6 months	A1/A2 if Germany route remains real
6–12 months	A2/B1
12–24 months	B1/B2
Germany relocation active	prioritize A1 immediately

9.5 German and technical roles

German is especially useful for:

- digital twin;
- industrial visualization;
- simulation;
- CAD/engineering workflows;
- automotive suppliers;
- manufacturing;
- med-tech;
- energy;
- defense-adjacent visualization;
- public-sector-funded R&D;
- non-English SMEs.

German is less critical for:

- global remote startups;
- US/UK/Canada contractor roles;
- Unity WebGL agencies;
- game studios using English internally;
- AI tooling startups.

9.6 Practical German recommendation

Do **not** attempt B2 German as the main 3–6 month project. That would create opportunity cost.

Recommended approach:

- Start A1 immediately but lightly.
 - Study consistently 3–5 hours/week.
 - Increase to 8–10 hours/week only if relocation becomes concrete.
 - Use partner interaction for daily vocabulary, pronunciation, and household phrases.
 - Target A1 certificate if marriage/Germany paperwork becomes active.
 - Target B1/B2 only after core portfolio/job-search assets are finished.
-

6.1.11 10. Portuguese strategy

10.1 Strategic role of Portuguese

Portuguese has three values:

1. **Portugal integration value:** bureaucracy, housing, healthcare, taxes, local services.
2. **Brazil/lusophone market value:** potential clients and companies.
3. **EU identity value:** aligns with future Portuguese citizenship, but does not create the citizenship itself.

Important distinction:

The passport creates EU work rights. Portuguese language improves living and local employability but is not the legal basis of EU work authorization once citizenship is granted.

10.2 Portuguese and Portugal

If Alexander becomes a Portuguese citizen, Portuguese language will be useful for:

- NIF/NISS/tax processes;
- renting an apartment;
- dealing with public services;
- healthcare;
- local accountants;
- social integration;
- Portuguese SMEs;
- local networking;
- long-term residence stability.

For English-first tech roles in Lisbon/Porto or remote EU work, Portuguese may be secondary.

10.3 Portuguese and Brazil

Portuguese can help with Brazil-facing roles, but Brazil does not currently appear to be the highest-compensation route compared with:

- US remote contracting;

- EU technical visualization;
- Germany industrial/simulation roles;
- global Unity/WebGL agencies.

Therefore, Brazilian Portuguese should not drive the main strategy unless a strong Brazil-linked opportunity appears.

10.4 Certification correction: CAPLE vs CELPE-Bras

The Deep Research output emphasized **CELPE-Bras**. That is useful for Brazilian Portuguese, but for Portugal the more relevant system is **CAPLE**, associated with European Portuguese certification.

Certificate	Variant	Best use
CAPLE CIPLE	European Portuguese A2	Portugal bureaucracy/basic proof
CAPLE DEPLE	European Portuguese B1	daily/local integration
CAPLE DIPLE	European Portuguese B2	professional/local proof
CELPE-Bras	Brazilian Portuguese	Brazil-facing roles

For Alexander:

- if Portugal is the main scenario: prioritize European Portuguese and CAPLE;
- if Brazil work becomes relevant: CELPE-Bras can be considered;
- if no formal requirement exists: a certificate is optional.

10.5 Target by time horizon

Horizon	Portuguese target
0–3 months	Light exposure
3–6 months	A1/A2
6–12 months	A2/B1
12–24 months	B1/B2 if Portugal relocation likely
After passport	B1 minimum for life; B2 for local work

Portuguese has a better time-to-benefit ratio than German because Spanish gives a strong starting advantage. However, false confidence is a risk: comprehension may advance quickly, but writing and pronunciation require deliberate practice.

6.1.12 11. Time-to-proficiency model

11.1 English

Alexander is not starting from zero. The realistic target is not “English learning,” but professional polishing.

Target	Estimated work
Interview-ready C1	40–80 focused hours
Technical demo narration	10–20 hours
IELTS/TOEFL prep	40–100 hours
Professional writing polish	20–40 hours

11.2 German

German requires much more time. Goethe/CEFR-related estimates commonly place German B1/B2 in the hundreds of instructional hours.

Target	Approximate cumulative effort
A1	80–200 class hours
A2	200–350 class hours
B1	350–650 class hours
B2	600–800 class hours
C1	800–1,000 class hours

For Alexander, this means:

- A1 is realistic in 2–4 months with consistent study.
- A2 is realistic in 4–8 months.
- B1 is realistic in 9–18 months.
- B2 is a 18–30 month target unless studied intensively.

11.3 Portuguese

Portuguese is structurally closer to Spanish, so time-to-functional use is lower. But exact CEFR progress depends on pronunciation, listening, and active production.

Target	Estimated effort
A1/A2	40–100 hours
B1	120–250 hours
B2	250–450 hours
C1	500+ hours

These estimates are strategic planning estimates, not official guarantees.

6.1.13 12. Certification ROI

12.1 English certification

Recommended options:

Exam	Use
IELTS Academic/General	immigration, study, general proof
TOEFL iBT	study, US/Canada academic contexts
Cambridge C1 Advanced	long-term C1 proof
Duolingo English Test	quick academic proof if accepted

For job applications, certification has lower value than:

- interview performance;
- portfolio clarity;
- GitHub writing;
- demo video;
- recruiter communication.

Recommendation:

Certify English only after job-search assets are built, unless a specific program requires it.

12.2 German certification

Recommended options:

Exam	Use
Goethe A1	spouse-route paperwork risk
Goethe/telc B1	integration and local credibility
Goethe/telc B2	local technical job credibility
TestDaF	academic route

If Germany route becomes active, German A1 certification has practical value. If Germany remains optional, certification can wait.

12.3 Portuguese certification

Recommended options:

Exam	Use
CAPLE CIPLE A2	Portugal basic proof
CAPLE DEPLE B1	Portugal daily/life proof
CAPLE DIPLE B2	professional/local proof
CELPE-Bras	Brazil-facing proof

For a Portugal path, CAPLE is more aligned than CELPE-Bras.

6.1.14 13. Language sequencing plan

13.1 First 3 months

Primary objective: job-search execution.

Language	Action
English	Use daily for portfolio, CV, LinkedIn, demo script.
German	Start A1, 3–5 hours/week.
Portuguese	Light exposure, 1–2 hours/week.

Deliverables:

- English TwinSight pitch.
- English demo script.
- English LinkedIn/CV.
- English GitHub README.
- 20 interview answers in English.
- German A1 foundation if Germany route is active.

13.2 Months 3–6

Primary objective: applications and interviews.

Language	Action
English	Mock interviews and negotiation scripts.
German	A1/A2 if Germany still likely.
Portuguese	A1/A2 if Portugal path remains likely.

Deliverables:

- 10 mock interviews.
- 5 recorded project explanations.
- 1 polished English demo video.
- German A1 certificate only if administrative need appears.

13.3 Months 6–12

Primary objective: consolidate job and prepare EU route.

Language	Action
English	Maintain professional C1.
German	Move toward A2/B1.
Portuguese	Move toward A2/B1.

Deliverables:

- English negotiation fluency.
- German practical life vocabulary.
- Portuguese bureaucracy/life vocabulary.
- Decide if German or Portuguese gets accelerated.

13.4 Months 12–24

Primary objective: mobility readiness.

Scenario	Language priority
Germany via marriage	German B1/B2
Portugal via passport	Portuguese B1/B2
EU remote from any base	English + local language B1
Staying in Colombia	English + technical specialization

6.1.15 14. Recommended weekly allocation

Before first international job

Activity	Weekly hours
Portfolio/job search	20–30
English professional output	4–6
German	3–5
Portuguese	1–2
Total language load	8–13

After portfolio is publishable

Activity	Weekly hours
Applications/interviews	15–25
English interview practice	3–5
German	4–6
Portuguese	2–4
Total language load	9–15

If Germany relocation becomes concrete

Activity	Weekly hours
German	8–12
English	2–4
Portuguese	0–2

If Portugal relocation becomes concrete

Activity	Weekly hours
Portuguese	6–10
English	2–4
German	1–3

6.1.16 15. Recruiter-facing language statements

15.1 Current safe statements

Use:

Spanish: native. English: professional working proficiency / C1 self-assessed; comfortable in technical interviews and written documentation.

Avoid:

Certified C1 English.

unless there is a certificate.

Use:

Currently learning German for potential Germany relocation.

Avoid:

German professional proficiency.

Use:

Portuguese is planned as part of medium-term EU/Portugal mobility.

Avoid:

Portuguese professional proficiency.

15.2 After English proof exists

Use:

English: C1, certified by [exam], comfortable with technical interviews, documentation, and client communication.

15.3 After German A1/B1/B2

Use:

German: A1/B1/B2 certified by [Goethe/telc], currently improving toward professional use.

15.4 After Portuguese B1/B2

Use:

Portuguese: B1/B2, focused on Portugal/EU mobility and professional communication.

15.5 Work-authorization language is separate

Language claims must not be mixed with legal work authorization.

Do not write:

German speaker with EU work authorization.

unless EU work authorization exists independently.

Do write, currently:

Colombia-based; available for remote contractor/B2B work with international clients.

After Portuguese passport:

Portuguese/EU citizen; authorized to work in the EU, subject to local registration and tax requirements.

After German residence permit through marriage:

Authorized to work in Germany under residence status, subject to employer verification.

6.1.17 16. Strategic contradictions to avoid

16.1 “German is urgent” vs “remote-first from Colombia”

German is not urgent for remote-first roles. It becomes urgent only if Germany relocation becomes concrete.

16.2 “Portuguese passport soon” vs “Portuguese language creates EU rights”

The passport creates EU rights. Portuguese language improves integration but does not create work authorization.

16.3 “C1 English” vs weak technical explanation

A C1 claim is only credible if Alexander can explain projects clearly, handle interviews, write docs, and negotiate.

16.4 “Three languages at once” vs job-search execution

Studying three languages intensely while building a portfolio and applying to jobs creates execution risk.

16.5 “Certificates matter more than proof of work”

For tech hiring, project evidence normally matters more than language certificates.

6.1.18 17. Final priority matrix

Priority	Language action	Why
1	Professionalize English output	Immediate job ROI
2	English mock interviews	Hiring bottleneck
3	German A1 foundation	Germany route risk control
4	Portuguese A1/A2 exposure	Future passport path
5	German B1/B2	Germany relocation/local jobs
6	Portuguese B1/B2	Portugal residence/local jobs
7	English certification	Useful but not first bottleneck
8	Portuguese/German certification	Only if scenario requires

6.1.19 18. Decision rules

Rule 1 — If the goal is a job in 3–6 months

Prioritize:

1. English job materials.
2. English interviews.
3. Portfolio.
4. Applications.
5. Technical skill gaps.

Do not prioritize:

- German B2;
- Portuguese B2;
- generic language certificates;
- language study that delays applying.

Rule 2 — If Germany relocation becomes concrete

Prioritize:

1. German A1 certification check.
2. German A2/B1 practical ability.
3. English tech employability.
4. German labor-market vocabulary.

Rule 3 — If Portuguese passport arrives first

Prioritize:

1. EU work authorization messaging.
2. English-first EU jobs.
3. Portuguese A2/B1 for living.
4. German only if Germany is the actual destination.

Rule 4 — If income maximization dominates

Prioritize:

1. English.
 2. technical proof;
 3. US/global remote;
 4. high-value role positioning;
 5. local language only when it unlocks specific market access.
-

6.1.20 19. Open questions for later sections

The following should be validated later, preferably during the company/job-board research section:

1. What percentage of German technical visualization / digital twin jobs accept English-only candidates?
2. How many Unity/Technical Artist postings in Germany explicitly require German B2/C1?
3. How many Portugal tech roles require Portuguese vs English-only?
4. Are there specific EU employers that value Spanish + Portuguese for LATAM/EU bridge roles?
5. Is there a repeat pattern of German industrial clients requiring German even when the job post is in English?
6. Are there remote-first companies where German or Portuguese create a measurable salary premium?
7. Which language appears more often in target job descriptions: German, Portuguese, or neither?

These questions are better suited for **agent-based job posting sampling** in a later company/job-board section, not for this language framework file.

6.1.21 20. Final recommendation

The recommended language strategy is:

Treat English as a professional delivery system, German as a Germany-route risk reducer and long-term market unlock, and Portuguese as a medium-term EU/Portugal integration asset linked to the expected passport.

Practical order:

1. Build all job-search materials in English.
2. Practice technical interviews in English.

3. Start German A1 now at low intensity.
 4. Start Portuguese lightly, then accelerate if Portugal becomes the base.
 5. Certify only when the certificate has a concrete use.
 6. Do not allow language study to delay the TwinSight case study, portfolio, GitHub cleanup, or applications.
-

6.2 Appendix A — Source notes

This file uses the following evidence classes:

- **CEFR and test bodies:** Cambridge, IELTS/TOEFL, Goethe-Institut, telc, CAPLE/Camões, CELPE-Bras.
- **Mobility logic:** EU free movement, Germany spouse/family reunification, future Portuguese citizenship scenario.
- **Market logic:** role-market fit from sections 02 and 03.
- **Weak signals removed or downgraded:** Reddit, generic job-board blogs, unsourced salary claims, LinkedIn posts.

Important source-quality correction:

Any claim about exact language requirements for German spouse/family reunification must be verified with the German embassy/consulate at the time of application. This file treats A1 German as a risk-reduction target, not as individualized legal advice.

Chapter 7

06 — Education, Courses, Rebelway, Master's and PhD Strategy

Fuente: 06_education_courses_rebelway_masters_phd.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 740

Ultima modificacion: 2026-06-11 13:27

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Candidate: Alexander Woodcock Salomón **Current status:** Ingeniería Multimedia thesis completed; defense pending **Primary employment target:** Unity / Real-Time 3D / Technical Visualization / CAD-to-realtime / WebGL **Time horizon:** 3–6 months for first improved job; 12–24 months for stronger compensation and EU mobility **Document purpose:** define which courses, certifications, master's programs or PhD routes have rational ROI.

Contenido completo depurado

Candidate: Alexander Woodcock Salomón

Current status: Ingeniería Multimedia thesis completed; defense pending

Primary employment target: Unity / Real-Time 3D / Technical Visualization / CAD-to-realtime / WebGL

Time horizon: 3–6 months for first improved job; 12–24 months for stronger compensation and EU mobility

Document purpose: define which courses, certifications, master's programs or PhD routes have rational ROI.

7.0.1 1. Executive decision

The education strategy should not be degree-first. It should be **portfolio-first, course-assisted, certification-light, master's-optional and PhD-negative** for the next 24 months.

The highest ROI path is:

1. Finish thesis defense.
2. Convert TwinSight X500 into a recruiter-grade case study.
3. Use owned courses only when they produce visible portfolio artifacts.
4. Use Rebelway selectively for Houdini / procedural / realtime FX capabilities.
5. Consider Unity certification only after the portfolio is already usable.
6. Consider a master's only if it aligns with EU relocation, specialization, scholarships or a clear simulation/XR/digital-twin transition.

7. Do not pursue a PhD unless the goal changes toward research, academia or R&D-heavy computer graphics.

The reason is structural. For Unity Technical Artist, Real-Time 3D Developer, Interactive 3D Developer and Technical Visualization roles, **proof of capability beats course accumulation**. Courses without certificates are still valuable, but only if converted into artifacts: demos, breakdowns, GitHub repos, reels, case studies, shaders, tools or optimized assets.

7.0.2 2. Audit of Deep Research result

The Deep Research output is usable as a first pass, but not complete enough as a final section.

What it got right

- It identified Rebelway, Unity certifications, Coursera, butic and U-tad as relevant education signals.
- It understood that non-certified courses have weak formal signaling but can support skill acquisition.
- It correctly treated master's programs as longer-term and higher-cost options.
- It mentioned that projects and portfolio outputs matter more than passive course completion.

What was weak or incomplete

- It did not fully integrate the actual Excel course inventory.
- It over-emphasized Rebelway alumni testimonials as employability evidence. These are useful signals, not placement statistics.
- It mixed income estimates from salary research into the education section. Salary targets belong mainly to section 03.
- It did not distinguish enough between **certificate**, **course completion**, **portfolio artifact** and **academic credential**.
- It treated some Unreal/VFX training as directly relevant to Unity Technical Artist roles without enough qualification.
- It did not provide a clear rule for when a no-certificate course is worth doing.
- It did not separate short-term job-search ROI from long-term specialization ROI.
- It did not adequately classify which parts of the Excel course library should be ignored, postponed or converted into portfolio pieces.

Corrective direction

The final strategy must treat education as a support system for employability, not as an independent goal. A course is valuable only if it moves one of these constraints:

- stronger TwinSight case study;
 - better Unity/WebGL runtime proof;
 - better technical art proof;
 - better procedural/optimization proof;
 - better demo reel quality;
 - stronger secondary tooling/automation route;
 - future EU academic or immigration pathway.
-

7.0.3 3. Current education and signaling power

Formal education

Credential	Signal
Ingeniería Multimedia, UNAD	Primary degree signal
Thesis completed	Strong project signal
UNAL Electronic Engineering coursework	Technical/math background
Non-certified courses	Skill support only
Rebelway access	Portfolio accelerator
Future master's	Optional relocation/specialization signal

The strongest formal signal is still the completed Ingeniería Multimedia degree once defended. The strongest practical signal is TwinSight X500.

The UNAL electronic engineering coursework should not be over-sold as a degree, but it is useful evidence of math, physics, systems thinking and technical discipline.

7.0.4 4. Inventory audit of Courses 2025.xlsx

The spreadsheet contains approximately **157 usable course rows**. Sensitive file names, archives and access keys were intentionally excluded from this section.

Course inventory by area

Area	Count
3D Art	78
VFX	32
Game Development	21
Compositing	10
AI	7
Storytelling	6
Web Development	4
Video Editing	4
2D / Painting	6

Course inventory by software

Software	Count
Blender	41
Houdini	30
Unreal Engine	27
Maya	21
ZBrush	13
Nuke	13
Photoshop	13
After Effects	9
Substance	8
Ziva	6
Mari	6

Course inventory by provider

Provider	Count
Rebelway	20
Coloso	11
The Gnomon Workshop	10

Provider	Count
CG Circuit	8
Wingfox	6
Compositing Academy	5
Blender Bros	4
FlippedNormals	4
CG Cookie	3
Creative Shrimp	3

Interpretation

The course library is broad but dangerous if used without prioritization. It can easily create strategic dispersion across character art, CFX, VFX, stylized art, compositing, Unreal, Houdini, web, AI and filmmaking.

For the current career goal, only a subset is immediately useful. The course library should be treated as a **production resource library**, not as a curriculum to complete sequentially.

7.0.5 5. Credential hierarchy for this profile

Strongest signals

1. Shipped or live interactive demo.
2. Case study with metrics.
3. Technical breakdown.
4. GitHub repo with clear README.
5. Demo reel or walkthrough video.
6. Recognized certification.
7. Non-certified course list.

Weakest signals

- Long course lists without output.
- Unfinished course libraries.
- Screenshots without explanation.
- "I studied X" without proof.
- Certificates unrelated to target roles.

Rule

A course without a certificate is still relevant if it produces a stronger artifact than the certificate would have produced.

Examples:

- A non-certified Houdini course that produces a clean VAT-to-Unreal FX breakdown is useful.
- A non-certified Blender optimization course that improves TwinSight assets is useful.
- A non-certified character anatomy course that produces only practice sculpts is low ROI for this target.
- A certificate with no project behind it is weak.

7.0.6 6. Non-certified courses: are they worth doing?

Yes, but selectively.

Non-certified courses from the Excel sheet are worth doing when they meet at least one condition:

1. They directly improve TwinSight.
2. They create a new portfolio piece in less than 2–4 weeks.
3. They fill a specific job-posting skill gap.
4. They strengthen technical art, optimization, procedural workflows or interactivity.
5. They help build a demo reel, not just passive knowledge.

They are not worth doing when:

- they expand into unrelated art directions;
- they require months before visible output;
- they focus on CFX/film pipelines unrelated to Unity/WebGL;
- they produce no public artifact;
- they delay CV, LinkedIn, GitHub or applications.

7.0.7 7. Rebelway strategy

Evidence base

Rebelway's Realtime FX course is positioned around Houdini and Unreal, is 100% online, lists 25+ hours of training, 8 weeks, intermediate skill level and Houdini + Unreal as the software stack. Rebelway also states certificate availability, lifetime access, homework critique and a 5–10 hour weekly commitment for typical courses. Source: <https://www.rebelway.net/realtime-fx-for-games-and-cinematics>

Rebelway's Houdini Fundamentals course lists 35+ hours of training, 12 weeks, beginner skill level, Houdini as the software and 100% online delivery. Source: <https://www.rebelway.net/houdini-fundamentals>

Strategic interpretation

Rebelway is relevant, but not because it gives a certificate. Its value is that it can create high-value portfolio artifacts in procedural FX, Houdini, realtime FX and Unreal/Houdini workflows.

For Alexander's current target, Rebelway is **not the first priority before employment materials**. It becomes high ROI after TwinSight is packaged, because Houdini/procedural workflows can support:

- technical artist credibility;
- simulation-adjacent positioning;
- procedural asset workflows;
- realtime VFX literacy;
- digital twin / technical visualization differentiation;
- long-term technical art specialization.

Rebelway priority from the Excel sheet

Priority	Courses
High	Houdini Fundamentals
High	Introduction to Houdini for 3D Artists
High	Stylized Realtime FX for Games
High	Python for Houdini Artists
Medium	VEX for Houdini Artists
Medium	Advanced Asset Creation
Medium	City Creation in Houdini
Medium	Studio FX Workflows
Medium	Intro to Houdini FX
Low now	Advanced Water FX
Low now	Fluid FX
Low now	Advanced Compositing
Low now	Rigging in Houdini
Low now	Fantasy FX
Low now	Machine Learning in Houdini

Why this ordering

The immediate goal is not to become a film FX artist. The goal is to become employable around Unity, interactive 3D, technical visualization and technical art. Therefore:

- Houdini fundamentals are valuable.
- Procedural asset creation is valuable.
- Python/VEX are valuable if connected to tools/pipeline.
- Realtime FX is valuable if converted into an Unreal/Unity technical breakdown.
- Long fluid/pyro/CFX tracks are lower priority unless the strategy shifts toward VFX/FX TD.

7.0.8 8. Internal course strategy from Excel

Tier 0 — Already useful; convert to portfolio evidence

Course	Use
CG Cookie HUMAN	Blender portrait breakdown
Blender Bros Hard Surface Game Asset	Hard-surface asset workflow
Interactive Content Creation: Blender to Web	3D web/WebGL support
DaVinci Resolve Masterclass	Demo reel editing

These should not be treated as “courses completed” first. They should be converted into output:

- technical breakdown images;
- before/after topology;
- material passes;
- optimization notes;
- final rendered clips;
- short portfolio video.

Tier 1 — Short-term support for TwinSight and applications

Area	Courses
Hard surface	Blender Bros / Hard Ops / Box Cutter
Web 3D	Blender to Web / Spline basics
Demo video	DaVinci Resolve
Blender scripting	Python handlers
Procedural	Geometry Nodes Bootcamp
UE UI/logic	Advanced Widgets / Blueprints basics

These are relevant if used surgically. The goal is not to complete them end-to-end. The goal is to extract the parts that improve portfolio proof.

Tier 2 — Medium-term specialization

Area	Courses
Houdini	Rebelway fundamentals
Realtime FX	Stylized realtime FX / Niagara references
Procedural assets	Advanced Asset Creation / City Creation
Pipeline	Python for Houdini / VEX
Photogrammetry	Photogrammetry + RealityCapture

This tier becomes relevant after the first application cycle starts or after TwinSight is fully packaged.

Tier 3 — Long-term or optional

Area	Examples
CFX	Ziva / Houdini muscles
Character	ZBrush anatomy / creature courses
Compositing	Nuke deep comp / CG integration
Storytelling	Visual storytelling / filmmaking
AI image/video	Creative support only

These are not bad courses. They are simply lower ROI for the current employment target.

7.0.9 9. Certification strategy

Unity certifications

Unity's Certified Associate: Game Developer is designed for future game developers seeking a first professional Unity role. Unity lists prerequisites such as building complex games in C#, having built a game for publication, understanding end-to-end production, debugging and solving programming challenges. The exam covers animation, asset management, lighting, materials/effects, physics, programming, project management and related topics. Source: <https://unity.com/products/unity-certifications/associate-game-developer>

Unity also frames Associate certification as suitable for creators with a portfolio of Unity projects ready to apply for a first professional Unity role.

Recommendation

Unity certification is **optional, not urgent**.

It becomes useful if:

- the portfolio is already public;
- TwinSight has a polished case study;
- the exam can be prepared in 2–4 weeks;
- the cost is acceptable;
- it supports recruiter filtering for Unity roles.

It is not useful if:

- it delays the portfolio;
- it is used to compensate for missing demo evidence;
- it becomes a study rabbit hole;
- it replaces job applications.

Other certifications

Certification area	ROI
Unity Associate	Medium
Unity Professional	Later
Unreal certificates	Low/medium
Blender certificates	Low
Houdini course certificates	Low/medium
Google Data Analytics	Low now
Generic AI certificates	Low unless ARA is polished
Cloud certs	Low unless full-stack route grows

Practical rule

Do not pay for a certification unless it does one of these:

1. Helps pass HR screening.
2. Is commonly requested in target postings.
3. Can be earned quickly.
4. Adds credibility to an already strong project.
5. Does not delay portfolio or applications.

7.0.10 10. Portfolio vs certification tradeoff

Option	Short-term ROI
TwinSight case study	Very high
TwinSight demo video	Very high
GitHub cleanup	High
LinkedIn rewrite	High
Unity certification	Medium
Rebelway course	Medium
Coursera certificate	Low/medium
Master's degree	Low short-term
PhD	Negative short-term

Interpretation

In the next 3–6 months, the highest return comes from packaging existing proof, not from adding more education. The current problem is not absence of learning material. The current problem is **insufficient public evidence of capability**.

7.0.11 11. Coursera and MOOCs

Coursera's Michigan State Unity specialization is a 5-course series, beginner level, with a recommended completion time of about 2 months at 10 hours per week. It teaches Unity/C# and includes 2D and 3D game projects, a capstone and a shareable certificate. Source: <https://www.coursera.org/specializations/game-design-and-development>

This is useful only if Alexander needs structured Unity fundamentals or a small certificate for presentation. It is not more important than TwinSight.

MOOC decision

Scenario	Decision
Need Unity basics	Use selectively
Need C# practice	Use selectively
Need certificate	Coursera acceptable
Already building TwinSight	Lower priority
Need job in 3–6 months	Portfolio first

7.0.12 12. Master's degree analysis

Strategic value of a master's

A master's may be valuable if it provides:

- EU relocation path;
- recognized degree signal;
- visa/residence bridge;
- industry network;
- internship access;
- research credibility;
- specialization in simulation, XR, graphics or digital twins.

A master's is not valuable if it only delays employment without materially improving target-role access.

Evaluated programs

Program	Signal
U-tad M.S. Graphics/VR/Simulation	Strong technical fit
Saarland M.Sc. Visual Computing	Strong research/graphics fit
Utrecht Game & Media Technology	Strong game/media tech fit
butic MBVRIA	Strong Unreal visualization fit

U-tad — M.S. in Graphic Computing, VR and Simulation

U-tad's program is on campus, 60 ECTS, Spanish language, and includes a simulation itinerary and VR itinerary. The study plan includes mathematical fundamentals, numerical methods, dynamic systems simulation, 3D APIs, VR solutions, rendering, effects simulation, data display, internships and a master's thesis. U-tad states that graduates can work as simulation experts, immersive environment developers, computer graphics programmers or researchers in VR/AI. Source: <https://u-tad.com/en/studies/masters-degree-in-computational-graphics-and-simulation/>

Fit: high for simulation/XR/computer graphics.

Weakness: Spanish/on-campus, not necessarily remote-compatible, and may delay job search.

Saarland University — M.Sc. Visual Computing

Saarland University's M.Sc. Visual Computing is taught in English, has a standard period of four semesters, requires 120 CP, and does not charge tuition fees beyond semester fees. The program covers processing visual information, image generation, image synthesis, image analysis, geometry, mathematics, informatics, physics and related disciplines. Admission requires an equivalent related bachelor's degree and proof of advanced English proficiency at C1 level. Source: <https://www.uni-saarland.de/en/study/programmes/master/visual-computing.html>

Fit: high for research-grade computer graphics, simulation and visual computing.

Weakness: academic/research-heavy; stronger for long-term R&D than immediate Unity employment.

Utrecht University — Game and Media Technology

Utrecht's Game and Media Technology master is taught in English, full-time, 2 years, and lists 2026–2027 tuition of €2,694 for Dutch/EU/EEA students and €25,306 for non-EU/EEA students. It emphasizes collaborative projects and links to the Dutch game region. Source: <https://www.uu.nl/en/masters/game-and-media-technology>

Fit: good for EU/post-passport phase.

Weakness: expensive as non-EU before Portuguese passport; 2-year opportunity cost.

butic — MBVRIA

butic's MBVRIA is an online program with official Unreal Engine title, 12 months, 430 hours, 2025–2026 call, and a start date listed for 11 November 2026. It is oriented to Unreal Engine visualization, XR, cinematic design, visual development and realtime 3D. Source: <https://butic.es/master-especialidad-visualizacion-3d-tiempo-real-oficial-unreal-engine-ia/>

Fit: medium/high for Unreal visualization and ArchViz/XR.

Weakness: less aligned with Unity/WebGL unless the strategy shifts toward Unreal realtime visualization.

7.0.13 13. Master's ROI decision matrix

Path	Decision
No master, portfolio-first	Best now
Online specialization	Good if cheap
Spain master	Consider later
Germany master	Good after EU plan
Netherlands master	Good after passport
Portugal master	Needs agent research
PhD	Do not pursue now

When a master's becomes rational

A master's becomes rational if at least three conditions are true:

1. Portuguese passport is obtained or nearly obtained.
2. Tuition drops due to EU status.
3. Program is in English or Spanish and directly tied to graphics/XR/simulation.
4. There is internship access.
5. There is relocation value.
6. Employment search has plateaued.
7. The program produces a portfolio/research project relevant to digital twins or simulation.

When a master's is irrational

A master's is irrational if:

- it delays job applications;
- it is pursued to compensate for insecurity;
- it is unaffordable without income;
- it has weak industry placement;
- it is purely artistic and not technical;
- it does not create EU or network advantage.

7.0.14 14. PhD viability

A PhD is not recommended for the current 3–24 month plan.

When it would strengthen the profile

- Goal shifts to computer graphics research.
- Goal shifts to simulation R&D.

- Goal shifts to university/research labs.
- There is funded admission.
- There is a specific advisor/project in graphics, XR, digital twins or human-computer interaction.

When it weakens the profile

- Goal is international employability in 3–6 months.
- Goal is USD 3k–6k/month income growth.
- Portfolio is not yet public.
- Industry experience is still limited.
- The PhD is used as an avoidance strategy.

Decision

Do not pursue PhD now. Re-evaluate only after 2–3 years of industry experience or after a strong funded research opportunity appears.

7.0.15 15. Time allocation model

0–6 months

Focus:

- thesis defense;
- TwinSight case study;
- demo video;
- GitHub cleanup;
- LinkedIn/CV;
- job applications;
- one or two small support courses only.

Recommended course use:

- DaVinci Resolve for demo editing;
- Blender-to-Web if it improves WebGL/Web portfolio;
- Blender Python handlers if it creates a tool;
- no long Rebelway track unless applications are already running.

6–12 months

Focus:

- targeted specialization;
- first paid work;
- secondary portfolio piece;
- light certification if useful.

Recommended course use:

- Houdini Fundamentals;
- Introduction to Houdini for 3D Artists;
- Python for Houdini Artists;
- Stylized Realtime FX for Games;
- Unity certification only if portfolio already exists.

12–24 months

Focus:

- higher compensation;
- EU/passport planning;
- advanced specialization;
- master's decision.

Recommended course use:

- VEX for Houdini Artists;
 - Advanced Asset Creation;
 - City Creation in Houdini;
 - Unreal/XR visualization if market pull exists;
 - evaluate Saarland/Utrecht/U-tad/butic or Portugal programs.
-

7.0.16 16. Decision rules: course vs certificate vs master**Choose a no-certificate course when**

- it is already owned;
- it fills one specific skill gap;
- it produces a portfolio artifact;
- it can be completed or mined in under 2–4 weeks;
- the result can be shown publicly.

Choose a certificate when

- recruiters actually recognize it;
- preparation time is short;
- it supports an already strong project;
- it appears in job filters;
- it is not expensive relative to current runway.

Choose a master's when

- it provides relocation leverage;
- it provides EU tuition advantage;
- it provides internship access;
- it is strongly aligned with simulation/XR/computer graphics;
- job search results show degree/visa/network as bottlenecks.

Reject or postpone when

- the course expands scope;
 - the certificate is decorative;
 - the master is unaffordable;
 - the PhD delays employability;
 - the activity does not produce public evidence.
-

7.0.17 17. Education roadmap

Immediate queue

1. Finish thesis defense.
2. Package TwinSight.
3. Edit 60–120 second demo video.
4. Clean GitHub README.
5. Create portfolio MVP.
6. Use short support courses only.

Short course queue

1. DaVinci Resolve.
2. Interactive Blender to Web.
3. Blender Python Handlers.
4. Hard Surface Game Asset.
5. Geometry Nodes basics.

Medium specialization queue

1. Houdini Fundamentals.
2. Introduction to Houdini for 3D Artists.
3. Stylized Realtime FX for Games.
4. Python for Houdini Artists.
5. VEX for Houdini Artists.

Long specialization queue

1. Advanced Asset Creation.
 2. City Creation in Houdini.
 3. Studio FX Workflows.
 4. Advanced Water/Fluid FX.
 5. XR / Unreal visualization if market demand appears.
-

7.0.18 18. How to present non-certified courses

Do not write

- “Certified in Houdini” if no recognized certification exists.
- “Certified Technical Artist” if only a course was completed.
- Long lists of Udemy/Gnomon/Wingfox courses.
- “Mastered” unless there is public proof.

Acceptable wording

- “Additional training in Houdini procedural workflows, Blender hard-surface asset production and realtime FX.”
- “Completed project-based training in Blender portrait production and technical asset workflows.”
- “Self-directed study supported by Rebelway, CG Cookie, Gnomon, Blender Bros and CG Circuit course material.”
- “Coursework used to produce portfolio artifacts: [project name].”

Best practice

Attach each course cluster to an output:

- course → artifact;
 - artifact → skill;
 - skill → role;
 - role → portfolio section.
-

7.0.19 19. Education risks

Main risks

- Course hoarding.
- Certification chasing.
- Delaying applications.
- Fragmented identity.
- Over-indexing on Unreal while positioning as Unity.
- Becoming a generalist 3D artist instead of technical visualization developer.
- Spending months on CFX/character while the job target is interactive 3D.

Control mechanisms

- No course without a deliverable.
 - No more than one main course at a time.
 - No long course before TwinSight is public.
 - No certificate before portfolio MVP.
 - No master before first job-search cycle.
 - Monthly review: course output must become public evidence.
-

7.0.20 20. Final recommendation

The optimal education strategy is:

1. **Do not start a master's now.**
2. **Do not start a PhD.**
3. **Do not complete the whole Excel course library.**
4. **Use courses as targeted production support.**
5. **Use Rebelway later for Houdini/procedural/realtime FX specialization.**
6. **Use Unity certification only after TwinSight is public.**
7. **Convert existing courses into visible artifacts.**
8. **Re-evaluate master's programs after Portuguese passport status, first job-search cycle and income situation are clearer.**

The current bottleneck is not education. The bottleneck is **public proof, positioning and application execution.**

7.0.21 21. Evidence gaps and whether agent is needed

Agent is not required to finalize this strategic section.

Agent may be useful later for a separate annex:

```
Bloque de codigo: text
06A_eu_masters_program_catalog.md
```

That annex would collect 30–50 concrete master's programs across Portugal, Germany, Netherlands, Nordics and Spain with tuition, language, application deadlines, admission requirements, scholarships and links.

For the current section, the decision is already clear enough without spending an agent use.

7.0.22 Source links used

- Rebelway — Realtime FX in Houdini & Unreal Engine: <https://www.rebelway.net/realtime-fx-for-games-and-cinematics>
- Rebelway — Houdini Fundamentals: <https://www.rebelway.net/houdini-fundamentals>
- Unity — Certified Associate: Game Developer: <https://unity.com/products/unity-certifications/associate-game-developer>
- Coursera — Game Design and Development with Unity Specialization: <https://www.coursera.org/specializations/game-design-and-development>
- U-tad — M.S. in Graphic Computing, VR and Simulation: <https://u-tad.com/en/studies/masters-degree-in-computational-graphics-and-simulation/>
- butic — MBVRIA, Realtime 3D Visualization + Unreal Engine + IA: <https://butic.es/master-especialidad-visualizacion-3d-tiempo-real-oficial-unreal-engine-ia/>
- Saarland University — M.Sc. Visual Computing: <https://www.uni-saarland.de/en/study/programmes/master/visual-computing.html>
- Utrecht University — Game and Media Technology: <https://www.uu.nl/en/masters/game-and-media-technology>

Chapter 8

07 — Portfolio Strategy and Project Architecture

Fuente: 07_portfolio_strategy_and_project_architecture.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 776

Ultima modificacion: 2026-06-11 13:30

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This file defines the portfolio architecture for Alexander Woodcock Salomón's international job-search strategy.

Contenido completo depurado

8.0.1 0. Purpose of this section

This file defines the portfolio architecture for Alexander Woodcock Salomón's international job-search strategy.

It does **not** write the final portfolio copy yet. It defines what the portfolio must prove, which projects should be included, which projects should be downplayed, what evidence each project must show, and how the portfolio should be structured for the target role families defined in 02_market_role_fit_and_positioning.md.

The core principle is simple:

The portfolio must make TwinSight X500 look like a credible professional technical visualization project, not just an academic thesis or a personal experiment.

The portfolio must be optimized for:

- remote international hiring from Colombia;
- Unity / Real-Time 3D / Interactive 3D roles;
- technical visualization, digital twin, simulation and CAD-to-realtime roles;
- Unity Technical Artist roles focused on optimization, runtime systems and tool-adjacent workflows;
- secondary Python / automation / AI tooling credibility without diluting the main profile.

8.0.2 1. Current source-of-truth constraints

1.1 Correct current state

Area	Portfolio implication
Thesis	Completed; pending defense. Do not describe it as early-stage or in progress.
Flagship asset	TwinSight X500. This must dominate the portfolio.
Main role axis	Real-Time 3D / Unity / Technical Visualization / WebGL.
Secondary axis	Python / AI-assisted tools / research automation.
Current risk	Portfolio fragmentation and over-positioning.
Short-term goal	Be employable within 3–6 months.
Medium-term goal	Build stronger international leverage in 12–24 months.
EU status	No current EU work authorization. Future Portuguese passport expected in ~2 years.

1.2 Claims that must not appear

The portfolio must avoid the following claims unless future evidence supports them:

- “Senior Technical Artist” as a primary title.
- “Digital Twin Engineer” as if there is formal industry experience.
- “AI Engineer” or “Machine Learning Engineer.”
- “Full-stack Developer” as the main identity.
- “EU work authorized” before Portuguese passport or German residence authorization is actually obtained.
- “Certified” for courses that do not issue formal certificates.
- “Production-ready enterprise AI system” for ARA.
- “Commercial shipped Unity product” for TwinSight unless deployed publicly and documented as such.

8.0.3 2. Portfolio objective by hiring audience

Different audiences will inspect the same portfolio differently. The architecture must allow each audience to find the relevant proof quickly.

Audience	What they need to verify
Recruiter Unity lead	Role fit, technologies, English clarity, availability, portfolio quality. Systems thinking, C# structure, Unity implementation, debugging maturity.
Technical artist lead Simulation / digital twin company	Optimization, shader/material workflow, asset pipeline, visual modes. CAD-to-realtime, technical UI, spatial explanation, technical documentation.
Web / interactive 3D company AI tooling startup	WebGL deployment, performance, UI, browser constraints. Python, LangGraph, automation logic, system design, documentation discipline.
Game studio	Unity fluency, technical art foundation, visual polish, shader/FX potential.

The portfolio must not try to satisfy all audiences equally on the first screen. It must lead with the strongest overlap:

Real-time 3D technical visualization using Unity, WebGL, CAD optimization and interactive systems.

8.0.4 3. Recommended portfolio positioning

3.1 Primary portfolio title

Recommended title:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist
Interactive Technical Visualization • CAD-to-Realtime • Unity WebGL
```

3.2 Secondary subtitle

Bloque de código: text
I build interactive 3D systems that transform complex technical products into usable real-time visual experiences.

3.3 Alternative variants by target role

Target role family	Portfolio title variant
Interactive 3D Unity WebGL	Real-Time 3D Developer focused on interactive technical visualization. Unity WebGL Developer building optimized browser-based 3D applications.
Technical Visualization	Technical Visualization Developer specializing in CAD-to-realtime workflows.
Unity Technical Artist	Unity Technical Artist focused on optimization, shaders, UI and runtime interaction systems.
Tools / AI secondary	Technical Artist with Python automation and AI-assisted production tooling.

3.4 Positioning rule

Use **one primary identity** and **one secondary modifier**.

Good:

Bloque de código: text
Real-Time 3D Developer / Unity Technical Artist

Too diluted:

Bloque de código: text
Technical Artist / Unity Developer / AI Engineer / Full-stack Developer / 3D Artist / XR Developer

8.05 4. Project hierarchy

4.1 Project priority ranking

Priority	Project	Portfolio role
1	TwinSight X500	Flagship case study.
2	ARA Framework	Secondary technical tooling case study.
3	Hyperrealistic Blender portrait	Visual/asset pipeline proof.
4	ai-news-aggregator	Optional secondary software proof if functional.
5	FitApp-Free	Mostly omit unless product/system logic is needed.
6	Older/lower-quality repos	Hide, archive, or omit from recruiter-facing portfolio.

4.2 What each project proves

Project	Strong proof	Weak proof
TwinSight X500	Unity, C#, WebGL, CAD-to-realtime, technical UI, optimization, validation.	Commercial shipping, team production experience.
ARA Framework	Python, LangGraph, automation, multi-agent logic, research tooling.	Production AI engineering, ML research.

Project	Strong proof	Weak proof
Blender portrait	Blender, topology, materials, grooming, lighting, persistence.	Runtime optimization unless documented.
ai-news-aggregator	FastAPI, React, PostgreSQL, Redis, full-stack logic.	Main technical art fit.
FitApp-Free	Product logic, systems thinking, domain modeling.	Unity / 3D / technical art relevance.

4.3 Project inclusion decision

Project	Include in MVP?	Include in full portfolio?	Condition
TwinSight X500	Yes	Yes	Must have demo video, screenshots, case study and technical breakdown.
ARA Framework	No / optional	Yes	Only if cleaned, documented and presented as prototype.
Blender portrait	Optional	Yes	Must be presented as pipeline breakdown, not just render.
ai-news-aggregator	No	Maybe	Only if functional and visually presentable.
FitApp-Free	No	Maybe	Only if used as product/system-design proof.
Old GitHub tutorials	No	No	Hide or leave unfeatured.

8.0.6 5. Portfolio architecture

5.1 MVP portfolio architecture

The MVP portfolio should be small, polished and role-aligned. It does not need to contain every project.

Recommended MVP structure:

```
Bloque de codigo: text
Home
├── Hero / Positioning
├── Featured Project: TwinSight X500
├── Technical Breakdown Preview
├── Skills / Toolchain
├── Secondary Proof
│   ├── ARA Framework preview OR Blender portrait preview
├── Contact / Links
│   ├── GitHub
│   ├── LinkedIn
│   ├── ArtStation
│   └── Email
```

5.2 Full portfolio architecture

The full version can expand after applications start.

```
Bloque de codigo: text
Home
├── About / Positioning
├── Case Studies
│   ├── TwinSight X500
│   ├── ARA Framework
│   ├── Blender Portrait Pipeline
│   └── Optional: AI News Aggregator
├── Technical Breakdowns
│   ├── CAD-to-Realtime Pipeline
│   ├── Unity WebGL Optimization
│   ├── Visual Modes / Shader Systems
│   └── UI Toolkit / Interaction Systems
├── Projects Archive
├── Resume / CV
└── Contact
```

5.3 Recommended first-release format

Do not wait for a perfect custom website.
Priority order:

- 1. One public TwinSight case study page.
- 2. One 60–120 second demo video.
- 3. One GitHub README recruiter version.
- 4. One ArtStation or Behance-style technical breakdown.
- 5. Personal site only after those assets exist.

8.0.7 6. TwinSight X500 portfolio architecture

TwinSight is the core proof asset. Its case study must be treated as a professional project.

6.1 Required sections

```
Bloque de codigo: text
TwinSight X500 Case Study
├─ 1. One-line summary
├─ 2. Problem
├─ 3. Role and contribution
├─ 4. Technical constraints
├─ 5. CAD-to-realtime pipeline
├─ 6. Unity implementation
├─ 7. Interaction systems
├─ 8. Visual modes
├─ 9. WebGL / performance considerations
├─ 10. Evaluation and metrics
├─ 11. AI-assisted development disclosure
├─ 12. Results
├─ 13. Limitations
└─ 14. What I would improve next
```

6.2 What TwinSight must prove

Proof target	Evidence needed
Unity competence	Screenshots of scene, systems, UI, inspector organization.
C# implementation	Code snippets, architecture diagram, interaction flow.
CAD optimization	Before/after triangle counts, retopology examples.
WebGL readiness	Build notes, browser/mobile constraints, performance decisions.
Technical art	Materials, visual modes, X-Ray, wireframe, ghosted, blueprint.
Product thinking	User tasks, usability testing, SUS/NASA-TLX summary.
Communication	Clear diagrams and concise technical explanations.

6.3 TwinSight must not overclaim

Avoid:

- “Digital twin platform” unless framed carefully as a prototype for technical visualization.
- “Thermal simulation” if thermal mode is qualitative, not physics-based FEA.
- “Production-ready industrial system” unless validated as such.
- “Fully original CAD model” if manufacturer CAD or base geometry was used.
- “Built without assistance” if AI was materially used in programming.

Better wording:

Bloque de codigo: text
Unity WebGL technical visualization prototype for drone assembly inspection, built as a completed thesis project using
↔ CAD-to-realtime optimization, interactive systems, technical UI and usability evaluation.

8.0.8 7. ARA Framework portfolio architecture

ARA is useful, but it must not dilute the main identity.

7.1 Recommended positioning

Bloque de codigo: text
Secondary project: Python-based research automation prototype using LangGraph-style multi-agent orchestration.

7.2 Include only if the following are ready

Requirement	Status needed before publishing
README	Clear, honest and runnable or semi-runnable.
Demo	Short video or terminal walkthrough.
Architecture	Agent flow diagram.
Scope	Explicitly marked as prototype.
Claims	No enterprise-grade AI claims.
Security	No API keys, secrets, private prompts or unstable dependencies exposed.

7.3 What ARA can prove

- Python architecture.
- Workflow automation.
- Research tooling.
- LLM orchestration literacy.
- Technical documentation.
- Systems thinking.

7.4 What ARA does not prove yet

- ML engineering.
- Model training.
- Production AI deployment.
- Scalable SaaS product.
- Enterprise reliability.

8.0.9 8. Blender portrait portfolio architecture

The Blender portrait should be included only if presented as a technical breakdown.

8.1 Correct framing

Good:

Bloque de codigo: text
High-fidelity Blender character pipeline study focused on topology, materials, grooming, lighting and render presentation.

Weak:

Bloque de codigo: text
Portrait render.

8.2 Required breakdown assets

Asset	Purpose
Final render	Visual quality.
Clay render	Form and modeling.
Topology screenshots	Technical mesh quality.
UV layout	Asset pipeline proof.
Material node screenshots	Shader/material understanding.
Grooming breakdown	Character detail.
Lighting setup	Presentation skill.
Before/after stages	Process discipline.

8.3 Role relevance

Role	Relevance
Unity Technical Artist	Medium if connected to optimization/materials.
Real-Time 3D Developer	Low/medium.
3D Artist	High, but this is not the primary target.
Technical Visualization	Low.
Game Technical Artist	Medium if adapted to real-time constraints.

8.0.10 9. ai-news-aggregator and FitApp-Free

9.1 ai-news-aggregator

This project can support a secondary software/AI tooling narrative only if it is functional.

Include only if:

- there is a working demo;
- the README explains the architecture;
- the UI is presentable;
- the project does not look abandoned;
- it supports Python/FastAPI/React/PostgreSQL evidence.

Do not pin it above TwinSight or ARA.

9.2 FitApp-Free

FitApp-Free is probably not relevant to the main portfolio.

Include only if used to show:

- product logic;
- progression systems;
- routine planning;
- user-state modeling;
- database design.

Otherwise omit from recruiter-facing materials.

8.0.11 10. GitHub portfolio architecture

GitHub must support the portfolio, not expose disorder.

10.1 Pinned repository strategy

Recommended pinned repos:

Priority	Repo	Action
1	WebGL-Thesis-Proposal / TwinSight	Clean README, add images, video, metrics.
2	ARA Framework	Publish only after cleanup.
3	ai-news-aggregator	Publish only if functional.
4	Portfolio site	Publish once stable.
5	Small Unity/WebGL demo	Create if needed.
6	Blender breakdown repo/page	Optional.

10.2 Repos to avoid featuring

- skills-introduction-to-github
- unfinished tutorials;
- old experimental repos without README;
- abandoned client projects;
- incomplete portfolio attempts;
- anything with secrets, poor commit hygiene or broken setup.

10.3 README standards

Every featured repo needs:

```
Bloque de código: text
- Project title
- One-line summary
- Role and contribution
- Tech stack
- Features
- Screenshots or GIFs
- Demo link or video
- Setup instructions if public
- Architecture notes
- Limitations
- Status
```

8.0.12 11. ArtStation / Behance / visual platform strategy

ArtStation is useful for visual credibility, but it must not reposition the profile as a pure 3D artist.

11.1 Recommended ArtStation projects

Project	Format
TwinSight X500	Technical breakdown.
Blender portrait	Character pipeline breakdown.
Future Rebelway/Houdini project	FX breakdown if completed.

11.2 ArtStation structure for TwinSight

```
Bloque de código: text
Title: TwinSight X500 – Unity WebGL Technical Visualization Prototype
Images:
1. Final interactive viewer screenshot
2. Exploded view
3. X-Ray / ghosted mode
4. Clipping / cross-section
5. CAD-to-realtime before/after
6. UI interaction screens
7. Optimization metrics graphic
8. Architecture diagram
Description:
- Problem
- Role
- Pipeline
- Technical systems
- Metrics
- Limitations
```

11.3 Avoid

- posting only beauty renders;
- no explanation;
- unclear role contribution;
- mixing unrelated fan art or unfinished sketches in the main profile;
- presenting course outputs as original production work without context.

8.0.13 12. Demo video strategy

12.1 MVP video requirement

A portfolio without video is weaker for interactive 3D roles. TwinSight needs a short demo.

Recommended duration:

```
Bloque de código: text
60-120 seconds
```

12.2 Video structure

```
Bloque de código: text
0-10s: Problem and final result
10-25s: Interactive selection and component inspection
25-40s: Exploded view
40-55s: Cross-section / clipping
55-70s: Visual modes
70-90s: CAD optimization and WebGL constraints
90-110s: Validation metrics and outcome
110-120s: Tech stack and links
```

12.3 Video rules

- Show the application immediately.
- Avoid long intros.
- Use captions.
- Use clean screen capture.
- Include before/after CAD optimization if visually understandable.
- Include mobile/browser footage if WebGL/mobile-first is claimed.
- Do not overstate thermal or digital twin claims.

8.0.14 13. Screenshot and asset shot list

13.1 TwinSight screenshots

Screenshot	Purpose
Hero final viewer	Immediate visual proof.
Component selection	Interaction proof.
Exploded view	Spatial relationship proof.
Cross-section clipping	Technical visualization proof.
X-Ray / ghost mode	Technical art proof.
Blueprint / wireframe	Visual mode proof.
UI panels	UX and UI Toolkit proof.
Before/after mesh optimization	CAD-to-realtime proof.
Mobile/browser view	WebGL deployment proof.
Evaluation chart	Academic validation proof.

13.2 ARA screenshots

Screenshot	Purpose
Architecture diagram	Multi-agent system proof.
Terminal/demo output	Functionality proof.
Generated report example	End-to-end value.
Config/checkpointing	Engineering structure.

13.3 Blender portrait screenshots

Screenshot	Purpose
Final render	Visual quality.
Clay render	Form.
Topology	Technical modeling.
UVs	Production pipeline.
Materials	Shading.
Grooming	Detail process.

8.0.15 14. Portfolio MVP checklist

The MVP is ready to send to recruiters only when the following exist:

```
Bloque de código: text
[ ] TwinSight 60-120s demo video
[ ] TwinSight case study page
[ ] TwinSight README recruiter version
[ ] 8-12 strong screenshots
[ ] clear role/contribution statement
[ ] metrics table: triangles, SUS, NASA-TLX, participant count
[ ] AI-assisted development disclosure wording
[ ] GitHub profile cleaned
[ ] LinkedIn featured section updated
[ ] ArtStation or visual breakdown published
[ ] one-page CV aligned with portfolio
```

Do not wait for:

- full custom website perfection;
 - all secondary projects polished;
 - all courses finished;
 - perfect visual identity;
 - a complete demo reel with many projects.
-

8.0.16 15. Portfolio quality bar

15.1 Minimum professional standard

A recruiter or lead should understand within 30 seconds:

- who Alexander is;
- what role he targets;
- what TwinSight is;
- what technologies were used;
- what he personally contributed;
- why the project is relevant to the job;
- where to see more.

15.2 Strong professional standard

A technical lead should understand within 3–5 minutes:

- the core system architecture;
 - CAD-to-realtime decisions;
 - Unity/WebGL constraints;
 - interaction flow;
 - performance/optimization logic;
 - validation method;
 - limitations;
 - next improvements.
-

8.0.17 16. AI-assisted development disclosure

16.1 Disclosure principle

Do not hide AI use if asked. Do not foreground it unnecessarily in the first sentence.

16.2 Recommended wording

Bloque de código: text

I used AI tools as coding and documentation assistants during implementation, while retaining responsibility for architecture decisions, integration, debugging, testing, Unity implementation, technical judgment and final delivery.

16.3 Avoid

Bloque de código: text

AI built the project.

Bloque de código: text

I coded everything manually without assistance.

Bloque de código: text

Vibe-coded prototype.

16.4 Where to include disclosure

Location	Include?
CV bullet	Usually no.
Portfolio case study	Yes, in process/limitations section.
Interview	Yes, if asked.
GitHub README	Optional.
LinkedIn headline	No.

8.0.18 17. Portfolio risks and mitigations

Risk	Mitigation
Looks academic, not professional	Use case-study structure, demo video and technical metrics.
Too many identities	Lead with Real-Time 3D / Unity Technical Visualization.
ARA dilutes portfolio	Keep it secondary and clearly separated.
Blender portrait shifts profile to pure artist	Present it as technical pipeline evidence.
GitHub looks messy	Pin only relevant repos and clean READMEs.
AI use reduces credibility	Explain AI as assistant, not author.
Manufacturer/CAD asset ambiguity	State source and transformation clearly.
WebGL performance unproven	Include performance notes and device/browser constraints.
Thermal mode overclaim	Label as qualitative visual mode unless physically simulated.

8.0.19 18. Application-stage portfolio variants

18.1 Variant A — Technical Visualization / Digital Twin

Lead with:

- TwinSight X500;
- CAD-to-realtime pipeline;
- assembly inspection problem;
- clipping, exploded view and component understanding;
- usability metrics.

Downplay:

- Blender portrait;
- game art;
- FitApp.

18.2 Variant B — Unity WebGL / Interactive 3D

Lead with:

- WebGL deployment;
- interaction systems;
- UI Toolkit;
- browser constraints;
- optimization.

Downplay:

- academic evaluation unless relevant;
- AI tooling.

18.3 Variant C — Unity Technical Artist

Lead with:

- optimization;
- visual modes;
- shaders/materials;
- asset pipeline;
- Blender-to-Unity workflow.

Include:

- Blender portrait pipeline if polished.

18.4 Variant D — Tools / Python Automation

Lead with:

- ARA Framework;
- Python automation;
- workflow orchestration;
- documentation generation.

Still mention TwinSight as 3D systems proof.

8.0.20 19. Recommended build order

Order	Task	Reason
1	TwinSight demo video	Highest proof density.
2	TwinSight case study	Core portfolio asset.
3	GitHub README cleanup	Recruiter and technical lead support.
4	LinkedIn Featured update	Distribution channel.
5	ArtStation technical breakdown	Visual/technical credibility.
6	ARA cleanup	Secondary route.
7	Blender portrait breakdown	Supplemental technical art proof.
8	Custom portfolio site polish	Useful, but not first bottleneck.

8.0.21 20. Definition of done

This section is considered implemented when the following exist:

```
Bloque de codigo: text
[ ] TwinSight has one polished case study.
[ ] TwinSight has one short demo video.
[ ] TwinSight has a recruiter-facing README.
[ ] GitHub pinned repos are curated.
[ ] LinkedIn Featured section points to TwinSight first.
[ ] ArtStation contains at least one technical breakdown.
[ ] Secondary projects are clearly labeled as secondary.
[ ] No outdated "thesis in progress" language remains.
[ ] No false certification or EU authorization claim exists.
```

8.0.22 21. Open questions before implementation

These questions must be resolved during the next sections:

1. Is there a public WebGL build of TwinSight ready to share?
2. Are all manufacturer/CAD assets legally safe to show publicly?
3. Which screenshots are already available?
4. Is there a current GitHub branch that should become the public recruiter version?
5. Should the portfolio be hosted on GitHub Pages, a custom domain, Webflow, Notion, or another static site?
6. Can ARA be safely made public without exposing secrets or unstable code?
7. Does the Blender portrait have enough breakdown images to justify inclusion?
8. Are ai-news-aggregator and FitApp-Free functional enough to include, or should they remain private?

8.0.23 22. Final portfolio direction

The portfolio should not be a broad archive of everything Alexander has touched.

It should be a targeted proof system built around one main claim:

Alexander can design, optimize and implement interactive real-time 3D technical visualization systems using Unity, WebGL, Blender and CAD-to-realtime workflows.

Everything else is secondary.

Chapter 9

08 — TwinSight X500 Case Study

Fuente: 08_twinsight_x500_case_study.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 893

Ultima modificacion: 2026-06-11 13:32

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This document defines the complete portfolio case study structure for **TwinSight X500 / WebGL-Thesis-Proposal**, the flagship project for Alexander Woodcock Salomón's international job-search strategy.

Contenido completo depurado

9.0.1 Document purpose

This document defines the complete portfolio case study structure for **TwinSight X500 / WebGL-Thesis-Proposal**, the flagship project for Alexander Woodcock Salomón's international job-search strategy.

The objective is to convert the thesis project into a recruiter-readable and technically credible case study for these role families:

1. Real-Time 3D / Interactive Technical Visualization Developer.
2. Unity WebGL / Interactive 3D Developer.
3. Technical Visualization / Digital Twin / Simulation Developer.
4. Unity Technical Artist — optimization, runtime interaction, tools, technical UI.
5. CAD-to-Realtime Technical Artist / Optimization Specialist.

This document is not the final public portfolio page yet. It is the **source case-study blueprint** from which the website page, GitHub README, LinkedIn project entry, ArtStation post, demo video, interview talking points and CV bullets should be derived.

9.0.2 Source-of-truth status

Field	Current status
Project name	TwinSight X500 / WebGL-Thesis-Proposal
Academic context	Ingeniería Multimedia thesis project, UNAD
Thesis status	Complete, pending defense / sustentación
Public repo	https://github.com/delarge95/WebGL-Thesis-Proposal

Field	Current status
Candidate role	Unity integration, runtime systems, 3D optimization, UI, documentation, evaluation
Project type	Unity WebGL interactive technical visualization prototype
Subject	Holybro X500 V2 drone assembly visualization
Primary value	Convert complex 2D/CAD assembly documentation into interactive 3D understanding
Primary portfolio use	Flagship project for international technical art / real-time 3D / visualization roles

9.0.3 Correct positioning

Recommended one-line positioning

TwinSight X500 is a Unity WebGL interactive technical visualization prototype that transforms complex drone assembly documentation into a real-time 3D inspection experience, combining CAD-to-realtime optimization, runtime interaction systems, technical UI and usability validation.

Short portfolio summary

TwinSight X500 is an interactive 3D technical visualization system built in Unity WebGL for inspecting the assembly structure of a Holybro X500 V2 drone. The project converts manufacturer CAD and technical documentation into an optimized real-time model with component selection, exploded view, clipping, technical information panels and multiple visualization modes.

The project was developed as an academic thesis and validated with usability and workload instruments. Its purpose is not only to show parts, but to help users understand spatial relationships in a complex technical assembly.

Recruiter-facing framing

Use this project to signal:

- Unity / C# implementation capacity.
- Real-time 3D interaction design.
- CAD-to-realtime asset optimization.
- WebGL deployment awareness.
- Technical UI and component information systems.
- Ability to connect engineering documentation, 3D production and user evaluation.
- Ability to document and defend technical design decisions.

What this project should not claim

Do **not** present TwinSight as:

- a commercial shipped product;
- a production digital twin connected to live telemetry;
- an engineering-grade simulation;
- an FEA / CFD / thermodynamic simulator;
- a certified aerospace training platform;
- an original CAD design of the drone;
- a full-stack SaaS product;
- a solo-coded project without AI assistance;
- evidence of senior-level industry production experience.

9.0.4 Core problem

Traditional assembly documentation often relies on 2D diagrams, exploded drawings, text instructions and static CAD references. For spatially complex systems such as drones, users must mentally reconstruct relationships between parts, fasteners, modules, vertical layers and internal assemblies.

TwinSight addresses this problem by creating an interactive 3D representation where the user can inspect components, isolate assemblies, visualize relationships and understand how parts connect in space.

Core thesis statement

Seeing parts is not enough. Users need to understand spatial relationships.

Problem dimensions

1. Spatial reconstruction load

Static documentation requires the user to infer 3D relationships from 2D views.

1. Component relationship ambiguity

Assemblies contain parent parts, subparts, fasteners, mounts and nested modules that are difficult to understand from text alone.

1. Scale and density

Imported CAD data can be too dense for real-time web deployment and must be optimized.

1. Web performance constraints

A browser-based viewer must balance fidelity, interactivity and performance.

1. Instructional clarity

Users need technical information, not just visual inspection.

9.0.5 Project objectives

General objective

Design and implement an interactive Unity WebGL 3D visualization prototype that supports the inspection and understanding of the Holybro X500 V2 drone assembly.

Specific objectives

1. Convert CAD/manufacturer geometry and technical references into a real-time optimized 3D asset pipeline.
 2. Implement runtime interaction systems for component selection, isolation, exploded view and technical inspection.
 3. Design a technical UI that exposes part identity, specifications and assembly information.
 4. Add visual modes that support different inspection needs: realistic, X-Ray, blueprint, solid color, wireframe, ghosted and qualitative thermal visualization.
 5. Deploy the experience through Unity WebGL with web/mobile performance considerations.
 6. Evaluate the prototype using usability, workload and qualitative feedback instruments.
-

9.0.6 Project context and constraints

Technical context

TwinSight is not a conventional game project. It sits between real-time 3D development, technical art, CAD optimization, digital product visualization and instructional interface design.

The central challenge is not only visual quality. The central challenge is making a complex technical assembly understandable, performant and navigable in a browser environment.

Constraints

Constraint	Impact
CAD density	Required polygon reduction and selective reconstruction
WebGL target	Required performance-aware asset decisions
Academic scope	Prototype quality, not commercial deployment quality
Manufacturer assets	Requires careful attribution and portfolio risk mitigation
Time constraints	Some features may be prototype-level rather than production-hardened
AI-assisted coding	Requires transparent explanation of implementation process

9.0.7 Technical stack

Area	Tools / technologies
Engine	Unity
Runtime language	C#
Target platform	WebGL / browser
Rendering	URP, real-time visualization modes
UI	Unity UI Toolkit / technical interface systems
3D pipeline	Blender, CAD cleanup, retopology, optimization
CAD source	Holybro X500 V2 manufacturer geometry / technical references
Optimization	CAD-to-realtime reduction, modular assets, high-to-low workflow
Evaluation	SUS, NASA-TLX Raw, Think-Aloud, task records
Versioning	Git / GitHub
AI assistance	Used as implementation support; final integration/debugging/technical judgment retained by candidate

9.0.8 Feature inventory

Core interaction features

Feature	Status	Portfolio value
Component selection	Implemented / expected	Shows runtime interaction design
Technical part panels	Implemented / expected	Shows UI + documentation integration
Exploded view	Implemented / expected	Shows assembly relationship visualization
Cross-section / clipping	Implemented / expected	Shows technical inspection capability
Isolation / focus	Implemented / expected	Shows user-centered navigation
Multiple visual modes	Implemented / expected	Shows technical art and rendering modes
Mobile-first UI thinking	Implemented / expected	Shows web deployment awareness
Performance optimization	Implemented / expected	Shows CAD-to-realtime competence

Visual modes

The public case study should show each visual mode with a screenshot or short clip:

1. Realistic.
2. X-Ray.
3. Blueprint.
4. Solid color.
5. Wireframe.
6. Ghosted.
7. Qualitative thermal visualization.

Technical data panels

Recommended information hierarchy:

1. Identification.
2. Specifications.
3. Assembly relationship.
4. Parent / child relationship.
5. Notes / warnings / interpretation.

9.0.9 Metrics and validation

Reported quantitative metrics

Metric	Reported value	Interpretation
CAD source complexity	Above 6.5M triangles	Indicates need for optimization
Optimized result	95,617 triangles	Core evidence for CAD-to-realtime workflow
Validation participants	12	Formative usability validation; not large-scale statistical validation
SUS average	91.88	Strong usability signal, but sample size should be stated
NASA-TLX Raw — 3D viewer	8.69	Lower perceived workload condition
NASA-TLX Raw — 2D support	19.89	Higher perceived workload comparator
Task-condition records	96	Useful evaluation dataset size for task-level observations

How to present the metrics

Use precise language:

In a formative academic evaluation with 12 participants and 96 task-condition records, the prototype obtained a SUS average of 91.88 and lower NASA-TLX Raw workload scores for the 3D viewer condition compared with the 2D support condition.

Avoid overclaiming:

- Do not claim universal proof of superiority.
- Do not claim statistical generalization unless the thesis includes formal statistical testing.
- Do not claim commercial validation.
- Do not claim training certification validity.

Suggested visualization for portfolio

Use a compact before/after metric panel:

Before / comparator	After / result
Dense CAD above 6.5M triangles 2D support NASA-TLX Raw 19.89 Static assembly references	Optimized model at 95,617 triangles 3D viewer NASA-TLX Raw 8.69 Interactive component inspection

9.0.10 Candidate contribution map

Strongly defensible contribution areas

Contribution	Evidence needed
Unity integration	repo structure, scripts, demo video
Runtime interaction systems	selection, exploded view, clipping, visual modes
Technical UI	screenshots, UI hierarchy, data panel structure
CAD-to-realtime workflow	before/after topology, polycount table, optimization notes
Blender optimization	screenshots of cleanup/retopo/high-low workflow
Propeller modeling	modeling breakdown / wireframe
Academic evaluation	SUS/TLX tables, methodology summary
Documentation	thesis excerpts, diagrams, README
AI-assisted implementation management	explanation of architecture, debugging, integration decisions

Carefully worded contribution statement

I developed TwinSight X500 as a thesis prototype, leading the Unity integration, runtime interaction systems, technical UI, CAD-to-realtime optimization workflow, documentation and evaluation. AI tools were used as implementation support, but I was responsible for the architecture, integration, debugging, design decisions, validation and final project ownership.

Claims that require evidence before publication

Claim	Evidence required
"Optimized from 6.5M+ to 95,617 triangles"	polycount screenshots / thesis table
"Unity WebGL deployment"	live build or recorded browser demo
"Mobile-first interface"	mobile screenshots / responsive demo
"Clipping system"	technical video or code snippet
"Visual modes"	screenshot grid
"SUS 91.88"	thesis evaluation section
"NASA-TLX reduction"	evaluation table
"AI-assisted but personally integrated"	interview explanation + code review readiness

9.0.11 CAD-to-realtime pipeline narrative

Recommended pipeline section

The case study should include a dedicated **CAD-to-Realtime Optimization Pipeline** section.

Suggested structure:

1. Source

Manufacturer CAD / technical references from the Holybro X500 V2 assembly.

1. Import / conversion

CAD imported or converted into polygonal geometry suitable for Blender and Unity.

1. Assessment

Original geometry was too dense for WebGL deployment and contained parts requiring cleanup or reconstruction.

1. Optimization

Reduced, rebuilt, retopologized or simplified geometry while preserving visual and instructional fidelity.

1. Modularization

Organized the drone into inspectable component entities and scene nodes.

1. Bake / material planning

High-to-low workflow used or planned for retaining detail on optimized meshes.

1. Unity integration

Imported optimized assets into Unity, assigned metadata, interaction states and visual modes.

1. Evaluation

Tested usability and cognitive workload in comparison with 2D support materials.

Suggested asset breakdown

Asset class	Portfolio treatment
Drone frame	show CAD-to-realtime conversion
Motors	show imported/manufacturer source + optimized version
Propellers	show original modeling contribution
Fasteners	show modular strategy, not every screw individually
Cables	show carefully if final model includes them; otherwise mark as scope limitation
Battery / GPS / telemetry modules	show if final and visually strong

9.0.12 Runtime systems breakdown

System 1 — Component selection

Public explanation:

The selection system allows users to inspect individual drone components and access part-specific technical information. Each selectable component is associated with metadata and visual feedback states.

Evidence to show:

- selection highlight screenshot;
- selected part panel;
- hierarchy example;
- code snippet only if clean and safe.

System 2 — Exploded view

Public explanation:

The exploded view reveals assembly relationships by separating components spatially while preserving their logical structure. This supports inspection of layered assemblies and internal connections.

Evidence to show:

- normal vs exploded view comparison;
- short animation clip;
- note on limitations for cables/flexible elements if relevant.

System 3 — Cross-section / clipping

Public explanation:

The clipping system enables sectional inspection, allowing users to reveal internal structures and understand how components relate across vertical or horizontal planes.

Evidence to show:

- clipping plane screenshot;
- UI control screenshot;
- simple diagram if available.

System 4 — Visual modes

Public explanation:

Multiple visual modes support different inspection needs: realistic rendering for recognition, ghosted/X-Ray for internal relationships, wireframe for structure and blueprint mode for technical interpretation.

Evidence to show:

- visual mode grid;
- explain purpose of each mode in one sentence.

System 5 — Technical UI

Public explanation:

The UI connects visual selection with structured component information. It is designed as an inspection interface rather than a game HUD.

Evidence to show:

- panel layout;
- part information hierarchy;
- mobile/desktop variants if available.

9.0.13 Suggested public case study structure

Use this order for the website / portfolio page.

1. Hero section

Content:

- Project title.
- One-line summary.
- 15–30 second autoplay muted video or high-quality render.
- Tags: Unity, WebGL, C#, Technical Visualization, CAD-to-Realtime, Blender, UI Toolkit.

Suggested hero copy:

Unity WebGL technical visualization prototype for inspecting the assembly structure of a Holybro X500 V2 drone.

2. Project snapshot

Compact facts:

Category	Detail
Type	Thesis / interactive technical visualization prototype
Role	Unity developer, technical artist, 3D optimization, UI, evaluation
Tools	Unity, C#, WebGL, Blender, GitHub
Result	Optimized drone viewer with interaction, visual modes and usability validation

3. Problem

Explain the documentation-to-understanding gap.

4. Solution

Explain the interactive viewer and main features.

5. Technical pipeline

Show CAD-to-realtime reduction and asset preparation.

6. Runtime interaction systems

Selection, exploded view, clipping, visual modes, UI.

7. Evaluation results

Show SUS, NASA-TLX Raw and task records.

8. My contribution

Explain what Alexander did, what came from manufacturer CAD and what was AI-assisted.

9. Constraints and limitations

Show maturity and honesty.

10. What I would improve next

Tie to employability and production mindset.

9.0.14 ArtStation / technical breakdown structure

ArtStation should not be the main project home, but it can host a visual technical breakdown.

Recommended ArtStation post title:

TwinSight X500 — CAD-to-Realtime Drone Visualization Breakdown

Recommended sections:

1. Final viewer screenshots.
2. CAD source vs optimized model.
3. Wireframe / topology views.
4. Propeller modeling breakdown.
5. Material / visual mode grid.
6. Exploded view sequence.
7. Clipping / X-Ray inspection screenshots.
8. UI panels.
9. Performance/optimization summary.
10. Link to portfolio case study and GitHub.

Avoid making ArtStation the only evidence source because recruiters for Unity/WebGL roles need technical explanation, not only renders.

9.0.15 GitHub README structure

The GitHub README should be recruiter-readable and not thesis-only.

Recommended README sections:

1. Project name and one-line summary.
2. Live demo link or demo video link.
3. Screenshots.
4. Features.
5. Technical stack.
6. Architecture overview.
7. CAD-to-realtime optimization.
8. Evaluation metrics.
9. Installation / local run instructions.
10. Known limitations.
11. AI-assisted development disclosure.
12. Academic and asset attribution.
13. License / usage notes.

README warning

If the repository contains messy commits, generated code, large assets, broken branches or academic clutter, do not overexpose it before cleanup. A polished README with curated screenshots is more important than making every internal artifact public.

9.0.16 Demo video plan

Recommended duration

60–120 seconds.

Target style

Technical, direct, visual, no cinematic overproduction. The video should prove the system works.

Structure

Time	Content
0–10s	Final viewer hero shot; title overlay
10–25s	Problem: dense drone assembly / static documentation
25–45s	Component selection + technical panel
45–60s	Exploded view
60–75s	Clipping / cross-section
75–90s	Visual modes grid
90–105s	Optimization metric: 6.5M+ to 95,617 triangles
105–120s	Evaluation metrics + role statement

Suggested narration

TwinSight X500 is a Unity WebGL technical visualization prototype developed for my Multimedia Engineering thesis. The goal was to transform complex drone assembly documentation into an interactive 3D inspection experience. I optimized manufacturer CAD data for real-time use, built runtime interaction systems for selection, exploded view and clipping, and designed technical UI panels for component-level information. In a formative evaluation with 12 participants, the prototype achieved a SUS average of 91.88 and lower perceived workload than the 2D support condition. The project demonstrates my focus on Unity, WebGL, CAD-to-realtime optimization and interactive technical visualization.

Demo video warnings

Do not include:

- long thesis explanation;
- raw editor footage unless useful;
- unpolished UI states;
- claims not visible in the video;
- excessive AI/process discussion.

9.0.17 Screenshot and asset shot list

Mandatory screenshots

1. Hero full drone view.
2. Selected component with technical panel.
3. Exploded view.
4. Cross-section / clipping.
5. X-Ray or ghosted mode.
6. Wireframe / blueprint mode.

7. Before/after CAD optimization if available.
8. UI close-up.
9. Mobile layout if available.
10. Evaluation metrics panel.

Optional screenshots

1. Blender optimization workspace.
2. High-poly vs low-poly comparison.
3. UV layout.
4. Bake result.
5. Unity scene hierarchy.
6. Code architecture diagram.
7. Interaction flow diagram.

Avoid screenshots

- messy Unity editor without explanation;
 - unfinished placeholder assets;
 - low-resolution mobile captures;
 - screenshots that reveal private notes, credentials or irrelevant folders;
 - copyrighted documentation pages unless clearly allowed and attributed.
-

9.0.18 Interview explanation

30-second answer

TwinSight X500 is my thesis project: a Unity WebGL technical visualization prototype for inspecting a Holybro X500 V2 drone assembly. I focused on converting dense CAD/manufacture geometry into a real-time optimized model, building interaction systems such as component selection, exploded view and clipping, and validating the result with usability and workload instruments. It is the clearest evidence of my fit for Unity, WebGL, technical art and interactive visualization roles.

90-second answer

The project started from a documentation problem: complex assemblies are hard to understand through static 2D references alone. I built a Unity WebGL viewer that lets users inspect the Holybro X500 V2 drone in 3D, select components, view technical information, activate exploded view, use clipping and switch visual modes such as X-Ray, blueprint, wireframe and ghosted. The technical challenge was converting dense CAD-style geometry into a performant browser-ready experience while preserving the assembly meaning. I also evaluated the prototype with 12 participants using SUS, NASA-TLX Raw and Think-Aloud. The strongest result was that the 3D viewer condition showed lower perceived workload than the 2D support condition. AI tools helped with implementation support, but I handled architecture, integration, debugging, final design decisions and validation.

Answer to “How much did you build yourself?”

I used AI tools as implementation support, mainly to accelerate coding, explore approaches and debug issues. I do not present the project as hand-coded line by line without assistance. My contribution was defining the architecture, integrating the Unity systems, adapting and debugging the code, building the interaction flow, preparing the 3D assets, connecting the UI and data structure, running the evaluation and making the final technical decisions. In interviews, I can explain the systems, trade-offs and limitations because I was responsible for the final integration and project outcome.

Answer to “Is this a commercial product?”

No. It is an academic prototype developed as a thesis project. I treat it as a portfolio-quality technical visualization case study, not as a commercial training platform. The value is that it demonstrates an end-to-end workflow: CAD-to-realtime optimization, Unity WebGL implementation, technical UI and user evaluation.

Answer to “Is it a digital twin?”

It is adjacent to digital twin visualization, but I would not call it a full digital twin because it does not include live data, bidirectional synchronization or operational telemetry. It is more accurately described as an interactive technical visualization prototype or assembly inspection viewer.

9.0.19 AI-assisted development disclosure

Recommended public wording

Use this only in GitHub or extended case study, not necessarily in the hero section:

AI tools were used as implementation support during development. Final architecture, integration, debugging, interaction design, asset preparation, evaluation and technical decisions were handled by the author.

When to disclose

Context	Recommended level
CV bullet	Usually omit unless asked
LinkedIn project	Usually omit or mention “AI-assisted tooling” only if relevant
GitHub README	Include short disclosure
Interview	Be transparent if asked
Portfolio case study	Include in “Process / Implementation” section, not hero

Risk control

The disclosure must not sound like:

- “AI made the project.”
- “I only prompted.”
- “I cannot explain the code.”
- “I copied generated systems without understanding.”

It must sound like:

- AI accelerated implementation.
- Candidate owned architecture and integration.
- Candidate can explain trade-offs.
- Candidate validated the result.

9.0.20 Asset and IP risk mitigation

Risk

The project uses manufacturer geometry or technical references related to the Holybro X500 V2. This is acceptable as academic/portfolio context only if handled carefully, but it requires attribution and avoidance of misleading ownership claims.

Required public wording

This project uses publicly available/manufacturer reference material for academic and portfolio purposes. The drone product and original manufacturer design belong to their respective owners. The portfolio case study focuses on the real-time visualization, optimization, interaction design and academic evaluation work.

Do not do

- Do not imply ownership of the original Holybro drone design.
- Do not redistribute proprietary CAD files if licensing is unclear.
- Do not present manufacturer assets as original modeling work.
- Do not include confidential documentation.
- Do not use official logos in a way that implies endorsement.

Safer publication strategy

1. Publish optimized screenshots and demo video.
 2. Publish code if clean and legally safe.
 3. Avoid redistributing raw CAD sources.
 4. Include attribution and academic-use note.
 5. If uncertain, keep heavy assets out of the public repo.
-

9.0.21 Role-specific positioning variants

Variant A — Real-Time 3D / Interactive 3D Developer

Emphasis:

- Unity runtime systems.
- WebGL deployment.
- Interaction design.
- Component selection.
- Technical UI.

Suggested description:

Built an interactive Unity WebGL 3D inspection viewer for a drone assembly, implementing component selection, exploded view, clipping and visual modes for technical understanding.

Variant B — Unity Technical Artist

Emphasis:

- CAD-to-realtime pipeline.
- Optimization.
- Visual modes.

- Technical rendering states.
- Bridge between assets and runtime.

Suggested description:

Developed a CAD-to-realtime optimization and visualization pipeline for a Unity WebGL drone assembly viewer, reducing dense source geometry into a browser-ready inspectable model.

Variant C — Technical Visualization / Digital Twin-adjacent

Emphasis:

- Assembly inspection.
- Component relationships.
- Engineering documentation.
- Evaluation.

Suggested description:

Created an interactive technical visualization prototype that helps users understand drone assembly relationships through 3D inspection, exploded view, clipping and structured component information.

Variant D — Unity WebGL Developer

Emphasis:

- Browser deployment.
- Performance constraints.
- UI.
- Runtime optimization.

Suggested description:

Implemented a Unity WebGL technical viewer designed for browser-based 3D inspection, balancing interaction, visual fidelity and optimized asset delivery.

Variant E — Game Technical Artist

Emphasis:

- Technical art systems.
- Visual modes.
- Real-time rendering.
- Optimization.

Warning: this should be secondary, because the project is not a game.

Suggested description:

Built real-time visualization systems in Unity, including X-Ray, wireframe, ghosted and blueprint modes, with an optimized 3D asset pipeline for interactive inspection.

9.0.22 CV bullet bank

Use only after the CV strategy section confirms the target role.

Strong bullets

- Developed **TwinSight X500**, a Unity WebGL technical visualization prototype for inspecting Holybro X500 V2 drone assembly relationships.
- Optimized dense CAD/manufacture geometry from **6.5M+ triangles to 95,617 triangles** for real-time browser-based inspection.
- Implemented runtime interaction systems including component selection, exploded view, cross-section/clipping and multiple visualization modes.
- Designed technical UI panels linking selected components to identification, specification and assembly information.
- Conducted formative usability evaluation with **12 participants, 96 task-condition records**, SUS and NASA-TLX Raw workload measures.
- Achieved a **SUS average of 91.88** and lower perceived workload for the 3D viewer condition compared with 2D support material.
- Built a CAD-to-realtime workflow using Blender and Unity to support optimized interactive technical visualization.

Safer bullets if evidence is not yet public

- Developed a Unity WebGL technical visualization thesis prototype for interactive drone assembly inspection.
- Built component selection, exploded view, clipping and visual mode systems to support spatial understanding of a complex 3D assembly.
- Prepared and optimized CAD-derived 3D assets for real-time visualization in a browser environment.
- Evaluated the prototype through formative usability and workload testing.

Avoid bullets

- “Shipped a commercial digital twin platform.”
- “Built an aerospace simulation system.”
- “Created the original Holybro X500 V2 CAD model.”
- “Developed a production-ready training platform.”
- “Implemented AI-powered drone diagnostics.”

9.0.23 LinkedIn project description

Recommended version

TwinSight X500 — Unity WebGL Technical Visualization Prototype

Developed a Unity WebGL interactive 3D visualization prototype for inspecting the assembly structure of a Holybro X500 V2 drone. The project converts dense CAD/manufacture geometry and technical documentation into an optimized real-time viewer with component selection, exploded view, cross-section/clipping, technical information panels and multiple visualization modes.

The prototype was developed as a Multimedia Engineering thesis project and evaluated with usability and workload instruments including SUS, NASA-TLX Raw and Think-Aloud. The project demonstrates CAD-to-realtime optimization, Unity/C# runtime systems, technical UI design and browser-based interactive 3D visualization.

Technologies: Unity, C#, WebGL, URP, UI Toolkit, Blender, CAD-to-Realtime Optimization, Technical Visualization.

9.0.24 Portfolio page copy draft

Opening paragraph

TwinSight X500 is a Unity WebGL technical visualization prototype designed to help users inspect and understand the assembly structure of a Holybro X500 V2 drone. The project addresses a common limitation of static techni-

cal documentation: users must mentally reconstruct complex 3D relationships from 2D diagrams, text and CAD references.

The prototype converts dense CAD/manufacture geometry into an optimized real-time model and adds interaction systems for component selection, exploded view, clipping and visual mode switching. It was developed as a Multimedia Engineering thesis project and evaluated through a formative usability study.

Problem paragraph

Drone assemblies contain stacked frames, motors, fasteners, electronics, supports and internal relationships that are difficult to understand from static documentation alone. Traditional instructions show the parts, but they do not always make the spatial relationships clear. TwinSight focuses on making these relationships visible, inspectable and easier to reason about.

Technical challenge paragraph

The main technical challenge was converting dense CAD-style source geometry into a browser-ready interactive 3D experience. This required reducing geometric complexity, organizing components into inspectable entities, designing runtime interaction states and balancing visual clarity with WebGL performance constraints.

Evaluation paragraph

The prototype was evaluated with 12 participants using SUS, NASA-TLX Raw and Think-Aloud. The results indicated strong usability and lower perceived workload in the 3D viewer condition compared with the 2D support condition. These findings should be interpreted as formative academic validation rather than commercial-scale product validation.

9.0.25 Evidence checklist before publication

Minimum publication-ready evidence

Item	Required before public launch
60–120 sec demo video	Yes
Hero screenshot	Yes
Feature screenshot grid	Yes
Optimization before/after metric	Yes
SUS / NASA-TLX summary	Yes
Attribution note	Yes
GitHub README cleanup	Yes
AI-assisted disclosure	Yes, at least in GitHub/process
Broken branches removed/hidden	Yes
Live WebGL build	Strongly recommended

Optional but high-value evidence

Item	Value
Live browser demo	Very high
Interaction architecture diagram	High
CAD-to-realtime pipeline diagram	High
Technical UI flow diagram	Medium-high
Code snippet of clean system	Medium
Mobile performance notes	Medium
Thesis PDF excerpt	Medium, if clean

9.0.26 Known limitations section

The public case study should include limitations to avoid overclaiming.
Recommended wording:

TwinSight X500 is an academic prototype, not a commercial training product. It does not include live telemetry, physics simulation, certified maintenance workflows or production-grade backend infrastructure. The project focuses on interactive assembly visualization, CAD-to-realtime optimization, technical UI and formative usability evaluation.

Potential limitations:

- WebGL performance may vary by device.
- Some geometry may be simplified for real-time use.
- Some flexible elements such as cables may be simplified or excluded.
- Evaluation sample size is formative.
- Manufacturer assets require careful attribution.
- The system is not a full digital twin.

9.0.27 Final case study decision

TwinSight X500 should be the **primary flagship project** across:

1. Portfolio homepage.
2. GitHub pinned repositories.
3. LinkedIn Featured section.
4. CV project section.
5. Demo reel.
6. Interview narrative.

It is the project with the strongest alignment between Alexander's real profile and the target labor market: Unity, WebGL, CAD-to-realtime, real-time interaction, technical visualization, UI, Blender optimization and academic evaluation.
ARA, Blender portrait and other projects should support the profile, but they should not compete with TwinSight for attention in the first 10 seconds of recruiter review.

9.0.28 Completion status

Section	Status
Positioning	Complete
Problem framing	Complete
Technical stack	Complete
Feature inventory	Complete
Metrics framing	Complete
Contribution map	Complete
Demo plan	Complete
Screenshot plan	Complete
GitHub README outline	Complete
LinkedIn project draft	Complete
Risk mitigation	Complete
Evidence checklist	Complete

9.0.29 Next actions for this project

1. Record a 60–120 second demo video.
2. Capture the mandatory screenshot set.
3. Produce a visual mode grid.
4. Capture CAD-to-realtime before/after evidence.
5. Clean the GitHub README.
6. Add a public attribution and AI-assisted implementation disclosure.
7. Build the portfolio page using the case study structure above.
8. Prepare the 30-second and 90-second interview explanations.

Chapter 10

08B_twinsight_case_study_final_structure

Fuente: 08B_twinsight_case_study_final_structure.md

Seccion: root

Estado: canonical

Rol: Modulo final complementario

Lineas originales: 1111

Ultima modificacion: 2026-06-11 19:42

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Use this framing across portfolio, LinkedIn, GitHub and ArtStation:

Contenido completo depurado

10.0.1 0. Module status

```
Bloque de codigo: text
Module: 08B_twinsight_case_study_final_structure.md
Date generated: 2026-06-12
Status: Complete v1
Purpose: Final portfolio case-study structure for TwinSight X500
Inputs: 08, 17, 19, 20, 21, 21B, 29, 29B, 29C, B1, B5
Use: Portfolio page, GitHub README, LinkedIn Featured, ArtStation breakdown, interview follow-up
```

10.1 1. Final positioning

10.1.1 1.1 Primary project identity

```
Bloque de codigo: text
TwinSight X500
Unity WebGL Technical Visualization for Drone Assembly Inspection
```

10.1.2 1.2 One-line description

```
Bloque de codigo: text
An interactive browser-based 3D viewer that helps users inspect and understand the assembly of a Holybro X500 V2 drone through
↩ component selection, exploded view, cross-section tools, visual modes and technical information panels.
```

10.1.3 1.3 Professional framing

Use this framing across portfolio, LinkedIn, GitHub and ArtStation:

Bloque de codigo: text

TwinSight X500 is a Unity WebGL technical visualization prototype for drone assembly inspection. It transforms CAD-derived
↳ drone assets into an optimized browser-based 3D viewer with component-level interaction, exploded assembly visualization,
↳ cross-section inspection, visual modes and formative usability/workload evaluation.

10.1.4 1.4 Role-aligned keywords

Bloque de codigo: text

Unity
C#
Unity WebGL
Real-Time 3D
Technical Visualization
Technical Art
CAD-to-Realtime
Interactive 3D
URP
UI Toolkit
Shader Graph
Blender
Optimization
Exploded View
Clipping
Cross-Section
SUS
NASA-TLX
Think-Aloud

10.1.5 1.5 What this project should prove

Proof area	What TwinSight must communicate
Real-time 3D development	interactive browser-based Unity application
Technical art	visual modes, optimization, rendering constraints
CAD-to-realtime	heavy CAD data converted into optimized runtime asset
Technical visualization	assembly understanding, inspection, spatial relationships
UI/UX	technical panels, bottom-sheet interaction, mobile-first logic
Evaluation	SUS, NASA-TLX and Think-Aloud validation
Portfolio maturity	clear case study, video, metrics, limitations, links

10.2 2. Page hierarchy

10.2.1 2.1 Recommended case-study page order

Bloque de codigo: text

01. Hero
02. Short value proposition
03. Demo video
04. Problem
05. Solution
06. Key features
07. Technical pipeline
08. CAD-to-realtime optimization
09. Unity WebGL architecture
10. Interaction systems
11. Visual modes
12. Evaluation results
13. Role and contribution
14. AI-assisted workflow disclosure
15. Constraints and limitations
16. Screenshots / media gallery
17. Links
18. Next steps

10.2.2 2.2 Recruiter-first logic

The first screen must answer:

Bloque de codigo: text

What is it?
What role does it support?
What technologies were used?
What can the user do?
Why is it credible?

Do not start with academic methodology.

10.3 3. Hero section

10.3.1 3.1 Hero title

Bloque de codigo: text

TwinSight X500

10.3.2 3.2 Hero subtitle

Bloque de codigo: text

Unity WebGL technical visualization for drone assembly inspection.

10.3.3 3.3 Hero paragraph

Bloque de codigo: text

TwinSight X500 is an interactive browser-based 3D viewer for inspecting the assembly of a Holybro X500 V2 drone. It was built
↪ in Unity WebGL with C#, URP and UI Toolkit, using a CAD-to-realtime pipeline to transform complex source geometry into an
↪ optimized real-time visualization.

10.3.4 3.4 Hero metadata

Field	Value
Role	Real-Time 3D Developer / Unity Technical Artist
Stack	Unity, C#, WebGL, URP, UI Toolkit, Blender
Domain	Technical visualization / CAD-to-realtime
Project type	Academic/portfolio prototype
Platform	Browser / WebGL
Evaluation	SUS, NASA-TLX Raw, Think-Aloud

10.3.5 3.5 Hero CTA buttons

Bloque de codigo: text

Watch demo
View GitHub
Read technical breakdown
Open WebGL prototype

Only include “Open WebGL prototype” if the build is public and stable.

10.4 4. Demo video placement

10.4.1 4.1 Primary video

Embed:

```
Bloque de codigo: text
twinsight_x500_demo_90s.mp4
```

or hosted link:

```
Bloque de codigo: text
YouTube / Vimeo / portfolio-hosted video
```

10.4.2 4.2 Caption under video

```
Bloque de codigo: text
90-second overview showing browser-based interaction, component selection, exploded view, clipping, visual modes, CAD
↪ optimization and evaluation results.
```

10.4.3 4.3 Secondary video

Optional:

```
Bloque de codigo: text
twinsight_x500_technical_breakdown_5min.mp4
```

Caption:

```
Bloque de codigo: text
Technical breakdown covering CAD preparation, Unity WebGL architecture, runtime interaction systems, visual modes, UI Toolkit
↪ and evaluation methodology.
```

10.5 5. Problem section

10.5.1 5.1 Section title

```
Bloque de codigo: text
Problem: 2D documentation makes spatial relationships harder to understand
```

10.5.2 5.2 Final copy

```
Bloque de codigo: text
Drone assembly documentation is often presented through static 2D diagrams, part lists and written instructions. This forces
↪ users to mentally reconstruct spatial relationships between components, orientations, fasteners and hidden structures.

For complex assemblies such as a Holybro X500 V2 drone, this increases cognitive effort and makes it harder to understand how
↪ parts relate to each other in 3D space.
```

10.5.3 5.3 Problem bullets

```
Bloque de codigo: text
- Static diagrams do not allow direct inspection.
- Hidden structures require mental reconstruction.
- Part relationships are difficult to infer from 2D views.
- Assembly hierarchy is not always visually obvious.
- Users cannot easily isolate, cut through or compare components.
```

10.5.4 5.4 Visuals for problem section

Visual	Purpose
2D diagram / manual-style screenshot	show limitation
dense part list	show complexity
drone exploded diagram reference	show assembly complexity
side-by-side 2D vs 3D	communicate transition

Use only visuals you have rights to use or self-created diagrams.

10.6 6. Solution section

10.6.1 6.1 Section title

Bloque de codigo: text
Solution: interactive technical visualization in the browser

10.6.2 6.2 Final copy

Bloque de codigo: text
TwinSight X500 converts CAD-derived drone geometry into an optimized Unity WebGL viewer. Instead of reading static
↪ documentation, users can navigate the assembly, select components, inspect technical information, activate exploded view,
↪ cut through the model and switch between visual modes.

The goal is not only to show the drone, but to make the relationships between its parts easier to understand.

10.6.3 6.3 Solution bullets

Bloque de codigo: text
- Browser-based Unity WebGL viewer.
- Component-level selection and highlighting.
- Technical part information panels.
- Exploded view for assembly hierarchy.
- Cross-section / clipping for hidden structures.
- Multiple visual modes for inspection.
- CAD-to-realtime optimization pipeline.
- Formative usability and workload evaluation.

10.6.4 6.4 Solution diagram

Recommended diagram:

Bloque de codigo: text
2D documentation
↓
CAD-derived 3D source
↓
Blender optimization / cleanup
↓
Unity WebGL interactive viewer
↓
User inspection + evaluation

10.7 7. Key features section

10.7.1 7.1 Feature cards

Feature	Description	Visual
Component selection	Select individual parts and inspect related data	selected arm / motor / frame part
Exploded view	Separate assemblies to reveal hierarchy and placement	full exploded drone
Cross-section tools	Cut through geometry to inspect hidden structures	clipping plane screenshot
Visual modes	Switch between realistic, X-ray, ghosted, blueprint, wireframe and thermal-style views	visual modes grid
Technical UI	Access part information through contextual panels	UI panel close-up
CAD optimization	Convert heavy CAD-derived assets into WebGL-ready geometry	before/after wireframe
Evaluation	Validate usability and workload with structured methods	metrics card

10.7.2 7.2 Feature copy blocks

Component selection

Bloque de código: text
Users can select components directly in the 3D viewer. The selected part is highlighted and linked to contextual technical
↪ information, reducing the need to search through separate lists or diagrams.

Exploded view

Bloque de código: text
The exploded view separates assemblies while preserving spatial logic. This helps users understand how components relate to
↪ each other and where each part belongs in the full structure.

Cross-section / clipping

Bloque de código: text
The clipping tool cuts through the model to reveal internal structures and hidden relationships. It supports technical
↪ inspection without requiring users to disassemble the visual context.

Visual modes

Bloque de código: text
Multiple visual modes support different inspection tasks: realistic view for material context, X-ray and ghosted modes for
↪ internal visibility, blueprint and wireframe modes for technical reading, and a thermal-style qualitative mode for visual
↪ exploration.

Technical UI

Bloque de código: text
The interface provides technical part information while preserving the 3D inspection context. The UI was designed with a
↪ mobile-first bottom-sheet logic and runtime interaction in mind.

10.8 8. Technical pipeline section

10.8.1 8.1 Section title

Bloque de codigo: text
 Technical pipeline: CAD-to-realtime optimization

10.8.2 8.2 Final copy

Bloque de codigo: text
 The source drone model came from CAD-derived geometry that was not suitable for direct WebGL deployment. The pipeline focused
 ↪ on preserving technical readability while reducing geometry complexity, preparing assets for real-time interaction and
 ↪ organizing the model for runtime selection and visualization.

10.8.3 8.3 Pipeline stages

Bloque de codigo: text
 01. CAD source inspection
 02. STEP / tessellation route exploration
 03. Blender cleanup and reconstruction
 04. Modular fastener handling
 05. High-poly / low-poly preparation
 06. UV and bake preparation
 07. Material and visual-mode setup
 08. Unity import and scene organization
 09. Runtime interaction systems
 10. WebGL build and evaluation

10.8.4 8.4 Pipeline table

Stage	Tool	Purpose
CAD source	STEP / CAD route	source geometry
Tessellation	Mol3D / Stepper route	generate usable mesh
Cleanup	Blender	remove/repair/rebuild geometry
Optimization	Blender	reduce geometry for WebGL
UV/Bake	RizomUV / Marmoset / Blender	prepare textures if used
Runtime	Unity	interaction and visualization
UI	UI Toolkit	technical panels and controls
Deployment	Unity WebGL	browser delivery
Evaluation	SUS / NASA-TLX / Think-Aloud	usability/workload validation

Only list tools actually used in final public copy. If RizomUV or Marmoset were planned but not used, mark them as planned or omit them.

10.9 9. Optimization section

10.9.1 9.1 Section title

Bloque de codigo: text
 Optimization: from heavy CAD data to WebGL-ready geometry

10.9.2 9.2 Final copy

Bloque de codigo: text
 A central technical challenge was converting complex CAD-derived geometry into a model that could run in a browser while
 ↪ preserving the visual information needed for assembly inspection.

10.9.3 9.3 Metrics card

Metric	Value
Optimized geometry	95,617 triangles
Source CAD routes	6.5M+ triangles
Canonical research parts	28
Scene nodes / anchors	30
Technical elements	257
Evaluation participants	12
Task-condition records	96

10.9.4 9.4 Safe metric language

Use:

Bloque de código: text
 Reduced CAD-derived source routes exceeding 6.5M triangles into a 95,617-triangle optimized model for browser-based real-time inspection.

Avoid:

Bloque de código: text
 Optimized the model by 98.5% and proved production scalability.

The first is factual. The second overclaims.

10.9.5 9.5 Required visuals

Visual	Required
high-poly / source route screenshot	Yes
optimized model screenshot	Yes
wireframe comparison	Yes
triangle metric card	Yes
WebGL browser proof	Yes
Blender cleanup screenshot	Recommended
Unity profiler screenshot	Recommended

10.10 10. Unity WebGL architecture section

10.10.1 10.1 Section title

Bloque de código: text
 Unity WebGL architecture

10.10.2 10.2 Final copy

Bloque de código: text
 The Unity application coordinates runtime selection, visualization states, UI panels and inspection tools inside a WebGL build. The architecture separates model organization, user interaction, visual states and interface logic so that the viewer can support multiple inspection modes without becoming a static 3D model viewer.

10.10.3 10.3 Architecture components

Bloque de codigo: text

- Model hierarchy and selectable parts
- Selection and highlight system
- Camera and navigation logic
- Exploded-view state controller
- Clipping / cross-section controller
- Visual-mode switching system
- Technical metadata / part information
- UI Toolkit panels
- WebGL deployment configuration

10.10.4 10.4 Architecture visual

Recommended diagram:

Bloque de codigo: text

```
graph TD
    A[User input] --> B[Interaction controller]
    B --> C[Selection / camera / clipping / exploded states]
    C --> D[Visualization systems]
    D --> E[UI Toolkit panels + 3D model feedback]
    E --> F[WebGL runtime]
```

10.11 11. Visual modes section

10.11.1 11.1 Section title

Bloque de codigo: text

Visual modes for technical inspection

10.11.2 11.2 Visual mode table

Mode	Purpose	Claim safety
Realistic	material and form readability	safe
X-ray	internal visibility	safe if implemented
Ghosted	semi-transparent inspection	safe
Blueprint	technical drawing aesthetic	safe
Wireframe	structure/topology reading	safe
Solid	neutral inspection	safe
Thermal-style	qualitative visual exploration	must clarify not simulation

10.11.3 11.3 Required disclaimer

Bloque de codigo: text

The thermal-style mode is a qualitative visual mode. It is not a physical thermal simulation, FEA result or sensor-driven thermal analysis.

10.11.4 11.4 Visual layout

Use a grid:

Bloque de codigo: text
Realistic | X-ray | Ghosted
Blueprint | Wireframe | Solid
Thermal-style | Clipping | Exploded

10.12 12. Evaluation section

10.12.1 12.1 Section title

Bloque de codigo: text
Evaluation: usability and perceived workload

10.12.2 12.2 Final copy

Bloque de codigo: text
TwinSight X500 was evaluated as an academic/portfolio prototype using usability, workload and qualitative observation methods.
↔ The goal was to compare how users interacted with the 3D viewer and traditional 2D support during assembly-understanding tasks.

10.12.3 12.3 Evaluation methods

Method	Purpose
SUS	usability perception
NASA-TLX Raw	perceived workload
Think-Aloud	qualitative insight
Task-condition records	structured interaction data

10.12.4 12.4 Evaluation metrics

Metric	Value
Participants	12
SUS average	91.88
NASA-TLX Raw, 3D viewer	8.69
NASA-TLX Raw, 2D support	19.89
Task-condition records	96

10.12.5 12.5 Safe interpretation

Use:

Bloque de codigo: text
The formative evaluation suggested high perceived usability and lower perceived workload for the 3D viewer condition compared
↔ with 2D support in the tested tasks.

Avoid:

Bloque de codigo: text
TwinSight proves that 3D documentation is always better than 2D documentation.

10.12.6 12.6 Evaluation visuals

Visual	Use
SUS metric card	recruiter-readable proof
NASA-TLX comparison	workload evidence
participant/task count	credibility
Think-Aloud quote snippets	qualitative support
evaluation flow diagram	academic grounding

10.13 13. Role and contribution section

10.13.1 13.1 Section title

Bloque de codigo: text

My role

10.13.2 13.2 Final copy

Bloque de codigo: text

I worked across the complete prototype pipeline: CAD-to-realtime asset preparation, Unity WebGL implementation, C# runtime
↔ interaction systems, technical UI, visual modes, documentation and formative evaluation.

10.13.3 13.3 Contribution bullets

Bloque de codigo: text

- Prepared and optimized CAD-derived drone geometry for real-time WebGL use.
- Organized selectable components and scene hierarchy.
- Implemented runtime interaction systems in Unity/C#.
- Built component selection, highlighting and information panels.
- Developed exploded view and clipping/cross-section interaction.
- Implemented multiple visual inspection modes.
- Designed the technical UI and mobile-first interaction logic.
- Prepared evaluation workflow using SUS, NASA-TLX Raw and Think-Aloud.
- Documented constraints, findings and future improvements.

10.13.4 13.4 Role clarity note

If the thesis project involved AI assistance, external assets or source CAD files, clarify the boundary.

Use:

Bloque de codigo: text

The source CAD model was used as the technical basis for the drone geometry. My work focused on transforming, optimizing,
↔ organizing and implementing it as an interactive real-time WebGL visualization prototype.

10.14 14. AI-assisted workflow disclosure

10.14.1 14.1 Why include this

AI assistance can become a trust risk if hidden and discovered later.

The disclosure should be clear, controlled and not self-sabotaging.

10.14.2 14.2 Recommended disclosure

Bloque de código: text
 AI assistance was used as an implementation and iteration support tool during parts of the development process. Final
 ↪ architecture decisions, debugging, Unity integration, asset preparation, evaluation design, documentation and project
 ↪ ownership remained my responsibility.

10.14.3 14.3 Short version

Bloque de código: text
 AI-assisted implementation support; final integration, validation and technical ownership by Alexander Woodcock Salomón.

10.14.4 14.4 What not to say

Avoid:

Bloque de código: text
 AI built the project.
 I used ChatGPT for everything.
 I do not know how parts of it work.

If asked in interview, answer precisely and separate:

Bloque de código: text
 assisted coding
 debugging
 architecture
 integration
 validation
 final ownership

10.15 15. Limitations section

10.15.1 15.1 Section title

Bloque de código: text
 Constraints and limitations

10.15.2 15.2 Final copy

Bloque de código: text
 TwinSight X500 is an academic/portfolio prototype, not a deployed industrial product. It does not include live sensor
 ↪ integration, predictive maintenance, WebAR, multi-user collaboration or production monitoring. The current focus is
 ↪ technical visualization and assembly understanding through browser-based 3D interaction.

10.15.3 15.3 Limitation table

Limitation	Status
Not a production digital twin	true
No live IoT data	true
No predictive maintenance	true
No WebAR placement	true
No real thermal simulation	true
Academic participant sample	true
WebGL constraints	true
Prototype scope	true

10.15.4 15.4 Why this helps

Limitations increase credibility. They prevent overclaiming and make the project easier to defend in interviews.

10.16 16. Next steps section

10.16.1 16.1 Section title

Bloque de codigo: text
Future improvements

10.16.2 16.2 Safe future improvements

Bloque de codigo: text

- Add guided assembly sequences.
- Improve metadata depth for each component.
- Add search and part filtering.
- Add WebGPU or performance-focused rendering experiments.
- Add optional XR inspection mode.
- Add sensor/state data as a digital-twin extension.
- Expand to other drones or technical assemblies.
- Improve mobile interaction and loading performance.

10.16.3 16.3 Do not present future work as current functionality

Use:

Bloque de codigo: text
future extension
planned improvement
possible next step

Do not use:

Bloque de codigo: text
includes
supports
delivers

unless implemented.

10.17 17. Required screenshots and media

10.17.1 17.1 Minimum media package

Bloque de codigo: text

- 01_hero_final_view.jpg
- 02_browser_webgl_view.jpg
- 03_component_selection.jpg
- 04_technical_panel.jpg
- 05_exploded_view.jpg
- 06_cross_section.jpg
- 07_visual_modes_grid.jpg
- 08_cad_optimization_before_after.jpg
- 09_wireframe_comparison.jpg
- 10_metrics_card.jpg
- 11_pipeline_diagram.jpg
- 12_architecture_diagram.jpg

10.17.2 17.2 Optional media

Bloque de código: text
13_blender_cleanup.jpg
14_unity_editor_scene.jpg
15_ui_toolkit_panel.jpg
16_shader_graph.jpg
17_github_readme_preview.jpg
18_mobile_view.jpg
19_profiler_screenshot.jpg
20_think_aloud_quote_card.jpg

10.17.3 17.3 Video assets

Bloque de código: text
twinsight_x500_teaser_30s.mp4
twinsight_x500_demo_90s.mp4
twinsight_x500_technical_breakdown_5min.mp4

10.18 18. Portfolio page copy blocks

10.18.1 18.1 Project card

Bloque de código: text
TwinSight X500
Unity WebGL technical visualization for drone assembly inspection.

Interactive browser-based 3D viewer with component selection, exploded view, clipping tools, visual modes, CAD-to-realtime
↪ optimization and SUS/NASA-TLX evaluation.

10.18.2 18.2 Short homepage feature

Bloque de código: text
TwinSight X500 converts CAD-derived drone geometry into an optimized Unity WebGL viewer for browser-based technical
↪ inspection.

10.18.3 18.3 Detailed intro

Bloque de código: text
TwinSight X500 is my flagship real-time 3D project: a Unity WebGL technical visualization prototype designed to help users
↪ understand the assembly of a Holybro X500 V2 drone. The project combines CAD-to-realtime optimization, interactive runtime
↪ systems, technical UI, multiple visual modes and academic usability/workload evaluation.

10.18.4 18.4 Technical summary

Bloque de código: text
Built with Unity, C#, WebGL, URP, UI Toolkit and Blender. The prototype includes component-level selection, exploded view,
↪ clipping/cross-section tools, technical information panels, multiple visual modes and optimized CAD-derived geometry for
↪ browser-based interaction.

10.18.5 18.5 Validation summary

Bloque de código: text
The prototype was evaluated with 12 participants using SUS, NASA-TLX Raw and Think-Aloud protocols. The formative results
↪ suggested high perceived usability and lower perceived workload for the 3D viewer condition compared with 2D support in
↪ the tested tasks.

10.19 19. GitHub README integration

10.19.1 19.1 README opening

Bloque de código: markdown
TwinSight X500

Unity WebGL technical visualization prototype for drone assembly inspection.

TwinSight X500 transforms CAD-derived drone geometry into an optimized browser-based 3D viewer with component selection,
↪ exploded view, cross-section tools, visual modes and technical information panels.

10.19.2 19.2 README demo block

Bloque de código: markdown
Demo

[![TwinSight X500 Demo](./media/twinsight-demo-thumbnail.jpg)](https://youtube.com/...)

- Live demo: [link]
- Portfolio case study: [link]
- Technical breakdown: [link]

10.19.3 19.3 README highlights

Bloque de código: markdown
Highlights

- Unity WebGL deployment
- CAD-to-realtime optimization
- Component-level selection
- Exploded assembly view
- Cross-section / clipping inspection
- Multiple visual modes
- UI Toolkit technical panels
- SUS + NASA-TLX evaluation

10.20 20. ArtStation integration

10.20.1 20.1 ArtStation title

Bloque de código: text
TwinSight X500 – Unity WebGL Technical Visualization

10.20.2 20.2 ArtStation description

Bloque de código: text

A browser-based technical visualization prototype for inspecting the assembly of a Holybro X500 V2 drone. Built in Unity WebGL
↪ with C#, URP, UI Toolkit and Blender, the project focuses on CAD-to-realtime optimization, component selection, exploded
↪ view, cross-section tools and visual inspection modes.

10.20.3 20.3 ArtStation post order

Bloque de código: text
01. Cover image
02. 30-60 s video
03. Short description
04. Final screenshots
05. Feature breakdown
06. CAD optimization
07. Visual modes grid
08. UI and interaction
09. Evaluation metrics
10. Tools

11. Links

10.20.4 20.4 Tags

```
Bloque de codigo: text
Unity
Unity WebGL
Technical Art
Real-Time 3D
Technical Visualization
CAD Optimization
Drone
Interactive 3D
CSharp
URP
Blender
```

Use Digital Twin only if the description says:

```
Bloque de codigo: text
digital-twin-adjacent visualization prototype
```

not full digital twin.

10.21 21. LinkedIn integration

10.21.1 21.1 Featured item title

```
Bloque de codigo: text
TwinSight X500 – Unity WebGL Technical Visualization
```

10.21.2 21.2 Featured item description

```
Bloque de codigo: text
Interactive browser-based 3D viewer for drone assembly inspection, built with Unity, C#, WebGL, URP, UI Toolkit and Blender.
↪ Includes CAD-to-realtime optimization, component selection, exploded view, clipping tools, visual modes and SUS/NASA-TLX
↪ evaluation.
```

10.21.3 21.3 Launch post

```
Bloque de codigo: text
I built TwinSight X500, a Unity WebGL technical visualization prototype for drone assembly inspection.

The project explores how interactive 3D can support spatial understanding when static 2D documentation is not enough.

Core features:
- component selection
- exploded assembly view
- cross-section inspection
- technical part panels
- multiple visual modes
- CAD-to-realtime optimization
- SUS / NASA-TLX evaluation

Stack:
Unity, C#, WebGL, URP, UI Toolkit, Blender.
```

10.22 22. Interview defense notes

10.22.1 22.1 Explain in 30 seconds

Bloque de código: text

TwinSight X500 is a Unity WebGL technical visualization prototype for inspecting a Holybro X500 V2 drone assembly. I built it
 ↳ to address the limitations of static 2D documentation by allowing users to inspect components, activate exploded view, cut
 ↳ through geometry, switch visual modes and access technical information directly in the browser. The project also includes
 ↳ a CAD-to-realtime optimization pipeline and formative evaluation using SUS, NASA-TLX and Think-Aloud.

10.22.2 22.2 Explain in 2 minutes

Bloque de código: text

The project started from the problem that 2D assembly documentation requires users to mentally reconstruct spatial
 ↳ relationships between parts. I used a CAD-derived drone model as the basis and transformed it into a Unity WebGL
 ↳ prototype.

The technical work included optimizing geometry for browser use, organizing selectable components, implementing runtime
 ↳ interaction systems in C#, building a technical UI with UI Toolkit, adding exploded view and clipping tools, and creating
 ↳ several visual modes for inspection.

The project was evaluated with 12 participants using SUS, NASA-TLX Raw and Think-Aloud. The formative results suggested high
 ↳ perceived usability and lower perceived workload for the 3D viewer condition compared with 2D support in the tested tasks.

10.22.3 22.3 If asked about AI assistance

Bloque de código: text

I used AI as an implementation support and iteration tool, mainly to accelerate coding, debugging and structuring parts of the
 ↳ work. I was responsible for the architecture, integration, testing, Unity implementation decisions, asset preparation,
 ↳ evaluation design and final project ownership.

10.22.4 22.4 If asked whether it is a digital twin

Bloque de código: text

I would describe it as digital-twin-adjacent, not a full production digital twin. It represents a real physical drone assembly
 ↳ in interactive 3D, but it does not currently include live sensor data, operational state synchronization or predictive
 ↳ maintenance.

10.23 23. Quality checklist before publishing

10.23.1 23.1 Content checklist

Bloque de código: text

- [] title is role-aligned
- [] first paragraph is understandable to recruiters
- [] technical stack visible
- [] demo video embedded
- [] problem clearly stated
- [] solution clearly stated
- [] features shown visually
- [] CAD optimization explained
- [] metrics included
- [] role clearly stated
- [] AI assistance disclosed if relevant
- [] limitations included
- [] links working

10.23.2 23.2 Visual checklist

Bloque de código: text

- [] hero image strong
- [] UI readable
- [] visual modes grid clear

```
[ ] screenshots consistent
[ ] no debug artifacts
[ ] no unfinished materials visible
[ ] captions concise
[ ] diagrams readable
[ ] video thumbnail clear
```

10.23.3 23.3 Claim-safety checklist

```
Bloque de codigo: text
[ ] does not claim production deployment
[ ] does not claim full digital twin
[ ] does not claim live IoT
[ ] does not claim real thermal simulation
[ ] does not overstate evaluation causality
[ ] does not claim senior industry experience
[ ] does not hide prototype status
```

10.24 24. Final recommended page skeleton

```
Bloque de codigo: markdown
# TwinSight X500
## Unity WebGL Technical Visualization for Drone Assembly Inspection

[Hero image]
[Demo video]

TwinSight X500 is an interactive browser-based 3D viewer...

## Problem
2D documentation makes spatial relationships harder to understand...

## Solution
TwinSight converts CAD-derived drone geometry into...

## Key Features
- Component selection
- Exploded view
- Cross-section tools
- Visual modes
- Technical UI
- CAD-to-realtime optimization
- Evaluation

## Technical Pipeline
CAD → Blender → Unity → WebGL → Evaluation

## Optimization
6.5M+ source CAD routes → 95,617 optimized triangles

## Unity WebGL Architecture
Runtime interaction systems, UI Toolkit panels, visual states...

## Visual Modes
Realistic, X-ray, ghosted, blueprint, wireframe, thermal-style...

## Evaluation
SUS 91.88
NASA-TLX Raw: 8.69 viewer vs 19.89 2D support
12 participants

## My Role
Real-time 3D development, Unity WebGL integration...

## AI-Assisted Workflow
AI assistance was used as implementation support...

## Limitations
Academic/portfolio prototype. Not a full production digital twin...

## Links
- Live demo
- GitHub
- Demo video
- ArtStation breakdown
```

10.25 25. Next module

Proceed with:

Bloque de codigo: text
20B_portfolio_page_implementation_copy.md

10.25.1 Tool recommendation

Tool	Use
Agent mode	No
Deep Research	No

Reason:

The structure is now fixed. The next file should translate it into final portfolio implementation copy: homepage, project card, project page, CTA, metadata, SEO and section-by-section text.

Chapter 11

09 — ARA, AI Tools and Secondary Projects Strategy

Fuente: 09_ara_ai_tools_and_secondary_projects.md
Seccion: root
Estado: canonical
Rol: Modulo principal
Lineas originales: 840
Ultima modificacion: 2026-06-11 13:34
Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This file defines how Alexander Woodcock Salomón should use ARA Framework, ai-news-aggregator, FitApp-Free, the hyperrealistic Blender portrait and minor repositories inside the broader international job-search strategy.

Contenido completo depurado

11.0.1 0. Purpose of this section

This file defines how Alexander Woodcock Salomón should use ARA Framework, ai-news-aggregator, FitApp-Free, the hyperrealistic Blender portrait and minor repositories inside the broader international job-search strategy. The goal is not to expand every project. The goal is to decide which projects support the primary positioning, which projects should remain secondary, which require cleanup before being public, and which should be omitted to avoid profile dilution.

The primary positioning remains:

Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization, and WebGL deployment.

Secondary projects must support that positioning. They must not replace it.

11.0.2 1. Source-of-truth assumptions

Item	Current status
Thesis	Complete, pending defense
Flagship project	TwinSight X500
Primary market route	Unity / Real-Time 3D / Technical Visualization
Secondary route	Tools / Pipeline / Python automation

Item	Current status
AI route	Supporting route, not primary yet
Art route	Supporting evidence, not main positioning
Full-stack route	Fallback only
Fitness/product route	Low priority for current job search

The practical implication is simple: secondary projects are useful only if they strengthen one of these claims:

- technical systems thinking;
- automation mindset;
- ability to build tools;
- ability to document technical workflows;
- 3D production literacy;
- ability to finish and present a coherent product.

If a project does not support those claims, it should not be highlighted.

11.0.3 2. Project hierarchy

Tier	Project	Public priority
1	TwinSight X500	Mandatory
2	ARA Framework	Conditional
2	Hyperrealistic Blender portrait	Conditional
3	ai-news-aggregator	Low / conditional
3	FitApp-Free	Low
4	deus-ex-machina	Audit before showing
4	skills-introduction-to-github	Do not highlight
4	old web/client projects	Only if relevant and clean

The portfolio should not look like a random archive of experiments. It should look like a technical professional profile with a main axis and controlled secondary proof.

11.0.4 3. ARA Framework

3.1 Current positioning

ARA Framework is an academic research automation system / multi-agent research workflow prototype. It uses Python, LangGraph, LangChain, Groq LLaMA 3.3-70B, Redis, optional Supabase, Semantic Scholar API, Playwright MCP, MarkItDown MCP, PDF processing, web scraping, checkpointing, budget management and Mark-down report generation.

It should be described as:

AI-assisted research automation prototype for structured academic and technical report generation.

It should not be described as:

- enterprise AI platform;
- production-ready research operating system;
- autonomous research agent comparable to commercial products;
- ML/AI engineering proof;
- peer-reviewed research infrastructure;
- finished SaaS.

3.2 Strategic value

ARA is useful because it shows:

- Python systems thinking;
- workflow decomposition;
- agent orchestration awareness;
- API integration;
- retrieval/research workflow design;
- automation mindset;
- technical documentation;
- ability to use AI beyond casual prompting.

ARA is not yet strong enough to be the main employment hook unless it has:

- working demo;
- clear README;
- installation instructions;
- example output;
- architecture diagram;
- limitations;
- clean environment setup;
- no secrets in repo;
- controlled claims.

3.3 Relevance by target role

Role family	ARA relevance
Real-Time 3D / Interactive 3D	Low / indirect
Unity Technical Artist	Medium if framed as tooling mindset
Technical Visualization	Medium if framed as documentation automation
Unity WebGL	Low
Tools / Pipeline Developer	High
Python Automation Developer	High
LLM Application Developer	Medium / conditional
Full-stack Developer	Medium if UI/API exists
Game Technical Artist	Low / indirect

3.4 Public/private decision

ARA should become public only after a cleanup pass.

Recommended status now:

Keep private or semi-private until it has a recruiter-safe README and one reproducible demo.

Minimum public release requirements:

- no API keys;
- no private paths;
- no broken scripts in root;
- README.md explaining purpose, architecture, setup and limitations;
- .env.example;
- example input and output;
- architecture diagram;
- screenshots or terminal demo;
- short demo video or GIF;

- license decision;
- explicit “prototype” label.

3.5 README structure for ARA

Recommended README sections:

```
Bloque de código: text
# ARA Framework

AI-assisted research automation prototype for structured academic and technical report generation.

## Problem
Manual academic research workflows are repetitive, fragmented and hard to reproduce.

## What it does
- niche analysis
- literature discovery
- technical architecture draft
- implementation research
- report synthesis

## Architecture
- Niche Analyst
- Literature Researcher
- Technical Architect
- Implementation Specialist
- Content Synthesizer

## Tech stack
Python, LangGraph, LangChain, Redis, Semantic Scholar API, Playwright MCP, MarkItDown MCP, Groq LLaMA.

## Current status
Prototype. Functional core under cleanup. Not a production system.

## Demo
Input example, output example, screenshots.

## Limitations
- depends on external APIs
- outputs require human review
- source quality varies
- not a substitute for academic judgment

## Future work
- stronger evaluation
- better source validation
- UI layer
- cost controls
```

3.6 Claims allowed

Allowed:

- “Built a Python prototype for structured research automation.”
- “Designed a multi-agent workflow using LangGraph.”
- “Integrated external research and document-processing tools.”
- “Implemented checkpointing, report generation and modular agent responsibilities.”
- “Used AI assistance while retaining architecture, debugging and integration responsibility.”

Not allowed unless evidence exists:

- “Production-ready.”
- “Autonomous research platform.”
- “Enterprise-grade.”
- “AI engineer.”
- “Machine learning engineer.”
- “Deployed SaaS.”
- “Scientifically validated.”

3.7 ARA action plan

Priority	Task	Purpose
1	Remove secrets and private config	Security
2	Verify reproducible run	Credibility
3	Add example input/output	Recruiter comprehension
4	Add architecture diagram	Technical proof
5	Add limitations	Trust
6	Add demo video/GIF	Fast review
7	Decide public/private	Exposure control

ARA should not receive major development time before TwinSight portfolio assets are published.

11.0.5 4. ai-news-aggregator

4.1 Current positioning

ai-news-aggregator is a private full-stack AI news aggregation and analysis platform. It reportedly uses FastAPI, React, PostgreSQL, Redis, Python backend and an analysis pipeline.

It is useful only if it can be demonstrated as a working system.

Current recommended status:

Keep private until a minimal demo works.

4.2 Strategic value

Potential value:

- FastAPI proof;
- React proof;
- PostgreSQL / Redis proof;
- product logic;
- data ingestion;
- AI summarization pipeline;
- deployment potential.

Current risk:

- it dilutes the 3D/technical-art positioning;
- it makes the profile look like unfocused full-stack/AI exploration;
- if incomplete, it weakens credibility;
- if public but broken, it is worse than invisible.

4.3 Role relevance

Role family	Relevance
Real-Time 3D / Interactive 3D	Low
Unity Technical Artist	Low
Technical Visualization	Low / indirect
Tools / Pipeline Developer	Medium
Python Automation Developer	Medium / high
LLM Application Developer	Medium
Full-stack Developer	High

4.4 Publication decision

Do not pin this repo unless it has:

- working local setup;
- live demo or screenshots;
- seeded sample data;
- clean README;
- architecture diagram;
- no broken dependencies;
- no private API keys;
- clear explanation of what AI does and does not do.

4.5 Minimum viable public version

The minimum version does not need to be impressive. It needs to be reliable.

Required:

- article ingestion from one or two sources;
- dashboard with list and detail view;
- summary or categorization pipeline;
- backend API documented;
- local setup command;
- screenshots;
- known limitations.

Optional:

- user accounts;
- saved topics;
- PostgreSQL deployment;
- Docker compose;
- evaluation metrics;
- LangGraph integration.

4.6 Recommendation

Do not prioritize ai-news-aggregator in the first 30 days of job-search preparation.

Use it later only if applying to:

- Python automation roles;
- LLM application roles;
- full-stack roles;
- internal tooling roles.

It should not appear above TwinSight or ARA.

11.0.6 5. FitApp-Free

5.1 Current positioning

FitApp-Free is a fitness / calisthenics / multi-discipline app in development. It focuses on progressions, injury considerations, routine tracking, workout tracking, weight progression, repetitions, user progress and exercise programming logic.

5.2 Strategic value

FitApp may show:

- product thinking;
- domain modeling;
- routine/progression logic;
- UX/system design;
- persistence/data structures;
- long-term product initiative.

But it does not strongly support the main profile.

5.3 Risk

FitApp is the highest dilution risk among the named projects because it shifts the reader toward:

- generic app development;
- fitness product management;
- personal-product hobby;
- incomplete startup idea.

For the current strategy, this is mostly noise.

5.4 Recommended status

Keep private or omit from recruiter-facing materials for now.

Use only if:

- applying to full-stack/product roles;
- it has a clean demo;
- it shows strong system design;
- it does not displace TwinSight.

5.5 Minimum public threshold

Before showing FitApp publicly, it needs:

- working user flow;
- exercise/progression model;
- routine builder;
- tracking UI;
- screenshots;
- README explaining system design;
- no incomplete placeholder state.

5.6 Recommendation

Do not spend significant job-search time on FitApp during the 0–6 month employment sprint.

11.0.7 6. Hyperrealistic Blender portrait

6.1 Current positioning

The Blender portrait was made 100% in Blender following CG Cookie HUMAN. It demonstrates character pipeline, topology, materials, grooming, lighting, rendering and production discipline.

It is useful only if presented as a technical breakdown, not as a generic render.

6.2 Strategic value

The portrait can support:

- Blender fluency;
- topology awareness;
- material setup;
- grooming;
- lighting and rendering;
- ability to follow a high-end character pipeline;
- visual quality standards.

It is less relevant for:

- Unity WebGL;
- technical visualization;
- CAD optimization;
- simulation;
- tools development.

6.3 Recommended framing

Recommended title:

Hyperrealistic Character Pipeline Study — Blender

Not recommended:

Senior Character Artist Portfolio Piece

6.4 ArtStation structure

Recommended ArtStation page:

1. Final render.
2. Clay render.
3. Wireframe / topology.
4. UV layout.
5. Material breakdown.
6. Grooming breakdown.
7. Lighting setup.
8. Reference / course context.
9. What was learned.
10. Limitations.

6.5 Disclosure

Because the work followed a course, the page should say:

Character pipeline study completed in Blender following CG Cookie HUMAN training material. Final execution, scene setup, grooming/material work and presentation prepared independently as a portfolio learning piece.

This avoids implying it was a fully original character art commission or production asset.

6.6 Role relevance

Role family	Relevance
Unity Technical Artist	Medium
Real-Time 3D Developer	Low / medium
Technical Visualization	Low
3D Artist / Generalist	High
Game Technical Artist	Medium
Tools / Pipeline	Low

6.7 Recommendation

Include it in ArtStation and portfolio as a secondary piece after TwinSight.
It should not be the first portfolio item.

11.0.8 7. deus-ex-machina

7.1 Current status

Public project. Exact current quality must be audited before use.

7.2 Potential value

Depending on content, it may show:

- older game development work;
- GameMaker/GML experience;
- design iteration;
- early programming interest;
- narrative/gameplay prototyping.

7.3 Risk

If immature or messy, it may signal:

- outdated tools;
- low polish;
- unfinished habits;
- lack of focus.

7.4 Recommendation

Do not pin by default.

Use only if:

- it has screenshots or gameplay video;
- it has a clean README;
- it supports a game-dev application;
- it is framed as “early independent game prototype,” not as a flagship technical project.

11.0.9 8. skills-introduction-to-github

8.1 Current status

Public training / tutorial repository.

8.2 Recommendation

Do not highlight. Do not pin.

It can remain public, but it should not be part of the recruiter-facing portfolio.

Reason:

- it does not demonstrate differentiated capability;
- it may make the GitHub profile look beginner-level if pinned;
- it consumes attention without adding evidence.

11.0.10 9. Other lower-priority repositories

Named repositories include:

- Chocolateria;
- Bosque de Sombras Manuales;
- alexander-fullstack-portfolio;
- alexander-3d-portfolio;
- delarge95.github.io;
- 3DWeb;
- job-search-strategy;
- ludex_framework;
- WoodSleiman;
- WebGL_tesis.

These require a repository-level audit before publication decisions.

9.1 Audit criteria

Each repository should be scored against:

Criterion	Pass condition
Relevance	Supports target role
Completeness	Has functional output
README	Explains purpose and setup
Visual proof	Screenshots/video/demo
Code hygiene	No broken or private content
Recruiter value	Understandable in 60 seconds
Dilution risk	Does not confuse positioning

9.2 Default decisions

Repository type	Default action
Old portfolio repos	Merge/replace with current portfolio
Client web projects	Show only if professional and permission-safe
Job-search repo	Keep private
Manual/document repos	Omit unless relevant

Repository type	Default action
Duplicate thesis repos	Consolidate or archive
Experimental prototypes	Keep private unless polished

9.3 GitHub visibility rule

Public does not mean featured.

Pinned repositories should be fewer and stronger. A recruiter should see a coherent technical profile within the first 30–60 seconds.

11.0.11 10. Recommended GitHub pinned repositories

10.1 Immediate target state

Pin order	Repository	Status
1	WebGL-Thesis-Proposal / TwinSight X500	Mandatory
2	ARA Framework	Conditional after cleanup
3	Portfolio site	Mandatory once active
4	Blender portrait breakdown repo or page	Optional
5	ai-news-aggregator	Conditional
6	one small polished tool/demo	Optional

10.2 Repos to avoid pinning

- skills-introduction-to-github;
- incomplete FitApp;
- old duplicated thesis repos;
- unmaintained tests;
- unclear client projects;
- any repo with missing README or broken setup.

11.0.12 11. Secondary project case-study formats

11.1 ARA Framework case study

Recommended structure:

1. Problem.
2. System goal.
3. Workflow architecture.
4. Agent roles.
5. Tech stack.
6. Input/output example.
7. Integration challenges.
8. Human validation layer.
9. Limitations.
10. Future work.

Key message:

ARA demonstrates automation and systems design, not production AI authority.

11.2 ai-news-aggregator case study

Recommended structure:

1. Problem.
2. Data ingestion.
3. Backend architecture.
4. Frontend UI.
5. Analysis pipeline.
6. Storage model.
7. Deployment status.
8. Limitations.
9. Future work.

Key message:

Shows full-stack and AI-product experimentation; secondary to real-time 3D.

11.3 FitApp-Free case study

Recommended structure if ever public:

1. Problem.
2. Exercise progression model.
3. Routine logic.
4. Tracking architecture.
5. UI flow.
6. Data model.
7. Limitations.
8. Future work.

Key message:

Shows product thinking, not target-role proof.

11.4 Blender portrait case study

Recommended structure:

1. Final result.
2. Goal.
3. Reference/training context.
4. Modeling/topology.
5. Materials.
6. Grooming.
7. Lighting/rendering.
8. Lessons learned.
9. Limitations.

Key message:

Demonstrates 3D production literacy and visual standards.

11.0.13 12. AI-assisted development disclosure across projects

12.1 General rule

Do not hide AI assistance if directly asked. Do not over-emphasize it unnecessarily.

The safest formula:

I used AI assistance as part of the development workflow for ideation, code scaffolding, debugging and documentation support. I was responsible for architecture, integration, testing, technical decisions and final implementation.

12.2 Project-specific disclosure

Project	Disclosure level
TwinSight	Moderate if asked
ARA	Explicit; AI is part of the system
ai-news-aggregator	Moderate
FitApp	Low unless asked
Blender portrait	Mention course context, not AI unless used

12.3 Claims to avoid

Avoid:

- “I coded everything manually” if AI was involved;
- “AI built it” because that undermines ownership;
- “fully autonomous” unless evaluated;
- “production AI system” unless deployed and monitored;
- “research-grade” unless validated.

11.0.14 13. Project-to-role mapping

Project	Best-supported roles	Weak roles
TwinSight X500	Real-Time 3D, Unity WebGL, Technical Visualization	AI Tools
ARA Framework	Python Automation, Tools, LLM Apps	Unity WebGL
ai-news-aggregator	Full-stack, AI Apps	Technical Art
FitApp-Free	Product / Full-stack	Technical Visualization
Blender portrait	3D Generalist, Technical Art support	AI / Full-stack
deus-ex-machina	Game prototype	Technical Visualization

11.0.15 14. Short-term project time allocation

For the 0–6 month job-search sprint:

Area	Recommended effort
TwinSight case study	45%
Portfolio site	20%
LinkedIn/CV/GitHub cleanup	15%

Area	Recommended effort
ARA cleanup	10%
Blender portrait presentation	5%
Other projects	5%

This allocation is intentionally strict. The main risk is over-dispersing into too many unfinished projects.

11.0.16 15. Decision gates

Gate 1 — Before public release

A project can be public-facing only if it has:

- clear purpose;
- clean README;
- screenshots or video;
- no secrets;
- no broken critical path;
- accurate status label;
- no exaggerated claims.

Gate 2 — Before pinning

A project can be pinned only if it supports the target role within 60 seconds.

Gate 3 — Before including in CV

A project can enter the CV only if it supports the specific job target.

Examples:

- Unity Technical Artist CV: TwinSight + Blender portrait + ARA lightly.
- Python Automation CV: ARA + ai-news-aggregator + TwinSight lightly.
- Full-stack CV: ai-news-aggregator + FitApp + ARA.
- Technical Visualization CV: TwinSight first; ARA only if relevant.

11.0.17 16. Recommended final project portfolio order

Recommended public order:

1. TwinSight X500 — flagship case study.
2. ARA Framework — tools/automation case study after cleanup.
3. Hyperrealistic Blender Portrait — technical 3D breakdown.
4. ai-news-aggregator — optional technical appendix.
5. FitApp-Free — omit unless targeting product/full-stack.
6. Older repos — archive or omit unless cleaned.

11.0.18 17. Risks and mitigations

Risk	Mitigation
Profile dilution	Keep TwinSight first
Overclaiming AI skill	Label ARA as prototype
Broken public repos	Audit before pinning
Too many unfinished projects	Limit active cleanup to 2 projects
Course-project ambiguity	Disclose course context
Recruiter confusion	Use role-specific portfolio variants

11.0.19 18. Immediate action checklist

18.1 ARA

- Confirm current runnable state.
- Remove secrets.
- Add `.env.example`.
- Add architecture diagram.
- Add example input/output.
- Add limitations.
- Add 60-second demo.
- Decide public/private.

18.2 ai-news-aggregator

- Verify whether it still runs.
- Fix minimum broken dependencies.
- Capture screenshots.
- Decide if worth publishing.
- Keep private if not demonstrable.

18.3 FitApp-Free

- Do not prioritize now.
- Keep private.
- Revisit only for full-stack/product route.

18.4 Blender portrait

- Export final render.
- Export clay render.
- Export topology screenshots.
- Export UV/material/groom breakdown.
- Publish as ArtStation technical study.

18.5 GitHub

- Pin TwinSight first.
- Remove beginner/tutorial repos from pinned area.
- Add README to every visible repo.
- Archive duplicate thesis/portfolio repos.
- Keep job-search strategy private.

11.0.20 19. Final decision

ARA should be used as a secondary differentiator, not as the main identity. It strengthens a tools/pipeline/automation narrative but should not compete with TwinSight as the flagship.

ai-news-aggregator and FitApp-Free should not receive major attention during the initial job-search sprint unless the strategy pivots toward full-stack or AI application roles.

The Blender portrait is worth presenting, but only as a technical 3D breakdown and clearly below TwinSight.

The portfolio must communicate one core message:

Alexander builds interactive real-time 3D technical systems, and also has supporting automation/tooling and 3D production skills.

Chapter 12

10 — GitHub, LinkedIn and CV Strategy

Fuente: 10_github_linkedin_cv_strategy.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1210

Ultima modificacion: 2026-06-11 13:38

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This file defines how Alexander Woodcock Salomón should structure GitHub, LinkedIn and CV materials for an international job-search strategy focused on:

Contenido completo depurado

12.0.1 0. Purpose of this section

This file defines how Alexander Woodcock Salomón should structure GitHub, LinkedIn and CV materials for an international job-search strategy focused on:

1. Real-Time 3D / Interactive Technical Visualization Developer.
2. Unity WebGL / Interactive 3D Developer.
3. Technical Visualization / Digital Twin / Simulation Developer.
4. Unity Technical Artist focused on optimization, runtime interaction and tools.
5. Tools / Pipeline / Python Automation Developer as a controlled secondary route.

The goal is not cosmetic optimization. The goal is to make the public professional profile coherent, defensible and aligned with the strongest evidence: **TwinSight X500**.

This file uses the source-of-truth decisions from:

- 01_source_of_truth_profile.md
- 02_market_role_fit_and_positioning.md
- 07_portfolio_strategy_and_project_architecture.md
- 08_twinsight_x500_case_study.md
- 09_ara_ai_tools_and_secondary_projects.md

It does not create final application-specific CVs for individual job postings. It defines the base strategy, reusable text, variants and quality gates.

12.0.2 1. Core positioning rule

All public materials must converge on one primary identity:

Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization and WebGL deployment.

Secondary identity:

Python / AI-assisted tooling developer for research and production workflows.

This secondary identity must never overpower the primary identity. A recruiter should understand in less than 10 seconds that the main employable axis is Unity / real-time 3D / technical visualization, not generic full-stack, not pure 3D art, and not AI research.

12.0.3 2. Non-negotiable constraints

2.1 Claims that are safe

The following claims are consistent with the current evidence:

- Unity / C# implementation.
- Unity WebGL interactive 3D prototype.
- CAD-to-realtime asset optimization.
- Blender optimization, retopology, UV and baking workflows.
- Technical UI and runtime interaction systems.
- Academic usability validation using SUS, NASA-TLX Raw and Think-Aloud.
- Python automation prototypes.
- AI-assisted development with human ownership of integration, debugging and technical decisions.
- English professional working proficiency, if interview performance supports it.
- Colombia-based remote contractor availability.

2.2 Claims that must be avoided

Do not use these claims unless future evidence changes:

- “Senior Technical Artist.”
- “Senior Unity Developer.”
- “AI Engineer.”
- “Machine Learning Engineer.”
- “Digital Twin Engineer” as a formal job-title identity.
- “EU work authorized” before Portuguese passport or German residence authorization exists.
- “Certified” for non-certified courses.
- “Production-ready AI platform” for ARA.
- “Commercial shipped Unity product” for TwinSight.
- “Original drone CAD design.”
- “Full-stack developer” as the primary headline.

2.3 Correct work authorization language

Current valid language:

Bloque de código: text
Based in Colombia. Available for remote contractor/B2B work with international teams.

Do not state:

Bloque de codigo: text
Authorized to work in the EU.

Future language after Portuguese passport is issued:

Bloque de codigo: text
Portuguese / EU citizen. Authorized to work across the EU/EEA.

Future language if legally resident in Germany through marriage/family reunification and work authorization is issued:

Bloque de codigo: text
Based in Germany with work authorization.

Until that status exists, it must remain a future mobility possibility, not a current claim.

12.0.4 3. Public profile architecture

3.1 Recommended public hierarchy

The public profile should show this order:

1. TwinSight X500.
2. ARA Framework, only if cleaned and presented as tooling.
3. Hyperrealistic Blender portrait, only as a technical art / asset pipeline breakdown.
4. Optional ai-news-aggregator, only if functional and documented.
5. Older repositories only if cleaned, contextualized or hidden.

3.2 What each platform should prove

Platform	Main proof
CV	Fit, keywords, quantified evidence.
LinkedIn	Recruiter discoverability and credibility.
GitHub	Technical implementation and documentation.
Portfolio	Project quality and visual/technical storytelling.
ArtStation	Visual pipeline credibility.

The CV should not carry everything. GitHub should not be treated as a storage dump. LinkedIn should not be a biography. Each platform has a distinct job.

12.1 Part A — GitHub Strategy

12.1.1 4. GitHub objective

GitHub must show that Alexander can structure, document and maintain technical projects, not merely upload code.

For the target roles, GitHub should prove:

- Unity project organization.
- C# system structure.

- technical documentation quality;
- reproducible project setup;
- understanding of asset pipelines;
- clear ownership and limitations;
- controlled use of AI assistance;
- ability to communicate technical tradeoffs.

A messy GitHub weakens the profile. A smaller but curated GitHub is better than many incomplete repositories.

12.1.2 5. GitHub profile cleanup

5.1 Profile bio

Recommended GitHub bio:

Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization
↔ and WebGL deployment.

Alternative shorter version:

Bloque de codigo: text
Unity • WebGL • Real-Time 3D • Technical Visualization • CAD-to-Realtime

5.2 Profile links

The GitHub profile should link to:

1. Portfolio website.
2. LinkedIn.
3. ArtStation, only after meaningful content exists.
4. Email or contact method, only if professional and stable.

Do not link to unfinished portfolio branches as the main link unless the page is presentable.

5.3 Pinned repositories

Recommended pinned repository order:

1. WebGL-Thesis-Proposal / TwinSight-X500
2. ARA-Framework, only after cleanup.
3. Blender-Technical-Art-Breakdown or equivalent, only if converted into a portfolio repo.
4. ai-news-aggregator, only if functional.
5. Unity-WebGL-Demo or future smaller polished project.
6. Optional: portfolio website repository.

If GitHub allows only six pinned repositories, do not pin low-signal repos such as tutorial repos, broken experiments or unclean client work.

12.1.3 6. Repository decision matrix

Repository	Decision
WebGL-Thesis-Proposal	Public flagship. Mandatory cleanup.
ARA Framework	Conditional public release.
ai-news-aggregator	Keep private until functional.

Repository	Decision
FitApp-Free	Keep private or low priority.
Hyperrealistic portrait	Publish as portfolio breakdown, not code repo.
skills-introduction-to-github	Do not pin. Archive if needed.
old client/web repos	Hide unless clean and relevant.
deus-ex-machina	Audit before showing.
delarge95.github.io	Use only if portfolio-ready.

12.1.4 7. TwinSight GitHub README structure

The TwinSight README must be recruiter-readable first and technically detailed second.

Recommended structure:

```
Bloque de codigo: text
# TwinSight X500 – Unity WebGL Technical Visualization Prototype

## Overview
## Live Demo / Video Demo
## Why This Project Exists
## Core Features
## Technical Stack
## CAD-to-Realtime Pipeline
## Runtime Systems
## Optimization Results
## Usability Evaluation
## My Role and Contributions
## AI-Assisted Development Disclosure
## Limitations
## Repository Structure
## Setup Notes
## Screenshots
## Related Links
```

7.1 README opening paragraph

Recommended opening:

```
Bloque de codigo: text
TwinSight X500 is a Unity WebGL interactive technical visualization prototype for inspecting the assembly structure of a
↳ Holybro X500 V2 drone. It converts CAD/manufacturer geometry and technical documentation into an optimized real-time 3D
↳ experience with component selection, exploded view, clipping, technical information panels and multiple visualization
↳ modes.
```

7.2 Required metrics block

Include a compact metric block:

```
Bloque de codigo: text
Key metrics:
- CAD routes reduced from >6.5M triangles to 95,617 optimized triangles.
- 12 validation participants.
- SUS average: 91.88.
- NASA-TLX Raw: 8.69 for 3D viewer vs 19.89 for 2D support.
- 96 task-condition records.
```

Add a note that these are academic evaluation metrics, not commercial product KPIs.

7.3 AI disclosure block

Recommended wording:

```
Bloque de codigo: text
Development note: AI tools were used as coding and documentation assistants during implementation. Final architecture,
↳ integration, debugging, Unity scene assembly, technical decisions, asset optimization, evaluation design and project
↳ ownership remained my responsibility.
```

This is direct enough to be honest but avoids weakening the project unnecessarily.

12.1.5 8. ARA GitHub README structure

ARA must be positioned as tooling, not as a production AI company product.

Recommended title:

```
Bloque de codigo: text
ARA Framework – AI-Assisted Research Automation Prototype
```

Recommended README structure:

```
Bloque de codigo: text
# ARA Framework

## Overview
## Problem
## Architecture
## Agent Workflow
## Tech Stack
## Example Output
## Current Status
## Limitations
## Setup
## Roadmap
## AI and API Disclosure
```

8.1 ARA positioning statement

```
Bloque de codigo: text
ARA Framework is a Python-based research automation prototype that explores structured academic and technical report
↪ generation using LangGraph, LangChain, LLM APIs, retrieval tools, checkpointing and Markdown output workflows.
```

8.2 ARA claims to avoid

Do not describe ARA as:

- autonomous research scientist;
 - production-grade AI platform;
 - enterprise-ready agent system;
 - replacement for academic review;
 - evidence of ML engineering.
-

12.1.6 9. GitHub quality gates before applying

Before sending applications, GitHub must satisfy these conditions:

- Profile bio aligned with target role.
- TwinSight pinned first.
- TwinSight README complete.
- Broken or irrelevant repos unpinned.
- Private repos hidden if unfinished.
- At least one demo video linked.
- At least one screenshot set linked.
- No secret keys, API keys or credentials.
- No unprofessional commit messages in visible history.
- No overclaiming in README files.
- AI assistance disclosed where relevant.

If these conditions are not met, LinkedIn and CV may generate interest but technical reviewers may reject the profile after checking GitHub.

12.2 Part B — LinkedIn Strategy

12.2.1 10. LinkedIn objective

LinkedIn must make Alexander discoverable for international recruiters and technical leads searching for:

- Unity Developer;
- Technical Artist;
- Real-Time 3D Developer;
- Interactive 3D Developer;
- Unity WebGL Developer;
- Technical Visualization Developer;
- Digital Twin Visualization;
- Simulation Developer;
- CAD-to-realtime;
- Technical Art;
- Python automation;
- AI-assisted tooling.

The profile must be written in English as the primary international job-search language.

12.2.2 11. LinkedIn headline options

Recommended final headline

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist | Unity WebGL, C#, CAD-to-Realtime Optimization, Technical Visualization
```

This is the best balance between clarity, ATS keywords and defensibility.

Alternative headlines

1. Unity Technical Artist / Real-Time 3D Developer | WebGL, C#, Blender, CAD-to-Realtime Workflows
2. Interactive 3D Developer | Unity WebGL, Technical Visualization, CAD Optimization, C#
3. Real-Time 3D Developer | Unity, WebGL, Technical UI, CAD-to-Realtime Visualization
4. Unity WebGL Developer | Interactive Technical Visualization, C#, Blender, Real-Time 3D
5. Technical Artist & Real-Time 3D Developer | Unity URP, WebGL, Optimization, Visualization Systems
6. Unity Developer for Interactive 3D Visualization | C#, WebGL, CAD-to-Realtime Pipelines
7. Real-Time Technical Visualization Developer | Unity, WebGL, Blender, C#
8. Unity Technical Artist | Asset Optimization, Technical UI, WebGL, C#
9. Interactive 3D / Unity Developer | Technical Visualization, WebGL Deployment, CAD Optimization
10. Technical Visualization Developer | Unity WebGL, Real-Time 3D, C#, Blender
11. Unity / WebGL Developer | Real-Time 3D, Technical Art, Interactive Visualization
12. CAD-to-Realtime Technical Artist | Unity, Blender, WebGL, Interactive 3D Systems

13. Multimedia Engineer | Unity WebGL, Real-Time 3D, Technical Visualization, Python Automation

14. Unity Technical Artist & Tools Developer | Real-Time 3D, WebGL, Python Automation

15. Technical Artist in Training | Unity WebGL, C#, Real-Time 3D, Technical Visualization

Avoid “Senior” in the headline. Avoid leading with Python or AI unless targeting a Python-specific role.

12.2.3 12. LinkedIn About section

Recommended About:

Bloque de código: text

I am a Multimedia Engineering student completing my thesis defense, focused on real-time 3D, Unity WebGL and interactive
↪ technical visualization.

My flagship project, TwinSight X500, is a Unity WebGL prototype for inspecting the assembly structure of a Holybro X500 V2
↪ drone. The project transforms CAD/manufacture geometry and technical documentation into an optimized real-time 3D
↪ experience with component selection, exploded view, clipping, technical information panels and multiple visualization
↪ modes.

My strongest areas are Unity, C#, WebGL deployment, CAD-to-realtime optimization, Blender asset preparation, technical UI,
↪ real-time interaction systems and technical documentation. I also build Python-based automation prototypes, including
↪ AI-assisted research tooling with LangGraph and LLM workflows.

I am currently based in Colombia and available for remote contractor/B2B opportunities with international teams. I am
↪ primarily targeting roles in real-time 3D, Unity technical art, interactive 3D visualization, technical visualization,
↪ simulation, XR and digital twin visualization.

Core stack: Unity, C#, WebGL, URP, UI Toolkit, Shader Graph, Blender, Python, Git, technical documentation.

12.1 About section rationale

This version does four things:

1. Leads with current role identity.
2. Makes TwinSight visible immediately.
3. Includes relevant keywords without keyword stuffing.
4. States Colombia remote availability without pretending to have EU work authorization.

12.2.4 13. LinkedIn Experience section

13.1 Recommended current role title

Use:

Bloque de código: text

Independent Technical Artist & Real-Time 3D Developer

Alternative:

Bloque de código: text

Independent Unity / Real-Time 3D Developer

Avoid:

Bloque de código: text

Senior Technical Artist
AI Engineer
Full-Stack Developer
Founder

13.2 Recommended experience entry

```
Bloque de codigo: text
Independent Technical Artist & Real-Time 3D Developer
Freelance / Portfolio Projects
Mar 2024 - Present
Colombia · Remote / Hybrid

- Developed TwinSight X500, a Unity WebGL interactive technical visualization prototype for inspecting the assembly structure
  ↳ of a Holybro X500 V2 drone.
- Built runtime interaction systems including component selection, exploded view, clipping/cross-section controls,
  ↳ visualization modes and technical information panels.
- Optimized CAD/manufacturer geometry for real-time WebGL deployment, reducing high-density CAD routes from more than 6.5M
  ↳ triangles to 95,617 optimized triangles.
- Prepared assets through Blender-based cleanup, retopology, UV and high-to-low workflow planning.
- Designed and documented an academic validation process using SUS, NASA-TLX Raw and Think-Aloud methods.
- Explored Python automation and AI-assisted research tooling through ARA Framework, a LangGraph-based academic research
  ↳ automation prototype.
```

13.3 Notes

The experience entry should not read like traditional employment if it was primarily academic/portfolio work. “Freelance / Portfolio Projects” is clearer than pretending it was formal industry employment.

12.2.5 14. LinkedIn Education section

14.1 UNAD entry

Recommended:

```
Bloque de codigo: text
Universidad Nacional Abierta y a Distancia - UNAD
Bachelor of Engineering, Multimedia Engineering
Aug 2021 - Jul 2026

Relevant areas: computer graphics, software engineering, mathematics, physics, real-time 3D, Unity development, technical
  ↳ visualization and interactive systems.

Thesis: TwinSight X500 - Unity WebGL interactive technical visualization prototype for drone assembly inspection.
```

Important correction: do not describe ARA Framework as the thesis unless formally true.

14.2 UNAL entry

Recommended:

```
Bloque de codigo: text
Universidad Nacional de Colombia - UNAL
Advanced Coursework in Electronic Engineering
Mar 2013 - Nov 2018

Completed advanced coursework in electronic engineering, including mathematics, physics, circuits, systems, simulation and
  ↳ algorithmic problem-solving. This background supports technical visualization, engineering-oriented 3D work and
  ↳ spatial/analytical reasoning.
```

Alternative shorter wording:

```
Bloque de codigo: text
Coursework toward B.Eng. in Electronic Engineering
```

Avoid saying “graduated” from UNAL if no degree was obtained.

12.2.6 15. LinkedIn Projects section

15.1 TwinSight X500

Bloque de codigo: text
TwinSight X500 – Unity WebGL Technical Visualization Prototype

Interactive 3D technical visualization system for inspecting the assembly structure of a Holybro X500 V2 drone. Built with
↪ Unity WebGL, C#, URP, UI Toolkit and Blender-based CAD-to-realtime optimization.

Features include component selection, exploded view, clipping/cross-section interaction, technical information panels and
↪ multiple visualization modes. The project reduced high-density CAD routes from more than 6.5M triangles to 95,617
↪ optimized triangles and was evaluated using SUS, NASA-TLX Raw and Think-Aloud methods.

15.2 ARA Framework

Bloque de codigo: text
ARA Framework – AI-Assisted Research Automation Prototype

Python-based prototype for structured academic and technical research automation using LangGraph, LangChain, LLM APIs, Redis,
↪ Semantic Scholar integration, Playwright-based retrieval and Markdown report generation.

Positioned as a tooling and automation project, not a production AI platform.

15.3 Blender portrait

Bloque de codigo: text
Hyperrealistic Blender Portrait – Technical Art Breakdown

Blender character/portrait study focused on topology, materials, grooming, lighting and rendering. Used as evidence of 3D
↪ production literacy and asset pipeline understanding.

Only add this after there is a public ArtStation or portfolio breakdown.

12.2.7 16. LinkedIn Featured section

Recommended order:

1. TwinSight case study page.
2. TwinSight demo video.
3. TwinSight GitHub repository.
4. Portfolio homepage.
5. ARA Framework, only after cleanup.
6. ArtStation breakdown, only after cleanup.

Do not feature unfinished branches, old plans or weak repositories.

12.2.8 17. LinkedIn skills order

Recommended top skills:

1. Unity
2. C#
3. Real-Time 3D
4. Technical Art
5. WebGL
6. 3D Optimization
7. Blender
8. Computer Graphics
9. Technical Visualization

10. Shader Graph
11. UI Toolkit
12. Python
13. Git
14. Technical Documentation
15. CAD-to-Realtime Workflows
16. Asset Optimization
17. Interactive 3D
18. XR Foundations
19. Research
20. Software Engineering

If LinkedIn does not allow custom skills such as “CAD-to-Realtime Workflows,” use the closest available equivalent and include the phrase in About, Experience and Projects.

12.2.9 18. LinkedIn activity strategy

Post only when there is technical substance. Do not post motivational content or generic job-search posts.

Recommended post types:

1. TwinSight technical breakdown.
2. CAD optimization before/after.
3. Unity WebGL performance note.
4. Clipping/cross-section feature demo.
5. Technical UI breakdown.
6. Lessons from converting CAD to real-time assets.
7. Short demo video with role-targeted caption.
8. ARA architecture note, only after cleanup.
9. Blender topology/material breakdown.

Suggested cadence:

- 1 strong post per week for 8 weeks.
- 3–5 thoughtful comments per week on posts from Unity, XR, technical art, simulation and visualization professionals.
- 5–10 recruiter or technical lead connection requests per week after profile cleanup.

Avoid daily posting unless the quality is high.

12.2.10 19. LinkedIn connection scripts

19.1 Recruiter connection

```
Bloque de codigo: text
Hi [Name], I'm a Colombia-based Real-Time 3D / Unity developer focused on interactive technical visualization, Unity WebGL and
↳ CAD-to-realtime optimization. I'm currently preparing my thesis project, TwinSight X500, as a portfolio case study and I'm
↳ exploring remote international roles. Open to connecting.
```

19.2 Technical lead connection

```
Bloque de codigo: text
Hi [Name], I'm working on Unity WebGL technical visualization and CAD-to-realtime optimization. My flagship project is
↳ TwinSight X500, an interactive 3D drone assembly inspection prototype. I'd like to follow your work in real-time 3D /
↳ visualization.
```

19.3 Referral request

```
Bloque de codigo: text
Hi [Name], I saw the [Role] opening at [Company]. My strongest fit is Unity/WebGL technical visualization, CAD-to-realtime
↪ optimization and interactive 3D systems through my TwinSight X500 project. Would you be open to pointing me to the right
↪ recruiter or sharing whether this role accepts remote contractors from LATAM?
```

12.3 Part C — CV Strategy

12.3.1 20. CV objective

The CV must function as an ATS-readable proof document. It should not be a life story.

The CV must answer:

1. What role is he applying for?
2. What is the strongest evidence?
3. What technologies can he defend?
4. Is he available remotely from Colombia?
5. What are the quantified results?
6. What should the recruiter click next?

The default CV should be one page. A two-page version is acceptable only for highly technical roles where project detail is requested.

12.3.2 21. Base CV structure

Recommended order:

```
Bloque de codigo: text
Name + Title
Location / Remote availability / Links
Professional Summary
Technical Skills
Selected Projects
Experience
Education
Additional Training
Languages
```

If applying to roles that heavily value formal experience, place Experience before Projects. For Alexander's current stage, **Selected Projects should come before Experience** because TwinSight is the strongest proof.

12.3.3 22. Base CV header

```
Bloque de codigo: text
Alexander Woodcock Salomón
Real-Time 3D Developer / Unity Technical Artist
Colombia • Remote Contractor/B2B • English C1 • Spanish Native
LinkedIn: [link] • GitHub: [link] • Portfolio: [link] • ArtStation: [link if ready]
```

Do not include full address. City/country is enough.

12.3.4 23. Base CV professional summary

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, C#, CAD-to-realtime optimization and interactive
↳ technical visualization. Built TwinSight X500, a Unity WebGL drone assembly visualization prototype with component
↳ selection, exploded view, clipping, technical UI and multiple visualization modes. Strong background in Blender asset
↳ optimization, technical documentation, computer graphics fundamentals and Python automation prototypes. Based in Colombia
↳ and available for remote contractor/B2B work with international teams.
```

This is the default version. It can be adapted by role family.

12.3.5 24. Technical skills section

Recommended structure:

```
Bloque de codigo: text
Real-Time 3D: Unity, C#, Unity WebGL, URP, UI Toolkit, Shader Graph, runtime interaction systems
3D Pipeline: Blender, retopology, UVs, baking, CAD-to-realtime optimization, asset cleanup
Visualization: technical visualization, exploded views, clipping/cross-section interaction, technical UI
Programming / Tools: Python, Git/GitHub, LangGraph, LangChain, FastAPI basics, React basics
Documentation / Evaluation: technical documentation, SUS, NASA-TLX Raw, Think-Aloud, academic validation
```

Only include secondary technologies like React, PostgreSQL, Redis or FastAPI if the target role values them and there is a project that demonstrates them.

12.3.6 25. Selected projects — CV entries

25.1 TwinSight X500

```
Bloque de codigo: text
TwinSight X500 – Unity WebGL Technical Visualization Prototype
Academic thesis project · Unity, C#, WebGL, URP, UI Toolkit, Blender

- Built an interactive Unity WebGL prototype for inspecting the assembly structure of a Holybro X500 V2 drone through
↳ component selection, exploded view, clipping and technical information panels.
- Converted CAD/manufacture geometry into optimized real-time assets, reducing high-density CAD routes from more than 6.5M
↳ triangles to 95,617 optimized triangles.
- Implemented multiple visualization modes including realistic, X-Ray, blueprint, solid color, wireframe, ghosted and
↳ thermal-style views.
- Designed technical UI and runtime interaction systems for component inspection and spatial relationship understanding.
- Evaluated the prototype with 12 participants using SUS, NASA-TLX Raw and Think-Aloud methods; recorded 96 task-condition
↳ observations.
```

25.2 ARA Framework

```
Bloque de codigo: text
ARA Framework – AI-Assisted Research Automation Prototype
Python, LangGraph, LangChain, Redis, Semantic Scholar API, Playwright MCP

- Developed a Python prototype for structured academic and technical report generation using a multi-agent workflow with
↳ LangGraph and LLM APIs.
- Designed agent roles for niche analysis, literature research, technical architecture, implementation planning and content
↳ synthesis.
- Integrated retrieval, checkpointing, budget management and Markdown report generation workflows.
- Positioned as a research automation prototype and tooling experiment, not as a production AI platform.
```

Use ARA only in CV variants where Python/tooling helps the target job.

25.3 Blender portrait

```
Bloque de codigo: text
Hyperrealistic Blender Portrait – Technical Art Study
Blender, topology, materials, grooming, lighting, rendering

- Completed a Blender character/portrait study focused on topology, material setup, grooming, lighting and rendering.
- Structured the project as evidence of 3D production literacy and technical art pipeline understanding.
```

Only include if the portfolio/ArtStation page exists.

12.3.7 26. Experience section

Recommended entry:

```
Bloque de codigo: text
Independent Technical Artist & Real-Time 3D Developer
Freelance / Portfolio Projects · Colombia · Mar 2024 – Present

- Developed Unity, WebGL, Python and 3D portfolio projects focused on interactive visualization, technical documentation and
  ↔ production workflows.
- Built and documented TwinSight X500 as a thesis-based technical visualization prototype.
- Explored AI-assisted research automation through Python and LangGraph-based tooling.
- Completed freelance web and digital projects for small clients, strengthening client communication, delivery discipline and
  ↔ technical troubleshooting.
```

If formal freelance projects are weak or unrelated, keep this section short and let Projects dominate.

12.3.8 27. Education section

27.1 UNAD

```
Bloque de codigo: text
Universidad Nacional Abierta y a Distancia – UNAD
B.Eng. Multimedia Engineering · Expected 2026
Thesis: TwinSight X500 – Unity WebGL interactive technical visualization prototype for drone assembly inspection.
Relevant areas: computer graphics, software engineering, mathematics, physics, real-time 3D, Unity development and technical
  ↔ visualization.
```

27.2 UNAL

```
Bloque de codigo: text
Universidad Nacional de Colombia – UNAL
Advanced Coursework in Electronic Engineering · 2013–2018
Completed advanced coursework in mathematics, physics, electronics, circuits, systems, simulation and algorithmic
  ↔ problem-solving.
```

This wording is honest and internationally understandable.

12.3.9 28. Additional training section

Because many courses are non-certified, they should not be presented as formal credentials.

Recommended heading:

```
Bloque de codigo: text
Additional Training and Self-Directed Study
```

Recommended format:

```
Bloque de codigo: text
- CG Cookie HUMAN – High-Fidelity Character Pipeline & Topology; completed as self-directed technical art training.
- Alive! – 3D Animation Mechanics & Dynamic Motion; completed as self-directed animation study.
- Rebelway / Houdini / Unreal FX coursework; used for portfolio-directed skill development. [Only after meaningful progress.]
```

Do not write:

```
Bloque de codigo: text
Certified in CG Cookie HUMAN
Certified Rebelway FX Artist
```

unless a formal certificate exists and can be verified.

12.3.10 29. Languages section

Bloque de código: text
 Spanish – Native
 English – Full professional proficiency / C1 self-assessed
 German – Beginner / A1 target if Germany route becomes active
 Portuguese – Future strategic language for Portugal/EU mobility; not current work language unless studied

For a one-page CV, include only Spanish and English unless German or Portuguese becomes relevant to the job.

12.4 Part D — CV Variants

12.4.1 30. Variant 1 — Unity Technical Artist / Real-Time 3D Developer

Summary

Bloque de código: text
 Unity Technical Artist / Real-Time 3D Developer focused on interactive technical visualization, Unity WebGL, asset
 ↪ optimization and technical UI. Built TwinSight X500, a drone assembly visualization prototype combining CAD-to-realtime
 ↪ optimization, runtime interaction systems, clipping, exploded view and usability validation.

Emphasize

- Unity.
- C#.
- WebGL.
- Blender optimization.
- Shader Graph.
- Technical UI.
- CAD-to-realtime.

Downplay

- ARA.
- Full-stack.
- FitApp-Free.

12.4.2 31. Variant 2 — Unity WebGL / Interactive 3D Developer

Summary

Bloque de código: text
 Unity WebGL / Interactive 3D Developer focused on browser-based real-time visualization, C# interaction systems and optimized
 ↪ 3D assets. Built TwinSight X500, a Unity WebGL prototype for technical drone assembly inspection with selection, exploded
 ↪ view, clipping and technical information panels.

Emphasize

- WebGL deployment.
- runtime performance constraints;
- interactive UI;
- browser-accessible 3D;
- mobile-first considerations.

Add if true

- live demo URL;
 - build size;
 - load time;
 - FPS / frame-time data.
-

12.4.3 32. Variant 3 — Technical Visualization / Digital Twin / Simulation**Summary**

Bloque de codigo: text
 Technical Visualization Developer focused on real-time 3D systems for explaining complex assemblies and spatial relationships.
 ↪ Built TwinSight X500, a Unity WebGL drone assembly inspection prototype combining CAD-to-realtime optimization, technical
 ↪ UI, exploded view, clipping and structured usability evaluation.

Emphasize

- engineering-oriented visualization;
- CAD/documentation conversion;
- technical communication;
- component hierarchy;
- spatial relationships;
- usability evaluation.

Avoid

Do not claim live sensor data, simulation physics, IoT integration or true digital twin functionality unless added later.

12.4.4 33. Variant 4 — Tools / Pipeline / Python Automation**Summary**

Bloque de codigo: text
 Tools-oriented Unity / Python developer focused on technical workflows, automation and structured documentation. Built
 ↪ TwinSight X500 as a real-time 3D visualization prototype and ARA Framework as a Python/LangGraph research automation
 ↪ prototype for structured academic and technical report generation.

Emphasize

- Python;
- LangGraph;
- workflow automation;
- documentation;
- reproducibility;
- tooling mindset;
- technical process design.

Downplay

- pure art;
- game-only framing;

- incomplete full-stack projects.

12.5 Part E — Keyword Strategy

12.5.1 34. Primary ATS keyword cluster

Use these consistently across CV, LinkedIn, GitHub and portfolio:

```
Bloque de codigo: text
Unity
C#
Unity WebGL
Real-Time 3D
Technical Artist
Interactive 3D
Technical Visualization
CAD-to-Realtime
3D Optimization
Blender
URP
UI Toolkit
Shader Graph
Runtime Interaction Systems
Technical UI
Exploded View
Clipping / Cross-Section
WebGL Deployment
Computer Graphics
Technical Documentation
```

12.5.2 35. Secondary keyword cluster

Use only where relevant:

```
Bloque de codigo: text
Python
LangGraph
LangChain
LLM Applications
AI-Assisted Tooling
Research Automation
FastAPI
React
PostgreSQL
Redis
Playwright
Semantic Scholar API
Markdown Automation
```

12.5.3 36. Industry keyword cluster

Use selectively by job type:

```
Bloque de codigo: text
Digital Twin Visualization
Simulation
XR
AR/VR
Product Visualization
Industrial Visualization
Technical Training
Interactive Documentation
Drone Assembly
Engineering Visualization
Manufacturing Visualization
```

12.5.4 37. Spanish keyword equivalents

Use in LATAM searches or Spanish CV variants:

```
Bloque de código: text
Desarrollador Unity
Artista Técnico
Desarrollador 3D en Tiempo Real
Visualización Técnica
Optimización 3D
WebGL
Gemelo Digital
Simulación
XR / Realidad Extendida
Automatización con Python
Documentación Técnica
```

12.6 Part F — Quality Control

12.6.1 38. Consistency checklist

Before applying, verify that:

- LinkedIn headline matches CV title.
- CV title matches portfolio title.
- GitHub pinned repo supports the headline.
- TwinSight appears first everywhere.
- ARA appears secondary everywhere.
- No document says the thesis is ARA if the thesis is TwinSight.
- No document says thesis is still in early development.
- No course without certificate is labeled “certified.”
- No EU work authorization is claimed prematurely.
- GitHub does not expose weak repos above strong ones.
- LinkedIn Featured does not show unfinished pages.
- CV does not look like a generic full-stack profile.

12.6.2 39. Overclaiming risk map

High-risk claims

```
Bloque de código: text
Senior Technical Artist
AI Engineer
Digital Twin Engineer
Production AI Platform
EU Work Authorized
Certified Houdini / Unity / Rebelway without proof
```

These should be removed.

Medium-risk claims

```
Bloque de código: text
Technical Artist
Tools Developer
Simulation Developer
XR Developer
Full-stack Developer
```

These are acceptable only when supported by project evidence.

Low-risk claims

Bloque de código: text
 Unity WebGL
 Real-Time 3D
 Interactive Technical Visualization
 CAD-to-Realtime Optimization
 Python Automation Prototype
 Technical Documentation

These align with current evidence.

12.6.3 40. Recruiter-facing work availability

Current version

Bloque de código: text
 Based in Colombia. Available for remote contractor/B2B roles with international teams.

For U.S. companies

Bloque de código: text
 I am based in Colombia and available for remote contractor/B2B collaboration. I do not currently require U.S. payroll
 ↪ employment.

For EU companies before Portuguese passport

Bloque de código: text
 I am currently based in Colombia and available for remote contractor/B2B work. I do not currently hold EU work authorization,
 ↪ but I expect Portuguese citizenship/passport in approximately two years.

Use the Portuguese citizenship line only if strategically relevant and only after the conversation reaches relocation/work authorization.

After Portuguese passport

Bloque de código: text
 Portuguese / EU citizen. Authorized to work across the EU/EEA.

If Germany residence/work authorization through marriage exists

Bloque de código: text
 Based in Germany with valid residence and work authorization.

12.6.4 41. Recommended immediate edits

GitHub

1. Rewrite GitHub bio.
2. Pin TwinSight first.
3. Rewrite TwinSight README.
4. Hide/unpin tutorial and weak repos.
5. Add demo video and screenshots.
6. Add AI-assistance disclosure.
7. Add clear limitations section.

LinkedIn

1. Replace headline.
2. Rewrite About.
3. Correct UNAD thesis description.
4. Reframe UNAL as advanced coursework.
5. Add TwinSight as project.
6. Reorder skills.
7. Add Featured links only after assets are ready.
8. Set Open to Work carefully.

CV

1. Create one-page base CV.
2. Create four variants.
3. Use TwinSight metrics.
4. Remove unsupported claims.
5. Include Colombia remote contractor availability.
6. Add portfolio and GitHub links.
7. Keep courses in Additional Training, not Certifications.

12.6.5 42. Final recommended profile package

The minimum public package before serious applications:

```
Bloque de codigo: text
1. One-page CV: Unity / Real-Time 3D variant.
2. LinkedIn rewritten and corrected.
3. GitHub profile cleaned.
4. TwinSight README complete.
5. TwinSight case-study page live.
6. 60-120 second TwinSight demo video.
7. 8-12 high-quality screenshots.
8. ARA hidden or cleaned as secondary.
9. ArtStation updated only if Blender portrait has a technical breakdown.
```

Applications before this package is ready may still work, but conversion probability will be lower.

12.6.6 43. Open questions before final CV writing

These must be answered before producing the final CV files:

1. Is there a live TwinSight WebGL demo URL?
2. Is the GitHub repository clean enough to show now?
3. Are there screenshots ready?
4. Is the demo video recorded?
5. Which metrics are final and defensible after thesis defense?
6. Which courses from the Excel were actually completed?
7. Does ARA run reliably enough to show publicly?
8. Does ai-news-aggregator currently work?
9. Should the CV mention freelance web work or omit it?
10. What email/contact method should appear on public materials?

12.6.7 44. Decision for next document

This section defines the public profile strategy. The next section should move from profile assets to external market execution:

```
Bloque de codigo: text  
11_company_targets_job_boards_recruiters.md
```

That next file should use external research and structured data collection. It requires company lists, job boards, recruiters, communities, role-title searches and links.

Recommended tool use for the next section:

- Deep Research: yes.
- Agent mode: yes, if available.

Chapter 13

11_company_targets_job_boards_recruiters.m

Fuente: 11_company_targets_job_boards_recruiters.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 879

Ultima modificacion: 2026-06-11 15:30

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This file consolidates the Deep Research output, the partial agent collection, and manual cleanup into a structured target ecosystem for Alexander Woodcock Salomón.

Contenido completo depurado

13.0.1 Scope

This file consolidates the Deep Research output, the partial agent collection, and manual cleanup into a structured target ecosystem for Alexander Woodcock Salomón.

The target profile is:

- Colombia-based.
- Spanish native; English C1 self-assessed.
- Portuguese passport expected around 2028; not current EU work authorization.
- Possible Germany relocation through legal marriage with German partner.
- Main proof asset: TwinSight X500, a Unity WebGL technical visualization project.
- Primary role families: Real-Time 3D, Unity WebGL, Technical Visualization, Digital Twin, Simulation, Unity Technical Artist, XR.
- Secondary route: Python automation / AI tooling.

This is not a final application plan. It is a target ecosystem map.

13.0.2 Data quality rules

The previous agent result was useful but incomplete. It collected some strong targets and job boards, but it did not verify 120 companies one by one. This final version therefore uses three verification labels:

- **Verified:** official website/careers page or direct evidence checked.
- **Partial:** official website known, but current openings/remote rules not checked.
- **Watchlist:** plausible fit, but must be verified before application.

Remote labels are conservative. **Remote does not mean worldwide** unless the company states it.

13.0.3 Current operational interpretation

Use this file as an ecosystem map, not as a direct application list.

- The company identity table is the broad universe of possible targets.
- The hiring feasibility table is a working hypothesis about role fit, remote posture, and authorization friction.
- The scoring table is a prioritization seed, not proof that a company has a current opening.
- A target only becomes application-ready after a current opening, remote scope, contractor/payroll route, and work authorization requirements are checked.

Historic notes should not override current profile data. If a newer non-historic file conflicts with older historic context, use the newer non-historic data.

13.0.4 Scoring definitions

Score	Meaning
5	Direct TwinSight fit or high salary route.
4	Strong adjacent fit.
3	Plausible but needs adaptation.
2	Weak fit or high friction.
1	Watchlist only.

Priority:

Priority	Meaning
A	Apply/research first.
B	Good second wave.
C	Keep for later.
Watchlist	Do not prioritize yet.

Operational layers override the broad A/B/C label when planning weekly execution:

Layer	Use	Action
A1	Highest practical first wave	Verify live openings and contacts first.
A2	High strategic fit, higher friction	Research after A1 or when relocation/passport status changes.
B1	Good fallback / volume source	Use for steady pipeline, especially LATAM contractor roles.
C	Later exploration	Check monthly or when portfolio changes.
Watchlist	Do not spend active search time yet	Revisit only if a role is unusually aligned.

13.0.5 Company identity table

#	Company	Region	Category	Website / careers
1	Treeview Studio	USA / LATAM	Unity / XR / digital twin	https://treeview.studio/
2	Unity Technologies	Global	Engine / RT3D platform	https://unity.com/careers
3	realvirtual.io	Germany	Unity simulation / digital twin	https://realvirtual.io/

#	Company	Region	Category	Website / careers
4	Capgemini Engineering	Global / EU	Digital twin consulting	https://www.capgemini.com/careers/
5	E Tech Group	USA	Industrial simulation	https://www.etechgroup.com/careers/
6	Holo-Light	Germany	XR / industrial visualization	https://holo-light.com/careers/
7	Innoactive	Germany	VR training / enterprise XR	https://www.innoactive.io/
8	Virtualis	UK	VR / simulation	https://www.virtualis.com/
9	FundamentalVR	UK / USA	Medical simulation XR	https://fundamentalvr.com/
10	Osso VR	USA	Medical VR training	https://www.ossovr.com/careers
11	Talespin	USA	VR training / enterprise	https://www.talespin.com/careers
12	EON Reality	Global	XR education / simulation	https://eonreality.com/careers/
13	Arch Virtual	USA	Enterprise VR / training	https://archvirtual.com/
14	Varjo	Finland	XR hardware / simulation	https://varjo.com/company/careers/
15	Magic Leap	USA	AR hardware	https://www.magicleap.com/careers/
16	Niantic	USA / global	AR / game tech	https://nianticlabs.com/jobs/
17	Snap AR	USA / global	AR platform	https://www.snap.com/jobs
18	Meta Reality Labs	USA / global	XR platform	https://www.metacareers.com/
19	Active Theory	USA	WebGL / creative tech	https://activetheory.net/careers/
20	UNIT9	UK / global	Creative tech / XR	https://www.unit9.com/careers
21	Resn	NZ / Netherlands	WebGL creative tech	https://resn.co.nz/
22	Monogrid	Italy	WebGL / interactive 3D	https://www.monogrid.com/
23	14islands	Sweden	WebGL / creative tech	https://www.14islands.com/
24	Dogstudio	Belgium / USA	Creative tech	https://dogstudio.co/
25	Media.Monks	Global	Creative tech / production	https://media.monks.com/careers
26	Demodern	Germany	Creative tech / XR	https://demodern.com/
27	Elastique.	Germany	Interactive media	https://elastique.de/
28	Moment Factory	Canada / global	Immersive experiences	https://momentfactory.com/careers
29	VTPRODesign	USA	Experiential tech	https://vtprodesign.com/
30	BUCK	Global	Design / motion / tech	https://buck.co/careers
31	Threkit	USA	3D product configurator	https://www.threkit.com/company/careers
32	VNTANA	USA	3D commerce / product viz	https://www.vntana.com/careers/
33	Emersya	France	3D configurators	https://emersya.com/
34	Expivi	Netherlands	3D configurators	https://www.expivi.com/
35	Vection Technologies	Australia / Italy	XR / digital twin	https://vection-technologies.com/careers/
36	Eagle 3D Streaming	USA / global	Pixel streaming / Unreal	https://www.eagle3dstreaming.com/career/
37	Cesium	USA	3D geospatial	https://cesium.com/careers/
38	Matterport	USA	3D capture / digital twin	https://matterport.com/careers
39	DroneDeploy	USA	Drone mapping / digital twin	https://www.dronedeploy.com/careers/
40	Pix4D	Switzerland	Drone photogrammetry	https://www.pix4d.com/careers/
41	Skydio	USA	Drones / autonomy	https://www.skydio.com/careers
42	Zipline	USA / global	Drones / logistics	https://www.flyzipline.com/careers
43	Auterion	Switzerland / USA	Drone software	https://auterion.com/careers/
44	Esri	USA / global	GIS / 3D visualization	https://www.esri.com/en-us/about/careers/main
45	Autodesk	Global	CAD / digital twin	https://www.autodesk.com/careers
46	NVIDIA Omniverse	USA / global	RT3D / simulation	https://www.nvidia.com/en-us/about-nvidia/careers/
47	Epic Games	USA / global	Engine / real-time 3D	https://www.epicgames.com/site/en-US/careers

#	Company	Region	Category	Website / careers
48	Bentley Systems	USA / global	Infrastructure digital twin	https://www.bentley.com/company/careers/
49	Ansys	USA / global	Simulation software	https://www.ansys.com/careers
50	MathWorks	USA / global	Simulation tools	https://www.mathworks.com/company/jobs.html
51	Siemens Digital Industries	Germany / global	Industrial digital twin	https://www.siemens.com/global/en/company/jobs.html
52	Dassault Systèmes	France / global	CAD / digital twin	https://careers.3ds.com/
53	PTC	USA / global	IoT / AR / digital twin	https://www.ptc.com/en/careers
54	Hexagon AB	Sweden / global	Industrial digital reality	https://hexagon.com/company/careers
55	Schneider Electric	France / global	Industrial automation	https://www.se.com/ww/en/about-us/careers/overview.jsp
56	Rockwell Automation	USA / global	Industrial automation	https://www.rockwellautomation.com/en-us/company/careers.html
57	ABB	Switzerland / global	Robotics / automation	https://global.abb/group/en/careers
58	Bosch	Germany / global	Engineering / simulation	https://www.bosch.com/careers/
59	Honeywell	USA / global	Industrial / aerospace	https://careers.honeywell.com/
60	GE Aerospace	USA / global	Aerospace digital twin	https://jobs.gecareers.com/aviation/global/en
61	Critical TechWorks	Portugal	BMW software / simulation	https://www.criticaltechworks.com/careers/
62	Critical Software	Portugal / UK	Safety-critical software	https://criticalsoftware.com/careers/
63	Xpand IT	Portugal	Software consulting	https://www.xpand-it.com/careers/
64	Imaginary Cloud	Portugal / UK	Product software	https://www.imaginarycloud.com/careers/
65	xgeeks	Portugal	Software engineering	https://xgeeks.io/careers/
66	LOAD	Portugal	Product / software	https://www.load.digital/careers/
67	YAGER	Germany	Game studio	https://www.yager.de/career
68	Crytek	Germany	Game studio	https://www.crytek.com/career
69	Keen Games	Germany	Game studio	https://www.keengames.com/jobs
70	InnoGames	Germany	Game studio	https://www.innogames.com/career/
71	Kolibri Games	Germany	Mobile games	https://www.kolibrigames.com/careers/
72	GameDuell	Germany	Games	https://inside.gameduell.com/jobs/
73	Scopely	USA / global	Mobile games	https://www.scopely.com/en/careers
74	King	UK / Sweden / global	Mobile games	https://careers.king.com/
75	Gameloft	Global / LATAM	Games	https://www.gameloft.com/corporate/en/jobs
76	Wildlife Studios	Brazil	Mobile games	https://wildlifestudios.com/careers/
77	Teravision Games	Colombia / USA	Game development	https://teravisiongames.com/careers/
78	Globant Gaming	LATAM / global	Game tech / outsourcing	https://www.globant.com/careers
79	Amber Studio	Romania / global	Game outsourcing	https://amberstudio.com/careers/
80	Keywords Studios	Global	Game outsourcing	https://www.keywordsstudios.com/en/careers/
81	Sperasoft	Global	Game co-dev	https://sperasoft.com/careers/
82	Room 8 Studio	Global	Game art / co-dev	https://room8studio.com/careers/
83	Virtuos	Global	Game co-dev	https://www.virtuosgames.com/en/careers/
84	BairesDev	LATAM / USA	Nearshore software	https://www.bairesdev.com/careers/
85	Gorilla Logic	LATAM / USA	Nearshore software	https://gorillalogic.com/careers/

#	Company	Region	Category	Website / careers
86	Lean Tech	Colombia / USA	Nearshore software	https://leangroup.com/careers/
87	MAS Global	Colombia / USA	Nearshore software	https://masglobalconsulting.com/careers/
88	Endava	Global / LATAM	Software consulting	https://www.endava.com/careers
89	EPAM	Global	Software consulting	https://www.epam.com/careers
90	SoftServe	Global	Software consulting / XR	https://www.softserveinc.com/en-us/careers
91	Wizeline	Mexico / global	Nearshore software	https://www.wizeline.com/careers/
92	Encora	Global / LATAM	Nearshore software	https://www.encora.com/careers
93	Perficient LATAM	LATAM / USA	Nearshore software	https://www.perficient.com/careers
94	HatchWorks AI	USA / LATAM	AI nearshore	https://hatchworks.com/careers/
95	LangChain	USA / remote	AI tooling / agents	https://www.langchain.com/careers
96	Humanloop	UK	LLM tooling	https://humanloop.com/careers
97	Replit	USA / remote	Developer tools / AI	https://replit.com/site/careers
98	Sourcegraph	Global / remote	Code intelligence / AI	https://sourcegraph.com/jobs
99	Roboflow	USA / remote	Computer vision tools	https://roboflow.com/careers
100	Labelbox	USA	AI data platform	https://labelbox.com/careers/
101	Kaedim	UK / remote	AI 3D assets	https://www.kaedim3d.com/careers
102	Avataar	USA / India	3D commerce / AI	https://www.avataar.ai/careers
103	8th Wall / Niantic	USA	WebAR platform	https://www.8thwall.com/
104	ShapesXR	Denmark / remote	XR creation tools	https://www.shapesxr.com/careers
105	ShapesXR-adjacent startups	EU / US	XR tooling	not found
106	Moth+Flame	USA	VR training	https://www.mothandflamevr.com/careers
107	Strivr	USA	VR training	https://www.strivr.com/careers
108	Virti	UK	VR/AR training	https://www.virti.com/careers
109	Oxford Medical Simulation	UK	Healthcare simulation	https://oxfordmedicalsimulation.com/careers/
110	FundamentalVR-adjacent medsim	UK / US	Medical simulation	not found
111	Virbela	USA / remote	Virtual worlds	https://www.virbela.com/careers
112	Spatial	USA / remote	3D social / web	https://www.spatial.io/careers
113	Decentraland Foundation	Remote	Web3 / 3D virtual world	https://decentraland.org/careers
114	Roblox	USA	3D platform	https://corp.roblox.com/careers/
115	Linden Lab	USA / remote	Virtual worlds	https://www.lindenlab.com/careers
116	Khronos Group member companies	Global	WebGL / graphics ecosystem	https://www.khronos.org/jobs/
117	Babylon.js / Microsoft ecosystem	Global	Web3D / engine	https://jobs.careers.microsoft.com/
118	PlayCanvas	UK	WebGL engine	https://playcanvas.com/
119	Sketchfab / Epic	France / USA	3D platform	https://www.epicgames.com/site/en-US/careers
120	Chaos / Cyllindo	Global	3D commerce / rendering	https://www.chaos.com/careers

13.0.6 Hiring feasibility table

#	Typical roles	Remote	Contractor	Language	Salary tier	Auth note
1	Unity dev; XR dev; real-time 3D	Hybrid / unknown	Medium	English / Spanish	Medium	Unknown
2	Technical Artist; Unity dev; tools; simulation	US/EU/Canada mostly	Medium	English	High	Local auth likely
3	Unity simulation; PLC; digital twin	EU / unknown	Medium	English + German	Medium	EU/Germany useful
4	Unity digital twin; simulation; consultant	Hybrid / EU-heavy	Medium	English	Medium	EU useful
5	Unity simulation; controls; digital twin	US / hybrid	Low	English	Medium	US auth likely
6	XR dev; Unity dev; streaming	EU / hybrid	Medium	English + German	Medium	EU/Germany useful
7	Unity XR; training simulation	EU / hybrid	Medium	English + German	Medium	Germany useful
8	Unity/Unreal XR; simulation	UK/EU / hybrid	Medium	English	Medium	UK/EU auth likely
9	Unity XR; simulation; tools	UK/US / hybrid	Medium	English	High	UK/US auth likely
10	Unity XR; tech art; simulation	US / remote some	Medium	English	High	US auth likely
11	Unity dev; XR dev	US / remote some	Medium	English	Medium	US auth likely
12	XR dev; 3D dev	Global / unknown	Medium	English	Medium	Unknown
13	Unity XR; simulation	US / remote unknown	Medium	English	Medium	Unknown
14	XR engineer; graphics; simulation	EU / hybrid	Low	English	High	EU useful
15	AR dev; Unity; tools	US / hybrid	Low	English	High	US auth likely
16	AR dev; Unity; tools	US/EU / hybrid	Medium	English	High	US/EU likely
17	AR engineer; Lens Studio	US/EU / hybrid	Low	English	High	US/EU likely
18	XR dev; technical artist	US/EU / hybrid	Low	English	High	US/EU likely
19	WebGL dev; creative technologist	Remote / hybrid	High	English	Medium	Unknown
20	Creative technologist; XR dev	Remote / hybrid	Medium	English	High	UK/EU helpful
21	WebGL dev; creative technologist	Hybrid / remote possible	Medium	English	Medium	Unknown
22	WebGL dev; 3D dev	Hybrid / remote possible	Medium	English / Italian	Medium	EU useful
23	WebGL dev; creative dev	EU / remote unknown	Medium	English	Medium	EU useful
24	WebGL dev; creative dev	Hybrid / remote	Medium	English	Medium	EU useful
25	Creative technologist; 3D dev	Global / hybrid	Medium	English	Medium	Unknown
26	Unity dev; creative technologist	Germany / hybrid	Medium	English + German	Medium	Germany useful
27	Creative dev; 3D dev	Germany / hybrid	Medium	English + German	Medium	Germany useful
28	Interactive dev; real-time 3D	Hybrid / on-site	Medium	English / French	Medium	Canada auth likely
29	Unity dev; creative technologist	US / hybrid	Medium	English	Medium	US auth likely
30	Interactive dev; creative tech	Global / hybrid	Medium	English	Medium	Unknown
31	3D dev; WebGL; product viz	US / remote some	Medium	English	Medium	US auth likely
32	3D pipeline; web 3D	US / remote some	Medium	English	Medium	US auth likely
33	3D web dev; configurator	EU / hybrid	Medium	English / French	Medium	EU useful
34	3D web dev; product viz	EU / hybrid	Medium	English	Medium	EU useful
35	Unity/XR dev; visualization	Hybrid / EU/AUS	Medium	English	Medium	EU useful
36	3D streaming; Web3D	Remote unknown	Medium	English	Medium	Unknown
37	WebGL; 3D geospatial	US / remote some	Medium	English	High	US auth likely
38	3D visualization; web 3D	US / hybrid	Medium	English	High	US auth likely
39	3D visualization; mapping	US / remote some	Medium	English	High	US auth likely
40	3D visualization; mapping	EU / hybrid	Medium	English	Medium	EU useful
41	Simulation; visualization; tools	US / hybrid	Low	English	High	US auth likely
42	Simulation; tools; visualization	US / hybrid	Low	English	High	US auth likely
43	Simulation; visualization; tools	EU/US / hybrid	Medium	English	High	EU/US likely
44	3D visualization; web GIS	US / global offices	Medium	English	High	Local auth likely
45	3D tools; visualization; simulation	Global / hybrid	Medium	English	High	Local auth likely
46	Simulation; graphics; tools	US/EU / hybrid	Low	English	High	US/EU likely
47	Tools; tech art; engine	US/EU / hybrid	Medium	English	High	US/EU likely
48	3D visualization; simulation	Global / hybrid	Medium	English	High	Local auth likely
49	Simulation; visualization	Global / hybrid	Low	English	High	Local auth likely
50	Simulation; tools; visualization	Global / hybrid	Low	English	High	Local auth likely
51	Simulation; digital twin; UX 3D	EU/global / hybrid	Low	English + German	High	EU/Germany useful
52	3D visualization; simulation	EU/global / hybrid	Low	English / French	High	EU useful
53	AR/IoT; visualization	US/EU / hybrid	Medium	English	High	Local auth likely
54	3D visualization; simulation	EU/global / hybrid	Medium	English	High	EU useful
55	Simulation; digital twin	Global / hybrid	Low	English	High	Local auth likely
56	Simulation; HMI; visualization	US/global / hybrid	Low	English	High	Local auth likely
57	Simulation; robotics; digital twin	Global / hybrid	Low	English	High	Local auth likely
58	Simulation; tools; visualization	EU/global / hybrid	Low	English + German	High	Germany useful
59	Simulation; visualization	US/global / hybrid	Low	English	High	US auth likely
60	Simulation; digital twin	US/global / hybrid	Low	English	High	US auth likely
61	Software; simulation; visualization	Portugal / hybrid	Medium	English / Portuguese	Medium	Future PT passport useful
62	Simulation; tools; software	Portugal / hybrid	Medium	English / Portuguese	Medium	Future PT passport useful
63	Web dev; data; tools	Portugal / hybrid	Medium	English / Portuguese	Medium	Future PT passport useful
64	Full-stack; Python	Portugal / remote	Medium	English	Medium	Future PT passport useful
65	Full-stack; cloud; Python	Portugal / remote	Medium	English / Portuguese	Medium	Future PT passport useful
66	Web dev; product dev	Portugal / hybrid	Medium	English / Portuguese	Medium	Future PT passport useful
67	Technical artist; tools; Unity?	Germany / hybrid	Low	English + German	Medium	Germany route useful
68	Technical artist; tools	Germany / hybrid	Low	English + German	High	Germany route useful
69	Technical artist; tools	Germany / hybrid	Low	English + German	Medium	Germany route useful
70	Unity? tools; technical artist	Germany / hybrid	Low	English + German	Medium	Germany route useful
71	Unity dev; tech artist	Germany / hybrid	Low	English + German	Medium	Germany route useful

#	Typical roles	Remote	Contractor	Language	Salary tier	Auth note
72	Unity dev; tools	Germany / hybrid	Low	English + German	Medium	Germany route useful
73	Unity dev; tech artist	US/EU / hybrid	Medium	English	High	Local auth likely
74	Unity dev; technical artist	EU / hybrid	Low	English	High	EU useful
75	Unity dev; technical artist	Global / hybrid	Medium	English / Spanish	Medium	Local offices
76	Unity dev; tools	LATAM / hybrid	Medium	English / Portuguese	Medium	LATAM possible
77	Unity dev; tech artist	LATAM / hybrid	Medium	English / Spanish	Medium	Colombia-friendly
78	Unity dev; tools; XR	LATAM / remote/hybrid	Medium	English / Spanish	Medium	LATAM-friendly
79	Unity dev; tech artist	Global / hybrid	Medium	English	Medium	EU useful
80	Technical artist; tools; QA	Global / hybrid	Medium	English	Medium	Local offices
81	Technical artist; tools	Global / hybrid	Medium	English	Medium	Local auth likely
82	Tech artist; 3D; tools	Global / remote/hybrid	Medium	English	Medium	Unknown
83	Technical artist; tools	Global / hybrid	Medium	English	Medium	Local auth likely
84	Unity dev; Python; full-stack	LATAM / remote	High	English / Spanish	Medium	LATAM-friendly
85	Python; web; mobile	LATAM / remote	High	English	Medium	LATAM-friendly
86	Python; full-stack	LATAM / remote	High	English / Spanish	Medium	Colombia-friendly
87	Python; web; data	LATAM / remote	High	English / Spanish	Medium	Colombia-friendly
88	Software; simulation?	Global / hybrid	Medium	English	Medium	Local offices
89	Software; XR?	Global / hybrid	Medium	English	Medium	Local auth likely
90	XR; software; AI	Global / hybrid	Medium	English	Medium	EU/US likely
91	Python; web; AI	LATAM / remote	Medium	English / Spanish	Medium	LATAM-friendly
92	Python; full-stack	LATAM / remote	Medium	English	Medium	LATAM-friendly
93	Software; UX; cloud	LATAM / hybrid	Medium	English / Spanish	Medium	LATAM-friendly
94	Python; AI tools	LATAM / remote	Medium	English	Medium	LATAM-friendly
95	Python; LLM apps	Remote / US-heavy	Medium	English	High	Unknown
96	LLM app dev; Python	UK / hybrid	Medium	English	High	UK auth likely
97	AI tools; Python	US / remote some	Medium	English	High	US likely
98	AI tools; dev tools	Remote / global	Medium	English	High	Open remote more likely
99	Python; tools; CV	Remote / US-heavy	Medium	English	High	US likely
100	Python; tooling	US / hybrid	Low	English	High	US auth likely
101	3D pipeline; AI tools	Remote / hybrid	Medium	English	Medium	UK/EU likely
102	3D web; AI; product viz	Hybrid / remote unknown	Medium	English	Medium	Unknown
103	WebAR; creative dev	US / hybrid	Medium	English	High	US likely
104	XR dev; Unity	EU / remote	Medium	English	Medium	EU useful
105	XR dev; 3D tools	Unknown	Unknown	English	Medium	Unknown
106	Unity XR; training	US / remote unknown	Medium	English	Medium	US likely
107	Unity XR; training	US / hybrid	Low	English	Medium	US likely
108	XR dev; simulation	UK / remote unknown	Medium	English	Medium	UK auth likely
109	Unity XR; simulation	UK / hybrid	Medium	English	Medium	UK auth likely
110	Unity XR; simulation	Unknown	Unknown	English	Medium	Unknown
111	Unity dev; virtual worlds	Remote / US-heavy	Medium	English	Medium	US likely
112	Web 3D; Unity?	Remote / US-heavy	Medium	English	Medium	US likely
113	3D web; tools	Remote / global	Medium	English	Medium	Open remote possible
114	Tools; 3D platform	US / hybrid	Low	English	High	US likely
115	3D tools; client dev	US / remote some	Medium	English	Medium	US likely
116	Graphics; WebGL; tools	Global / unknown	Medium	English	Medium	Unknown
117	WebGL/WebGPU; tools	Global / hybrid	Low	English	High	Local auth likely
118	WebGL engine dev	UK / unknown	Medium	English	Medium	UK auth likely
119	3D platform; tools	EU/US / hybrid	Low	English	High	EU/US likely
120	3D visualization; rendering	EU/global / hybrid	Medium	English	Medium	EU useful

13.0.7 Scoring table

#	Fit	Prob.	Comp.	Portfolio	Priority
1	5	3	4	5	A
2	5	3	5	5	A
3	5	2	3	5	A
4	5	3	4	5	A
5	5	2	4	5	B
6	5	2	3	5	B
7	5	2	3	5	B
8	4	2	3	4	B
9	5	2	4	5	B
10	4	2	4	4	B
11	4	2	4	4	B
12	4	3	3	4	B
13	4	2	3	4	B
14	4	2	5	4	B
15	3	1	5	3	Watchlist
16	4	2	5	4	B
17	3	1	5	3	Watchlist
18	4	1	5	4	Watchlist
19	5	4	4	5	A
20	4	3	4	4	B
21	4	3	3	4	B
22	4	3	3	4	B
23	4	3	3	4	B
24	4	3	3	4	B
25	3	3	3	3	B
26	4	2	3	4	B
27	3	2	3	3	C
28	4	2	3	4	B
29	3	2	3	3	C
30	3	2	3	3	C
31	5	3	4	5	B
32	4	3	4	4	B
33	5	3	3	5	B
34	5	3	3	5	B
35	4	2	3	4	B
36	3	2	3	3	C
37	4	2	4	4	B
38	4	2	4	4	B
39	4	2	4	4	B
40	4	2	3	4	B
41	3	1	5	3	Watchlist
42	3	1	5	3	Watchlist
43	3	2	4	3	C
44	3	2	4	3	B
45	4	2	5	4	B
46	4	1	5	4	Watchlist
47	4	2	5	4	Watchlist
48	4	2	4	4	B
49	3	1	4	3	C
50	3	1	4	3	C
51	4	2	5	4	B
52	3	1	4	3	C
53	4	2	4	4	B
54	4	2	4	4	B
55	3	1	4	3	C
56	3	1	4	3	C
57	3	1	4	3	C
58	3	1	4	3	C
59	2	1	4	2	Watchlist
60	2	1	5	2	Watchlist
61	3	3	3	3	B
62	3	3	3	3	B
63	2	3	3	2	C
64	2	3	3	2	C
65	2	3	3	2	C
66	2	3	3	2	C
67	3	1	3	3	Watchlist
68	3	1	4	3	Watchlist
69	3	1	3	3	Watchlist
70	3	2	3	3	C
71	3	2	3	3	C
72	3	2	3	3	C
73	3	2	4	3	B
74	3	2	4	3	B
75	3	3	3	3	B
76	3	3	3	3	B
77	4	4	3	4	A

#	Fit	Prob.	Comp.	Portfolio	Priority
78	4	4	3	4	A
79	3	3	3	3	B
80	3	3	3	3	B
81	3	2	3	3	C
82	2	2	3	2	C
83	3	2	3	3	C
84	3	5	3	3	A
85	2	4	3	2	B
86	2	4	3	2	B
87	2	4	3	2	B
88	2	3	3	2	C
89	2	3	3	2	C
90	3	3	3	3	B
91	2	3	3	2	C
92	2	3	3	2	C
93	2	3	3	2	C
94	3	3	3	3	B
95	3	2	5	3	Watchlist
96	3	1	4	3	Watchlist
97	2	1	5	2	Watchlist
98	2	2	4	2	C
99	2	1	4	2	Watchlist
100	1	1	4	1	Watchlist
101	3	2	3	3	C
102	4	2	3	4	B
103	4	2	4	4	B
104	4	2	3	4	B
105	3	1	3	3	Watchlist
106	4	2	3	4	C
107	3	1	3	3	C
108	3	2	3	3	C
109	4	2	3	4	C
110	4	1	3	4	Watchlist
111	3	2	3	3	C
112	3	2	3	3	C
113	2	2	3	2	C
114	3	1	5	3	Watchlist
115	3	2	3	3	C
116	3	1	3	3	Watchlist
117	3	1	5	3	Watchlist
118	4	2	3	4	C
119	3	1	4	3	Watchlist
120	3	2	3	3	C

13.0.8 First-wave verification queue

This queue is the next useful pass before applications. It does not claim that openings exist today; it defines where to verify first.

A1 - verify first

These targets combine portfolio fit with a more plausible remote, contractor, LATAM, or WebGL/interactive route.

#	Company	Why first	Verification focus
1	Treeview Studio	Direct Unity/XR/digital twin fit with LATAM signal.	Current roles, contractor route, portfolio contact.
19	Active Theory	WebGL creative tech fit; strong TwinSight narrative bridge.	Remote scope, freelance/contract openings, technical art roles.
20	UNIT9	XR/WebGL agency with project-based work pattern.	Contract openings, creative technology recruiter/contact.
21	Resn	WebGL studio; strong interactive 3D fit.	Current hiring, remote/country limits, portfolio submission path.
22	Monogrid	Interactive 3D and WebGL adjacent.	EU contractor possibility and English-only feasibility.
23	14islands	WebGL/creative technology fit.	Contract route, remote restrictions, Three.js vs Unity emphasis.
24	Dogstudio	Creative tech agency; possible WebGL/experience work.	Current roles, contractor/vendor path, region constraints.
25	Media.Monks	Large creative production network.	Unity/WebGL openings, LATAM hubs, contractor route.

#	Company	Why first	Verification focus
31	Threekit	3D product configurator; direct technical visualization fit.	Unity/WebGL requirement, remote scope, customer-facing role fit.
33	Emersya	3D configurators; strong product visualization angle.	EU work authorization limits, contractor possibility.
34	Expivi	3D configurators; strong product visualization angle.	Remote scope, technical role stack, authorization limits.
77	Teravision Games	Colombia/US game development path with Unity relevance.	Unity/technical artist openings, seniority, remote/hybrid.
78	Globant Gaming	LATAM/global volume source.	Unity/XR roles, contractor/payroll structure, English requirements.
84	BairesDev	High-probability LATAM remote software pipeline.	Unity/Python openings, contract model, compensation range.
85	Gorilla Logic	LATAM remote fallback with software route.	Python/tooling roles and any Unity/3D-adjacent openings.
86	Lean Tech	Colombia-friendly nearshore fallback.	Python automation/tooling roles and remote contract path.
87	MAS Global	Colombia-friendly nearshore fallback.	Python/tooling roles, remote scope, salary band.
90	SoftServe	Global consulting with XR/AI adjacency.	XR/Unity units, LATAM eligibility, contractor model.
94	HatchWorks AI	AI nearshore route if ARA becomes portfolio-ready.	AI tooling roles, portfolio fit, LATAM remote scope.
102	Avataar	3D commerce/AI; product visualization overlap.	3D/WebGL openings, remote constraints, contractor possibility.

A2 - high fit, higher friction

These are strategically valuable but likely to require stronger authorization, seniority, relocation, or company-specific timing.

#	Company	Main upside	Main friction to verify
2	Unity Technologies	Brand and engine fit.	Local authorization, seniority, competition.
3	realvirtual.io	Direct Unity simulation/digital twin fit.	Germany/EU authorization and German-market expectations.
4	Capgemini Engineering	Digital twin consulting route.	EU-heavy hiring and consulting seniority.
6	Holo-Light	Industrial XR fit.	Germany/EU authorization.
7	Innoactive	Enterprise XR training fit.	Germany/EU authorization and language.
9	FundamentalVR	Medical simulation fit.	UK/US authorization and seniority.
10	Osso VR	Medical VR training fit.	US authorization and competition.
14	Varjo	High-end XR/simulation signal.	Finland/EU authorization and hardware specialization.
38	Matterport	Digital twin/3D capture brand.	US-heavy hiring and product specialization.
39	DroneDeploy	Drone mapping/digital twin route.	US-heavy hiring and domain requirements.
45	Autodesk	CAD/digital twin platform relevance.	Seniority and high competition.
48	Bentley Systems	Infrastructure digital twin.	Domain specificity and authorization.
51	Siemens Digital Industries	Industrial digital twin route.	Germany/EU authorization and enterprise hiring bar.
53	PTC	IoT/AR/digital twin route.	Enterprise seniority and location limits.
54	Hexagon AB	Industrial digital reality.	Domain specialization and EU/global location constraints.
104	ShapesXR	XR tooling fit.	Smaller team timing and remote scope.

Application disqualifiers

Do not apply until resolved when a posting has any of these flags:

- Requires current US/EU/UK work authorization and has no contractor/B2B path.
- Requires security clearance.
- Requires shipped AAA titles or senior commercial game credits as a hard requirement.

- Is a pure art production role with no technical systems, tooling, scripting, optimization, or interactive component.
- Is local Germany/Portugal payroll-only before the relevant residence/work authorization path is active.
- Has no credible bridge to TwinSight, Unity WebGL, CAD-to-realtime optimization, simulation, XR, technical art, Python automation, or ARA.

13.0.9 Evidence and source notes

#	Evidence / note	Date checked
1	Agent + DR; enterprise Unity, simulation, digital twin	2026-06-11
2	Official careers; direct engine fit	2026-06-11
3	Agent; Unity-based simulation platform	2026-06-11
4	DR; acquired Unity digital twin services arm	2026-06-11
5	Agent; Unity industrial simulation	2026-06-11
6	Agent; industrial AR/XR fit	2026-06-11
7	Enterprise XR; strong TwinSight fit	2026-06-11
8	Industrial VR simulation	2026-06-11
9	Medical XR simulation	2026-06-11
10	VR medical training	2026-06-11
11	Enterprise XR training	2026-06-11
12	XR education and simulation	2026-06-11
13	Enterprise VR	2026-06-11
14	High-end XR hardware	2026-06-11
15	Hardware; high competition	2026-06-11
16	AR platform and games	2026-06-11
17	AR platform; less Unity-centric	2026-06-11
18	High comp; hard entry	2026-06-11
19	Agent; WebGL creative tech	2026-06-11
20	Agent; XR/WebGL agency	2026-06-11
21	Agent; WebGL studio	2026-06-11
22	Agent; interactive 3D	2026-06-11
23	Creative WebGL	2026-06-11
24	Interactive websites	2026-06-11
25	Large creative tech agency	2026-06-11
26	German creative tech	2026-06-11
27	German interactive studio	2026-06-11
28	Immersive installations	2026-06-11
29	Experiential tech	2026-06-11
30	Creative studio; not always 3D dev	2026-06-11
31	Product configurators	2026-06-11
32	3D commerce pipeline	2026-06-11
33	3D product visualization	2026-06-11
34	3D configurator platform	2026-06-11
35	XR + digital twin	2026-06-11
36	Web 3D streaming; Unreal-heavy	2026-06-11
37	Web 3D/geospatial; not Unity-centric	2026-06-11
38	Built-environment digital twins	2026-06-11
39	Drone + 3D mapping fit	2026-06-11
40	Drone/3D reconstruction fit	2026-06-11
41	Defense/security possible	2026-06-11
42	Drones; not Unity explicit	2026-06-11
43	Drone software; robotics	2026-06-11
44	3D GIS; web visualization	2026-06-11
45	CAD-to-realtime adjacent	2026-06-11
46	High comp; very competitive	2026-06-11
47	Unreal-heavy; real-time relevance	2026-06-11
48	Infrastructure digital twins	2026-06-11
49	Physics simulation; less Unity	2026-06-11
50	Simulation/math; less 3D portfolio	2026-06-11
51	Agent; large industrial digital twin	2026-06-11
52	Agent; 3DEXPERIENCE	2026-06-11
53	Agent; Vuforia/ThingWorx	2026-06-11
54	Agent; digital reality	2026-06-11
55	Industrial digital twin	2026-06-11
56	Industrial automation	2026-06-11
57	Robotics/digital twin	2026-06-11
58	German industrial path	2026-06-11
59	Security-sensitive roles possible	2026-06-11
60	Aerospace; auth constraints	2026-06-11
61	Portugal target	2026-06-11
62	Portugal target	2026-06-11
63	Fallback software route	2026-06-11
64	AI/full-stack fallback	2026-06-11
65	Fallback tech route	2026-06-11
66	Portugal fallback	2026-06-11

#	Evidence / note	Date checked
67	Game studio; likely Unreal	2026-06-11
68	AAA; hard entry	2026-06-11
69	Game studio	2026-06-11
70	Mobile/game path	2026-06-11
71	Unity mobile games	2026-06-11
72	German games	2026-06-11
73	Remote/hybrid roles; Unity mobile	2026-06-11
74	Candy Crush; mobile TA	2026-06-11
75	Mobile game roles	2026-06-11
76	LATAM mobile games	2026-06-11
77	Colombia game studio; realistic	2026-06-11
78	Large LATAM tech; game/XR units	2026-06-11
79	Game outsourcing	2026-06-11
80	Game services; technical roles	2026-06-11
81	Game co-dev	2026-06-11
82	More art-focused	2026-06-11
83	AAA services; competitive	2026-06-11
84	LATAM remote contractor path	2026-06-11
85	Fallback software route	2026-06-11
86	Colombia remote path	2026-06-11
87	Colombia-friendly fallback	2026-06-11
88	Fallback software	2026-06-11
89	Fallback software	2026-06-11
90	XR and AI units	2026-06-11
91	Fallback AI/software	2026-06-11
92	Fallback software	2026-06-11
93	Fallback software	2026-06-11
94	AI tooling secondary	2026-06-11
95	ARA-adjacent; competitive	2026-06-11
96	AI tooling; not Unity	2026-06-11
97	Competitive AI tools	2026-06-11
98	AI tooling secondary	2026-06-11
99	AI/CV secondary	2026-06-11
100	Secondary only	2026-06-11
101	3D + AI bridge	2026-06-11
102	3D product visualization	2026-06-11
103	WebAR; TwinSight web angle	2026-06-11
104	XR tools; EU	2026-06-11
105	Find via XR job boards	2026-06-11
106	Enterprise VR training	2026-06-11
107	VR training	2026-06-11
108	Healthcare XR training	2026-06-11
109	Medical training simulation	2026-06-11
110	Use medical XR boards	2026-06-11
111	Unity virtual worlds	2026-06-11
112	Web/3D virtual spaces	2026-06-11
113	Web3 not core	2026-06-11
114	High comp; different stack	2026-06-11
115	3D platform	2026-06-11
116	Use Khronos jobs and members	2026-06-11
117	Web3D; high bar	2026-06-11
118	WebGL engine; small team	2026-06-11
119	3D asset platform; Epic-owned	2026-06-11
120	3D commerce pipeline	2026-06-11

13.0.10 Job boards

Name	URL	Category	Search terms	Remote	Contract	Region	Signal	Noise	Frequency	Notes
LinkedIn Jobs	https://www.linkedin.com/jobs/	General tech	Unity Technical Artist; Digital Twin Unity	Yes	No	Yes	High	Medium	2-3/week	Best universal source; remote scope.
Indeed	https://www.indeed.com/	General tech	Remote Unity Developer; Technical Artist	Yes	No	Yes	Medium	High	Weekly	High volume; duplicate.
Glassdoor	https://www.glassdoor.com/Job/	General tech	Unity Developer remote	Yes	No	Yes	Medium	High	Weekly	Useful salary + company reviews.
Wellfound	https://wellfound.com/jobs	Startup remote	Unity; WebGL; AI tools	Yes	Startup jobs	Yes	Medium	Medium	Weekly	Good for startups; remote scope varies.
RemoteOK	https://remoteok.com/	Remote tech	WebGL; Python; Unity	Yes	No	Limited	Medium	High	Daily	Check worldwide vs. region-only.
Remotive	https://remotive.com/remote-jobs	Remote tech	Unity; XR; Python	Yes	No	Limited	Medium	Medium	Weekly	Lower volume, clear postings.
We Work Remotely	https://weworkremotely.com/	Remote tech	Unity; 3D; Python	Yes	No	Limited	Medium	Medium	Weekly	Good global source.
Arc.dev	https://arc.dev/remote-jobs	Remote contractor	Unity; Python; WebGL	Yes	Yes	Yes	Medium	Medium	Weekly	Contractor-friendly.
Toptal	https://www.toptal.com/freelance-jobs	Freelance	Unity; WebGL; Python	Yes	Yes	Yes	Medium	Medium	Weekly	Screened network; high rate.
Upwork	https://www.upwork.com/freelance-jobs/unity-3d/	Freelance	Unity WebGL; 3D viewer	Yes	Yes	Yes	Medium	High	Daily	Useful for first contract; rate pressure.
Contra	https://contra.com/	Freelance	3D developer; Unity	Yes	Yes	Yes	Low	Medium	Weekly	Creative freelancer network.
Get on Board	https://www.getonboard.com/jobs-remote	LATAM remote	Unity; Python; WebGL	Yes	Yes	LATAM	High	Medium	2-3/week	Strong LATAM signal.
Torre	https://torre.ai/roles	LATAM remote	Unity; Python; full-stack	Yes	Yes	LATAM	Medium	High	Weekly	Useful but noisy.
BairesDev Careers	https://www.bairesdev.com/careers/	LATAM remote	Unity Developer; Python	Yes	Contract	LATAM	Medium	Medium	Weekly	Frequent LATAM openings.
Remote Game Jobs	https://remotegamejobs.com/	Game industry	Technical Artist; Unity	Yes	Mixed	Global	High	Low	Weekly	Best dedicated remote game jobs.
Work With Indies	https://www.workwithindies.com/	Indie games	Unity Technical Artist	Yes	Mixed	Global	High	Medium	Weekly	Good indie/contractor focus.
Hitmarker	https://hitmarker.net/	Games / esports	Technical Artist; Unity	Yes	Mixed	Yes	Medium	Medium	Weekly	Game roles; filter noise.
Games Jobs Direct	https://www.gamesjobsdirect.com/	Game industry	Technical Artist; Tools	Yes	No	Yes	Medium	Medium	Weekly	EU-heavy game roles.
CreativeHeads	https://www.creativeheads.net/	VFX / games	Technical Artist; Pipeline	Yes	Mixed	Yes	Medium	Medium	Weekly	Art/tech art jobs.
ArtStation Jobs	https://www.artstation.com/jobs	3D / game art	Technical Artist; 3D Artist	Limited	No	Yes	Medium	High	Weekly	More art than dev.
Polycount Jobs	https://polycount.com/categories/c44/jobs	3D / game art	Technical Artist	No	No	Limited	Low	Medium	Weekly	Community job board.
80 Level Jobs	https://80.lv/jobs/	Games / 3D	Technical Artist; Unity	Yes	Mixed	Yes	Medium	Medium	Weekly	Good industry signal.
ARVR Jobs	https://www.arvrjobs.dev/	XR	Unity XR; AR Developer	Yes	No	Yes	High	Low	Weekly	Focused XR board.
XR Bootcamp Jobs	https://xrbootcamp.com/	XR community	Unity XR developer	Community	Unknown	Unknown	Medium	Low	Monthly	Useful community/school.
Digital Twin Consortium	https://www.digitaltwinconsortium.org/	Digital twin	Digital Twin Engineer	No	No	Global	Low	Low	Monthly	More networking than jobs.
Three.js Jobs	https://discourse.threejs.org/c/jobs/10	WebGL / Web3D	Three.js; WebGL dev	No	Mixed	Global	Medium	Medium	Weekly	Good for contract work.
Khronos Jobs	https://www.khronos.org/jobs/	Graphics standards	WebGL; OpenXR	No	No	Global	Low	Low	Monthly	Specialized graphics.
EU Startup Jobs	https://www.eu-startups.com/jobs/	EU startups	Unity; XR; Python	Yes	Mixed	EU	Medium	Medium	Weekly	EU work rights matters.
StepStone Germany	https://www.stepstone.de/	Germany	Unity Entwickler; Technical Artist	Yes	No	Germany	Medium	Medium	Weekly	German market; German useful.
ITJobs.pt	https://www.itjobs.pt/	Portugal tech	Unity; Python; XR	Yes	Mixed	Portugal	Medium	Medium	Weekly	Future of Portugal routing.
Net-Empregos	https://www.net-empregos.com/	Portugal general	Unity; 3D; Python	Yes	Mixed	Portugal	Medium	High	Weekly	Portuguese language helpful.

13.0.11 Recruiters, agencies, and contractor platforms

Name	URL	Region	Specialization	Relevant roles	Contractor	Remote	Notes
Aardvark Swift	https://aswift.com/	UK/EU	Games recruiting	Technical artist; tools programmer	Medium	Low	EU/UK games; relocation roles.
Datascope Recruitment	https://datascope.co.uk/	UK/EU	Games / interactive	Technical artist; Unity programmer	Medium	Medium	Games + interactive roles.
Skillsearch	https://www.skillsearch.com/	UK/EU	Games / tech	Unity; technical artist; tools	Medium	Medium	Contracts and permanent roles.
Interactive Selection	https://interactive.com/	UK/EU	Games recruiting	Technical artist; programmer	Medium	Medium	Games and emerging tech.
Games Factory Talent	https://gamesfactorytalent.com/	Global	Game jobs	Technical artist; Unity	Medium	Medium	Game industry recruiting.
CreativeHeads	https://www.creativeheads.net/	US/global	VFX / games	Technical artist; pipeline	Medium	Medium	Job board + talent.
Toptal	https://www.toptal.com/	Global	Freelance network	Unity; WebGL; Python	High	High	High bar; contractor path.
Lemon.io	https://lemon.io/	Global	Freelance devs	Unity; Python	High	High	Good contractor signal.
Arc.dev	https://arc.dev/	Global	Remote devs	Unity; Python; full-stack	High	High	Remote developer marketplace.
Gun.io	https://www.gun.io/	US/global	Freelance devs	Software; Python; Unity possible	High	High	Contractor path.
Braintrust	https://www.usebraintrust.com/	Global	Talent network	Software; AI; creative tech	High	High	Remote contractor network.
Torre	https://torre.ai/	LATAM/global	Remote talent	Python; Unity; full-stack	High	High	LATAM remote path.
GeekHunter	https://geekhunter.com/	Brazil/LATAM	Tech recruiting	Software; Unity possible	Medium	High	LATAM tech recruiting.
BairesDev	https://www.bairesdev.com/	LATAM/US	Nearshore staffing	Unity; Python; software	High	High	Frequent LATAM remote roles.
Deel Talent	https://www.deel.com/	Global	EOR / hiring	Global tech roles	High	Medium	Compliance layer, not recruiter only.
Remote.com Talent	https://remote.com/	Global	EOR / hiring	Global tech roles	Medium	Medium	EOR + hiring support.
Oyster	https://www.oysterhr.com/	Global	EOR	Global tech roles	Medium	Medium	EOR; useful for employer concerns.
Hays Germany	https://www.hays.de/	Germany	Tech recruiting	Software; simulation	Low	Medium	German market; local language useful.
Michael Page Portugal	https://www.michaelpage.pt/	Portugal	Tech recruiting	Software; engineering	Low	Medium	Portugal route after passport.
Landing.Jobs	https://landing.jobs/	Portugal/EU	Tech jobs	Software; Python; Unity possible	Medium	Medium	Portugal/EU roles.

Biblia completa

CHAPTER 13. 11_COMPANY_TARGETS_JOB_BOARDS_RECRUITERS.MD

13.0.12 Communities

Name	URL	Platform	Category	Why useful	Jobs	Feedback	Networking
Unity Discussions	https://discussions.unity.com/	Forum	Unity	Official Q&A; jobs occasionally	Yes	Yes	Medium
Unity Discord	https://discord.gg/unity	Discord	Unity	Technical help and networking	Yes	Yes	Medium
Tech-Artists.Org	https://tech-artists.org/	Forum/Slack	Technical Art	Core TA community	Yes	Yes	High
Polycount	https://polycount.com/	Forum	3D / Tech Art	Portfolio feedback	Yes	Yes	Medium
RealtimeVFX	https://realtimevfx.com/	Forum	Realtime VFX	VFX/Unity/Houdini feedback	Yes	Yes	High
The Rookies	https://www.therookies.co/	Portfolio/community	3D / games	Portfolio exposure	Yes	Yes	Medium
ArtStation	https://www.artstation.com/	Portfolio	3D / Tech Art	Portfolio + jobs	Yes	Limited	Medium
80 Level	https://80.lv/	News/jobs	Games / 3D	Industry jobs and articles	Yes	No	Medium
Three.js Discourse	https://discourse.threejs.org/	Forum	WebGL	Web3D jobs and help	Yes	Yes	Medium
WebGL Dev List / Khronos	https://www.khronos.org/webgl/	Standards/community	WebGL	Standards, ecosystem	Limited	No	Low
OpenXR / Khronos	https://www.khronos.org/openxr/	Standards	XR	XR ecosystem	Limited	No	Low
ARVR Jobs	https://www.arvrjobs.dev/	Job board/community	XR	XR jobs by skill	Yes	No	High
XR Bootcamp	https://xrbootcamp.com/	Training/community	XR	Networking and jobs	Yes	Yes	Medium
Digital Twin Consortium	https://www.digitaltwinconsortium.org/	Association	Digital Twin	Events and ecosystem	Limited	No	Medium
Industrial Metaverse Community	https://industrialmetaverse.org/	Community	Industrial 3D	Industrial visualization	Limited	No	Low
Houdini Discord	https://www.sidefx.com/community/	Community	Houdini/VFX	Procedural skills network	Yes	Yes	Medium
SideFX Forum	https://www.sidefx.com/forum/	Forum	Houdini	Technical VFX help	Limited	Yes	Medium
LangChain Community	https://www.langchain.com/community	Community	AI tools	ARA-adjacent networking	Yes	No	Medium
Hugging Face Discord	https://huggingface.co/join/discord	Discord	AI tooling	AI/LLM community	Yes	No	Medium
MLOps Community	https://mlops.community/	Slack/events	AI tooling	Tools and MLOps networking	Yes	No	Medium
Colombia.dev	https://colombia.dev/	Community	LATAM tech	Local tech network	Yes	No	Medium
Medellin Tech / Ruta N	https://www.rutanmedellin.org/	Local ecosystem	Colombia tech	Local startups and events	Limited	No	Medium
German Game Dev Network	https://www.game.de/en/	Association	Germany games	German games network	Yes	No	Medium
Games Germany	https://www.gamesgermany.de/	Association	Germany games	Studios and jobs	Yes	No	Medium
Portugal Game Dev	https://www.apvideogames.pt/	Association	Portugal games	Portuguese games ecosystem	Limited	No	Medium
Landing.Jobs Community	https://landing.jobs/	Jobs/community	Portugal/EU tech	Portugal/EU network	Yes	No	Medium

13.0.13 Search strings to reuse

Use these exactly in LinkedIn, Google, job boards, and company career pages:

```
Bloque de codigo: text
Unity Technical Artist remote
Unity Developer remote contractor LATAM
Unity WebGL Developer remote
WebGL 3D Developer remote
Interactive 3D Developer remote
Real-Time 3D Developer remote
Technical Visualization Developer Unity
Digital Twin Unity Developer
Simulation Developer Unity
XR Developer Unity remote
AR VR Developer Unity contractor
CAD to real-time visualization developer
3D product configurator developer WebGL
Unity Tools Developer remote
Technical Artist jobs Europe remote
Technical Artist Germany Unity
Unity Developer Portugal jobs
Digital Twin Engineer Germany
Simulation Engineer Germany 3D visualization
Python automation developer remote LATAM
LLM tools developer remote
AI tooling engineer remote
remote game jobs technical artist
industrial visualization Unity
enterprise 3D visualization company
"Unity WebGL" contractor remote
"technical visualization" Unity remote
"digital twin" Unity developer remote
"3D configurator" WebGL developer remote
"interactive 3D" developer contractor
"Unity Technical Artist" LATAM remote
"Unity Developer" "Colombia" remote
"Unity WebGL" "LinkedIn" "Remote"
site:linkedin.com/jobs "Unity WebGL" "Remote"
site:linkedin.com/jobs "Technical Artist" "Unity" "Remote"
site:linkedin.com/jobs "Digital Twin" "Unity"
Unity Entwickler Simulation
Technical Artist Unity Deutschland
Digital Twin Unity Entwickler
Unity Developer Portugal remote
desenvolvedor Unity remoto
desenvolvedor 3D WebGL remoto
```

13.0.14 Targeting logic by category

Highest-fit categories

1. Unity / Real-Time 3D agencies.
2. Technical visualization and digital twin firms.
3. XR / simulation training companies.
4. WebGL / Web 3D creative technology studios.
5. Product configurator / 3D commerce companies.
6. LATAM nearshore companies when they have Unity or Python roles.

Lower-priority categories

1. AAA game studios requiring shipped commercial titles.
2. Defense/aerospace companies requiring clearance.
3. Pure 3D artist roles.
4. Pure full-stack roles not connected to 3D, tooling, or visualization.
5. AI tooling startups unless ARA is polished enough to show.

13.0.15 Work authorization flags

Scenario	Recruiter-facing implication
Current Colombia-based status	Best framed as remote contractor/B2B from Colombia.
Future Portuguese passport	Expected around 2028; future EU work authorization, not current authorization.
Germany via marriage/residence	Possible future German work authorization after residence process.
US roles	Assume US authorization needed unless explicitly contractor/worldwide.
EU-only remote roles	Not viable now unless contractor accepted; viable later with Portuguese passport.
Germany roles	German helps; residence/work rights matter more than language alone.

13.0.16 Recommended update process

This list should be maintained as a living spreadsheet.

Suggested columns:

```

Bloque de código: text
Company
Category
Country
Website
Careers URL
LinkedIn URL
Current opening found?
Opening URL
Role title
Remote scope
Contractor accepted?
Work authorization
Language
Salary signal
Fit score
Probability score
Compensation score
Portfolio-fit score
Priority
Last checked
Next action

```

Minimum fields required before applying:

- Opening URL or named recruiter/contact.
- Exact role title.
- Remote scope: worldwide, LATAM, Colombia, US-only, EU-only, hybrid, or unclear.
- Contractor/B2B accepted: yes, no, or unclear.
- Work authorization requirement.
- Seniority requirement and hard disqualifiers.
- Portfolio angle: TwinSight, Unity WebGL, CAD-to-realtime, simulation/XR, technical art, Python automation, or ARA.
- Next action: apply, contact recruiter, save for later, or discard.

Update frequency:

- A targets: weekly.
- B targets: every two weeks.
- C targets: monthly.
- Watchlist: only when portfolio or work authorization changes.

13.0.17 Gaps requiring future agent pass

A later agent pass would still be useful for:

1. Verifying LinkedIn URLs for all 120 companies.
2. Checking live openings for A/B priority companies.
3. Tagging remote scope precisely.
4. Extracting recruiter names from LinkedIn.

5. Creating a CSV-ready tracker.

Do not spend agent runs on Watchlist targets before A and B targets are cleaned.

13.0.18 Future agent prompt for top-30 verification

Use this prompt when live verification is ready:

```
Bloque de código: text
Verify the first 30 practical targets from 11_company_targets_job_boards_recruiters.md, prioritizing A1 first and then A2.

For each company, check the official careers page and LinkedIn Jobs. Do not treat "remote" as worldwide unless the posting
↪ says so. Do not assume EU work authorization before the Portuguese passport is active around 2028.

Return a CSV-ready table with these columns:
Company, Company #, Careers URL, LinkedIn URL, Current opening found, Opening URL, Role title, Remote scope, Contractor/B2B
↪ accepted, Work authorization requirement, Seniority, Language, Salary signal, Portfolio angle, Priority layer, Last
↪ checked, Recommendation, Notes.

Use these recommendations only:
Apply now, Contact recruiter, Track weekly, Track monthly, Discard for now.
```

Chapter 14

12_application_system_and_tracker.md

Fuente: 12_application_system_and_tracker.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1327

Ultima modificacion: 2026-06-11 14:23

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This file defines the operating system for Alexander Woodcock Salomón's international job search.

Contenido completo depurado

14.1 Application System and Tracker

14.1.1 0. Purpose

This file defines the operating system for Alexander Woodcock Salomón's international job search.

The goal is not to "apply more." The goal is to run a controlled pipeline that converts the existing portfolio assets, especially TwinSight X500, into interviews, offers, contractor opportunities, and market feedback.

This system assumes the previous strategic decisions are already fixed:

- Primary target: Real-Time 3D / Interactive Technical Visualization / Unity WebGL.
- Secondary target: Unity Technical Artist focused on optimization, runtime systems, tools, shaders, and CAD-to-realtime workflows.
- Expansion target: Digital Twin / Simulation / XR / Industrial Visualization.
- Controlled secondary route: Tools / Python Automation / AI-assisted production tooling.
- Fallback routes: Full-stack and 3D generalist only when strategically useful.
- Current priority: remote from Colombia.
- Future mobility: Portuguese passport expected in approximately two years; Germany route possible through legal marriage and residence with German partner.

This section is operational. It does not redefine the positioning. It translates the strategy into execution.

14.1.2 1. Core operating principle

The job search must be treated as a pipeline, not as scattered applications.

Every target should move through a controlled sequence:

Bloque de código: text
 Target identified → Fit scored → Evidence selected → Application customized → Outreach sent → Follow-up → Interview → Test →
 ↩ Offer → Decision → Retrospective

The system must prevent four failure modes:

1. Applying to roles without enough fit.
2. Spending too much time customizing low-probability applications.
3. Over-positioning into roles where the evidence is weak.
4. Losing track of companies, conversations, follow-ups, and feedback.

The tracker is the source of operational truth.

14.1.3 2. Application categories

All opportunities must be classified before applying.

14.1.4 2.1 Category A — High-fit targets

These are the highest-priority opportunities.

They match TwinSight directly and do not require major repositioning.

Examples:

- Interactive 3D Developer.
- Real-Time 3D Developer.
- Unity WebGL Developer.
- Technical Visualization Developer.
- Digital Twin Visualization Developer.
- Simulation Visualization Developer.
- Unity Developer for enterprise 3D.
- XR Developer for industrial or training applications.

A Category A opportunity should satisfy most of these conditions:

- Unity, WebGL, real-time 3D, simulation, digital twin, XR, or technical visualization is central to the role.
- A technical portfolio case study is relevant.
- TwinSight can be used as the primary proof.
- Remote, contractor, LATAM, or global work is plausible.
- The role does not require senior commercial experience as a hard gate.
- The company works in industrial, education, simulation, XR, training, product visualization, or enterprise 3D.

Application effort: high.

Customization level: strong.

Expected weekly volume: 5–8.

14.1.5 2.2 Category B — Adjacent targets

These are plausible but require adaptation.

Examples:

- Unity Technical Artist in games.
- Game Technical Artist.
- Shader Technical Artist.

- Tools Developer.
- Pipeline Developer.
- 3D product configurator developer.
- WebGL / Three.js Developer.
- Python Automation Developer.
- AI Tools Developer.

A Category B opportunity is valid when:

- The role is related to the portfolio but not a direct TwinSight match.
- ARA, Blender portrait, or shader work may support the application.
- Some missing skills exist, but the gap is not fatal.
- The application can be made credible without exaggeration.

Application effort: medium.

Customization level: moderate.

Expected weekly volume: 8–12.

14.1.6 2.3 Category C — Volume targets

These are lower-fit but useful for market testing, interview practice, or income fallback.

Examples:

- Junior Unity Developer.
- C# Developer with light Unity.
- Full-stack developer.
- Python automation support.
- Freelance web projects.
- 3D generalist roles.
- Simple contractor gigs.

A Category C opportunity is acceptable when:

- It can produce income quickly.
- It does not consume excessive customization time.
- It does not force a false professional identity.
- It can be applied to with a lighter version of the CV.

Application effort: low.

Customization level: minimal.

Expected weekly volume: 10–20.

14.1.7 2.4 Watchlist targets

These are not immediate targets but should be monitored.

Examples:

- Senior Technical Artist AAA.
- Senior Simulation Engineer.
- Digital Twin Architect.
- Research Engineer.
- Graphics Programmer.
- AI Engineer.
- EU-only roles requiring current EU work authorization.

- Germany-only roles requiring German language.

Watchlist targets should not consume active application time unless a specific opening unusually matches the profile.

14.1.8 3. Weekly application mix

A balanced weekly target should avoid both over-customization and indiscriminate spam.

Recommended baseline after the thesis defense:

Category	Weekly target
A	5–8
B	8–12
C	10–20
Speculative outreach	10–15
Recruiter contacts	5–10

Total weekly outbound actions:

Bloque de codigo: text
 Minimum controlled pace: 30 actions/week
 Strong pace: 45–55 actions/week
 Aggressive pace: 60–75 actions/week

An outbound action means one of the following:

- Submitted application.
- Recruiter message.
- Founder / technical lead outreach.
- Referral request.
- Follow-up.
- Portfolio feedback request.
- Community post with job-search relevance.

Do not count passive browsing as execution.

14.1.9 4. Daily execution blocks

The job search should be split into repeatable blocks.

14.1.10 4.1 Minimum daily block

Use this when time or energy is limited.

Bloque de codigo: text
 Duration: 90 minutes

1. Review tracker: 10 min
2. Find opportunities: 20 min
3. Apply or message: 40 min
4. Update tracker: 10 min
5. Next-action planning: 10 min

Expected output:

- 2–4 applications or messages.
- Tracker updated.
- No loose tabs left open.

14.1.11 4.2 Standard daily block

Use this as the default after defense.

```
Bloque de codigo: text
Duration: 3 hours

1. Pipeline review: 15 min
2. Target search: 30 min
3. Custom applications: 75 min
4. Outreach/follow-up: 30 min
5. Tracker update: 15 min
6. Portfolio/CV adjustment: 15 min
```

Expected output:

- 4–8 outbound actions.
- 1–2 high-quality customized applications.
- Follow-ups sent on schedule.

14.1.12 4.3 Deep work block

Use this 2–3 times per week for strategic assets.

```
Bloque de codigo: text
Duration: 2–4 hours

Allowed work:
- TwinSight case study refinement.
- Demo video update.
- GitHub README polish.
- Portfolio section improvement.
- Technical test preparation.
- Interview rehearsal.
```

This block is not for browsing job boards.

14.1.13 5. Weekly schedule

14.1.14 5.1 Recommended weekly structure

```
Bloque de codigo: text
Monday:
- Review metrics.
- Refresh target list.
- Apply to new Category A roles.

Tuesday:
- Category B applications.
- Recruiter outreach.
- Portfolio refinement.

Wednesday:
- Category A applications.
- Speculative outreach to companies.
- Follow-ups.

Thursday:
- Category B/C applications.
- Community networking.
- Interview preparation.

Friday:
- Follow-ups.
- Tracker cleanup.
- Weekly retrospective.

Saturday:
- Optional portfolio or technical skill sprint.

Sunday:
- No active applications unless urgent.
- Light planning only.
```

14.1.15 5.2 Weekly minimum deliverables

At the end of each week, the following must exist:

- All new applications logged.
- All follow-up dates assigned.
- All responses categorized.
- All rejections tagged with reason if known.
- At least one portfolio or profile asset improved.
- Weekly metrics written down.

If the tracker is not updated, the week is not operationally complete.

14.1.16 6. Tracker architecture

The tracker should be built in Google Sheets, Notion, Airtable, or a local spreadsheet.

Preferred structure:

```
Bloque de codigo: text
1. Dashboard
2. Applications
3. Companies
4. Contacts
5. Follow-ups
6. Interviews
7. Technical tests
8. Offers
9. Metrics
10. Lessons learned
```

A spreadsheet is sufficient. Notion is optional.

14.1.17 7. Applications table

14.1.18 7.1 Required columns

Use these fields for every application.

Field	Values
ID	APP-0001
Date applied	YYYY-MM-DD
Company	Name
Role title	Exact title
Category	A / B / C / Watchlist
Source	LinkedIn / Job board / Referral
Location	Country / Remote
Remote scope	WW / LATAM / US / EU / Hybrid
Contract type	Employee / Contractor / Unknown
Salary range	Stated / Estimated
Status	Pipeline status
Next action	Specific action
Follow-up date	YYYY-MM-DD

14.1.19 7.2 Fit columns

Use these scoring fields to prevent bad applications.

Field	Scale
Role fit	1–5
Portfolio fit	1–5
Remote feasibility	1–5
Compensation potential	1–5
Probability	1–5
Priority	A / B / C

Scoring rule:

Bloque de código: text

Total score = Role fit + Portfolio fit + Remote feasibility + Compensation potential + Probability

Interpretation:

Score	Meaning
21–25	Strong target
17–20	Apply selectively
13–16	Low effort only
<13	Do not apply

14.1.20 7.3 Evidence columns

Each application should record which proof was used.

Field	Examples
Primary proof	TwinSight
Secondary proof	ARA / Blender portrait
Portfolio link used	URL
GitHub link used	URL
Demo link used	URL
CV variant	Unity / WebGL / Digital Twin
Cover note	Yes / No

This prevents using the same generic proof for every role.

14.1.21 8. Company table

The company table prevents duplicate research.

Required fields:

Field	Values
Company	Name
Website	URL
Careers page	URL
LinkedIn	URL
Category	Digital Twin / XR / WebGL
Region	US / EU / LATAM
Language	English / German / Portuguese
Remote scope	WW / LATAM / EU / Unknown

Field	Values
Contractor likelihood	High / Medium / Low
Priority	A / B / C / Watchlist
Last checked	YYYY-MM-DD

Additional notes:

- Keep company notes short.
- Do not research the same company repeatedly.
- If no suitable opening exists, mark it as Watchlist.
- Review Watchlist companies every 30 days.

14.1.22 9. Contacts table

Any person contacted must be logged.

Required fields:

Field	Values
Name	Contact name
Role	Recruiter / Lead / Founder
Company	Company
LinkedIn	URL
Email	If public
Relationship	Cold / Warm / Referral
Last contact	YYYY-MM-DD
Next follow-up	YYYY-MM-DD
Status	No reply / Replied / Call

Rules:

- Do not message the same person repeatedly without a reason.
- Follow up once after 5–7 business days.
- A second follow-up is only justified for high-fit targets.
- No third follow-up unless they previously replied.

14.1.23 10. Status pipeline

Use a controlled status list.

```
Bloque de código: text
Identified
Scored
Applied
Outreach sent
Follow-up due
Follow-up sent
Recruiter screen
Technical interview
Portfolio review
Technical test
Final interview
Offer
Rejected
Ghosted
Withdrawn
Archived
```

Do not create custom statuses for every case. Too many statuses make the tracker unusable.

14.1.24 11. Follow-up rules

14.1.25 11.1 Application follow-up

For ordinary applications:

```
Bloque de codigo: text
Day 0: Apply
Day 7: Follow up if contact exists
Day 14: Final follow-up if high priority
Day 21: Archive if no response
```

For high-fit Category A applications:

```
Bloque de codigo: text
Day 0: Apply + targeted outreach
Day 5-7: Follow up
Day 14: Second follow-up or new contact
Day 30: Archive or move to Watchlist
```

14.1.26 11.2 Recruiter follow-up

```
Bloque de codigo: text
Day 0: Contact recruiter
Day 5-7: Follow up
Day 14: Archive if no answer
```

If a recruiter replies but has no current role, set reminder for 30–45 days.

14.1.27 11.3 Interview follow-up

```
Bloque de codigo: text
Same day: thank-you note
Day 5: follow up if no timeline was given
Day 10: follow up if timeline expired
```

Do not over-message after interviews.

14.1.28 12. Customization rules

Not every application deserves the same effort.

14.1.29 12.1 Category A customization

Category A applications require:

- CV variant selected intentionally.
- Short customized opening message.
- TwinSight case study link.
- Relevant bullet reordered near the top.
- Company-specific reason.
- Follow-up contact identified.

Expected time per application:

```
Bloque de codigo: text
30-60 minutes
```


14.1.30 12.2 Category B customization

Category B applications require:

- Correct CV variant.
- One paragraph tailored to the role.
- Relevant project link.
- Basic tracker entry.

Expected time per application:

Bloque de codigo: text
15-30 minutes

14.1.31 12.3 Category C customization

Category C applications require:

- Correct CV variant.
- No long cover letter.
- No excessive research.
- Quick tracker entry.

Expected time per application:

Bloque de codigo: text
5-15 minutes

Category C should not steal time from portfolio and Category A roles.

14.1.32 13. CV variant selection

Use the correct CV variant for each role.

Role type	CV variant
Interactive 3D	Real-Time 3D
Unity WebGL	WebGL / Interactive 3D
Digital Twin	Technical Visualization
Simulation	Technical Visualization
XR	XR / Unity
Technical Artist	Unity Technical Artist
Tools	Tools / Python
AI automation	Python / AI Tools
Full-stack	Full-stack fallback
3D generalist	3D / Technical Art fallback

Do not send the AI Tools CV to Unity roles unless the job explicitly values automation.

Do not send a 3D Artist CV to technical visualization roles.

14.1.33 14. Portfolio link selection

Each application should include only the most relevant links.

14.1.34 14.1 Default link order

For most roles:

```
Bloque de codigo: text
1. Portfolio homepage
2. TwinSight case study
3. GitHub profile
4. LinkedIn
```

For technical roles:

```
Bloque de codigo: text
1. TwinSight case study
2. GitHub repository
3. Demo video
4. Portfolio homepage
```

For tools / automation roles:

```
Bloque de codigo: text
1. ARA case study
2. GitHub repository
3. TwinSight as secondary proof
4. LinkedIn
```

For art-adjacent roles:

```
Bloque de codigo: text
1. TwinSight case study
2. Blender portrait technical breakdown
3. ArtStation
4. Portfolio homepage
```

14.1.35 15. Application quality gates

Before applying, check these gates.

14.1.36 15.1 Mandatory gate

Do not apply if all are true:

- Requires current US/EU work authorization.
- Does not allow remote or contractor work.
- Requires 5+ years formal industry experience.
- Has no connection to Unity, 3D, simulation, XR, WebGL, tools, or Python.
- Would require a false claim.

14.1.37 15.2 Apply anyway gate

Apply despite imperfect fit if at least three are true:

- TwinSight is directly relevant.
 - Remote scope is worldwide, LATAM, or contractor-friendly.
 - Requirements are flexible.
 - Role is junior, mid-junior, associate, contractor, or project-based.
 - Company works in simulation, digital twin, XR, education, product visualization, or industrial 3D.
 - The application can be customized in less than 45 minutes.
-

14.1.38 16. Outreach system

This section defines cadence only. Full outreach scripts belong in a later file.

14.1.39 16.1 Who to contact

Priority contacts:

- 1. Technical Artist Lead.
- 2. Unity Lead.
- 3. 3D / XR / Simulation Lead.
- 4. Founder of small studio.
- 5. Technical recruiter.
- 6. Creative technology director.
- 7. Engineering manager.
- 8. Former UNAD / UNAL / project contacts.

Avoid contacting random HR profiles without relevance when better technical contacts exist.

14.1.40 16.2 Outreach pairing

Use this pattern:

Bloque de codigo: text
Application + targeted message

Do not rely only on job portals.
For Category A roles, always attempt at least one direct contact.
For Category B roles, contact if the company is small or the role is highly relevant.
For Category C roles, direct outreach is optional.

14.1.41 17. Metrics dashboard

Track weekly metrics.

14.1.42 17.1 Activity metrics

Metric	Target
Applications sent	20–40
Category A applications	5–8
Direct outreach messages	10–15
Follow-ups sent	5–15
Recruiters contacted	5–10
Portfolio improvements	1–2

14.1.43 17.2 Conversion metrics

Metric	Good signal
Response rate	>8%
Recruiter screen rate	>3%
Interview rate	>2%
Technical test rate	>1%
Offer rate	>0.3%

These numbers are directional. They are not guarantees.

If response rate is below 3% after 100 applications, the issue is likely one of these:

- Positioning is too broad.
- CV is weak.
- Portfolio is not accessible enough.
- Wrong roles are being targeted.
- Applications are going to local-only jobs.
- Outreach is missing.

14.1.44 18. Weekly review

Every Friday, complete a short review.

14.1.45 18.1 Review questions

Bloque de código: text

1. How many applications were sent?
2. How many were Category A?
3. How many direct messages were sent?
4. How many responses arrived?
5. Which roles responded?
6. Which roles ignored the application?
7. Which portfolio link was used most?
8. Which objections appeared?
9. What should change next week?
10. What should be stopped?

14.1.46 18.2 Weekly decision rules

If there are no responses after 50 applications:

- Review CV headline.
- Review portfolio homepage.
- Add demo video thumbnail.
- Improve TwinSight summary.
- Increase direct outreach.

If there are responses but no interviews:

- Check salary expectation.
- Check work authorization wording.
- Check whether the role requires experience not shown.
- Improve project explanation.

If there are interviews but no progress:

- Improve verbal explanation.
- Prepare technical breakdown.

- Practice AI-assisted development disclosure.
- Prepare code and architecture discussion.

If there are technical tests but poor performance:

- Create a skill sprint.
- Reduce application volume temporarily.
- Focus on Unity/C#/WebGL fundamentals.

14.1.47 19. Monthly review

At the end of every month, classify the job search.

14.1.48 19.1 Outcome categories

```
Bloque de codigo: text
Healthy pipeline:
- Responses arriving.
- Some recruiter screens.
- Some interviews.
- No major positioning contradiction.

Weak pipeline:
- Low responses.
- Few interviews.
- Applications mostly ignored.

Wrong market:
- Roles require US/EU authorization.
- Remote is not actually worldwide.
- Seniority mismatch is frequent.

Wrong evidence:
- Recruiters ask for commercial experience.
- Portfolio does not show enough proof.
- TwinSight not immediately understood.
```

14.1.49 19.2 Monthly adjustment rules

After 30 days:

- Keep applying if response rate is acceptable.
- If no response, revise CV and portfolio front page.
- If roles reject due to authorization, shift toward LATAM contractor and global contractor platforms.
- If roles reject due to experience, increase project-based outreach.
- If roles reject due to skill gaps, schedule a focused sprint.

After 60 days:

- Double down on the top two responding role families.
- Stop applying to low-response categories.
- Use feedback to rewrite positioning.
- Increase direct outreach and referral attempts.

After 90 days:

- Reassess salary target.
- Consider contractor/freelance bridge work.
- Consider temporary fallback route if no traction.
- Decide whether to invest more in ARA, WebGL, XR, or Unity Technical Art.

14.1.50 20. Bad-offer rejection criteria

Reject or negotiate offers with these red flags.

14.1.51 20.1 Structural red flags

- No written contract.
 - Unclear payment terms.
 - Payment only after vague milestones.
 - Full-time exclusivity for low contractor pay.
 - Fixed schedule but no employee benefits.
 - Required unpaid test longer than 4–6 hours.
 - No clarity on IP ownership.
 - No clarity on termination terms.
 - Salary below minimum target without learning value.
-

14.1.52 20.2 Remote-work red flags

- “Remote” but requires local work authorization.
 - “Remote” but demands relocation soon without support.
 - “Contractor” but requires employee-like control.
 - No mention of payment currency.
 - No invoicing process.
 - No tax documentation.
 - No equipment clarity.
-

14.1.53 20.3 Portfolio/IP red flags

- Requests full project files before contract.
 - Asks to recreate TwinSight-like work for free.
 - Requires confidential unpaid work.
 - Claims ownership of test work without compensation.
 - Requests manufacturer/CAD assets without legal clarity.
-

14.1.54 21. Offer evaluation framework

Evaluate every offer using six dimensions.

Dimension	Weight
Net income	30%
Role fit	20%
Portfolio leverage	15%
International leverage	15%
Stability	10%
Learning value	10%

Use this scoring:

```
Bloque de codigo: text
5 = excellent
4 = strong
3 = acceptable
2 = weak
1 = unacceptable
```

Any offer below 3/5 in both net income and role fit should be rejected unless there is an urgent financial need.

14.1.55 22. Gross-to-net tracking

Track compensation as gross and estimated net.

Required fields:

Field	Example
Gross monthly	USD 3,000
Payment currency	USD
Platform fee	0–5%
Conversion spread	0–3%
Social security reserve	Estimate
Tax reserve	Estimate
Net estimate	Conservative

Do not compare offers only by gross salary.

A USD 3,000 contractor offer may be weaker than a lower employee offer if it has unstable payments, no benefits, and high compliance cost.

14.1.56 23. Tracker templates

14.1.57 23.1 Applications CSV schema

```
Bloque de codigo: csv
id,date_applied,company,role_title,category,source,location,remote_scope,contract_type,salary_range,status,next_action,follow
↩ _up_date,role_fit,portfolio_fit,remote_feasibility,compensation_potential,probability,total_score,primary_proof,secondary
↩ _proof,cv_variant,portfolio_link,github_link,contact_name,contact_url,notes
```

14.1.58 23.2 Companies CSV schema

```
Bloque de codigo: csv
company,website,careers_url,linkedin_url,category,region,language,remote_scope,contractor_likelihood,priority,last_checked,no
↩ tes
```

14.1.59 23.3 Contacts CSV schema

```
Bloque de codigo: csv
name,role,company,linkedin,email,relationship,last_contact,next_follow_up,status,notes
```

14.1.60 23.4 Interviews CSV schema

```
Bloque de codigo: csv
company,role,interview_stage,date,interviewer,next_step,deadline,questions_asked,weak_points,follow_up_sent,notes
```

14.1.61 24. File and folder system

Create a job-search folder with this structure:

```
Bloque de codigo: text
/job_search/
/cv/
  alexander_woodcock_realtime_3d_cv.pdf
  alexander_woodcock_unity_technical_artist_cv.pdf
  alexander_woodcock_webgl_interactive_3d_cv.pdf
  alexander_woodcock_ai_tools_cv.pdf
/cover_notes/
/applications/
/interviews/
/technical_tests/
/company_research/
/portfolio_assets/
/screenshots/
/offers/
```

Never overwrite submitted CVs. Keep dated versions.

Example:

```
Bloque de codigo: text
alexander_woodcock_realtime_3d_cv_2026-06-15.pdf
```

14.1.62 25. Naming conventions

Use consistent names.

```
Bloque de codigo: text
Company_Role_Date_Status
```

Example:

```
Bloque de codigo: text
Treeview_UnityDeveloper_2026-06-18_Applied
```

For interview notes:

```
Bloque de codigo: text
Company_Role_InterviewStage_Date.md
```

Example:

```
Bloque de codigo: text
Treeview_UnityDeveloper_RecruiterScreen_2026-06-24.md
```


14.1.63 26. Interview feedback capture

After every interview, write notes within 30 minutes.

Template:

```
Bloque de codigo: text
Company:
Role:
Date:
Stage:
Interviewers:

What they asked:

What I answered well:

What I answered poorly:

Technical gaps exposed:

Portfolio questions:

AI-assisted development questions:

Work authorization questions:

Salary questions:

Next action:

Adjustment needed:
```

This creates feedback loops.

14.1.64 27. Technical test handling

Before accepting a test, record:

- Expected time.
- Deadline.
- Compensation.
- IP ownership.
- Scope.
- Tools required.
- Evaluation criteria.

Reject or negotiate unpaid tests that require excessive time or production-quality deliverables.

Suggested rule:

```
Bloque de codigo: text
Accept unpaid test: ≤4 hours
Question unpaid test: 4–8 hours
Reject or negotiate: >8 hours
```

Exception: a high-fit, high-salary Category A opportunity may justify a longer test if the scope is clearly bounded.

14.1.65 28. Market feedback tags

Tag every rejection or weak response.

Useful tags:

```
Bloque de codigo: text
No EU authorization
No US authorization
Too junior
Too senior
Needs shipped game
```

```
Needs commercial Unity
Needs WebGL proof
Needs shader proof
Needs C++
Needs Unreal
Needs German
Needs portfolio polish
Salary mismatch
No response
Role closed
```

After 30–50 responses or rejections, these tags reveal the actual bottleneck.

14.1.66 29. Skill sprint trigger system

Do not start new courses randomly.

Start a skill sprint only when the tracker shows repeated evidence.

Trigger	Sprint
Needs WebGL proof	Unity WebGL build polish
Needs shader proof	Shader Graph/HLSL mini-breakdown
Needs tools proof	Unity editor tool demo
Needs XR proof	Small XR prototype
Needs Python proof	ARA cleanup
Needs German	A1/A2 German routine
Needs commercial polish	TwinSight case study upgrade

A skill sprint should last 1–2 weeks and produce a visible artifact.

14.1.67 30. Anti-dispersion rules

The main operational risk is not lack of opportunities. It is fragmentation.

Rules:

1. Do not create a new portfolio project unless it supports an observed market gap.
2. Do not rewrite LinkedIn every week.
3. Do not change positioning after a few rejections.
4. Do not apply to roles that require false claims.
5. Do not study German, Houdini, AI, WebGL, React, and Unreal at the same time.
6. Do not polish low-priority projects before TwinSight is publicly strong.
7. Do not treat browsing as work.
8. Do not measure progress by tabs opened.

14.1.68 31. Execution phases

14.1.69 31.1 Phase 1 — Setup

Duration:

```
Bloque de codigo: text
1–2 weeks
```

Deliverables:

- Final CV variants.

- LinkedIn updated.
- GitHub pinned repos fixed.
- TwinSight case study published.
- Demo video uploaded.
- Tracker created.
- Company list imported.
- First 30 targets scored.

14.1.70 31.2 Phase 2 — Controlled launch

Duration:

Bloque de código: text
Weeks 3-6

Actions:

- Apply to 20–40 roles per week.
- Prioritize Category A and B.
- Contact recruiters.
- Run follow-up system.
- Record responses.
- Update assets based on objections.

14.1.71 31.3 Phase 3 — Optimization

Duration:

Bloque de código: text
Weeks 7-12

Actions:

- Stop low-response categories.
- Double down on responding markets.
- Improve weak proof points.
- Increase direct outreach.
- Prepare negotiation.
- Consider fallback contractor work if needed.

14.1.72 32. Minimum viable tracker setup

If time is limited, create only one sheet with these columns:

Bloque de código: csv
Date,Company,Role,Category,Source,Remote Scope,Contract Type,Salary,Status,Follow-up Date,Fit Score,Portfolio Link,CV
↔ Variant,Contact,Notes

This is enough to start.

Do not delay applications because the tracker is too complex.

14.1.73 33. Quality control checklist

Before each weekly review, check:

- No duplicate applications.
 - No missing follow-up dates.
 - No applications without category.
 - No high-fit company without direct outreach attempt.
 - No untracked recruiter conversation.
 - No interview without notes.
 - No offer without gross-to-net estimate.
 - No technical test without scope evaluation.
-

14.1.74 34. Final operational recommendation

The initial target is not to find the perfect job. The initial target is to generate enough market signal to know which version of the profile is converting.

The first 100 outbound actions should be treated as a calibration period.

After 100 outbound actions, evaluate:

```
Bloque de codigo: text
1. Which role family responds?
2. Which market responds?
3. Which salary range responds?
4. Which project gets attention?
5. Which requirement blocks progress?
6. Which claim needs adjustment?
7. Which skill gap appears repeatedly?
```

Then adjust the strategy based on evidence, not intuition.

14.1.75 35. Next file dependency

This file feeds directly into:

```
Bloque de codigo: text
13_interview_negotiation_and_offer_risk.md
14_30_60_90_execution_plan.md
15_final_integrated_strategy.md
```

The tracker defines the execution loop. The interview and negotiation file will define how to convert interviews into offers and how to reject bad opportunities.

Chapter 15

13 — Interview, Negotiation and Offer-Risk Framework

Fuente: 13_interview_negotiation_and_offer_risk.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 816

Ultima modificacion: 2026-06-11 14:40

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This file defines a practical interview, negotiation and offer-risk framework for a Colombia-based Unity / Real-Time 3D / Technical Visualization candidate.

Contenido completo depurado

15.0.1 1. Purpose and scope

This file defines a practical interview, negotiation and offer-risk framework for a Colombia-based Unity / Real-Time 3D / Technical Visualization candidate.

It is built around the source-of-truth profile established in previous files:

- Flagship project: **TwinSight X500**, a Unity WebGL technical-visualization thesis project.
- Positioning: Real-Time 3D / Unity Technical Artist / Technical Visualization / WebGL / CAD-to-realtime.
- Current legal position: Colombia-based remote contractor or employee through compliant hiring structure.
- Future mobility: Portuguese passport expected in approximately two years; Germany route possible through marriage and residence processing.
- Experience profile: strong portfolio evidence, limited formal industry experience.
- Languages: Spanish native; English C1 self-assessed; German not currently usable professionally.

This document is not legal, tax or immigration advice. It is an operational framework for interviews, negotiation, contract review and offer evaluation.

15.0.2 2. Source-quality audit of the Deep Research output

The Deep Research output was useful as a raw input, but not acceptable as a final section without correction.

What was useful

- It identified common interview formats: screening, technical interview, take-home test, hiring-manager interview and offer stage.
- It surfaced representative Unity interview topics: GameObjects, components, Update, FixedUpdate, prefabs and colliders.
- It correctly separated employee and contractor negotiation issues.
- It highlighted red flags in remote offers.
- It connected salary anchoring to section 03 benchmarks.

What required correction

- It repeated salary data that belongs primarily in section 03.
- It used some weak sources for interview patterns and red flags.
- It overstated some tax points, especially around IVA/exported services, which must be handled cautiously.
- It did not adequately separate role-family interview expectations.
- It did not provide a rigorous take-home test risk model.
- It did not give a structured offer-scoring model.
- It did not sufficiently address AI-assisted development disclosure.

Corrective direction

This final version keeps the useful structure, removes weak generalizations, and organizes the section around the actual job-search process:

1. Interview evidence package.
2. Interview preparation by role family.
3. Technical proof expectations.
4. Take-home and unpaid-labor risk.
5. Negotiation logic.
6. Contractor and compliance risks.
7. AI-disclosure handling.
8. Offer evaluation.
9. Accept / counter / reject rules.

15.0.3 3. Candidate-specific interview positioning

The candidate should not enter interviews as a generic Unity developer, generic 3D artist or generalist full-stack developer.

The most defensible interview frame is:

Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization, WebGL deployment and technical UI systems.

Defensible claims

- Built a Unity WebGL interactive 3D technical visualization prototype for drone assembly understanding.
- Worked across CAD-to-realtime optimization, Unity integration, interaction systems, UI, visual modes and technical documentation.
- Used academic usability and workload metrics to evaluate the tool.
- Can explain asset optimization trade-offs, interaction design, WebGL constraints and validation methodology.
- Used AI as an implementation assistant while retaining technical judgment, debugging, integration and final ownership.

Claims to avoid

- Do not claim senior industry experience.
- Do not claim shipped commercial product experience unless a client/product actually shipped.
- Do not claim current EU work authorization.
- Do not claim pure AI engineer / ML engineer profile.
- Do not claim Houdini FX TD ability unless a public reel proves it.
- Do not imply that all TwinSight code was handcrafted line-by-line without AI assistance if challenged directly.

15.0.4 4. Interview evidence package

Before applying or interviewing, the candidate should have a compact evidence package.

Minimum evidence package

Asset	Required state
Portfolio page	TwinSight case study public
Demo video	60–120 seconds
GitHub README	Recruiter-readable
Screenshots	Clear UI and features
Metrics table	Triangles, SUS, NASA-TLX
Technical diagram	CAD-to-realtime pipeline
Role statement	Exact contribution stated
AI disclosure	Prepared but not overemphasized

Strong evidence package

The stronger version adds:

- WebGL live build or controlled video demo.
- Before/after optimization visuals.
- Interaction-system architecture diagram.
- WebGL constraints section.
- Mobile-performance discussion.
- One clean C# code sample.
- One editor-tool or automation sample.
- One concise technical breakdown on ArtStation or portfolio site.

15.0.5 5. Interview pipeline by role family

5.1 Unity Technical Artist

Likely interview focus:

- Unity rendering pipeline: URP, materials, Shader Graph, basic HLSL.
- Asset optimization: meshes, materials, textures, batching, draw calls, LODs.
- Art-to-engine integration.
- Debugging visual bugs.
- Technical communication between artists and programmers.
- Tooling or editor scripts.

Expected proof:

- Visual breakdown of TwinSight modes.
- CAD-to-realtime before/after examples.

- Explanation of optimization decisions.
- One shader or visual-mode breakdown.
- One example of bridging asset pipeline and runtime constraints.

Risk:

The profile is viable for optimization/runtime/integration technical art, but weaker for shader-heavy, Houdini-heavy or AAA pipeline roles.

5.2 Real-Time 3D Developer

Likely interview focus:

- C# architecture.
- Runtime interaction systems.
- Object selection, highlighting, isolation, camera controls.
- UI integration.
- Data-driven component panels.
- Performance profiling.
- WebGL deployment constraints.

Expected proof:

- TwinSight interaction system explanation.
- Technical UI architecture.
- Build constraints and optimization notes.
- Clear answer to “what did you program yourself?”

Risk:

The candidate must avoid sounding like a pure artist. The value proposition is technical implementation, interaction design and 3D systems.

5.3 Unity WebGL / Web 3D Developer

Likely interview focus:

- Unity WebGL build constraints.
- Memory limits and load time.
- Browser compatibility.
- Asset compression.
- Asynchronous loading.
- Deployment workflows.
- Front-end integration if the role involves React/Three.js.

Expected proof:

- WebGL build or video evidence.
- Explanation of WebGL limitations encountered.
- Asset-size and performance trade-offs.
- Understanding of mobile browser constraints.

Risk:

If the role is actually Three.js/WebGPU-heavy, TwinSight helps conceptually but does not fully prove the stack. The candidate should present Three.js/WebGPU as an adjacent skill only if supported by a small demo.

5.4 Technical Visualization / Digital Twin / Simulation Developer

Likely interview focus:

- Translating physical systems into interactive 3D representations.
- CAD-to-realtime workflow.
- Scene hierarchy and component relationships.
- Technical UI and metadata.
- Simulation vs visualization distinction.
- Data integration concepts.
- Usability for technical users.

Expected proof:

- TwinSight as the primary case study.
- Explanation of how 2D technical documentation becomes interactive 3D.
- Metrics and user-evaluation results.
- Clear limitation statement: TwinSight is not a physics-accurate digital twin unless real-time sensor/data integration exists.

Risk:

Do not overclaim “digital twin” if the system does not connect live operational data. Use “technical visualization” or “digital-twin-adjacent prototype” when necessary.

5.5 XR / AR / VR Developer

Likely interview focus:

- Unity XR stack.
- Spatial interaction.
- Performance constraints.
- Input systems.
- Comfort and usability.
- Platform SDKs.

Expected proof:

- TwinSight proves 3D interaction and technical visualization, not XR experience by itself.
- A small XR prototype would significantly improve credibility.
- If no XR artifact exists, position XR as adjacent, not primary.

Risk:

XR roles may require headset-specific experience. Without Meta Quest / OpenXR / AR Foundation artifacts, probability is lower.

5.6 Tools / Pipeline Developer

Likely interview focus:

- Automation mindset.
- Python scripting.
- Unity editor tooling.
- Asset processing.
- Naming, validation, import rules.
- Batch operations.
- Pipeline documentation.

Expected proof:

- ARA as secondary tooling evidence.
- Any Unity editor tools or scripts used in TwinSight.
- Before/after workflow reduction.

- Small public tool repo if possible.

Risk:

ARA supports tooling logic, but does not prove production art-pipeline engineering unless made stable, documented and demoable.

5.7 Python Automation / AI Tools / LLM Application Developer

Likely interview focus:

- Python architecture.
- API use.
- LangGraph or LLM orchestration.
- Error handling.
- Prompt/tool reliability.
- Data flow.
- Cost control.
- Evaluation.

Expected proof:

- ARA demo or README.
- Clear architecture diagram.
- Known limitations.
- No overclaiming as ML engineer.

Risk:

This route can dilute the main profile. It should be used for tools, research automation and technical workflow roles, not as a full AI engineer identity.

15.0.6 6. Technical question bank

This section is not intended for memorization. It defines categories that should be practiced with precise answers.

Unity fundamentals

- What is a GameObject and how do components relate to it?
- Explain the Unity lifecycle: Awake, Start, Update, FixedUpdate, LateUpdate.
- When should FixedUpdate be used?
- What is a prefab and why does it matter in production?
- What are colliders and rigidbodies?
- How does Unity serialization work at a basic level?
- How do ScriptableObjects differ from MonoBehaviours?
- How would you structure object selection and highlighting?

Performance and optimization

- How do you diagnose low FPS in Unity?
- What metrics do you check first: CPU, GPU, draw calls, batches, memory, textures, triangles?
- How would you optimize a CAD-imported scene?
- What is the difference between static batching, dynamic batching, GPU instancing and SRP batching?
- When should you use LODs?
- How do texture size and material count affect WebGL performance?
- What causes long load times in a browser build?
- How do you reduce memory pressure in Unity WebGL?

Rendering and technical art

- What is URP and why use it?
- What can Shader Graph solve without custom HLSL?
- When is HLSL necessary?
- How would you implement X-Ray, ghosted, wireframe or blueprint visual modes?
- What are common shader performance problems?
- What makes a material setup maintainable?

Technical visualization

- How do you transform technical documentation into an interactive 3D experience?
- What is the difference between product visualization, technical visualization, simulation and a digital twin?
- How would you organize part metadata?
- How would you design a component hierarchy for a complex assembly?
- How would you present uncertainty or non-physical visualization modes such as qualitative thermal overlays?

WebGL / web deployment

- What are Unity WebGL constraints?
- How do browsers affect memory and performance?
- How would you optimize first-load experience?
- What is the difference between Unity WebGL and Three.js for web-based 3D?
- What should be tested on mobile browsers?

Tools / pipeline

- What manual process would you automate first in a 3D pipeline?
- How would you validate imported assets?
- How would you batch-process hundreds of meshes or textures?
- How would you design a naming convention or asset audit tool?
- How would you document a pipeline for artists and engineers?

AI tooling

- What problem does ARA solve?
- What are the failure modes of LLM-based automation?
- How do you evaluate the output of an AI tool?
- How do you avoid hallucinated citations or unverifiable claims?
- How do you control cost and retries in LLM workflows?

15.0.7 7. TwinSight interview narrative

60-second version

TwinSight X500 is my thesis project: a Unity WebGL interactive 3D technical-visualization prototype for understanding the assembly of a Holybro X500 V2 drone. The goal was to move beyond static 2D documentation and help users understand spatial relationships between components. I worked on the Unity integration, interaction systems, UI, visual modes, CAD-to-realtime optimization and evaluation documentation. The project reduced heavy CAD geometry into a realtime-ready asset and was evaluated with usability and workload instruments.

2-minute version

The extended explanation should cover:

1. Problem: 2D documentation requires mental reconstruction.
2. Solution: interactive 3D inspection and assembly understanding.
3. Technical stack: Unity, WebGL, C#, URP, UI Toolkit, Blender pipeline.
4. Pipeline: CAD/STEP to optimized real-time asset.
5. Features: selection, isolation, exploded view, clipping, visual modes, part data.
6. Evaluation: SUS, NASA-TLX Raw, Think-Aloud.
7. Contribution: integration, systems, pipeline, documentation, debugging, AI-assisted implementation.
8. Limits: prototype, not shipped commercial product; not live IoT digital twin.

Evidence statements

Use quantified claims only when they are in the thesis or portfolio material:

- CAD route above 6.5 million triangles reduced to 95,617 optimized triangles.
- 12 validation participants.
- SUS average: 91.88.
- NASA-TLX Raw: 8.69 for 3D viewer vs 19.89 for 2D support.
- 96 task-condition records.

15.0.8 8. Take-home test risk analysis

Take-home tests can be useful, but they carry unpaid-labor and scope-risk problems.

Acceptable take-home test

Criterion	Acceptable form
Scope	2–4 hours maximum
Objective	Skill demonstration
Dataset	Toy or anonymized data
IP	Candidate retains portfolio rights if generic
Feedback	Some review or next step
Deadline	Reasonable, not overnight

Risky take-home test

Risk	Signal
Free labor	Production-ready feature requested
Unbounded scope	"Spend as much time as needed"
IP capture	Company keeps all code with no compensation
No feedback	Auto-rejection after work
Vague evaluation	No criteria
Excessive urgency	24-hour turnaround for complex task

Refusal or boundary script

Use a firm but neutral boundary:

I'm comfortable completing a focused technical assessment. To keep it fair, I can spend up to 3–4 hours and provide a short README explaining decisions and trade-offs. If the task requires production-ready implementation beyond that scope, I'd prefer to treat it as a paid trial or reduce the deliverable.

Paid-trial preference

For complex real company work:

This looks close to a production task. I can do it as a paid trial with a defined scope, deliverables, IP terms and timeline.

15.0.9 9. Negotiation framework

9.1 Core negotiation principle

For international remote work from Colombia, the candidate should negotiate based on **gross contractor compensation**, not local Colombian employee salary.

A contractor must cover:

- taxes;
- social-security contributions;
- health insurance or EPS obligations;
- retirement/pension contributions;
- unpaid vacation;
- unpaid sick days;
- equipment;
- software subscriptions;
- currency conversion and transfer fees;
- accountant/legal support;
- bench time between contracts.

Therefore, the contractor rate must be higher than an equivalent employee salary.

9.2 Salary anchoring by market

Use section 03 as the primary salary source. This file should not replace it.

Practical anchor bands

Target	Gross monthly	Typical route
Floor	USD 1.5k	local/LATAM junior role
Base target	USD 3k	LATAM contractor / junior-strong remote
Strong target	USD 4k–6k	mid-junior contractor / niche Unity-WebGL
High target	USD 6k+	US/EU contractor, digital twin, simulation, XR

9.3 When asked “What are your salary expectations?”

Do not give a single number too early if the scope is unclear.

Recommended answer:

It depends on scope, employment model and benefits. For a contractor arrangement, I usually evaluate the monthly gross range based on responsibilities, time commitment and whether the role is closer to Unity development, technical art, simulation or WebGL deployment. For a full-time contractor role, I would expect a range aligned with international remote Unity/real-time 3D roles rather than local Colombian payroll rates.

If pressed:

For contractor work, my current target range is USD [X–Y] gross per month, depending on scope, exclusivity, payment terms and whether the work includes production ownership or exploratory prototyping.

9.4 Anchor ranges by priority

Priority	Use case	Starting anchor
A	US/EU technical visualization	USD 5k–7k+
B	LATAM/global contractor	USD 3k–5k
C	local or fallback	USD 1.5k–3k
Paid trial	limited test	hourly/day rate

These anchors must be adjusted after validating scope, seniority, hours and hiring model.

15.0.10 10. Contractor vs employee decision matrix

Model	Advantages	Risks	Best use
Colombia contractor	fastest setup	taxes, no benefits	foreign remote clients
EOR employee	compliant payroll	employer cost	serious foreign employer
Local employee	stability	lower pay	bridge role
Freelance platform	fast entry	low trust, fees	small projects
EU employee	benefits, EU rights	future only	after passport/residence
German family route	work authorization path	paperwork	if relocation occurs

Contractor checklist

Before accepting a contractor role, verify:

- legal entity and signatory;
- payment currency;
- payment schedule;
- late payment clause;
- invoice requirements;
- IP ownership;
- confidentiality scope;
- non-compete scope;
- termination notice;
- exclusivity;
- equipment/software responsibility;
- allowed public portfolio mention;
- governing law;
- dispute resolution;
- tax documents required.

15.0.11 11. Colombia-specific remote-contracting risks

This section is high-level only. It must be confirmed with a Colombian accountant.

Risk categories

Risk	What to check
RUT	correct activity codes
Invoicing	DIAN electronic invoicing
IVA	export-service treatment
Retención	local vs foreign client
Social security	EPS, pension, ARL base
Currency	exchange and documentation
Tax residency	Colombian obligations
Misclassification	contractor vs employee behavior

Misclassification risk

A contractor relationship becomes riskier if the company controls:

- fixed schedule;
- daily supervision;
- mandatory exclusivity;
- company equipment only;
- internal hierarchy;
- long-term dependency;
- employee-like benefits without formal employment.

If the role behaves like employment, the company may need an EOR or local employment structure.

Practical mitigation

Use a contract that states:

- independent contractor status;
- deliverables or services;
- invoicing procedure;
- autonomy of execution;
- no automatic employment benefits;
- clear payment terms;
- IP transfer only for paid deliverables.

15.0.12 12. EU move implications on compensation

Current Colombia position

Current strongest route:

- remote contractor;
- foreign client;
- USD or EUR gross compensation;
- tax/social-security planning in Colombia.

After Portuguese passport

Once the passport is issued, the candidate can legally state EU work authorization, but compensation depends on location and employment model.

Possible outcomes:

- EU employee salary may be lower gross than US contractor compensation.
- Benefits, healthcare and stability improve.
- Cost of living and tax burden may increase.
- EU work authorization improves probability for Germany, Portugal, Netherlands and EU-only remote roles.

Germany through marriage/residence

If legally married and resident in Germany under the correct residence title, Germany becomes a realistic work market.

Implications:

- more local/hybrid opportunities;
- German language matters more;
- health insurance and tax obligations become significant;
- employer credibility improves because work authorization is clearer;
- net income may be lower than gross headline salary suggests.

Recruiter-facing wording by state

State	Accurate wording
Now	Colombia-based; available as remote contractor
Passport pending	EU citizenship expected in approximately two years, not current work authorization
Passport issued	EU citizen; authorized to work in EU/EEA
Germany residence	Authorized to work in Germany, if residence title grants it
Blue Card path	Would require sponsorship if used before EU citizenship

15.0.13 13. AI-assisted development disclosure

AI use is not automatically disqualifying. Poor disclosure is the risk.

Bad disclosure

Avoid:

I used AI to build the project.

This sounds like lack of ownership.

Better disclosure

Use:

I used AI as an implementation assistant for iteration, code generation support and debugging. I was responsible for architecture decisions, integration, testing, debugging, technical judgment and final behavior.

When asked “How much did you build yourself?”

Answer structure:

1. State ownership.
2. Separate AI assistance from engineering responsibility.

- 3. Explain a concrete decision you made.
- 4. Mention debugging and validation.
- 5. Offer to walk through code or architecture.

Example:

The project was AI-assisted, but I owned the system design, Unity integration, debugging, interaction behavior, asset workflow decisions and final validation. I used AI the way many developers use autocomplete, documentation search or pair-programming support: as an accelerator, not as a replacement for technical judgment. I can walk through the interaction architecture and explain the decisions behind the pipeline.

Evidence needed

To make the disclosure credible, prepare:

- architecture diagram;
- one code walkthrough;
- one debugging story;
- one performance trade-off;
- one limitation you identified independently;
- one feature you revised after testing.

15.0.14 14. Offer red flags

Critical red flags

Red flag	Risk
No written offer	unenforceable terms
Salary changes after interview	bait-and-switch
unpaid production task	free labor
asks for money	scam
unclear company identity	fraud
vague IP terms	portfolio/legal risk
unlimited non-compete	career restriction
60+ day payment terms	cash-flow risk
crypto-only payment	compliance risk
“remote” but no country scope	work-authorization risk

Moderate red flags

- no technical interviewer;
- no clarity on reporting manager;
- unrealistic timeline;
- “we are like a family” language replacing contract clarity;
- no clear definition of done;
- no tool/license budget;
- no test device or hardware plan for XR/WebGL;
- “equity instead of pay” without legal documentation.

15.0.15 15. Offer evaluation scorecard

Use this scoring model before accepting any offer.

Core scoring table

Category	Weight	Score 1–5
Role fit	20%	
Compensation	20%	
Remote legality	15%	
Portfolio leverage	15%	
Contract safety	15%	
Growth potential	10%	
Lifestyle/time-zone fit	5%	

Decision interpretation

Weighted result	Decision
4.2–5.0	strong accept / negotiate upward
3.5–4.1	viable if terms are clean
3.0–3.4	negotiate or use as bridge
2.5–2.9	accept only if urgent
below 2.5	reject

Minimum acceptance gates

Do not accept if any of these fail:

- no written contract;
- payment terms unclear;
- work authorization impossible;
- unpaid production task required;
- non-compete blocks future technical art/Unity work broadly;
- IP terms prevent portfolio use entirely without justification;
- expected compensation below survival/runway needs.

15.0.16 16. Counteroffer logic

When to counter

Counter if:

- the role fit is strong;
- compensation is below market;
- contract terms are negotiable;
- company is responsive;
- there is no hard legal blocker.

What to counter

Rank negotiation variables:

1. base cash compensation;
2. payment schedule;
3. scope/exclusivity;
4. paid trial terms;
5. contract length;

6. notice period;
7. portfolio permission;
8. equipment/software budget;
9. learning budget;
10. relocation or EOR support.

Counteroffer template

Thank you for the offer. The role is strongly aligned with my Unity and technical-visualization background, especially the work around [specific area]. Based on the scope, contractor structure and market ranges for remote Unity/real-time 3D work, I would be comfortable moving forward at USD [X] gross per month / USD [Y] per hour. I'm also open to discussing structure if there are constraints around salary, such as a paid trial, review after 60–90 days, or a software/equipment budget.

15.0.17 17. Interview preparation calendar

Before first applications

- Prepare TwinSight 60-second pitch.
- Prepare TwinSight 2-minute technical explanation.
- Prepare AI-disclosure answer.
- Prepare salary-expectation answer.
- Prepare work-authorization answer.
- Prepare one C# code walkthrough.
- Prepare one optimization story.

Weekly technical practice

Area	Frequency
C# / Unity fundamentals	2× week
Unity profiling / optimization	1× week
WebGL constraints	1× week
Technical art / rendering	1× week
Portfolio walkthrough	2× week
Mock interview in English	1× week

Before each interview

- Re-read job description.
- Map role requirements to TwinSight evidence.
- Prepare three project-specific examples.
- Check company remote scope.
- Prepare compensation range.
- Prepare five questions for interviewer.

15.0.18 18. Questions to ask employers

Technical role clarity

- What are the main technical problems this role needs to solve in the first 90 days?
- Is the role more runtime development, technical art, pipeline tooling or visualization?
- What platforms are targeted: desktop, web, mobile, XR or embedded?

- What are the current performance constraints?
- What is the current asset pipeline?

Remote and contract clarity

- Is the role open to Colombia-based contractors?
- Do you hire through direct B2B, contractor platform or EOR?
- What countries are eligible for remote work?
- Are working hours fixed or overlap-based?
- What is the payment schedule?

Offer safety

- What is the governing law of the contract?
- Are there exclusivity or non-compete clauses?
- Who owns tools or reusable scripts created during the engagement?
- Can selected non-confidential work be shown in a portfolio?
- What is the notice period?

15.0.19 19. Role-specific proof checklist

Role	Must show
Unity Technical Artist	optimization, visual modes, materials, pipeline
Real-Time 3D Developer	interaction architecture, C#, runtime systems
Unity WebGL Developer	WebGL build, memory/load constraints
Technical Visualization	CAD-to-realtime, metadata, user tasks
Digital Twin / Simulation	physical-system reasoning, data limitations
XR Developer	XR prototype or strong adjacent proof
Tools Developer	automation artifact, editor tooling, Python
AI Tools Developer	ARA demo, architecture, reliability limits

15.0.20 20. Final operational rules

- Use TwinSight as the interview anchor.
- Use ARA only when the role includes tooling, automation or AI workflows.
- Do not lead with Blender portrait unless applying to 3D-art-adjacent roles.
- Do not negotiate against Colombian local rates for international contractor roles.
- Do not accept undefined “remote” terms.
- Do not complete production-grade unpaid tests.
- Do not sign broad non-competes without review.
- Do not overstate EU work authorization before the Portuguese passport or German residence status is real.
- Do not hide AI assistance if directly asked; frame it as engineering acceleration with ownership.

15.0.21 21. Source reference list

Primary source categories used in this section:

- Internal Revenue Service — guidance on employee vs independent-contractor tax treatment: <https://www.irs.gov/businesses/small-businesses-self-employed/independent-contractor-self-employed-or-employee>
- Lemon.io Unity developer hourly-rate report: <https://lemon.io/rate-calculator/unity-developers/>

- Freelancermap Unity developer role and salary reference: <https://www.freelancermap.com/blog/es/que-hace-programador-unity/>
- Adaface Unity interview question examples: <https://www.adaface.com/es/blog/unity-preguntas-entrevista/>
- Tokio School interview-format reference: <https://www.tokioschool.com/noticias/entrevista-tecnica/>
- Colombia contractor and remote-hiring administrative context from section 03.
- EU / Portugal / Germany authorization context from section 04.
- Role-family prioritization from section 02.
- Portfolio and project-evidence structure from sections 07–10.

Chapter 16

14_30_60_90_execution_plan

Fuente: 14_30_60_90_execution_plan.md
Seccion: root
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Resumen de orientacion

This section converts the prior strategy files into a 90-day execution system.

Contenido completo depurado

16.0.1 Purpose

This section converts the prior strategy files into a 90-day execution system.
It is not a market research document. It is an operating plan for moving from thesis completion and portfolio preparation into structured international applications, interviews, negotiation, and offer evaluation.

16.0.2 Baseline assumptions

Item	Current state
Location	Colombia
Time zone	America/Bogota
Thesis	Complete
Thesis status	Pending defense
Primary asset	TwinSight X500
Main target	Remote international work
Preferred model	Contractor / B2B
Minimum target	USD 1.5k/month
6-month target	USD 3k/month
12–24 month target	USD 6k/month
EU status now	No current EU work authorization
EU future	Portuguese passport expected
Germany route	Possible via marriage/residence
Primary roles	Real-Time 3D / Unity / Technical Visualization
Secondary roles	Tools / Python automation
Main risk	Strategic dispersion

16.0.3 Operating principle

The plan prioritizes **visible proof of skill** over additional passive learning.

The critical path is:

Bloque de código: text
TwinSight proof → portfolio MVP → LinkedIn/CV/GitHub alignment → targeted applications → interviews → offer evaluation

The plan should not be interrupted by:

- starting a new major project;
- rewriting the thesis unnecessarily;
- taking long courses without output;
- polishing low-relevance repositories;
- targeting unrelated full-stack jobs too early;
- waiting for Portuguese citizenship before applying;
- treating Germany as the immediate employment base.

16.0.4 90-day outcome targets

By the end of 90 days, the expected state should be:

1. A public-facing portfolio MVP exists.
2. TwinSight X500 has a recruiter-ready case study.
3. GitHub no longer looks unfocused.
4. LinkedIn and CV match the same positioning.
5. At least 120 companies have been classified.
6. At least 80–150 targeted applications or outreach actions have been made.
7. At least 8–15 recruiter or hiring-manager conversations have been attempted.
8. At least 3–6 interview processes have been initiated.
9. Offer evaluation criteria are operational.
10. A 12-month skill roadmap is active.

These targets are not guarantees. They are execution metrics.

16.1 1. Strategic focus for the first 90 days

16.1.1 Primary role families

The first 90 days should target only these role families:

1. **Real-Time 3D / Interactive Technical Visualization Developer**
2. **Unity WebGL / Interactive 3D Developer**
3. **Technical Visualization / Digital Twin / Simulation Developer**
4. **Unity Technical Artist — runtime, optimization, tooling**
5. **Tools / Pipeline / Python Automation Developer**

16.1.2 Roles to deprioritize

Role	Status
Pure 3D Artist	Downplay
Blender Generalist	Secondary
Pure Full-stack Developer	Fallback
AI/ML Engineer	Reject for now

Role	Status
Senior Shader TA	Future path
AAA Game TA	Long-shot
Houdini FX TD	Later specialization
German local-only roles	Future

16.1.3 Project hierarchy

Project	Use
TwinSight X500	Flagship
ARA Framework	Secondary tooling
Blender portrait	Technical art support
ai-news-aggregator	Optional
FitApp-Free	Defer
Old repos	Hide/archive unless relevant

16.2 2. Phase 1 — Days 1–30

16.2.1 Objective

Build minimum credible market presence.

This phase is not primarily for applying at scale. It is for removing blockers that would make applications weak.

16.2.2 Day 1–3: asset inventory and decision lock

Tasks

- Confirm final public name: TwinSight X500.
- Confirm final positioning statement.
- Decide which repositories remain public.
- Decide which projects appear on portfolio MVP.
- Collect all existing screenshots and screen recordings.
- Create folder structure for job-search assets.

Output

```
Bloque de codigo: text
/job_search_assets
/cv
/linkedin
/portfolio
/twinsight
/ara
/screenshots
/demo_video
/applications
/company_research
```

Approval criteria

Item	Requirement
Role target	Locked
Project order	Locked
Public claims	Controlled
EU statements	Accurate
AI disclosure	Prepared

16.2.3 Day 4–10: TwinSight case study package

Tasks

Create the public proof package for TwinSight.

Required assets:

1. Portfolio case study draft.
2. GitHub README recruiter version.
3. Demo video script.
4. 8–12 screenshots.
5. 1–2 technical diagrams.
6. Optimization metrics table.
7. Contribution statement.
8. AI-assisted development explanation.
9. Asset/manufacturer disclaimer.
10. Interview explanation.

Minimum screenshots

Screenshot	Purpose
Full drone view	First impression
Component selection	Interaction
Exploded view	Spatial reasoning
Cross-section	Technical feature
Visual modes	Rendering systems
Mobile layout	Web/mobile proof
CAD-to-low-poly	Optimization proof
Metrics/result screen	Validation proof

Required case study sections

Bloque de codigo: text

1. Problem

2. Context

3. My role

4. Constraints

5. Pipeline

6. Systems

7. Optimization

8. Evaluation

9. Results

10. Limitations

11. What I would improve

Approval criteria

TwinSight can be understood in under 90 seconds by a recruiter and in under 5 minutes by a technical reviewer.

16.2.4 Day 11–15: GitHub cleanup

Tasks

- Pin only relevant repositories.
- Archive or hide weak repos.
- Add README to TwinSight.
- Add README skeleton to ARA.
- Add clear repo descriptions.
- Remove unclear old portfolio attempts from front-page visibility.
- Add disclaimers where needed.

Recommended pinned repositories

Slot	Repository
1	TwinSight X500
2	ARA Framework
3	Blender technical breakdown
4	ai-news-aggregator only if functional
5	Portfolio site
6	Optional tools/scripts repo

Repos to downplay

Repo type	Action
Tutorial repos	Unpin
Unfinished unrelated apps	Private
Old portfolio attempts	Archive
Weak experiments	Private
Client-sensitive repos	Private

16.2.5 Day 16–20: LinkedIn and CV alignment

Tasks

- Apply the headline from section 10.
- Rewrite About section.
- Rewrite freelance/self-employed experience.
- Correct thesis description.
- Correct UNAL wording.
- Add TwinSight as project.
- Add ARA as secondary project.
- Add Featured links.
- Reorder skills.
- Build base CV.
- Build 2 role-specific CV variants.

Required CV variants first

CV	Priority
Real-Time 3D / Unity	First
Technical Visualization / Digital Twin	Second
Tools / Python Automation	Later
Full-stack fallback	Later

Approval criteria

A recruiter should see the same profile in:

- CV;
- LinkedIn;
- GitHub;
- portfolio;
- email pitch.

No channel should imply a different career direction.

16.2.6 Day 21–25: portfolio MVP

Tasks

Launch a minimal portfolio, even if not final.

Minimum structure:

```
Bloque de codigo: text
Home
Work
  TwinSight X500
  ARA Framework
  Blender portrait
About
Contact
```

Home page must communicate

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist
Interactive technical visualization
CAD-to-realtime optimization
Unity WebGL / C# / Blender / Python tooling
Available for remote contract work from Colombia
```

MVP rule

Do not wait for a perfect custom website. A clean static page, GitHub Pages, Notion, Webflow, Framer, or simple HTML is acceptable.

16.2.7 Day 26–30: first application batch

Tasks

- Create application tracker.
- Select 25 A-priority companies.
- Select 25 B-priority companies.
- Prepare 3 message templates.
- Send first 15–25 applications/outreach messages.
- Apply only where portfolio fit is real.

First-month application volume

Type	Count
A-priority applications	5–10
B-priority applications	10–15
Recruiter messages	10
Community posts	1–2
Portfolio feedback requests	3–5

Phase 1 success criteria

Deliverable	Status needed
TwinSight case study	Public or near-public
Demo video	Draft or published
GitHub	Cleaned
LinkedIn	Updated
CV	Ready
Portfolio MVP	Live
Tracker	Active
First applications	Sent

16.3 3. Phase 2 — Days 31–60

16.3.1 Objective

Move from setup to market contact.

This phase tests whether positioning generates responses.

16.3.2 Week 5: application cadence begins

Weekly minimum

Action	Minimum
Targeted applications	15
Direct outreach	15
Recruiter contacts	10
Follow-ups	10
Portfolio improvements	2
Interview practice blocks	3

Weekly maximum

Do not exceed:

Action	Maximum
Applications	40
Direct outreach	40
New project work	5 hrs

Action	Maximum
Course consumption	4 hrs

High volume is not useful if quality collapses.

16.3.3 Week 6: feedback loop

Review signals

Signal	Interpretation
No replies	Positioning weak
Recruiter views only	CV/LinkedIn mismatch
Replies but no interviews	Portfolio weak
Interviews but no progress	Technical prep gap
Low offers only	Targeting too low
Strong interviews	Continue

Corrective actions

Problem	Fix
No replies	Rewrite headline + opener
Weak fit	Narrow role target
No technical interest	Improve TwinSight breakdown
Too much full-stack pull	Remove fallback emphasis
Too junior perception	Add metrics and systems proof
AI concern	Prepare disclosure answer

16.3.4 Week 7: ARA and secondary proof

Tasks

- Decide whether ARA becomes public.
- If public, clean README.
- Create architecture diagram.
- Add controlled AI disclosure.
- Avoid overstating production readiness.
- Build one short demo or screenshots.

ARA should prove

Bloque de código: text
 Python automation
 LLM orchestration awareness
 Research workflow tooling
 Structured systems thinking
 Technical documentation

ARA should not claim:

Bloque de código: text
 Production AI platform
 Enterprise AI system
 ML engineering expertise
 Autonomous research replacement
 Senior AI developer proof

16.3.5 Week 8: interview preparation sprint

Technical preparation blocks

Role	Prep focus
Unity TA	Profiling, rendering, shaders
Real-Time 3D	C#, architecture, optimization
WebGL	build size, memory, browser limits
Digital Twin	data flow, CAD, simulation
XR	interaction, performance, devices
Tools	editor scripts, automation
Python/AI	APIs, LangGraph, reliability

Interview artifacts

Prepare short answers for:

1. Why this role?
2. Why remote from Colombia?
3. What exactly did you build in TwinSight?
4. How much AI assistance was used?
5. What was the hardest technical problem?
6. What would you improve?
7. Why no formal industry experience yet?
8. What salary range are you targeting?
9. Contractor or employee?
10. What is your availability?

16.3.6 Week 9: salary and negotiation readiness

Tasks

- Define minimum acceptable gross.
- Define minimum acceptable net.
- Define contractor rate.
- Define hourly/day/month equivalents.
- Prepare response for salary expectations.
- Prepare offer scorecard.
- Prepare red-flag checklist.

Compensation anchors

Scenario	Target
Minimum survival	USD 1.5k/mo
Acceptable starter	USD 2k–3k/mo
Strong 6-month target	USD 3k–4.5k/mo
High-value contract	USD 5k–6k+/mo
Below rejection line	Context-specific

Phase 2 success criteria

Metric	Target
Total applications	50–90
Direct contacts	40–80
Recruiter conversations	3–8
Interview processes	2–5
Portfolio iterations	4–8
Case study feedback	3+ sources

16.4 4. Phase 3 — Days 61–90

16.4.1 Objective

Scale what works, reject weak paths, and convert interviews into offers.

16.4.2 Week 10: target refinement

Segment all companies into

Segment	Meaning
A1	High fit, active hiring
A2	High fit, no active role
B1	Good fit, realistic
B2	Good fit, weak signal
C	Fallback
Reject	Distraction

Actions

- Stop applying to weak C targets unless cashflow requires it.
- Increase outreach to A1/A2.
- Use speculative outreach for high-fit studios.
- Reuse TwinSight in technical messages.
- Ask for referrals where possible.

16.4.3 Week 11: interview conversion

Tasks

- Prepare per-company notes.
- Build role-specific TwinSight explanation.
- Prepare 5 technical stories.
- Prepare whiteboard/system design answers.
- Prepare code walkthrough.
- Prepare negotiation stance.

Technical stories to prepare

Story	Evidence
CAD optimization	Triangle reduction
Web deployment	WebGL constraints
Interaction system	Selection/exploded/cut
UI system	UI Toolkit
Validation	SUS / NASA-TLX
AI assistance	Judgment/debugging
Limitation	Honest tradeoffs

16.4.4 Week 12: offer and pipeline management

Offer evaluation model

Score each offer from 1–5:

Criterion	Weight
Gross compensation	20
Net compensation	15
Role fit	20
Portfolio leverage	15
Contract safety	10
Remote stability	10
Learning ROI	5
EU leverage	5

Reject or renegotiate if:

- total score is low;
- payment terms are vague;
- unpaid trial is large;
- exclusivity is broad;
- IP terms are abusive;
- salary is below minimum;
- role dilutes the portfolio.

16.4.5 Week 13: strategic decision point

At day 90, decide between four paths:

Path	Trigger
Continue same strategy	Interviews active
Increase volume	Replies low
Improve portfolio	Technical rejection
Add fallback roles	Cashflow pressure
Pause for skill sprint	Repeated test failures

Day 90 review questions

1. Which role family generated the most replies?
2. Which portfolio asset got technical attention?
3. Which job boards produced interviews?
4. Which companies ignored the profile?

5. Which salary ranges were accepted?
6. Which questions caused weak answers?
7. Which technical tests failed?
8. Is ARA helping or distracting?
9. Is the portfolio sufficient?
10. Is cashflow becoming urgent?

16.5 5. Weekly operating rhythm

16.5.1 Default weekly schedule

Day	Focus
Monday	Targeting + applications
Tuesday	Outreach + GitHub/portfolio
Wednesday	Applications + interview prep
Thursday	Recruiters + follow-ups
Friday	Portfolio polish + metrics
Saturday	Skill sprint
Sunday	Review + planning

16.5.2 Daily block model

Block	Duration
Applications	60–90 min
Outreach	45–60 min
Portfolio/GitHub	60–120 min
Interview prep	45–90 min
Learning	30–60 min
Admin/tracker	20 min

16.5.3 Minimum viable day

On low-energy days, do only:

```
Bloque de código: text
1 application
1 outreach message
1 tracker update
20 minutes TwinSight/portfolio polish
```

This prevents zero-output days.

16.6 6. Application volume model

16.6.1 Recommended weekly range

Category	Weekly target
A-priority roles	5–10

Category	Weekly target
B-priority roles	10–20
C/fallback roles	0–5
Recruiter outreach	10–20
Speculative outreach	5–10
Follow-ups	10–20

16.6.2 Quality rules

A-priority

Requires:

- tailored CV;
- tailored message;
- correct portfolio link;
- company-specific reason;
- follow-up;
- role-specific TwinSight angle.

B-priority

Requires:

- role-specific CV;
- short customized opener;
- portfolio link;
- tracker entry.

C-priority

Requires:

- fast application;
- no heavy customization;
- no deep portfolio rewrite.

16.7 7. Outreach cadence

16.7.1 Follow-up schedule

Moment	Action
Day 0	Apply / message
Day 3–5	First follow-up
Day 10–14	Second follow-up
Day 21	Final note
Day 30+	Archive or reclassify

16.7.2 Stop conditions

Stop following up if:

- company rejects;
- no response after three messages;
- role is filled;
- role requires unavailable work authorization;
- compensation is below minimum;
- company shows red flags.

16.8 8. Interview preparation system

16.8.1 Weekly practice

Area	Frequency
TwinSight explanation	3×/week
Unity/C# questions	3×/week
Optimization scenarios	2×/week
WebGL constraints	2×/week
Behavioral answers	2×/week
Salary scripts	1×/week
AI disclosure	1×/week

16.8.2 Mock interview prompts

1. Explain TwinSight in 90 seconds.
2. Explain the CAD-to-realtime pipeline.
3. Explain how you reduced geometry.
4. Explain WebGL performance constraints.
5. Explain how you would debug low FPS.
6. Explain how you used AI responsibly.
7. Explain a system you architected.
8. Explain what you would rebuild differently.
9. Explain your current work authorization.
10. Explain your salary expectations.

16.9 9. Skill sprint model

16.9.1 Skill priorities

Priority	Skill
1	Unity profiling
2	WebGL optimization
3	C# architecture
4	Shader Graph / basic HLSL
5	UI Toolkit
6	Git/GitHub polish
7	Python tooling
8	Three.js/WebGPU
9	XR basics

Priority	Skill
10	German A1/A2

16.9.2 Rule

Every learning block must produce one artifact:

Learning type	Required artifact
Unity profiling	Before/after metric
Shader learning	Small demo
WebGL learning	Build-size note
C# architecture	Refactor snippet
Python tooling	Script/demo
German	CEFR progress
Rebelway	Reel fragment

No artifact = low ROI.

16.10 10. Risk management

16.10.1 Main risks

Risk	Mitigation
Strategic dispersion	Weekly focus lock
Portfolio perfectionism	MVP deadline
Low applications	Minimum viable day
Weak replies	Message iteration
Low salary offers	Company targeting
AI credibility concern	Transparent disclosure
Legal confusion	Authorization statements
Burnout	Fixed weekly review
Course hoarding	Artifact rule
Cashflow pressure	Controlled fallback

16.10.2 Early warning signs

Signal	Meaning
Many tabs, no output	Planning loop
New project idea	Avoidance
Course binge	Delay tactic
Website redesign	Perfectionism
Applying only easy jobs	Fear of rejection
No tracker updates	System failure
Ignoring follow-ups	Pipeline leakage

16.11 11. Decision gates

16.11.1 Gate 1 — Day 30

Proceed to scaled applications only if:

- CV ready;
- LinkedIn updated;
- TwinSight visible;
- GitHub acceptable;
- portfolio MVP live or shareable;
- tracker active.

If not, extend Phase 1 by one week.

16.11.2 Gate 2 — Day 60

Continue current strategy only if:

- replies exist;
- interviews exist or seem close;
- portfolio feedback is not negative;
- target roles still fit.

If not, adjust:

Failure	Adjustment
No replies	Rewrite positioning
Wrong replies	Change target list
Low offers	Raise company tier
Technical failures	Skill sprint
Portfolio confusion	Simplify projects

16.11.3 Gate 3 — Day 90

Decide:

Bloque de codigo: text
Continue / Pivot / Add fallback / Skill sprint / Accept offer

The decision must be based on tracker evidence, not mood.

16.12 12. Output checklist by day

16.12.1 Day 30

Output	Required
Base CV	Yes
CV variant 1	Yes
CV variant 2	Yes
LinkedIn update	Yes
GitHub cleanup	Yes
TwinSight case study	Yes

Output	Required
Demo video draft	Yes
Portfolio MVP	Yes
Tracker	Yes
First applications	Yes

16.12.2 Day 60

Output	Required
50+ applications	Yes
40+ outreach actions	Yes
ARA decision	Yes
Interview scripts	Yes
Salary scripts	Yes
Portfolio iteration	Yes
Recruiter list active	Yes

16.12.3 Day 90

Output	Required
80–150 applications	Yes
80+ outreach actions	Yes
Interview pipeline	Target
Offer scorecard	Active
Weak paths removed	Yes
Skill roadmap updated	Yes
Next 90-day plan	Yes

16.13 13. What not to do during the 90 days

Do not:

- rebuild the portfolio from scratch more than once;
- start a new flagship project;
- present ARA as the thesis;
- claim EU work authorization before it exists;
- apply broadly as full-stack unless necessary;
- spend more time on courses than portfolio;
- publish incomplete weak repos;
- overexplain AI usage defensively;
- accept vague unpaid trials;
- accept broad non-competes;
- ignore gross-to-net calculations;
- wait passively for recruiters.

16.14 14. Final 90-day operating statement

The operational target for the next 90 days is:

```
Bloque de codigo: text
Use TwinSight X500 as the proof asset to enter real-time 3D, Unity WebGL, technical visualization, digital twin, simulation,
↪ and Unity Technical Artist conversations, while using ARA and Python tooling only as secondary evidence of automation
↪ capability.
```

The first offer does not need to be perfect. It must satisfy:

1. compensation above the minimum threshold;
2. role fit with the portfolio direction;
3. acceptable contract risk;
4. remote feasibility from Colombia;
5. ability to build toward USD 3k–6k/month over 12–24 months.

If an offer fails these criteria, it is not automatically a strategic win even if it is international.

Chapter 17

15_final_integrated_strategy

Fuente: 15_final_integrated_strategy.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 981

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Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This is the integrated strategy document that connects the previous sections into one coherent operating model.

Contenido completo depurado

17.0.1 Purpose

This is the integrated strategy document that connects the previous sections into one coherent operating model.

It does not replace the detailed files. It functions as the final synthesis layer:

- what the profile is;
- which roles to target;
- how to position the portfolio;
- how to apply;
- how to negotiate;
- how to avoid strategic dilution;
- what to do in the next 90 days;
- what to build over 12–24 months.

This document assumes the thesis is complete and pending defense.

17.1 1. Strategic diagnosis

17.1.1 Current position

Alexander Woodcock Salomón is best positioned as a **Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization, Unity WebGL deployment, and Python-assisted tooling.**

The strongest evidence is not formal industry experience. The strongest evidence is the **TwinSight X500** thesis project, supported by secondary technical projects such as ARA Framework, Blender portrait work, and selected software/tooling experiments.

17.1.2 Core asymmetry

The profile has a strong project but limited formal industry history.

That creates a specific strategy:

Bloque de código: text
 Do not compete as a generic junior developer.
 Do not compete as a pure 3D artist.
 Do not compete as a generic full-stack developer.
 Compete as a niche candidate with unusually relevant proof in Unity WebGL, technical visualization and CAD-to-realtime.

17.1.3 Main market advantage

TwinSight creates a defensible bridge between:

- Unity;
- WebGL;
- interactive 3D;
- CAD-to-realtime asset optimization;
- technical UI;
- inspection/assembly visualization;
- usability validation;
- technical documentation.

This combination is more aligned with **technical visualization, simulation, digital twins, XR, product visualization and industrial 3D** than with pure game art.

17.1.4 Main weakness

The main weakness is not lack of potential. It is **lack of packaged proof**.

The profile must be converted into recruiter-readable evidence:

- case study;
- demo video;
- GitHub README;
- portfolio;
- screenshots;
- CV bullets;
- LinkedIn project entry;
- interview explanation.

Until that exists, the job search will underperform.

17.2 2. Source-of-truth profile

17.2.1 Current factual profile

Area	Correct statement
Location	Colombia
Main language	Spanish

Area	Correct statement
English	C1 self-assessed
German	Beginner / none
Portuguese	Relevant for future EU/Portugal route
Degree	Ingeniería Multimedia, UNAD
Thesis	Complete, pending defense
Prior studies	Advanced coursework in Electronic Engineering, UNAL
Immediate work authorization	Colombia-based contractor / employment depending on structure
EU work authorization now	No
Future EU route	Portuguese passport expected in ~2 years
Germany route	Possible through marriage/residence with German partner
Current main proof	TwinSight X500
Primary career identity	Real-Time 3D / Unity Technical Visualization

17.2.2 Claims that should be avoided

Do not claim:

- current EU work authorization;
- German residence;
- Portuguese passport already issued;
- senior industry experience;
- formal Unity certification unless obtained;
- formal Rebelway/CG Cookie certification unless issued;
- production AI engineering seniority;
- that ARA is the thesis if TwinSight is the thesis;
- that AI wrote the project without your technical ownership;
- that all projects are production-ready.

17.2.3 Claims that are safe if phrased accurately

Safe claims:

- thesis project completed, pending defense;
- Unity WebGL technical visualization prototype;
- CAD-to-realtime optimization;
- C# runtime systems;
- interactive 3D UI;
- component selection, exploded view, clipping/cross-section;
- Blender optimization and asset preparation;
- academic validation using SUS / NASA-TLX / Think-Aloud if supported by final thesis data;
- AI-assisted implementation with human technical judgment, debugging and ownership;
- Python automation and LLM orchestration prototype through ARA if shown as prototype.

17.3 3. Recommended positioning

17.3.1 Primary positioning statement

Bloque de código: text
 Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization,
 ↩ and Unity WebGL deployment.

17.3.2 Expanded version

Bloque de código: text

I build interactive real-time 3D systems for technical visualization using Unity, C#, WebGL and Blender. My work focuses on
 ↪ converting complex CAD or technical assets into optimized, browser-accessible 3D experiences with inspection tools,
 ↪ technical UI, visual modes, and performance-conscious deployment.

17.3.3 Spanish version

Bloque de código: text

Desarrollador 3D en tiempo real / Technical Artist Unity enfocado en visualización técnica interactiva, optimización
 ↪ CAD-to-realtime y despliegue WebGL.

17.3.4 Do not position primarily as

Positioning	Reason
Full-stack developer	Dilutes strongest proof
3D artist generalist	Lowers compensation and weakens technical angle
AI engineer	ARA is not enough proof for ML/AI engineering
Game artist	TwinSight is stronger for technical/industrial visualization
Pure XR developer	XR is adjacent, not current strongest proof
Senior technical artist	Formal industry evidence is insufficient

17.4 4. Role targeting model

17.4.1 Primary role families

Priority	Role family	Reason
1	Real-Time 3D / Interactive 3D Developer	Strongest fit with TwinSight
2	Unity WebGL / Web 3D Developer	Direct fit with deployment medium
3	Technical Visualization / Digital Twin / Simulation Developer	Best strategic niche
4	Unity Technical Artist	Viable if framed around optimization/tools/runtime
5	Tools / Pipeline / Python Automation Developer	Secondary route supported by ARA

17.4.2 Secondary role families

Role	Use
XR / AR / VR Developer	Apply when Unity + technical visualization overlap
Product Visualization Developer	Good fit if WebGL/configurator oriented
Creative Technology Developer	Good if interactive 3D/WebGL strong
Technical 3D Generalist	Use only if technical systems are emphasized

17.4.3 Fallback role families

Role	Condition
Full-stack developer	Only under cashflow pressure
Python automation developer	Use when ARA is polished
3D artist / Blender generalist	Only if tied to technical art
Local web freelance	Cashflow only, not core strategy

17.5 5. Salary and work-model synthesis

17.5.1 Practical salary targets

Target	Timeframe	Realism	Best path
USD 1.5k/month	0–3 months	High	LATAM contractor, local remote, small Unity/WebGL contract
USD 3k/month	3–6 months	Medium	International contractor, LATAM nearshore, Unity/WebGL project
USD 6k/month	12–24 months	Low short-term / medium long-term	US/EU contractor, digital twin, simulation, XR, seniorizing path

17.5.2 Correct interpretation

USD 3k/month is not unrealistic. It requires:

- working portfolio;
- strong case study;
- consistent applications;
- contractor readiness;
- ability to interview in English;
- avoiding low-value local roles.

USD 6k/month is not impossible, but it should not be treated as the immediate baseline. It requires one or more of:

- better portfolio maturity;
- stronger interview performance;
- international contract access;
- niche technical visualization positioning;
- EU work authorization;
- stronger professional network;
- proven work history.

17.5.3 Preferred work model now

Model	Current usefulness
Foreign contractor / B2B	Highest short-term ROI
LATAM nearshore contractor	Realistic
Colombian employee	Stability, lower ceiling
EOR employee	Good if company supports it
Direct foreign employee	Usually difficult
Freelance marketplace	Useful for cashflow, noisy
EU employee	Future, after passport/residence

17.5.4 Gross-to-net warning

All compensation targets must be evaluated as gross and net.

Contractor income must account for:

- taxes;
- social security;
- payment platform fees;
- currency conversion;
- unpaid vacation;
- sick days;
- equipment/software;
- accountant/legal support.

A USD 3k/month contractor agreement is not the same as USD 3k/month net.

17.6 6. Colombia remote-readiness checklist

Before taking international contractor work, prepare:

Item	Status needed
RUT	Required
Electronic invoicing	Verify with accountant
CIIU activity	Choose correctly
Export-service treatment	Verify case-by-case
VAT/IVA responsibility	Verify
Social security	EPS, pension, ARL
Payment method	Wise/Payoneer/SWIFT/etc.
Contract template	Reviewable
Tax reserve	Monthly
Accountant	Strongly recommended

Do not treat payment platforms as tax compliance. They only move money.

17.7 7. EU / Portugal / Germany mobility synthesis

17.7.1 Current legal position

At present, the candidate should say:

Bloque de código: text
I am based in Colombia and available for remote contractor/B2B work. I do not currently hold EU work authorization.

17.7.2 Portuguese passport route

The Portuguese passport expected in ~2 years is strategically important, but it is not current authorization.

Once issued, it may unlock:

- right to live and work in Portugal;
- right to work across EU/EEA subject to local registration rules;
- easier EU applications;
- stronger recruiter response for EU roles;

- relocation without employer-sponsored visa in many cases.

Until issued, it should be described carefully:

Bloque de código: text
 I expect to obtain Portuguese citizenship/passport in approximately two years, but I am currently Colombia-based and available
 ↩ as a contractor.

17.7.3 Germany route through marriage/residence

Marriage to a German citizen can be a real route if relocation happens legally and the corresponding residence title is granted.

It can potentially unlock:

- residence in Germany;
- work authorization;
- local employment applications;
- access to German/EU employer market;
- more relevance for German language.

It should not be presented as current work authorization.

17.7.4 Blue Card

The EU Blue Card is an alternative route, but not the primary plan.

It is more relevant if:

- a German/EU employer offers a qualifying salary;
- the degree/qualification is accepted;
- the role fits shortage or skilled-worker requirements;
- relocation is planned.

17.7.5 Recruiter-facing work authorization statements

Scenario	Statement
Now	Colombia-based, available for remote contractor work
Applying to US remote	Available as independent contractor from Colombia
Applying to EU now	No current EU work authorization; remote contractor possible
After Portuguese passport	EU citizen, authorized to work in EU/EEA
After German residence	Authorized to work in Germany, if residence title grants work rights
German marriage not completed	Do not claim German work authorization

17.8 8. Language strategy synthesis

17.8.1 Priority order

Language	Priority	Reason
English	Immediate	Main language for remote roles
German	Conditional strategic	Useful if Germany route activates
Portuguese	Medium-term	Useful for Portugal/Brazil/integration, not main job lever
Spanish	Native advantage	LATAM, Spain, client communication

17.8.2 English

English should remain the highest immediate language priority.

Short-term focus:

- interview fluency;
- technical explanations;
- salary negotiation;
- written CV/LinkedIn/portfolio;
- demo narration;
- recruiter calls.

A formal English certificate is not urgent unless a visa, university or employer explicitly requests it.

17.8.3 German

German is not the highest ROI for the next 3–6 months if the goal is remote income from Colombia.

German becomes high ROI if:

- relocation to Germany becomes concrete;
- marriage/residence route is activated;
- local German employers become targets;
- integration and bureaucracy become immediate.

Recommended path:

```
Bloque de codigo: text
0-6 months: A1 survival if feasible
6-12 months: A2/B1 if Germany route remains likely
12-24 months: B1/B2 if relocating
```

17.8.4 Portuguese

Portuguese helps with:

- Portugal integration;
- Brazil market;
- bureaucracy;
- social life;
- future EU base.

It does not by itself create EU work authorization before the passport exists.

For Portugal, prioritize European Portuguese if relocation to Portugal is likely.

17.9 9. Education and course strategy synthesis

17.9.1 Main principle

Courses without certificates are still useful if they generate public evidence.

They are weak as credentials but useful as skill acquisition.

17.9.2 Correct rule

```
Bloque de codigo: text
Course value = output artifact, not course completion.
```

17.9.3 Short-term course policy

During the first 90 days, courses must not displace:

- TwinSight case study;
- portfolio MVP;
- CV/LinkedIn/GitHub cleanup;
- applications;
- interviews.

17.9.4 Rebelway

Rebelway is valuable if used for:

- technical art specialization;
- Houdini/procedural thinking;
- VFX reel fragments;
- real-time FX;
- future technical artist differentiation.

It should not become the main short-term path unless the target shifts toward VFX/FX technical art.

17.9.5 Certifications

Certification	Usefulness
Unity certification	Moderate if affordable and fast
English certificate	Only if needed
Blender certificate	Low compared to portfolio
Rebelway non-certified	Useful via artifacts
Google/Data certs	Low unless pivoting
Master's	Medium/long-term, not immediate
PhD	Not recommended now

17.9.6 Master's

A master's can be useful if it supports:

- EU relocation;
- computer graphics;
- simulation;
- XR;
- digital twins;
- research/industry hybrid;
- structured immigration.

It is not the fastest path to income.

17.9.7 PhD

A PhD should be avoided for now unless the target becomes research, academia, or advanced simulation/graphics R&D.

For the current goal, it delays income and creates opportunity cost.

17.10 10. Portfolio architecture synthesis

17.10.1 Portfolio hierarchy

Level	Project
Flagship	TwinSight X500
Secondary technical	ARA Framework
Supporting art/TA	Blender portrait
Optional	ai-news-aggregator
Defer	FitApp-Free
Hide/archive	Weak old repos

17.10.2 Portfolio MVP structure

```
Bloque de codigo: text
Home
Work
  TwinSight X500
  ARA Framework
  Blender portrait
About
Contact
```

17.10.3 Home-page message

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist
Interactive technical visualization
CAD-to-realtime optimization
Unity WebGL / C# / Blender / Python tooling
Available for remote contract work from Colombia
```

17.10.4 TwinSight must be visible first

TwinSight should have:

- hero image or video;
- short problem statement;
- technical role;
- pipeline;
- feature list;
- metrics;
- screenshots;
- demo video;
- code/architecture overview;
- limitations;
- AI disclosure.

17.10.5 ARA must not compete with TwinSight

ARA should be positioned as:

```
Bloque de codigo: text
AI-assisted research automation prototype / Python tooling project
```

Not as:

```
Bloque de codigo: text
main thesis
production AI platform
ML engineering proof
enterprise AI system
```

17.11 11. TwinSight case study synthesis

17.11.1 Public positioning

```
Bloque de codigo: text
TwinSight X500 is a Unity WebGL interactive 3D technical visualization prototype for inspecting and understanding the assembly
↳ of a Holybro X500 V2 drone. It converts complex technical documentation and CAD-derived assets into an optimized real-time
↳ 3D experience with component inspection, exploded view, cross-section tools, visual modes and academic usability
↳ evaluation.
```

17.11.2 Core case study narrative

```
Bloque de codigo: text
Seeing parts is not enough. Users need to understand spatial relationships.
```

17.11.3 Must-show evidence

Evidence	Purpose
Triangle reduction	Optimization proof
WebGL build	Deployment proof
Selection system	Interaction proof
Exploded view	Spatial reasoning proof
Cross-section	Technical visualization proof
Visual modes	Rendering/UI proof
SUS/NASA-TLX	Evaluation proof
Screenshots	Recruiter proof
Demo video	Fast communication
GitHub README	Technical proof

17.11.4 Interview explanation

The strongest answer structure:

```
Bloque de codigo: text
1. Problem
2. Constraint
3. Pipeline
4. Interaction systems
5. Optimization
6. Evaluation
7. Tradeoffs
8. What I would improve
```

17.11.5 AI disclosure

The correct stance:

```
Bloque de codigo: text
I used AI as an implementation assistant and debugging accelerator, but I was responsible for architecture decisions,
↳ integration, testing, technical judgment, validation and final ownership.
```

Avoid:

```
Bloque de codigo: text
AI built everything.
```

Also avoid:

```
Bloque de codigo: text
I wrote every line manually.
```

17.12 12. GitHub / LinkedIn / CV synthesis

17.12.1 GitHub

GitHub should communicate:

- technical seriousness;
- project clarity;
- readable READMEs;
- relevant pins;
- no clutter;
- no weak tutorial repos on the front page.

Pinned repos should prioritize:

1. TwinSight X500.
2. ARA Framework.
3. Portfolio site.
4. Blender technical breakdown.
5. ai-news-aggregator only if functional.
6. Optional tools/scripts repo.

17.12.2 LinkedIn headline

Recommended:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist | Unity WebGL, C#, CAD-to-Realtime Optimization, Technical Visualization
```

17.12.3 LinkedIn About structure

1. What you build.
2. Flagship project.
3. Technical stack.
4. What roles you target.
5. Availability.

17.12.4 CV structure

One-page base CV:

```
Bloque de codigo: text
Header
Professional Summary
Technical Skills
```

Selected Projects
Experience
Education
Courses / Training
Languages
Availability

17.12.5 CV variants

Create these first:

CV variant	Use
Real-Time 3D / Unity	Main
Technical Visualization / Digital Twin	Main
Unity WebGL / Interactive 3D	Main
Tools / Python Automation	Secondary

Do not create the full-stack fallback CV first.

17.13 13. Company targeting synthesis

17.13.1 Target categories

Use the section 11 target ecosystem as source.

Highest-priority categories:

1. Unity / real-time 3D studios.
2. Technical visualization / digital twin / simulation companies.
3. XR / AR / VR companies.
4. WebGL / Web 3D creative technology companies.
5. Product visualization / configurator companies.
6. LATAM nearshore companies.
7. Remote-first technical agencies.
8. Tools / Python automation companies as secondary.

17.13.2 Priority logic

Priority	Meaning
A	Strong fit, role plausible, portfolio relevant
B	Good fit, requires adaptation
C	Lower fit or weaker compensation
Watchlist	Future, EU, senior, or low probability

17.13.3 Application logic

A-priority applications require:

- tailored CV;
- tailored message;
- correct project link;
- company-specific reason;

- follow-up.

B-priority applications require:

- role-specific CV;
- short custom opener;
- portfolio link.

C-priority applications should not consume deep customization time.

17.14 14. Application system synthesis

17.14.1 Weekly target

Action	Weekly target
A-priority applications	5–10
B-priority applications	10–20
Recruiter outreach	10–20
Speculative outreach	5–10
Follow-ups	10–20
Portfolio improvement blocks	2–4

17.14.2 Tracker must include

- company;
- role;
- priority;
- location;
- remote scope;
- contractor option;
- salary range;
- work authorization notes;
- CV variant used;
- portfolio link used;
- application date;
- follow-up dates;
- response;
- interview stage;
- offer score;
- rejection reason.

17.14.3 Feedback loop

Signal	Meaning
No replies	Positioning or target problem
Replies but no interviews	Portfolio/CV weak
Interviews but rejection	Technical prep gap
Low salary offers	Wrong market
EU-only blockers	Authorization issue
Full-stack replies only	Profile diluted

17.15 15. Interview and negotiation synthesis

17.15.1 Preparation priorities

Prepare hardest for:

1. explaining TwinSight;
2. Unity optimization;
3. WebGL constraints;
4. CAD-to-realtime pipeline;
5. C# architecture;
6. technical UI;
7. AI-assisted development disclosure;
8. contractor availability;
9. salary expectations.

17.15.2 Take-home test policy

Accept only if:

- scope is limited;
- time requirement is reasonable;
- no production work is being extracted;
- ownership/IP terms are clear;
- the company is legitimate;
- the role is high enough priority.

Reject or renegotiate if:

- unpaid test exceeds reasonable scope;
- it resembles client work;
- it requires full feature delivery;
- no feedback is promised;
- company avoids contract clarity.

17.15.3 Negotiation stance

For contractor work:

Bloque de código: text
My expected range depends on scope, duration, time-zone overlap, and whether the role is contractor or employee. For
↪ international contractor work, I usually evaluate monthly retainers or hourly rates based on project complexity and
↪ expected availability.

Do not reveal minimum too early.

17.15.4 Offer scorecard

Score offers on:

- gross compensation;
- net compensation;
- role fit;
- portfolio leverage;
- contract safety;
- remote stability;

- learning ROI;
 - EU leverage;
 - payment reliability;
 - IP and exclusivity terms.
-

17.16 16. 30/60/90 execution synthesis

17.16.1 Days 1–30

Main objective: **build credible market presence.**

Deliverables:

- TwinSight case study;
- demo video draft;
- GitHub cleanup;
- LinkedIn rewrite;
- base CV;
- two CV variants;
- portfolio MVP;
- tracker;
- first applications.

17.16.2 Days 31–60

Main objective: **test market response.**

Actions:

- 50–90 total applications;
- 40–80 direct contacts;
- recruiter outreach;
- ARA cleanup decision;
- portfolio iteration;
- interview preparation;
- salary scripts.

17.16.3 Days 61–90

Main objective: **convert pipeline into interviews/offers.**

Actions:

- double down on role families generating replies;
 - remove weak paths;
 - prepare technical stories;
 - negotiate carefully;
 - evaluate offers with scorecard;
 - define next 90-day plan.
-

17.17 17. Strategic risks

17.17.1 Highest risks

Risk	Why it matters	Control
Strategic dispersion	Too many roles/projects	Role lock
Course hoarding	Delays portfolio	Artifact rule
Portfolio perfectionism	Delays applications	MVP deadline
Overclaiming	Interview credibility risk	Claim discipline
Underclaiming	Low salary offers	Metrics and proof
AI disclosure mishandling	Trust risk	Prepared answer
EU misstatement	Legal/recruiter risk	Accurate authorization text
Low-quality jobs	Career dilution	Offer scorecard
Weak GitHub	Technical credibility	Repo cleanup
No follow-up	Lost pipeline	Tracker cadence

17.17.2 Primary control mechanism

The tracker is the control system. Decisions must be based on data, not emotion.

17.18 18. What to do next

17.18.1 Immediate next actions

1. Finish thesis defense preparation.
2. Create TwinSight case study assets.
3. Record 60–120 second demo.
4. Clean GitHub.
5. Publish portfolio MVP.
6. Update LinkedIn.
7. Build base CV and two variants.
8. Start A/B company applications.
9. Track every action.
10. Review after 30 days.

17.18.2 Do not do next

Do not:

- start a new flagship project;
- overbuild ARA before TwinSight is public;
- spend weeks choosing portfolio tech;
- apply broadly as full-stack;
- start intensive German before job assets exist;
- wait for Portuguese passport before applying;
- make FitApp a portfolio priority;
- publish weak repos;
- accept vague unpaid trials.

17.19 19. Final recommended strategy

17.19.1 Best path

The best path is a **balanced but focused international contractor strategy**:

Bloque de código: text
 Use TwinSight X500 as proof for Unity / Real-Time 3D / Technical Visualization roles.
 Prioritize remote work from Colombia now.
 Build toward EU employability through future Portuguese passport and possible Germany residence.
 Use ARA and Python tooling only as secondary differentiation.
 Avoid profile dilution.

17.19.2 Core execution sentence

Bloque de código: text
 For the next 90 days, every action must either strengthen TwinSight as proof, improve recruiter-facing credibility, create
 ↪ interview opportunities, or increase contractor readiness.

17.19.3 Final target profile

Bloque de código: text
 Real-Time 3D Developer / Unity Technical Artist with a focus on interactive technical visualization, CAD-to-realtime
 ↪ optimization, Unity WebGL deployment, and Python-assisted production tooling.

17.19.4 Final role targets

1. Real-Time 3D / Interactive Technical Visualization Developer.
2. Unity WebGL / Interactive 3D Developer.
3. Technical Visualization / Digital Twin / Simulation Developer.
4. Unity Technical Artist focused on runtime, optimization and tooling.
5. Tools / Pipeline / Python Automation Developer as secondary path.

17.19.5 Final rule

The first job does not need to be the final destination.

It must satisfy:

- compensation above the minimum threshold;
- remote feasibility from Colombia;
- defensible contract terms;
- role fit with the real-time 3D direction;
- ability to build toward USD 3k–6k/month;
- no major strategic dilution.

If a role provides money but destroys positioning, treat it as a temporary cashflow decision, not as the main career strategy.

Chapter 18

16_final_consistency_audit_and_gap_register

Fuente: 16_final_consistency_audit_and_gap_register.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 740

Ultima modificacion: 2026-06-11 15:02

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This document is the final consistency audit before converting the research system into application materials.

Contenido completo depurado

18.0.1 Purpose

This document is the final consistency audit before converting the research system into application materials.

It checks whether the strategy files from 00 to 15 are coherent, actionable, and safe to use publicly. It does not rewrite the CV, LinkedIn, GitHub, portfolio, or outreach messages yet.

The goal is to identify:

- what is already closed;
- what is still uncertain;
- what requires external verification;
- what requires user confirmation;
- what should be produced next;
- what should not be claimed publicly;
- what could create legal, credibility, or execution risk.

18.1 1. Current document system

18.1.1 Completed strategic files

File	Status
00_master_audit_blueprint.md	Complete
01_source_of_truth_profile.md	Complete

File	Status
02_market_role_fit_and_positioning.md	Complete
03_salary_benchmark_and_remote_colombia.md	Complete, but requires future tax/accounting verification
04_eu_portugal_germany_mobility_strategy.md	Complete, but legal facts should be rechecked before relocation
05_language_strategy_and_roi.md	Complete
06_education_courses_rebelway_masters_phd.md	Complete
07_portfolio_strategy_and_project_architecture.md	Complete
08_twinsight_x500_case_study.md	Complete
09_ara_ai_tools_and_secondary_projects.md	Complete
10_github_linkedin_cv_strategy.md	Complete
11_company_targets_job_boards_recruiters.md	Complete as target ecosystem; needs live verification before application
12_application_system_and_tracker.md	Complete
13_interview_negotiation_and_offer_risk.md	Complete
14_30_60_90_execution_plan.md	Complete
15_final_integrated_strategy.md	Complete

18.1.2 Remaining operational files

These still need to be generated:

File	Purpose
17_cv_base_and_role_variants.md	CV base + role-specific variants
18_linkedin_final_rewrite.md	LinkedIn profile copy
19_github_readme_templates.md	GitHub profile + repo README templates
20_portfolio_copy_and_site_structure.md	Portfolio copy and structure
21_twinsight_demo_video_script_and_shotlist.md	Demo scripts and shot list
22_outreach_templates_english_spanish.md	Recruiter and company messages
23_interview_answer_bank.md	Interview answers and technical stories
24_offer_evaluation_scorecard.md	Offer scoring and decision system
25_application_tracker_template.xlsx	Spreadsheet tracker
26_weekly_execution_dashboard.md	Weekly execution dashboard

18.2 2. Core strategy consistency check

18.2.1 Final positioning

Current approved positioning:

Bloque de código: text
 Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization, ↔ and Unity WebGL deployment.

This is consistent across:

- role-market fit;
- portfolio strategy;
- TwinSight case study;
- GitHub/LinkedIn/CV strategy;
- company targeting;
- application plan;
- final integrated strategy.

18.2.2 Consistency result

Area	Status
Primary role identity	Consistent
Project hierarchy	Consistent
Salary targets	Mostly consistent
EU mobility	Consistent
Language strategy	Consistent
Course strategy	Consistent
Application system	Consistent
Interview/negotiation	Consistent
Company targeting	Useful but verification-limited

18.2.3 Main strategic thesis

The strategy is coherent if the profile is not diluted.

The strongest path is:

Bloque de codigo: text
 TwinSight X500 → Real-Time 3D / Unity WebGL / Technical Visualization → international contractor or remote role → later EU
 ↔ leverage through Portuguese passport or German residence.

This must remain the core execution line.

18.3 3. Contradiction audit

18.3.1 Resolved contradictions

Previous contradiction	Resolution
Thesis described as in-progress	Corrected: thesis complete, pending defense
ARA described as thesis	Corrected: TwinSight is the thesis/flagship
EU authorization implied	Corrected: no current EU work authorization
German relocation treated as immediate	Corrected: conditional future route
Portuguese passport treated as current	Corrected: expected future route
Courses treated as credentials	Corrected: skills/portfolio input, not certifications
Full-stack route overemphasized	Corrected: fallback only
3D artist route overemphasized	Corrected: supporting signal, not primary
AI tooling overclaimed	Corrected: secondary tooling route
CAD-to-realtime treated as role title	Corrected: differentiator/keyword, not main title

18.3.2 Remaining possible contradictions

Issue	Risk	Required action
Exact thesis metrics	Medium	Verify final thesis numbers
Public repo state	Medium	Confirm what is actually visible
Portfolio MVP availability	High	Build or publish it
GitHub cleanup status	High	Execute repo cleanup
Company list freshness	Medium	Verify top 30 targets before applying
Legal contractor setup	High	Consult accountant
EU legal route timing	Medium	Reconfirm before acting
Portuguese passport certainty	Medium	Treat as high probability, not current fact

18.4 4. Claims audit

18.4.1 Safe public claims

These can be used in CV, LinkedIn, GitHub, portfolio, and interviews if factually true:

- Unity WebGL technical visualization prototype.
- Real-time 3D drone assembly inspection.
- CAD-to-realtime asset optimization.
- C# runtime interaction systems.
- Component selection and technical UI.
- Exploded view and cross-section/clipping.
- Visual modes for inspection.
- Blender-based optimization and asset preparation.
- Thesis completed, pending defense.
- Academic usability evaluation if final data is confirmed.
- AI-assisted implementation under human technical ownership.
- Python automation prototype through ARA.

18.4.2 Conditional claims

Use only if confirmed:

Claim	Condition
Triangle reduction from 6.5M+ to 95,617	Verify exact source and final number
SUS 91.88	Verify final report
NASA-TLX 8.69 vs 19.89	Verify final report
96 task-condition records	Verify methodology
12 validation participants	Verify final sample
Live WebGL demo	Confirm public URL
Public GitHub code	Confirm repo state
ARA almost functional	Confirm reproducible run
Blender portrait complete	Confirm final renders/breakdown
English C1	Use as self-assessed unless certified
Portuguese passport in ~2 years	Phrase as expected, not guaranteed

18.4.3 Claims to avoid

Do not claim:

- “Senior Technical Artist.”
- “EU work authorization” before passport/residence exists.
- “German work authorization” before residence title exists.
- “Certified Unity Developer” unless certified.
- “Certified Rebelway / CG Cookie” if no certificate exists.
- “Production AI engineer.”
- “Enterprise-ready AI platform.”
- “Commercially shipped Unity product” unless true.
- “All code written manually” if AI assistance was used.
- “AI built the project” because it undermines technical ownership.
- “Full-stack specialist” as main positioning.

18.5 5. Source reliability audit

18.5.1 High-confidence internal facts

Fact	Confidence
Thesis status: complete, pending defense	High
TwinSight is flagship	High
UNAD Multimedia Engineering	High
UNAL Electronic Engineering coursework	High
Colombia-based remote preference	High
English C1 self-assessed	Medium-high
Portuguese passport expected	Medium-high
German partner route possible	Medium
ARA secondary project	High
Courses mostly non-certified	High

18.5.2 Medium-confidence research facts

Area	Reason
Salary ranges	Market varies by source
Contractor rates	Platform/vendor dependent
Company targets	Needs live verification
Job board quality	Changes over time
EU routes	Legal rules change
Portugal tax regimes	Changes frequently
German Blue Card thresholds	Annual updates
Colombian tax/admin rules	Needs accountant

18.5.3 Low-confidence or future-verification areas

Area	Reason
Exact remote policy per company	Changes constantly
Whether a company accepts Colombia contractors	Must verify per opening
Whether a role is worldwide remote	Must verify posting
Whether a salary applies to Colombia-based contractor	Usually not direct
Whether export-service VAT treatment applies	Case-specific
Whether ARA is public-ready	Requires technical audit
Whether portfolio is strong enough	Requires external review

18.6 6. Legal and administrative gap register

18.6.1 Colombia contractor setup

Item	Current status	Risk	Action
RUT	Unknown	High	Confirm/update
CIU code	Unknown	Medium	Select with accountant
Electronic invoicing	Unknown	High	Confirm requirement
IVA responsibility	Unknown	High	Verify with accountant
Export services treatment	Unknown	High	Verify per contract
Social security	Unknown	High	Define monthly process

Item	Current status	Risk	Action
ARL	Unknown	Medium	Verify by contract type
Payment method	Unknown	Medium	Choose Wise/Payoneer/SWIFT
Tax reserve	Unknown	High	Define percentage
Contract review process	Unknown	High	Create checklist

18.6.2 Required action

Before accepting international contractor work, consult a Colombian accountant or tax professional.

The strategy can proceed before that, but any signed contract must be checked.

18.6.3 Germany / Portugal

Item	Status	Action
Portuguese passport	Expected future	Track documentation/timeline
EU work rights	Not current	Do not claim
Germany marriage route	Possible	Verify legal requirements if activated
German residence	Not current	Do not claim
German tax/health system	Future	Research when relocation is real
Portugal tax status	Future	Research when move is planned

18.7 7. Portfolio readiness gap register

18.7.1 TwinSight

Asset	Status	Gap
Case study text	Planned	Needs final public version
Demo video	Planned	Needs recording/editing
Screenshots	Needed	Must capture final build
WebGL demo	Unknown	Public URL needed if available
GitHub README	Planned	Needs final version
Metrics table	Available in draft	Verify final numbers
Contribution statement	Defined	Needs public wording
AI disclosure	Defined	Needs placement
Asset disclaimer	Needed	Confirm CAD/manufacture permissions
Technical diagrams	Needed	Create pipeline/system diagrams

18.7.2 ARA

Asset	Status	Gap
Public repo	Unknown	Decide public/private
README	Needed	Draft controlled README
Demo	Unknown	Create screenshots or short run
Architecture diagram	Needed	Add if published
Claims	Risky	Avoid production/enterprise claims

18.7.3 Blender portrait

Asset	Status	Gap
Final renders	Unknown	Confirm
Breakdown	Needed	Topology/materials/grooming
ArtStation entry	Needed	Publish only if polished
Technical relevance	Medium	Frame as pipeline, not pure art

18.7.4 ai-news-aggregator

Asset	Status	Gap
Functional state	Unknown	Test
Public value	Unclear	Only include if demo works
Main profile relevance	Low-medium	Do not prioritize

18.7.5 FitApp-Free

Asset	Status	Gap
Functional state	Unknown	Defer
Portfolio relevance	Low	Do not prioritize
Strategy role	Distraction risk	Keep private for now

18.8 8. GitHub readiness gap register

18.8.1 Required GitHub actions

Action	Priority
Pin TwinSight first	Critical
Write TwinSight README	Critical
Add project images/GIFs	High
Hide weak tutorial repos	High
Archive old portfolio attempts	High
Clean repo descriptions	High
Add ARA README if public	Medium
Add GitHub profile README	Medium
Remove misleading thesis references	Critical
Add license/disclaimer where needed	Medium

18.8.2 GitHub risk

A weak GitHub front page can reduce credibility faster than having fewer public repos.

Quality beats quantity.

18.9 9. LinkedIn and CV gap register

18.9.1 LinkedIn gaps

Section	Required
Headline	Replace with final positioning
About	Rewrite around TwinSight and target roles
Experience	Reframe freelance/self-employed
Education	Correct thesis description
UNAL	Phrase as advanced coursework
Projects	Add TwinSight
Featured	Add portfolio/GitHub/demo
Skills	Reorder around Unity/C#/WebGL/Blender
Languages	English as professional/self-assessed
Work authorization	Colombia contractor now, EU future carefully

18.9.2 CV gaps

Asset	Required
Base CV	Needed
Unity/Real-Time 3D CV	Needed
Technical Visualization CV	Needed
WebGL/Interactive 3D CV	Needed
Tools/Python CV	Later
Full-stack fallback CV	Only if needed
Bullet bank	Needed
One-page format	Required

18.10 10. Company targeting gap register

18.10.1 Current status

Section 11 provides a useful target ecosystem but should not be treated as a fully verified live database.

18.10.2 Main limitations

Limitation	Risk
Careers pages may change	Medium
Remote policies may be outdated	High
LinkedIn URLs may need verification	Medium
Some companies may not hire juniors	High
Some roles may require EU/US authorization	High
Some companies are too senior/enterprise	Medium
Salary tiers are estimates	Medium

18.10.3 Future agent task

Use mode agent only for a narrow verification task:

```
Bloque de codigo: text
Verify top 30 A-priority companies:
- active roles;
- careers URL;
- remote scope;
- contractor possibility;
- seniority;
- work authorization;
- relevant contact/recruiter if public.
```

Do not ask the agent to rebuild the whole strategy.

18.11 11. Application system readiness

18.11.1 Ready

The application system is structurally ready.

It includes:

- A/B/C/Watchlist categories;
- tracker fields;
- weekly cadence;
- follow-up rules;
- customization depth;
- metrics;
- rejection criteria.

18.11.2 Missing operational artifact

Need actual spreadsheet:

```
Bloque de codigo: text
25_application_tracker_template.xlsx
```

Suggested sheets:

1. Applications.
2. Companies.
3. Contacts.
4. Interviews.
5. Offers.
6. Weekly Metrics.
7. Skill Sprint.
8. Content Assets.

This should be generated after CV/LinkedIn/portfolio materials are drafted.

18.12 12. Interview and negotiation readiness

18.12.1 Ready

The interview and negotiation framework is ready.

It covers:

- role-specific technical preparation;

- take-home test risk;
- contractor vs employee;
- salary expectation;
- offer scorecard;
- AI disclosure;
- red flags.

18.12.2 Missing operational artifact

Need:

```
Bloque de codigo: text
23_interview_answer_bank.md
```

This should include actual prepared answers for:

- “Tell me about yourself.”
- “Explain TwinSight.”
- “What did you build yourself?”
- “How did you use AI?”
- “Why this role?”
- “Why no formal industry experience?”
- “What salary do you expect?”
- “Are you authorized to work in the EU?”
- “Can you work as contractor from Colombia?”
- “What would you improve in TwinSight?”

18.13 13. Execution risk audit

18.13.1 Highest execution risks

Risk	Severity	Control
Not publishing portfolio	Critical	MVP deadline
Over-polishing TwinSight	High	80/20 case study
Applying before assets ready	Medium	Day-30 gate
Never applying	Critical	Minimum viable day
Course distraction	High	Artifact rule
New project distraction	High	Project freeze
Weak legal setup	High	Accountant
Weak interview answers	High	Answer bank
Low-quality offers	Medium	Offer scorecard
Company list decay	Medium	Top-30 verification

18.13.2 Main behavioral risk

The largest practical risk is **remaining in research/planning mode**.

The research base is sufficient. The next phase must produce public-facing assets.

18.14 14. Missing user information

18.14.1 Required before CV finalization

Data	Needed
Exact full name format	Yes
Preferred professional email	Yes
Phone/WhatsApp inclusion	Decide
City/country wording	Yes
GitHub URL	Confirm
LinkedIn URL	Confirm
Portfolio URL	If available
ArtStation URL	Confirm
Thesis defense date	If known
Graduation date	Expected
Freelance work dates	Exact
Client/project names	If public
YMCA/call center dates	If included
English proof	Certificate or self-assessed

18.14.2 Required before portfolio

Data	Needed
TwinSight public URL	If available
Repo final URL	Yes
Screenshots	Yes
Demo video	Yes
Final metrics	Yes
CAD/asset permission limits	Yes
AI disclosure preference	Yes
Portfolio platform	Choose

18.14.3 Required before contractor setup

Data	Needed
RUT status	Yes
Facturación electrónica status	Yes
Accountant	Yes/no
Payment method	Decide
Minimum net monthly	Define
Acceptable contract types	Define
Tax reserve estimate	Define with accountant

18.15 15. Recommended next production order

Do not generate all assets at once. Use this sequence:

18.15.1 Step 1

```
Bloque de codigo: text  
17_cv_base_and_role_variants.md
```

Why first:

- defines public claims;
- feeds LinkedIn;
- feeds outreach;
- feeds interview answers.

18.15.2 Step 2

```
Bloque de codigo: text  
18_linkedin_final_rewrite.md
```

Why second:

- converts CV logic into public profile;
- aligns recruiter discovery.

18.15.3 Step 3

```
Bloque de codigo: text  
19_github_readme_templates.md
```

Why third:

- makes technical proof readable;
- supports applications.

18.15.4 Step 4

```
Bloque de codigo: text  
20_portfolio_copy_and_site_structure.md
```

Why fourth:

- creates public destination.

18.15.5 Step 5

```
Bloque de codigo: text  
21_twinsight_demo_video_script_and_shotlist.md
```

Why fifth:

- strengthens portfolio and LinkedIn.

18.15.6 Step 6

```
Bloque de codigo: text  
22_outreach_templates_english_spanish.md
```

Why sixth:

- outreach should reference actual CV/portfolio.

18.15.7 Step 7

Bloque de codigo: text
23_interview_answer_bank.md

Why seventh:

- prepares conversion after outreach starts.

18.15.8 Step 8

Bloque de codigo: text
24_offer_evaluation_scorecard.md
25_application_tracker_template.xlsx
26_weekly_execution_dashboard.md

Why last:

- operational controls.

18.16 16. Tool recommendations for next files

File	Deep Research	Agent	Reason
17_cv_base_and_role_variants.md	No	No	Uses internal strategy
18_linkedin_final_rewrite.md	No	No	Uses internal strategy
19_github_readme_templates.md	No	No	Uses project details
20_portfolio_copy_and_site_structure.md	No	No	Uses project hierarchy
21_twinsight_demo_video_script_and_shotlist.md	No	No	Uses TwinSight details
22_outreach_templates_english_spanish.md	No	No	Uses company strategy
23_interview_answer_bank.md	No	No	Uses interview framework
24_offer_evaluation_scorecard.md	No	No	Uses section 13
25_application_tracker_template.xlsx	No	No	Use spreadsheet generation
26_weekly_execution_dashboard.md	No	No	Operational planning
Top-30 company verification	No	Yes	Narrow live verification

18.17 17. Go / no-go checklist before applying

18.17.1 Go if

- CV base ready.
- One role-specific CV ready.
- LinkedIn updated.
- TwinSight case study shareable.
- GitHub cleaned.
- Portfolio MVP or case-study page exists.
- Application tracker ready.
- Work authorization statement clear.
- Salary expectation prepared.
- AI disclosure prepared.

18.17.2 No-go if

- thesis still described incorrectly.
 - ARA is still presented as thesis.
 - GitHub front page looks unfocused.
 - portfolio has no TwinSight case study.
 - no demo/screenshots exist.
 - EU authorization is overstated.
 - contractor setup is misunderstood.
 - CV targets too many unrelated roles.
-

18.18 18. Final audit conclusion

The strategy is ready to move from research to asset production.

The remaining work is not more broad research. The remaining work is:

```
Bloque de codigo: text
public proof + application materials + execution system
```

The next file should be:

```
Bloque de codigo: text
17_cv_base_and_role_variants.md
```

No Deep Research. No agent.

The key constraint is claim discipline. Every public document must preserve the same message:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization,
↔ and Unity WebGL deployment.
```

The execution should now shift from “planning the plan” to producing the materials that allow applications to start.

Chapter 19

17_cv_base_and_role_variants

Fuente: 17_cv_base_and_role_variants.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1152

Ultima modificacion: 2026-06-11 15:06

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This document converts the approved career strategy into CV-ready material.

Contenido completo depurado

19.0.1 Purpose

This document converts the approved career strategy into CV-ready material.

It includes:

- a base one-page CV structure;
- role-specific CV variants;
- reusable bullet banks;
- project descriptions;
- skills blocks;
- education wording;
- training/course wording;
- language and work-authorization wording;
- claims to avoid;
- editing rules.

This is not the final designed PDF CV. It is the source text for producing one-page ATS-friendly resumes.

19.1 1. CV positioning rules

19.1.1 Core identity

Use this as the main professional identity:

Bloque de código: text
Real-Time 3D Developer / Unity Technical Artist

Expanded version:

Bloque de código: text
Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization, ↔ and Unity WebGL deployment.

19.1.2 Primary CV target families

Use separate variants for:

1. Real-Time 3D / Unity Developer.
2. Technical Visualization / Digital Twin / Simulation Developer.
3. Unity WebGL / Interactive 3D Developer.
4. Unity Technical Artist.
5. Tools / Python Automation Developer.

19.1.3 Do not use one generic CV for every job

The base CV should be adapted per role family. The same facts can be reorganized, but the claims must remain consistent.

19.1.4 Critical claim discipline

Do not claim:

- Senior Technical Artist.
- EU work authorization now.
- German residence now.
- Portuguese passport already issued.
- Unity certification if not obtained.
- Rebelway / CG Cookie / Alive certification if no certificate exists.
- Production AI engineer.
- Enterprise AI platform developer.
- Commercially shipped Unity product unless true.
- Full-stack specialist as primary identity.

19.2 2. Information still requiring user confirmation

Before creating a final PDF CV, confirm:

Field	Current placeholder
Professional email	[email]
Phone / WhatsApp	[phone]
Location line	Colombia or Medellín, Colombia
GitHub URL	'https://github.com/delarge95'
LinkedIn URL	'https://www.linkedin.com/in/alexander-woodcock-0132382a6/'
Portfolio URL	[portfolio URL]
ArtStation URL	'https://www.artstation.com/alexanderwoodcocksalomon3'
Thesis defense date	[pending defense date]
UNAD expected graduation date	[expected date]
Freelance project dates	[dates]

Field	Current placeholder
Client names	[public / private]
Exact TwinSight metrics	verify final report
Public WebGL demo URL	[demo URL if available]

19.3 3. Base CV — ATS-friendly one-page version

19.3.1 Header

Bloque de codigo: text

Alexander Woodcock Salomón
Real-Time 3D Developer / Unity Technical Artist
Colombia | Open to remote contractor roles
Email: [email] | Phone: [phone]
LinkedIn: <https://www.linkedin.com/in/alexander-woodcock-0132382a6/>
GitHub: <https://github.com/delarge95>
Portfolio: [portfolio URL]
ArtStation: <https://www.artstation.com/alexanderwoodcocksalomon3>

19.3.2 Professional summary

Bloque de codigo: text

Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization, and Unity WebGL deployment. Background in Multimedia Engineering with advanced Electronic Engineering coursework, combining Unity, C#, Blender, WebGL, technical UI, asset optimization, and Python-assisted tooling. Built TwinSight X500, a Unity WebGL technical visualization prototype for inspecting drone assembly through component selection, exploded views, cross-sections, visual modes, and usability evaluation.

Alternative shorter version:

Bloque de codigo: text

Unity-focused Real-Time 3D Developer specializing in interactive technical visualization, CAD-to-realtime optimization, WebGL deployment, and technical UI. Built TwinSight X500, a thesis project that converts drone assembly documentation and CAD-derived assets into an optimized interactive 3D inspection experience.

19.3.3 Technical skills

Bloque de codigo: text

Unity: C#, Unity WebGL, URP, UI Toolkit, Shader Graph, runtime interaction systems, component selection, visual modes, performance-conscious deployment

Real-Time 3D: CAD-to-realtime optimization, high-poly to low-poly workflow, retopology, UVs, baking, mesh optimization, technical visualization, WebGL constraints

3D Tools: Blender, Marmoset Toolbag, RizomUV, CAD-derived asset cleanup, hard-surface modeling, technical asset preparation

Programming & Tooling: C#, Python, Git/GitHub, AI-assisted coding workflows, basic FastAPI, LangGraph prototype experience, web automation concepts

Design & Evaluation: technical documentation, usability evaluation, SUS, NASA-TLX Raw, Think-Aloud, multimedia engineering, visual communication

Foundations: computer graphics, linear algebra, physics, electronics coursework, systems thinking

19.3.4 Selected projects

TwinSight X500 — Unity WebGL Technical Visualization Prototype

Bloque de codigo: text

Thesis project | Unity, C#, WebGL, URP, UI Toolkit, Blender

Bullets:

Bloque de codigo: text

- Built an interactive Unity WebGL 3D visualization prototype for inspecting and understanding the assembly of a Holybro X500 V2 drone.
- Developed runtime interaction systems for component selection, exploded view, cross-section/clipping, visual modes, and technical part information panels.
- Implemented a CAD-to-realtime asset workflow using Blender-based cleanup, optimization, retopology, UV preparation, and high-to-low asset processing.
- Optimized CAD-derived geometry from over 6.5M triangles to approximately 95,617 triangles. [verify final number]
- Designed a browser-accessible inspection experience focused on spatial understanding, technical clarity, and mobile/web performance constraints.
- Evaluated the prototype through usability and workload instruments, including SUS, NASA-TLX Raw, and Think-Aloud methodology. [verify final metrics]
- Used AI as an implementation and debugging assistant while retaining responsibility for architecture, integration, validation, technical judgment, and final ownership.

Shorter version for one-page CV:

Bloque de codigo: text

- Built TwinSight X500, a Unity WebGL technical visualization prototype for drone assembly inspection, including component selection, exploded view, cross-section tools, visual modes, and technical UI.
- Created a CAD-to-realtime pipeline using Blender optimization and asset preparation, reducing CAD-derived geometry from over 6.5M triangles to approximately 95,617 triangles. [verify final number]
- Evaluated the prototype using SUS, NASA-TLX Raw, and Think-Aloud methodology, connecting technical implementation with user-centered validation. [verify final metrics]

ARA Framework — AI-Assisted Research Automation Prototype

Bloque de codigo: text

Prototype | Python, LangGraph, LangChain, Redis, Semantic Scholar API, Playwright/MCP concepts

Bullets:

Bloque de codigo: text

- Developed a Python-based research automation prototype using LangGraph-style agent orchestration for niche analysis, literature research, technical architecture planning, implementation guidance, and report synthesis.
- Designed a multi-agent workflow with checkpointing, budget awareness, external research APIs, PDF/document processing concepts, and structured Markdown report generation.
- Positioned the system as a research-tooling prototype, not as a production AI platform, emphasizing automation architecture, workflow design, and technical documentation.

Shorter version:

Bloque de codigo: text

- Built ARA Framework, a Python/LangGraph research automation prototype for multi-agent literature analysis, technical planning, and structured report generation.

Hyperrealistic Blender Portrait — Technical Art Study

Bloque de codigo: text

Personal project | Blender, topology, materials, grooming, lighting, rendering

Bullets:

Bloque de codigo: text

- Completed a high-fidelity Blender portrait study focused on topology, material definition, grooming, lighting, and realistic rendering.
- Used the project as technical art evidence for 3D pipeline understanding rather than as a primary character-artist positioning asset.

ai-news-aggregator — Full-stack AI News Prototype

Bloque de codigo: text

Private / optional | FastAPI, React, PostgreSQL, Redis, Python

Bullets:

Bloque de codigo: text

- Developed a prototype full-stack AI news aggregation and analysis platform using a Python backend, database-backed storage, and a web frontend. [include only if functional/demo-ready]

19.3.5 Experience

Independent Technical Artist & Developer

Bloque de código: text
Freelance / Self-Directed Projects | Colombia | Mar 2024 – Present

Bullets:

Bloque de código: text
- Built portfolio and thesis projects combining Unity, C#, Blender, WebGL, Python automation, and technical visualization workflows.
- Developed interactive 3D systems, real-time visualization prototypes, and technical documentation for web-based and multimedia applications.
- Used AI-assisted development workflows to accelerate implementation, debugging, research, and documentation while maintaining final technical ownership.
- Completed web/freelance projects for small clients, covering implementation, design support, deployment, and client communication. [adjust with actual client details]

Earlier Professional Experience

Use only if space allows.

Bloque de código: text
Customer Service / English-speaking roles | [Company / YMCA / Call Center] | [dates]

Possible wording:

Bloque de código: text
- Developed professional English communication through customer-facing roles in international and bilingual environments.
- Built experience in client communication, service delivery, problem solving, and cross-cultural collaboration.

19.3.6 Education

Universidad Nacional Abierta y a Distancia — UNAD

Bloque de código: text
B.Eng. / Ingeniería Multimedia | Expected [date]

Bullets:

Bloque de código: text
- Thesis: TwinSight X500 – Unity WebGL interactive technical visualization prototype for drone assembly inspection.
- Relevant focus: computer graphics, multimedia systems, software development, technical documentation, usability evaluation, and interactive 3D applications.

Universidad Nacional de Colombia — UNAL

Recommended wording:

Bloque de código: text
Advanced Coursework in Electronic Engineering | 2013 – 2018

Bullets:

Bloque de código: text
- Completed advanced coursework in mathematics, physics, electronics, systems, simulation, programming logic, and engineering fundamentals.
- Academic background supports technical reasoning in real-time systems, optimization, hardware-aware thinking, and technical visualization.

Avoid:

Bloque de código: text
Bachelor's degree in Electronic Engineering

unless the degree was completed.

19.3.7 Courses and training

Use this section only if space allows.

Bloque de código: text
Selected non-certified training:
- CG Cookie HUMAN – High-Fidelity Character Pipeline & Topology
- Alive! – 3D Animation Mechanics & Dynamic Motion
- Rebelway courses – Houdini / real-time FX / procedural workflows [if completed or in progress]

Required wording:

Bloque de código: text
Non-certified training used for skill development and portfolio production.

Do not list them as “certifications.”

19.3.8 Languages

Bloque de código: text
Spanish – Native
English – Professional working proficiency / C1 self-assessed
German – Beginner / planned study
Portuguese – Planned / strategically relevant for future Portugal/EU route

If no certificate exists, avoid:

Bloque de código: text
English C1 certified

Use:

Bloque de código: text
English – C1 self-assessed; comfortable with technical interviews and documentation

19.3.9 Availability and work authorization

Current safe wording:

Bloque de código: text
Available for remote contractor/B2B work from Colombia.

Expanded version:

Bloque de código: text
Currently based in Colombia and available for remote contractor/B2B work. Expected future Portuguese citizenship/passport may
↪ support EU work authorization in the medium term, but current applications should be treated as Colombia-based
↪ remote/contractor availability.

Do not include all legal nuance in a one-page CV unless relevant.

19.4 4. CV Variant 1 — Real-Time 3D / Unity Developer

19.4.1 Target roles

Use for:

- Real-Time 3D Developer.
- Unity Developer.
- Interactive 3D Developer.
- Unity C# Developer.
- Unity Technical Developer.
- Interactive Visualization Developer.

19.4.2 Header title

Bloque de código: text

Real-Time 3D Developer | Unity, C#, WebGL, Technical Visualization

19.4.3 Summary

Bloque de código: text

Real-Time 3D Developer focused on Unity, C#, WebGL deployment, interactive technical visualization, and optimized 3D asset workflows. Built TwinSight X500, a Unity WebGL prototype for drone assembly inspection using component selection, exploded views, cross-section tools, visual modes, technical UI, and CAD-to-realtime optimization.

19.4.4 Skills emphasis

Prioritize:

Bloque de código: text

Unity, C#, Unity WebGL, URP, UI Toolkit, runtime systems, technical UI, asset optimization, Blender, Git/GitHub, performance constraints

De-emphasize:

Bloque de código: text

Python AI tooling, full-stack, Houdini, character art

19.4.5 TwinSight bullets for this variant

Bloque de código: text

- Built runtime interaction systems in Unity/C# for component selection, exploded views, cross-section/clipping, visual modes, and technical information panels.
- Designed the application for WebGL deployment, accounting for browser constraints, interaction clarity, asset optimization, and performance-conscious delivery.
- Integrated optimized 3D assets, UI Toolkit interfaces, and inspection workflows into a cohesive real-time technical visualization prototype.

19.4.6 ARA bullet for this variant

Bloque de código: text

- Developed Python automation prototypes to support research, documentation, and technical workflow acceleration.

19.4.7 Best keywords

```
Bloque de codigo: text
Unity
C#
Real-Time 3D
Interactive 3D
WebGL
URP
UI Toolkit
Runtime Systems
Technical Visualization
3D Optimization
CAD-to-Realtime
Blender
Git
```

19.5 5. CV Variant 2 — Technical Visualization / Digital Twin / Simulation

19.5.1 Target roles

Use for:

- Technical Visualization Developer.
- Digital Twin Visualization Developer.
- Simulation Developer.
- Industrial Visualization Developer.
- CAD Visualization Developer.
- Training Simulation Developer.
- Product/Assembly Visualization Developer.

19.5.2 Header title

```
Bloque de codigo: text
Technical Visualization Developer | Unity WebGL, CAD-to-Realtime, Simulation-Oriented 3D
```

19.5.3 Summary

```
Bloque de codigo: text
Technical Visualization Developer focused on transforming complex technical assets and documentation into interactive
↪ real-time 3D systems. Built TwinSight X500, a Unity WebGL drone assembly visualization prototype that combines
↪ CAD-to-realtime optimization, component inspection, exploded views, cross-sections, visual modes, and usability/workload
↪ evaluation.
```

19.5.4 Skills emphasis

Prioritize:

```
Bloque de codigo: text
Technical visualization, CAD-to-realtime, Unity WebGL, Blender optimization, component inspection, assembly visualization,
↪ cross-section tools, exploded view, usability evaluation, SUS, NASA-TLX
```

De-emphasize:

```
Bloque de codigo: text
Game development, pure 3D art, full-stack
```

19.5.5 TwinSight bullets for this variant

Bloque de código: text

- Converted CAD-derived drone assembly assets into an optimized Unity WebGL visualization prototype for inspection and spatial understanding.
- Built inspection features including component selection, exploded view, cross-section/clipping, visual modes, and technical part information panels.
- Connected technical implementation with evaluation methods, using SUS, NASA-TLX Raw, and Think-Aloud to compare user experience and perceived workload. [verify exact metrics]

19.5.6 ARA bullet for this variant

Bloque de código: text

- Built a Python-based research automation prototype supporting structured analysis, documentation, and technical report generation.

19.5.7 Best keywords

Bloque de código: text

Technical Visualization
Digital Twin
Simulation
CAD-to-Realtime
Unity
Unity WebGL
Interactive 3D
Industrial Visualization
Assembly Visualization
Cross-Section
Exploded View
Usability Evaluation
Technical Documentation

19.6 6. CV Variant 3 — Unity WebGL / Interactive 3D Developer

19.6.1 Target roles

Use for:

- Unity WebGL Developer.
- Interactive 3D Developer.
- Web 3D Developer.
- 3D Web Developer.
- WebGL Visualization Developer.
- Interactive Product / Technical Visualization Developer.

19.6.2 Header title

Bloque de código: text

Unity WebGL / Interactive 3D Developer | Browser-Based Technical Visualization

19.6.3 Summary

Bloque de código: text

Unity WebGL / Interactive 3D Developer focused on browser-accessible technical visualization, optimized 3D assets, and real-time interaction systems. Built TwinSight X500, a WebGL-based drone assembly inspection prototype with component selection, exploded view, cross-sections, visual modes, and technical UI.

19.6.4 Skills emphasis

Prioritize:

Bloque de código: text
 Unity WebGL, browser deployment, optimized assets, UI Toolkit, runtime interaction, technical UI, Blender optimization,
 ↪ WebAssembly/WebGL concepts, GitHub

De-emphasize:

Bloque de código: text
 XR, pure game development, offline rendering, full-stack unless role requires it

19.6.5 TwinSight bullets for this variant

Bloque de código: text
 - Built a Unity WebGL interactive 3D prototype designed for browser-based inspection of drone assembly components.
 - Developed interaction systems and technical UI for selecting, isolating, inspecting, and visually analyzing 3D components.
 - Prepared optimized 3D assets for web deployment, balancing visual clarity, geometry reduction, and interaction performance.

19.6.6 Optional web/full-stack bullet

Use only if relevant:

Bloque de código: text
 - Built supporting web prototypes using Python/FastAPI, React, PostgreSQL, and Redis concepts in private experimental
 ↪ projects. [include only if demonstrable]

19.6.7 Best keywords

Bloque de código: text
 Unity WebGL
 WebGL
 WebAssembly
 Interactive 3D
 Web 3D
 Browser-Based 3D
 Technical UI
 Unity C#
 3D Optimization
 Blender
 Product Visualization
 3D Configurator

19.7 7. CV Variant 4 — Unity Technical Artist

19.7.1 Target roles

Use for:

- Unity Technical Artist.
- Technical Artist.
- Technical Art Generalist.
- Runtime Technical Artist.
- Technical Artist — Optimization.
- Technical Artist — Tools.

19.7.2 Header title

Bloque de codigo: text
Unity Technical Artist | Real-Time 3D Optimization, Runtime Systems, Technical Visualization

19.7.3 Summary

Bloque de codigo: text
Unity Technical Artist focused on bridging 3D asset production, real-time optimization, runtime interaction systems, and
↪ technical visualization. Built TwinSight X500, a Unity WebGL drone assembly inspection prototype involving CAD-to-realtime
↪ asset preparation, technical UI, visual modes, component interaction, and performance-conscious deployment.

19.7.4 Skills emphasis

Prioritize:

Bloque de codigo: text
Unity, C#, Blender, technical art, optimization, retopology, UVs, baking, Shader Graph, visual modes, UI Toolkit, runtime
↪ interaction, Python tooling

De-emphasize:

Bloque de codigo: text
Senior shader specialization, Houdini FX unless completed, pure character art

19.7.5 TwinSight bullets for this variant

Bloque de codigo: text
- Built technical art workflows connecting CAD-derived geometry, Blender optimization, Unity integration, visual modes, and
↪ runtime interaction systems.
- Developed real-time inspection tools including exploded view, cross-section/clipping, component selection, and technical UI
↪ panels.
- Created optimized asset workflows for WebGL deployment, balancing geometry reduction, visual fidelity, and usability
↪ requirements.

19.7.6 Blender portrait bullets for this variant

Bloque de codigo: text
- Completed a Blender portrait study focused on topology, materials, grooming, lighting, and rendering, used as supporting
↪ evidence of 3D pipeline literacy.

19.7.7 ARA bullet for this variant

Bloque de codigo: text
- Built Python-based workflow automation prototypes that support research, technical documentation, and production-tooling
↪ thinking.

19.7.8 Best keywords

Bloque de codigo: text
Technical Artist
Unity Technical Artist
Real-Time 3D
Unity
C#
Shader Graph
URP
Blender
Optimization
Retopology
UVs
Baking

Runtime Tools
Technical UI
Python Tools

19.8 8. CV Variant 5 — Tools / Python Automation Developer

19.8.1 Target roles

Use selectively for:

- Python Automation Developer.
- Tools Developer.
- Pipeline Developer.
- Research Automation Developer.
- LLM Application Developer.
- AI Tools Developer.

This is a secondary route, not the main CV.

19.8.2 Header title

Bloque de codigo: text
Tools / Python Automation Developer | Research Automation, Technical Workflows, Real-Time 3D Support

19.8.3 Summary

Bloque de codigo: text
Tools / Python Automation Developer with a background in real-time 3D, technical visualization, and AI-assisted production
↪ workflows. Built ARA Framework, a Python/LangGraph research automation prototype, and TwinSight X500, a Unity WebGL
↪ technical visualization project demonstrating systems thinking, documentation, and technical integration.

19.8.4 Skills emphasis

Prioritize:

Bloque de codigo: text
Python, LangGraph, LangChain, Redis, APIs, automation, research tooling, technical documentation, Git/GitHub, AI-assisted
↪ workflows, Unity/C# as domain context

De-emphasize:

Bloque de codigo: text
Deep ML, model training, MLOps, data science, senior AI engineering

19.8.5 ARA bullets for this variant

Bloque de codigo: text
- Built ARA Framework, a Python-based research automation prototype using agent-style workflow orchestration for niche
↪ analysis, literature research, technical planning, and report synthesis.
- Designed a modular multi-agent architecture with checkpointing, structured outputs, API integrations, document-processing
↪ concepts, and Markdown report generation.
- Used LLM orchestration tools as workflow infrastructure while avoiding claims of ML model training or production AI
↪ engineering.

19.8.6 TwinSight bullet for this variant

Bloque de codigo: text
 - Applied automation-oriented technical thinking to a Unity WebGL visualization system, integrating runtime features,
 ↳ documentation, validation, and AI-assisted development workflows.

19.8.7 Best keywords

Bloque de codigo: text
 Python
 Automation
 LangGraph
 LangChain
 LLM Applications
 Research Automation
 Workflow Automation
 Technical Documentation
 APIs
 Redis
 Playwright
 GitHub
 AI-Assisted Development

19.9 9. Optional fallback CV — Full-stack Developer

Use only if cashflow pressure requires it.

19.9.1 Header title

Bloque de codigo: text
 Full-Stack / Python Developer with Real-Time 3D and Automation Background

19.9.2 Warning

This CV should not be used as the default because it weakens the main market positioning.

19.9.3 Summary

Bloque de codigo: text
 Python and web developer with experience building prototypes across automation, research tooling, and interactive 3D systems.
 ↳ Background includes FastAPI/React/PostgreSQL/Redis prototypes, Python automation, Git/GitHub, and Unity WebGL technical
 ↳ visualization.

19.9.4 Include only if demonstrable

- ai-news-aggregator;
- FastAPI;
- React;
- PostgreSQL;
- Redis;
- deployment;
- API design.

If these projects are not functional, do not use this variant aggressively.

19.10 10. Bullet bank by competency

19.10.1 Unity / C#

Bloque de código: text

- Developed runtime interaction systems in Unity/C# for component selection, inspection, exploded views, cross-section tools, ↩ and visual modes.
- Integrated technical UI panels using Unity UI Toolkit to support component-level inspection and assembly understanding.
- Built WebGL-oriented Unity systems with attention to browser constraints, asset weight, usability, and performance-conscious ↩ deployment.
- Structured interactive 3D features around inspection workflows rather than passive model viewing.

19.10.2 Technical visualization

Bloque de código: text

- Transformed technical assembly documentation into an interactive 3D visualization workflow focused on spatial understanding.
- Designed inspection modes that allow users to isolate, analyze, and understand relationships between drone components.
- Built visual analysis tools such as exploded view, ghosted/X-ray modes, wireframe/blueprint modes, and ↩ cross-section/clipping views.
- Connected real-time visualization features with usability and workload evaluation.

19.10.3 CAD-to-realtime

Bloque de código: text

- Processed CAD-derived geometry into optimized real-time assets for Unity WebGL deployment.
- Reduced dense technical geometry while preserving visual clarity and component identity. [insert verified metric]
- Used Blender-based cleanup, retopology, UV preparation, and high-to-low workflows to prepare assets for real-time rendering.
- Balanced technical fidelity, browser performance, and inspection clarity in the asset pipeline.

19.10.4 WebGL / browser deployment

Bloque de código: text

- Built a browser-accessible 3D prototype using Unity WebGL.
- Designed real-time interaction features while accounting for WebGL deployment constraints.
- Prepared optimized assets and UI for web/mobile-first inspection scenarios.
- Structured the experience to avoid requiring desktop-only CAD software or specialized viewers.

19.10.5 Evaluation / research

Bloque de código: text

- Evaluated the prototype using SUS, NASA-TLX Raw, and Think-Aloud methodology. [verify final data]
- Designed task-based evaluation conditions comparing interactive 3D visualization with conventional 2D support. [verify ↩ wording]
- Documented results, limitations, and technical tradeoffs as part of an academic thesis process.

19.10.6 AI-assisted development

Bloque de código: text

- Used AI-assisted development workflows to accelerate implementation, debugging, research, and documentation.
- Maintained responsibility for architecture decisions, integration, testing, validation, and final technical ownership.
- Applied AI as a productivity tool rather than as a replacement for technical judgment.

19.10.7 Python / tooling

Bloque de código: text

- Built Python automation prototypes for structured research, report generation, and technical workflow support.
- Designed modular agent-style workflows using LangGraph/LangChain concepts and structured Markdown outputs.
- Integrated research APIs, document-processing concepts, and checkpointed workflows into an automation prototype.

19.10.8 Communication

Bloque de codigo: text
 - Produced technical documentation connecting implementation decisions, usability constraints, evaluation methods, and project limitations.
 - Communicated complex technical workflows through structured documentation, visual breakdowns, and project case studies.
 - Developed professional English communication through international and bilingual work environments.

19.11 11. Education wording bank

19.11.1 UNAD

Bloque de codigo: text
 Universidad Nacional Abierta y a Distancia – UNAD
 B.Eng. / Ingeniería Multimedia
 Expected [date]

Description:

Bloque de codigo: text
 Relevant focus: computer graphics, interactive systems, multimedia engineering, software development, technical documentation, usability evaluation, and real-time 3D applications.

Thesis line:

Bloque de codigo: text
 Thesis: TwinSight X500 – Unity WebGL interactive technical visualization prototype for drone assembly inspection.

19.11.2 UNAL

Best wording:

Bloque de codigo: text
 Universidad Nacional de Colombia – UNAL
 Advanced Coursework in Electronic Engineering
 2013 – 2018

Description:

Bloque de codigo: text
 Completed advanced coursework in mathematics, physics, electronics, programming logic, systems thinking, and engineering fundamentals.

Avoid:

Bloque de codigo: text
 Bachelor of Electronic Engineering

unless degree completed.

19.12 12. Course/training wording bank

19.12.1 Safe format

Bloque de codigo: text
 Selected non-certified training:
 - CG Cookie HUMAN – High-Fidelity Character Pipeline & Topology
 - Alive! – 3D Animation Mechanics & Dynamic Motion
 - Rebelway – Houdini / Real-Time FX / Procedural Workflows [if completed or in progress]

19.12.2 Expanded wording

Bloque de código: text
Completed or currently using selected non-certified courses as skill-development inputs for portfolio production. These are
↔ not listed as formal certifications.

19.12.3 Avoid

Bloque de código: text
Certified in CG Cookie HUMAN
Certified in Rebelway Houdini
Certified 3D Character Artist

unless a verifiable certificate exists.

19.13 13. Work authorization wording

19.13.1 CV-safe short line

Bloque de código: text
Location: Colombia | Available for remote contractor/B2B roles

19.13.2 Expanded line for international applications

Bloque de código: text
Currently based in Colombia and available for remote contractor/B2B work. Future Portuguese citizenship/passport is expected
↔ in the medium term, but current availability should be treated as Colombia-based remote contractor status.

19.13.3 Do not write

Bloque de código: text
Authorized to work in the EU

until Portuguese passport or relevant residence authorization exists.

19.13.4 Germany-specific future wording

Only after legal status exists:

Bloque de código: text
Authorized to work in Germany under residence status as spouse of a German citizen.

Do not use before it is true.

19.14 14. CV length and formatting rules

19.14.1 One-page target

The default CV should be one page.

If needed, use a two-page technical resume only for:

- highly technical roles;
- technical artist roles requiring detailed portfolio breakdown;
- senior reviewers who explicitly ask for detail.

19.14.2 Recommended section order

```
Bloque de codigo: text
Header
Professional Summary
Technical Skills
Selected Projects
Experience
Education
Training
Languages / Availability
```

19.14.3 ATS formatting rules

Use:

- plain section headers;
- no icons;
- no columns if ATS compatibility is critical;
- simple bullets;
- standard file name;
- PDF for submission;
- DOCX only if requested.

File naming:

```
Bloque de codigo: text
Alexander_Woodcock_RealTime3D_Unity_CV.pdf
Alexander_Woodcock_Technical_Visualization_CV.pdf
Alexander_Woodcock_Unity_WebGL_CV.pdf
Alexander_Woodcock_Tools_Python_CV.pdf
```

19.15 15. CV customization rules

19.15.1 For A-priority applications

Customize:

- title;
- summary;
- top skills;
- first project bullet;
- keywords from job description;
- cover note / application message.

19.15.2 For B-priority applications

Customize:

- title;
- summary;
- first skills row;
- first 1–2 bullets.

19.15.3 For C-priority applications

Use closest existing variant.

Do not spend more than 10 minutes.

19.16 16. Red-flag edits to avoid

Do not:

- add too many unrelated technologies;
 - list every course;
 - list every repo;
 - exaggerate years of experience;
 - overemphasize full-stack;
 - overemphasize AI tooling;
 - overemphasize character art;
 - describe unfinished projects as complete;
 - hide the fact that TwinSight is academic;
 - make the CV read like a thesis abstract;
 - use dense paragraphs;
 - use vague phrases like “passionate,” “dynamic,” or “hardworking.”
-

19.17 17. Recommended final CV variants to produce first

19.17.1 Produce now

1. Alexander_Woodcock_RealTime3D_Unity_CV
2. Alexander_Woodcock_Technical_Visualization_CV
3. Alexander_Woodcock_Unity_WebGL_CV

19.17.2 Produce second

1. Alexander_Woodcock_Unity_Technical_Artist_CV
2. Alexander_Woodcock_Tools_Python_Automation_CV

19.17.3 Produce only if needed

1. Alexander_Woodcock_FullStack_Fallback_CV
-

19.18 18. Final CV readiness checklist

Before exporting final PDFs:

- Confirm email.
- Confirm phone.
- Confirm LinkedIn URL.
- Confirm GitHub URL.
- Confirm portfolio URL.
- Confirm thesis metrics.
- Confirm thesis defense/graduation wording.
- Confirm which freelance work can be public.
- Confirm ARA public/private status.
- Confirm if WebGL demo is live.
- Confirm exact language wording.
- Confirm work authorization line.

Proofread for overclaiming.
 Keep each variant one page.
 Export PDF.

19.19 19. Final recommended base CV text

Use this as the first draft for the main CV.

```
Bloque de código: text
Alexander Woodcock Salomón
Real-Time 3D Developer / Unity Technical Artist
Colombia | Available for remote contractor/B2B roles
[email] | [phone]
LinkedIn: https://www.linkedin.com/in/alexander-woodcock-0132382a6/
GitHub: https://github.com/delarge95
Portfolio: [portfolio URL]

PROFESSIONAL SUMMARY
Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization,
↪ and Unity WebGL deployment. Background in Multimedia Engineering with advanced Electronic Engineering coursework,
↪ combining Unity, C#, Blender, WebGL, technical UI, asset optimization, and Python-assisted tooling. Built TwinSight X500,
↪ a Unity WebGL technical visualization prototype for inspecting drone assembly through component selection, exploded views,
↪ cross-sections, visual modes, and usability evaluation.

TECHNICAL SKILLS
Unity / Real-Time 3D: Unity, C#, Unity WebGL, URP, UI Toolkit, Shader Graph, runtime interaction systems, component selection,
↪ exploded view, cross-section tools, visual modes.
3D Pipeline: Blender, CAD-to-realtime optimization, retopology, UVs, baking, high-poly to low-poly workflows, technical asset
↪ preparation.
Programming / Tools: Python, Git/GitHub, AI-assisted coding workflows, LangGraph prototype experience, basic
↪ FastAPI/React/PostgreSQL concepts.
Evaluation / Documentation: SUS, NASA-TLX Raw, Think-Aloud, technical documentation, usability evaluation, multimedia
↪ engineering.

SELECTED PROJECTS
TwinSight X500 – Unity WebGL Technical Visualization Prototype
Thesis project | Unity, C#, WebGL, URP, UI Toolkit, Blender
- Built an interactive Unity WebGL 3D visualization prototype for inspecting and understanding the assembly of a Holybro X500
↪ V2 drone.
- Developed runtime interaction systems for component selection, exploded view, cross-section/clipping, visual modes, and
↪ technical part information panels.
- Created a CAD-to-realtime workflow using Blender-based cleanup, optimization, retopology, UV preparation, and real-time
↪ asset processing.
- Reduced CAD-derived geometry from over 6.5M triangles to approximately 95,617 triangles. [verify final number]
- Evaluated the prototype through usability and workload instruments, including SUS, NASA-TLX Raw, and Think-Aloud
↪ methodology. [verify final metrics]

ARA Framework – AI-Assisted Research Automation Prototype
Python, LangGraph, LangChain, Redis, Semantic Scholar API, Playwright/MCP concepts
- Built a Python/LangGraph research automation prototype for multi-agent literature analysis, technical planning, and
↪ structured report generation.
- Designed a modular workflow for niche analysis, literature research, architecture planning, implementation guidance, and
↪ synthesis.
- Positioned the system as a research-tooling prototype, not as a production AI platform.

Independent Technical Artist & Developer
Freelance / Self-Directed Projects | Colombia | Mar 2024 – Present
- Built portfolio and thesis projects combining Unity, C#, Blender, WebGL, Python automation, and technical visualization
↪ workflows.
- Developed interactive 3D systems, real-time visualization prototypes, and technical documentation for web-based and
↪ multimedia applications.
- Used AI-assisted development workflows to accelerate implementation, debugging, research, and documentation while
↪ maintaining final technical ownership.

EDUCATION
Universidad Nacional Abierta y a Distancia – UNAD
B.Eng. / Ingeniería Multimedia | Expected [date]
- Thesis: TwinSight X500 – Unity WebGL interactive technical visualization prototype for drone assembly inspection.

Universidad Nacional de Colombia – UNAL
Advanced Coursework in Electronic Engineering | 2013 – 2018
- Completed advanced coursework in mathematics, physics, electronics, programming logic, systems thinking, and engineering
↪ fundamentals.

SELECTED TRAINING
- CG Cookie HUMAN – High-Fidelity Character Pipeline & Topology. Non-certified training.
- Alive! – 3D Animation Mechanics & Dynamic Motion. Non-certified training.
- Rebelway – Houdini / Real-Time FX / Procedural Workflows. [if completed/in progress]

LANGUAGES
Spanish – Native
English – Professional working proficiency / C1 self-assessed
German – Beginner
Portuguese – Planned / strategically relevant for future Portugal/EU route
```

19.20 20. Next file dependency

The next file should be:

```
Bloque de codigo: text  
18_linkedin_final_rewrite.md
```

It should reuse:

- the main positioning;
- the CV summary;
- the TwinSight bullets;
- the work-authorization wording;
- the education wording;
- the course wording.

No Deep Research or agent required.

Chapter 20

18_linkedin_final_rewrite

Fuente: 18_linkedin_final_rewrite.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1285

Ultima modificacion: 2026-06-11 15:08

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This document converts the approved career strategy and CV positioning into a complete LinkedIn profile rewrite.

Contenido completo depurado

20.0.1 Purpose

This document converts the approved career strategy and CV positioning into a complete LinkedIn profile rewrite.

It includes:

- final headline recommendation;
- 15 headline alternatives;
- About section;
- Experience rewrite;
- Education rewrite;
- Projects section;
- Featured section order;
- Skills ordering;
- Creator/activity strategy;
- recruiter keyword strategy;
- availability and work-authorization wording;
- future EU/Portugal/Germany wording;
- sections to avoid or downplay.

This document is written in English because the target market is international remote work.

20.1 1. LinkedIn positioning rule

The LinkedIn profile must communicate one coherent identity:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization,
↔ and Unity WebGL deployment.
```

It must not present separate unrelated identities such as:

```
Bloque de codigo: text
Full-stack developer + AI engineer + 3D artist + game developer + researcher + designer
```

Those areas can exist as supporting evidence, but not as equal headline identities.

20.2 2. Final recommended headline

Use this as the primary headline:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist | Unity WebGL, C#, CAD-to-Realtime Optimization, Technical Visualization
```

20.2.1 Why this headline works

It balances:

- role identity;
- engine/tool keywords;
- technical visualization niche;
- recruiter-search keywords;
- portfolio alignment with TwinSight X500;
- enough specificity without overclaiming seniority.

It avoids:

- “Senior”;
 - “AI Engineer” as primary identity;
 - “Full-stack” as primary identity;
 - vague “multimedia engineer” wording;
 - generic “3D artist” positioning.
-

20.3 3. Headline alternatives

Use one depending on the target market.

20.3.1 Best overall

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist | Unity WebGL, C#, CAD-to-Realtime Optimization, Technical Visualization
```

20.3.2 More technical visualization oriented

```
Bloque de codigo: text
Technical Visualization Developer | Unity WebGL, C#, Interactive 3D, CAD-to-Realtime Optimization
```

20.3.3 More Unity developer oriented

Bloque de código: text
Unity Developer / Real-Time 3D Developer | WebGL, C#, URP, Interactive Technical Visualization

20.3.4 More Technical Artist oriented

Bloque de código: text
Unity Technical Artist | Real-Time 3D Optimization, Blender, Shader Graph, Technical Visualization

20.3.5 More WebGL oriented

Bloque de código: text
Unity WebGL / Interactive 3D Developer | Browser-Based Technical Visualization, C#, Blender

20.3.6 More digital twin oriented

Bloque de código: text
Real-Time 3D Developer | Unity, WebGL, CAD-to-Realtime, Digital Twin & Simulation Visualization

20.3.7 More tools oriented

Bloque de código: text
Unity Technical Artist & Tools Developer | C#, Python Automation, Real-Time 3D Workflows

20.3.8 More portfolio oriented

Bloque de código: text
Real-Time 3D Developer | Building Unity WebGL Technical Visualization Systems

20.3.9 More international recruiter oriented

Bloque de código: text
Unity / Real-Time 3D Developer | Technical Visualization, WebGL, C# | Remote Contractor

20.3.10 More concise

Bloque de código: text
Real-Time 3D Developer | Unity WebGL, C#, Technical Visualization

20.3.11 More industrial visualization oriented

Bloque de código: text
Interactive 3D Developer | Unity, CAD-to-Realtime, Industrial & Technical Visualization

20.3.12 More XR-adjacent

Bloque de código: text
Unity Developer | Real-Time 3D, WebGL, XR-Ready Technical Visualization

20.3.13 More academic-thesis oriented

Bloque de codigo: text
Unity WebGL Developer | Technical Visualization Thesis: Interactive 3D Drone Assembly Inspection

20.3.14 More AI/tooling secondary

Bloque de codigo: text
Real-Time 3D Developer | Unity WebGL, Technical Visualization, Python-Assisted Tooling

20.3.15 More broad but still controlled

Bloque de codigo: text
Technical Artist / Real-Time 3D Developer | Unity, Blender, WebGL, Interactive Visualization

20.4 4. LinkedIn About section

20.4.1 Recommended final version

Bloque de codigo: text
I build interactive real-time 3D systems for technical visualization using Unity, C#, WebGL and Blender.

My main project is TwinSight X500, a Unity WebGL technical visualization prototype for inspecting and understanding the
↪ assembly of a Holybro X500 V2 drone. The project converts complex technical documentation and CAD-derived assets into an
↪ optimized interactive 3D experience with component selection, exploded views, cross-section tools, visual modes, technical
↪ UI, and usability/workload evaluation.

My strongest areas are:

- Unity / C# runtime interaction systems
- Unity WebGL deployment
- CAD-to-realtime asset optimization
- Blender-based technical asset preparation
- Interactive technical visualization
- Real-time inspection workflows
- Technical UI and documentation
- Python-assisted tooling and automation

I am currently completing my Multimedia Engineering degree at UNAD. I also completed advanced coursework in Electronic
↪ Engineering at Universidad Nacional de Colombia, which supports my technical foundation in mathematics, physics, systems
↪ thinking and engineering logic.

I am based in Colombia and available for remote contractor/B2B roles in real-time 3D, Unity, WebGL, technical visualization,
↪ digital twin visualization, simulation, XR-adjacent development, and technical art workflows.

I also build secondary tooling prototypes with Python and LLM orchestration, including ARA Framework, a research automation
↪ prototype. I treat AI as a productivity and implementation assistant, while maintaining responsibility for architecture,
↪ integration, testing, technical judgment and final ownership.

20.4.2 Shorter About version

Use if a more concise profile is preferred:

Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization,
↪ and Unity WebGL deployment.

My flagship project is TwinSight X500, a Unity WebGL drone assembly visualization prototype that transforms technical
↪ documentation and CAD-derived assets into an optimized interactive 3D inspection experience. It includes component
↪ selection, exploded views, cross-section tools, visual modes, technical UI, and academic usability/workload evaluation.

Core stack: Unity, C#, Unity WebGL, URP, UI Toolkit, Shader Graph, Blender, CAD-to-realtime optimization, Git/GitHub,
↪ Python-assisted tooling.

I am based in Colombia and available for remote contractor/B2B roles in Unity, real-time 3D, technical visualization, digital
↪ twin/simulation visualization, and interactive WebGL experiences.

20.4.3 More recruiter-optimized About version

Bloque de codigo: text

I am a Colombia-based Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, C#, CAD-to-realtime optimization and interactive technical visualization.

My strongest project is TwinSight X500, a Unity WebGL technical visualization prototype for drone assembly inspection. The project includes real-time component selection, exploded view, cross-section/clipping tools, visual inspection modes, technical UI panels, optimized CAD-derived assets, and usability/workload evaluation.

Relevant areas:
Unity • C# • Unity WebGL • URP • UI Toolkit • Shader Graph • Blender • CAD-to-Realtime • Technical Visualization • Digital Twin Visualization • Simulation • Interactive 3D • Python Automation • Git/GitHub

I am open to remote contractor/B2B opportunities from Colombia, especially roles involving Unity, real-time 3D, WebGL, technical visualization, product/industrial visualization, simulation interfaces, XR-adjacent development, and technical art workflows.

20.5 5. Experience section rewrite

20.5.1 Current role

Use this as the main experience entry:

Bloque de codigo: text

Independent Technical Artist & Developer
Freelance / Self-Directed Projects
Mar 2024 - Present
Colombia • Remote / Hybrid

20.5.2 Description version 1 — best general version

Bloque de codigo: text

Developing real-time 3D, Unity WebGL, technical visualization and automation projects focused on interactive systems, optimized 3D assets and technical documentation.

Key work:

- Built TwinSight X500, a Unity WebGL technical visualization prototype for inspecting and understanding the assembly of a Holybro X500 V2 drone.
- Developed runtime interaction systems in Unity/C# for component selection, exploded view, cross-section/clipping, visual modes and technical information panels.
- Created a CAD-to-realtime asset workflow using Blender-based cleanup, optimization, retopology, UV preparation and real-time asset processing.
- Designed the project for browser-accessible inspection, balancing visual clarity, technical accuracy, interaction design and WebGL performance constraints.
- Evaluated the prototype using usability and workload methods including SUS, NASA-TLX Raw and Think-Aloud methodology.
- [verify final metrics before publishing]
- Built ARA Framework, a Python/LangGraph research automation prototype for multi-agent literature analysis, technical planning and structured report generation.
- Used AI-assisted development workflows to accelerate implementation, debugging, documentation and research while retaining final responsibility for architecture, integration, validation and technical judgment.

20.5.3 Description version 2 — more concise

Bloque de codigo: text

Working on real-time 3D, Unity WebGL, technical visualization and automation projects.

Main project: TwinSight X500, a Unity WebGL drone assembly visualization prototype with component selection, exploded view, cross-section tools, visual modes, technical UI, CAD-to-realtime optimization and usability/workload evaluation.

Additional work includes Python-based research automation prototypes, Blender technical art studies and selected web/software projects.

20.5.4 Description version 3 — contractor-friendly

Bloque de codigo: text

Independent developer focused on Unity, real-time 3D, WebGL, technical visualization, 3D optimization and Python-assisted tooling.

Available for remote contractor/B2B projects involving:

- Unity/C# development
- Unity WebGL deployment
- Technical visualization
- CAD-to-realtime asset workflows
- Interactive 3D UI
- Real-time inspection tools
- Blender optimization pipelines
- Python automation and technical documentation

20.6 6. Project entries

LinkedIn Projects should be used because they allow the flagship work to become visible outside the Experience section.

20.6.1 Project 1 — TwinSight X500

Project name

Bloque de código: text
TwinSight X500 – Unity WebGL Technical Visualization Prototype

Associated with

Bloque de código: text
Universidad Nacional Abierta y a Distancia – UNAD

or

Bloque de código: text
Independent Technical Artist & Developer

If LinkedIn allows both, associate with UNAD if emphasizing thesis; associate with self-employed role if emphasizing portfolio.

Date

Bloque de código: text
[Start date] – [Completion date / Present if still polishing public version]

Project URL

Bloque de código: text
[Portfolio case study URL]
[GitHub URL]
[Demo URL if available]

Description

Bloque de código: text
TwinSight X500 is a Unity WebGL interactive technical visualization prototype for inspecting and understanding the assembly of
↪ a Holybro X500 V2 drone.

The project transforms complex technical documentation and CAD-derived assets into an optimized real-time 3D inspection
↪ experience. It includes component selection, exploded view, cross-section/clipping tools, visual modes, technical
↪ information panels, WebGL deployment and academic usability/workload evaluation.

My work included Unity/C# implementation, runtime interaction systems, technical UI, CAD-to-realtime asset optimization, ↩ Blender-based preparation, documentation, evaluation design and final integration.

Key areas:
Unity WebGL · C# · URP · UI Toolkit · Blender · CAD-to-Realtime · Technical Visualization · 3D Optimization · Usability
↩ Evaluation

Shorter description

Bloque de codigo: text
Unity WebGL technical visualization prototype for drone assembly inspection. Includes component selection, exploded view, ↩ cross-section tools, visual modes, technical UI, CAD-to-realtime optimization and usability/workload evaluation.

20.6.2 Project 2 — ARA Framework

Project name

Bloque de codigo: text
ARA Framework – AI-Assisted Research Automation Prototype

Project URL

Bloque de codigo: text
[GitHub URL if public]
[Demo/screenshots if available]

Description

Bloque de codigo: text
ARA Framework is a Python-based research automation prototype exploring multi-agent workflows for niche analysis, literature ↩ research, technical planning, implementation guidance and structured report generation.

The system uses LangGraph/LangChain-style orchestration concepts, external research APIs, document-processing concepts, ↩ checkpointing and Markdown report generation.

Positioned as a tooling and automation prototype, not as a production AI platform.

Key areas:
Python · LangGraph · LangChain · LLM Orchestration · Research Automation · Technical Documentation · Workflow Automation

20.6.3 Project 3 — Blender Portrait Technical Study

Project name

Bloque de codigo: text
Hyperrealistic Blender Portrait – Technical Art Study

Description

Bloque de codigo: text
High-fidelity Blender portrait study focused on topology, material definition, grooming, lighting and rendering. Used as ↩ supporting technical art evidence for 3D pipeline understanding rather than as a primary character-artist positioning ↩ asset.

Key areas:
Blender · Topology · Materials · Grooming · Lighting · Rendering · Technical Art

20.7 7. Education rewrite

20.7.1 UNAD

School

Bloque de código: text
Universidad Nacional Abierta y a Distancia – UNAD

Degree

Bloque de código: text
Ingeniería Multimedia / Multimedia Engineering

Dates

Bloque de código: text
Aug 2021 – [Expected graduation date]

Description

Bloque de código: text
Focus on multimedia engineering, interactive systems, computer graphics, software development, technical documentation,
↔ usability evaluation and real-time 3D applications.

Thesis: TwinSight X500 – Unity WebGL interactive technical visualization prototype for drone assembly inspection.

The thesis explores how interactive real-time 3D visualization can support spatial understanding of technical assembly
↔ processes through component inspection, exploded views, cross-section tools, visual modes and usability/workload
↔ evaluation.

20.7.2 Important correction

Do not describe ARA Framework as the thesis if TwinSight X500 is the thesis.

20.7.3 UNAL

School

Bloque de código: text
Universidad Nacional de Colombia

Degree / wording

Recommended:

Bloque de código: text
Advanced Coursework in Electronic Engineering

or:

Bloque de código: text
Coursework toward B.Eng. in Electronic Engineering

Dates

Bloque de código: text
Mar 2013 – Nov 2018

Description

Bloque de código: text
Completed advanced coursework in mathematics, physics, electronics, programming logic, systems thinking, simulation and
↪ engineering fundamentals.

This background supports technical reasoning in real-time systems, optimization, hardware-aware thinking, technical
↪ visualization and structured problem solving.

20.7.4 Avoid

Bloque de código: text
Bachelor of Electronic Engineering

unless the degree was completed.

20.8 8. Volunteering / research incubator

If keeping the ARCADE Research Incubator entry:

20.8.1 Title

Bloque de código: text
Student Researcher

20.8.2 Organization

Bloque de código: text
ARCADE Research Incubator, UNAD Colombia

20.8.3 Description

Bloque de código: text
Participation in academic research and technical exploration activities related to multimedia engineering, interactive
↪ systems, technical documentation and applied research workflows.

If the work is not very relevant or not active, keep it short.

20.9 9. Featured section order

LinkedIn Featured should make the profile readable in under 30 seconds.

20.9.1 Recommended order

1. TwinSight X500 portfolio case study.
2. TwinSight X500 demo video.
3. TwinSight X500 GitHub repository.
4. ARA Framework GitHub / demo.
5. Blender portrait ArtStation breakdown.
6. Personal portfolio homepage.

20.9.2 Do not feature

- weak repos;
 - old portfolio attempts;
 - unfinished FitApp;
 - tutorial repositories;
 - unrelated low-quality freelance websites;
 - generic certificates without relevance.
-

20.10 10. Skills ordering

LinkedIn skills should reflect the target roles.

20.10.1 Top 20 recommended skills

1. Unity
2. C#
3. Real-Time 3D
4. Technical Art
5. Unity WebGL
6. Technical Visualization
7. WebGL
8. Blender
9. 3D Optimization
10. CAD-to-Realtime
11. UI Toolkit
12. Shader Graph
13. URP
14. Python
15. Git
16. GitHub
17. Computer Graphics
18. Interactive 3D
19. Technical Documentation
20. Usability Evaluation

20.10.2 Secondary skills

```
Bloque de codigo: text
LangGraph
LangChain
LLM Applications
Automation
FastAPI
React
```

```
PostgreSQL
Redis
XR
AR/VR
Retopology
UV Mapping
Baking
Hard-Surface Modeling
```

20.10.3 Skills to avoid overemphasizing

Skill	Reason
Full-stack Development	Dilutes main target
Graphic Design	Not core
Social Media	Not relevant
General Game Development	Too broad
AI Engineering	Overclaim risk
Data Analytics	Not current target
Character Art	Supporting only

20.11 11. Services section

If LinkedIn allows services, use only focused services.

20.11.1 Recommended services

```
Bloque de codigo: text
Unity Development
Technical Art
3D Visualization
Interactive 3D
WebGL Development
3D Asset Optimization
Technical Documentation
Python Automation
```

Do not include:

```
Bloque de codigo: text
Graphic design
Social media design
Generic web design
Data analytics
AI consulting
```

unless intentionally pursuing those as paid side work.

20.12 12. Open to work settings

20.12.1 Recommended job titles

Use a controlled list:

```
Bloque de codigo: text
Real-Time 3D Developer
Unity Developer
Unity Technical Artist
Technical Artist
Technical Visualization Developer
```

```
Interactive 3D Developer
Unity WebGL Developer
Simulation Developer
XR Developer
Tools Developer
Python Automation Developer
```

Do not add too many unrelated titles.

20.12.2 Recommended locations

```
Bloque de codigo: text
Remote
Colombia
Latin America
United States remote
Canada remote
Europe remote
Germany future
Portugal future
```

If LinkedIn forces specific locations, prioritize:

```
Bloque de codigo: text
Remote
Colombia
United States
Canada
Germany
Portugal
Netherlands
Spain
United Kingdom
```

20.12.3 Employment types

```
Bloque de codigo: text
Contract
Full-time
Freelance
Temporary
```

Contract should be included because it is the most realistic immediate international path.

20.13 13. Work authorization wording

20.13.1 For LinkedIn About

Use:

```
Bloque de codigo: text
I am based in Colombia and available for remote contractor/B2B roles.
```

20.13.2 For recruiter messages

Use:

```
Bloque de codigo: text
I am currently based in Colombia and can work remotely as a contractor/B2B provider. I do not currently hold EU work
↪ authorization, but I expect a future Portuguese passport in the medium term and may also have a Germany relocation route
↪ through family residence.
```

20.13.3 For profile itself

Do not over-detail the legal situation in the headline or first line.

Keep it simple:

```
Bloque de codigo: text
Colombia-based · Open to remote contractor roles
```

20.13.4 Do not say

```
Bloque de codigo: text
Authorized to work in the EU
```

until legally true.

20.14 14. Future EU / Portugal / Germany wording

20.14.1 Safe wording

```
Bloque de codigo: text
Medium-term EU mobility is strategically relevant due to an expected Portuguese passport and a possible Germany relocation
↔ route through family residence. Current availability remains Colombia-based remote contractor work.
```

20.14.2 Shorter wording

```
Bloque de codigo: text
Remote from Colombia now; EU mobility expected to become relevant in the medium term.
```

20.14.3 Avoid

```
Bloque de codigo: text
I can work in Europe.
I have EU work authorization.
I am relocating to Germany.
I am a Portuguese citizen.
```

unless those are already legally true.

20.15 15. AI-assisted development disclosure on LinkedIn

Do not overexplain AI in the About section. Use it as a controlled secondary point.

20.15.1 Recommended phrasing

```
Bloque de codigo: text
I use AI-assisted workflows for implementation support, debugging, research and documentation, while retaining responsibility
↔ for architecture, integration, validation and final technical ownership.
```

20.15.2 Avoid

```
Bloque de codigo: text
AI built my project.
Vibe-coded with AI.
I am an AI engineer because I use ChatGPT.
```


20.15.3 Where to include it

Best places:

- About section, last paragraph.
- ARA project.
- Interview answers.
- GitHub README disclosure.

Not headline.

20.16 16. Activity strategy

The goal is to look active and technically credible, not inspirational.

20.16.1 Posting cadence

Type	Frequency
TwinSight progress/breakdown	1× per week
Technical note	1× per week
Portfolio update	1× every 2 weeks
Job-search/open-to-work post	1× after assets are ready
Commenting on relevant posts	3–5× per week

20.16.2 Post categories

Category 1 — TwinSight technical breakdown

Examples:

```
Bloque de codigo: text
- CAD-to-realtime optimization: what changed from CAD mesh to WebGL-ready asset.
- Why exploded views help understand assembly.
- Cross-section tools for technical inspection.
- How I structured component selection in Unity.
- WebGL constraints for technical visualization.
```

Category 2 — Technical visualization

```
Bloque de codigo: text
- Why technical visualization is different from showing a nice 3D model.
- What interactive 3D can add to technical documentation.
- Lessons from building a drone assembly viewer.
```

Category 3 — Portfolio updates

```
Bloque de codigo: text
- Published TwinSight X500 case study.
- Added demo video.
- Added GitHub README.
- Added ARA Framework as secondary tooling project.
```

Category 4 — Learning artifacts

Bloque de código: text
- Short Unity profiling note.
- Shader Graph experiment.
- WebGL optimization note.
- Blender optimization before/after.

20.16.3 Avoid posting

- emotional job-search posts before materials are ready;
- vague motivational content;
- political arguments;
- broad AI hype;
- unrelated personal content;
- unfinished weak screenshots.

20.17 17. Suggested first LinkedIn posts

20.17.1 Post 1 — TwinSight launch

Bloque de código: text
I recently completed TwinSight X500, a Unity WebGL technical visualization prototype for inspecting the assembly of a Holybro
↪ X500 V2 drone.

The project explores how interactive real-time 3D can support spatial understanding of technical assembly processes.

Core features:

- component selection;
- exploded view;
- cross-section/clipping;
- visual inspection modes;
- technical information panels;
- CAD-to-realtime optimization;
- WebGL deployment;
- usability/workload evaluation.

The main idea behind the project:

Seeing parts is not enough. Users need to understand spatial relationships.

[portfolio link]

20.17.2 Post 2 — CAD-to-realtime note

Bloque de código: text
One of the main challenges in TwinSight X500 was converting CAD-derived drone geometry into assets suitable for real-time
↪ WebGL inspection.

The goal was not to preserve every raw CAD detail. The goal was to preserve component identity, assembly relationships and
↪ inspection clarity while reducing geometry enough for browser deployment.

That required:

- cleanup;
- retopology decisions;
- UV preparation;
- asset grouping;
- real-time material planning;
- Unity integration;
- interaction-aware optimization.

This is the kind of technical problem I want to keep working on: real-time 3D systems that make complex technical information
↪ easier to understand.

20.17.3 Post 3 — AI disclosure / workflow

Bloque de código: text
I used AI-assisted workflows during parts of the TwinSight X500 development process.

The useful part was not replacing technical judgment. It was accelerating implementation, debugging, documentation and ↔ research.

The responsibility still remained human:

- architecture decisions;
- integration;
- testing;
- validation;
- technical tradeoffs;
- final ownership.

For me, AI is most useful when it works as an engineering assistant, not as a substitute for understanding the system.

20.18 18. Recruiter keyword strategy

20.18.1 Core keyword cluster

```
Bloque de codigo: text
Unity
C#
Unity WebGL
Real-Time 3D
Technical Artist
Technical Visualization
Interactive 3D
CAD-to-Realtime
3D Optimization
Blender
WebGL
URP
UI Toolkit
Shader Graph
Technical UI
Simulation
Digital Twin
XR
Python Automation
GitHub
```

20.18.2 Recruiter search phrases

Recruiters may search:

```
Bloque de codigo: text
Unity Technical Artist
Unity WebGL Developer
Real-Time 3D Developer
Technical Visualization Developer
Interactive 3D Developer
Unity Developer C#
Digital Twin Developer Unity
Simulation Developer Unity
3D Visualization Developer
CAD Visualization Developer
Unity Tools Developer
Technical Artist WebGL
```

These terms should appear naturally across:

- headline;
- About;
- Experience;
- Projects;
- Skills;
- Featured descriptions.

20.19 19. Boolean search strings for recruiters and self-search

20.19.1 Unity / Technical Artist

```
Bloque de código: text
("Unity Technical Artist" OR "Technical Artist") AND (Unity OR C#) AND (remote OR contractor)
```

20.19.2 Real-Time 3D

```
Bloque de código: text
("Real-Time 3D Developer" OR "Interactive 3D Developer") AND (Unity OR WebGL) AND remote
```

20.19.3 Technical visualization

```
Bloque de código: text
("Technical Visualization" OR "Digital Twin" OR Simulation) AND (Unity OR "real-time 3D") AND remote
```

20.19.4 Unity WebGL

```
Bloque de código: text
("Unity WebGL" OR "WebGL Developer" OR "3D Web Developer") AND (interactive OR visualization) AND remote
```

20.19.5 Tools / Python

```
Bloque de código: text
("Tools Developer" OR "Pipeline Developer" OR "Python Automation") AND (Unity OR technical OR 3D) AND remote
```

20.19.6 Germany future search

```
Bloque de código: text
(Unity OR "Technical Artist" OR "Digital Twin") AND (Germany OR Berlin OR Munich) AND English
```

20.19.7 Portugal future search

```
Bloque de código: text
(Unity OR "Technical Artist" OR "3D Developer") AND (Portugal OR Lisbon OR Porto) AND English
```

20.20 20. Profile photo guidance

Use a clean professional photo.

Requirements:

- neutral background;
- face clearly visible;
- no heavy filters;
- no fantasy/over-stylized image;
- good lighting;
- simple clothing;
- crop around head and shoulders;

- expression neutral or slight smile.

Avoid:

- mirror selfies;
 - gaming/fantasy aesthetics;
 - overly edited AI portraits;
 - distracting backgrounds;
 - low-resolution images.
-

20.21 21. Banner concept

The banner should communicate real-time 3D / technical visualization without looking like generic gaming art.

20.21.1 Recommended banner content

Use a clean dark or neutral layout with:

- TwinSight screenshot;
- wireframe overlay;
- CAD-to-realtime visual;
- small technical labels;
- no dense text.

20.21.2 Possible banner text

```
Bloque de codigo: text
Real-Time 3D · Unity WebGL · Technical Visualization
```

or:

```
Bloque de codigo: text
Interactive Technical Visualization with Unity, WebGL and Blender
```

Avoid:

- too many keywords;
 - illegible text;
 - generic AI art;
 - fantasy/game character art unless applying to games;
 - logos you do not own.
-

20.22 22. LinkedIn URL and public profile cleanup

20.22.1 Recommended URL

If available:

```
Bloque de codigo: text
linkedin.com/in/alexander-woodcock
```

or:

```
Bloque de codigo: text
linkedin.com/in/alexander-woodcock-salomon
```

If the current URL must remain:

```
Bloque de codigo: text
linkedin.com/in/alexander-woodcock-0132382a6
```

it is acceptable, but a cleaner URL is better.

20.22.2 Public profile checks

Before applying:

- profile visible publicly;
 - headline updated;
 - About updated;
 - Featured links working;
 - project links working;
 - GitHub link working;
 - portfolio link working;
 - no outdated thesis description;
 - no misleading certifications.
-

20.23 23. Sections to remove or downplay

Downplay:

- unrelated web freelance projects;
- low-quality client websites;
- tutorial repos;
- old game prototypes;
- unfinished FitApp;
- weak design/social media references;
- generic full-stack claims;
- broad AI hype.

Keep only if tied to target:

- Blender portrait;
 - ARA Framework;
 - ai-news-aggregator;
 - YMCA/call center roles.
-

20.24 24. LinkedIn consistency checklist

Before publishing:

Headline matches CV.
About matches portfolio.
TwinSight is described as thesis/flagship.
ARA is not described as thesis.

- No current EU authorization claim.
- No false certification claim.
- GitHub link works.
- Portfolio link works.
- Featured section ordered correctly.
- Skills reflect Unity/real-time 3D.
- English wording is concise.
- AI disclosure is controlled.
- Work availability is clear.
- No full-stack dilution.

20.25 25. Final LinkedIn profile draft

20.25.1 Headline

Bloque de codigo: text

Real-Time 3D Developer / Unity Technical Artist | Unity WebGL, C#, CAD-to-Realtime Optimization, Technical Visualization

20.25.2 About

Bloque de codigo: text

I build interactive real-time 3D systems for technical visualization using Unity, C#, WebGL and Blender.

My main project is TwinSight X500, a Unity WebGL technical visualization prototype for inspecting and understanding the

- ↪ assembly of a Holybro X500 V2 drone. The project converts complex technical documentation and CAD-derived assets into an
- ↪ optimized interactive 3D experience with component selection, exploded views, cross-section tools, visual modes, technical
- ↪ UI, and usability/workload evaluation.

My strongest areas are:

- Unity / C# runtime interaction systems
- Unity WebGL deployment
- CAD-to-realtime asset optimization
- Blender-based technical asset preparation
- Interactive technical visualization
- Real-time inspection workflows
- Technical UI and documentation
- Python-assisted tooling and automation

I am currently completing my Multimedia Engineering degree at UNAD. I also completed advanced coursework in Electronic

- ↪ Engineering at Universidad Nacional de Colombia, which supports my technical foundation in mathematics, physics, systems
- ↪ thinking and engineering logic.

I am based in Colombia and available for remote contractor/B2B roles in real-time 3D, Unity, WebGL, technical visualization,

- ↪ digital twin visualization, simulation, XR-adjacent development, and technical art workflows.

I also build secondary tooling prototypes with Python and LLM orchestration, including ARA Framework, a research automation

- ↪ prototype. I treat AI as a productivity and implementation assistant, while maintaining responsibility for architecture,
- ↪ integration, testing, technical judgment and final ownership.

20.25.3 Experience

Bloque de codigo: text

Independent Technical Artist & Developer

Freelance / Self-Directed Projects

Mar 2024 - Present

Colombia · Remote / Hybrid

Developing real-time 3D, Unity WebGL, technical visualization and automation projects focused on interactive systems,

- ↪ optimized 3D assets and technical documentation.

Key work:

- Built TwinSight X500, a Unity WebGL technical visualization prototype for inspecting and understanding the assembly of a
 - ↪ Holybro X500 V2 drone.
- Developed runtime interaction systems in Unity/C# for component selection, exploded view, cross-section/clipping, visual
 - ↪ modes and technical information panels.
- Created a CAD-to-realtime asset workflow using Blender-based cleanup, optimization, retopology, UV preparation and real-time
 - ↪ asset processing.
- Designed the project for browser-accessible inspection, balancing visual clarity, technical accuracy, interaction design and
 - ↪ WebGL performance constraints.

- Evaluated the prototype using usability and workload methods including SUS, NASA-TLX Raw and Think-Aloud methodology.
↳ [verify final metrics before publishing]
- Built ARA Framework, a Python/LangGraph research automation prototype for multi-agent literature analysis, technical
↳ planning and structured report generation.
- Used AI-assisted development workflows to accelerate implementation, debugging, documentation and research while retaining
↳ final responsibility for architecture, integration, validation and technical judgment.

20.25.4 Education — UNAD

Bloque de codigo: text
Universidad Nacional Abierta y a Distancia – UNAD
Ingeniería Multimedia / Multimedia Engineering
Aug 2021 – [Expected graduation date]

Focus on multimedia engineering, interactive systems, computer graphics, software development, technical documentation,
↳ usability evaluation and real-time 3D applications.

Thesis: TwinSight X500 – Unity WebGL interactive technical visualization prototype for drone assembly inspection.

The thesis explores how interactive real-time 3D visualization can support spatial understanding of technical assembly
↳ processes through component inspection, exploded views, cross-section tools, visual modes and usability/workload
↳ evaluation.

20.25.5 Education — UNAL

Bloque de codigo: text
Universidad Nacional de Colombia
Advanced Coursework in Electronic Engineering
Mar 2013 – Nov 2018

Completed advanced coursework in mathematics, physics, electronics, programming logic, systems thinking, simulation and
↳ engineering fundamentals.

This background supports technical reasoning in real-time systems, optimization, hardware-aware thinking, technical
↳ visualization and structured problem solving.

20.25.6 Project — TwinSight X500

Bloque de codigo: text
TwinSight X500 – Unity WebGL Technical Visualization Prototype

TwinSight X500 is a Unity WebGL interactive technical visualization prototype for inspecting and understanding the assembly of
↳ a Holybro X500 V2 drone.

The project transforms complex technical documentation and CAD-derived assets into an optimized real-time 3D inspection
↳ experience. It includes component selection, exploded view, cross-section/clipping tools, visual modes, technical
↳ information panels, WebGL deployment and academic usability/workload evaluation.

My work included Unity/C# implementation, runtime interaction systems, technical UI, CAD-to-realtime asset optimization,
↳ Blender-based preparation, documentation, evaluation design and final integration.

Key areas:
Unity WebGL · C# · URP · UI Toolkit · Blender · CAD-to-Realtime · Technical Visualization · 3D Optimization · Usability
↳ Evaluation

20.25.7 Project — ARA Framework

Bloque de codigo: text
ARA Framework – AI-Assisted Research Automation Prototype

ARA Framework is a Python-based research automation prototype exploring multi-agent workflows for niche analysis, literature
↳ research, technical planning, implementation guidance and structured report generation.

The system uses LangGraph/LangChain-style orchestration concepts, external research APIs, document-processing concepts,
↳ checkpointing and Markdown report generation.

Positioned as a tooling and automation prototype, not as a production AI platform.

Key areas:
Python · LangGraph · LangChain · LLM Orchestration · Research Automation · Technical Documentation · Workflow Automation

20.26 26. Next file dependency

The next file should be:

```
Bloque de codigo: text  
19_github_readme_templates.md
```

It must produce:

- GitHub profile README;
- TwinSight README;
- ARA README;
- repo descriptions;
- pin order;
- asset/CAD disclaimer;
- AI-assisted development disclosure;
- project badges/sections.

No Deep Research or agent required.

Chapter 21

19_github_readme_templates

Fuente: 19_github_readme_templates.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1413

Ultima modificacion: 2026-06-11 15:13

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This document provides GitHub-facing templates and cleanup rules for Alexander Woodcock Salomón's portfolio repositories.

Contenido completo depurado

21.0.1 Purpose

This document provides GitHub-facing templates and cleanup rules for Alexander Woodcock Salomón's portfolio repositories.

It includes:

- GitHub profile README template;
- pinned repository order;
- repository naming rules;
- TwinSight X500 README template;
- ARA Framework README template;
- Blender technical breakdown README template;
- optional ai-news-aggregator README structure;
- repo descriptions;
- disclaimer language;
- AI-assisted development disclosure;
- asset/CAD/manufacturer disclaimer;
- commit and release hygiene;
- GitHub Pages / portfolio link guidance.

The objective is to make GitHub support the target profile:

Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization,
↔ and Unity WebGL deployment.

21.1 1. GitHub strategy

21.1.1 Function of GitHub in this job search

GitHub is not the portfolio by itself. It is technical evidence.

It should prove:

- the project exists;
- technical scope is real;
- documentation is readable;
- code/project structure is not chaotic;
- the candidate can communicate implementation decisions;
- the candidate understands limitations and tradeoffs;
- AI assistance was used responsibly, not hidden or overclaimed.

21.1.2 GitHub should not look like

- a dump of unfinished experiments;
- a student account with tutorial repos pinned;
- a full-stack profile if the target is real-time 3D;
- an AI engineer profile if ARA is not production-ready;
- a 3D artist profile without code or systems;
- a repository graveyard with no README.

21.1.3 Main GitHub message

Every visible repo should support this message:

Bloque de código: text
 I build technical real-time 3D systems, optimized assets, interactive visualization tools, and supporting automation
 ↪ workflows.

21.2 2. Public repository decision matrix

Repository / project	Public status	Action
WebGL-Thesis-Proposal / TwinSight X500	Public	Clean and make flagship
ARA Framework	Public only if runnable or understandable	Add controlled README
Portfolio site	Public	Use as GitHub Pages or source repo
Blender portrait breakdown	Public if polished	Use README or ArtStation link
ai-news-aggregator	Public only if demo-ready	Otherwise keep private
FitApp-Free	Private / defer	Not relevant now
tutorial repos	Public okay, not pinned	Avoid front-page emphasis
old portfolio attempts	Archive/private	Avoid confusion
client-sensitive repos	Private	Do not expose
unrelated experiments	Private/archive	Reduce noise

21.3 3. Recommended pinned repository order

21.3.1 Default pins

Pin	Repository	Purpose
1	TwinSight-X500 / current thesis repo	Flagship technical proof
2	ARA-Framework	Secondary automation/tooling proof
3	portfolio / delarge95.github.io	Public portfolio site
4	blender-portrait-breakdown	Supporting technical art proof
5	ai-news-aggregator	Only if functional
6	unity-tools-or-scripts	Optional future tools repo

21.3.2 If only three repos are ready

Pin only:

1. TwinSight X500.
2. Portfolio site.
3. ARA Framework.

Do not pin weak filler repos just to fill all six slots.

21.4 4. Repository naming rules

21.4.1 Recommended naming

If renaming is possible without breaking thesis links:

```
Bloque de codigo: text
TwinSight-X500
ARA-Framework
alexander-woodcock-portfolio
blender-portrait-technical-study
ai-news-aggregator
unity-tools-prototypes
```

21.4.2 If keeping current thesis repo name

Current:

```
Bloque de codigo: text
WebGL-Thesis-Proposal
```

Add a clear description:

```
Bloque de codigo: text
TwinSight X500 – Unity WebGL technical visualization prototype for drone assembly inspection.
```

Add the project name in the README title:

```
Bloque de codigo: text
# TwinSight X500
```

Do not rely on the repository name alone.

21.5 5. GitHub profile README template

Create a repository named exactly:

```
Bloque de codigo: text
delarge95
```

Then add this README.

```
Bloque de codigo: markdown
# Alexander Woodcock Salomón

Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization,
↩ and Unity WebGL deployment.

I build browser-accessible and real-time 3D systems using Unity, C#, WebGL, Blender and Python-assisted tooling.

## Focus

- Unity / C# runtime interaction systems
- Unity WebGL deployment
- Interactive technical visualization
- CAD-to-realtime asset optimization
- Blender-based technical asset preparation
- Technical UI and inspection workflows
- Python automation and research tooling

## Flagship project

### TwinSight X500

Unity WebGL technical visualization prototype for inspecting and understanding the assembly of a Holybro X500 V2 drone.

The project includes:

- component selection;
- exploded view;
- cross-section / clipping tools;
- technical information panels;
- visual inspection modes;
- optimized CAD-derived assets;
- browser-accessible 3D interaction;
- usability/workload evaluation.

[Portfolio case study](PORTFOLIO_URL) · [Repository](REPO_URL) · [Demo](DEMO_URL)

## Secondary project

### ARA Framework

Python-based research automation prototype using LangGraph/LangChain-style workflow orchestration for literature analysis,
↩ technical planning and structured report generation.

[Repository](ARA_REPO_URL)

## Current availability

Based in Colombia. Available for remote contractor/B2B work in:

- Unity development;
- real-time 3D;
- Unity WebGL;
- technical visualization;
- CAD-to-realtime workflows;
- technical art;
- Python automation.

## Links

- Portfolio: PORTFOLIO_URL
- LinkedIn: LINKEDIN_URL
- ArtStation: ARTSTATION_URL
```

21.5.1 Notes

Replace all placeholders before publishing.

Do not include personal legal details about Portuguese passport or Germany on GitHub profile README. Keep GitHub technical.

21.6 6. TwinSight X500 README template

This is the most important repository README.

21.6.1 Recommended file name

```
Bloque de codigo: text
README.md
```

21.6.2 Template

```
Bloque de codigo: markdown
# TwinSight X500

Unity WebGL technical visualization prototype for drone assembly inspection.

## Overview

TwinSight X500 is an interactive real-time 3D visualization prototype built in Unity WebGL to inspect and understand the
↪ assembly of a Holybro X500 V2 drone.

The project transforms complex technical documentation and CAD-derived assets into an optimized browser-accessible 3D
↪ experience with component selection, exploded views, cross-section tools, visual modes and technical information panels.

Core idea:

> Seeing parts is not enough. Users need to understand spatial relationships.

## Project status

Thesis project completed. Public portfolio version in preparation / available at:

- Portfolio case study: PORTFOLIO_URL
- WebGL demo: DEMO_URL
- Demo video: VIDEO_URL

## My role

I was responsible for:

- Unity / C# implementation;
- runtime interaction systems;
- component selection workflows;
- exploded view behavior;
- cross-section / clipping tools;
- visual modes;
- technical UI and information panels;
- asset integration;
- CAD-to-realtime optimization workflow;
- Blender-based cleanup and preparation;
- documentation;
- usability/workload evaluation design and reporting;
- final integration and technical ownership.

AI-assisted tools were used as implementation, debugging, research and documentation support. Architecture decisions,
↪ integration, validation, technical judgment and final ownership remained my responsibility.

## Key features

- Interactive drone assembly inspection
- Component selection and highlighting
- Exploded view
- Cross-section / clipping tools
- Visual inspection modes
  - realistic
  - X-ray / ghosted
  - blueprint
  - wireframe
  - solid color
  - thermal-style qualitative mode
- Technical information panels
- Unity WebGL deployment
- Optimized CAD-derived assets
- Mobile/web-oriented interface considerations
- Academic usability and workload evaluation

## Technical stack

- Unity
- C#
- Unity WebGL
- URP
- UI Toolkit
- Shader Graph
```

```

- Blender
- Marmoset Toolbag
- RizomUV
- Git / GitHub

## CAD-to-realtime pipeline

The source model came from CAD/manufacturer-derived assets. The production workflow involved:

1. CAD conversion / import.
2. Geometry inspection.
3. Cleanup and simplification.
4. Blender-based optimization.
5. Retopology / low-poly preparation.
6. UV preparation.
7. High-to-low asset processing where relevant.
8. Material setup.
9. Unity import and integration.
10. WebGL-oriented optimization and testing.

## Optimization summary

| Metric | Before | After |
|---|---|---|
| Triangle count | 6.5M+ | ~95,617 |
| Deployment target | CAD/native tools | Unity WebGL |
| Interaction model | Static documentation | Real-time 3D inspection |

> Verify these values against the final thesis report before publishing.

## Evaluation

The project was evaluated using:

- SUS;
- NASA-TLX Raw;
- Think-Aloud methodology;
- task-based comparison between 3D visualization and conventional 2D support.

Summary metrics:

| Metric | Result |
|---|---|
| Participants | 12 |
| SUS average | 91.88 |
| NASA-TLX Raw, 3D viewer | 8.69 |
| NASA-TLX Raw, 2D support | 19.89 |
| Task-condition records | 96 |

> Verify final values before public release.

## Screenshots

Add images here:

```

[!TwinSight overview](#)

[!Component selection](#)

[!Exploded view](#)

[!Cross-section](#)

[!Visual modes](#)

Bloque de codigo

Demo video

Add link:

[Watch demo video](#)

Bloque de codigo

Repository structure

Assets/

Scripts/

UI/

Materials/

Models/
 Scenes/
 Docs/
 images/
 diagrams/
 metrics/
 README.md

```
Bloque de codigo

Adjust to actual repository structure.

## What this project demonstrates

- Real-time 3D technical visualization
- Unity/C# runtime systems
- Unity WebGL deployment
- CAD-to-realtime optimization
- Technical UI design
- Interactive inspection workflows
- Usability-aware engineering
- Documentation and evaluation

## Limitations

This is an academic prototype, not a commercial production product.

Known limitations:

- based on thesis scope and available time;
- public version may differ from internal development version;
- some CAD/manufacture-derived assets may require usage disclaimers;
- not a full engineering simulation or FEA system;
- thermal mode, if shown, is qualitative unless explicitly validated.

## Asset and manufacturer disclaimer

This project is an academic/portfolio prototype for technical visualization and assembly understanding.

Holybro X500 V2 references and CAD/manufacture-derived geometry are used for educational and technical visualization
↳ purposes. This project is not affiliated with, endorsed by, or sponsored by Holybro.

## AI-assisted development disclosure

AI tools were used during parts of development for implementation support, debugging, research and documentation assistance.

AI did not replace technical ownership. Architecture, integration, validation, testing, technical judgment and final project
↳ decisions were handled by the developer.

## License

Code license: LICENSE_TYPE

Asset license: see asset disclaimer and repository notes.

If license status is uncertain, do not add an open license until asset/code rights are clear.
```

21.7 7. TwinSight short README version

Use this if the repo should stay concise.

```
Bloque de codigo: markdown
# TwinSight X500

Unity WebGL technical visualization prototype for drone assembly inspection.

TwinSight X500 transforms CAD-derived drone assembly assets and technical documentation into an optimized interactive 3D
↳ inspection experience.

## Features

- Unity WebGL deployment
- Component selection
- Exploded view
- Cross-section / clipping
- Visual inspection modes
- Technical information panels
- CAD-to-realtime asset optimization
```



```
- Usability/workload evaluation

## Stack

Unity · C# · WebGL · URP · UI Toolkit · Shader Graph · Blender · Git

## My role

I handled Unity/C# implementation, runtime interaction systems, technical UI, CAD-to-realtime optimization, Blender asset
↔ preparation, documentation, evaluation and final integration.

## Links

- Portfolio case study: PORTFOLIO_URL
- Demo: DEMO_URL
- Video: VIDEO_URL

## Note

Academic thesis prototype. Not affiliated with or endorsed by Holybro.
```

21.8 8. ARA Framework README template

ARA must be framed carefully. It should not compete with TwinSight or overclaim production AI expertise.

```
Bloque de codigo: markdown
# ARA Framework

AI-assisted research automation prototype.

## Overview

ARA Framework is a Python-based research automation prototype exploring multi-agent workflows for niche analysis, literature
↔ research, technical planning, implementation guidance and structured report generation.

It is positioned as an experimental tooling project, not as a production AI platform.

## Goals

- Automate parts of academic and technical research workflows.
- Structure research into modular agent-style stages.
- Generate Markdown-based technical reports.
- Explore LLM orchestration for research support.
- Test workflow architecture using LangGraph/LangChain-style concepts.

## Architecture

Planned / implemented workflow stages:

1. Niche Analyst
2. Literature Researcher
3. Technical Architect
4. Implementation Specialist
5. Content Synthesizer

## Stack

- Python 3.12+
- LangGraph
- LangChain
- Redis
- Semantic Scholar API
- Playwright / MCP concepts
- MarkItDown / document processing concepts
- Markdown report generation

## Features

- Multi-stage research workflow
- Agent-style orchestration
- Checkpointing
- Budget-aware execution concepts
- External research API integration
- PDF/document processing concepts
- Structured Markdown output

## Project status

Prototype / experimental.

Current limitations:

- not production-ready;
- not a fully autonomous research system;
- requires configuration and testing;
- output quality depends on sources, prompts and validation;
```

```
- should not be used as a replacement for expert review.

## What this project demonstrates

- Python automation
- LLM orchestration awareness
- Workflow architecture
- Technical documentation
- Research tooling
- Structured output design
- API integration concepts

## AI disclosure

This project uses LLM tools as part of its core workflow. Outputs require human validation and should not be treated as
↳ authoritative without review.

## Setup
```

```
git clone REPO_URL
cd ara-framework
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

```
Bloque de codigo

Add real setup commands only after verifying they work.

## Usage
```

```
python main.py
```

```
Bloque de codigo

Replace with actual command.

## Environment variables
```

```
GROQ_API_KEY=
SEMANTIC_SCHOLAR_API_KEY=
REDIS_URL=
```

```
Bloque de codigo

Only list variables that are actually used.

## Roadmap

- Improve reliability.
- Add better source validation.
- Add clearer report templates.
- Add test cases.
- Add demo run.
- Add documentation.

## License

LICENSE_TYPE
```

21.9 9. Blender portrait README / project page template

This can be either a repository README or a portfolio/ArtStation text.

```
Bloque de codigo: markdown
# Hyperrealistic Blender Portrait – Technical Art Study
```

```

## Overview

This project is a high-fidelity Blender portrait study focused on topology, material definition, grooming, lighting and
↪ rendering.

It is presented as supporting technical art evidence, not as the primary career positioning asset.

## Goals

- Study realistic character pipeline fundamentals.
- Practice topology and facial structure.
- Build material and shading understanding.
- Practice grooming, lighting and rendering.
- Document the technical process.

## Tools

- Blender
- Sculpting/modeling tools
- Grooming tools
- Shader/material setup
- Lighting and rendering workflow

## Breakdown

Include images for:

- base mesh;
- topology;
- UVs;
- material passes;
- grooming;
- lighting setup;
- final render;
- before/after progression.

## What this project demonstrates

- 3D pipeline literacy
- topology awareness
- material and lighting understanding
- attention to visual detail
- ability to follow and complete a structured training pipeline

## Training note

This project was made while following CG Cookie HUMAN training material.

The course was used as non-certified skill development. It is not listed as a formal certification.

## Limitations

This is a study project, not a shipped production asset.

It should support technical art positioning, but the main portfolio focus remains TwinSight X500.

```

21.10 10. ai-news-aggregator README structure

Use only if the project is functional.

```

Bloque de código: markdown
# AI News Aggregator

Experimental full-stack AI news aggregation and analysis prototype.

## Overview

AI News Aggregator is a private/prototype project exploring news collection, analysis and structured presentation using a
↪ Python backend and web frontend.

## Stack

- Python
- FastAPI
- React
- PostgreSQL
- Redis
- AI-assisted analysis pipeline

## Features

- News/source collection
- Backend API
- Data storage
- Frontend display
- Analysis pipeline concepts

```

```
## Project status

Prototype.

Do not present as production-ready unless deployment, tests and reliability are confirmed.

## What this project demonstrates

- Python backend development
- API design
- Full-stack prototyping
- Data handling
- AI-assisted workflow experimentation

## Relevance

Secondary project only. Main portfolio focus remains real-time 3D / Unity technical visualization.
```

21.11 11. Repository descriptions

Use GitHub's short repository description field.

21.11.1 TwinSight

Bloque de código: text
Unity WebGL technical visualization prototype for drone assembly inspection, CAD-to-realtime optimization and interactive 3D analysis.

21.11.2 ARA

Bloque de código: text
Python/LangGraph research automation prototype for multi-agent literature analysis, technical planning and Markdown report generation.

21.11.3 Portfolio

Bloque de código: text
Personal portfolio for real-time 3D, Unity WebGL, technical visualization and Python-assisted tooling projects.

21.11.4 Blender portrait

Bloque de código: text
Blender technical art study focused on topology, materials, grooming, lighting and realistic rendering.

21.11.5 ai-news-aggregator

Bloque de código: text
Experimental FastAPI/React AI news aggregation prototype. Secondary full-stack/tooling project.

21.11.6 Unity tools repo

Bloque de código: text
Unity/C# tooling and technical art prototypes for real-time 3D workflows.

21.12 12. Repository topic tags

21.12.1 TwinSight topics

```
Bloque de codigo: text
unity
csharp
webgl
unity-webgl
real-time-3d
technical-visualization
cad-to-realtime
blender
interactive-3d
technical-art
drone
assembly-visualization
```

21.12.2 ARA topics

```
Bloque de codigo: text
python
langgraph
langchain
llm
research-automation
workflow-automation
multi-agent
semantic-scholar
markdown
technical-documentation
```

21.12.3 Portfolio topics

```
Bloque de codigo: text
portfolio
unity
technical-art
real-time-3d
webgl
technical-visualization
```

21.12.4 Blender portrait topics

```
Bloque de codigo: text
blender
technical-art
topology
materials
grooming
lighting
rendering
```

21.13 13. Badges

Use badges sparingly. Do not clutter the README.

21.13.1 TwinSight badges

```
Bloque de codigo: markdown
![Unity](https://img.shields.io/badge/Engine-Unity-black)
![C#](https://img.shields.io/badge/Language-C%23-blue)
![WebGL](https://img.shields.io/badge/Target-WebGL-orange)
![Status](https://img.shields.io/badge/Status-Thesis%20Prototype-lightgrey)
```

21.13.2 ARA badges

```
Bloque de codigo: markdown
![Python](https://img.shields.io/badge/Language-Python-blue)
![Status](https://img.shields.io/badge/Status-Prototype-lightgrey)
![LLM](https://img.shields.io/badge/Workflow-LLM%20Orchestration-purple)
```

Avoid fake coverage/build badges if no CI exists.

21.14 14. Image and media structure

Recommended folder:

```
Bloque de codigo: text
docs/
  images/
    overview.png
    component-selection.png
    exploded-view.png
    cross-section.png
    visual-modes.png
    cad-before-after.png
  diagrams/
    pipeline.png
    system-architecture.png
  videos/
    demo-preview.gif
  metrics/
    evaluation-summary.md
```

21.14.1 README media order

Use this order:

1. Hero screenshot.
2. Short description.
3. Demo video link.
4. Feature screenshots.
5. Pipeline diagram.
6. Metrics table.
7. Technical details.

21.14.2 GIF warning

Keep GIFs short and optimized.

Large GIFs slow down GitHub pages.

21.15 15. AI-assisted development disclosure templates

21.15.1 Short version

```
Bloque de codigo: markdown
## AI-assisted development note

AI tools were used for implementation support, debugging, research and documentation assistance. Architecture decisions,
↩ integration, testing, validation, technical judgment and final ownership remained my responsibility.
```

21.15.2 Detailed version

```
Bloque de codigo: markdown
## AI-assisted development note

This project used AI-assisted workflows during parts of development.

AI was used for:

- implementation support;
- debugging assistance;
- documentation drafting;
- research support;
- code explanation and iteration.

AI was not treated as a replacement for technical ownership. I remained responsible for architecture decisions, system
↪ integration, testing, validation, tradeoff analysis and final project delivery.
```

21.15.3 Avoid

```
Bloque de codigo: markdown
This was vibe coded.
ChatGPT made this project.
Mostly AI-generated.
```

These formulations reduce technical credibility.

21.16 16. Asset / manufacturer disclaimer templates

21.16.1 TwinSight disclaimer

```
Bloque de codigo: markdown
## Asset and manufacturer disclaimer

This project is an academic/portfolio prototype created for technical visualization and assembly-understanding purposes.

References to the Holybro X500 V2 platform and any CAD/manufacturer-derived assets are used for educational and technical
↪ visualization purposes only.

This project is not affiliated with, endorsed by, sponsored by, or officially supported by Holybro.
```

21.16.2 If asset rights are uncertain

```
Bloque de codigo: markdown
Some source assets are not redistributed in this repository. Public materials may include screenshots, derived low-poly
↪ representations or documentation-only references where appropriate.

If any asset usage must be restricted, the public repository should exclude those files and provide screenshots or diagrams
↪ instead.
```

21.16.3 If repo cannot include model files

```
Bloque de codigo: markdown
Note: Due to asset/source restrictions, this repository may not include all 3D model files used in the original academic
↪ prototype. The repository focuses on documentation, screenshots, architecture and selected implementation samples.
```

21.17 17. License guidance

21.17.1 If code can be public

Use:

```
Bloque de codigo: text
MIT License
```

or:

```
Bloque de codigo: text
Apache-2.0
```

MIT is simpler for portfolio projects.

21.17.2 If asset rights are mixed

Use:

```
Bloque de codigo: text
Code: MIT
Assets: All rights reserved / see asset disclaimer
```

README wording:

```
Bloque de codigo: markdown
## License

Code in this repository is licensed under the MIT License unless otherwise stated.

3D assets, CAD-derived models, images and third-party references may be subject to separate restrictions. See the asset
↪ disclaimer.
```

21.17.3 If uncertain

Do not add a permissive license yet.

Use:

```
Bloque de codigo: markdown
## License

License pending. This repository is shared for portfolio review and technical documentation purposes. Please contact the
↪ author before reusing code or assets.
```

21.18 18. Commit hygiene

21.18.1 Before sharing repos

Do:

- remove API keys;
- remove credentials;
- remove personal paths;
- remove large unnecessary files;
- remove broken builds if possible;
- clean README;
- add screenshots;
- add .gitignore;
- add license/disclaimer;
- check links.

21.18.2 Commit message style

Use simple descriptive messages:

```
Bloque de código: text
docs: add TwinSight project overview
docs: add screenshots and pipeline diagram
feat: add component selection system
fix: improve WebGL build notes
refactor: organize interaction scripts
```

Do not rewrite all history unless secrets were committed.

If secrets were committed, rotate them and purge history.

21.19 19. Security checklist

Before making any repo public:

- No API keys.
 - No passwords.
 - No .env files.
 - No private client data.
 - No personal documents.
 - No copyrighted course material.
 - No licensed assets that cannot be redistributed.
 - No large binary files unless needed.
 - No broken or misleading claims.
 - No hidden personal notes in comments.
 - No thesis drafts with private evaluator comments unless intended.
-

21.20 20. GitHub Pages / portfolio link guidance

If using GitHub Pages, recommended repo:

```
Bloque de código: text
delarge95.github.io
```

or:

```
Bloque de código: text
portfolio
```

21.20.1 Portfolio repo README

```
Bloque de código: markdown
# Alexander Woodcock – Portfolio

Portfolio for real-time 3D, Unity WebGL, technical visualization and Python-assisted tooling projects.

Live site: PORTFOLIO_URL

## Featured projects

- TwinSight X500
- ARA Framework
- Blender Portrait Technical Study

## Stack

- HTML/CSS/JS or chosen framework
```

```
- GitHub Pages
- Unity WebGL embeds or external demo links
```

```
## Notes
```

```
This repository contains the source for my public portfolio site.
```

21.21 21. README quality checklist

Each public repo should answer:

- What is this?
- Why does it exist?
- What did you build?
- What technologies did you use?
- What does it demonstrate?
- What is the current status?
- How can someone view it?
- What are the limitations?
- Are assets/code reusable?
- Was AI used?
- Is there a demo?

21.21.1 Minimum README sections

```
Bloque de codigo: text
# Title
Overview
Features
Tech Stack
My Role
Screenshots / Demo
Status
Limitations
License / Disclaimer
```

21.22 22. GitHub cleanup checklist

21.22.1 Immediate actions

Update GitHub profile bio.
Update profile links.
Pin correct repositories.
Unpin tutorial repos.
Archive old portfolio attempts.
Hide irrelevant unfinished repos.
Add TwinSight README.
Add TwinSight screenshots.
Add TwinSight disclaimer.
Add AI disclosure.
Add ARA README if public.
Add repo descriptions.
Add topic tags.
Check all public repo links.
Remove secrets/private files.

21.22.2 Suggested GitHub bio

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist · Unity WebGL · CAD-to-Realtime · Technical Visualization · Python Tooling
```

21.22.3 Suggested profile link

Set profile website to:

```
Bloque de codigo: text
PORTFOLIO_URL
```

If portfolio is not ready, use:

```
Bloque de codigo: text
TwinSight case study URL
```

or LinkedIn temporarily.

21.23 23. README anti-patterns

Avoid:

- overly academic abstracts as the first paragraph;
- long thesis sections before screenshots;
- no screenshots;
- no demo link;
- no role explanation;
- vague “this is a project for university” wording;
- overstating commercial readiness;
- hiding limitations;
- apologetic language;
- too many badges;
- walls of text;
- incomplete setup commands;
- unsupported technical claims;
- broken images.

21.24 24. Recruiter-first README ordering

For TwinSight, use this order:

```
Bloque de codigo: text
Title
One-line summary
Hero image
Demo links
Key features
My role
Tech stack
Visual screenshots
Optimization / pipeline
Evaluation
Limitations
AI disclosure
Asset disclaimer
Setup / repo structure
License
```

Do not start with installation instructions unless the repo is a library/tool.

21.25 25. Technical-reviewer README ordering

For deeper technical readers:

```
Bloque de codigo: text
Overview
Architecture
Runtime systems
Data/model pipeline
Optimization decisions
Code structure
Evaluation
Known limitations
Development notes
```

Consider adding:

```
Bloque de codigo: text
docs/technical-breakdown.md
docs/cad-to-realtime-pipeline.md
docs/evaluation-summary.md
```

This keeps the main README readable while providing depth.

21.26 26. Suggested additional docs for TwinSight

21.26.1 docs/cad-to-realtime-pipeline.md

Sections:

```
Bloque de codigo: text
Source asset
Conversion process
Cleanup
Retopology
UVs
Optimization decisions
Unity import
WebGL constraints
Tradeoffs
```

21.26.2 docs/runtime-systems.md

Sections:

```
Bloque de codigo: text
Component selection
Exploded view
Cross-section tools
Visual modes
UI Toolkit panels
State management
Performance concerns
```

21.26.3 docs/evaluation-summary.md

Sections:

```
Bloque de codigo: text
Evaluation design
Participants
Tasks
SUS
NASA-TLX Raw
Think-Aloud
Limitations
Interpretation
```

21.26.4 docs/ai-assisted-development.md

Sections:

```
Bloque de codigo: text
Where AI was used
Where AI was not used
Human technical ownership
Validation process
Limitations
```

21.27 27. Final recommended GitHub action order

21.27.1 Step 1

Clean profile:

```
Bloque de codigo: text
bio
links
pinned repos
weak repo visibility
```

21.27.2 Step 2

Fix TwinSight README.

21.27.3 Step 3

Add screenshots and diagrams.

21.27.4 Step 4

Add disclaimers.

21.27.5 Step 5

Add ARA README if public.

21.27.6 Step 6

Add portfolio site link.

21.27.7 Step 7

Review as a recruiter:

```
Bloque de codigo: text
Can I understand what this person does in 30 seconds?
```

21.27.8 Step 8

Review as a technical interviewer:

```
Bloque de codigo: text
Can I see evidence of implementation and tradeoffs?
```

21.28 28. Final GitHub readiness gate

GitHub is ready to include in applications only when:

- TwinSight is pinned first;
- TwinSight README is readable;
- screenshots exist;
- no misleading public repos dominate;
- repo descriptions are clear;
- links work;
- no secrets are public;
- AI disclosure is controlled;
- asset disclaimer is present;
- portfolio or demo link exists.

If this gate is not passed, use LinkedIn/portfolio first and delay GitHub emphasis.

21.29 29. Next file dependency

The next file should be:

```
Bloque de codigo: text
20_portfolio_copy_and_site_structure.md
```

It should use:

- TwinSight README;
- CV summary;
- LinkedIn About;
- project hierarchy;
- portfolio MVP from section 07;
- TwinSight case study from section 08.

No Deep Research or agent required.

Chapter 22

19B_github_readme_twinsight_final

Fuente: 19B_github_readme_twinsight_final.md
Seccion: root
Estado: canonical
Rol: Modulo final complementario
Lineas originales: 787
Ultima modificacion: 2026-06-11 19:52
Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This file provides a final professional README structure and copy for the GitHub repository associated with:

Contenido completo depurado

22.0.1 0. Module status

Bloque de codigo: text
Module: 19B_github_readme_twinsight_final.md
Date generated: 2026-06-12
Status: Complete v1
Purpose: Final GitHub README content for TwinSight X500 / WebGL-Thesis-Proposal
Inputs: 08B, 19, 20B, 21B, 29, 29B, 29C
Use: GitHub README, portfolio technical proof, interview review, recruiter validation

22.1 1. Purpose

This file provides a final professional README structure and copy for the GitHub repository associated with:

Bloque de codigo: text
TwinSight X500
Unity WebGL Technical Visualization for Drone Assembly Inspection

The README must support Alexander’s positioning as:

Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist

with evidence in:

Bloque de codigo: text
Unity · C# · WebGL · URP · UI Toolkit · Blender · CAD-to-Realtime · Technical Visualization

The README should be concise enough for recruiters and structured enough for technical reviewers.

22.2 2. README goals

22.2.1 2.1 Primary goals

```
Bloque de código: text
1. Explain the project in under 10 seconds.
2. Show visual proof immediately.
3. Provide demo/video/portfolio links.
4. Clarify technical stack.
5. Explain the problem and solution.
6. Show key features.
7. Document CAD-to-realtime optimization.
8. Present evaluation metrics.
9. State limitations honestly.
10. Clarify AI-assisted workflow if relevant.
```

22.2.2 2.2 What the README should not do

Avoid turning the README into:

```
Bloque de código: text
a full thesis document
a generic academic abstract
a long personal story
a marketing-only page
a code dump without visual proof
```

22.3 3. Final README skeleton

Use this structure in the actual repository:

```
Bloque de código: markdown
# TwinSight X500

Unity WebGL technical visualization prototype for drone assembly inspection.

[Demo thumbnail / video link]

## Overview
## Demo
## Key Features
## Screenshots
## Technical Stack
## Problem
## Solution
## Technical Pipeline
## CAD-to-Realtime Optimization
## Unity WebGL Architecture
## Evaluation
## My Role
## AI-Assisted Workflow
## Limitations
## Roadmap
## Repository Structure
## Setup / Running Locally
## Links
## License / Credits
```

22.4 4. Final README copy

The following block can be copied into the repository and adapted with final links.


```

Bloque de código: markdown
# TwinSight X500

**Unity WebGL technical visualization prototype for drone assembly inspection.**

TwinSight X500 is an interactive browser-based 3D viewer for inspecting the assembly of a Holybro X500 V2 drone. It transforms
↳ CAD-derived drone assets into an optimized real-time Unity WebGL experience with component selection, exploded view,
↳ cross-section tools, visual modes and technical information panels.

> From static 2D documentation to interactive spatial understanding.

---

## Demo

[![TwinSight X500 Demo](./media/twinsight-demo-thumbnail.jpg)](https://your-demo-video-link.com)

- **Live WebGL demo:** [add link]
- **Portfolio case study:** [add link]
- **Technical breakdown:** [add link]
- **ArtStation breakdown:** [add link]

---

## Highlights

- Unity WebGL technical visualization prototype.
- Browser-based drone assembly inspection.
- CAD-derived geometry optimized for real-time interaction.
- Component-level selection and highlighting.
- Technical part information panels.
- Exploded assembly view.
- Cross-section / clipping inspection.
- Multiple visual modes: realistic, X-ray, ghosted, blueprint, wireframe, solid and thermal-style.
- Formative usability and workload evaluation using SUS, NASA-TLX Raw and Think-Aloud.

---

## Screenshots

| Viewer | Exploded View | Cross-Section |
|---|---|---|
| [![TwinSight viewer](./media/01_viewer.jpg)] | [![Exploded view](./media/02_exploded_view.jpg)] |
↳ [![Cross-section](./media/03_cross_section.jpg)] |

| Visual Modes | Technical UI | CAD Optimization |
|---|---|---|
| [![Visual modes](./media/04_visual_modes_grid.jpg)] | [![Technical UI](./media/05_technical_ui.jpg)] | [![CAD
↳ optimization](./media/06_cad_optimization.jpg)] |

---

## Technical Stack

| Area | Tools / Technologies |
|---|---|
| Engine | Unity |
| Programming | C# |
| Deployment | Unity WebGL |
| Rendering | URP, Shader Graph |
| UI | UI Toolkit |
| 3D pipeline | Blender |
| Asset pipeline | CAD-derived geometry, mesh cleanup, optimization |
| Evaluation | SUS, NASA-TLX Raw, Think-Aloud |

---

## Problem

Drone assembly documentation is often presented through static 2D diagrams, part lists and written instructions. This forces
↳ users to mentally reconstruct spatial relationships between components, orientations, fasteners and hidden structures.

For complex assemblies such as a Holybro X500 V2 drone, this increases cognitive effort and makes it harder to understand how
↳ parts relate to each other in 3D space.

---

## Solution

TwinSight X500 converts CAD-derived drone geometry into an optimized Unity WebGL viewer. Instead of reading static
↳ documentation, users can navigate the assembly, select components, inspect technical information, activate exploded view,
↳ cut through the model and switch between visual modes.

The goal is not only to show the drone, but to make the relationships between its parts easier to understand.

---

## Key Features

### Component Selection

Users can select drone components directly in the 3D viewer. The selected part is highlighted and linked to contextual
↳ technical information.

```

```

### Exploded View

The exploded view separates assemblies while preserving spatial logic, helping users understand component hierarchy and
↪ placement.

### Cross-Section / Clipping

The clipping tool cuts through the model to reveal hidden structures and internal relationships.

### Visual Modes

TwinSight includes several inspection modes:

- Realistic
- X-ray
- Ghosted
- Blueprint
- Wireframe
- Solid
- Thermal-style

> The thermal-style mode is a qualitative visual mode. It is not a physical thermal simulation, FEA result or sensor-driven
↪ thermal analysis.

### Technical UI

The interface provides technical part information while preserving the 3D inspection context.

---

## Technical Pipeline

```

CAD-derived source

↓

Tessellation / conversion route exploration

↓

Blender cleanup and optimization

↓

Selectable component organization

↓

Unity import and scene setup

↓

Runtime interaction systems

↓

UI Toolkit interface

↓

Unity WebGL build

↓

Usability and workload evaluation

```

Bloque de código

The pipeline focused on preserving the visual information needed for technical inspection while reducing complexity enough for
↪ browser-based real-time interaction.

---

## CAD-to-Realtime Optimization

A central technical challenge was converting complex CAD-derived geometry into a model suitable for Unity WebGL.

| Metric | Value |
|---|---|
| Optimized geometry | 95,617 triangles |
| Source CAD routes | 6.5M+ triangles |
| Canonical research parts | 28 |
| Scene nodes / anchors | 30 |
| Technical elements | 257 |

The final optimized model was prepared for browser-based real-time inspection while preserving the component structure needed
↪ for selection, exploded view and technical visualization.

---

```

Unity WebGL Architecture

The Unity application coordinates runtime selection, visualization states, UI panels and inspection tools inside a WebGL ↪ build.

Main runtime systems:

- Model hierarchy and selectable parts.
- Selection and highlight system.
- Camera and navigation logic.
- Exploded-view state controller.
- Clipping / cross-section controller.
- Visual-mode switching system.
- Technical metadata / part information.
- UI Toolkit panels.
- WebGL deployment configuration.

Evaluation

TwinSight X500 was evaluated as an academic/portfolio prototype using usability, workload and qualitative observation methods.

Metric	Value
Participants	12
Task-condition records	96
SUS average	91.88
NASA-TLX Raw, 3D viewer	8.69
NASA-TLX Raw, 2D support	19.89

The formative evaluation suggested high perceived usability and lower perceived workload for the 3D viewer condition compared ↪ with 2D support in the tested tasks.

My Role

I worked across the complete prototype pipeline:

- Prepared and optimized CAD-derived drone geometry for real-time WebGL use.
- Organized selectable components and scene hierarchy.
- Implemented runtime interaction systems in Unity/C#.
- Built component selection, highlighting and information panels.
- Developed exploded view and clipping/cross-section interaction.
- Implemented multiple visual inspection modes.
- Designed technical UI and mobile-first interaction logic.
- Prepared evaluation workflow using SUS, NASA-TLX Raw and Think-Aloud.
- Documented constraints, findings and future improvements.

The source CAD model was used as the technical basis for the drone geometry. My work focused on transforming, optimizing, ↪ organizing and implementing it as an interactive real-time WebGL visualization prototype.

AI-Assisted Workflow

AI assistance was used as an implementation and iteration support tool during parts of the development process. Final ↪ architecture decisions, debugging, Unity integration, asset preparation, evaluation design, documentation and project ↪ ownership remained my responsibility.

Limitations

TwinSight X500 is an academic/portfolio prototype, not a deployed industrial product.

Current limitations:

- Not a production digital twin.
- No live IoT or sensor data.
- No predictive maintenance.
- No WebAR placement.
- No real thermal simulation.
- Formative academic evaluation.
- Prototype scope and WebGL constraints.

Roadmap

Possible future improvements:

- Guided assembly sequences.
- Deeper metadata for each component.
- Search and part filtering.
- WebGPU or performance-focused rendering experiments.
- Optional XR inspection mode.
- Sensor/state data as a digital-twin extension.
- Expansion to other drones or technical assemblies.
- Improved mobile interaction and loading performance.

```
## Repository Structure
```

```
.
├── Assets/
│   ├── Scripts/
│   ├── Materials/
│   ├── Models/
│   ├── UI/
│   └── Scenes/
├── ProjectSettings/
├── Packages/
├── media/
│   ├── twinsight-demo-thumbnail.jpg
│   ├── 01_viewer.jpg
│   ├── 02_exploded_view.jpg
│   ├── 03_cross_section.jpg
│   ├── 04_visual_modes_grid.jpg
│   ├── 05_technical_ui.jpg
│   └── 06_cad_optimization.jpg
└── README.md
```

Bloque de código

Adapt this structure to the actual repository before publishing.

Setup / Running Locally

> Add final setup instructions after confirming the repository structure.

Example:

git clone <https://github.com/your-user/your-repo.git>

Bloque de código

Open the project in Unity:

Unity version: [add version]

Target platform: WebGL

Render pipeline: URP

Bloque de código

Build instructions:

File → Build Settings → WebGL → Build

Bloque de código

If the repository does not include full source assets or large media files, state that clearly.

```
## Links
- Portfolio: [add link]
- Live WebGL demo: [add link]
- Demo video: [add link]
- ArtStation breakdown: [add link]
- LinkedIn: [add link]

---

## License and Credits

Source drone model:
```

Holybro X500 V2 source/reference model used as the basis for technical visualization.

```
Bloque de codigo

Add exact attribution and license terms before publishing.

Project:
```

TwinSight X500 developed by Alexander Woodcock Salomón as an academic/portfolio prototype.

```
Bloque de codigo
```

22.5 5. Required media folder

Create:

```
Bloque de codigo: text
/media
```

Recommended files:

```
Bloque de codigo: text
twinsight-demo-thumbnail.jpg
01_viewer.jpg
02_exploded_view.jpg
03_cross_section.jpg
04_visual_modes_grid.jpg
05_technical_ui.jpg
06_cad_optimization.jpg
07_pipeline_diagram.jpg
08_metrics_card.jpg
09_architecture_diagram.jpg
10_browser_webgl_view.jpg
```

22.5.1 5.1 Minimum acceptable README media

```
Bloque de codigo: text
[ ] demo thumbnail
[ ] final viewer screenshot
[ ] exploded view screenshot
[ ] cross-section screenshot
[ ] visual modes grid
[ ] optimization before/after
[ ] metrics card
```

22.5.2 5.2 Strong README media

```
Bloque de codigo: text
[ ] 90-second demo video thumbnail
[ ] 30-second GIF preview
[ ] WebGL browser screenshot
```

```
[ ] Unity editor screenshot
[ ] Blender cleanup screenshot
[ ] visual modes grid
[ ] SUS/NASA-TLX card
[ ] CAD-to-realtime pipeline diagram
```

22.6 6. Badge recommendations

Use badges sparingly.

Recommended badges:

```
Bloque de codigo: markdown
![Unity](https://img.shields.io/badge/Unity-000000?logo=unity&logoColor=white)
![C#](https://img.shields.io/badge/C%23-239120?logo=csharp&logoColor=white)
![WebGL](https://img.shields.io/badge/WebGL-990000?logo=webgl&logoColor=white)
![URP](https://img.shields.io/badge/URP-Unity-blue)
![Status](https://img.shields.io/badge/status-portfolio%20prototype-orange)
```

Avoid excessive badges.

Do not use badges that imply production deployment, enterprise use or certification.

22.7 7. GitHub repository description

Use this in the repository short description:

```
Bloque de codigo: text
Unity WebGL technical visualization prototype for drone assembly inspection, featuring CAD-to-realtime optimization, component
↪ selection, exploded view, clipping tools and visual modes.
```

If too long:

```
Bloque de codigo: text
Unity WebGL technical visualization prototype for drone assembly inspection.
```

22.8 8. GitHub topics

Recommended topics:

```
Bloque de codigo: text
unity
unity-webgl
csharp
webgl
technical-visualization
real-time-3d
technical-art
cad-optimization
interactive-3d
blender
drone
ui-toolkit
urp
```

Optional:

```
Bloque de codigo: text
digital-twin
simulation
xr
usability-testing
nasa-tlx
sus
```

Use digital-twin only if README clearly states that the project is digital-twin-adjacent, not a full production digital twin.

22.9 9. README visual order

Recommended order:

```
Bloque de codigo: text
1. Demo thumbnail
2. Highlights
3. Screenshot grid
4. Technical stack
5. Problem
6. Solution
7. Features
8. Pipeline
9. Optimization
10. Architecture
11. Evaluation
12. Role
13. Limitations
14. Setup
15. Links
```

Reason:

Recruiters and technical reviewers should see proof before reading details.

22.10 10. Claim-safety rules

22.10.1 10.1 Safe wording

Use:

```
Bloque de codigo: text
technical visualization prototype
browser-based 3D viewer
CAD-derived geometry
optimized for WebGL
formative evaluation
digital-twin-adjacent
qualitative thermal-style visual mode
AI-assisted implementation support
```

22.10.2 10.2 Avoid

```
Bloque de codigo: text
industrial-grade digital twin
production-ready drone twin
real thermal simulation
predictive maintenance system
IoT-integrated platform
enterprise deployment
senior-level Unity system
fully automated CAD pipeline
```

Unless those claims become true and are supported.

22.11 11. README variants

22.11.1 11.1 Recruiter-friendly shorter README

Use if the repository should be brief:

Bloque de codigo: markdown

```
# TwinSight X500

Unity WebGL technical visualization prototype for drone assembly inspection.

TwinSight X500 transforms CAD-derived drone geometry into an optimized browser-based 3D viewer with component selection,
↪ exploded view, cross-section tools, visual modes and technical information panels.

## Demo

[Demo video]
[Live WebGL build]
[Portfolio case study]

## Highlights

- Unity WebGL
- C#
- CAD-to-realtime optimization
- Component selection
- Exploded view
- Clipping tools
- Technical UI
- SUS / NASA-TLX evaluation

## Metrics

| Metric | Value |
| --- | ---: |
| Optimized triangles | 95,617 |
| Source CAD routes | 6.5M+ |
| SUS average | 91.88 |
| Participants | 12 |

## Stack

Unity, C#, WebGL, URP, UI Toolkit, Blender.

## My Role

I developed the Unity WebGL integration, runtime interaction systems, technical UI, visual modes, CAD-to-realtime optimization
↪ workflow, documentation and evaluation pipeline.

## Links

- Portfolio:
- Demo:
- Video:
```

22.11.2 11.2 Technical longer README

Use the full version from section 4.

22.12 12. README update checklist

Bloque de codigo: text

```
[ ] Repository description updated
[ ] Topics added
[ ] README title updated
[ ] Demo thumbnail added
[ ] Demo video linked
[ ] Live WebGL link added
[ ] Portfolio case study linked
[ ] Screenshot grid added
[ ] Tech stack table added
[ ] Feature list added
[ ] Pipeline diagram added
[ ] Optimization metrics added
[ ] Evaluation metrics added
[ ] Role/contribution clarified
[ ] AI-assisted workflow note added
[ ] Limitations added
[ ] Setup instructions verified
[ ] Credits/license clarified
[ ] Broken links checked
```


22.13 13. Missing information before final publication

Before publishing the README, fill:

```
Bloque de codigo: text
[ ] final repository URL
[ ] live WebGL build URL
[ ] YouTube/Vimeo demo URL
[ ] portfolio case-study URL
[ ] ArtStation breakdown URL
[ ] final Unity version
[ ] exact source model attribution/license
[ ] whether Marmoset/RizomUV were actually used
[ ] whether public build includes all assets
[ ] final setup instructions
```

22.14 14. Integration with portfolio

The README should link outward:

```
Bloque de codigo: text
GitHub → Portfolio case study
GitHub → Demo video
GitHub → Live WebGL build
GitHub → ArtStation breakdown
```

The portfolio should link inward:

```
Bloque de codigo: text
Portfolio → GitHub
Portfolio → Live demo
Portfolio → Video
Portfolio → ArtStation
```

This creates a proof loop.

22.15 15. Integration with LinkedIn

Use README as technical proof, but do not send recruiters only to GitHub.

Preferred recruiter flow:

```
Bloque de codigo: text
LinkedIn → Portfolio case study → Demo video → GitHub
```

Developer flow:

```
Bloque de codigo: text
LinkedIn → GitHub → Demo video → Technical breakdown
```

22.16 16. Next module

Proceed with:

```
Bloque de codigo: text
30_live_job_postings_market_snapshot.md
```

22.16.1 Tool recommendation

Tool	Use
Agent mode	No, unless adding more jobs
Deep Research	No

Reason:

B1 and B5 already exist. The next useful step is not more browsing; it is to integrate B1, B5 and the B2 partial capture into a cleaned market snapshot with gaps, role signals and portfolio implications.

Chapter 23

20_portfolio_copy_and_site_structure

Fuente: 20_portfolio_copy_and_site_structure.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1246

Ultima modificacion: 2026-06-11 15:15

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This document provides the copy, structure, page architecture and content rules for the public portfolio.

Contenido completo depurado

23.0.1 Purpose

This document provides the copy, structure, page architecture and content rules for the public portfolio.

It converts the strategy, CV, LinkedIn, GitHub and TwinSight materials into a web-ready structure.

The portfolio should communicate one clear professional identity:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization,
↔ and Unity WebGL deployment.
```

The portfolio is not a general personal website. It is a job-search conversion tool.

23.1 1. Portfolio objective

23.1.1 Primary objective

The portfolio must make a recruiter or technical reviewer understand in under 60 seconds:

1. what you do;
2. what your strongest project is;
3. why TwinSight is relevant to real jobs;
4. what technologies you can defend;
5. how to contact you;
6. whether you are available for remote work.

23.1.2 Secondary objective

The portfolio must give technical reviewers enough depth to evaluate:

- Unity/C# implementation;
- WebGL deployment;
- CAD-to-realtime workflow;
- Blender optimization;
- runtime interaction systems;
- technical UI;
- evaluation method;
- project limitations;
- AI-assisted development disclosure.

23.1.3 What the portfolio should not do

It should not:

- look like a generic 3D artist portfolio;
 - lead with FitApp, AI News or old repos;
 - overemphasize full-stack;
 - present ARA as the main project;
 - hide TwinSight behind too many pages;
 - use vague “creative technologist” language without technical proof;
 - rely on long paragraphs;
 - require the viewer to download files;
 - make the user guess what role you want.
-

23.2 2. Recommended portfolio structure

23.2.1 MVP version

Use this structure first:

```
Bloque de codigo: text
Home
Work
  TwinSight X500
  ARA Framework
  Blender Portrait Technical Study
About
Contact
```

23.2.2 Full version

Use this later:

```
Bloque de codigo: text
Home
Work
  TwinSight X500
  ARA Framework
  Blender Portrait Technical Study
  AI News Aggregator [optional]
Technical Notes
About
Contact
```

23.2.3 Do not include initially

Item	Reason
FitApp-Free	Low relevance
Old web freelance work	Dilutes positioning
weak full-stack projects	Lowers technical signal
unfinished repos	Credibility risk
personal blog	Only if technically relevant
unrelated graphic design	Different career track

23.3 3. Site navigation

23.3.1 Recommended top navigation

```
Bloque de codigo: text
Work
TwinSight
About
Contact
GitHub
LinkedIn
```

23.3.2 Alternative compact navigation

```
Bloque de codigo: text
Projects
About
Contact
```

23.3.3 Navigation rule

TwinSight should be accessible in one click from the homepage.

23.4 4. Home page structure

23.4.1 Section 1 — Hero

Hero headline

Recommended:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist
```

Hero subheadline

```
Bloque de codigo: text
I build interactive technical visualization systems with Unity, WebGL, C#, Blender and CAD-to-realtime optimization.
```

Supporting line

```
Bloque de codigo: text
Based in Colombia. Available for remote contractor/B2B roles.
```

Primary CTA

Bloque de código: text
View TwinSight X500

Secondary CTA

Bloque de código: text
View GitHub

Optional third CTA

Bloque de código: text
Download CV

Only include CV download if the CV is final.

23.4.2 Hero layout

Recommended visual:

- TwinSight screenshot;
- drone overview;
- subtle wireframe overlay;
- dark/neutral background;
- no clutter;
- no generic AI art.

Avoid:

- fantasy/game visuals unless applying to game art;
- unrelated portrait render as hero;
- animated background that hurts readability;
- too much text.

23.5 5. Home page copy

23.5.1 Full home page draft

Bloque de código: text
Real-Time 3D Developer / Unity Technical Artist

I build interactive technical visualization systems with Unity, WebGL, C#, Blender and CAD-to-realtime optimization.

My flagship project, TwinSight X500, is a Unity WebGL prototype for inspecting the assembly of a Holybro X500 V2 drone through ↩ component selection, exploded views, cross-section tools, visual modes and technical information panels.

I focus on real-time 3D systems that make complex technical information easier to inspect, understand and communicate.

Based in Colombia. Available for remote contractor/B2B roles.

23.5.2 Short version

Bloque de código: text
I build real-time 3D systems for technical visualization using Unity, C#, WebGL and Blender.

Flagship project: TwinSight X500 – a Unity WebGL drone assembly inspection prototype.

23.5.3 More recruiter-friendly version

Bloque de codigo: text

Unity / Real-Time 3D developer focused on technical visualization, WebGL deployment and optimized 3D workflows.

Available for remote contractor roles involving Unity, interactive 3D, technical visualization, CAD-to-realtime pipelines, ↔ simulation interfaces and WebGL experiences.

23.6 6. Featured projects section

23.6.1 Section title

Bloque de codigo: text

Selected Work

23.6.2 Intro line

Bloque de codigo: text

Projects focused on real-time 3D, technical visualization, Unity WebGL, automation and technical art workflows.

23.6.3 Project card order

1. TwinSight X500.
2. ARA Framework.
3. Blender Portrait Technical Study.
4. AI News Aggregator only if functional.

23.6.4 Project card template

Bloque de codigo: text

Project name

One-sentence description

Tags

Primary link

Secondary link

23.7 7. Project card copy

23.7.1 TwinSight X500 card

Title

Bloque de codigo: text

TwinSight X500

Subtitle

Bloque de codigo: text

Unity WebGL technical visualization prototype for drone assembly inspection.

Description

Bloque de codigo: text
Interactive 3D system for inspecting a Holybro X500 V2 drone assembly using component selection, exploded view, cross-section
↪ tools, visual modes, technical UI and CAD-to-realtime optimization.

Tags

Bloque de codigo: text
Unity · C# · WebGL · Blender · CAD-to-Realtime · Technical Visualization

CTAs

Bloque de codigo: text
View Case Study
View GitHub
Watch Demo

23.7.2 ARA Framework card

Title

Bloque de codigo: text
ARA Framework

Subtitle

Bloque de codigo: text
Python research automation prototype.

Description

Bloque de codigo: text
Experimental multi-agent research automation system using Python and LangGraph/LangChain-style workflow orchestration for
↪ literature analysis, technical planning and structured report generation.

Tags

Bloque de codigo: text
Python · LangGraph · LangChain · LLM Workflows · Automation · Research Tooling

CTAs

Bloque de codigo: text
View Project
View GitHub

23.7.3 Blender portrait card

Title

Bloque de codigo: text
Hyperrealistic Blender Portrait

Subtitle

Bloque de codigo: text
Technical art study in topology, materials, grooming and lighting.

Description

Bloque de codigo: text
High-fidelity Blender portrait study used as supporting evidence of 3D pipeline literacy, material work, grooming, lighting
↔ and rendering fundamentals.

Tags

Bloque de codigo: text
Blender · Topology · Materials · Grooming · Lighting · Rendering

CTAs

Bloque de codigo: text
View Breakdown
ArtStation

23.7.4 AI News Aggregator card

Include only if functional.

Title

Bloque de codigo: text
AI News Aggregator

Subtitle

Bloque de codigo: text
Full-stack AI news aggregation prototype.

Description

Bloque de codigo: text
Experimental FastAPI/React prototype for collecting, organizing and analyzing AI news with a Python backend, database-backed
↔ storage and web interface.

Tags

Bloque de codigo: text
Python · FastAPI · React · PostgreSQL · Redis · AI Tooling

CTAs

Bloque de codigo: text
View Project
View GitHub

23.8 8. TwinSight case study page

23.8.1 Page title

Bloque de codigo: text
TwinSight X500

23.8.2 Subtitle

Bloque de codigo: text
Unity WebGL technical visualization prototype for drone assembly inspection.

23.8.3 Opening paragraph

Bloque de codigo: text
TwinSight X500 is a Unity WebGL interactive technical visualization prototype for inspecting and understanding the assembly of
↩ a Holybro X500 V2 drone. The project transforms complex technical documentation and CAD-derived assets into an optimized
↩ real-time 3D inspection experience with component selection, exploded views, cross-section tools, visual modes and
↩ technical information panels.

23.8.4 Core idea block

Bloque de codigo: text
Seeing parts is not enough. Users need to understand spatial relationships.

23.8.5 Metadata block

Field	Value
Role	Unity/C# developer, technical artist, 3D optimization, documentation, evaluation
Type	Thesis / portfolio prototype
Stack	Unity, C#, WebGL, URP, UI Toolkit, Shader Graph, Blender
Focus	Technical visualization, CAD-to-realtime, interaction systems
Status	Thesis complete, pending defense
Links	GitHub, demo, video

23.9 9. TwinSight case study structure

23.9.1 Recommended sections

Bloque de codigo: text
1. Problem
2. Context
3. My role
4. Constraints
5. Solution overview
6. CAD-to-realtime pipeline
7. Interaction systems
8. Visual modes and technical UI
9. WebGL and optimization
10. Evaluation
11. Results
12. Limitations
13. What I would improve
14. AI-assisted development note
15. Asset/manufacturer disclaimer

23.10 10. TwinSight case study copy

23.10.1 Problem

Bloque de codigo: text

Drone assembly documentation often relies on 2D diagrams, manuals and static references. These materials can show individual ↵ parts, but they require users to mentally reconstruct spatial relationships between components, fasteners, subassemblies ↵ and assembly steps.

TwinSight X500 explores whether an interactive real-time 3D viewer can make those spatial relationships easier to inspect and ↵ understand.

23.10.2 Context

Bloque de codigo: text

The project was developed as a Multimedia Engineering thesis. It uses the Holybro X500 V2 drone as a technical assembly case ↵ and focuses on converting complex technical documentation and CAD-derived assets into a browser-accessible Unity WebGL ↵ experience.

23.10.3 My role

Bloque de codigo: text

I handled Unity/C# implementation, runtime interaction systems, technical UI, CAD-to-realtime asset optimization, ↵ Blender-based asset preparation, documentation, evaluation design and final integration.

AI-assisted tools were used as support for implementation, debugging, research and documentation. Architecture decisions, ↵ integration, testing, validation and final technical ownership remained my responsibility.

23.10.4 Constraints

Bloque de codigo: text

The project had to balance technical clarity, real-time interaction, WebGL deployment constraints, academic scope, asset ↵ optimization, usability evaluation and limited production time.

The goal was not to build a full engineering simulation. The goal was to create an interactive technical visualization ↵ prototype for assembly inspection and spatial understanding.

23.10.5 Solution overview

Bloque de codigo: text

TwinSight X500 provides an interactive 3D inspection environment where users can select components, view technical ↵ information, explore exploded views, use cross-section tools and switch between visual modes such as realistic, ghosted, ↵ blueprint, wireframe and qualitative thermal-style visualization.

23.10.6 CAD-to-realtime pipeline

Bloque de codigo: text

The source geometry came from CAD/manufacture-derived assets. The pipeline involved conversion, inspection, cleanup, ↵ optimization, Blender-based preparation, UV work, material planning, Unity integration and WebGL-conscious asset handling.

The goal was to reduce heavy CAD geometry while preserving component identity and assembly relationships.

23.10.7 Optimization copy

Bloque de codigo: text

The project reduced dense CAD-derived geometry from over 6.5 million triangles to approximately 95,617 optimized triangles. ↵ [verify final metric]

This reduction was necessary to make the model suitable for real-time browser-based visualization while preserving the visual ↵ information needed for technical inspection.

23.10.8 Interaction systems

Bloque de codigo: text
The runtime systems were designed around inspection rather than passive viewing.

Core interactions include:

- component selection;
- highlighting;
- technical information panels;
- exploded view;
- cross-section / clipping;
- visual mode switching;
- mobile/web-oriented UI behavior.

23.10.9 Visual modes

Bloque de codigo: text
The visual modes allow users to inspect the drone from different technical perspectives. Realistic mode supports general
↪ recognition, wireframe and blueprint modes emphasize structure, ghosted/X-ray modes help reveal hidden relationships, and
↪ qualitative thermal-style visualization explores an alternative analytical layer.

23.10.10 Evaluation

Bloque de codigo: text
The project was evaluated using usability and workload methods including SUS, NASA-TLX Raw and Think-Aloud methodology.
↪ [verify exact final metrics]

The evaluation was designed to compare interactive 3D support with conventional 2D reference material in assembly-inspection
↪ tasks.

23.10.11 Results

Use only after verifying final numbers:

Bloque de codigo: text
Reported evaluation data included:

- 12 validation participants;
- SUS average: 91.88;
- NASA-TLX Raw: 8.69 for the 3D viewer;
- NASA-TLX Raw: 19.89 for the 2D support condition;
- 96 task-condition records.

These results should be presented as academic prototype findings, not as commercial product validation.

23.10.12 Limitations

Bloque de codigo: text
TwinSight X500 is an academic prototype, not a commercial production product. It does not replace engineering CAD software,
↪ formal simulation tools, FEA analysis or manufacturer documentation.

Some visual modes are qualitative. Some asset usage may require public-display limitations. The public portfolio version may
↪ not include every model file from the original thesis project.

23.10.13 What I would improve

Bloque de codigo: text
Given more time, I would improve build profiling, formalize LOD handling, refine mobile interaction details, add more
↪ structured user analytics, improve the public WebGL deployment pipeline, expand technical documentation and separate the
↪ prototype into clearer runtime modules.

23.10.14 AI-assisted development note

Bloque de codigo: text

AI-assisted workflows were used during parts of development for implementation support, debugging, research and documentation.
↪ AI was not used as a replacement for technical ownership.

I remained responsible for architecture decisions, integration, validation, testing, tradeoff analysis and final project
↪ delivery.

23.10.15 Asset/manufacturer disclaimer

Bloque de codigo: text
This project is an academic/portfolio prototype created for technical visualization and assembly-understanding purposes.

References to the Holybro X500 V2 platform and any CAD/manufacturer-derived assets are used for educational and technical
↪ visualization purposes only.

This project is not affiliated with, endorsed by, sponsored by, or officially supported by Holybro.

23.11 11. ARA Framework project page

23.11.1 Page title

Bloque de codigo: text
ARA Framework

23.11.2 Subtitle

Bloque de codigo: text
AI-assisted research automation prototype.

23.11.3 Opening paragraph

Bloque de codigo: text
ARA Framework is a Python-based research automation prototype exploring multi-agent workflows for niche analysis, literature
↪ research, technical planning, implementation guidance and structured report generation.

23.11.4 Positioning note

Bloque de codigo: text
ARA is a secondary project. It supports the profile by showing automation, tooling, workflow architecture and technical
↪ documentation. It is not positioned as a production AI platform or as the main career direction.

23.11.5 Metadata

Field	Value
Type	Prototype
Stack	Python, LangGraph, LangChain, Redis, APIs
Focus	Research automation, LLM workflows, technical documentation
Status	Experimental / prototype
Role	Architecture, implementation, documentation

23.11.6 Case study sections

Bloque de codigo: text

1. Problem
2. Workflow concept
3. Architecture
4. Tools
5. What it demonstrates
6. Limitations
7. Next steps

23.11.7 Copy

Bloque de codigo: text

The goal of ARA Framework is to test how LLM-assisted workflows can structure parts of academic and technical research. The ↩ system explores modular agent-style stages such as niche analysis, literature research, technical architecture planning, ↩ implementation guidance and report synthesis.

The project demonstrates Python automation, workflow architecture, structured output design and technical documentation. It ↩ does not claim to replace expert review or operate as a production research system.

23.12 12. Blender portrait project page

23.12.1 Page title

Bloque de codigo: text

Hyperrealistic Blender Portrait

23.12.2 Subtitle

Bloque de codigo: text

Technical art study in topology, materials, grooming and lighting.

23.12.3 Opening paragraph

Bloque de codigo: text

This project is a high-fidelity Blender portrait study focused on realistic character pipeline fundamentals, including ↩ topology, material definition, grooming, lighting and rendering.

23.12.4 Positioning note

Bloque de codigo: text

This project supports the technical art side of the portfolio. It is not the main career-positioning project and should not ↩ displace TwinSight X500.

23.12.5 Sections

Bloque de codigo: text

1. Goal
2. Process
3. Topology
4. Materials
5. Grooming
6. Lighting
7. Final render
8. Lessons learned

23.12.6 Copy

Bloque de código: text

The project was made as a structured technical art study following non-certified training material. It demonstrates 3D
↪ pipeline literacy, attention to surface detail, topology awareness, material work and rendering fundamentals.

Its role in the portfolio is to support the technical artist positioning, especially when applying to Unity Technical Artist
↪ or real-time 3D roles that value asset-pipeline awareness.

23.13 13. About page

23.13.1 About headline

Bloque de código: text

I build interactive real-time 3D systems for technical visualization.

23.13.2 About body

Bloque de código: text

I am a Colombia-based Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, C#, CAD-to-realtime optimization
↪ and interactive technical visualization.

My strongest work is TwinSight X500, a Unity WebGL technical visualization prototype for inspecting drone assembly. The
↪ project combines real-time interaction systems, optimized CAD-derived assets, technical UI, visual modes and
↪ usability/workload evaluation.

My background combines Multimedia Engineering, advanced Electronic Engineering coursework, computer graphics, Blender-based
↪ asset workflows, Python-assisted tooling and technical documentation.

I am interested in roles involving:

- Unity development;
- real-time 3D;
- Unity WebGL;
- technical visualization;
- digital twin visualization;
- simulation interfaces;
- XR-adjacent development;
- technical art workflows;
- Python automation for production or research tooling.

I am based in Colombia and available for remote contractor/B2B work.

23.13.3 Short About version

Bloque de código: text

Colombia-based Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, C#, CAD-to-realtime optimization and
↪ interactive technical visualization.

Main project: TwinSight X500, a Unity WebGL drone assembly inspection prototype.

Available for remote contractor/B2B roles.

23.14 14. Contact page

23.14.1 Contact headline

Bloque de código: text

Available for remote Unity, real-time 3D and technical visualization work.

23.14.2 Contact copy

Bloque de codigo: text
 I am currently based in Colombia and available for remote contractor/B2B roles involving Unity, real-time 3D, Unity WebGL,
 ↔ technical visualization, CAD-to-realtime workflows, interactive 3D systems and Python-assisted tooling.
 For relevant opportunities, contact me through email or LinkedIn.

23.14.3 Contact links

Bloque de codigo: text
 Email: [email]
 LinkedIn: [LinkedIn URL]
 GitHub: [GitHub URL]
 ArtStation: [ArtStation URL]
 CV: [CV URL]

23.14.4 Optional work authorization note

Bloque de codigo: text
 Current availability: remote contractor/B2B from Colombia.
 Medium-term EU mobility may become relevant through expected Portuguese citizenship/passport, but current applications should
 ↔ be treated as Colombia-based remote availability.

Do not over-detail legal pathways on the contact page unless necessary.

23.15 15. Footer copy

Bloque de codigo: text
 © Alexander Woodcock Salomón
 Real-Time 3D • Unity WebGL • Technical Visualization • CAD-to-Realtime

Alternative:

Bloque de codigo: text
 Built for technical visualization, real-time 3D and Unity WebGL work.

23.16 16. Visual direction

23.16.1 Recommended style

Element	Direction
Theme	Dark or neutral
Visual language	Technical, clean, precise
Main visual	TwinSight screenshot
Accent	Wireframe / CAD lines
Typography	Clean sans-serif
Animation	Minimal
Layout	Fast scanning
Tone	Technical, direct

23.16.2 Avoid

- heavy decorative animations;
- visual noise;
- huge paragraphs;
- generic AI-generated backgrounds;
- too much gaming/fantasy art;
- unrelated profile photo as hero;
- over-stylized “creative portfolio” look.

23.16.3 Portfolio should feel like

Bloque de codigo: text
technical product case study + real-time 3D portfolio

not:

Bloque de codigo: text
graphic design portfolio

23.17 17. Case study media checklist

23.17.1 TwinSight media

Required:

- hero screenshot;
- 60–120 second demo video;
- component selection screenshot;
- exploded view screenshot;
- cross-section screenshot;
- visual modes screenshot;
- CAD-to-realtime before/after image;
- optimization table;
- UI screenshot;
- evaluation summary image/table.

Optional:

- short GIF loop;
- architecture diagram;
- pipeline diagram;
- mobile layout screenshot;
- code snippet screenshot.

23.17.2 ARA media

Required if published:

- architecture diagram;
- workflow screenshot;
- sample output screenshot;
- README screenshot or terminal screenshot.

Optional:

- short CLI demo;
- report generation preview.

23.17.3 Blender media

Required:

- final render;
 - topology screenshot;
 - material/lighting breakdown;
 - grooming close-up;
 - progression image.
-

23.18 18. Site copy consistency rules

Use the same phrasing across portfolio, CV, LinkedIn and GitHub.

23.18.1 Preferred terms

Use:

```
Bloque de codigo: text
Real-Time 3D
Unity WebGL
Technical Visualization
CAD-to-Realtime
Interactive 3D
Technical UI
Runtime Interaction Systems
Asset Optimization
Blender-based preparation
```

Avoid overusing:

```
Bloque de codigo: text
multimedia
creative
generalist
full-stack
AI expert
game developer
graphic design
```

23.18.2 Thesis wording

Correct:

```
Bloque de codigo: text
Thesis project completed, pending defense.
```

Avoid:

```
Bloque de codigo: text
Currently starting thesis development.
```

23.18.3 AI wording

Correct:

```
Bloque de codigo: text
AI-assisted workflows were used as implementation, debugging, research and documentation support.
```

Avoid:

```
Bloque de codigo: text
AI built this project.
```

23.19 19. SEO and recruiter keywords

Use naturally in page headings, project tags and descriptions.

23.19.1 Core keywords

```
Bloque de codigo: text
Unity Developer
Unity Technical Artist
Real-Time 3D Developer
Unity WebGL Developer
Technical Visualization Developer
Interactive 3D Developer
CAD-to-Realtime
Digital Twin Visualization
Simulation Visualization
3D Optimization
Blender
C#
WebGL
Technical UI
```

23.19.2 Secondary keywords

```
Bloque de codigo: text
Python Automation
LLM Workflows
Research Automation
Shader Graph
URP
UI Toolkit
XR
AR/VR
Product Visualization
Assembly Visualization
```

23.19.3 Do not keyword stuff

The text must remain readable.

23.20 20. Portfolio MVP build options

23.20.1 Acceptable MVP platforms

Platform	Use case
GitHub Pages	Developer-facing, free
Framer	Fast visual site

Platform	Use case
Webflow	Polished no-code site
Notion / Super	Very fast MVP
Custom HTML/CSS	Full control
WordPress	Only if already fast to deploy

23.20.2 Recommendation

Use the fastest option that allows:

- case study page;
- project cards;
- screenshots;
- video embeds;
- external links;
- readable design;
- easy updates.

Do not spend weeks building the portfolio tech stack.

The portfolio content is more important than the framework.

23.21 21. Portfolio MVP acceptance criteria

The MVP is ready to use when:

- homepage clearly states role;
- TwinSight appears first;
- TwinSight case study is readable;
- at least 6 TwinSight screenshots exist;
- GitHub and LinkedIn links work;
- contact email works;
- no false claims;
- no outdated thesis status;
- mobile layout is acceptable;
- page loads reasonably fast;
- CV link works if included.

23.21.1 Not required for MVP

- perfect animations;
 - full blog;
 - all projects complete;
 - custom CMS;
 - advanced scroll effects;
 - perfect bilingual version;
 - 3D hero embed;
 - multiple case studies.
-

23.22 22. Portfolio full version backlog

Add later:

- bilingual English/Spanish toggle;
 - technical notes section;
 - WebGL embedded demo;
 - more detailed pipeline diagrams;
 - downloadable case study PDF;
 - project filtering;
 - ARA demo;
 - updated Blender breakdown;
 - blog posts on Unity WebGL optimization.
-

23.23 23. Suggested portfolio file structure

If building custom/static:

```
Bloque de codigo: text
portfolio/
  index.html
  about.html
  contact.html
  work/
    twinsight-x500.html
    ara-framework.html
    blender-portrait.html
  assets/
    images/
      twinsight/
      ara/
      blender/
    video/
    css/
    js/
  docs/
  cv.pdf
```

If using React:

```
Bloque de codigo: text
src/
  pages/
    Home.jsx
    About.jsx
    Contact.jsx
  projects/
    TwinSight.jsx
    ARA.jsx
    BlenderPortrait.jsx
  components/
    ProjectCard.jsx
    Hero.jsx
    Section.jsx
  assets/
```

Do not overbuild unless there is time.

23.24 24. Portfolio review checklist

Before sharing publicly:

- No typos.
- English is clean.
- TwinSight metrics verified.

Links work.
Images load.
Mobile view works.
Contact works.
CV link works.
GitHub is not embarrassing.
No old thesis status.
ARA not overclaimed.
AI disclosure present.
Asset disclaimer present.
Site communicates target role in 5 seconds.

23.25 25. Final homepage draft

Use this as the first implementation copy.

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist

I build interactive technical visualization systems with Unity, WebGL, C#, Blender and CAD-to-realtime optimization.

My flagship project, TwinSight X500, is a Unity WebGL prototype for inspecting the assembly of a Holybro X500 V2 drone through
↪ component selection, exploded views, cross-section tools, visual modes and technical information panels.

I focus on real-time 3D systems that make complex technical information easier to inspect, understand and communicate.

Based in Colombia. Available for remote contractor/B2B roles.

[View TwinSight X500] [View GitHub] [Contact]
```

23.26 26. Final Work page draft

```
Bloque de codigo: text
Selected Work

Projects focused on real-time 3D, technical visualization, Unity WebGL, automation and technical art workflows.

TwinSight X500
Unity WebGL technical visualization prototype for drone assembly inspection.
Unity · C# · WebGL · Blender · CAD-to-Realtime · Technical Visualization
[View Case Study]

ARA Framework
Python research automation prototype using multi-agent workflow orchestration for literature analysis, technical planning and
↪ structured report generation.
Python · LangGraph · LangChain · LLM Workflows · Automation
[View Project]

Hyperrealistic Blender Portrait
Technical art study in topology, materials, grooming, lighting and rendering.
Blender · Topology · Materials · Grooming · Lighting
[View Breakdown]
```

23.27 27. Final About page draft

```
Bloque de codigo: text
I build interactive real-time 3D systems for technical visualization.

I am a Colombia-based Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, C#, CAD-to-realtime optimization
↪ and interactive technical visualization.

My strongest work is TwinSight X500, a Unity WebGL technical visualization prototype for inspecting drone assembly. The
↪ project combines real-time interaction systems, optimized CAD-derived assets, technical UI, visual modes and
↪ usability/workload evaluation.
```

My background combines Multimedia Engineering, advanced Electronic Engineering coursework, computer graphics, Blender-based
↳ asset workflows, Python-assisted tooling and technical documentation.

I am interested in roles involving Unity development, real-time 3D, Unity WebGL, technical visualization, digital twin
↳ visualization, simulation interfaces, XR-adjacent development, technical art workflows and Python automation for
↳ production or research tooling.

I am based in Colombia and available for remote contractor/B2B work.

23.28 28. Final Contact page draft

Bloque de codigo: text

Available for remote Unity, real-time 3D and technical visualization work.

I am currently based in Colombia and available for remote contractor/B2B roles involving Unity, real-time 3D, Unity WebGL,
↳ technical visualization, CAD-to-realtime workflows, interactive 3D systems and Python-assisted tooling.

Contact:

Email: [email]

LinkedIn: [LinkedIn URL]

GitHub: [GitHub URL]

ArtStation: [ArtStation URL]

CV: [CV URL]

23.29 29. Next file dependency

The next file should be:

Bloque de codigo: text

21_twinsight_demo_video_script_and_shotlist.md

It should define:

- 60-second demo script;
- 90-second demo script;
- 120-second demo script;
- shot list;
- capture checklist;
- captions;
- voiceover;
- LinkedIn cutdown;
- silent version;
- technical breakdown version.

No Deep Research or agent required.

Chapter 24

21_twinsight_demo_video_script_and_shotlist

Fuente: 21_twinsight_demo_video_script_and_shotlist.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 912

Ultima modificacion: 2026-06-11 15:18

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This document defines the video plan for presenting **TwinSight X500** as the flagship portfolio asset.

Contenido completo depurado

24.0.1 Purpose

This document defines the video plan for presenting **TwinSight X500** as the flagship portfolio asset.

It includes:

- 60-second demo script;
- 90-second demo script;
- 120-second demo script;
- silent version;
- LinkedIn cutdown;
- technical breakdown version;
- shot list;
- capture checklist;
- captions;
- voiceover text;
- editing structure;
- export formats;
- quality gate.

The goal is to make TwinSight understandable quickly to recruiters while still giving technical reviewers enough evidence to care.

24.1 1. Video objective

The demo video must communicate, in under two minutes:

1. what TwinSight X500 is;
2. why it exists;
3. what problem it solves;
4. what features were built;
5. what technical skills it demonstrates;
6. what the measurable outcomes were;
7. what role Alexander had in the project.

The video should not look like a thesis lecture. It should look like a concise technical product demo.

24.2 2. Primary audience

24.2.1 Recruiter audience

Recruiters need to understand:

- this is a real Unity/WebGL project;
- the candidate can build interactive systems;
- the project is relevant to technical visualization, simulation, XR, digital twin, Unity and WebGL roles;
- there is enough substance to justify an interview.

24.2.2 Technical reviewer audience

Technical reviewers need to see:

- Unity runtime interaction;
 - asset optimization;
 - inspection systems;
 - cross-section/clipping;
 - visual modes;
 - UI behavior;
 - WebGL/browser deployment;
 - evaluation metrics;
 - limitations handled honestly.
-

24.3 3. Core message

Use this as the video thesis:

Bloque de código: text
TwinSight X500 transforms drone assembly documentation and CAD-derived assets into an optimized Unity WebGL technical
↔ visualization prototype for interactive inspection and spatial understanding.

Shorter version:

Bloque de código: text
Seeing parts is not enough. Users need to understand spatial relationships.

24.4 4. Video versions to produce

24.4.1 Required versions

Version	Duration	Use
Recruiter demo	60 s	LinkedIn, quick applications
Portfolio demo	90 s	Portfolio hero/project page
Technical breakdown	120 s	GitHub, technical reviewers
Silent captioned version	60–90 s	LinkedIn autoplay
Vertical cutdown	30–45 s	Optional social post

24.4.2 Recommended production order

1. 90-second portfolio demo.
2. 60-second recruiter cut.
3. 120-second technical breakdown.
4. Silent LinkedIn version.

Do not start with the 120-second version. It can become too dense.

24.5 5. Visual style

24.5.1 Recommended style

Element	Direction
Background	Dark or neutral
Text	Minimal, large, readable
Pace	Fast but not chaotic
Transitions	Simple cuts / fades
Captions	Short phrases
Audio	Clean voiceover or no voice
Music	Optional, subtle, non-distracting
Visual priority	App footage first
Branding	Minimal

24.5.2 Avoid

- long intro animation;
- thesis defense tone;
- generic motivational music;
- excessive text;
- tiny UI text;
- shaky screen recording;
- uncompressed laggy capture;
- showing development errors;
- overexplaining AI;
- putting legal disclaimers in the main video.

24.6 6. Capture requirements

24.6.1 Minimum capture quality

Parameter	Recommendation
Resolution	1920×1080
Frame rate	30 fps minimum
Format	MP4 H.264
Audio	48 kHz if voiceover
Cursor	Visible only if useful
UI scale	Large enough
Browser	Clean window
Notifications	Off
Desktop clutter	Hidden

24.6.2 Capture tools

Use any stable screen recorder:

- OBS Studio;
- Nvidia ShadowPlay;
- Windows Game Bar;
- macOS screen recording;
- Unity Recorder if recording inside editor;
- browser capture if demo is live.

24.6.3 Capture rule

Record more footage than needed. Edit down later.

24.7 7. Shot list overview

24.7.1 Required shots

Shot	Purpose
Hero full drone	Immediate project recognition
Problem/2D reference	Explain why project exists
Orbit/inspect	Show real-time 3D
Component selection	Show interaction
Info panel	Show technical UI
Exploded view	Show spatial relationships
Cross-section/clipping	Show technical visualization
Visual modes	Show rendering/inspection systems
Optimization before/after	Show technical asset work
Metrics/evaluation	Show validation
Closing links	Call to action

24.7.2 Optional shots

Shot	Purpose
Mobile layout	Web/mobile proof
Wireframe overlay	Technical art proof
CAD mesh vs optimized mesh	Pipeline proof
Unity editor hierarchy	Implementation proof
Code snippet	Technical credibility
ARA not included	Avoid dilution

24.8 8. 60-second recruiter demo script

24.8.1 Structure

Time	Visual	Caption / Voice
0–5 s	Full drone hero	TwinSight X500 — Unity WebGL technical visualization
5–10 s	2D/manual vs 3D viewer	Drone assembly documentation can be hard to understand spatially
10–18 s	Orbit around drone	I built an interactive 3D inspection prototype in Unity
18–26 s	Component selection	Users can select components and inspect technical information
26–34 s	Exploded view	Exploded view reveals spatial relationships between parts
34–42 s	Cross-section	Cross-section tools support internal inspection
42–50 s	Visual modes	Visual modes help analyze structure, visibility and form
50–56 s	Optimization metric	CAD-derived geometry optimized for WebGL deployment
56–60 s	Closing screen	Unity · C# · WebGL · Blender · Technical Visualization

24.8.2 60-second voiceover

Bloque de código: text
 This is TwinSight X500, a Unity WebGL technical visualization prototype for inspecting the assembly of a Holybro X500 V2
 ↪ drone.

The project addresses a common problem in technical documentation: 2D diagrams can show parts, but users still have to
 ↪ reconstruct spatial relationships mentally.

I built an interactive 3D inspection experience with component selection, technical information panels, exploded view,
 ↪ cross-section tools and multiple visual modes.

The asset pipeline converts CAD-derived geometry into optimized real-time assets for browser deployment.

TwinSight was developed as my Multimedia Engineering thesis and evaluated using usability and workload methods including SUS,
 ↪ NASA-TLX Raw and Think-Aloud.

My role covered Unity/C# implementation, runtime interaction systems, technical UI, Blender-based asset preparation,
 ↪ documentation and final integration.

24.8.3 60-second captions

Use these captions:

Bloque de código: text
 TwinSight X500
 Unity WebGL technical visualization

Problem:
 2D documentation requires mental reconstruction

Interactive 3D inspection
 Unity · C# · WebGL

Component selection
 Technical information panels

Exploded view
 Spatial relationships

Cross-section tools
 Internal inspection

Visual modes
 Real-time analysis

CAD-to-realtime optimization
 Browser deployment

Built as Multimedia Engineering thesis
 Available for remote Unity / real-time 3D work

24.9 9. 90-second portfolio demo script

24.9.1 Structure

Time	Visual	Message
0–5 s	Title + drone hero	What the project is
5–15 s	2D problem / static diagram	Why it exists
15–25 s	Full interactive viewer	Unity WebGL prototype
25–35 s	Component selection	Interaction systems
35–45 s	Technical UI	Information architecture
45–55 s	Exploded view	Assembly relationships
55–65 s	Cross-section/clipping	Inspection tools
65–75 s	Visual modes	Rendering/analysis modes
75–82 s	Optimization stats	CAD-to-realtime proof
82–88 s	Evaluation metrics	Usability/workload proof
88–90 s	Links/CTA	Portfolio/GitHub/contact

24.9.2 90-second voiceover

Bloque de código: text
 TwinSight X500 is a Unity WebGL technical visualization prototype for inspecting and understanding the assembly of a Holybro
 ↪ X500 V2 drone.

The starting problem was technical documentation. Static diagrams and manuals can identify parts, but they often force users
 ↪ to mentally reconstruct how components relate in space.

TwinSight turns that assembly context into an interactive real-time 3D viewer.

Users can orbit the drone, select components, highlight parts and read technical information through the UI.

The exploded view separates components while preserving their spatial relationship, making the structure easier to inspect.

Cross-section and clipping tools allow users to analyze internal areas that would be difficult to understand from external
 ↪ views alone.

The prototype also includes multiple visual modes, including realistic, ghosted, blueprint, solid color and wireframe-style
 ↪ inspection modes.

A major part of the work was the CAD-to-realtime pipeline. CAD-derived geometry had to be cleaned, optimized and prepared in
 ↪ Blender before integration into Unity WebGL.

The project was evaluated using usability and workload methods such as SUS, NASA-TLX Raw and Think-Aloud.

My work covered Unity and C# implementation, runtime interaction systems, technical UI, Blender-based asset preparation,
 ↪ documentation, evaluation design and final integration.

The result is an academic prototype that demonstrates real-time 3D, WebGL deployment, technical visualization and
 ↪ CAD-to-realtime optimization.

24.9.3 90-second captions

Bloque de código: text

TwinSight X500
Unity WebGL drone assembly visualization

Problem:
2D diagrams show parts, but not always spatial relationships

Solution:
Interactive real-time 3D inspection

Feature:
Component selection + technical UI

Feature:
Exploded view for assembly structure

Feature:
Cross-section / clipping tools

Feature:
Visual inspection modes

Pipeline:
CAD-derived assets → optimized real-time model

Evaluation:
SUS · NASA-TLX Raw · Think-Aloud

Role:
Unity/C# · WebGL · Blender · Technical Visualization

Portfolio / GitHub / Contact

24.10 10. 120-second technical breakdown script

24.10.1 Structure

Time	Visual	Message
0–6 s	Title and hero	Project identity
6–18 s	Problem / documentation	Research motivation
18–30 s	App overview	Unity WebGL viewer
30–42 s	Selection system	Runtime interaction
42–54 s	UI panels	Technical data display
54–66 s	Exploded view	Assembly relation system
66–78 s	Cross-section	Inspection/clipping
78–90 s	Visual modes	Rendering modes
90–102 s	CAD pipeline	Asset optimization
102–112 s	Metrics	Reduction/evaluation
112–120 s	Role + closing	Candidate contribution

24.10.2 120-second voiceover

Bloque de codigo: text
This is TwinSight X500, a Unity WebGL technical visualization prototype built to inspect and understand the assembly of a
↪ Holybro X500 V2 drone.

The research problem comes from technical assembly documentation. Static diagrams and manuals identify components, but they
↪ often require users to mentally reconstruct spatial relationships between parts, subassemblies and fasteners.

TwinSight explores an interactive alternative: a browser-accessible real-time 3D viewer focused on inspection and spatial
↪ understanding.

The prototype allows users to orbit the drone, select components, highlight parts and access technical information panels.

The component selection system is designed around inspection workflows. The goal is not just to click a mesh, but to connect
↪ geometry, identity and technical information.

The exploded view separates the drone into readable groups while preserving assembly logic. This helps users understand how
↪ components relate to each other in space.

Cross-section and clipping tools support internal inspection, allowing hidden areas and layered components to be analyzed more
↪ clearly.

The project also includes visual modes such as realistic, ghosted, blueprint, solid color and wireframe-style views. These
↪ modes support different inspection tasks.

A major technical part of the project was the CAD-to-realtime pipeline. Dense CAD-derived geometry had to be converted, ↵ cleaned, optimized and prepared for Unity WebGL deployment.

The workflow included Blender-based cleanup, retopology and asset preparation, with the goal of preserving component identity ↵ while reducing the model enough for browser-based interaction.

Reported thesis metrics include geometry reduction from over 6.5 million triangles to approximately 95,617 optimized ↵ triangles, plus usability and workload evaluation using SUS, NASA-TLX Raw and Think-Aloud. These final values should be ↵ verified against the thesis report before public release.

My role covered Unity/C# implementation, runtime interaction systems, technical UI, CAD-to-realtime asset preparation, ↵ documentation, evaluation design and final integration.

TwinSight is an academic prototype, not a commercial engineering tool, but it demonstrates the kind of real-time 3D technical ↵ visualization work I want to continue building.

24.10.3 120-second captions

Bloque de codigo: text
TwinSight X500
Unity WebGL technical visualization prototype

Problem:
Assembly documentation can hide spatial relationships

Goal:
Interactive 3D inspection and understanding

Runtime system:
Component selection and highlighting

Technical UI:
Part information panels

Assembly analysis:
Exploded view

Inspection:
Cross-section / clipping

Visual analysis:
Realistic · ghosted · blueprint · wireframe

Pipeline:
CAD-derived geometry → optimized real-time assets

Optimization:
6.5M+ triangles → ~95,617 triangles
[verify final metric]

Evaluation:
SUS · NASA-TLX Raw · Think-Aloud

Role:
Unity/C# implementation · technical UI · asset optimization · evaluation

Built as Multimedia Engineering thesis

24.11 11. Silent LinkedIn version

LinkedIn often autoplays without sound. The silent version must work with captions only.

24.11.1 Silent version rules

- Use large captions.
- Keep each caption under 8 words when possible.
- Use strong visual changes.
- No tiny UI text.
- Add final call-to-action.
- Keep under 60–75 seconds.

24.11.2 Silent caption sequence

```

Bloque de código: text
TwinSight X500

Unity WebGL technical visualization

Problem:
2D assembly diagrams are limited

Goal:
Understand spatial relationships

Interactive drone inspection

Select components

View technical information

Exploded view

Cross-section tools

Visual inspection modes

CAD-to-realtime optimization

Built with:
Unity · C# · WebGL · Blender

Evaluated with:
SUS · NASA-TLX · Think-Aloud

Thesis complete, pending defense

Portfolio / GitHub / LinkedIn

```

24.12 12. LinkedIn 30–45 second cutdown

24.12.1 Use case

Use this as a first public LinkedIn post if the full demo is not ready.

24.12.2 Structure

Time	Visual	Caption
0–3 s	Hero	TwinSight X500
3–8 s	Orbit	Unity WebGL drone inspection
8–14 s	Selection	Component selection
14–20 s	Exploded	Exploded view
20–27 s	Cross-section	Cross-section tools
27–34 s	Visual modes	Multiple inspection modes
34–42 s	Closing	Unity · C# · WebGL · Blender

24.12.3 Caption text for post

```

Bloque de código: text
TwinSight X500 is my Unity WebGL technical visualization prototype for drone assembly inspection.

The project explores how interactive real-time 3D can support spatial understanding of technical assembly processes.

Built with Unity, C#, WebGL and Blender.

```


24.13 13. Technical-reviewer cut

24.13.1 Use case

Use when applying to technical visualization, simulation, digital twin or Unity technical artist roles.

24.13.2 Emphasis

This version should include more technical footage:

- Unity editor view;
- scene hierarchy;
- scripts briefly;
- asset before/after;
- triangle count;
- WebGL build;
- interaction architecture diagram;
- evaluation summary.

24.13.3 Structure

```
Bloque de codigo: text
1. App overview
2. Interaction systems
3. Asset pipeline
4. WebGL optimization
5. Evaluation
6. Limitations
7. Role/contribution
```

24.13.4 Voiceover focus

Use more technical phrases:

```
Bloque de codigo: text
runtime interaction systems
CAD-derived geometry
WebGL constraints
asset optimization
state management
technical UI
evaluation methodology
```

Do not overdo code screenshots unless they are readable.

24.14 14. Capture shot checklist

24.14.1 Scene preparation

- Use final or clean build.
- Hide debug UI unless needed.
- Set good camera starting position.
- Use readable UI scaling.
- Disable unnecessary console logs.
- Use stable frame rate.
- Prepare saved camera positions if possible.
- Ensure model materials look acceptable.

Prepare visual mode transitions.
Prepare exploded view state.
Prepare cross-section state.
Prepare selected component examples.

24.14.2 Recording checklist

Record full drone orbit.
Record component selection.
Record info panel.
Record exploded view activation.
Record exploded view rotation.
Record cross-section/clipping.
Record each visual mode.
Record CAD/optimized comparison if available.
Record metrics/evaluation slide.
Record closing screen.

24.14.3 Post-capture checklist

Cut dead time.
Remove errors.
Add captions.
Add title screen.
Add end card.
Normalize audio.
Export 1080p.
Create silent version.
Create thumbnail.

24.15 15. Suggested screen recording order

Record in this order to simplify editing:

1. full drone idle;
 2. camera orbit;
 3. component selection;
 4. info panel;
 5. isolate/highlight behavior;
 6. exploded view;
 7. exploded view + camera orbit;
 8. cross-section/clipping;
 9. visual modes;
 10. CAD/optimized before-after;
 11. metrics/evaluation screen;
 12. end card.
-

24.16 16. Thumbnail options

24.16.1 Thumbnail 1 — best general

Visual:

- drone full view;
- clean dark background;
- wireframe overlay;
- short text.

Text:

```
Bloque de codigo: text
TwinSight X500
Unity WebGL Technical Visualization
```

24.16.2 Thumbnail 2 — technical

Text:

```
Bloque de codigo: text
CAD-to-Realtime Drone Visualization
Unity · WebGL · Blender
```

24.16.3 Thumbnail 3 — recruiter

Text:

```
Bloque de codigo: text
Interactive 3D Drone Assembly Viewer
```

Avoid thumbnail text longer than 6–8 words.

24.17 17. End card

24.17.1 End card text

```
Bloque de codigo: text
TwinSight X500

Unity · C# · WebGL · Blender
Technical Visualization · CAD-to-Realtime

Portfolio: [URL]
GitHub: [URL]
LinkedIn: [URL]
```

24.17.2 Shorter end card

```
Bloque de codigo: text
TwinSight X500
Unity WebGL Technical Visualization

Portfolio · GitHub · LinkedIn
```

24.18 18. Music and voice

24.18.1 Voiceover

Recommended if English pronunciation is acceptable.

If voiceover is weak or noisy, use silent captioned version.

24.18.2 Music

Optional.

If used:

- low volume;
- no dramatic cinematic music;
- no lyrics;
- no distracting beats;
- royalty-free;
- lower than voice.

24.18.3 Audio priority

```
Bloque de codigo: text  
Clear voice > music
```

If voice is not clear, remove it.

24.19 19. Export formats

24.19.1 Portfolio video

```
Bloque de codigo: text  
MP4  
1920x1080  
30 fps  
H.264  
90 seconds
```

24.19.2 LinkedIn video

```
Bloque de codigo: text  
MP4  
1920x1080 or 1080x1080  
30 fps  
captioned  
30-75 seconds
```

24.19.3 GitHub video/GIF

Use:

- YouTube/Vimeo/Loom link;
- or small optimized GIF under docs/images/demo-preview.gif.

Avoid large video files inside GitHub repo.

24.19.4 File names

```
Bloque de codigo: text
twinsight-x500-demo-90s.mp4
twinsight-x500-recruiter-demo-60s.mp4
twinsight-x500-technical-breakdown-120s.mp4
twinsight-x500-linkedin-cutdown.mp4
```

24.20 20. Demo page embed copy

Use this under the embedded video:

```
Bloque de codigo: text
TwinSight X500 is a Unity WebGL technical visualization prototype for inspecting drone assembly. The demo shows component
↔ selection, exploded view, cross-section tools, visual modes and CAD-to-realtime optimization.
```

24.21 21. Common mistakes to avoid

Do not:

- start with a long thesis explanation;
 - spend 15 seconds on logos;
 - show UI text too small to read;
 - show a laggy or broken WebGL build;
 - overfocus on the drone model beauty;
 - underexplain the technical work;
 - mention every technology in the voiceover;
 - show code that is unreadable;
 - include unverified metrics;
 - use “AI-built” language;
 - show manufacturer assets without disclaimer if public usage is uncertain;
 - exceed 2 minutes for the main demo.
-

24.22 22. Quality gate

The video is ready when a viewer can answer:

1. What is TwinSight?
2. What problem does it solve?
3. What did Alexander build?
4. What technologies were used?
5. What makes it relevant to real-time 3D / Unity roles?
6. What features are visible?
7. What evidence exists beyond visuals?
8. Where can I see more?

If the answer to any of these is unclear, revise.

24.23 23. Editing checklist

24.23.1 Before exporting

Title is clear.
First 5 seconds show the project.
Captions are readable.
Video shows actual interaction.
Feature sequence is logical.
No dead time.
No broken UI.
No unverified metrics.
AI disclosure is not overdone.
End card includes links.
File is under reasonable size.
Audio levels are clean.
Silent version still works.

24.24 24. Final recommended production sequence

24.24.1 Step 1

Record raw footage.

24.24.2 Step 2

Build the 90-second portfolio demo.

24.24.3 Step 3

Cut the 60-second recruiter version.

24.24.4 Step 4

Add captions.

24.24.5 Step 5

Export silent LinkedIn version.

24.24.6 Step 6

Create thumbnail.

24.24.7 Step 7

Embed in portfolio.

24.24.8 Step 8

Add to LinkedIn Featured.

24.24.9 Step 9

Link in GitHub README.

24.24.10 Step 10

Use in outreach.

24.25 25. Next file dependency

The next file should be:

```
Bloque de codigo: text
22_outreach_templates_english_spanish.md
```

It should include:

- recruiter connection messages;
- hiring manager messages;
- technical director messages;
- speculative application;
- follow-up messages;
- referral requests;
- salary expectation responses;
- remote-from-Colombia explanation;
- contractor setup explanation;
- AI-assisted development explanation;
- low formal experience explanation;
- English and Spanish versions.

No Deep Research or agent required.

Chapter 25

21B_twinsight_demo_video_final_plan

Fuente: 21B_twinsight_demo_video_final_plan.md

Seccion: root

Estado: canonical

Rol: Modulo final complementario

Lineas originales: 896

Ultima modificacion: 2026-06-11 19:39

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Produce a polished video system for **TwinSight X500**, the flagship portfolio project.

Contenido completo depurado

25.0.1 0. Module status

```
Bloque de codigo: text
Module: 21B_twinsight_demo_video_final_plan.md
Date generated: 2026-06-12
Status: Complete v1
Purpose: Final production plan for TwinSight X500 demo videos
Input modules: 21, 29, 29B, 29C, B1, B5
Primary artifact: demo video system for portfolio, LinkedIn, GitHub and ArtStation
```

25.1 1. Objective

Produce a polished video system for **TwinSight X500**, the flagship portfolio project.

The video system must prove, in the shortest possible time, that Alexander can operate as:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist
Unity WebGL · C# · CAD-to-Realtime · Technical Visualization · Interactive 3D
```

The video must not be a generic thesis presentation. It must be a recruiter-readable and technical-lead-readable demonstration of:

- final product quality;
- Unity WebGL implementation;
- runtime interaction;
- technical UI;

- CAD-to-realtime optimization;
- real-time visualization modes;
- evaluation evidence;
- clarity of role and contribution.

25.2 2. Strategic reason

The market evidence from B1 and B5 reinforces that video proof is not optional.

Relevant hiring signals found:

Market signal	Implication for video
Unity / C# appears repeatedly	show Unity + C# clearly
Technical Art roles request pipeline and optimization	show visual modes and CAD optimization
Render / Graphic Programmer roles value shaders, math, GPU and performance	include visual mode and optimization evidence
VR / XR Unity roles value interaction and profiling	show runtime interaction and future XR-adjacent logic
LATAM remote roles ask for portfolio evidence	video must be portfolio-ready
Freelance/talent-pool roles ask for strong portfolio	video must be self-contained proof
Blender appears in some Unity/VR postings	show Blender pipeline briefly
Web platform appears in LATAM freelance roles	show browser/WebGL proof

Conclusion:

Bloque de código: text
The video must show both product value and technical implementation.

25.3 3. Output video package

The required package has three videos.

Asset	Duration	Purpose	Priority
Main demo	90–120 s	portfolio, LinkedIn Featured, case study	1
Short teaser	30–45 s	LinkedIn post, homepage, quick recruiter skim	2
Technical breakdown	4–7 min	interview follow-up, GitHub, technical validation	3

25.3.1 Minimum viable version

Before applications, at least produce:

Bloque de código: text
[] Main demo: 90–120 s
[] Short teaser: 30–45 s
[] 8–12 strong screenshots
[] Portfolio embed
[] GitHub README embed
[] LinkedIn Featured entry

The long technical breakdown can come after the main demo if time is limited.

25.4 4. Core positioning line

Use this exact positioning across the video package:

Bloque de codigo: text
TwinSight X500 is a Unity WebGL technical visualization prototype for drone assembly inspection.

Expanded version:

Bloque de codigo: text
It transforms CAD-derived drone assets into an optimized browser-based 3D viewer with component selection, exploded view, ↩ cross-section tools, visual modes, technical UI panels and usability/workload evaluation.

Do not lead with:

Bloque de codigo: text
This is my thesis.

The thesis context appears later as validation.

25.5 5. Audience hierarchy

25.5.1 5.1 Primary audience

Bloque de codigo: text
Technical hiring manager
Unity lead
Technical art lead
Interactive 3D studio founder
XR / visualization team
Nearshore recruiter screening portfolio

25.5.2 5.2 Secondary audience

Bloque de codigo: text
HR recruiter
LinkedIn viewer
ArtStation viewer
Academic evaluator
Freelance client

25.5.3 5.3 What each audience needs to see

Audience	Needs to understand
Recruiter	role fit and keywords fast
Technical lead	implementation credibility
Technical artist	visual modes, pipeline, optimization
Visualization company	industrial/technical communication value
XR/simulation company	interaction logic and future extensibility
Freelance client	deliverable clarity and browser access

25.6 6. Main demo — 90-second structure

25.6.1 6.1 Timeline

```
Bloque de código: text
00-05 s Hook / hero shot
05-12 s Problem: 2D documentation lacks spatial context
12-22 s Solution: browser-based Unity WebGL viewer
22-34 s Component selection and technical panels
34-46 s Exploded view
46-58 s Cross-section / clipping
58-68 s Visual modes
68-78 s CAD-to-realtime optimization
78-86 s Evaluation metrics
86-90 s Role, stack, links
```

25.6.2 6.2 Shot-by-shot plan

Time	Shot	Caption	Purpose
00-03	Final viewer hero	TwinSight X500	immediate visual proof
03-05	Orbit with UI visible	Unity WebGL technical visualization	position clearly
05-08	2D/manual support visual	Static 2D documentation hides spatial relationships	problem
08-12	transition to 3D viewer	Interactive spatial inspection in the browser	solution
12-17	browser frame + app load	Browser-based, no install required	WebGL proof
17-22	camera navigation	Real-time 3D drone assembly viewer	interaction
22-27	hover/select component	Component-level selection	runtime proof
27-34	info panel opens	Technical part information	UI proof
34-40	exploded view activation	Exploded assembly view	feature proof
40-46	exploded close-up	Understand hierarchy and placement	value proof
46-52	clipping plane movement	Cross-section inspection	technical inspection
52-58	internal structure exposed	Reveal hidden relationships	spatial reasoning
58-64	visual modes switch	X-ray · ghosted · blueprint · wireframe	technical art
64-68	visual modes grid	Multiple inspection modes	visual system
68-73	CAD high-poly / low-poly comparison	CAD-to-realtime optimization	pipeline proof
73-78	triangle count card	6.5M+ source routes → 95,617 triangles	metric
78-83	evaluation card	SUS 91.88 · NASA-TLX Raw 8.69 vs 19.89	validation
83-86	final viewer + metrics	Evaluated with 12 participants	credibility
86-90	end card	Unity · C# · WebGL · URP · UI Toolkit · Blender	stack

25.7 7. Main demo — 120-second version

Use this if the project needs more breathing room.

```
Bloque de código: text
00-05 s Hero shot
05-12 s Project title and positioning
12-22 s Problem context
22-32 s WebGL browser interaction
32-44 s Component selection
44-56 s Exploded view
56-68 s Cross-section
68-80 s Visual modes
80-92 s CAD optimization pipeline
92-102 s UI Toolkit / runtime systems
102-112 s Evaluation metrics
112-120 s Links / stack / role
```

25.7.1 120-second shot additions

Added section	Content
UI Toolkit	show bottom sheet / panel interaction
Runtime systems	show state transitions
Blender pipeline	brief viewport before/after
GitHub proof	one-second repo/README preview
Role clarity	explicit contribution card

25.8 8. Short teaser — 30–45 seconds

25.8.1 8.1 30-second version

```
Bloque de código: text
00-03 s Hero shot
03-07 s Select component
07-12 s Exploded view
12-17 s Clipping
17-22 s Visual modes
22-27 s Metrics card
27-30 s Portfolio link / title
```

25.8.2 8.2 45-second version

```
Bloque de código: text
00-04 s Hero shot + title
04-09 s Problem contrast
09-15 s Browser/WebGL proof
15-21 s Component selection
21-28 s Exploded view
28-34 s Clipping
34-39 s Visual modes
39-43 s Metrics
43-45 s CTA
```

25.8.3 8.3 Teaser caption script

```
Bloque de código: text
Static 2D assembly docs are hard to understand.

TwinSight X500 turns drone assembly into an interactive Unity WebGL viewer.

Select components.
Explode the assembly.
Cut through structures.
Switch visual modes.
Inspect technical data.

CAD-derived assets optimized for browser-based real-time 3D.
```

25.9 9. Technical breakdown — 4–7 minutes

25.9.1 9.1 Purpose

The long breakdown is not for recruiters. It is for technical leads and interviews.

It should prove that Alexander understands:

- CAD optimization constraints;

- Unity WebGL constraints;
- runtime systems;
- technical UI;
- visual modes;
- evaluation logic;
- limitations.

25.9.2 9.2 Structure

```
Bloque de codigo: text
00:00-00:30 Project overview
00:30-01:10 Problem and design goal
01:10-02:00 CAD source and optimization
02:00-02:50 Blender cleanup / asset preparation
02:50-03:40 Unity WebGL architecture
03:40-04:30 Interaction systems
04:30-05:10 Visual modes / shaders
05:10-05:50 UI Toolkit / technical panels
05:50-06:30 Evaluation results
06:30-07:00 Limitations and next steps
```

25.9.3 9.3 What to show

Segment	Visual proof
CAD source	raw geometry / high-poly route
Optimization	wireframe before/after
Blender	object cleanup, UV, bake, organization
Unity	scene hierarchy, prefabs, UI
WebGL	running browser build
Interaction	selection, exploded, clipping
Visual modes	shader/material switch
Evaluation	SUS/NASA-TLX result cards
GitHub	repo structure / README

25.10 10. Captions

25.10.1 10.1 Caption rules

Rule	Apply
Max length	6–10 words
Style	technical, direct
Frequency	every major feature
Placement	lower third or side label
Contrast	readable over footage
Avoid	paragraphs

25.10.2 10.2 Caption bank

```
Bloque de codigo: text
Unity WebGL technical visualization
Browser-based drone assembly inspection
CAD-derived asset optimization
Component-level selection
Technical part information
Exploded assembly view
Cross-section inspection
Multiple visual modes
```

Real-time interaction systems
 Optimized for browser deployment
 Evaluated with SUS + NASA-TLX
 Built with Unity, C#, URP, UI Toolkit, Blender

25.10.3 10.3 Metrics captions

Bloque de codigo: text
 6.5M+ CAD triangle routes → 95,617 optimized triangles
 12 validation participants
 SUS average: 91.88
 NASA-TLX Raw: 8.69 viewer vs 19.89 2D support
 96 task-condition records

25.11 11. Voiceover

25.11.1 11.1 Recommendation

Use:

Bloque de codigo: text
 captions + restrained music

for:

- main demo;
- short teaser.

Use:

Bloque de codigo: text
 voiceover + captions

only for:

- long technical breakdown.

25.11.2 11.2 Main demo voiceover option

If voiceover is used, keep it minimal:

Bloque de codigo: text
 TwinSight X500 is a Unity WebGL technical visualization prototype for drone assembly inspection. It converts CAD-derived
 ↪ assets into an optimized browser-based viewer with component selection, exploded view, cross-section tools, visual modes
 ↪ and technical UI panels. The prototype was evaluated with SUS, NASA-TLX and Think-Aloud protocols as part of an academic
 ↪ validation.

25.11.3 11.3 Avoid

Do not use an overly promotional voiceover.

Avoid:

Bloque de codigo: text
 revolutionary
 game-changing
 industry-leading
 next-generation
 metaverse

25.12 12. Required footage list

25.12.1 12.1 Must capture

Bloque de codigo: text

[] Full drone hero view
[] Browser/WebGL proof
[] Orbit/navigation
[] Component hover
[] Component selection
[] Technical info panel
[] Exploded view activation
[] Exploded view close-up
[] Clipping plane movement
[] Internal structure reveal
[] Realistic visual mode
[] X-ray / ghosted visual mode
[] Blueprint / wireframe mode
[] Thermal-style mode
[] CAD/high-poly vs optimized comparison
[] Blender cleanup screenshot/clip
[] Unity editor screenshot/clip
[] Evaluation metrics cards
[] Final stack card

25.12.2 12.2 Optional footage

Bloque de codigo: text

[] GitHub README scroll
[] UI Toolkit layout
[] Shader Graph
[] C# script snippet
[] mobile viewport
[] profiler
[] Marmoset bake preview
[] RizomUV layout

Only include optional footage if it is readable.

25.13 13. Editing style

25.13.1 13.1 Visual style

Element	Recommendation
Theme	dark technical / clean
Typography	modern sans-serif
Motion	simple cuts and subtle fades
Captions	consistent lower-third
Music	low-volume, neutral
Sound effects	minimal
Transitions	no flashy transitions
Color	consistent with portfolio

25.13.2 13.2 Do not use

Bloque de codigo: text

glitch transitions
cinematic trailer excess
slow 20-second intro
unreadable UI
random beauty shots
meme-style cuts
oversaturated music
large walls of text

25.14 14. Export specs

25.14.1 14.1 Main demo

Parameter	Value
Resolution	1920×1080
Frame rate	30 fps or 60 fps
Duration	90–120 s
Format	MP4 / H.264
Aspect	16:9
Captions	burned-in
Audio	optional
Use	portfolio, GitHub, LinkedIn Featured

25.14.2 14.2 Short teaser

Parameter	Value
Resolution	1920×1080 or 1080×1080
Duration	30–45 s
Format	MP4 / H.264
Captions	mandatory
Hook	first 2–3 seconds
Use	LinkedIn post, portfolio homepage

25.14.3 14.3 Technical breakdown

Parameter	Value
Resolution	1920×1080
Duration	4–7 min
Format	MP4 / H.264
Audio	voiceover recommended
Captions	recommended
Use	interview, GitHub, technical follow-up

25.15 15. Tooling workflow

25.15.1 15.1 Capture

Recommended tools:

Bloque de código: text
 OBS Studio
 Unity Recorder
 Screen recording from browser
 Blender viewport capture
 DaVinci Resolve / Premiere / CapCut Desktop

25.15.2 15.2 Capture settings

Setting	Recommendation
Capture resolution	1920×1080
Browser zoom	100% or 110%
UI scale	readable
Cursor	visible for interactions
Frame pacing	stable
Background apps	closed
Recording format	MKV or MP4
Audio	separate if using voiceover

25.15.3 15.3 Unity capture checklist

Bloque de código: text

[]

Disable editor gizmos unless needed

[]

Use clean camera path

[]

Verify UI readability

[]

Remove debug labels unless intentional

[]

Keep frame rate stable

[]

Use final materials

[]

Avoid half-finished elements

25.16 16. Production order

25.16.1 16.1 Recommended sequence

Bloque de código: text

1.

Capture clean hero shots

2.

Capture each feature separately

3.

Capture pipeline visuals

4.

Create metric cards

5.

Build 30-second teaser

6.

Build 90-second main demo

7.

Build long breakdown

8.

Export thumbnails

9.

Embed in portfolio

10.

Update GitHub and LinkedIn

25.16.2 16.2 Why teaser first

The teaser forces clarity. If the project cannot be explained in 30 seconds, the 90-second demo will also become unfocused.

25.17 17. Thumbnail plan

25.17.1 17.1 Main thumbnail

Use:

Bloque de código: text

Exploded drone + UI panel + title

Title:

Bloque de código: text

TwinSight X500

Unity WebGL Technical Visualization

25.17.2 17.2 Alternative thumbnails

Thumbnail	Use
Cross-section close-up	technical inspection
Visual modes grid	technical art
Browser screenshot	WebGL proof
Metrics card	validation proof

25.17.3 17.3 Avoid

Bloque de código: text
 plain drone render
 unreadable small text
 busy interface
 academic title slide

25.18 18. Distribution plan

25.18.1 18.1 Portfolio

Embed the 90–120 s main demo at the top of the TwinSight case study.

Then place:

Bloque de código: text
 Problem → Solution → Features → Pipeline → Metrics → Role → Links

25.18.2 18.2 GitHub

Use:

Bloque de código: markdown
 ## Demo

[![TwinSight X500 Demo](./media/twinsight-demo-thumbnail.jpg)](https://youtube.com/...)

Highlights

- Unity WebGL technical visualization
- CAD-to-realtime optimization
- Component selection and technical UI
- Exploded view and clipping tools
- Visual modes for inspection
- SUS + NASA-TLX evaluation

25.18.3 18.3 LinkedIn

Use:

- 30–45 s teaser as native upload;
- 90 s video in Featured section;
- project case-study link;
- concise post text.

25.18.4 18.4 ArtStation

Use:

- 30–60 s reel near the top;
- final screenshots;
- breakdown images;
- pipeline diagram;
- metrics card;
- tools and tags.

25.19 19. LinkedIn post options

25.19.1 19.1 Technical version

Bloque de código: text
I built TwinSight X500, a Unity WebGL technical visualization prototype for drone assembly inspection.

The goal was to move beyond static 2D assembly documentation and support spatial understanding through interactive 3D ↔ inspection.

Core features:

- component selection
- exploded assembly view
- cross-section tools
- multiple visual modes
- technical part panels
- CAD-to-realtime optimization
- SUS / NASA-TLX evaluation

Stack:
Unity, C#, WebGL, URP, UI Toolkit, Blender.

25.19.2 19.2 Short version

Bloque de código: text
TwinSight X500 – Unity WebGL technical visualization for drone assembly inspection.

CAD-derived assets optimized into an interactive browser-based viewer with component selection, exploded view, clipping tools, ↔ visual modes and usability/workload evaluation.

25.20 20. Risk controls

25.20.1 20.1 Overclaim risks

Risk	Control
calling it a full digital twin	say digital-twin-adjacent
implying production deployment	say prototype
implying IoT	state no live IoT
implying physical simulation	thermal-style is visual only
hiding AI assistance	disclose if relevant
overclaiming seniority	focus on project evidence
overclaiming XR	say future extension unless implemented

25.20.2 20.2 Exact safe wording

```
Bloque de código: text
academic/portfolio prototype
formative evaluation
technical visualization
digital-twin-adjacent
browser-based viewer
qualitative thermal-style visual mode
CAD-derived assets
optimized for WebGL
```

25.21 21. Acceptance criteria

The video package is acceptable only if a viewer understands within 20 seconds:

```
Bloque de código: text
1. What the project is.
2. What problem it solves.
3. What the user can do.
4. Why it is technically relevant.
```

25.21.1 21.1 Main demo checklist

```
Bloque de código: text
[ ] final result shown in first 5 seconds
[ ] project title visible
[ ] WebGL/browser proof visible
[ ] component selection shown
[ ] exploded view shown
[ ] clipping shown
[ ] visual modes shown
[ ] optimization shown
[ ] metrics shown
[ ] stack shown
[ ] role clear
[ ] no unsupported claims
```

25.21.2 21.2 Technical quality checklist

```
Bloque de código: text
[ ] UI readable
[ ] camera movement smooth
[ ] no debug artifacts
[ ] no stutter in captured footage
[ ] captions readable
[ ] video under target duration
[ ] file size manageable
[ ] thumbnail clear
[ ] links working
```

25.22 22. Immediate production checklist

25.22.1 22.1 Before recording

```
Bloque de código: text
[ ] final build opens reliably
[ ] UI layout stable
[ ] visual modes functional
[ ] exploded view functional
[ ] clipping functional
[ ] component metadata filled
[ ] browser performance acceptable
[ ] model centered and framed
[ ] final materials applied
[ ] metrics confirmed
```

25.22.2 22.2 During recording

Bloque de codigo: text

```
[ ] record each feature separately
[ ] record extra seconds before/after each action
[ ] keep cursor movement controlled
[ ] avoid rushed camera motion
[ ] avoid accidental UI glitches
[ ] capture at least 2 takes per feature
```

25.22.3 22.3 After editing

Bloque de codigo: text

```
[ ] export main video
[ ] export teaser
[ ] export thumbnail
[ ] export GIF or web preview if needed
[ ] upload video
[ ] embed in portfolio
[ ] update GitHub README
[ ] add to LinkedIn Featured
[ ] add to ArtStation post
```

25.23 23. Final recommended deliverables

25.23.1 Required

Bloque de codigo: text

```
twinsight_x500_demo_90s.mp4
twinsight_x500_teaser_30s.mp4
twinsight_x500_thumbnail.jpg
twinsight_x500_metrics_card.png
twinsight_x500_visual_modes_grid.png
twinsight_x500_pipeline_diagram.png
```

25.23.2 Recommended

Bloque de codigo: text

```
twinsight_x500_technical_breakdown_5min.mp4
twinsight_x500_artstation_breakdown_images.zip
twinsight_x500_github_media_folder
```

25.24 24. Next module

After this file, continue with:

Bloque de codigo: text

```
08B_twinsight_case_study_final_structure.md
```

25.24.1 Tool recommendation

Tool	Use
Agent mode	No
Deep Research	No

Reason:

The case-study structure should be generated from existing research and the final video plan, not from new browsing.

25.24.2 Prompt

```
Bloque de codigo: text
Proceed with 08B_twinsight_case_study_final_structure.md.

Create the final portfolio case-study structure for TwinSight X500 using the existing roadmap, 29, 29B, 29C and 21B.

It must include:
- title
- subtitle
- hero structure
- problem
- solution
- key features
- technical pipeline
- CAD-to-realtime optimization
- Unity/WebGL architecture
- visual modes
- evaluation results
- role and contribution
- AI-assisted disclosure
- limitations
- screenshots needed
- links
- final copy blocks ready for portfolio implementation

Do not browse.
Do not use agent.
Generate .md.
```

Chapter 26

22_outreach_templates_english_spanish

Fuente: 22_outreach_templates_english_spanish.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1164

Ultima modificacion: 2026-06-11 15:20

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This document provides reusable outreach templates for the international job-search system.

Contenido completo depurado

26.0.1 Purpose

This document provides reusable outreach templates for the international job-search system.

It includes English and Spanish versions for:

- recruiter connection messages;
- hiring manager messages;
- technical art lead messages;
- Unity technical director messages;
- XR / simulation / digital twin companies;
- creative technology studios;
- game studio recruiters;
- speculative applications;
- referral requests;
- professor / advisor recommendation requests;
- former freelance client testimonial requests;
- follow-ups;
- salary expectation responses;
- remote-from-Colombia explanations;
- contractor setup explanations;
- future Portuguese / EU mobility explanations;
- contract/payment questions;
- declining unstable offers;
- explaining low formal experience;
- explaining AI-assisted development;

- answering “Why should we hire you?”

The goal is to support consistent, direct, non-desperate, evidence-based communication.

26.1 1. Messaging principles

26.1.1 Core positioning

Every message should reinforce this identity:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization,
↵ and Unity WebGL deployment.
```

26.1.2 Primary proof

Use TwinSight X500 as the main proof asset.

```
Bloque de codigo: text
TwinSight X500 = Unity WebGL technical visualization prototype for drone assembly inspection.
```

26.1.3 Secondary proof

Use ARA only when relevant:

```
Bloque de codigo: text
ARA Framework = Python/LangGraph research automation prototype.
```

26.1.4 Tone

Use:

- direct;
- technically specific;
- concise;
- evidence-based;
- no begging;
- no exaggerated enthusiasm;
- no inflated seniority.

Avoid:

- “I am passionate about everything tech”;
 - “I would love any opportunity”;
 - “I can do anything”;
 - long autobiographical explanations;
 - overexplaining legal status;
 - overexplaining AI usage unless asked.
-

26.2 2. Personalization variables

Use these placeholders:


```
Bloque de código: text
[Name]
[Company]
[Role]
[Project / product]
[Why they fit]
[Portfolio URL]
[TwinSight URL]
[GitHub URL]
[LinkedIn URL]
[CV URL]
[Your email]
[Date]
[Salary range]
[Time zone overlap]
```

26.2.1 Minimum personalization per message

Every message should include at least:

1. company name;
2. relevant role or product;
3. why TwinSight is related;
4. portfolio or GitHub link.

26.3 3. Short pitch options

26.3.1 English — 1-line pitch

```
Bloque de código: text
I build Unity/WebGL real-time 3D systems for technical visualization, with a focus on CAD-to-realtime optimization and
↪ interactive inspection workflows.
```

26.3.2 English — 2-line pitch

```
Bloque de código: text
I am a Colombia-based Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, CAD-to-realtime optimization and
↪ interactive technical visualization. My flagship project, TwinSight X500, is a Unity WebGL drone assembly inspection
↪ prototype with component selection, exploded view, cross-section tools and technical UI.
```

26.3.3 Spanish — 1-line pitch

```
Bloque de código: text
Desarrollo sistemas 3D en tiempo real con Unity/WebGL para visualización técnica, optimización CAD-to-realtime e inspección
↪ interactiva.
```

26.3.4 Spanish — 2-line pitch

```
Bloque de código: text
Soy Real-Time 3D Developer / Unity Technical Artist en Colombia, enfocado en Unity WebGL, optimización CAD-to-realtime y
↪ visualización técnica interactiva. Mi proyecto principal, TwinSight X500, es un prototipo Unity WebGL para inspección de
↪ ensamblaje de un dron con selección de componentes, vista explotada, cortes y UI técnica.
```

26.4 4. Recruiter connection request

26.4.1 English — LinkedIn connection request

```
Bloque de codigo: text
Hi [Name], I'm a Colombia-based Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, technical
↪ visualization and CAD-to-realtime optimization. I'm exploring remote Unity / interactive 3D roles and would like to
↪ connect.
```

26.4.2 English — shorter

```
Bloque de codigo: text
Hi [Name], I build Unity/WebGL technical visualization systems and I'm exploring remote real-time 3D roles. I'd like to
↪ connect.
```

26.4.3 Spanish — LinkedIn connection request

```
Bloque de codigo: text
Hola [Name], soy Real-Time 3D Developer / Unity Technical Artist en Colombia, enfocado en Unity WebGL, visualización técnica y
↪ optimización CAD-to-realtime. Estoy explorando roles remotos de Unity / 3D interactivo y me gustaría conectar.
```

26.5 5. Recruiter first message after connection

26.5.1 English

```
Bloque de codigo: text
Hi [Name], thanks for connecting.

I'm currently looking for remote contractor or full-time opportunities in Unity, real-time 3D, Unity WebGL, technical
↪ visualization, simulation/digital twin visualization, or technical art workflows.

My main project is TwinSight X500, a Unity WebGL technical visualization prototype for drone assembly inspection. It includes
↪ component selection, exploded view, cross-section tools, visual modes, technical UI and CAD-to-realtime optimization.

Portfolio: [Portfolio URL]
GitHub: [GitHub URL]

If you work with roles involving Unity, interactive 3D, XR, simulation or technical visualization, I'd be open to discussing
↪ fit.
```

26.5.2 Spanish

```
Bloque de codigo: text
Hola [Name], gracias por conectar.

Actualmente estoy buscando oportunidades remotas como contratador o full-time en Unity, 3D en tiempo real, Unity WebGL,
↪ visualización técnica, simulación/digital twin o technical art.

Mi proyecto principal es TwinSight X500, un prototipo Unity WebGL de visualización técnica para inspección de ensamblaje de un
↪ dron. Incluye selección de componentes, vista explotada, herramientas de corte, modos visuales, UI técnica y optimización
↪ CAD-to-realtime.

Portafolio: [Portfolio URL]
GitHub: [GitHub URL]

Si trabajas con roles de Unity, 3D interactivo, XR, simulación o visualización técnica, me interesa revisar posibles encajes.
```

26.6 6. Technical Art Lead message

26.6.1 English

Bloque de código: text

Hi [Name],

I saw your work at [Company] around [project/product/area]. I'm a Real-Time 3D Developer / Unity Technical Artist focused on
↔ Unity WebGL, technical visualization and CAD-to-realtime optimization.

My flagship project is TwinSight X500, a Unity WebGL drone assembly inspection prototype. It includes component selection,
↔ exploded view, cross-section tools, visual modes, technical UI and Blender-based asset optimization for real-time use.

I'm looking for remote roles or contract work where this kind of Unity technical visualization / runtime tooling background is
↔ relevant.

TwinSight: [TwinSight URL]

GitHub: [GitHub URL]

Would it make sense to connect or for me to send a short technical breakdown?

26.6.2 Spanish

Bloque de código: text

Hola [Name],

Vi el trabajo de [Company] en [project/product/area]. Soy Real-Time 3D Developer / Unity Technical Artist, enfocado en Unity
↔ WebGL, visualización técnica y optimización CAD-to-realtime.

Mi proyecto principal es TwinSight X500, un prototipo Unity WebGL para inspección de ensamblaje de un dron. Incluye selección
↔ de componentes, vista explotada, cortes, modos visuales, UI técnica y optimización de assets en Blender para tiempo real.

Estoy buscando roles remotos o contratos donde este perfil de visualización técnica en Unity / runtime tooling sea relevante.

TwinSight: [TwinSight URL]

GitHub: [GitHub URL]

¿Tendría sentido conectar o enviarte un breakdown técnico corto?

26.7 7. Unity Technical Director / Engineering Manager message

26.7.1 English

Bloque de código: text

Hi [Name],

I'm reaching out because [Company] works on [Unity / XR / simulation / visualization area], which matches my current focus.

I'm a Unity-focused Real-Time 3D Developer with a technical visualization background. My main project, TwinSight X500, is a
↔ Unity WebGL drone assembly inspection prototype built around CAD-to-realtime optimization, component selection, exploded
↔ views, cross-section tools, technical UI and usability/workload evaluation.

I'm currently based in Colombia and available for remote contractor/B2B work.

Portfolio: [Portfolio URL]

TwinSight: [TwinSight URL]

GitHub: [GitHub URL]

If your team is hiring or open to remote Unity contractors, I'd be interested in discussing where this profile may fit.

26.7.2 Spanish

Bloque de código: text

Hola [Name],

Te escribo porque [Company] trabaja en [Unity / XR / simulación / visualización], que encaja con mi enfoque actual.

Soy Real-Time 3D Developer enfocado en Unity y visualización técnica. Mi proyecto principal, TwinSight X500, es un prototipo
↔ Unity WebGL para inspección de ensamblaje de un dron, construido alrededor de optimización CAD-to-realtime, selección de
↔ componentes, vista explotada, herramientas de corte, UI técnica y evaluación de usabilidad/carga percibida.

Actualmente estoy en Colombia y disponible para trabajo remoto como contractor/B2B.

Portafolio: [Portfolio URL]
TwinSight: [TwinSight URL]
GitHub: [GitHub URL]

Si tu equipo está contratando o abierto a contractors remotos de Unity, me interesa revisar posible encaje.

26.8 8. XR / Simulation / Digital Twin company message

26.8.1 English

Bloque de codigo: text
Hi [Name],

I'm interested in [Company] because of your work in [XR / simulation / digital twins / industrial visualization].

My background is in Unity WebGL, real-time 3D and technical visualization. My flagship project, TwinSight X500, is a drone
↪ assembly inspection prototype that converts CAD-derived assets and technical documentation into an optimized interactive
↪ 3D viewer with component selection, exploded view, cross-section tools, visual modes and technical UI.

This seems relevant to teams building training simulations, digital twin viewers, product visualization tools, XR interfaces
↪ or technical inspection systems.

Portafolio: [Portfolio URL]
TwinSight: [TwinSight URL]

I'm based in Colombia and available for remote contractor/B2B work.

26.8.2 Spanish

Bloque de codigo: text
Hola [Name],

Me interesa [Company] por su trabajo en [XR / simulación / digital twins / visualización industrial].

Mi perfil está enfocado en Unity WebGL, 3D en tiempo real y visualización técnica. Mi proyecto principal, TwinSight X500, es
↪ un prototipo para inspección de ensamblaje de un dron que convierte assets derivados de CAD y documentación técnica en un
↪ visor 3D interactivo optimizado, con selección de componentes, vista explotada, cortes, modos visuales y UI técnica.

Esto puede ser relevante para equipos que desarrollan simuladores de entrenamiento, visores de digital twins, herramientas de
↪ visualización de producto, interfaces XR o sistemas de inspección técnica.

Portafolio: [Portfolio URL]
TwinSight: [TwinSight URL]

Estoy en Colombia y disponible para trabajo remoto como contractor/B2B.

26.9 9. Creative technology studio founder message

26.9.1 English

Bloque de codigo: text
Hi [Name],

I came across [Company] and your work in interactive 3D / creative technology.

I'm a Colombia-based Real-Time 3D Developer focused on Unity WebGL, technical visualization and optimized 3D workflows. My
↪ main project, TwinSight X500, is a Unity WebGL drone assembly inspection prototype with component selection, exploded
↪ view, cross-section tools, visual modes and technical UI.

I'm interested in remote contract work where interactive 3D, technical visualization, WebGL or Unity-based experiences are
↪ needed.

Portafolio: [Portfolio URL]
GitHub: [GitHub URL]

If you collaborate with remote specialists or contractors, I'd be open to discussing possible fit.

26.9.2 Spanish

Bloque de código: text
Hola [Name],

Encontré [Company] y su trabajo en 3D interactivo / creative technology.

Soy Real-Time 3D Developer en Colombia, enfocado en Unity WebGL, visualización técnica y flujos optimizados de 3D. Mi proyecto ↪ principal, TwinSight X500, es un prototipo Unity WebGL para inspección de ensamblaje de un dron, con selección de ↪ componentes, vista explotada, cortes, modos visuales y UI técnica.

Me interesa trabajo remoto por contrato donde se necesiten experiencias interactivas 3D, visualización técnica, WebGL o Unity.

Portafolio: [Portfolio URL]
GitHub: [GitHub URL]

Si colaboran con especialistas o contractors remotos, me interesa revisar un posible encaje.

26.10 10. Game studio recruiter message

26.10.1 English

Bloque de código: text
Hi [Name],

I'm exploring Unity / Technical Artist roles and wanted to reach out.

My strongest project is TwinSight X500, a Unity WebGL technical visualization prototype. Although it is not a game project, it ↪ demonstrates Unity/C# runtime interaction systems, technical UI, visual modes, WebGL deployment, Blender-based ↪ optimization and CAD-to-realtime asset workflows.

I'm especially interested in Unity Technical Artist, runtime TA, tools, optimization or interactive 3D roles where a hybrid ↪ art/technical profile is useful.

Portfolio: [Portfolio URL]
GitHub: [GitHub URL]

Would this profile be relevant for any current or upcoming roles at [Company]?

26.10.2 Spanish

Bloque de código: text
Hola [Name],

Estoy explorando roles de Unity / Technical Artist y quería contactarte.

Mi proyecto más fuerte es TwinSight X500, un prototipo Unity WebGL de visualización técnica. Aunque no es un proyecto de ↪ juego, demuestra sistemas runtime en Unity/C#, UI técnica, modos visuales, despliegue WebGL, optimización en Blender y ↪ flujos CAD-to-realtime.

Me interesan especialmente roles de Unity Technical Artist, runtime TA, tools, optimización o 3D interactivo donde sea útil un ↪ perfil híbrido técnico/artístico.

Portafolio: [Portfolio URL]
GitHub: [GitHub URL]

¿Este perfil podría ser relevante para algún rol actual o futuro en [Company]?

26.11 11. Speculative application

26.11.1 English

Bloque de código: text
Subject: Unity / Real-Time 3D Developer – Technical Visualization Portfolio

Hi [Name / Hiring Team],

I'm reaching out because [Company] works in [specific area], and my profile may be relevant for Unity, real-time 3D, technical ↪ visualization or interactive 3D work.

I'm a Colombia-based Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, CAD-to-realtime optimization and ↪ technical visualization.

My main project is TwinSight X500, a Unity WebGL drone assembly inspection prototype. It includes:

- component selection;
- exploded view;
- cross-section/clipping tools;
- visual modes;
- technical information panels;
- optimized CAD-derived assets;
- usability/workload evaluation.

Portfolio: [Portfolio URL]
TwinSight: [TwinSight URL]
GitHub: [GitHub URL]
CV: [CV URL]

I'm available for remote contractor/B2B work and would be interested in any current or upcoming roles where this background ↪ fits.

Best,
Alexander Woodcock Salomón

26.11.2 Spanish

Bloque de código: text
Asunto: Unity / Real-Time 3D Developer – Portafolio de visualización técnica

Hola [Name / Hiring Team],

Te escribo porque [Company] trabaja en [specific area], y mi perfil puede ser relevante para trabajo en Unity, 3D en tiempo ↪ real, visualización técnica o 3D interactivo.

Soy Real-Time 3D Developer / Unity Technical Artist en Colombia, enfocado en Unity WebGL, optimización CAD-to-realtime y ↪ visualización técnica.

Mi proyecto principal es TwinSight X500, un prototipo Unity WebGL para inspección de ensamblaje de un dron. Incluye:

- selección de componentes;
- vista explotada;
- herramientas de corte/clipping;
- modos visuales;
- paneles de información técnica;
- assets optimizados derivados de CAD;
- evaluación de usabilidad/carga percibida.

Portafolio: [Portfolio URL]
TwinSight: [TwinSight URL]
GitHub: [GitHub URL]
CV: [CV URL]

Estoy disponible para trabajo remoto como contractor/B2B y me interesa revisar roles actuales o futuros donde este perfil ↪ encaje.

Saludos,
Alexander Woodcock Salomón

26.12 12. Follow-up after applying

26.12.1 English — first follow-up

Bloque de código: text
Hi [Name],

I applied for the [Role] position on [date] and wanted to follow up briefly.

My background is in Unity WebGL, real-time 3D and technical visualization. My flagship project, TwinSight X500, is an ↪ interactive drone assembly inspection prototype with component selection, exploded view, cross-section tools, visual modes ↪ and CAD-to-realtime optimization.

Portfolio: [Portfolio URL]
TwinSight: [TwinSight URL]

I'd be glad to discuss whether this background fits the role.

Best,
Alexander

26.12.2 Spanish — first follow-up

Bloque de código: text
Hola [Name],

Apliqué al rol de [Role] el [date] y quería hacer un seguimiento breve.

Mi perfil está enfocado en Unity WebGL, 3D en tiempo real y visualización técnica. Mi proyecto principal, TwinSight X500, es
↪ un prototipo interactivo de inspección de ensamblaje de un dron, con selección de componentes, vista explotada, cortes,
↪ modos visuales y optimización CAD-to-realtime.

Portafolio: [Portfolio URL]
TwinSight: [TwinSight URL]

Quedo atento si tiene sentido conversar sobre el encaje con el rol.

Saludos,
Alexander

26.12.3 English — second follow-up

Bloque de código: text
Hi [Name],

Just following up once more regarding the [Role] position.

If the role requires Unity, WebGL, technical visualization, optimized 3D assets or interactive inspection tools, TwinSight
↪ X500 may be a relevant reference project.

Portfolio: [Portfolio URL]

If the position has already moved forward, no problem. I'd still be open to future roles involving Unity / real-time 3D /
↪ technical visualization.

Best,
Alexander

26.12.4 Spanish — second follow-up

Bloque de código: text
Hola [Name],

Hago un último seguimiento sobre el rol de [Role].

Si el rol requiere Unity, WebGL, visualización técnica, assets 3D optimizados o herramientas interactivas de inspección,
↪ TwinSight X500 puede ser una referencia relevante.

Portafolio: [Portfolio URL]

Si el proceso ya avanzó, no hay problema. Quedo abierto a futuros roles relacionados con Unity / 3D en tiempo real /
↪ visualización técnica.

Saludos,
Alexander

26.13 13. Referral request

26.13.1 English

Bloque de código: text
Hi [Name],

I saw that you're connected with [Company] / work at [Company]. I'm exploring roles in Unity, real-time 3D, Unity WebGL and
↪ technical visualization.

My main project is TwinSight X500, a Unity WebGL drone assembly inspection prototype with component selection, exploded view,
↪ cross-section tools, visual modes and CAD-to-realtime optimization.

If you think the profile is relevant, would you be open to referring me or pointing me to the right person?

Portfolio: [Portfolio URL]
CV: [CV URL]

No pressure if it is not a fit.

26.13.2 Spanish

Bloque de código: text
Hola [Name],

Vi que tienes conexión con [Company] / trabajas en [Company]. Estoy explorando roles en Unity, 3D en tiempo real, Unity WebGL ↪ y visualización técnica.

Mi proyecto principal es TwinSight X500, un prototipo Unity WebGL para inspección de ensamblaje de un dron, con selección de ↪ componentes, vista explotada, cortes, modos visuales y optimización CAD-to-realtime.

Si crees que el perfil puede ser relevante, ¿estarías dispuesto/a a referirme o indicarme quién sería la persona adecuada?

Portafolio: [Portfolio URL]
CV: [CV URL]

Sin problema si no encaja.

26.14 14. Professor / thesis advisor recommendation request

26.14.1 English

Bloque de código: text
Subject: Recommendation request – TwinSight X500 / Unity technical visualization

Hi Professor [Name],

I hope you are doing well.

I am preparing my international job-search materials around TwinSight X500 and my profile as a Real-Time 3D Developer / Unity ↪ Technical Artist focused on technical visualization.

Would you be willing to provide a short recommendation or testimonial about my work on the thesis project?

The most useful points would be:

- technical complexity of the project;
- Unity/WebGL implementation;
- CAD-to-realtime workflow;
- documentation and evaluation work;
- independence and technical ownership.

A short paragraph would be enough. I can also send a draft for your review if helpful.

Thank you,
Alexander

26.14.2 Spanish

Bloque de código: text
Asunto: Solicitud de recomendación – TwinSight X500 / visualización técnica en Unity

Hola profesor/a [Name],

Espero que esté muy bien.

Estoy preparando mis materiales para búsqueda laboral internacional alrededor de TwinSight X500 y mi perfil como Real-Time 3D ↪ Developer / Unity Technical Artist enfocado en visualización técnica.

Quería preguntarle si estaría dispuesto/a a darme una recomendación o testimonio breve sobre mi trabajo en el proyecto de ↪ tesis.

Los puntos más útiles serían:

- complejidad técnica del proyecto;
- implementación en Unity/WebGL;
- flujo CAD-to-realtime;
- documentación y evaluación;
- autonomía y responsabilidad técnica.

Un párrafo corto sería suficiente. Si le resulta más fácil, también puedo enviarle un borrador para revisión.

Muchas gracias,
Alexander

26.15 15. Former freelance client testimonial request

26.15.1 English

Bloque de código: text
Subject: Short testimonial request

Hi [Name],

I hope you're doing well.

I'm updating my professional portfolio and wanted to ask if you would be willing to provide a short testimonial about the work
↪ I did for [project/client].

A brief paragraph is enough. Useful points would be:

- communication;
- reliability;
- technical execution;
- problem solving;
- delivery quality.

If easier, I can send a short draft for you to edit.

Thank you,
Alexander

26.15.2 Spanish

Bloque de código: text
Asunto: Solicitud de testimonio breve

Hola [Name],

Espero que estés muy bien.

Estoy actualizando mi portafolio profesional y quería preguntarte si estarías dispuesto/a a darme un testimonio breve sobre el
↪ trabajo que realicé para [project/client].

Un párrafo corto es suficiente. Serían útiles puntos como:

- comunicación;
- responsabilidad;
- ejecución técnica;
- resolución de problemas;
- calidad de entrega.

Si prefieres, puedo enviarte un borrador corto para que lo ajustes.

Gracias,
Alexander

26.16 16. Salary expectation response

26.16.1 English — early stage

Bloque de código: text
At this stage, I'd prefer to understand the scope, responsibilities, expected time commitment, contract structure and
↪ seniority level before giving a final number.

For remote contractor work, I usually evaluate compensation based on monthly retainer or hourly rate, depending on scope and
↪ availability.

26.16.2 English — with range

Bloque de código: text
For remote contractor work, depending on scope and expected availability, I'm generally targeting the USD [range] / month
↪ range or an equivalent hourly rate.

I'm open to discussing structure if the role is a strong fit in Unity, real-time 3D, WebGL or technical visualization.

26.16.3 Spanish — early stage

Bloque de código: text

En esta etapa prefiero entender primero el alcance, responsabilidades, dedicación esperada, tipo de contrato y nivel del rol
↪ antes de dar una cifra final.

Para trabajo remoto como contractor, suelo evaluar la compensación según retainer mensual o tarifa por hora, dependiendo del
↪ alcance y disponibilidad.

26.16.4 Spanish — with range

Bloque de código: text

Para trabajo remoto como contractor, dependiendo del alcance y la dedicación esperada, estoy apuntando generalmente a un rango
↪ de USD [range] mensuales o una tarifa horaria equivalente.

Estoy abierto a revisar la estructura si el rol encaja bien con Unity, 3D en tiempo real, WebGL o visualización técnica.

26.17 17. Remote-from-Colombia explanation

26.17.1 English

Bloque de código: text

I'm currently based in Colombia and available for remote contractor/B2B work.

I can collaborate with international teams using overlap with US or European time zones depending on the schedule. For

↪ payroll-style employment, the company would need to confirm whether it supports an EOR or local-compliance setup. For
↪ contractor work, I can operate through invoice-based arrangements subject to Colombian tax and administrative
↪ requirements.

26.17.2 Spanish

Bloque de código: text

Actualmente estoy en Colombia y disponible para trabajo remoto como contractor/B2B.

Puedo colaborar con equipos internacionales con solapamiento horario con EE. UU. o Europa dependiendo del horario. Para

↪ contratación tipo nómina, la empresa tendría que confirmar si maneja EOR o una estructura legal compatible. Para trabajo
↪ como contractor, puedo operar mediante facturación, sujeto a los requisitos tributarios y administrativos en Colombia.

26.18 18. Contractor setup explanation

26.18.1 English

Bloque de código: text

For contractor work, I can work under a B2B / independent contractor agreement with defined scope, payment terms,
↪ deliverables, confidentiality and IP clauses.

Before signing, I would review:

- payment method and currency;
- invoicing requirements;
- tax documentation;
- IP ownership;
- confidentiality;
- termination terms;
- expected availability;
- whether the relationship is contractor or employee-like.

This helps keep the structure clear for both sides.

26.18.2 Spanish

Bloque de código: text

Para trabajo como contractor, puedo trabajar bajo un acuerdo B2B / prestación de servicios independiente con alcance, términos
↪ de pago, entregables, confidencialidad y propiedad intelectual definidos.

Antes de firmar, revisaría:

- método y moneda de pago;
- requisitos de facturación;
- documentación tributaria;
- propiedad intelectual;
- confidencialidad;
- causales de terminación;
- disponibilidad esperada;
- si la relación es realmente contractor o se parece a empleo.

Esto ayuda a mantener clara la estructura para ambas partes.

26.19 19. Future Portuguese / EU mobility explanation

26.19.1 English

Bloque de código: text

Currently, I should be treated as Colombia-based and available for remote contractor/B2B work.

I expect Portuguese citizenship/passport to become available in the medium term, which may support future EU work
↪ authorization. However, I do not currently represent myself as EU work-authorized until that status is legally issued.

26.19.2 Spanish

Bloque de código: text

Actualmente debo ser considerado como candidato basado en Colombia y disponible para trabajo remoto como contractor/B2B.

Espero obtener ciudadanía/pasaporte portugués en el mediano plazo, lo cual podría habilitar autorización laboral en la UE en
↪ el futuro. Sin embargo, actualmente no me presento como autorizado para trabajar en la UE hasta que ese estatus exista
↪ legalmente.

26.20 20. Germany relocation / marriage route explanation

26.20.1 English

Bloque de código: text

Germany is a possible medium-term relocation route due to my relationship with a German citizen. If that route becomes active
↪ and the corresponding residence/work authorization is granted, I would be able to consider local German or EU-based roles.

For now, my current availability remains remote contractor work from Colombia.

26.20.2 Spanish

Bloque de código: text

Alemania es una posible ruta de reubicación a mediano plazo debido a mi relación con una ciudadana alemana. Si esa ruta se
↪ activa y se concede el permiso de residencia/trabajo correspondiente, podría considerar roles locales en Alemania o dentro
↪ de la UE.

Por ahora, mi disponibilidad actual sigue siendo trabajo remoto como contractor desde Colombia.

26.21 21. Contract / payment / benefits questions

26.21.1 English

Bloque de codigo: text
Before moving forward, I'd like to clarify a few contract details:

1. Is the role employee, contractor, B2B, or EOR-based?
2. What is the expected weekly availability?
3. What time-zone overlap is required?
4. What is the payment currency and method?
5. What are the payment terms?
6. Are there paid time off, holidays or sick-day provisions?
7. How is IP ownership handled?
8. Are there exclusivity or non-compete clauses?
9. What equipment or software is provided?
10. What is the termination notice period?

26.21.2 Spanish

Bloque de codigo: text
Antes de avanzar, me gustaría aclarar algunos puntos del contrato:

1. ¿El rol es empleado, contractor, B2B o vía EOR?
2. ¿Cuál es la disponibilidad semanal esperada?
3. ¿Qué solapamiento horario se requiere?
4. ¿Cuál es la moneda y método de pago?
5. ¿Cuáles son los términos de pago?
6. ¿Hay vacaciones, festivos o días de enfermedad pagos?
7. ¿Cómo se maneja la propiedad intelectual?
8. ¿Hay exclusividad o cláusulas de no competencia?
9. ¿La empresa proporciona equipo o software?
10. ¿Cuál es el periodo de preaviso para terminación?

26.22 22. Declining unstable offers

26.22.1 English

Bloque de codigo: text
Hi [Name],

Thank you for the offer and for the time spent during the process.

After reviewing the scope, compensation and contract structure, I don't think this is the right fit for me at this stage.

The main concerns are [payment terms / unclear scope / unpaid trial scope / contract risk / compensation level / role
↪ mismatch].

I appreciate the opportunity and would be open to reconnecting in the future if there is a role with clearer structure and
↪ stronger alignment with Unity, real-time 3D or technical visualization work.

Best,
Alexander

26.22.2 Spanish

Bloque de codigo: text
Hola [Name],

Gracias por la oferta y por el tiempo durante el proceso.

Después de revisar el alcance, la compensación y la estructura del contrato, considero que en esta etapa no es el mejor encaje
↪ para mí.

Los principales puntos de preocupación son [términos de pago / alcance poco claro / prueba no paga demasiado amplia / riesgo
↪ contractual / nivel de compensación / desalineación del rol].

Agradezco la oportunidad y quedo abierto a reconectar en el futuro si aparece un rol con una estructura más clara y mejor
↪ alineación con Unity, 3D en tiempo real o visualización técnica.

Saludos,
Alexander

26.23 23. Explaining limited formal industry experience

26.23.1 English

Bloque de código: text

My formal industry experience in technical art / Unity is still limited, but I've built a substantial thesis and portfolio
↪ project that demonstrates the kind of work I'm targeting.

TwinSight X500 required Unity/C# implementation, WebGL deployment, CAD-to-realtime optimization, runtime interaction systems,
↪ technical UI, documentation and usability/workload evaluation.

So I'm not positioning myself as a senior industry technical artist. I'm positioning myself as a junior-strong / early-mid
↪ candidate with unusually relevant project evidence in Unity technical visualization.

26.23.2 Spanish

Bloque de código: text

Mi experiencia formal en industria dentro de technical art / Unity todavía es limitada, pero construí un proyecto de tesis y
↪ portafolio sustancial que demuestra el tipo de trabajo al que estoy apuntando.

TwinSight X500 requirió implementación en Unity/C#, despliegue WebGL, optimización CAD-to-realtime, sistemas de interacción
↪ runtime, UI técnica, documentación y evaluación de usabilidad/carga percibida.

No me estoy presentando como senior technical artist. Me estoy posicionando como un candidato junior fuerte / early-mid con
↪ evidencia de proyecto muy relevante en visualización técnica con Unity.

26.24 24. Explaining AI-assisted development

26.24.1 English

Bloque de código: text

I used AI-assisted tools during parts of the development process, mainly for implementation support, debugging, research and
↪ documentation.

The important point is that AI did not replace technical ownership. I was responsible for the architecture decisions,
↪ integration, testing, validation, tradeoffs and final implementation choices.

I can walk through the system, explain the technical decisions and discuss what I would improve.

26.24.2 Spanish

Bloque de código: text

Usé herramientas de IA durante partes del proceso de desarrollo, principalmente como apoyo para implementación, debugging,
↪ investigación y documentación.

El punto importante es que la IA no reemplazó la responsabilidad técnica. Yo fui responsable de las decisiones de
↪ arquitectura, integración, pruebas, validación, tradeoffs y decisiones finales de implementación.

Puedo explicar el sistema, justificar las decisiones técnicas y discutir qué mejoraría.

26.25 25. Answering “Why should we hire you?”

26.25.1 English

Bloque de código: text

You should consider me if the role needs someone who can bridge Unity development, real-time 3D asset workflows and technical
↪ visualization.

My strongest evidence is TwinSight X500, where I built a Unity WebGL technical visualization prototype for drone assembly inspection. It required C# interaction systems, WebGL-oriented optimization, technical UI, CAD-to-realtime asset preparation, documentation and usability/workload evaluation.

I'm not claiming senior industry experience. The value I bring is a focused technical profile, strong project evidence, English communication, engineering-oriented thinking and a clear interest in building real-time 3D systems for technical use cases.

26.25.2 Spanish

Bloque de código: text

Deberían considerarme si el rol necesita a alguien que conecte desarrollo en Unity, flujos de assets 3D en tiempo real y visualización técnica.

Mi evidencia más fuerte es TwinSight X500, donde construí un prototipo Unity WebGL de visualización técnica para inspección de ensamblaje de un dron. El proyecto requirió sistemas de interacción en C#, optimización orientada a WebGL, UI técnica, preparación de assets CAD-to-realtime, documentación y evaluación de usabilidad/carga percibida.

No estoy afirmando tener experiencia senior de industria. El valor que apporto es un perfil técnico enfocado, evidencia fuerte de proyecto, comunicación en inglés, pensamiento ingenieril y un interés claro en construir sistemas 3D en tiempo real para casos técnicos.

26.26 26. Answering “What kind of roles are you looking for?”

26.26.1 English

Bloque de código: text

I'm mainly targeting roles involving Unity, real-time 3D, Unity WebGL, technical visualization, simulation/digital twin visualization, interactive 3D systems or technical art workflows.

The best fit would be a role where I can work on runtime interaction systems, technical UI, optimized 3D assets, WebGL deployment, inspection tools or visualization interfaces.

26.26.2 Spanish

Bloque de código: text

Estoy apuntando principalmente a roles relacionados con Unity, 3D en tiempo real, Unity WebGL, visualización técnica, simulación/digital twin, sistemas interactivos 3D o flujos de technical art.

El mejor encaje sería un rol donde pueda trabajar en sistemas de interacción runtime, UI técnica, assets 3D optimizados, despliegue WebGL, herramientas de inspección o interfaces de visualización.

26.27 27. Answering “Are you open to relocation?”

26.27.1 English

Bloque de código: text

My immediate priority is remote work from Colombia, especially contractor/B2B roles.

I'm open to relocation if the opportunity is strong enough and the legal structure is clear. Europe is strategically relevant for me in the medium term due to an expected Portuguese passport and a possible Germany route through family residence, but I do not currently claim EU work authorization.

26.27.2 Spanish

Bloque de código: text

Mi prioridad inmediata es trabajo remoto desde Colombia, especialmente roles contractor/B2B.

Estoy abierto a reubicación si la oportunidad lo justifica y la estructura legal es clara. Europa es estratégicamente relevante para mí a mediano plazo por un pasaporte portugués esperado y una posible ruta hacia Alemania por residencia familiar, pero actualmente no afirmo tener autorización laboral en la UE.

26.28 28. Answering “Can you work US hours?”

26.28.1 English

Bloque de código: text

I can work with reasonable overlap with US time zones from Colombia. I’m in the America/Bogota time zone, which aligns well
↔ with Eastern and Central time and can partially overlap with Pacific time depending on the schedule.

26.28.2 Spanish

Bloque de código: text

Puedo trabajar con un solapamiento razonable con horarios de Estados Unidos desde Colombia. Estoy en la zona horaria
↔ America/Bogota, que encaja bien con Eastern y Central time, y puede tener solapamiento parcial con Pacific time
↔ dependiendo del horario.

26.29 29. Answering “Can you work Europe hours?”

26.29.1 English

Bloque de código: text

I can support partial overlap with European time zones from Colombia, especially for meetings scheduled earlier in the
↔ European day. For a full Europe-aligned schedule, I would need to review expectations and sustainability.

26.29.2 Spanish

Bloque de código: text

Puedo manejar un solapamiento parcial con horarios europeos desde Colombia, especialmente para reuniones programadas temprano
↔ en el día europeo. Para una jornada completamente alineada con Europa, tendría que revisar expectativas y sostenibilidad.

26.30 30. Outreach tracking rules

Every outreach message must be logged.

Track:

Bloque de código: text

Date
Company
Person
Role
Message type
Template used
Customization notes
Links sent
Follow-up date
Response
Next action

26.30.1 Follow-up rule

Do not send more than three messages without a response.

Recommended sequence:

Bloque de código: text

Day 0: initial message
Day 4-5: first follow-up
Day 10-14: second follow-up
Day 21+: archive

26.31 31. Message quality checklist

Before sending, verify:

- Message names the company or person.
- Message mentions a relevant area.
- TwinSight is included if relevant.
- Portfolio link is included.
- Message is under 180 words unless email.
- No false claims.
- No desperate wording.
- No overexplaining legal status.
- No giant biography.
- No attachment unless requested.
- Clear next action.

26.32 32. Recommended first outreach sequence

26.32.1 Week 1 after assets are ready

Send:

Type	Count
Recruiter connection requests	10
Technical lead messages	5
Company speculative applications	5
Referral requests	3
Former client/testimonial requests	2
Professor/advisor recommendation requests	1–2

26.32.2 Week 2

Send:

Type	Count
Follow-ups	10
New recruiter messages	10
Company applications	15
Community posts/messages	3

26.33 33. Templates not to use

Avoid:

Bloque de código: text
I'm looking for any job.
I'm passionate and hardworking.
I can do web, AI, 3D, games, design and everything.
I don't have experience but please give me a chance.
I used AI to make everything.
I can work in Europe soon.
I'm almost an EU citizen.
I accept any salary.

These reduce credibility or create legal/positioning risk.

26.34 34. Next file dependency

The next file should be:

```
Bloque de código: text  
23_interview_answer_bank.md
```

It should provide ready-to-use answers for:

- Tell me about yourself.
- Explain TwinSight.
- What did you build yourself?
- How did you use AI?
- Why this role?
- Why no formal industry experience?
- What salary do you expect?
- Are you authorized to work in the EU?
- Can you work as contractor from Colombia?
- What would you improve in TwinSight?
- Technical questions by role family.

No Deep Research or agent required.

Chapter 27

23_interview_answer_bank

Fuente: 23_interview_answer_bank.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 902

Ultima modificacion: 2026-06-11 15:25

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This document provides prepared interview answers for Alexander Woodcock Salomón's target roles.

Contenido completo depurado

27.0.1 Purpose

This document provides prepared interview answers for Alexander Woodcock Salomón's target roles.

It converts the strategy, CV, LinkedIn, portfolio, TwinSight case study, negotiation framework and outreach documents into interview-ready language.

The answers are written primarily in English because the target market is international. Spanish versions are included where they are useful for LATAM or Spanish-speaking recruiters.

27.1 1. Interview positioning rule

Every interview answer should reinforce the same profile:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization,
↩ and Unity WebGL deployment.
```

Do not drift into:

- generic full-stack developer;
- pure 3D artist;
- AI engineer;
- game designer;
- graphic designer;
- general multimedia generalist.

27.2 2. Core interview narrative

27.2.1 30-second version

Bloque de código: text

I'm a Colombia-based Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, C#, technical visualization and
↳ CAD-to-realtime optimization. My strongest project is TwinSight X500, a Unity WebGL drone assembly inspection prototype
↳ with component selection, exploded view, cross-section tools, visual modes, technical UI and usability/workload
↳ evaluation. I'm now looking for remote roles where I can apply that background to interactive 3D, technical visualization,
↳ simulation, digital twin, XR-adjacent or Unity technical art work.

27.2.2 60-second version

Bloque de código: text

I'm a Real-Time 3D Developer / Unity Technical Artist based in Colombia. My focus is building interactive technical
↳ visualization systems with Unity, C#, WebGL and Blender.

My flagship project is TwinSight X500, a Unity WebGL prototype for inspecting the assembly of a Holybro X500 V2 drone. The
↳ project converts technical documentation and CAD-derived assets into an optimized real-time 3D experience with component
↳ selection, exploded view, cross-section tools, visual modes and technical information panels.

My background combines Multimedia Engineering, advanced Electronic Engineering coursework, real-time 3D, Blender-based asset
↳ preparation, technical documentation and Python-assisted tooling. I'm looking for remote contractor or full-time roles
↳ where Unity, interactive 3D, technical visualization, WebGL or simulation/digital twin interfaces are relevant.

27.2.3 90-second version

Bloque de código: text

I'm a Colombia-based Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization,
↳ CAD-to-realtime optimization and Unity WebGL deployment.

My strongest project is TwinSight X500, a Multimedia Engineering thesis project built in Unity WebGL to inspect and understand
↳ the assembly of a Holybro X500 V2 drone. The project addresses a common issue in technical documentation: 2D diagrams can
↳ show components, but users often have to mentally reconstruct spatial relationships.

TwinSight turns that into an interactive 3D viewer. Users can select components, inspect technical information, use exploded
↳ views, apply cross-section tools and switch between visual modes. A major part of the project was converting CAD-derived
↳ geometry into optimized real-time assets using Blender-based cleanup, preparation and Unity integration.

I also have secondary Python tooling work through ARA Framework, a research automation prototype. I treat that as supporting
↳ evidence of automation and systems thinking, while the core direction remains Unity, real-time 3D and technical
↳ visualization.

I'm not positioning myself as a senior industry technical artist. I'm positioning myself as a junior-strong / early-mid
↳ candidate with unusually relevant project evidence in Unity WebGL and technical visualization.

27.3 3. “Tell me about yourself”

27.3.1 English answer

Bloque de código: text

I'm a Real-Time 3D Developer / Unity Technical Artist based in Colombia, focused on Unity WebGL, C#, technical visualization
↳ and CAD-to-realtime optimization.

My strongest project is TwinSight X500, a Unity WebGL technical visualization prototype for drone assembly inspection. It
↳ includes component selection, exploded view, cross-section tools, visual modes, technical UI and an optimized
↳ CAD-to-realtime asset workflow.

My academic background is in Multimedia Engineering, with advanced previous coursework in Electronic Engineering, so I tend to
↳ approach 3D work from both a visual and engineering perspective. I'm especially interested in roles where real-time 3D is
↳ used to make technical systems easier to inspect, understand and communicate.

27.3.2 Spanish answer

Bloque de código: text

Soy Real-Time 3D Developer / Unity Technical Artist en Colombia, enfocado en Unity WebGL, C#, visualización técnica y
↳ optimización CAD-to-realtime.

Mi proyecto más fuerte es TwinSight X500, un prototipo Unity WebGL de visualización técnica para inspección de ensamblaje de un dron. Incluye selección de componentes, vista explotada, herramientas de corte, modos visuales, UI técnica y un flujo optimizado de assets derivados de CAD.

Mi formación es en Ingeniería Multimedia, con estudios avanzados previos en Ingeniería Electrónica, así que tiendo a abordar el 3D desde una perspectiva visual y también ingenieril. Me interesan especialmente roles donde el 3D en tiempo real ayude a inspeccionar, entender y comunicar sistemas técnicos.

27.3.3 What not to say

Bloque de código: text

I'm a multimedia engineer who does a bit of everything: 3D, AI, web, design, games, fitness apps and research.

This dilutes the positioning.

27.4 4. “Explain TwinSight X500”

27.4.1 30-second answer

Bloque de código: text

TwinSight X500 is a Unity WebGL technical visualization prototype for inspecting the assembly of a Holybro X500 V2 drone. It transforms technical documentation and CAD-derived assets into an interactive 3D viewer with component selection, exploded view, cross-section tools, visual modes and technical information panels. The goal is to help users understand spatial relationships in the assembly, not just look at individual parts.

27.4.2 60-second answer

Bloque de código: text

TwinSight X500 is my Multimedia Engineering thesis project. It is a Unity WebGL interactive 3D visualization prototype for inspecting and understanding the assembly of a Holybro X500 V2 drone.

The problem is that traditional assembly documentation is usually static. It can show parts and diagrams, but the user still has to mentally reconstruct how components relate in space.

TwinSight addresses that through an interactive real-time 3D viewer. Users can select components, inspect technical information, use exploded view to understand relationships, apply cross-section tools and switch between visual modes.

Technically, it involved Unity/C# runtime systems, UI Toolkit, WebGL deployment, Blender-based asset preparation and CAD-to-realtime optimization.

27.4.3 Technical answer

Bloque de código: text

TwinSight X500 is a browser-accessible Unity WebGL prototype focused on technical inspection of a drone assembly.

The system combines a CAD-to-realtime asset pipeline with runtime interaction systems. On the asset side, dense CAD-derived geometry had to be cleaned, simplified and prepared for real-time use. On the Unity side, the prototype includes component selection, highlighting, technical information panels, exploded view behavior, cross-section/clipping tools and multiple visual modes.

The project was also evaluated academically using usability and workload methods such as SUS, NASA-TLX Raw and Think-Aloud. The important point is that the project is not only a model viewer; it is an interaction system for technical understanding.

27.5 5. “What problem does TwinSight solve?”

27.5.1 Answer

Bloque de código: text

The project addresses the gap between static technical documentation and spatial understanding.

In assembly documentation, users can see parts in 2D diagrams or manuals, but they often have to mentally reconstruct how
↪ components relate in 3D space. That creates cognitive load, especially when there are overlapping parts, internal
↪ structures, fasteners or subassemblies.

TwinSight explores whether an interactive 3D viewer can reduce that burden by allowing users to directly inspect the assembly,
↪ select components, see exploded relationships, use cross-sections and switch visual modes.

27.5.2 Short version

Bloque de código: text

It helps users understand spatial relationships in a technical assembly instead of relying only on static 2D diagrams.

27.6 6. “What did you personally build?”

27.6.1 Answer

Bloque de código: text

My work covered Unity/C# implementation, runtime interaction systems, technical UI, asset integration, Blender-based asset
↪ preparation, CAD-to-realtime optimization, documentation, evaluation design and final integration.

Specifically, I worked on systems such as component selection, exploded view behavior, cross-section/clipping tools, visual
↪ modes and technical information panels.

I also handled the process of converting CAD-derived assets into something more appropriate for real-time Unity WebGL
↪ deployment. AI tools were used as support during parts of implementation and debugging, but I remained responsible for
↪ architecture decisions, integration, validation, technical judgment and final ownership.

27.6.2 If challenged

Bloque de código: text

I can walk through the project architecture, explain the feature systems, describe the asset pipeline and discuss what I would
↪ improve. My claim is not that I wrote every line without assistance; my claim is that I owned the technical integration,
↪ decisions and delivery of the system.

27.7 7. “How much of this did you build yourself?”

27.7.1 Answer

Bloque de código: text

I was responsible for the project direction, architecture decisions, Unity integration, feature implementation, debugging,
↪ asset workflow decisions, documentation, evaluation and final delivery.

I did use AI-assisted tools during parts of the process, mainly for implementation support, debugging, research and
↪ documentation. I treat that similarly to using a technical assistant: useful for acceleration, but not a replacement for
↪ understanding the system.

I can explain the project structure, the technical decisions, the tradeoffs, the limitations and the areas I would refactor.

27.7.2 Short version

Bloque de código: text

I used AI as a support tool, but I owned the architecture, integration, validation and final technical decisions.

27.7.3 What not to say

Bloque de código: text

AI did most of it.

or:

Bloque de codigo: text
I wrote every line manually.

Both are risky if inaccurate.

27.8 8. “How did you use AI?”

27.8.1 Answer

Bloque de codigo: text
I used AI-assisted tools mainly for implementation support, debugging, research, documentation drafting and exploring solution
↪ alternatives.

I did not use AI as a substitute for technical ownership. I was still responsible for choosing the architecture, integrating
↪ the systems, testing behavior, validating outputs and deciding what was technically appropriate for the project.

The best way to describe it is that AI helped me move faster, but I still had to understand, adapt, debug and own the result.

27.8.2 If interviewer is skeptical

Bloque de codigo: text
That concern is valid. That is why I do not present the project as proof that I manually authored every line in isolation. I
↪ present it as proof that I can define a technical problem, integrate Unity systems, prepare assets, validate behavior,
↪ document tradeoffs and use modern tools responsibly to deliver a working prototype.

27.9 9. “Why this role?”

27.9.1 Unity / Real-Time 3D answer

Bloque de codigo: text
This role fits because it combines Unity development, real-time interaction and technical problem solving. My strongest
↪ project, TwinSight X500, required Unity/C# runtime systems, WebGL deployment, technical UI, asset optimization and
↪ real-time inspection features.

I am specifically looking for roles where Unity is used to build interactive systems, not just passive visuals.

27.9.2 Technical Visualization answer

Bloque de codigo: text
This role fits because my strongest project is directly about turning technical information into an interactive 3D
↪ visualization system. TwinSight converts drone assembly documentation and CAD-derived assets into a real-time inspection
↪ experience with selection, exploded view, cross-section and visual modes.

That aligns with technical visualization, digital twins, simulation interfaces and product/assembly visualization.

27.9.3 Technical Artist answer

Bloque de codigo: text
This role fits because my work sits between 3D asset workflows and runtime systems. In TwinSight I had to prepare and optimize
↪ 3D assets, integrate them into Unity, build inspection tools, create visual modes and consider WebGL constraints.

I’m not positioning myself as a senior shader specialist, but as a Unity technical artist profile focused on optimization,
↪ runtime interaction, technical UI and visualization workflows.

27.9.4 WebGL answer

Bloque de codigo: text

This role fits because TwinSight was built for Unity WebGL deployment. The project required thinking about browser delivery,
↪ asset weight, interaction design and performance constraints.

I'm interested in building browser-accessible 3D tools where users can interact with technical or product data without needing
↪ native CAD or desktop tools.

27.10 10. "Why should we hire you?"

27.10.1 Main answer

Bloque de codigo: text

You should consider me if the role needs someone who can bridge Unity development, real-time 3D asset workflows and technical
↪ visualization.

My strongest evidence is TwinSight X500, where I built a Unity WebGL technical visualization prototype for drone assembly
↪ inspection. It required C# interaction systems, WebGL-oriented optimization, technical UI, CAD-to-realtime asset
↪ preparation, documentation and usability/workload evaluation.

I'm not claiming senior industry experience. The value I bring is a focused technical profile, strong project evidence,
↪ English communication, engineering-oriented thinking and a clear interest in building real-time 3D systems for technical
↪ use cases.

27.10.2 More concise version

Bloque de codigo: text

Because I have direct project evidence for the type of work this role needs: Unity, WebGL, real-time 3D interaction, technical
↪ visualization, asset optimization and documentation. I'm not a generic junior developer; I have a focused technical
↪ visualization project that is highly aligned with this kind of role.

27.11 11. "Why no formal industry experience in this field yet?"

27.11.1 Answer

Bloque de codigo: text

My formal industry experience in technical art and Unity is still limited because my recent focus has been completing my
↪ Multimedia Engineering thesis and building a portfolio around real-time 3D and technical visualization.

However, the thesis project is substantial and directly relevant. TwinSight X500 required Unity/C# implementation, WebGL
↪ deployment, CAD-to-realtime optimization, runtime interaction systems, technical UI, documentation and evaluation.

So I'm not positioning myself as a senior industry candidate. I'm positioning myself as a junior-strong / early-mid candidate
↪ with unusually relevant project evidence in Unity technical visualization.

27.11.2 Spanish version

Bloque de codigo: text

Mi experiencia formal de industria en technical art y Unity todavía es limitada porque mi foco reciente fue completar la tesis
↪ de Ingeniería Multimedia y construir un portafolio alrededor de 3D en tiempo real y visualización técnica.

Sin embargo, el proyecto de tesis es sustancial y directamente relevante. TwinSight X500 requirió implementación en Unity/C#,
↪ despliegue WebGL, optimización CAD-to-realtime, sistemas de interacción runtime, UI técnica, documentación y evaluación.

No me presento como candidato senior. Me posiciono como un perfil junior fuerte / early-mid con evidencia de proyecto muy
↪ relevante en visualización técnica con Unity.

27.12 12. “What are your strongest technical skills?”

27.12.1 Answer

Bloque de código: text

My strongest technical areas are Unity/C# runtime interaction systems, Unity WebGL deployment, technical visualization,
↪ CAD-to-realtime asset optimization and Blender-based asset preparation.

In practical terms, that means I can work on interactive 3D inspection tools, component selection systems, technical UI,
↪ exploded views, cross-section tools, visual modes and optimization workflows for real-time deployment.

27.12.2 Add if role is tools/Python

Bloque de código: text

I also have secondary experience with Python automation and LLM workflow prototyping through ARA Framework, which supports
↪ tooling and documentation workflows.

27.13 13. “What are your weaknesses or gaps?”

27.13.1 Safe answer

Bloque de código: text

My main gap is formal industry production experience in a professional Unity or technical art team. Most of my strongest
↪ evidence comes from thesis and portfolio work rather than shipped commercial products.

I’m addressing that by making my work public, documenting it clearly, preparing technical breakdowns and targeting roles where
↪ my project evidence is directly relevant.

Another gap is that I am not yet a deep shader/HLSL specialist. I have Shader Graph and basic HLSL familiarity, but I position
↪ that as a growth area rather than an expert claim.

27.13.2 Alternative

Bloque de código: text

I still need more professional team experience, code review exposure and production-scale collaboration. My current strength
↪ is project ownership and technical integration. The next step is applying that in a team environment with production
↪ standards.

27.14 14. “What would you improve in TwinSight?”

27.14.1 Answer

Bloque de código: text

I would improve five areas.

First, I would do deeper profiling and optimization, especially for WebGL build size, memory use and frame stability.

Second, I would separate the runtime systems more cleanly into modules, especially selection, exploded view, clipping and
↪ visual modes.

Third, I would improve the public WebGL deployment pipeline and add clearer build documentation.

Fourth, I would expand the evaluation with more participants and more controlled task design.

Fifth, I would refine the asset pipeline with more formal LODs and stricter asset validation.

27.14.2 Short version

Bloque de código: text
I would improve profiling, modular architecture, WebGL build pipeline, evaluation depth and LOD/asset validation.

27.15 15. “What was the hardest technical problem?”

27.15.1 Answer

Bloque de código: text
The hardest problem was balancing technical asset fidelity with real-time WebGL constraints.

CAD-derived geometry can be extremely dense and not appropriate for real-time browser deployment. The challenge was to
↪ preserve component identity and assembly relationships while reducing complexity enough for interactive performance.

That required asset inspection, cleanup, simplification, Blender-based preparation, Unity integration and constant tradeoffs
↪ between visual clarity, performance and scope.

27.15.2 Alternative: interaction-system answer

Bloque de código: text
Another difficult part was designing the runtime interaction model so that the viewer was not just a 3D model on screen. The
↪ features had to support inspection: component selection, highlighting, information panels, exploded view, cross-section
↪ and visual modes all needed to work as part of a coherent technical visualization workflow.

27.16 16. “How did you approach optimization?”

27.16.1 Answer

Bloque de código: text
I approached optimization from the asset pipeline first.

The source geometry was CAD-derived, so the initial priority was reducing unnecessary geometric density while preserving
↪ component identity and assembly readability. That involved cleanup, simplification, retopology decisions, UV preparation
↪ and Unity import decisions.

On the Unity side, I considered WebGL constraints, visual clarity, material complexity, interaction responsiveness and scene
↪ organization.

The goal was not just fewer polygons. The goal was a model that remained technically understandable while being suitable for
↪ browser-based real-time interaction.

27.16.2 If asked for metrics

Bloque de código: text
The reported reduction was from over 6.5 million triangles to approximately 95,617 optimized triangles, but I would verify the
↪ final number against the thesis report before publishing it as a fixed public claim.

27.17 17. “How would you debug low FPS in Unity WebGL?”

27.17.1 Answer

Bloque de código: text
I would start by separating whether the bottleneck is CPU, GPU, memory, loading or browser-specific overhead.

First, I would profile in Unity and then test the WebGL build in the browser because editor performance can be misleading. I
↳ would check draw calls, triangle count, material count, texture sizes, overdraw, scripts running per frame and garbage
↳ allocation.

For WebGL specifically, I would also check build size, memory configuration, compression, asset loading, shader complexity and
↳ browser console warnings.

Then I would apply fixes based on evidence: reduce geometry, combine or simplify materials, optimize textures, remove
↳ unnecessary update loops, avoid allocations, simplify shaders, use LODs where appropriate and reduce UI or rendering
↳ overhead.

27.18 18. “How would you structure an exploded view system?”

27.18.1 Answer

Bloque de codigo: text

I would treat exploded view as a controlled transformation system.

Each component or group needs an original position and an exploded target position. The system should store those transforms,
↳ interpolate between them, and preserve assembly relationships through grouping or hierarchy.

For a technical visualization project, I would avoid random outward movement. The exploded direction should communicate
↳ structure, so it should be based on assembly logic, axes, component groups or manually authored offsets.

I would also keep it reversible, expose a slider or state control, and ensure selection/highlighting still works in both
↳ assembled and exploded states.

27.19 19. “How would you implement component selection?”

27.19.1 Answer

Bloque de codigo: text

I would use colliders, raycasting and a mapping between scene objects and component metadata.

The selection system should not only identify a mesh. It should connect the selected object to a data model containing name,
↳ category, technical information, hierarchy, parent/child relationships and UI behavior.

I would also separate visual feedback from selection logic. That way highlighting, isolation, ghosting or UI updates can be
↳ changed without rewriting the selection core.

27.20 20. “How would you implement cross-section / clipping?”

27.20.1 Answer

Bloque de codigo: text

At a high level, I would use a clipping plane or shader-based clipping approach, depending on the rendering pipeline and
↳ platform constraints.

The system needs a plane position, orientation and user control. Materials or shaders then discard fragments on one side of
↳ the plane. For Unity WebGL, I would keep the shader approach as simple as possible and test compatibility carefully.

If the project requires capped cross-sections or engineering-grade sectioning, that becomes more complex and may require mesh
↳ processing or specialized rendering techniques. For a visualization prototype, shader-based clipping is often a practical
↳ starting point.

27.21 21. “What is CAD-to-realtime?”

27.21.1 Answer

Bloque de código: text

CAD-to-realtime is the process of converting CAD or engineering-derived geometry into assets that can run efficiently in a real-time engine like Unity.

CAD models are usually built for precision and manufacturing, not for real-time rendering. They can contain dense geometry, excessive detail, problematic topology, many small parts, or structures that are inefficient for WebGL or real-time engines.

The workflow usually involves conversion, cleanup, simplification, retopology where needed, UV preparation, material planning, hierarchy organization, optimization and engine integration.

The goal is to preserve the technical meaning of the object while making it usable in an interactive real-time environment.

27.22 22. “What is the difference between technical visualization and a 3D model viewer?”

27.22.1 Answer

Bloque de código: text

A model viewer primarily lets users look at a 3D object.

Technical visualization should help users understand technical information. That means the interaction, UI, visual modes and data presentation are designed around inspection, comparison, assembly logic or decision-making.

In TwinSight, the goal was not just to display a drone. The goal was to support understanding through component selection, exploded view, cross-section tools, technical information panels and evaluation of usability and workload.

27.23 23. “What is your experience with WebGL?”

27.23.1 Answer

Bloque de código: text

My direct WebGL experience is through Unity WebGL deployment in TwinSight X500.

The project required thinking about browser constraints, asset optimization, UI readability, interaction performance and deployment limitations. I would not claim to be a low-level WebGL graphics programmer, but I understand the practical constraints of delivering real-time 3D through the browser using Unity WebGL.

I’m interested in expanding that into Three.js/WebGPU where relevant.

27.24 24. “What is your experience with XR?”

27.24.1 Answer

Bloque de código: text

My strongest direct project is not an XR project, but it is adjacent to XR because it involves real-time 3D interaction, technical visualization, spatial understanding, optimized assets and inspection workflows.

For XR roles, I would be strongest where the role needs Unity, interactive 3D, technical UI, simulation interfaces or training/inspection workflows. I would still need to deepen platform-specific XR experience such as OpenXR, device interaction patterns and performance constraints for headsets.

27.25 25. “What is your experience with shaders?”

27.25.1 Answer

Bloque de codigo: text

I have experience with Shader Graph and basic HLSL concepts, especially in the context of visual modes and real-time visualization.

I would not position myself as a senior shader specialist yet. My current shader strength is practical integration: using shaders or material modes to support inspection, ghosting, wireframe-like views, blueprint-style visualization and similar technical visual modes.

Deep shader optimization and advanced HLSL are growth areas.

27.26 26. “What is ARA Framework?”

27.26.1 Answer

Bloque de codigo: text

ARA Framework is a secondary Python research automation prototype.

It explores multi-agent style workflows for niche analysis, literature research, technical planning, implementation guidance and structured report generation. It uses LangGraph/LangChain-style orchestration concepts, research APIs, document-processing ideas and Markdown output.

I do not present it as a production AI platform. I use it to demonstrate tooling, automation thinking, Python architecture and technical documentation.

27.27 27. “Are you an AI engineer?”

27.27.1 Answer

Bloque de codigo: text

No. I would not position myself as an AI engineer in the machine learning or model-training sense.

I have experience using AI-assisted development workflows and building LLM workflow prototypes like ARA Framework. That gives me practical experience with automation, orchestration and technical tooling, but my primary professional direction is still Unity, real-time 3D and technical visualization.

AI tooling is a secondary differentiator, not the main role identity.

27.28 28. “What salary are you expecting?”

27.28.1 Early-stage answer

Bloque de codigo: text

At this stage, I would prefer to understand the scope, responsibilities, expected availability, contract structure and seniority level before giving a final number.

For remote contractor work, I usually evaluate compensation based on monthly retainer or hourly rate, depending on scope and time commitment.

27.28.2 With range

Bloque de codigo: text

For remote contractor work, depending on scope and expected availability, I’m generally targeting the USD [range] per month range, or an equivalent hourly rate.

I’m open to discussing structure if the role is a strong fit in Unity, real-time 3D, WebGL or technical visualization.

27.28.3 If pressed for minimum

Bloque de codigo: text
For a full-time remote contractor arrangement, my minimum depends on scope and stability, but I'm primarily targeting roles
↳ that are meaningfully above local Colombian compensation and aligned with international Unity/technical visualization
↳ rates.

27.28.4 What not to say

Bloque de codigo: text
I can accept anything.

or:

Bloque de codigo: text
My minimum is USD 1,500.

Do not reveal the floor early unless necessary.

27.29 29. "Can you work remotely from Colombia?"

27.29.1 Answer

Bloque de codigo: text
Yes. I'm currently based in Colombia and available for remote contractor/B2B work.

I can work with reasonable overlap with US time zones, and partial overlap with European time zones depending on the schedule.
↳ For employee-style arrangements, the company would need to confirm whether it supports EOR or local compliance. For
↳ contractor work, I can operate through invoice-based arrangements subject to Colombian tax and administrative
↳ requirements.

27.30 30. "Are you authorized to work in the EU?"

27.30.1 Current answer

Bloque de codigo: text
Not currently. I am currently based in Colombia and should be treated as available for remote contractor/B2B work.

I expect a Portuguese passport in the medium term, which may support future EU work authorization, but I do not currently
↳ represent myself as EU-authorized until that status is legally issued.

27.30.2 After Portuguese passport

Only after it is legally true:

Bloque de codigo: text
Yes. I hold Portuguese citizenship and can work in the EU/EEA, subject to local registration and tax requirements.

27.31 31. “Are you open to relocation?”

27.31.1 Answer

Bloque de código: text

My immediate priority is remote work from Colombia, especially contractor/B2B roles.

I am open to relocation if the opportunity is strong enough and the legal structure is clear. Europe is strategically relevant
↪ for me in the medium term due to an expected Portuguese passport and a possible Germany route through family residence,
↪ but I do not currently claim EU work authorization.

27.32 32. “Can you work US hours?”

27.32.1 Answer

Bloque de código: text

Yes, I can work with reasonable overlap with US time zones. I’m in the America/Bogota time zone, which aligns well with
↪ Eastern and Central time and can partially overlap with Pacific time depending on the schedule.

27.33 33. “Can you work European hours?”

27.33.1 Answer

Bloque de código: text

I can support partial overlap with European time zones from Colombia, especially for meetings scheduled earlier in the
↪ European day. For a fully Europe-aligned schedule, I would need to review the expectations and sustainability.

27.34 34. “What type of contract are you open to?”

27.34.1 Answer

Bloque de código: text

I’m open to remote contractor/B2B arrangements, full-time roles where the company can support the appropriate legal structure,
↪ and project-based contracts if the scope is clear.

For international work from Colombia, contractor/B2B is usually the most practical short-term structure.

27.35 35. “How do you handle contractor work?”

27.35.1 Answer

Bloque de código: text

For contractor work, I prefer a clear agreement covering scope, deliverables, expected availability, payment terms, IP
↪ ownership, confidentiality, communication cadence and termination terms.

I also separate contractor arrangements from employee-like arrangements. If the company expects fixed full-time hours,
↪ exclusivity and direct supervision, then the legal structure needs to be reviewed carefully.

27.36 36. “What are your long-term goals?”

27.36.1 Answer

Bloque de codigo: text

My long-term goal is to specialize in real-time 3D systems for technical visualization, simulation, digital twins, XR-adjacent workflows and Unity technical art.

In the next 12-24 months, I want to strengthen my professional experience, improve my portfolio with production-level work, deepen Unity optimization and WebGL/XR knowledge, and position myself for higher-value international roles.

27.37 37. “Why not focus on games if you like videogames?”

27.37.1 Answer

Bloque de codigo: text

I am interested in games, but I’m being strategic about employability and compensation.

My strongest current evidence, TwinSight X500, is more aligned with technical visualization, simulation, digital twins, industrial/product visualization and Unity-based interactive systems than with pure game production.

Game technical art is still relevant, especially for Unity runtime, tools, shaders and optimization roles, but I do not want to force the portfolio into a pure game direction if the stronger market fit is technical visualization.

27.38 38. “What kind of team environment do you prefer?”

27.38.1 Answer

Bloque de codigo: text

I work best in teams where technical expectations are explicit, feedback is direct and work is organized around clear deliverables.

Because my background combines self-directed project work and technical documentation, I value written communication, task clarity, code/project organization and technical review. For remote work, I think clarity is more important than constant meetings.

27.39 39. “Describe a time you solved a difficult problem”

27.39.1 Answer using TwinSight

Bloque de codigo: text

A difficult problem in TwinSight was converting dense CAD-derived geometry into assets suitable for real-time WebGL inspection.

The raw assets were not designed for browser-based rendering. They had to preserve enough detail to remain technically meaningful, but also be simplified enough to support interaction.

I approached it by analyzing the geometry, identifying what detail was necessary for recognition and assembly understanding, simplifying or rebuilding parts where needed, preparing assets in Blender and integrating them into Unity.

The result was an optimized model that preserved the inspection value of the assembly while becoming more appropriate for real-time visualization.

27.40 40. “What would you do if you joined our team?”

27.40.1 Answer

Bloque de código: text

First, I would understand the project constraints: target platform, performance budget, asset pipeline, current architecture, ↵ user needs and delivery priorities.

Then I would look for where I can contribute fastest: Unity/C# interaction systems, technical UI, asset optimization, ↵ visualization tools, documentation or workflow automation.

I would avoid making assumptions too early. My first goal would be to understand how the team currently builds and ships, then ↵ contribute in a way that improves the existing workflow rather than fighting it.

27.41 41. “How do you learn new tools?”

27.41.1 Answer

Bloque de código: text

I learn best by connecting tools to a concrete output.

For example, I do not treat a course as valuable just because I completed it. I try to convert learning into an artifact: a ↵ prototype, a breakdown, a before/after optimization, a demo or documentation.

That is the same logic I use for Rebelway, Blender, Unity or Python automation. The value is not the course itself; it is the ↵ public evidence it helps produce.

27.42 42. Questions to ask interviewers

27.42.1 General

Bloque de código: text

What kind of real-time 3D or Unity systems would I be working on first?

How is the team structured between developers, artists, technical artists and designers?

What platforms are most important for this role?

What are the main technical constraints right now?

How do you evaluate success for this role in the first 3 months?

27.42.2 Unity / Technical Art

Bloque de código: text

What Unity version and render pipeline are you using?

How do you currently handle profiling and optimization?

How are technical artists involved in runtime systems?

How much of the role is tools, shaders, asset pipeline or implementation?

Do you expect this person to write production C#?

27.42.3 WebGL

Bloque de código: text

Is WebGL a primary deployment target or secondary?

What are the current build-size and performance constraints?

How do you handle asset loading and compression?

What browsers/devices are supported?

Do you use Unity WebGL, Three.js, custom WebGL or another stack?

27.42.4 Technical visualization / digital twin

Bloque de código: text
 What kinds of technical assets or CAD data do you work with?
 How do you process CAD or engineering data for real-time visualization?
 Is the system connected to live data, IoT, APIs or static assets?
 What level of simulation accuracy is expected?
 What parts are visualization vs engineering calculation?

27.42.5 Contractor

Bloque de código: text
 Is this role structured as contractor, employee, B2B or EOR?
 What is the expected weekly availability?
 What time-zone overlap is required?
 What are the payment terms and currency?
 How is IP ownership handled?
 Are there exclusivity or non-compete clauses?

27.43 43. Questions not to ask too early

Avoid in first interview unless they bring it up:

- vacation details;
- exact benefits if contractor structure is unclear;
- “how soon can I get promoted?”;
- relocation before role fit is clear;
- asking for visa sponsorship before explaining current contractor availability;
- deep salary negotiation before understanding role scope.

27.44 44. Red flag responses during interviews

Watch for:

Signal	Risk
“We pay after client approval”	Payment risk
“The test should take a weekend”	Free labor risk
“We need full-time but contractor only”	Misclassification risk
“No contract needed”	Legal/payment risk
“Equity instead of pay”	Cashflow risk
“We own all your side projects”	IP risk
“Remote, but must live in the US/EU”	Authorization blocker
“We need senior experience”	Misfit
“Can you do design, web, AI, 3D, marketing?”	Scope chaos

27.45 45. Final preparation checklist before interviews

- 30-second pitch memorized.
- 60-second TwinSight explanation memorized.
- AI-assisted development answer ready.
- Low formal experience answer ready.
- Remote Colombia answer ready.

EU authorization answer ready.
Salary expectation answer ready.
Portfolio link ready.
GitHub link ready.
Demo video ready.
Questions for interviewer ready.
Role-specific technical notes reviewed.
Contract red flags reviewed.

27.46 46. Next file dependency

The next file should be:

```
Bloque de codigo: text  
24_offer_evaluation_scorecard.md
```

It should include:

- offer scoring model;
- weighted criteria;
- gross-to-net model;
- contractor vs employee evaluation;
- red-flag checklist;
- acceptance / negotiation / rejection thresholds;
- compensation scenarios;
- EU mobility considerations;
- Colombia compliance checklist.

No Deep Research or agent required.

Chapter 28

24_offer_evaluation_scorecard

Fuente: 24_offer_evaluation_scorecard.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1050

Ultima modificacion: 2026-06-11 15:27

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This document provides a practical system for evaluating, negotiating, accepting or rejecting job offers.

Contenido completo depurado

28.0.1 Purpose

This document provides a practical system for evaluating, negotiating, accepting or rejecting job offers.

It is designed for a Colombia-based Real-Time 3D Developer / Unity Technical Artist targeting international remote work, contractor/B2B roles, Unity WebGL, technical visualization, simulation, digital twin, XR-adjacent work and Python tooling as a secondary route.

The system is not legal, tax or immigration advice. Any contract, tax setup or relocation decision should be reviewed with qualified professionals.

28.1 1. Core decision rule

Do not accept an offer only because it is international.

An offer is strategically useful only if it satisfies most of these conditions:

1. compensation is above the minimum threshold;
 2. net income is acceptable after taxes, fees and social security;
 3. the role strengthens the real-time 3D / Unity / technical visualization path;
 4. contract terms are clear;
 5. payment risk is controlled;
 6. IP and exclusivity terms are reasonable;
 7. remote work from Colombia is legally and operationally feasible;
 8. the role does not create major strategic dilution;
 9. the work can build toward USD 3k–6k/month over 12–24 months.
-

28.2 2. Offer categories

28.2.1 Offer type classification

Type	Description	Default stance
Strategic offer	Strong role fit + acceptable compensation + safe structure	Consider strongly
Cashflow offer	Pays acceptably but weak strategic fit	Accept only if needed
Portfolio leverage offer	Lower pay but strong proof/project value	Consider selectively
Risky offer	Unclear pay, scope or legal terms	Renegotiate or reject
Dilution offer	Pulls profile away from target path	Usually reject
Exploitative offer	Free labor, vague equity, abusive terms	Reject

28.2.2 Role-fit classification

Role	Strategic value
Real-Time 3D / Unity	High
Unity WebGL / Interactive 3D	High
Technical Visualization / Digital Twin	High
Unity Technical Artist	High if runtime/optimization/tools
XR / Simulation	Medium-high
Tools / Python Automation	Medium
Full-stack	Low-medium
3D Artist / Generalist	Low unless technical
Generic web design	Low
AI hype role without real scope	Risky

28.3 3. Offer scorecard

Score each criterion from 1 to 5.

Then multiply by weight.

28.3.1 Main scorecard

Criterion	Weight	Score
Gross compensation	15	1–5
Estimated net compensation	15	1–5
Role fit	20	1–5
Portfolio leverage	15	1–5
Contract clarity	10	1–5
Payment reliability	10	1–5
Remote feasibility	10	1–5
Growth / learning ROI	5	1–5
Total	100	—

28.3.2 Interpretation

Total score	Decision
85–100	Strong accept / negotiate upward
70–84	Acceptable if no hidden risks
55–69	Renegotiate or accept only for cashflow
40–54	Usually reject
Under 40	Reject

28.3.3 Automatic override

Reject even if score is high if there is:

- no written contract;
- unclear payment terms;
- broad unpaid test;
- abusive IP assignment;
- illegal or impossible work authorization assumption;
- non-payment history;
- requirement to misrepresent status;
- broad non-compete;
- no clarity on contractor vs employee;
- request for free production work.

28.4 4. Scoring definitions

28.4.1 Gross compensation

Score	Meaning
5	USD 6k+/month or equivalent high-value contract
4	USD 3k–6k/month
3	USD 1.5k–3k/month
2	Below USD 1.5k/month but acceptable short-term
1	Below minimum or unpaid/equity-only

28.4.2 Estimated net compensation

Score	Meaning
5	Strong net after taxes, fees, social security and reserves
4	Good net, manageable deductions
3	Acceptable net but not high leverage
2	Net barely above local alternatives
1	Net too low after obligations

28.4.3 Role fit

Score	Meaning
5	Direct Unity/WebGL/technical visualization/digital twin/simulation fit
4	Strong Unity/3D fit with one missing area

Score	Meaning
3	Adjacent technical role
2	Weakly related
1	Dilutes profile

28.4.4 Portfolio leverage

Score	Meaning
5	Strengthens TwinSight direction directly
4	Adds relevant production proof
3	Adds useful but adjacent proof
2	Little portfolio value
1	Creates irrelevant work history

28.4.5 Contract clarity

Score	Meaning
5	Clear written contract, scope, IP, payment, termination
4	Mostly clear, minor gaps
3	Usable but needs clarification
2	Significant ambiguity
1	No contract or vague agreement

28.4.6 Payment reliability

Score	Meaning
5	Established company, predictable payroll/payment
4	Good reputation, clear terms
3	Unknown but acceptable safeguards
2	Delayed/client-dependent payment
1	High non-payment risk

28.4.7 Remote feasibility

Score	Meaning
5	Explicitly remote worldwide or LATAM contractor
4	Remote possible with clear contractor structure
3	Remote allowed but legal setup unclear
2	Remote only with friction
1	Requires unavailable authorization or relocation

28.4.8 Growth / learning ROI

Score	Meaning
5	Strong mentorship, technical growth, production standards
4	Good learning exposure
3	Some growth
2	Repetitive or low-growth
1	Stagnant or harmful

28.5 5. Compensation thresholds

28.5.1 Current target bands

Band	Monthly gross	Interpretation
Minimum target	USD 1.5k	Entry acceptable
Starter international	USD 2k–3k	Good early target
Strong 6-month target	USD 3k–4.5k	Preferred
High-value contractor	USD 5k–6k+	Strong strategic offer
Long-term upside	USD 6k+	12–24 month target

28.5.2 Decision by compensation

Compensation	Default decision
Under USD 1.5k	Reject unless urgent cashflow or exceptional portfolio value
USD 1.5k–2k	Accept only if role fit is strong
USD 2k–3k	Viable early offer
USD 3k–4.5k	Strong 3–6 month target
USD 4.5k–6k	Strong offer
USD 6k+	High-value, evaluate legal/contract risk carefully

28.5.3 Important distinction

Always separate:

Bloque de código: text
Gross compensation
Net compensation
Cashflow timing
Long-term value

A high gross offer with delayed, risky or non-compliant payment can be worse than a lower but stable offer.

28.6 6. Gross-to-net checklist

Before accepting a contractor offer, estimate:

Item	Question
Gross monthly	What is the nominal amount?

Item	Question
Currency	USD, EUR, COP or other?
Payment fees	Wise, Payoneer, PayPal, SWIFT, bank spread?
Conversion loss	What exchange rate applies?
Taxes	What must be reserved?
Social security	EPS, pension, ARL?
Accountant	Monthly cost?
Software/equipment	Who pays?
Vacation	Unpaid or paid?
Sick days	Unpaid or paid?
Payment delay	7, 15, 30, 45, 60 days?
Contract termination	How much notice?

28.6.1 Net estimate formula

Use a conservative model:

```
Bloque de codigo: text
Estimated net = gross
- platform/payment fees
- currency conversion loss
- tax reserve
- social security
- accountant/admin costs
- unpaid time reserve
- software/equipment costs
```

28.6.2 Required reserve

For contractor income, keep reserves for:

- taxes;
- social security;
- slow payment;
- contract termination;
- equipment failure;
- medical/emergency buffer;
- unpaid vacation.

28.7 7. Colombia contractor compliance checklist

28.7.1 Before signing

Confirm:

Item	Status
RUT	Confirmed / pending
CIIU activity	Confirmed with accountant
Electronic invoicing	Confirmed / pending
IVA responsibility	Confirmed with accountant
Export-service treatment	Confirmed per contract
EPS/pension/ARL	Process defined
Payment method	Confirmed
Contract language	Reviewed
IP terms	Reviewed
Tax reserve	Defined
Accountant	Selected

28.7.2 Ask accountant

Questions to ask:

1. What CIIU code should be used for Unity / software / technical visualization services?
 2. Does this service qualify as export of services?
 3. How should invoices be issued to foreign clients?
 4. Is IVA applicable in this specific scenario?
 5. What social security base should be used?
 6. How should foreign currency payments be documented?
 7. What deductions or costs can be supported?
 8. What monthly tax reserve is prudent?
 9. What documents should be kept for DIAN?
 10. Are there risks if the relationship looks employee-like?
-

28.8 8. Contractor vs employee evaluation

28.8.1 Contractor advantages

Advantage	Meaning
International access	Easier for foreign companies
Higher gross potential	Often better than local salary
Flexibility	Can work with global clients
Portfolio control	Project-based leverage
Faster entry	No sponsorship needed

28.8.2 Contractor disadvantages

Disadvantage	Meaning
No benefits	No paid vacation unless negotiated
Tax/admin burden	Must manage compliance
Payment risk	Late or non-payment possible
No job security	Contract can end quickly
Misclassification risk	If relationship is employee-like
Equipment/software	Often self-funded

28.8.3 Employee advantages

Advantage	Meaning
Stability	More predictable
Benefits	Health, vacation, severance depending on jurisdiction
Lower admin	Payroll handled
Team integration	Better mentorship
Career signaling	Stronger formal experience

28.8.4 Employee disadvantages

Disadvantage	Meaning
Lower flexibility	Less control
Authorization barriers	Foreign employment may require local setup
Lower local salary	Colombia salaries may be lower
EOR dependency	Employer must support structure

28.8.5 EOR evaluation

EOR is useful if:

- company wants employee relationship;
- role is long-term;
- compensation justifies EOR cost;
- employer is willing to pay platform fees;
- compliance matters.

EOR is less likely for:

- small studios;
- short contracts;
- low-budget roles;
- trial projects.

28.9 9. Contract terms checklist

28.9.1 Must be clear

Term	Required clarity
Parties	Legal names
Role	Scope and title
Start date	Exact
End date	If fixed-term
Compensation	Amount and currency
Payment schedule	Weekly/monthly/milestones
Payment method	Bank/Wise/Payoneer/etc.
Late payment	Consequence or remedy
Work hours	Expected availability
Time zone	Required overlap
Deliverables	Defined
IP ownership	Clear
Confidentiality	Reasonable
Exclusivity	Limited or absent
Non-compete	Avoid broad terms
Termination	Notice period
Dispute law	Jurisdiction
Equipment/software	Who pays
Taxes	Contractor responsibility if B2B
Data/security	Reasonable requirements

28.9.2 Dangerous contract terms

Clause	Risk
Broad non-compete	Blocks future work
Broad exclusivity	Prevents other clients
Work-for-hire over all past/future work	IP abuse
Payment only after client approval	Cashflow risk
Unlimited unpaid revisions	Scope creep
No termination notice	Instability
No late-payment remedy	Non-payment risk
Mandatory unpaid trial	Exploitation
Confusing employee/contractor language	Misclassification
Jurisdiction impossible to enforce	Legal risk

28.10 10. Take-home test evaluation

28.10.1 Acceptable test

A take-home test is acceptable if:

- it is small;
- it has a clear time limit;
- it is relevant to the role;
- it does not produce usable client work;
- it has clear evaluation criteria;
- it does not require unpaid multi-day labor;
- IP terms are clear;
- the company is credible.

28.10.2 Reasonable test examples

Role	Reasonable test
Unity Developer	Small interaction prototype
Technical Artist	Shader/material or optimization task
WebGL	Small build/performance task
Tools	Small editor script
Python Automation	Small workflow script
Digital Twin	Conceptual architecture task

28.10.3 Unreasonable test examples

Test	Reason
Full feature implementation	Free labor
Client-ready prototype	Exploitation
Multi-day unpaid build	Excessive
Unclear ownership	IP risk
No feedback	Low value
Requires private assets without agreement	Legal risk
Looks like production backlog	Red flag

28.10.4 Test negotiation script

Bloque de código: text
I'm open to a focused technical exercise. To keep the scope reasonable, could we define an expected time limit and clarify
↔ whether the test is only for evaluation and not production use?

28.10.5 Test refusal script

Bloque de código: text
I'm not comfortable completing a large unpaid production-style task. I would be open to a smaller scoped exercise, a paid
↔ test, a portfolio walkthrough, or a live technical discussion instead.

28.11 11. Red flags

28.11.1 Critical red flags

Reject or pause if:

- no written contract;
- payment terms unclear;
- company refuses to define scope;
- asks for free production work;
- broad non-compete;
- broad IP claim over unrelated work;
- payment only after client approval;
- “remote” but requires unavailable work authorization;
- asks to misrepresent legal status;
- insists on employee-like control under contractor structure;
- salary below minimum with no strategic upside;
- company has public non-payment complaints.

28.11.2 Moderate red flags

Negotiate or clarify if:

- vague role title;
- unclear reporting structure;
- unclear timezone expectations;
- no onboarding process;
- vague “startup equity” promises;
- unclear tool/software reimbursement;
- all communication only by chat;
- no technical interviewer;
- constant urgency;
- trial period without clear compensation.

28.11.3 Low-level warning signs

Monitor if:

- delayed recruiter responses;
- overly broad job description;
- role mixes too many unrelated skills;

- no clear product/project;
 - old job posting;
 - compensation hidden until late;
 - vague remote policy.
-

28.12 12. Offer-specific questions

28.12.1 Compensation

Ask:

Bloque de codigo: text
What is the gross compensation?
What currency is used?
Is payment monthly, biweekly or milestone-based?
What are the payment terms?
Are there late-payment protections?
Is compensation fixed, hourly or retainer-based?

28.12.2 Contractor structure

Ask:

Bloque de codigo: text
Is this B2B/contractor, employee, EOR or freelance?
Who is the legal contracting entity?
What invoices or tax forms are required?
Is there a required platform?
Is there a trial period?

28.12.3 Scope

Ask:

Bloque de codigo: text
What are the first 30/60/90 day deliverables?
What does success look like?
What is the expected weekly availability?
How much meeting time is expected?
What timezone overlap is required?

28.12.4 IP and confidentiality

Ask:

Bloque de codigo: text
Who owns the deliverables?
Does IP assignment apply only to paid work?
Are personal/portfolio projects excluded?
Can I show non-confidential work in my portfolio?
Are there confidentiality restrictions?

28.12.5 Technical role

Ask:

Bloque de codigo: text
What engine/platform is used?
What Unity version and render pipeline?
Is WebGL/XR/mobile/desktop the target?
How is profiling handled?
How is asset optimization handled?
Will I write production C#?
Will I work with artists/designers/engineers?

28.13 13. Acceptance thresholds

28.13.1 Strong accept

Accept if:

- score above 80;
- compensation meets target;
- contract is clear;
- role fit is strong;
- remote structure is feasible;
- payment risk is low;
- no major red flags;
- portfolio leverage is high.

28.13.2 Conditional accept

Accept if:

- score 65–80;
- compensation is acceptable;
- role fit is useful;
- risks can be negotiated;
- cashflow or formal experience value is meaningful.

28.13.3 Reject

Reject if:

- score below 55;
 - compensation below minimum;
 - role dilutes profile severely;
 - payment risk is high;
 - contract is unclear;
 - unpaid work is excessive;
 - authorization assumptions are false;
 - IP/exclusivity terms are abusive.
-

28.14 14. Negotiation strategy

28.14.1 General rule

Negotiate based on structure, not just salary.

Consider:

- rate;
- hours;
- payment timing;
- trial period;
- deliverables;
- IP;
- exclusivity;
- scope;

- timezone;
- renewal terms;
- portfolio usage;
- software/equipment.

28.14.2 If compensation is low but role is strong

Negotiate:

Bloque de codigo: text
 higher rate after trial;
 shorter trial period;
 reduced hours;
 clear portfolio permission;
 defined scope;
 milestone payment;
 review after 30/60 days.

28.14.3 If role is good but contractor risk is high

Negotiate:

Bloque de codigo: text
 advance payment;
 short payment cycle;
 written contract;
 clear termination notice;
 paid test;
 no broad exclusivity;
 limited IP assignment.

28.14.4 If company wants full-time contractor

Clarify:

Bloque de codigo: text
 expected hours;
 exclusivity;
 benefits;
 paid time off;
 tools;
 whether relationship is truly contractor;
 tax responsibility;
 EOR possibility.

28.15 15. Negotiation scripts

28.15.1 Ask for higher compensation

Bloque de codigo: text
 Based on the scope, expected availability and the technical requirements around Unity, WebGL and real-time 3D visualization, I
 ↪ would be more comfortable at [target range].
 If that is above the current budget, we could also discuss adjusting scope, hours, trial length or review timing.

28.15.2 Ask for paid test

Bloque de codigo: text
 I'm open to a technical evaluation. Since the task appears close to production work and may take several hours, would you be
 ↪ open to making it a paid test or reducing the scope to a shorter evaluation exercise?

28.15.3 Ask for clearer scope

Bloque de código: text

Before accepting, I would like to clarify the first deliverables, expected weekly availability, decision-making process and
↪ success criteria for the first 30 days.

28.15.4 Ask about IP

Bloque de código: text

Could you clarify whether IP assignment applies only to paid deliverables created under this contract, and whether my previous
↪ work, portfolio projects and independent tools are excluded?

28.15.5 Ask about contractor structure

Bloque de código: text

Since I am based in Colombia, I want to confirm whether this would be structured as an independent contractor/B2B agreement,
↪ EOR employment or another setup.

28.15.6 Decline politely

Bloque de código: text

Thank you for the offer. After reviewing the compensation, scope and contract structure, I do not think this is the right fit
↪ at this stage.

I appreciate the opportunity and would be open to reconnecting in the future if there is a role with clearer alignment around
↪ Unity, real-time 3D or technical visualization.

28.16 16. Offer comparison model

When comparing two offers, do not choose only by gross pay.

28.16.1 Comparison table

Criterion	Offer A	Offer B
Gross monthly	—	—
Estimated net	—	—
Role fit	—	—
Portfolio value	—	—
Payment stability	—	—
Contract safety	—	—
Timezone fit	—	—
Learning value	—	—
Future salary leverage	—	—
EU relevance	—	—
Red flags	—	—
Total score	—	—

28.16.2 Tie-breaker

If two offers are similar, prefer the one that:

1. strengthens TwinSight-aligned positioning;
2. gives formal production experience;
3. has better payment reliability;

4. avoids legal ambiguity;
 5. improves future compensation;
 6. allows portfolio-safe work;
 7. does not consume all time for low leverage.
-

28.17 17. Strategic offer examples

28.17.1 Offer A — Low pay, strong fit

Example:

```
Bloque de codigo: text
USD 1.8k/month contractor
Unity WebGL technical visualization
Clear contract
Good portfolio permission
```

Decision:

```
Bloque de codigo: text
Possible accept if cashflow is needed and role creates strong production proof.
Negotiate review after 60 days.
```

28.17.2 Offer B — Higher pay, weak fit

Example:

```
Bloque de codigo: text
USD 3k/month
Generic WordPress/web support
No Unity/3D work
```

Decision:

```
Bloque de codigo: text
Accept only if cashflow pressure is high.
Do not treat as main career path.
Limit duration if possible.
```

28.17.3 Offer C — High pay, high risk

Example:

```
Bloque de codigo: text
USD 5k/month
Startup contractor
No contract
Payment after client approval
Broad IP terms
```

Decision:

```
Bloque de codigo: text
Do not accept without fixing contract/payment/IP risk.
```

28.17.4 Offer D — Strong strategic offer

Example:

```
Bloque de codigo: text
USD 3.5k-5k/month
Unity technical visualization
Remote contractor from Colombia
Clear contract
Defined deliverables
Portfolio-safe non-confidential work
```

Decision:

```
Bloque de codigo: text
Strong accept/negotiation candidate.
```

28.18 18. EU mobility and offer evaluation

28.18.1 Current state

Currently, evaluate offers as:

```
Bloque de codigo: text
Colombia-based remote contractor
```

Do not evaluate as:

```
Bloque de codigo: text
EU-authorized candidate
```

until legal status exists.

28.18.2 After Portuguese passport

The offer pool changes.

Higher priority can shift toward:

- EU employee roles;
- Germany/Portugal/Netherlands/Spain roles;
- hybrid EU roles;
- EOR or payroll roles in Europe;
- stronger long-term compensation paths.

28.18.3 Germany route

If German residence/work authorization becomes active, evaluate:

- German salary vs Colombian contractor net;
- health insurance;
- taxes;
- cost of living;
- local language;
- career leverage;
- long-term stability.

28.18.4 Do not relocate for weak offers

Relocation should require:

- strong role fit;
- legal clarity;
- sufficient compensation;
- housing feasibility;
- health insurance clarity;
- long-term career value.

28.19 19. Role dilution test

Before accepting, ask:

```
Bloque de codigo: text
Will this role make my next application stronger for Unity / real-time 3D / technical visualization?
```

28.19.1 If yes

Likely strategic.

28.19.2 If no

Treat as cashflow only.

28.19.3 If actively negative

Reject unless survival requires it.

28.19.4 Dilution examples

Offer	Dilution risk
Generic social media design	High
WordPress maintenance only	Medium-high
Data entry / AI labeling	High
Pure call center	High
Low-end 3D modeling only	Medium
Full-stack CRUD only	Medium
Unity mobile game support	Low-medium
WebGL product configurator	Low
Digital twin Unity role	Very low

28.20 20. Offer go/no-go checklist

28.20.1 Go

Proceed if:

- role fit is clear;

- compensation meets threshold;
- net estimate works;
- contract is written;
- payment method is clear;
- scope is defined;
- IP terms are reasonable;
- remote setup is feasible;
- no false work authorization assumption;
- no major red flags.

28.20.2 No-go

Do not proceed if:

- no written agreement;
- payment terms are vague;
- compensation below minimum;
- unpaid work is excessive;
- company refuses scope clarity;
- IP terms capture unrelated work;
- remote status is misleading;
- requires unavailable EU/US authorization;
- role damages the portfolio path;
- company pressure feels manipulative.

28.21 21. Final offer decision worksheet

Use this before accepting.

```

Bloque de codigo: text
Company:
Role:
Offer date:
Employment type:
Location:
Remote scope:
Contractor/EOR/employee:
Gross compensation:
Currency:
Payment cycle:
Estimated net:
Expected hours:
Timezone overlap:
Start date:
Trial period:
Termination notice:
IP terms:
Exclusivity:
Portfolio permission:
Main responsibilities:
Main risks:
Score:
Decision:
Negotiation points:
Final answer deadline:

```

28.22 22. Final recommendation

The safest decision model is:

```
Bloque de codigo: text
Do not optimize only for the first paycheck.
Optimize for the intersection of compensation, contract safety and real-time 3D positioning.
```

The ideal first international offer is not necessarily the highest gross number.

The best offer is the one that:

1. pays enough;
 2. is legally and operationally clear;
 3. strengthens the Unity / real-time 3D / technical visualization path;
 4. creates credible professional experience;
 5. does not trap the profile in unrelated work;
 6. can become a stepping stone toward USD 3k–6k/month.
-

28.23 23. Next file dependency

The next file should be:

```
Bloque de codigo: text
25_application_tracker_template.xlsx
```

This should be a real spreadsheet with sheets for:

1. Applications.
2. Companies.
3. Contacts.
4. Interviews.
5. Offers.
6. Weekly Metrics.
7. Skill Sprint.
8. Content Assets.
9. Settings / scoring definitions.

Spreadsheet generation should follow spreadsheet-specific tooling rules.

Chapter 29

26_weekly_execution_dashboard

Fuente: 26_weekly_execution_dashboard.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1095

Ultima modificacion: 2026-06-11 15:42

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This document defines the weekly operating system for executing the international job-search strategy.

Contenido completo depurado

29.0.1 Purpose

This document defines the weekly operating system for executing the international job-search strategy.

It explains how to use:

```
Bloque de codigo: text
25_application_tracker_template.xlsx
```

and how to make weekly decisions based on evidence instead of mood, anxiety, perfectionism or endless planning.

This is the final operational control layer of the system.

29.1 1. Core operating rule

The job search should now be treated as a measured pipeline.

The weekly question is not:

```
Bloque de codigo: text
Do I feel ready?
```

The weekly question is:

```
Bloque de codigo: text
What did the data show this week, and what should change next week?
```

The tracker exists to prevent:

- applying randomly;
 - forgetting follow-ups;
 - over-researching;
 - avoiding applications;
 - over-polishing the portfolio;
 - chasing low-value roles;
 - accepting bad offers;
 - misreading silence as failure;
 - making decisions without evidence.
-

29.2 2. Weekly review cadence

29.2.1 Recommended review time

Use one fixed weekly review block:

```
Bloque de codigo: text
Sunday evening or Monday morning
Duration: 45-75 minutes
```

Do not review strategy every day. Daily review creates noise.

29.2.2 Weekly review inputs

Before starting the review, open:

1. 25_application_tracker_template.xlsx
2. Portfolio / TwinSight case study
3. LinkedIn profile
4. GitHub profile
5. Calendar
6. Notes from interviews or recruiter calls

29.2.3 Weekly review outputs

At the end of the review, define:

- next week's application target;
 - next week's outreach target;
 - follow-ups due;
 - portfolio improvements;
 - interview prep focus;
 - skill sprint focus;
 - red flags;
 - one strategic adjustment maximum.
-

29.3 3. Weekly dashboard workflow

29.3.1 Step 1 — Update raw data

Update these sheets first:

```

Bloque de codigo: text
Applications
Companies
Contacts
Interviews
Offers
Weekly Metrics
Skill Sprint
Content Assets

```

Do not interpret anything until the tracker is updated.

29.3.2 Step 2 — Check Dashboard

Open the Dashboard sheet and check:

- total applications;
- A-priority applications;
- active pipeline;
- interviews started;
- offers;
- follow-ups due;
- average application score;
- average offer score;
- applications by status;
- weekly activity.

29.3.3 Step 3 — Identify bottleneck

Classify the week into one of these bottlenecks:

Bottleneck	Meaning
Asset bottleneck	Portfolio/CV/GitHub not ready
Volume bottleneck	Too few applications/outreach
Targeting bottleneck	Applying to wrong companies
Positioning bottleneck	Profile not generating replies
Conversion bottleneck	Replies but no interviews
Interview bottleneck	Interviews but no progress
Offer bottleneck	Offers weak or risky
Execution bottleneck	Tracker not maintained

29.3.4 Step 4 — Choose one correction

Only choose one primary correction per week.

Examples:

- improve TwinSight case study;
- increase A-priority applications;
- rewrite recruiter opener;
- verify top 10 companies;
- practice WebGL interview questions;
- adjust salary response;
- stop applying to C-priority roles.

Do not redesign the whole system weekly.

29.4 4. Weekly metrics

29.4.1 Required metrics

Track these every week:

Metric	Target
Applications sent	15–30
A-priority applications	5–10
Outreach messages	10–25
Follow-ups sent	10–20
Recruiter contacts	5–15
Interviews started	0–3
Technical tests	0–2
Offers	0–1
Portfolio blocks	2–4
Skill blocks	2–4

29.4.2 Minimum viable week

If energy or circumstances are bad, still complete:

Action	Minimum
Applications	5
Outreach	5
Follow-ups	5
Portfolio improvement	1
Tracker update	1
Interview prep	1

A low-output week is acceptable. A zero-tracking week is not.

29.5 5. Weekly decision thresholds

29.5.1 Application volume

Result	Interpretation	Action
0–5 applications	Too low	Reduce friction
6–14 applications	Below target	Add scheduled blocks
15–30 applications	Good	Maintain
31–50 applications	High	Check quality
50+ applications	Risk of low quality	Reduce volume

29.5.2 Response rate

Use a rolling 2–4 week view, not one week.

Response rate	Interpretation
0–2%	Positioning or targeting problem
3–5%	Weak but usable
6–10%	Acceptable
10%+	Strong for cold applications

29.5.3 Interview conversion

Signal	Interpretation
Replies but no interviews	CV/portfolio mismatch
Interviews but rejection	Technical/interview gap
Tests but no progress	Test execution gap
Offers but weak terms	Targeting or negotiation issue

29.6 6. Application quality system

29.6.1 A-priority application checklist

Before sending an A-priority application:

- Role matches one of the main role families.
- Remote scope is plausible.
- Work authorization does not immediately block it.
- CV variant is correct.
- TwinSight link is included.
- Message references company-specific fit.
- Salary range is not obviously below target.
- Follow-up date is entered.
- Tracker row is complete.

29.6.2 B-priority application checklist

- Role is adjacent but plausible.
- Correct CV variant used.
- Portfolio link included.
- Follow-up date entered.
- No major red flags.

29.6.3 C-priority application checklist

- Fast application only.
- No deep customization.
- Do not spend more than 10 minutes.
- Use only if volume or cashflow requires it.

29.7 7. Weekly review questions

29.7.1 Pipeline questions

Answer these every week:

1. How many A-priority applications were sent?
2. How many B-priority applications were sent?
3. Which roles produced replies?
4. Which role family produced no signal?
5. Which job boards produced useful roles?
6. Which companies need follow-up?
7. Which recruiters replied?
8. Which applications are stale?
9. Which offers or interviews require preparation?
10. Which target category should be reduced?

29.7.2 Portfolio questions

1. Did TwinSight become easier to understand this week?
2. Did the demo video improve?
3. Did GitHub become cleaner?
4. Did LinkedIn become more consistent?
5. Did the portfolio gain a visible proof asset?
6. Did any improvement become perfectionism?

29.7.3 Interview questions

1. Which interview question was weakest?
2. Which technical concept needs practice?
3. Can TwinSight be explained in 60 seconds?
4. Can AI-assisted development be explained cleanly?
5. Can salary expectations be stated without revealing the floor?
6. Can remote-from-Colombia status be explained accurately?

29.7.4 Strategic questions

1. Is the role target still coherent?
 2. Is full-stack creeping back in?
 3. Is AI tooling over-dominating the profile?
 4. Is 3D art over-dominating the profile?
 5. Are applications matching the salary target?
 6. Is the strategy moving toward USD 3k/month?
 7. Is the 12–24 month path to USD 6k/month being strengthened?
-

29.8 8. Weekly adjustment rules

29.8.1 Rule 1 — Do not pivot after one bad week

A bad week is not enough data.

Wait at least:

```
Bloque de codigo: text
2-4 weeks
50+ applications/outreach actions
```

before making major positioning changes.

29.8.2 Rule 2 — Change one variable at a time

Do not change:

- CV;
- headline;
- role target;
- portfolio structure;
- salary range;
- outreach message;

all in the same week.

Change one or two variables, then measure.

29.8.3 Rule 3 — Preserve the core identity

Every adjustment must preserve:

```
Bloque de codigo: text
Real-Time 3D / Unity / Technical Visualization
```

If an adjustment moves away from that, label it as fallback, not strategy.

29.9 9. Bottleneck diagnosis

29.9.1 Asset bottleneck

Symptoms

- avoiding applications because materials are not ready;
- portfolio link not shareable;
- GitHub not clean;
- no TwinSight demo;
- CV still uncertain.

Fix

Choose one asset and finish it to usable quality.

Priority order:

1. CV.
2. LinkedIn.
3. TwinSight case study.
4. GitHub README.
5. Demo video.
6. Portfolio polish.

Do not improve all assets at once.

29.9.2 Volume bottleneck

Symptoms

- fewer than 10 applications per week;
- many saved jobs but few sent;
- overcustomization;
- too much research.

Fix

Use application blocks:

```
Bloque de codigo: text  
3 blocks/week  
90 minutes each  
5 applications per block
```

Target:

```
Bloque de codigo: text  
15 applications/week minimum
```

29.9.3 Targeting bottleneck

Symptoms

- roles require senior-only experience;
- roles require EU/US authorization;
- roles are pure full-stack;
- roles are pure 3D art;
- roles are low-paid local jobs.

Fix

Increase proportion of:

- Unity WebGL;
- technical visualization;
- interactive 3D;
- digital twin visualization;
- simulation interfaces;
- product visualization;
- LATAM nearshore contractor roles.

Reduce:

- generic full-stack;
 - generic 3D art;
 - AAA senior roles;
 - EU-only local roles before authorization.
-

29.9.4 Positioning bottleneck

Symptoms

- profile views but no replies;
- recruiters misunderstand the profile;
- replies are for unrelated roles;
- people ask “what exactly do you do?”

Fix

Tighten public wording:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist
Unity WebGL
Technical Visualization
CAD-to-Realtime
TwinSight X500
Remote contractor from Colombia
```

Remove distracting emphasis from:

- full-stack;
 - AI engineer;
 - graphic design;
 - generic multimedia;
 - unfinished apps.
-

29.9.5 Conversion bottleneck

Symptoms

- replies but no interviews;
- recruiters ask for CV then disappear;
- portfolio not opened;
- no technical follow-up.

Fix

Improve:

- first paragraph of CV;
 - TwinSight case study top section;
 - demo video;
 - GitHub README;
 - role-specific application message;
 - portfolio card clarity.
-

29.9.6 Interview bottleneck

Symptoms

- interviews happen but no next stage;
- weak technical explanations;

- AI questions create discomfort;
- salary discussion feels unstable;
- WebGL/Unity questions expose gaps.

Fix

Use:

```
Bloque de código: text  
23_interview_answer_bank.md
```

Practice:

- TwinSight 60-second explanation;
 - AI-assisted development answer;
 - CAD-to-realtime explanation;
 - WebGL performance answer;
 - salary expectation response;
 - contractor-from-Colombia answer.
-

29.9.7 Offer bottleneck

Symptoms

- offers too low;
- contracts vague;
- unpaid tests;
- equity-only;
- role too broad;
- full-time contractor with employee-like control.

Fix

Use:

```
Bloque de código: text  
24_offer_evaluation_scorecard.md
```

Negotiate:

- rate;
- scope;
- paid test;
- payment timing;
- IP terms;
- portfolio permission;
- review after 30/60 days.

Reject if risk remains high.

29.10 10. Weekly work blocks

29.10.1 Default schedule

Day	Primary work
Monday	Target list + applications
Tuesday	Outreach + contacts
Wednesday	Applications + interview prep
Thursday	Follow-ups + recruiters
Friday	Portfolio/GitHub improvement
Saturday	Skill sprint
Sunday	Weekly review

29.10.2 Daily minimum

On low-energy days:

```
Bloque de codigo: text
1 application
1 outreach message
1 tracker update
20 minutes on one asset
```

29.10.3 Deep work blocks

Use 90-minute blocks for:

- portfolio case study;
- demo video editing;
- GitHub README;
- technical interview prep;
- skill sprint artifact.

Use 30-minute blocks for:

- follow-ups;
- tracker cleanup;
- recruiter replies;
- job board scanning.

29.11 11. Weekly plan template

Copy this into a weekly note.

```
Bloque de codigo: text
Week of: [date]

Primary goal:
[one main goal]

Targets:
- Applications:
- A-priority applications:
- Outreach:
- Follow-ups:
- Recruiter contacts:
- Portfolio blocks:
- Skill blocks:

Top target roles:
1.
2.
3.

Top target companies:
1.
2.
3.
4.
```


5.

Asset priority:
[CV / LinkedIn / TwinSight / GitHub / Demo / Portfolio]

Interview prep focus:
[topic]

Skill sprint:
[skill + artifact]

Main risk this week:
[risk]

Control action:
[action]

End-of-week result:
[filled during review]

29.12 12. Weekly review template

Use at the end of the week.

Bloque de codigo: text
Week reviewed: [date]

Metrics:

- Applications sent:
- Outreach sent:
- Follow-ups sent:
- Responses:
- Interviews:
- Tests:
- Offers:
- Rejections:
- Portfolio blocks:
- Skill blocks:

Best signal:
[what worked]

Worst signal:
[what failed]

Role family with strongest response:
[role family]

Role family with weakest response:
[role family]

Most useful job board/source:
[source]

Most useful message:
[message/template]

Main bottleneck:
[asset / volume / targeting / positioning / conversion / interview / offer / execution]

One correction for next week:
[correction]

Do not do next week:
[anti-distraction rule]

Next week target:
[measurable target]

29.13 13. Monthly review

Every 4 weeks, do a deeper review.

29.13.1 Monthly questions

1. Which role family generated the most responses?

2. Which role family generated the best interviews?
3. Which role family generated low-quality offers?
4. Which companies were realistic?
5. Which regions were most responsive?
6. Which job boards were useful?
7. Which outreach message worked best?
8. Which portfolio asset got attention?
9. What technical gap appeared repeatedly?
10. What should be stopped?

29.13.2 Monthly decision options

Evidence	Decision
Good response from Unity WebGL	Double down
Good response from technical visualization	Double down
Only full-stack replies	Fix positioning
No replies from EU	Deprioritize EU until authorization
Low salary LATAM only	Move higher in company tiers
Technical interview failures	Skill sprint
No interviews after 100 actions	Rewrite CV/portfolio
Recruiters confused	Simplify headline/about

29.14 14. 30-day gate

At day 30, verify:

- Base CV ready.
- At least one role-specific CV ready.
- LinkedIn updated.
- GitHub cleaned.
- TwinSight case study shareable.
- Portfolio MVP live or shareable.
- Tracker active.
- First applications sent.
- Outreach started.
- Demo video at least drafted.

29.14.1 If not complete

- Do not scale applications yet.
- Extend setup by 7 days.

29.14.2 If complete

- Move to volume + feedback phase.

29.15 15. 60-day gate

At day 60, verify:

Metric	Target
Applications	50–90
Outreach actions	40–80
Recruiter conversations	3–8
Interview processes	2–5
Portfolio iterations	4–8
Skill artifacts	2–4

29.15.1 If no replies

Likely issue:

- targeting;
- positioning;
- CV;
- portfolio top section.

Do not assume personal failure.

29.15.2 If replies but no interviews

Likely issue:

- portfolio clarity;
- role mismatch;
- weak proof link;
- CV not specific enough.

29.15.3 If interviews but no offers

Likely issue:

- technical explanation;
- salary mismatch;
- AI disclosure;
- role seniority mismatch;
- interview preparation.

29.16 16. 90-day gate

At day 90, choose one:

Path	Trigger
Continue	Interviews/offers active
Improve portfolio	Repeated proof gap
Increase volume	Low application volume
Refine targeting	Wrong roles replying
Add fallback	Cashflow pressure
Skill sprint	Repeated technical failures
Accept offer	Offer meets threshold
Reject/pause	Offers risky or dilutive

29.16.1 Day-90 decision rule

Use tracker evidence only.

Do not decide based on:

- one rejection;
- mood;
- comparison with other people;
- anxiety about age;
- perfectionism;
- fear of applying;
- dislike of one interview.

29.17 17. Anti-dispersion rules

29.17.1 Project freeze

During the active job-search phase, do not start:

- new flagship project;
- new game;
- new app;
- new full-stack product;
- new AI system;
- new portfolio redesign;
- new course path without artifact.

29.17.2 Allowed work

Allowed:

- TwinSight improvement;
- ARA README/demo cleanup;
- portfolio MVP;
- GitHub cleanup;
- demo video;
- interview prep;
- small skill artifacts;
- targeted company verification.

29.17.3 Course rule

No course is allowed unless it produces one public artifact.

Examples:

Course work	Required artifact
Unity profiling	before/after note
Shader Graph	small visual mode demo
Rebelway	FX clip or breakdown
German	CEFR checkpoint
Python automation	script/demo
WebGL	optimization note

29.18 18. Response interpretation rules

29.18.1 Silence

Silence usually means:

- timing;
- volume;
- mismatch;
- ATS filtering;
- hiring freeze;
- role closed;
- not enough proof;
- no referral.

It does not automatically mean the profile is invalid.

29.18.2 Rejection

A rejection is useful only if categorized:

Rejection reason	Action
Too junior	Add proof / target lower
Not authorized	Filter region
Not enough Unity	Add Unity proof
Not enough WebGL	Add WebGL proof
Not enough portfolio	Improve case study
Salary mismatch	Adjust target
Role filled	Ignore
No reason	Track only

29.18.3 Positive signal

A positive signal is:

- recruiter asks for CV;
- hiring manager responds;
- portfolio is opened;
- interview scheduled;
- technical question asked;
- request for demo;
- referral offered;
- compensation discussed.

Track all positive signals.

29.19 19. Tool usage recommendations

29.19.1 Do not use Deep Research for

- weekly planning;
- CV tweaks;
- LinkedIn tweaks;

- outreach messages;
- interview answers;
- offer evaluation;
- portfolio copy.

29.19.2 Use Deep Research only for

- legal updates;
- immigration rule changes;
- tax rule changes;
- salary benchmarking refresh;
- market shifts;
- new role-market research.

29.19.3 Use mode agent only for

- live company verification;
- active job openings;
- recruiter/contact discovery;
- job board scraping-like collection;
- top-30 target validation.

29.19.4 Do not use agent for

- strategy synthesis;
- personal positioning;
- CV writing;
- interview answers;
- legal interpretation.

29.20 20. Top-30 company verification prompt

Use this later with mode agent.

```
Bloque de codigo: text
Verify the current live hiring status for the top 30 A-priority companies in my target list.

For each company, check:
- careers URL;
- active relevant roles;
- role title;
- location;
- remote scope;
- contractor/B2B possibility if stated;
- work authorization requirement;
- required seniority;
- language;
- salary if posted;
- application URL;
- recruiter or hiring contact if publicly available;
- date checked.

Relevant roles:
Unity Developer, Unity WebGL Developer, Real-Time 3D Developer, Technical Artist, Technical Visualization Developer,
↳ Simulation Developer, Digital Twin Visualization Developer, XR Developer, Tools Developer, Python Automation Developer.

Do not write strategy.
Return tables only.
Do not infer worldwide remote unless explicitly stated.
Mark unavailable information as "not found".
```

29.21 21. Weekly dashboard anti-patterns

Do not:

- update the tracker only when motivated;
 - leave applications without follow-up dates;
 - count saved jobs as applications;
 - count portfolio polishing as applications;
 - count course watching as skill progress without artifact;
 - re-plan the whole strategy every week;
 - change role positioning after one rejection;
 - use low-quality volume to feel productive;
 - ignore low compensation red flags;
 - accept unclear contractor terms.
-

29.22 22. Final weekly operating statement

Every week must produce measurable movement in at least three of these areas:

1. applications;
2. outreach;
3. follow-ups;
4. portfolio/GitHub proof;
5. interviews;
6. skill artifacts;
7. offer evaluation.

If a week produces only research, planning or course consumption, it is not a productive job-search week.

29.23 23. System completion status

With this file, the strategic and operational document system is complete.

Completed blocks:

```
Bloque de codigo: text
00-16: research, strategy, audit and planning
17-24: application materials and decision systems
25: spreadsheet tracker
26: weekly execution dashboard
```

The remaining work is execution:

```
Bloque de codigo: text
1. Fill missing personal data.
2. Finalize CV.
3. Update LinkedIn.
4. Clean GitHub.
5. Publish TwinSight case study.
6. Record demo video.
7. Start applications.
8. Track weekly.
9. Adjust based on evidence.
```

29.24 24. Immediate next actions

29.24.1 Next 48 hours

1. Fill contact details for CV.
2. Verify TwinSight final metrics.
3. Decide portfolio platform.
4. Choose CV variant 1.
5. Update LinkedIn headline/About.
6. Clean GitHub pins.
7. Create or update TwinSight README.
8. Add 5–8 screenshots to TwinSight materials.
9. Open the tracker and delete/replace sample rows.
10. Define the first 25 A-priority applications.

29.24.2 Next 7 days

1. Publish portfolio MVP or shareable TwinSight case study.
 2. Export base CV PDF.
 3. Export first role-specific CV PDF.
 4. Record raw TwinSight footage.
 5. Send first 10–20 applications/outreach messages.
 6. Track every action.
 7. Run first weekly review.
-

29.25 25. Final rule

The roadmap is now safe enough to execute.

Do not keep expanding the research before producing public assets.

The next bottleneck is not information.

The next bottleneck is publication and outbound action.

Chapter 30

27_strategy_scenarios_conservative_balanced

Fuente: 27_strategy_scenarios_conservative_balanced_aggressive.md
Seccion: root
Estado: canonical
Rol: Modulo principal
Lineas originales: 1089
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Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This file defines three execution scenarios for the international job-search roadmap:

Contenido completo depurado

30.0.1 Purpose

This file defines three execution scenarios for the international job-search roadmap:

- 1. **Conservative**
- 2. **Balanced**
- 3. **Aggressive**

The goal is to decide how much risk, volume, salary ambition, contractor exposure, portfolio pressure, and international targeting should be used during the next 3–6 months.

This document does not replace the main strategy. It turns it into selectable operating modes.

30.1 1. Baseline profile

30.1.1 Current state

Area	Status
Location	Colombia
Main target	Remote international work
Preferred model	Contractor / B2B
Thesis	Complete, pending defense
Flagship project	TwinSight X500
Formal industry experience	Limited in Unity / Technical Art
Main strength	Strong project evidence
Main weakness	Limited packaged public proof

Area	Status
Current EU work authorization	No
Future EU route	Portuguese passport expected
Germany route	Possible through marriage/residence
Immediate income target	USD 1.5k+/month
6-month target	USD 3k/month
12–24 month target	USD 6k/month

30.1.2 Fixed positioning across all scenarios

All scenarios preserve the same core identity:

Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization,
↔ and Unity WebGL deployment.

The scenarios change execution intensity, not professional identity.

30.2 2. Strategic scenario overview

Scenario	Risk	Speed	Income upside	Stability	Best for
Conservative	Low	Slow-medium	Low-medium	High	Stability and first remote role
Balanced	Medium	Medium-fast	Medium-high	Medium	Best default strategy
Aggressive	High	Fast	High	Low-medium	High-value international contracts

30.2.1 Recommended default

The recommended route is:

Bloque de codigo: text
Balanced strategy with aggressive targeting for A-priority roles and conservative controls for contract risk.

This means:

- apply aggressively to high-fit international roles;
- negotiate carefully;
- do not accept unsafe contracts;
- keep fallback options controlled;
- do not dilute the profile.

30.3 3. Conservative strategy

30.3.1 Definition

The conservative strategy prioritizes stability, lower risk, realistic first employment, and controlled skill growth. It aims to secure a first relevant role without overexposing the candidate to risky contractor structures, high-pressure international interviews, or overly senior roles.

30.3.2 Best use case

Use conservative mode if:

- cashflow pressure becomes high;
- portfolio is still incomplete;
- interviews reveal technical gaps;
- confidence in English interviews is low;
- legal/admin contractor setup is not ready;
- there is strong need for stable routine;
- thesis defense or graduation tasks still consume time.

30.3.3 Primary target roles

Priority	Role
1	Unity Developer, junior/early-mid
2	Interactive 3D Developer
3	Technical Visualization Developer, junior
4	Unity WebGL Developer, small team
5	Python automation role, secondary
6	Local/LATAM software role, fallback

30.3.4 Target markets

Market	Priority
Colombia remote/hybrid	Medium
LATAM nearshore	High
US small remote companies	Medium
Global contractor platforms	Medium
EU local roles	Low until EU authorization
Germany local roles	Low until legal route active
AAA games	Low
Large digital twin enterprise	Watchlist

30.3.5 Salary target

Timeframe	Target
First 3 months	USD 1.5k–2.5k/month
3–6 months	USD 2k–3k/month
12 months	USD 3k–4k/month

30.3.6 Application volume

Action	Weekly target
Applications	10–20
A-priority applications	3–6
Outreach	5–10
Recruiter contacts	5–10
Follow-ups	5–10
Portfolio blocks	2–3

Action	Weekly target
Skill blocks	2–4

30.3.7 Portfolio standard

Minimum required:

- base CV;
- LinkedIn updated;
- TwinSight case study draft;
- GitHub cleaned enough;
- 5–8 screenshots;
- no major public contradictions.

Demo video can be draft-quality.

30.3.8 Contractor stance

Conservative mode prefers:

- clear written contracts;
- monthly retainer;
- stable payment terms;
- LATAM-friendly clients;
- smaller but safer roles;
- shorter commitments.

Avoid:

- unpaid tests;
- equity-only;
- payment-after-client-approval;
- ambiguous offshore contracts;
- broad IP/exclusivity clauses.

30.3.9 Advantages

- Lower probability of severe contract/payment problems.
- Easier first win.
- Less interview pressure.
- More room to continue improving portfolio.
- Lower risk of burnout.

30.3.10 Disadvantages

- Slower compensation growth.
- Higher chance of accepting roles below long-term potential.
- More risk of drifting into generic software/web work.
- Less exposure to high-value US/EU opportunities.

30.3.11 Conservative red lines

Reject if:

- below USD 1.5k/month unless survival requires it;
- role is unrelated and consumes full-time capacity;
- no written contract;
- no portfolio value;
- broad non-compete;
- payment risk high.

30.3.12 Conservative success definition

Success after 90 days:

Bloque de código: text
 A stable role or active pipeline around USD 1.5k–3k/month that preserves the Unity / real-time 3D / technical visualization
 ↳ direction.

30.4 4. Balanced strategy

30.4.1 Definition

The balanced strategy combines realistic application volume, strong portfolio positioning, international contractor targeting, and risk control.

It aims to reach the USD 3k/month target within 3–6 months while building toward higher-value roles over 12–24 months.

30.4.2 Best use case

Use balanced mode if:

- TwinSight case study is publishable;
- LinkedIn/CV/GitHub are coherent;
- English interviews are viable;
- contractor setup can be completed;
- there is no immediate survival-level cashflow pressure;
- the goal is better compensation without reckless risk.

30.4.3 Primary target roles

Priority	Role
1	Real-Time 3D / Interactive 3D Developer
2	Unity WebGL Developer
3	Technical Visualization Developer
4	Digital Twin / Simulation Visualization Developer
5	Unity Technical Artist
6	XR / AR / VR Developer
7	Tools / Python Automation Developer

30.4.4 Target markets

Market	Priority
LATAM nearshore	High
US remote contractor	High
Global remote studios	High
WebGL / creative tech agencies	High
Technical visualization companies	High
Product visualization/configurator companies	High
EU remote-friendly companies	Medium
Germany/Portugal	Medium future
Colombia local	Fallback
AAA games	Selective only

30.4.5 Salary target

Timeframe	Target
First 3 months	USD 2k–3k/month
3–6 months	USD 3k–4.5k/month
12–24 months	USD 4.5k–6k+/month

30.4.6 Application volume

Action	Weekly target
Applications	15–30
A-priority applications	5–10
Outreach	10–25
Recruiter contacts	10–15
Follow-ups	10–20
Portfolio blocks	2–4
Skill blocks	2–4

30.4.7 Portfolio standard

Required:

- polished CV;
- LinkedIn fully updated;
- TwinSight case study public or shareable;
- GitHub cleaned;
- TwinSight README usable;
- 8–12 screenshots;
- 60–90 second demo video or strong draft;
- ARA positioned correctly;
- Blender portrait included only if polished.

30.4.8 Contractor stance

Balanced mode accepts contractor work if:

- compensation is above minimum;
- payment terms are clear;
- scope is defined;

- IP is limited to paid work;
- remote from Colombia is acknowledged;
- no illegal work authorization assumptions;
- gross-to-net estimate works.

30.4.9 Advantages

- Best balance of upside and realism.
- Keeps the profile focused.
- Builds market feedback quickly.
- Increases chance of USD 3k/month.
- Preserves 12–24 month growth path.

30.4.10 Disadvantages

- Requires consistent execution.
- Requires more outreach and rejection tolerance.
- Requires contractor admin readiness.
- Requires stronger interview preparation.

30.4.11 Balanced red lines

Reject if:

- role is full-stack-only with no 3D/Unity relevance;
- compensation below USD 1.5k/month;
- unpaid production test;
- vague contractor terms;
- company cannot clarify remote/legal structure;
- role consumes full-time capacity while damaging positioning.

30.4.12 Balanced success definition

Success after 90 days:

Bloque de código: text
A live pipeline of interviews and/or offers in Unity, real-time 3D, technical visualization or WebGL roles, with realistic
↔ compensation moving toward USD 3k/month.

30.5 5. Aggressive strategy

30.5.1 Definition

The aggressive strategy prioritizes high-upside international opportunities, direct outreach, speculative applications, contractor positioning, and higher salary negotiation.

It assumes more rejection, more competition, and higher contract risk.

30.5.2 Best use case

Use aggressive mode if:

- TwinSight case study is strong;

- demo video is polished;
- GitHub and LinkedIn are clean;
- interview answers are rehearsed;
- contractor/tax setup is ready;
- cashflow pressure is manageable;
- the candidate can tolerate rejection and uncertainty;
- target is USD 4k–6k+/month as soon as possible.

30.5.3 Primary target roles

Priority	Role
1	Technical Visualization / Digital Twin Developer
2	Unity WebGL / Interactive 3D Developer
3	Real-Time 3D Developer
4	Unity Technical Artist — optimization/tools/runtime
5	XR / Simulation Developer
6	Product Visualization / Configurator Developer
7	Python Automation / AI Tools Developer, secondary

30.5.4 Target markets

Market	Priority
US remote contractor	Very high
Global remote-first companies	Very high
Technical visualization firms	Very high
Digital twin / simulation companies	Very high
XR/AR/VR companies	High
Creative technology agencies	High
Product visualization/configurator	High
EU companies open to contractors	Medium
Germany/Portugal future targets	Medium
LATAM nearshore	Medium
Colombia local	Low

30.5.5 Salary target

Timeframe	Target
First 3 months	USD 3k–5k/month
3–6 months	USD 4k–6k/month
12–24 months	USD 6k+/month

30.5.6 Application volume

Action	Weekly target
Applications	30–50
A-priority applications	10–20
Direct outreach	25–50
Recruiter contacts	15–30
Follow-ups	20–40
Speculative applications	10–20
Portfolio/content posts	2–3

Action	Weekly target
Technical interview practice	4–6 blocks

30.5.7 Portfolio standard

Required before aggressive mode:

- polished TwinSight case study;
- polished demo video;
- strong GitHub README;
- portfolio MVP live;
- LinkedIn Featured configured;
- CV variants ready;
- interview answer bank rehearsed;
- offer scorecard ready;
- contractor setup in progress;
- top 30 companies verified.

30.5.8 Contractor stance

Aggressive mode actively targets:

- higher monthly retainers;
- USD hourly rates;
- specialized project contracts;
- US/EU clients;
- remote-first companies;
- studios needing Unity/WebGL/interactive 3D;
- technical visualization contracts.

It also requires stricter risk control.

30.5.9 Advantages

- Highest income upside.
- More exposure to strong companies.
- Faster market feedback.
- Better chance of reaching USD 3k–6k/month.
- Stronger professional positioning if successful.

30.5.10 Disadvantages

- More rejection.
- More unpaid-test attempts.
- More legal/contract complexity.
- More senior-role mismatch.
- Higher emotional load.
- Greater chance of wasting effort if portfolio is not ready.

30.5.11 Aggressive red lines

Reject or pause if:

- no contract;
- no clear payment terms;
- broad IP assignment;
- unpaid test over reasonable scope;
- equity-only;
- role requires false work authorization;
- company refuses to define remote structure;
- role demands senior claims you cannot defend;
- role is outside core positioning.

30.5.12 Aggressive success definition

Success after 90 days:

Bloque de código: text

Multiple active interviews or one high-quality offer in the USD 3k-6k/month range, or strong evidence that the market responds ↪ to the Unity/WebGL technical visualization positioning.

30.6 6. Scenario comparison

30.6.1 Strategic comparison

Factor	Conservative	Balanced	Aggressive
Primary goal	Stability	Strong fit + income growth	Maximum upside
Best first offer	USD 1.5k–3k	USD 2k–4.5k	USD 3k–6k
Application quality	Medium	High	High
Application volume	Low-medium	Medium	High
Outreach intensity	Low	Medium	High
Portfolio requirement	Usable	Strong	Polished
Risk tolerance	Low	Medium	High
Contractor complexity	Low	Medium	High
Rejection volume	Low	Medium	High
Profile dilution risk	Medium	Low	Medium
Burnout risk	Low	Medium	High

30.6.2 Market focus comparison

Market	Conservative	Balanced	Aggressive
Colombia	Medium	Fallback	Low
LATAM nearshore	High	High	Medium
US remote	Medium	High	Very high
EU remote	Low-medium	Medium	Medium-high
Germany	Future	Future selective	Future + selective
Portugal	Future	Future selective	Future + selective
Global remote agencies	Medium	High	Very high
AAA games	Low	Selective	Selective
Digital twin	Medium	High	Very high
Product visualization	Medium	High	High

30.7 7. Scenario triggers

30.7.1 Switch to conservative if

- portfolio is not ready after 30 days;
- cashflow pressure increases;
- interview rejection shows strong technical gaps;
- contractor setup is not legally/admin ready;
- thesis defense demands attention;
- mental load becomes unsustainable;
- application volume collapses.

30.7.2 Stay balanced if

- applications are being sent consistently;
- some recruiter responses appear;
- portfolio is improving weekly;
- technical gaps are manageable;
- salary conversations are near target;
- rejection is normal but not total silence.

30.7.3 Switch to aggressive if

- TwinSight gets strong feedback;
 - portfolio/demo is polished;
 - recruiter response rate is above 6–10%;
 - interviews are happening;
 - contractor setup is clear;
 - USD 3k+ conversations are plausible;
 - top A-priority companies show fit.
-

30.8 8. Hybrid recommended strategy

30.8.1 Best practical model

The optimal path is not purely conservative, balanced or aggressive.

Recommended hybrid:

```
Bloque de codigo: text
Balanced strategy as default.
Aggressive targeting for A-priority companies.
Conservative contract-risk controls.
```

30.8.2 What this means

Use **balanced volume**:

```
Bloque de codigo: text
15–30 applications/week
10–25 outreach messages/week
```

Use **aggressive targeting** for:

- technical visualization companies;
- Unity WebGL roles;
- digital twin visualization;
- simulation interfaces;
- product configurators;
- XR-adjacent Unity roles;
- global remote studios.

Use **conservative rules** for:

- contract review;
- payment terms;
- legal status;
- IP ownership;
- unpaid tests;
- relocation;
- contractor setup.

30.8.3 Hybrid operating statement

Bloque de código: text
 Apply like an ambitious international contractor.
 Negotiate like a risk-aware professional.
 Position like a focused Unity technical visualization specialist.

30.9 9. Scenario-specific weekly plans

30.9.1 Conservative week

Day	Action
Monday	3–5 applications
Tuesday	portfolio/GitHub improvement
Wednesday	3–5 applications
Thursday	5 outreach/follow-ups
Friday	interview prep
Saturday	skill sprint
Sunday	review

30.9.2 Balanced week

Day	Action
Monday	5–8 applications
Tuesday	10 outreach messages
Wednesday	5–8 applications + follow-ups
Thursday	recruiter contacts + interview prep
Friday	portfolio/GitHub improvement
Saturday	skill sprint + demo polish
Sunday	review + next targets

30.9.3 Aggressive week

Day	Action
Monday	10 applications + 10 outreach
Tuesday	10 recruiter/hiring manager contacts
Wednesday	10 applications + follow-ups
Thursday	speculative outreach + portfolio posts
Friday	interviews/tests + technical prep
Saturday	top-company verification + skill sprint
Sunday	metrics review + next A-target batch

30.10 10. Scenario-specific portfolio requirements

30.10.1 Conservative portfolio

Minimum:

- CV ready;
- LinkedIn coherent;
- TwinSight case study draft;
- GitHub not damaging;
- screenshots visible.

30.10.2 Balanced portfolio

Minimum:

- public portfolio MVP;
- TwinSight case study ready;
- demo video draft or published;
- GitHub cleaned;
- README complete;
- ARA controlled;
- CV variants ready.

30.10.3 Aggressive portfolio

Minimum:

- polished portfolio;
 - polished TwinSight demo;
 - strong case study;
 - GitHub technical docs;
 - screenshots and diagrams;
 - LinkedIn Featured optimized;
 - interview walkthrough ready;
 - top 30 companies verified.
-

30.11 11. Scenario-specific salary stance

30.11.1 Conservative salary stance

Use when:

- role is first relevant job;
- company is LATAM/local;
- project fit is strong but budget moderate.

Language:

```
Bloque de codigo: text
For remote contractor work, I'm open to discussing compensation based on scope, expected availability and contract structure.
↩ My priority is a role with strong fit in Unity, real-time 3D or technical visualization.
```

Do not anchor too high if the role is clearly lower-budget.

30.11.2 Balanced salary stance

Use by default.

Language:

```
Bloque de codigo: text
For remote contractor work, depending on scope and expected availability, I'm generally targeting the USD 2k-4.5k/month range
↩ or an equivalent hourly rate.
```

Adjust per role.

30.11.3 Aggressive salary stance

Use only for strong-fit US/EU/global roles.

Language:

```
Bloque de codigo: text
For a full-time remote contractor arrangement involving Unity, WebGL, technical visualization and real-time 3D systems, I
↩ would generally evaluate compensation in the USD 4k-6k/month range depending on scope, ownership and contract structure.
```

Do not use aggressive anchors for low-budget LATAM roles.

30.12 12. Scenario-specific red flags

30.12.1 Conservative red flags

- low pay plus no portfolio value;
- local job consuming full-time capacity;
- unclear contract;
- unrelated responsibilities.

30.12.2 Balanced red flags

- role slowly becoming generic full-stack;
- technical visualization not actually part of the work;
- compensation below USD 1.5k;
- no contractor clarity;
- excessive unpaid tests.

30.12.3 Aggressive red flags

- overtargeting senior roles;
 - too much rejection without feedback;
 - unpaid production tasks;
 - high salary but bad contract;
 - US/EU authorization blockers;
 - burnout from high volume.
-

30.13 13. Decision tree

30.13.1 Start here

Bloque de código: text
Is the public portfolio ready enough to explain TwinSight clearly?

If no:

Bloque de código: text
Use conservative mode for 1–2 weeks.
Finish assets.

If yes:

Bloque de código: text
Use balanced mode.

Next question:

Bloque de código: text
Are recruiters/hiring managers responding after 50+ applications/outreach actions?

If no:

Bloque de código: text
Fix positioning, target list or portfolio top section.
Do not immediately switch roles.

If yes:

Bloque de código: text
Increase A-priority targeting.

Next question:

Bloque de código: text
Are salary conversations reaching USD 3k/month or higher?

If no:

Bloque de código: text
Improve target market and company tier.
Reduce low-value local applications.

If yes:

Bloque de codigo: text
Introduce aggressive mode for selected roles.

Next question:

Bloque de codigo: text
Are offers safe and strategically aligned?

If no:

Bloque de codigo: text
Use offer scorecard and renegotiate/reject.

If yes:

Bloque de codigo: text
Accept or negotiate upward.

30.14 14. What each scenario must not do

30.14.1 Conservative must not

- become passive;
- over-focus on local low-pay roles;
- delay applications forever;
- accept unrelated work as main strategy;
- hide from international interviews.

30.14.2 Balanced must not

- try to optimize every asset endlessly;
- apply to too many unrelated roles;
- ignore contractor/legal setup;
- treat USD 3k as impossible;
- delay outreach.

30.14.3 Aggressive must not

- overclaim seniority;
 - ignore red flags;
 - accept risky contracts for higher gross;
 - pretend current EU authorization exists;
 - burn out through excessive volume;
 - treat rejection as personal failure.
-

30.15 15. Recommended 90-day scenario plan

30.15.1 Days 1–30

Use:

Bloque de código: text
Conservative setup + balanced preparation

Meaning:

- finish CV;
- update LinkedIn;
- publish TwinSight case study;
- clean GitHub;
- create demo;
- send initial applications;
- do not over-scale before assets exist.

30.15.2 Days 31–60

Use:

Bloque de código: text
Balanced execution

Meaning:

- 15–30 applications/week;
- 10–25 outreach/week;
- A/B prioritization;
- interview prep;
- recruiter conversations;
- weekly metrics.

30.15.3 Days 61–90

Use:

Bloque de código: text
Balanced + selective aggressive targeting

Meaning:

- double down on high-response role families;
- verify top A-priority companies;
- negotiate stronger;
- pursue USD 3k–6k roles selectively;
- reject weak or dilutive offers.

30.16 16. Scenario-specific KPIs

30.16.1 Conservative KPIs

Metric	Target
Applications/week	10–20
Outreach/week	5–10
Portfolio blocks/week	3
Interviews by day 90	1–3

Metric	Target
Offer target	USD 1.5k–3k

30.16.2 Balanced KPIs

Metric	Target
Applications/week	15–30
Outreach/week	10–25
Portfolio blocks/week	2–4
Interviews by day 90	3–6
Offer target	USD 2k–4.5k

30.16.3 Aggressive KPIs

Metric	Target
Applications/week	30–50
Outreach/week	25–50
Portfolio/content posts/week	2–3
Interviews by day 90	5–10
Offer target	USD 3k–6k+

30.17 17. Risk controls by scenario

30.17.1 Conservative controls

- avoid low-quality full-time traps;
- preserve time for portfolio;
- do not accept below minimum;
- do not drop Unity/3D direction.

30.17.2 Balanced controls

- maintain tracker weekly;
- measure response rates;
- validate contractor terms;
- refine based on data;
- avoid overcustomization.

30.17.3 Aggressive controls

- cap weekly hours;
- reject exploitative tests;
- keep legal claims accurate;
- maintain sleep/routine;
- review contract risk before accepting;
- use scorecard objectively.

30.18 18. Recommended scenario right now

30.18.1 Current best choice

Based on the current profile and document system, the best path is:

```
Bloque de codigo: text
Balanced default strategy.
```

But the first 1–2 weeks should behave like:

```
Bloque de codigo: text
Conservative setup sprint.
```

After the portfolio, GitHub and TwinSight case study are public, shift to:

```
Bloque de codigo: text
Balanced execution.
```

After evidence of market response, selectively apply:

```
Bloque de codigo: text
Aggressive A-priority targeting.
```

30.18.2 Practical sequence

```
Bloque de codigo: text
Week 1-2:
Conservative setup sprint

Week 3-8:
Balanced applications and outreach

Week 9-12:
Balanced execution + aggressive targeting for strongest role families
```

30.19 19. Final recommendation

Do not choose a scenario based on fear or motivation.

Choose based on readiness:

Readiness state	Scenario
Assets incomplete	Conservative setup
Assets usable	Balanced
Assets polished + interviews active	Aggressive selected targeting
Cashflow pressure high	Conservative + fallback
Strong responses	Balanced/aggressive
No responses after 50+ actions	Fix positioning
Repeated technical failures	Skill sprint
Good offer	Score objectively

30.19.1 Final operating rule

Bloque de código: text
Use the balanced strategy as the main roadmap.
Use conservative controls to avoid bad contracts.
Use aggressive tactics only where TwinSight creates a strong match.

This is the safest path to move from the current profile toward:

Bloque de código: text
USD 1.5k/month minimum → USD 3k/month realistic target → USD 6k/month 12–24 month upside.

30.20 20. Integration with the tracker

Add a field to weekly review:

Bloque de código: text
Current scenario:
Conservative / Balanced / Aggressive / Hybrid

Add another field:

Bloque de código: text
Reason for scenario:
Assets incomplete / market response / cashflow / strong pipeline / technical gap / offer stage

This prevents switching modes based on emotion.

30.21 21. Immediate action

The current mode should be:

Bloque de código: text
Hybrid:
Conservative setup + balanced execution preparation.

Immediate priority:

1. finalize CV;
2. update LinkedIn;
3. clean GitHub;
4. publish TwinSight case study;
5. prepare demo video;
6. start first A/B applications;
7. track results;
8. move into balanced execution.

No additional broad research is needed before this.

Chapter 31

28_competitor_profile_benchmark

Fuente: 28_competitor_profile_benchmark.md
Seccion: root
Estado: canonical
Rol: Modulo principal
Lineas originales: 889
Ultima modificacion: 2026-06-11 16:29
Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

The source data was collected from publicly visible search snippets and LinkedIn-result metadata. LinkedIn profile pages were not fully accessible without login, so this benchmark must be treated as a **surface-level positioning benchmark**, not as a complete profile audit.

Contenido completo depurado

31.0.1 Scope

This module integrates the result of:

Bloque de codigo: text
A1_linkedin_profiles_benchmark

The source data was collected from publicly visible search snippets and LinkedIn-result metadata. LinkedIn profile pages were not fully accessible without login, so this benchmark must be treated as a **surface-level positioning benchmark**, not as a complete profile audit.

31.0.2 Data limitations

Limitation	Impact
LinkedIn profile pages restricted	About, full experience, education and featured projects are often unavailable
Search snippets only	Data is incomplete and may omit important context
Most portfolio links not visible	Cannot judge project quality directly
Seniority often inferred from title	Confidence varies
Remote/contractor status rarely visible	Must not assume remote eligibility
Search result snippets may change	Recheck important profiles manually before copying patterns

31.0.3 Correct use

Use this module to extract:

- headline patterns;
- role title patterns;
- market vocabulary;
- seniority patterns;
- niche clusters;
- signals to copy carefully;
- signals to avoid;
- profile gaps in Alexander's current positioning.

Do **not** use it to claim direct equivalence with senior profiles.

31.1 1. Executive findings

31.1.1 Main finding

The public profile landscape confirms that Alexander's positioning is coherent if expressed as:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist
```

with strong supporting keywords:

```
Bloque de codigo: text
Unity · C# · Unity WebGL · Technical Visualization · CAD-to-Realtime · Blender · Interactive 3D · Simulation · XR · Digital
↳ Twin
```

The strongest comparable profiles tend to cluster around one of five families:

1. **Unity Technical Artist**
2. **Real-Time 3D / Interactive 3D Developer**
3. **Unity WebGL Developer**
4. **XR / Simulation / Digital Twin Developer**
5. **Unity Tools / Pipeline Developer**

31.1.2 Key implication

Alexander should not market himself as a generic Unity developer. The benchmark shows that more differentiated profiles combine Unity with a domain:

```
Bloque de codigo: text
Unity + Technical Art
Unity + WebGL
Unity + XR
Unity + Simulation
Unity + Digital Twin
Unity + Tools
Unity + Optimization
```

TwinSight gives a stronger niche than generic gameplay.

31.1.3 Strongest benchmark cluster for Alexander

Cluster	Fit
Real-Time 3D Developer	Very high
Unity WebGL Developer	Very high
Technical Visualization / Digital Twin Developer	Very high
Unity Technical Artist	High
XR / Simulation Developer	Medium-high
Tools / Pipeline Developer	Medium
Pure game Technical Artist	Medium
Pure 3D Generalist	Low unless technical

31.2 2. Headline pattern extraction

31.2.1 Strong headline formulas found

Formula	Example pattern	Usefulness
Role + engine + domain	Real-Time 3D Developer Unity & Unreal	High
Role + specialization	Unity Technical Artist VFX & Shader Specialist	High
Role + platform	Unity WebGL Developer	High
Role + industry	Unity Developer Digital Twin Applications	High
Role + stack	XR Developer Unity · VR · AR	Medium-high
Seniority + role	Senior Unity Developer / Technical Artist	Use only if true
Multi-engine profile	Unreal & Unity Technical Artist	Useful only if both defensible
Freelance + role	Freelance Real-Time 3D Developer	Useful for contractor route

31.2.2 Recommended adaptation for Alexander

Best formula:

Bloque de código: text

Real-Time 3D Developer / Unity Technical Artist | Unity WebGL, C#, CAD-to-Realtime Optimization, Technical Visualization

Alternative for specific markets:

Bloque de código: text

Unity WebGL / Interactive 3D Developer | Technical Visualization, C#, Blender, CAD-to-Realtime

Alternative for digital twin / simulation:

Bloque de código: text

Technical Visualization Developer | Unity WebGL, CAD-to-Realtime, Interactive 3D

31.3 3. Role family benchmark

31.3.1 Unity Technical Artist

Observed patterns

Profiles in this cluster commonly emphasize:

- Unity;

- shaders;
- VFX;
- optimization;
- technical art;
- pipeline;
- performance;
- VR/AR;
- game studio or mobile game experience.

Relevant examples from public snippets

Name	Region	Title / Signal	Seniority	Relevance
Rachel Tal	Israel	Unity Technical Artist / FX Artist	Unknown	Unity + FX/TA positioning
Aldo Vazquez	Mexico	Unity Technical Artist, VFX, shaders, optimization	Senior	LATAM + strong TA niche
Omri Ovadia	Israel	Unity Technical Artist at Beach Bum	Unknown	Game studio TA
Scarlet Fairweather	UK	Unity Technical Artist at Scopely	Unknown	Studio TA benchmark
Daniel Peri	Israel	Unity TA with VR optimization	Unknown	TA + VR optimization
Aakash Upadhyay	India	3D Generalist & Unity Technical Artist	Unknown	Hybrid 3D + TA
Wiktorja Fruba	Poland	Unity Technical Artist at ASTEK Polska	Unknown	EU TA role
J Tuason	USA	Senior Technical Artist	Senior	Aspirational only
Aela W.	Unknown	Unity Technical Artist, non-traditional background	Unknown	Non-traditional path signal

Takeaways for Alexander

Copy:

- Unity + technical art combination;
- optimization vocabulary;
- shader/visual mode vocabulary if defensible;
- pipeline vocabulary;
- runtime + art bridge.

Do not copy:

- seniority if not defensible;
- heavy VFX/shader specialization unless portfolio supports it;
- AAA framing without shipped game experience;
- pure game art emphasis.

Recommended stance

Alexander can target Unity Technical Artist roles, but should qualify the specialization as:

Bloque de codigo: text
runtime systems • technical visualization • optimization • technical UI • CAD-to-realtime

not:

Bloque de codigo: text
senior shader specialist • AAA game TA • VFX specialist

31.3.2 Real-Time 3D / Interactive 3D Developer

Observed patterns

Profiles in this cluster often combine:

- Unity;
- Unreal;
- optimization;
- VR/mobile/web;
- interactive systems;
- simulation;
- full-stack or application design.

Relevant examples

Name	Region	Title / Signal	Seniority	Relevance
Matthew Cresswell	South Africa	Real-Time 3D Developer, Unity/Unreal, optimization	Unknown	Very relevant title pattern
Matthew Barrett	Australia	Senior Unity Developer / Technical Artist	Senior	Aspirational
Arunkumar M	India	Senior Unity 3D Developer, Automotive HMI	Senior	Real-time + industrial/HMI
Antony Allwin Joshuva	India	Real-Time 3D Developer, VR simulations	Unknown	Interactive environments
Abhilash Dandhare	Germany	Real-Time 3D Developer, Unity/Unreal, XR	Unknown	EU + XR route
Rafael Boceanu	UK	Interactive Real-Time 3D Developer	Unknown	Exact title pattern
Nicolas Augustin	Martinique	Freelance Real-Time 3D Developer	Unknown	Contractor route

Takeaways for Alexander

Copy:

- “Real-Time 3D Developer” as a primary market-readable title;
- optimization language;
- web/mobile/VR platform language;
- interactive systems language;
- freelance/contractor wording if used carefully.

Avoid:

- claiming Unreal unless there is real evidence;
- claiming full-stack systems as primary unless applying to hybrid roles;
- overclaiming automotive/HMI/simulation unless framed as target domain, not current experience.

Recommended stance

This is one of Alexander’s strongest role families.

Best headline adaptation:

```
Bloque de codigo: text
Real-Time 3D Developer | Unity WebGL, C#, Interactive Technical Visualization
```

31.3.3 Unity WebGL / Web 3D Developer

Observed patterns

This cluster is smaller but highly relevant.

Common signals:

- Unity WebGL;
- browser delivery;
- AR/WebGL experiences;
- casual games;
- web + Unity hybrid;
- React/web background in some profiles.

Relevant examples

Name	Region	Title / Signal	Seniority	Relevance
Toby Jackson	UK	Unity WebGL Developer at Imvizar	Unknown	Direct WebGL + AR
Andrei Liutynski	Poland	Unity WebGL Developer, self-employed	Unknown	Self-employed WebGL
Muhammad Ali Qureshi	UAE	Website / Unity WebGL Developer	Unknown	Web + Unity bridge
Pratik Meshram	India	Ex-Unity WebGL Developer	Senior	WebGL career path
Nikita Rasputin	Russia	Unity Teacher / WebGL Developer	Unknown	Teaching + WebGL
Miroslav Popugajev	Lithuania	ML Engineer / Unity-WebGL Developer	Unknown	Hybrid AI/web/3D
Arthur Novaes	Brazil	Unity WebGL / Full-Stack Game Engineer	Unknown	LATAM + WebGL

Takeaways for Alexander

Copy:

- explicit “Unity WebGL” keyword;
- browser-accessible 3D positioning;
- web + Unity bridge;
- self-employed/contractor evidence if true;
- interactive 3D wording.

Avoid:

- becoming a generic web developer;
- overemphasizing React/full-stack unless relevant;
- casual game WebGL positioning if targeting technical visualization.

Recommended stance

This is one of the highest-fit clusters because TwinSight is already Unity WebGL.

Best adaptation:

Bloque de código: text
Unity WebGL / Interactive 3D Developer | Browser-Based Technical Visualization

31.3.4 Digital Twin / Simulation / XR Developer

Observed patterns

Profiles in this cluster often combine:

- Unity;
- simulation;
- digital twin;
- XR/VR/AR;
- industrial metaverse;
- IoT;
- automotive/HMI;
- B2B projects.

Relevant examples

Name	Region	Title / Signal	Seniority	Relevance
William Wood	USA	Senior Unity Developer / Simulation Engineer	Senior	Real-time + digital twin
Vaibhav Rathod	India	Unity, Omniverse, Digital Twin	Unknown	Digital twin + Unity
ALI Haider	Pakistan	Unity Developer, Digital Twins, Tool Chain	Unknown	Game simulation + digital twin
Juan Puppín	Portugal	Unity Developer / Digital Twin Applications Specialist	Unknown	Portugal/EU benchmark
Logesh S	India	XR Developer / IoT / Digital Twins	Unknown	XR + IoT
Muhammad Faizan Khan	Pakistan	Technical Lead XR/3D, digital twin platform	Senior	SaaS digital twin
Kavin Chandar R. S.	India	Industrial Metaverse / XR / Digital Twin	Unknown	Industrial visualization
Aman Chaudhary	India	Unity XR / VR Simulation Developer	Unknown	XR simulation
Bhoopendra Mishra	India	XR Software Engineer / Real-Time 3D Developer	Unknown	XR + real-time
Volodymyr Kalinichenko	Germany	AR/VR & Real-Time 3D Developer Tools	Senior	Developer tools + XR

Takeaways for Alexander

Copy:

- digital twin / technical visualization wording;
- simulation visualization language;
- XR-adjacent language;
- industrial metaverse only if used carefully;
- real-time systems vocabulary.

Avoid:

- claiming actual digital twin production experience;
- claiming IoT integration unless added;
- claiming XR headset production experience unless built;
- using “industrial metaverse” as hype without evidence.

Recommended stance

Alexander can target this market as:

```
Bloque de codigo: text
technical visualization / digital twin visualization candidate
```

not as:

```
Bloque de codigo: text
senior digital twin engineer
```

TwinSight should be framed as a **digital-twin-adjacent technical visualization prototype**.

31.3.5 Colombia / LATAM Unity profiles

Observed patterns

Colombian and LATAM profiles in snippets mostly emphasize:

- Unity Developer;
- gameplay programmer;
- mobile/WebGL/console;
- optimization;
- open to remote;
- multimedia engineering.

Relevant examples

Name	Region	Title / Signal	Seniority	Relevance
Sergio U.	Colombia	Unity Developer	Unknown	Local benchmark
Diego Grajales	Colombia	Multimedia Engineer / Unity Developer	Mid/Senior signal	Very comparable education label
Jorge Bueno	Colombia	Senior Unity Developer, open to remote	Senior	Remote Colombia benchmark
Danny López	Colombia	Gameplay Programmer, Unity/C#	Unknown	Local game dev benchmark
Arthur Novaes	Brazil	Unity WebGL / Full-Stack Game Engineer	Unknown	LATAM WebGL benchmark
Aldo Vazquez	Mexico	Unity Technical Artist / Shader Specialist	Senior	LATAM TA aspirational

Takeaways for Alexander

Copy:

- Colombia-based remote openness;
- Unity + C# clarity;
- optimization vocabulary;
- multimedia engineering as supporting context;
- remote contractor orientation.

Avoid:

- competing only as gameplay programmer;
- presenting as pure game developer;
- hiding the technical visualization niche.

Recommended stance

Alexander’s profile can stand out in LATAM by not being generic:

Bloque de codigo: text
Unity WebGL + technical visualization + CAD-to-realtime

is more differentiated than:

Bloque de codigo: text
Unity Developer / Gameplay Programmer

31.4 4. Full benchmark table

31.4.1 Identity and role table

#	Name	Region	Current title	Company	Role family	Seniority	Confidence
1	Rachel Tal	Israel	Unity Technical Artist / FX Artist	not found	Unity Technical Artist	unknown	medium
2	Aldo Vazquez	Mexico	Unity Technical Artist	not found	Unity Technical Artist	senior	medium
3	Omri Ovadia	Israel	Unity Technical Artist	Beach Bum	Unity Technical Artist	unknown	medium
4	Travis Over	Canada	Unreal & Unity Technical Artist	PIXOMONDO	Technical Artist	senior	medium
5	Scarlet Fairweather	UK	Unity Technical Artist	Scopely	Unity Technical Artist	unknown	medium
6	Daniel Peri	Israel	Unity Technical Artist	not found	Unity Technical Artist	unknown	medium
7	Aakash Upadhyay	India	3D Generalist & Unity Technical Artist	not found	Unity Technical Artist	unknown	medium
8	Wiktorja Fruba	Poland	Unity Technical Artist	ASTEK Polska	Unity Technical Artist	unknown	medium
9	J Tuason	USA	Senior Technical Artist	Santa Monica Studio	Technical Artist	senior	medium
10	Aela W.	unknown	Unity Technical Artist	not found	Unity Technical Artist	unknown	medium
11	Matthew Cresswell	South Africa	Real-Time 3D Developer	not found	Real-Time 3D	unknown	medium
12	Matthew Barrett	Australia	Senior Unity Developer / Technical Artist	Light & Wonder	Unity / TA	senior	medium
13	Arunkumar M	India	Senior Unity 3D Developer / Automotive HMI Engineer	not found	Real-Time 3D / Simulation	senior	medium
14	Aman Chaudhary	India	Unity XR / VR Simulation Developer	not found	XR / Simulation	unknown	medium
15	Antony Allwin Joshua	India	Real-Time 3D Developer	not found	Real-Time 3D	unknown	medium
16	Abhilash Dandhare	Germany	Real-Time 3D Developer	not found	Real-Time 3D / XR	unknown	medium
17	Rafael Boceanu	UK	Interactive Real-Time 3D Developer	INFINITY27	Real-Time 3D	unknown	medium
18	Volodymyr Kalinichenko	Germany	Senior Software Quality Engineer	Apple	AR/VR Tools	senior	medium
19	Bhoopendra Mishra	India	XR Software Engineer / Real-Time 3D Developer	not found	XR / Real-Time 3D	unknown	medium
20	Nicolas Augustin	Martinique	Freelance Real-Time 3D Developer	DevOxyz	Real-Time 3D	unknown	medium
21	Toby Jackson	UK	Unity WebGL Developer	Imvizar	Unity WebGL	unknown	medium
22	Andrei Liutynski	Poland	Unity WebGL Developer	Self-employed	Unity WebGL	unknown	medium
23	Muhammad Ali Qureshi	UAE	Website / Unity WebGL Developer	QTech	Unity WebGL / Web	unknown	medium
24	Pratik Meshram	India	Senior Unity Game Developer	Capital Rush	Unity WebGL	senior	medium
25	Nikita Rasputin	Russia	Unity Developer / Teacher	not found	Unity WebGL	unknown	medium
26	Miroslav Popugajev	Lithuania	Machine Learning Engineer / Unity-WebGL Developer	not found	WebGL / AI hybrid	unknown	medium
27	Arthur Novaes	Brazil	Unity Game Developer	Appixar	Unity WebGL	unknown	medium
28	William Wood	USA	Senior Unity Developer / Simulation Engineer	not found	Simulation / Digital Twin	senior	medium
29	Vaibhav Rathod	India	Unity / Omniverse / Digital Twin Developer	not found	Digital Twin	unknown	medium
30	ALI Haider	Pakistan	Unity Developer / Digital Twins / Tool Chain	The Location Lab	Digital Twin	unknown	medium
31	Juan Puppín	Portugal	Unity Developer / Digital Twin Applications Specialist	not found	Digital Twin	unknown	medium
32	Logesh S	India	XR Developer / IoT / Digital Twins Developer	not found	XR / Digital Twin	unknown	medium
33	Muhammad Faizan Khan	Pakistan	Technical Lead XR/3D	not found	XR / Digital Twin	senior	medium
34	Kavin Chandar R. S.	India	Industrial Metaverse / XR Developer	not found	XR / Digital Twin	unknown	medium
35	Stanly Chen	UK	Unity Developer	not found	Unity Developer	unknown	medium
36	Sergio U.	Colombia	Unity Developer	not found	Unity Developer	unknown	medium
37	Diego Grajales	Colombia	Multimedia Engineer / Unity Developer	not found	Unity Developer	mid/senior signal	medium
38	Jorge Bueno	Colombia	Senior Unity Developer	not found	Unity Developer	senior	medium
39	Danny López	Colombia	Gameplay Programmer	not found	Unity Developer	unknown	medium

31.5 5. Profile signal table

Signal	Frequency in benchmark	Relevance
Unity	Very high	Must be front-loaded
C#	High	Must be explicit
WebGL	Medium	Differentiator
Technical Artist	High	Valid target title
Real-Time 3D	High	Strong primary title
XR / AR / VR	High	Adjacent target
Simulation	Medium	Strategic niche
Digital Twin	Medium	Strategic niche
Optimization	Medium-high	Core differentiator
Shader / VFX	Medium	Useful but do not overclaim
Tools / Pipeline	Medium	Secondary route
Blender	Medium	Strong support skill
Full-stack	Low-medium	Only secondary
Freelance / self-employed	Low but relevant	Contractor signal
Colombia/LATAM remote	Limited visible data	Needs separate recruiter/company search

31.6 6. What Alexander should copy

31.6.1 Copy these patterns

Pattern	Application
Role-first headline	Start with Real-Time 3D Developer / Unity Technical Artist
Engine + domain	Unity WebGL + Technical Visualization
Optimization language	CAD-to-realtime, asset optimization, WebGL constraints
Platform clarity	WebGL, browser-accessible, real-time
Technical art bridge	Blender + Unity + runtime systems
Simulation/digital twin adjacency	Use as target domain
Contractor clarity	Colombia-based remote contractor
Project proof	TwinSight as flagship, not just skill list
Non-traditional path normalization	Multimedia + electronics + technical project

31.6.2 Example adapted headline

Bloque de código: text
 Real-Time 3D Developer / Unity Technical Artist | Unity WebGL, C#, CAD-to-Realtime Optimization, Technical Visualization

31.6.3 Example adapted About opening

Bloque de código: text
 I build interactive real-time 3D systems for technical visualization using Unity, C#, WebGL and Blender. My flagship project, ↪ TwinSight X500, is a Unity WebGL drone assembly inspection prototype focused on component selection, exploded view, ↪ cross-section tools, visual modes and CAD-to-realtime optimization.

31.7 7. What Alexander should not copy

Pattern to avoid	Reason
Senior title	Not supported by formal experience
Principal / Lead language	Overclaim risk
AAA Technical Artist positioning	Portfolio does not yet prove it
Pure VFX/shader specialist	Not current strongest proof
Pure gameplay programmer	Dilutes TwinSight advantage
Generic Unity developer	Less differentiated
Full-stack headline	Secondary route only
AI engineer headline	ARA does not justify it as primary
Omniverse/digital twin production claims	Unless project evidence exists
XR specialist claim	Use XR-adjacent unless built headset projects
Industrial metaverse hype	Too broad/hype-heavy
Multi-engine claim with Unreal	Unless Unreal evidence exists

31.8 8. Positioning implications

31.8.1 Current validated position

The benchmark supports this public identity:

Bloque de código: text

Real-Time 3D Developer / Unity Technical Artist

31.8.2 Strongest differentiator

Bloque de código: text

Unity WebGL technical visualization with CAD-to-realtime optimization

This is more distinctive than:

Bloque de código: text

Unity Developer

or:

Bloque de código: text

3D Artist

31.8.3 Best immediate role cluster

Rank	Role cluster	Reason
1	Real-Time 3D / Interactive 3D Developer	Direct match with benchmark and TwinSight
2	Unity WebGL Developer	Direct project evidence
3	Technical Visualization / Digital Twin Visualization	Strong niche fit
4	Unity Technical Artist	Good fit if framed around runtime/optimization/tools
5	XR / Simulation Developer	Adjacent, needs more XR proof
6	Tools / Pipeline Developer	Secondary via ARA/Python
7	Full-stack	Fallback only

31.9 9. LinkedIn implications

31.9.1 Headline should include

Must include:

Bloque de codigo: text
Real-Time 3D Developer
Unity Technical Artist
Unity WebGL
C#
Technical Visualization
CAD-to-Realtime

Do not include too many secondary labels.

31.9.2 About section should emphasize

- 1. what he builds;
- 2. TwinSight;
- 3. Unity/C#/WebGL/Blender;
- 4. technical visualization;
- 5. CAD-to-realtime;
- 6. remote contractor availability.

31.9.3 Featured section should include

- 1. TwinSight case study;
- 2. TwinSight demo video;
- 3. GitHub repo;
- 4. ARA only after TwinSight;
- 5. Blender portrait only as support.

31.10 10. CV implications

31.10.1 CV title options

Target	CV title
General	Real-Time 3D Developer / Unity Technical Artist
WebGL	Unity WebGL / Interactive 3D Developer
Technical visualization	Technical Visualization Developer
TA	Unity Technical Artist
Digital twin	Real-Time 3D / Digital Twin Visualization Developer

31.10.2 Bullet emphasis

The benchmark supports emphasizing:

- Unity/C# implementation;
- real-time 3D;
- WebGL deployment;
- optimization;

- technical UI;
 - CAD-derived asset preparation;
 - simulation/digital twin adjacency;
 - Blender pipeline.
-

31.11 11. Portfolio implications

31.11.1 TwinSight should be shown as

Bloque de codigo: text
Unity WebGL technical visualization prototype

not just:

Bloque de codigo: text
drone viewer

or:

Bloque de codigo: text
thesis project

31.11.2 Portfolio tags should include

Bloque de codigo: text
Unity
C#
WebGL
Real-Time 3D
Technical Visualization
CAD-to-Realtime
Blender
Interactive 3D
Simulation
Digital Twin
Technical Art

31.11.3 Case study should be recruiter-readable

The benchmark suggests technical profiles often rely on keywords and titles. Alexander's advantage will come from stronger case-study packaging than average.

31.12 12. Gaps revealed by benchmark

31.12.1 Gap 1 — Public portfolio links

Most comparable profiles likely have visible projects, but the search snippets did not expose them.

Alexander needs direct visible links:

- portfolio;
- TwinSight demo;
- GitHub;
- ArtStation if used.

31.12.2 Gap 2 — Industry wording

The market uses short titles. Avoid academic phrasing.
Use:

```
Bloque de codigo: text
Unity WebGL technical visualization prototype
```

instead of:

```
Bloque de codigo: text
Design Science Research artifact for cognitive support in drone assembly documentation
```

Use the academic framing deeper in the case study, not in the headline.

31.12.3 Gap 3 — Specialized proof

For Technical Artist roles, current proof is strongest in:

- optimization;
- pipeline;
- visual modes;
- Unity runtime;
- UI;
- technical visualization.

It is weaker in:

- advanced shaders;
- Houdini;
- VFX;
- shipped game pipelines;
- studio production tools.

31.12.4 Gap 4 — XR

XR appears frequently in comparable profiles. Alexander can target XR-adjacent roles, but should add one small XR or spatial-interaction artifact if XR becomes a serious target.

31.12.5 Gap 5 — Digital twin

Digital twin appears as a strong adjacent market. Alexander can use TwinSight as a **digital-twin-adjacent technical visualization** project, but should not claim production digital twin experience.

31.13 13. Practical benchmark conclusions

31.13.1 Safe conclusions

Conclusion	Confidence
Real-Time 3D Developer is a valid title	High
Unity Technical Artist is a valid target family	High
Unity WebGL is a valuable differentiator	High
Digital twin / simulation is a strategic adjacent niche	Medium-high
XR appears frequently with Unity profiles	High

Conclusion	Confidence
Contractor/freelance language can be useful	Medium
Seniority must be handled carefully	High
Full-stack should remain secondary	High

31.13.2 Uncertain conclusions

Area	Why uncertain
Exact portfolio structures	LinkedIn snippets did not expose enough
Education patterns	Mostly not visible
Remote eligibility	Rarely visible
Contractor availability	Rarely visible
Current job market demand	Needs live job posting research
Compensation	Not available from profile snippets

31.14 14. Updates to existing roadmap

31.14.1 No change

The benchmark does not invalidate the current roadmap.

Keep:

```
Bloque de código: text
Real-Time 3D / Unity Technical Visualization as the central strategy.
```

31.14.2 Strengthen

Increase emphasis on:

- Unity WebGL;
- Real-Time 3D title;
- technical visualization;
- digital twin/simulation keywords;
- optimization;
- CAD-to-realtime.

31.14.3 Reduce

Reduce emphasis on:

- generic Unity Developer;
- pure gameplay;
- full-stack;
- AI tooling headline;
- pure Blender artist;
- academic-only language.

31.15 15. Recommended next benchmark tasks

The A1 task produced useful surface data but not enough depth on project presentation.
Next tasks should continue in this order:

```
Bloque de codigo: text
A2_public_cv_resume_benchmark
A3_github_profiles_benchmark
A4_artstation_portfolio_benchmark
A5_technical_visualization_case_studies
```

31.15.1 Tool recommendation for next task

```
Bloque de codigo: text
A2_public_cv_resume_benchmark
```

Tool	Use
Agent mode	Yes
Deep Research	No

Reason:
Public CV/resume benchmark needs web navigation and structured extraction. It does not require long-form research synthesis.

31.16 16. Prompt for next agent task

Use this next:

```
Bloque de codigo: text
Execute A2_public_cv_resume_benchmark.

Objective:
Collect public CV/resume examples for profiles similar to Alexander: Real-Time 3D Developer, Unity Technical Artist, Unity
↪ WebGL Developer, Technical Visualization Developer, XR Developer, Simulation Developer, Tools/Pipeline Developer.

Output structured tables only.

Fields:
- Name or anonymous label
- Resume/CV URL
- Role target
- Country/region if visible
- Seniority
- Resume structure
- Headline/title used
- Summary style
- Skills organization
- Project section structure
- Metrics used
- Education wording
- Portfolio/GitHub links if visible
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:
- Use only publicly available resumes/CVs.
- Do not invent missing data.
- If unavailable, write "not found".
- Prefer PDF, portfolio resume pages, GitHub resume READMEs, personal websites, and public resume examples from real
↪ candidates.
- Avoid generic resume advice articles unless they include a concrete resume sample.
- Do not write strategy.
- Do not write narrative recommendations.
- Tables only.
- Include source URLs.
```

Chapter 32

28B_public_cv_resume_benchmark.md

Fuente: 28B_public_cv_resume_benchmark.md

Seccion: root

Estado: canonical

Rol: Modulo final complementario

Lineas originales: 916

Ultima modificacion: 2026-06-11 17:14

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This module does **not** reopen the full labor-market strategy. It only audits and synthesizes the A2-Retry CV/resume benchmark.

Contenido completo depurado

32.0.1 0. Module status

```
Bloque de codigo: text
Module: 28B_public_cv_resume_benchmark
Task: A2_retry_public_cv_resume_benchmark
Generated: 2026-06-11
Project: Laboral / Alexander Woodcock Salomón
Purpose: Convert public CV/resume benchmark results into actionable updates for Alexander's CV strategy.
```

This module does **not** reopen the full labor-market strategy. It only audits and synthesizes the A2-Retry CV/resume benchmark.

Target positioning remains:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist
focused on Unity WebGL, CAD-to-realtime optimization, and technical visualization.
```

32.0.2 1. Quality audit of A2-Retry

1.1 Result quality

The A2-Retry result is usable.

It collected **17 examples**, which satisfies the requested range of **12–20** examples. The sources include real public PDFs, personal resume pages, GitHub resume content, recruiter blind CVs, and sample resumes.

However, the source quality is mixed. The strongest examples are personal websites and public PDFs. The weakest examples are generic sample resumes and one disordered Scribd document.

1.2 Source-quality classification

Class	Sources	Usefulness
Strong	11	High
Medium	5	Partial
Weak	1	Limited

1.3 Main limitation

Several examples are not exact matches for Alexander's niche. Many are closer to game development, XR, technical art, or 3D visualization than to **Unity WebGL technical visualization** specifically.

This is acceptable for A2 because the goal is CV structure benchmarking, not job-market validation.

32.0.3 2. Source inventory

2.1 Strong examples

Label	Source type	Role	Region	Confidence
ORLOVA-TA	Personal page	Technical Artist	Canada	High
MURZAK-GITHUB	GitHub README	Unity Developer	USA	High
DEMIRJIAN-UNITY	PDF	Unity Developer	USA	High
SEITZ-XR	PDF	XR Developer	USA	High
ARLEND-CAMB	Personal page	3D Visualization	USA	High
DELARA-STA	Personal page	Senior Technical Artist	USA	High
TORRES-TA	Personal page	Technical Artist	USA	High
GARCIA-TA	Personal page	Technical Artist	Uruguay	High
SZALAI-GAMEDEV	PDF	Unity Expert	Not found	High
MAYANK-UNITY	PDF	Senior Unity Developer	India	High
KENKEL-ENGINEER	PDF	Lead Software Engineer	USA	High

2.2 Medium examples

Label	Source type	Role	Region	Confidence
DEVSDATA-UNITY-SAMPLE	Blind CV	Unity Developer	Poland	Medium
HIREIT-UNITY	Resume database	Unity Developer	Not found	Medium
RESUMEKRAFT-UNITY-SAMPLE	Sample CV	Unity Developer	Not found	Medium
SPANN-GAMEDEV	PDF	Gameplay Programmer	USA	High
BEAUMONT-GAMEDEV	PDF	Game Developer	Canada	High

SPANN-GAMEDEV and BEAUMONT-GAMEDEV are real public PDFs, but they are less aligned with technical visualization than the strongest examples.

2.3 Weak example

Label	Source type	Role	Region	Confidence
KAISER-SIM	Scribd	3D Simulation Developer	Argentina	Low

KAISER-SIM is only useful as a negative formatting reference. The extracted content appears disordered and should not influence Alexander's CV structure.

32.0.4 3. Source URLs

Label	URL
ORLOVA-TA	https://tamaraorlova.com/resume
DEVSDATA-UNITY-SAMPLE	https://devsdata.com/resumes/unity3d/unity-developer-resume-sample/
HIREIT-UNITY	https://www.hireitpeople.com/resume-database/63-net-developers-architects-resumes/200613-unity-developer-resume
RESUMEKRAFT-UNITY-SAMPLE	https://resumekraft.com/unity-developer-cv-sample/
MURZAK-GITHUB	https://raw.githubusercontent.com/IvanMurzak/IvanMurzak/main/Resume.md
DEMIRJIAN-UNITY	https://fortynina.github.io/portfolio/assets/documents/NinaDemirjianResumeApril23.pdf
SEITZ-XR	https://www.josetteseitz.com/_files/ugd/d34654_c1180d17a7364855a9a3e18f51c49398.pdf
KAISER-SIM	https://www.scribd.com/document/951035904/Rodrigo-L-Kaiser-Eng-Resume-Text-ATS
ARLEND-CAMB	https://the3drenderguy.com/resume
DELARA-STA	https://www.samdelara.com/resume
TORRES-TA	https://www.carthtorres.com/resume
GARCIA-TA	https://iamtheroo.com/resume
SZALAI-GAMEDEV	https://peterszalai.com/resumes/peter-szalai-resume.pdf
SPANN-GAMEDEV	https://www.noahspann.com/resume.pdf
MAYANK-UNITY	https://agrmayank.com/cv/Resume%20-%20Mayank%20Raj.pdf
BEAUMONT-GAMEDEV	https://www.jalexbeaumont.com/resume.pdf
KENKEL-ENGINEER	https://nonathaj.github.io/Jon%20Kenkel%20Resume.pdf

32.0.5 4. Benchmark patterns

4.1 Strong CVs use a narrow headline

The strongest examples do not use broad titles like “creative developer” or “3D generalist” as the main positioning. They use direct role labels:

```
Bloque de código: text
Technical Artist
Senior Technical Artist
Senior Unity Developer
XR Developer
Unity Expert
3D Visualization Artist
```

Alexander should use a narrow headline, not a broad identity label.

Recommended primary headline:

```
Bloque de código: text
Real-Time 3D Developer / Unity Technical Artist
```

Alternative headline for Unity-heavy applications:

```
Bloque de código: text
Unity WebGL Developer focused on Technical Visualization
```

Alternative headline for 3D/technical art applications:

```
Bloque de código: text
Unity Technical Artist focused on CAD-to-Realtime Visualization
```


4.2 Strong CVs show proof before personality

The best examples prioritize evidence:

```
Bloque de codigo: text
role → project → tools → responsibility → result
```

Weak examples rely on generic traits:

```
Bloque de codigo: text
passionate
creative
hard-working
team player
problem solver
```

Alexander's CV should minimize soft claims and increase visible proof.

4.3 Strong CVs expose project scale

The best CVs include scale markers such as:

```
Bloque de codigo: text
5 shipped titles
40,000 downloads
150k+ downloads
500k users
100+ hours saved
2,500+ assets
33 ms frame-time budget
4 GB memory budget
```

Alexander should not invent metrics. He should add measured or verifiable project metrics once the TwinSight evaluation is complete.

Usable current metrics from the thesis context:

```
Bloque de codigo: text
28 canonical research parts
30 scene nodes
257 technical elements
Unity WebGL target
30 FPS target = 33 ms frame-time budget
CAD-to-realtime pipeline
mobile-first interface
```

Metrics still pending:

```
Bloque de codigo: text
actual FPS
actual frame time
load time
memory use
SUS score
NASA-TLX Raw score
participant count
```

4.4 Strong CVs organize technical skills by function

The better examples group skills by category. They do not dump every tool into one long list.

Recommended skill grouping for Alexander:

```
Bloque de codigo: text
Real-time development
3D production pipeline
Technical visualization
Web deployment
Automation and tooling
Version control
```

Bad pattern to avoid:

```
Bloque de codigo: text
Unity, Blender, Photoshop, Python, WebGL, Substance, Git, C#, AI, UX, UI, AR, VR, etc.
```

That kind of list dilutes positioning.

4.5 Strong CVs connect portfolio and resume

The strongest examples include links to one or more of:

```
Bloque de codigo: text
portfolio
GitHub
project pages
PDF resume
LinkedIn
live demos
store pages
itch.io
```

Alexander's CV should link directly to:

```
Bloque de codigo: text
portfolio site
TwinSight live demo
TwinSight GitHub repository
TwinSight case study
LinkedIn
ArtStation or 3D portfolio
```

The CV should not depend on the reader searching manually.

32.0.6 5. Recommended CV structure for Alexander

5.1 Primary structure

```
Bloque de codigo: text
1. Header
2. Role headline
3. 2-3 line professional summary
4. Core technical stack
5. Selected project: TwinSight X500
6. Additional projects
7. Technical skills
8. Education
9. Languages
10. Links
```

5.2 Why this structure

Alexander has limited formal industry experience in Unity/Technical Art. Therefore, the CV should be **project-led**, not employment-led.

The CV should make TwinSight function as the main proof of capability.

32.0.7 6. CV content recommendations

6.1 Header

Recommended fields:

```
Bloque de codigo: text
Alexander Woodcock Salomón
Real-Time 3D Developer / Unity Technical Artist
Medellín or Colombia
Email
LinkedIn
Portfolio
GitHub
```

Do not include:

```
Bloque de codigo: text
full street address
ID number
personal photo
marital status
birth date
unrelated social media
```

6.2 Summary

Recommended summary style:

```
Bloque de codigo: text
2-3 lines
specific role target
specific stack
specific proof project
no inflated seniority
```

Draft baseline:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist focused on interactive technical visualization, CAD-to-realtime optimization,
↔ and Unity WebGL deployment. Built TwinSight X500, a Unity WebGL prototype for inspecting and understanding the assembly of
↔ a Holybro X500 V2 drone through optimized 3D assets, technical UI, and mobile-first interaction design.
```

For a one-page CV, this can be shortened to:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, CAD-to-realtime optimization, and interactive
↔ technical visualization. Creator of TwinSight X500, a web-based 3D drone assembly inspection prototype.
```

6.3 Core technical stack

Recommended stack block:

```
Bloque de codigo: text
Unity | C# | WebGL | URP | UI Toolkit | Shader Graph
Blender | CAD-to-Realtime | UVs | Texture Baking | Optimization
Python | Git/GitHub | Technical Documentation | Usability Testing
```

This is stronger than a long mixed list because it mirrors the job family.

6.4 TwinSight X500 section

Recommended section title:

```
Bloque de codigo: text
Selected Technical Project – TwinSight X500
```

Recommended role line:

```
Bloque de codigo: text
Unity WebGL Technical Visualization Prototype | Thesis Project
```

Recommended bullets:

```
Bloque de codigo: text
- Designed and developed a Unity WebGL prototype for inspecting the assembly of a Holybro X500 V2 drone through interactive 3D
  ↳ visualization.
- Built a CAD-to-realtime pipeline using Blender, mesh cleanup, modular fasteners, UV preparation, texture baking, and Unity
  ↳ optimization.
- Structured the model into 28 canonical research parts, 30 scene nodes, and 257 technical elements for inspection and
  ↳ interaction.
- Designed a mobile-first interface with Unity UI Toolkit for part selection, isolation, transparency, and assembly
  ↳ understanding.
- Prepared technical evaluation around frame time, load time, memory use, usability, perceived workload, and qualitative
  ↳ feedback.
```

Do not claim final performance metrics until measured.

When measured, add one bullet like:

```
Bloque de codigo: text
- Achieved [X] FPS / [Y] ms frame time / [Z] MB memory use in WebGL testing on [device/browser].
```

6.5 Additional project sections

If the CV has space, add 1–2 more projects. These should reinforce the same positioning.

Useful project categories:

```
Bloque de codigo: text
Blender technical asset study
Unity interaction prototype
Python automation/tooling script
real-time shader or visualization experiment
WebGL deployment experiment
```

Avoid unrelated projects unless they prove technical art, Unity, or pipeline ability.

6.6 Education wording

Recommended wording:

```
Bloque de codigo: text
B.Eng. Multimedia Engineering – UNAD
Thesis: TwinSight X500, Unity WebGL technical visualization prototype for drone assembly inspection
```

If needed:

```
Bloque de codigo: text
Former Electronic Engineering coursework – Universidad Nacional de Colombia
```

The UNAL background should be secondary. It can support technical credibility, but it should not distract from the current target role.

6.7 Languages

Recommended wording:

```
Bloque de codigo: text
Spanish – Native
English – C1 working proficiency
German – A1 / beginner
Portuguese – planned strategic learning
```

For job applications, include only languages relevant to the role or region.

Do not overstate German or Portuguese.

32.0.8 7. What Alexander should adapt

7.1 From ORLOVA-TA

Adapt:

```
Bloque de codigo: text
summary + software stack + productions/projects
```

Do not copy:

```
Bloque de codigo: text
long narrative paragraphs
senior AAA framing
mentor awards framing
```

7.2 From MURZAK-GITHUB

Adapt:

```
Bloque de codigo: text
GitHub as resume
project metrics
tooling emphasis
visible links
```

Do not copy:

```
Bloque de codigo: text
image-heavy resume layout
informal phrasing
HTML-heavy structure
```

7.3 From DEMIRJIAN-UNITY

Adapt:

```
Bloque de codigo: text
project-first experience
museum/interactive media framing
public project metrics
```

Do not copy:

```
Bloque de codigo: text
lack of summary
minimal technical categorization
```

7.4 From SEITZ-XR

Adapt:

```
Bloque de codigo: text
hardware/software categories
training application framing
XR project detail
```

Do not copy:

```
Bloque de codigo: text
generic career objective
excess hardware listing
```

7.5 From DELARA-STA

Adapt:

```
Bloque de codigo: text
hours saved
asset counts
pipeline tooling
cross-functional work
```

Do not copy:

```
Bloque de codigo: text
senior title without equivalent experience
large-team claims
```

7.6 From GARCIA-TA

Adapt:

```
Bloque de codigo: text
optimization language
frame-time budget
memory budget
profiling tools
```

Do not copy:

```
Bloque de codigo: text
AAA console positioning
Unreal-heavy stack
```

7.7 From MAYANK-UNITY

Adapt:

```
Bloque de codigo: text
summary with metrics
AR/VR + Unity positioning
awards/projects section
```

Do not copy:

```
Bloque de codigo: text
overloaded project list
crowded layout
```

32.0.9 8. What Alexander should avoid

8.1 Avoid generic Unity Developer positioning

Weak headline:

```
Bloque de codigo: text
Unity Developer passionate about games and 3D
```

Stronger headline:

```
Bloque de codigo: text
Unity WebGL Developer focused on Technical Visualization
```

8.2 Avoid overclaiming seniority

Alexander should not present himself as:

```
Bloque de codigo: text
Senior Unity Developer
Senior Technical Artist
XR Lead
Pipeline TD
```

unless a specific application requires aspirational wording and the CV clearly frames the proof as project-based.

Safer positioning:

```
Bloque de codigo: text
Real-Time 3D Developer
Unity Technical Artist
Technical Visualization Developer
Unity WebGL Developer
```

8.3 Avoid generic sample-CV language

Do not use phrases like:

```
Bloque de codigo: text
excellent communication skills
highly motivated professional
proven track record
team player
creative problem solver
```

Replace with concrete evidence:

```
Bloque de codigo: text
Built a Unity WebGL prototype for drone assembly inspection.
Structured 257 technical elements for interactive visualization.
Prepared usability and performance evaluation using SUS, NASA-TLX Raw, FPS, frame time, load time, and memory.
```

8.4 Avoid software dumping

A long unsorted skills list weakens positioning.

Use categories instead:

```
Bloque de codigo: text
Realtime: Unity, C#, WebGL, URP, UI Toolkit
3D Pipeline: Blender, CAD cleanup, UVs, bake, optimization
Tooling: Python, Git/GitHub, technical documentation
```

32.0.10 9. Concrete updates for 17_cv_base_and_role_variants.md

9.1 Add a project-led CV variant

New variant title:

```
Bloque de codigo: text
Project-Led CV – Unity WebGL Technical Visualization
```

Purpose:

```
Bloque de codigo: text
For remote roles where portfolio proof matters more than formal employment history.
```

Use for:

```
Bloque de codigo: text
Unity WebGL Developer
Interactive 3D Developer
Technical Visualization Developer
Junior/Mid Technical Artist
XR/Simulation Developer
```

9.2 Replace broad summary with role-specific summary

Current summary should be rewritten toward:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, CAD-to-realtime optimization, and interactive
↔ technical visualization. Creator of TwinSight X500, a web-based 3D drone assembly inspection prototype built with Unity,
↔ C#, Blender, and a mobile-first interface architecture.
```

9.3 Add a “Selected Technical Project” block

Recommended insertion:

```
Bloque de codigo: text
SELECTED TECHNICAL PROJECT
TwinSight X500 – Unity WebGL Technical Visualization Prototype
```

Include 4–5 bullets:

```
Bloque de codigo: text
- Unity WebGL prototype for inspecting Holybro X500 V2 drone assembly.
- CAD-to-realtime pipeline using Blender, mesh cleanup, UV preparation, texture baking, and Unity optimization.
- 28 canonical parts, 30 scene nodes, and 257 technical elements prepared for interactive inspection.
- Mobile-first interaction design using Unity UI Toolkit.
- Evaluation framework covering FPS, frame time, load time, memory, SUS, NASA-TLX Raw, and Think-Aloud feedback.
```

9.4 Add metric placeholders

Until final testing is complete, use placeholders internally:

```
Bloque de codigo: text
[actual FPS]
[frame time]
[load time]
[memory]
[participant count]
[SUS]
[NASA-TLX Raw]
```

Do not publish placeholders in final applications.

9.5 Create three role-targeted CV headers

Variant	Header
Unity/WebGL	Unity WebGL Developer — Technical Visualization
Technical Art	Unity Technical Artist — CAD-to-Realtime Pipeline
Simulation/XR	Real-Time 3D Developer — XR/Simulation Visualization

9.6 Reorder sections for early applications

Recommended order:

```
Bloque de codigo: text
Header
Summary
Core Stack
Selected Technical Project
Additional Projects
Education
Languages
Links
```

Once Alexander has formal role experience, switch to:

```
Bloque de codigo: text
Header
Summary
Experience
Projects
Technical Skills
Education
Languages
Links
```

32.0.11 10. Immediate CV action list

Priority 1

```
Bloque de codigo: text
Rewrite CV headline and summary.
```

Priority 2

```
Bloque de codigo: text
Add TwinSight as the main project block.
```

Priority 3

```
Bloque de codigo: text
Group skills into functional categories.
```

Priority 4

```
Bloque de codigo: text
Add live links to portfolio, GitHub, demo, and case study.
```

Priority 5

Bloque de codigo: text

Insert measured performance/usability metrics after final testing.

32.0.12 11. Impact on roadmap

This module does not change the strategic direction.
It reinforces the existing strategy:

Bloque de codigo: text

remote-first
portfolio-led
Unity WebGL
technical visualization
CAD-to-realtime
contractor/B2B friendly

The main change is tactical:

Bloque de codigo: text

Alexander’s CV should become more project-led and more metric-driven.

32.0.13 12. Next task

Next task:

Bloque de codigo: text

A3_github_profiles_benchmark

Mode:

Bloque de codigo: text

Agent: Yes
Deep Research: No

Expected output:

Bloque de codigo: text

28C_github_profiles_benchmark.md

32.0.14 13. Prompt for A3

Bloque de codigo: text

Execute A3_github_profiles_benchmark.

Objective:
Collect public GitHub profile examples relevant to Alexander’s positioning:
Real-Time 3D Developer, Unity Technical Artist, Unity WebGL Developer, Technical Visualization Developer, XR Developer,
↔ Simulation Developer, Tools/Pipeline Developer, Python automation/tooling developer.

Find 20-35 GitHub profiles or repositories.

Prioritize:
- Unity projects with strong README;
- WebGL / Three.js / WebGPU projects;
- technical visualization projects;
- simulation or digital twin projects;
- Unity tools/editor tools;
- technical artist tools;
- Python automation / LLM tooling repos;

- portfolios using GitHub Pages.

Output structured tables only.

Fields:

- GitHub profile/repo label
- URL
- Source type: profile / repo / GitHub Pages / organization
- Role relevance
- Main technologies
- README structure
- Pinned repos if profile
- Project media used
- Badges/topics used
- Demo links
- Portfolio links
- License/disclaimer handling
- Documentation quality
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:

- Use only public GitHub data.
- Do not clone private repos.
- Do not invent unavailable data.
- If unavailable, write "not found".
- Tables only.
- No narrative strategy.
- Include source URLs.

Chapter 33

29_github_profiles_benchmark.md

Fuente: 28C_github_profiles_benchmark.md

Seccion: root

Estado: reference

Rol: Benchmark o complemento

Lineas originales: 737

Ultima modificacion: 2026-06-11 17:36

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This module does **not** reopen the base labor-market strategy.

Contenido completo depurado

33.0.1 0. Module status

Bloque de codigo: text
Module: 29_github_profiles_benchmark
Task: A3_github_profiles_benchmark
Generated: 2026-06-11
Project: Laboral / Alexander Woodcock Salomón
Purpose: Convert GitHub benchmark results into actionable updates for Alexander's public GitHub surface.

This module does **not** reopen the base labor-market strategy.

Target positioning remains:

Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist
focused on Unity WebGL, CAD-to-realtime optimization, technical visualization, and tooling.

33.0.2 1. Quality audit of A3

1.1 Result quality

A3 is usable.

It collected **21 GitHub examples**, which satisfies the requested range of **20–35** profiles/repositories.

The dataset is strongest for:

Bloque de codigo: text
Unity tooling

```

Unity technical art utilities
Unity WebGL deployment
XR samples
robotics / digital twin visualization
build automation
LLM-assisted Unity tooling

```

The dataset is weaker for:

```

Bloque de codigo: text
GitHub profile READMEs
Three.js / WebGL portfolios
general-purpose WebGL visualization
GitHub Pages portfolio examples

```

This is acceptable because Alexander's immediate public proof should be Unity/WebGL/tooling-led, not generic WebGL-led.

33.0.3 2. Source-quality classification

Class	Count	Usefulness
Strong	14	Directly adaptable
Medium	5	Partial pattern
Weak	2	Limited pattern

2.1 Strong examples

Label	Why strong
RAZLUTA-Profile	Strong pinned tooling surface
SPLATVFX	Clear experimental graphics README
UNITY-BUILDER	Strong tooling docs
REACT-UNITY-WEBGL	Strong WebGL package docs
UNITY-EDITOR-TOOLBOX	Deep editor-tool docs
UNITY-MCP	Strong AI/tooling positioning
PY-BUILD-AUTOMATION	Clear Python automation docs
UNITY-DATA-TOOLS	Strong analysis-tool README
ROS-TCP-CONNECTOR	Strong robotics integration docs
XR-INTERACTION-EXAMPLES	Strong XR sample structure
TEXTURE-PACKER	Strong technical-art tool docs
PACKAGE-MAKER	Strong package-tool docs
LIGHTS-LOD	Strong optimization-tool docs
ICON-GENERATOR	Strong production-tool docs

2.2 Medium examples

Label	Limitation
VFXGRAPH-TESTBED	Good media, sparse explanation
REACHY2-DT	Relevant, but hardware-specific
PDT-UNITY	Relevant, academic/research-heavy
RUNTIME-UNITY-EDITOR	Strong tool, modding-specific
LIGHTS-AUDIT	Useful, narrower scope

2.3 Weak examples

Label	Limitation
VR-PROJECT-TEMPLATE VR-GAME-JAM-TEMPLATE	Template-heavy README too sparse

33.0.4 3. Corrected source inventory

Some A3 rows contained citation artifacts or uncertain license wording. This inventory normalizes the usable sources.

Label	URL	Type	Role relevance	Quality
RAZLUTA-Profile	https://github.com/razluta	Profile	Unity tools / technical art	Strong
VFXGRAPH-TESTBED	https://github.com/keijiro/VfxGraphTestbed	Repo	Unity VFX	Medium
SPLATVFX	https://github.com/keijiro/SplatVFX	Repo	VFX / splats	Strong
UNITY-BUILDER	https://github.com/game-ci/unity-builder	Repo	Build pipeline	Strong
REACT-UNITY-WEBGL	https://github.com/jeffreylanters/react-unity-webgl	Repo	Unity WebGL	Strong
REACHY2-DT	https://github.com/pollen-robotics/Reach2-UnityDigitalTwin	Repo	XR / digital twin	Medium
PDT-UNITY	https://github.com/programming-digital-twins/LabBenchStudios-PDT-Unity	Repo	Digital twin	Medium
RUNTIME-UNITY-EDITOR	https://github.com/ManlyMarco/RuntimeUnityEditor	Repo	Runtime tooling	Medium
UNITY-EDITOR-TOOLBOX	https://github.com/arimger/Unity-Editor-Toolbox	Repo	Editor tooling	Strong
UNITY-MCP	https://github.com/CoplayDev/unity-mcp	Repo	AI + Unity tooling	Strong
PY-BUILD-AUTOMATION	https://github.com/angrysharkstudio/Unity-Python-Build-Automation	Repo	Python build automation	Strong
UNITY-DATA-TOOLS	https://github.com/Unity-Technologies/UnityDataTools	Repo	Build analysis	Strong
ROS-TCP-CONNECTOR	https://github.com/Unity-Technologies/ROS-TCP-Connector	Repo	Robotics visualization	Strong
XR-INTERACTION-EXAMPLES	https://github.com/Unity-Technologies/XR-Interaction-Toolkit-Examples	Repo	XR samples	Strong
VR-PROJECT-TEMPLATE	https://github.com/SamuelAsherRivello/unity-project-template-vr	Repo	VR template	Weak
VR-GAME-JAM-TEMPLATE	https://github.com/ValemVR/VR-Game-Jam-Template	Repo	VR template	Weak
TEXTURE-PACKER	https://github.com/razluta/UnityTextureRgpPacker	Repo	Technical art utility	Strong
PACKAGE-MAKER	https://github.com/razluta/UnityPackageManager	Repo	Package tooling	Strong
LIGHTS-LOD	https://github.com/razluta/UnityLightsLodSystem	Repo	Lighting optimization	Strong
LIGHTS-AUDIT	https://github.com/razluta/UnityLightsAuditTool	Repo	Lighting audit	Medium
ICON-GENERATOR	https://github.com/razluta/UnityIconGenerationFromModels	Repo	Asset pipeline utility	Strong

33.0.5 4. Benchmark patterns

4.1 Strong GitHub surfaces show value immediately

The strongest repositories do not start with long academic explanations.

They show:

```
Bloque de codigo: text
title
one-line value proposition
badges
hero image / GIF
live demo or docs link
quick-start path
```

For Alexander, the top of the README should answer:

```
Bloque de codigo: text
What is this?
What does it demonstrate?
Can I open it?
Can I see screenshots?
Can I understand the technical stack?
```

4.2 Strong READMEs separate user-facing value from implementation

Good structure:

```
Bloque de codigo: text
What it does
Live demo
Key features
Screenshots / GIFs
Technical stack
Architecture
Setup
Validation / metrics
Scope limits
License
```

Weak structure:

```
Bloque de codigo: text
long thesis summary
unfiltered internal notes
unclear scope
no media
no demo path
no license boundary
```

4.3 Strong technical-art tools show before/after or workflow

For technical art and tooling, the benchmark consistently favors:

```
Bloque de codigo: text
GIF of tool in use
input/output example
installation path
manual usage
API usage
limitations
license
```

Alexander should use this pattern for any future utility repo, such as:

```
Bloque de codigo: text
Blender batch export helper
Unity WebGL profiling helper
texture/material preparation script
fastener placement or validation tool
UI Toolkit prototype utility
```

4.4 Strong repositories document limitations explicitly

Good repos openly state:

```
Bloque de codigo: text
experimental status
unsupported scope
hardware requirements
Unity version
known limitations
license constraints
```

This matches Alexander's thesis repo well because the current project must avoid overclaiming.

The public repo should keep clear boundaries around:

```
Bloque de codigo: text
visual product twin
heuristic thermal reading
academic validation
unsupported metrics
hidden/internal managers
non-final prototype modules
```

4.5 Badges help, but only when restrained

Useful badges for Alexander:

```
Bloque de codigo: text
Unity version
WebGL
C#
Academic thesis
GitHub Pages
License / All rights reserved
```

Avoid:

```
Bloque de codigo: text
badge spam
fake quality badges
unused package-manager badges
npm badges unless a package exists
CI badges unless CI exists
```

33.0.6 5. Current repository audit

Repository inspected:

```
Bloque de codigo: text
delarge95/WebGL-Thesis-Proposal
Visibility: public
Default branch: master
Size: ~41 MB
README status: maintained
```

5.1 Current strengths

The current README already has several benchmark-aligned features:

```
Bloque de codigo: text
clear project title
one-line product description
Unity / WebGL / C# / academic badges
public demo link
canonical repository link
```



```
report and manual links
public scope disclaimer
canonical folder structure
Unity setup
runtime systems table
validation package explanation
public repository boundary
academic scope
```

This is better than many public Unity academic repos.

5.2 Current gaps

The current README is technically solid but still reads more like a thesis archive than a recruiter-facing GitHub proof page.

Main gaps:

```
Bloque de codigo: text
no hero screenshot or GIF near the top
no compact "what this demonstrates" section
no visual feature gallery
no short case-study link block
no technical proof bullets above the fold
no GitHub topics listed in the README
no separate portfolio-facing repo profile
```

These are not fatal. They are presentation gaps, not structural failures.

33.0.7 6. Recommended updates for WebGL-Thesis-Proposal

6.1 Add hero media after badges

Insert after the badge block:

```
Bloque de codigo: md
![TwinSight X500 WebGL demo](docs/media/twinsight-hero.gif)

Live demo: https://delarge95.github.io/WebGL-Thesis-Proposal/
```

Acceptable alternatives:

```
Bloque de codigo: text
animated GIF of selection flow
short MP4 converted to GIF
static 16:9 screenshot if GIF is too heavy
```

Required visual states:

```
Bloque de codigo: text
full drone view
part selection
bottom sheet
exploded view
cross-section
Studio visual mode
```

6.2 Add a compact proof section

Recommended section:

Bloque de código: md
What this demonstrates

- Unity WebGL deployment of an interactive technical visualization prototype.
- CAD-to-realtime asset preparation for a complex drone assembly.
- Mobile-first technical UI built around part inspection and spatial understanding.
- Runtime systems for selection, visibility, exploded view, cross-section and visual modes.
- Academic validation package with technical profiling, SUS, NASA-TLX Raw and Think-Aloud protocols.

This should appear before the long thesis summary.

6.3 Add visual feature gallery

Recommended structure:

Bloque de código: md
Feature gallery

Feature	Preview	Purpose
Selection	image	Identify parts
Inspect	image	Isolate components
Analyze	image	Explode / cut assembly
Studio	image	Compare visual readings
Technical sheets	image	Read specifications

Keep cells short. Use 5 rows maximum.

6.4 Add repository topics on GitHub

Recommended GitHub topics:

Bloque de código: text

```
unity
webgl
unity-webgl
technical-visualization
technical-art
real-time-3d
cad-to-realtime
drone
ui-toolkit
academic-thesis
```

Optional later:

Bloque de código: text

```
webassembly
interactive-3d
visual-product-twin
```

6.5 Keep the scope disclaimer

Do **not** remove the current visual product twin disclaimer.

It protects against overclaiming. It is also aligned with strong open-source documentation practice.

Keep explicit wording around:

Bloque de código: text

```
not live telemetry
not calibrated thermal analysis
metrics only when validated
hidden/internal systems not final public features
```

6.6 Do not over-convert the thesis repo into a personal portfolio

The repository should remain the canonical academic/project repo.

Alexander's personal GitHub surface should instead be improved through:

```
Bloque de código: text
profile README
pinned repositories
separate portfolio landing page
TwinSight case-study page
small tooling repos
```

Do not mix personal biography, job-seeking language and thesis archive text too aggressively inside the thesis repo.

33.0.8 7. Recommended GitHub profile README

Create or update:

```
Bloque de código: text
delarge95/delarge95/README.md
```

Recommended structure:

```
Bloque de código: md
# Alexander Woodcock Salomón

Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, CAD-to-realtime optimization and technical
↔ visualization.

## Selected work

- TwinSight X500 – Unity WebGL technical visualization prototype for Holybro X500 V2 drone assembly.
- CAD-to-realtime pipeline – Blender cleanup, UVs, baking, optimization and Unity integration.
- Technical UI systems – selection, bottom sheet, exploded view, cross-section and visual modes.

## Stack

Unity · C# · WebGL · URP · UI Toolkit · Blender · CAD cleanup · Git/GitHub · Python tooling

## Links

- Portfolio: [pending]
- TwinSight demo: https://delarge95.github.io/WebGL-Thesis-Proposal/
- TwinSight repo: https://github.com/delarge95/WebGL-Thesis-Proposal
- LinkedIn: [pending]
```

Avoid:

```
Bloque de código: text
student-only framing
long personal history
all tools ever used
unverified metrics
AI-generated badges without function
```

33.0.9 8. Recommended pinned repositories

Target pinned set:

Slot	Repo	Purpose
1	WebGL-Thesis-Proposal	Main proof
2	Portfolio site	Recruiter entry
3	Unity tool repo	Tooling proof
4	Blender/Python automation	Pipeline proof

Slot	Repo	Purpose
5	WebGL demo experiment	Deployment proof
6	Shader/visualization study	Technical art proof

Current available repo:

```
Bloque de codigo: text
WebGL-Thesis-Proposal
```

Missing repos to create later:

```
Bloque de codigo: text
portfolio site
small Unity tool
small Python automation tool
small shader/VFX or WebGL visualization demo
```

33.0.10 9. Recommended README template for future repos

Use this template for any future tooling or demo repository:

```
Bloque de codigo: md
# Project Name

One-line description of the problem solved.

![demo](docs/media/demo.gif)

## What it does

- Feature 1
- Feature 2
- Feature 3

## Why it matters

Short technical rationale.

## Tech stack

Unity · C# · WebGL · Blender · Python

## Quick start

1. Requirement
2. Installation
3. Run

## Workflow

Input -> Process -> Output

## Screenshots

| Step | Preview |
| --- | --- |
| Input | image |
| Output | image |

## Limitations

- Known limitation 1
- Known limitation 2

## License

State the license or rights boundary clearly.
```

33.0.11 10. Concrete updates for existing strategy files

10.1 Update GitHub strategy module

If a module similar to `18_github_profile_and_repo_strategy.md` exists, update it with:

```
Bloque de codigo: text
- GitHub profile README is required.
- TwinSight repo remains academic/project canonical.
- Add hero media to TwinSight README.
- Add "What this demonstrates" section.
- Add feature gallery.
- Add GitHub topics.
- Create at least 2 small utility repos after thesis delivery.
```

10.2 Update CV module

In `17_cv_base_and_role_variants.md`, add GitHub proof links:

```
Bloque de codigo: text
TwinSight demo
TwinSight repository
TwinSight case study
GitHub profile
```

Add a line under the main project:

```
Bloque de codigo: text
Public GitHub repository includes WebGL build, final report package, manuals, validation instruments and runtime architecture
↔ documentation.
```

10.3 Update portfolio module

The portfolio should not duplicate the GitHub README.

Portfolio case study should show:

```
Bloque de codigo: text
problem
process
screenshots
technical decisions
final demo
results / validation
```

GitHub README should show:

```
Bloque de codigo: text
how to inspect the repo
how to run / understand the demo
technical stack
scope limits
```

33.0.12 11. Priority order after A3

Do not start job applications yet.

Recommended order:

```
Bloque de codigo: text
1. Convert A3 into GitHub public-surface checklist.
2. Add missing visual/media assets to TwinSight README.
3. Create GitHub profile README.
4. Benchmark ArtStation / portfolio / case-study pages.
5. Build portfolio case study.
6. Then move into live jobs and companies.
```

33.0.13 12. Next task

Next task:

```
Bloque de codigo: text
A4_portfolio_artstation_case_study_benchmark
```

Mode:

```
Bloque de codigo: text
Agent: Si
Deep Research: No
```

Purpose:

```
Bloque de codigo: text
Collect public portfolio, ArtStation, personal-site and case-study examples relevant to real-time 3D, Unity technical art,
↳ technical visualization, XR/simulation and tool/pipeline development.
```

33.0.14 13. Prompt for A4

```
Bloque de codigo: text
Execute A4_portfolio_artstation_case_study_benchmark.

Objective:
Collect 20-30 public portfolio or case-study examples relevant to Alexander’s positioning:
Real-Time 3D Developer, Unity Technical Artist, Unity WebGL Developer, Technical Visualization Developer, XR Developer,
↳ Simulation Developer, Tools/Pipeline Developer, Python automation/tooling developer.

Prioritize:
- personal portfolio websites;
- ArtStation project pages;
- Behance case studies if technical enough;
- GitHub Pages portfolios;
- itch.io project pages with technical breakdowns;
- Unity/WebGL live demo pages;
- technical art breakdowns;
- XR/simulation/visualization case studies;
- portfolio pages with strong project cards and media.

Output structured tables only.

Fields:
- Source label
- URL
- Source type
- Role relevance
- Country/region if visible
- Portfolio structure
- Hero section
- Project card structure
- Case study depth
- Media used
- Demo links
- GitHub/source links
- Technical breakdown quality
- Metrics/results shown
- Contact/CTA handling
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:
- Public sources only.
- Do not invent unavailable data.
- If unavailable, write “not found”.
- Keep cells short.
- Include source URLs.
- Tables only.
- No narrative strategy.
```

Chapter 34

28D — ArtStation / Portfolio Benchmark

Fuente: 28D_artstation_portfolio_benchmark (1).md

Seccion: root

Estado: duplicate

Rol: Version parcial / duplicada

Lineas originales: 271

Ultima modificacion: 2026-06-11 18:06

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

— tipo: modulo_benchmark modulo: 28D_artstation_portfolio_benchmark estado: corregido fecha: 2026-06-11
fuente_base: A4_artstation_portfolio_benchmark perfil_objetivo: Alexander Woodcock — Real-Time 3D / Unity
WebGL / Technical Visualization revision: 2 nota: reemplaza la versión anterior porque A4 no investigó ArtStation
de forma válida —

Contenido completo depurado

tipo: modulo_benchmark

modulo: 28D_artstation_portfolio_benchmark

estado: corregido

fecha: 2026-06-11

fuente_base: A4_artstation_portfolio_benchmark

perfil_objetivo: Alexander Woodcock — Real-Time 3D / Unity WebGL / Technical Visualization

revision: 2

nota: reemplaza la versión anterior porque A4 no investigó ArtStation de forma válida

34.0.1 1. Corrección crítica

La versión anterior de este módulo no era válida como **ArtStation benchmark**.

El archivo A4 usado como base contenía portafolios externos, blogs técnicos y GitHub Pages, pero no una muestra real de perfiles o proyectos de ArtStation. Por tanto, cualquier conclusión presentada como patrón de ArtStation queda retirada.

Este módulo corregido separa tres cosas:

1. lo que sí sirve del intento anterior;
2. lo que no debe usarse;
3. la tarea necesaria para completar ArtStation correctamente.

34.0.2 2. Estado de evidencia

Bloque	Evidencia	Estado
ArtStation perfiles	0	inválido
ArtStation proyectos	0–1	insuficiente
Blogs técnicos	3+	usable
Portfolios externos	7+	usable
GitHub Pages	4+	usable
Benchmark ArtStation	no	pendiente
Benchmark portfolio general	parcial	usable

34.0.3 3. Lo que queda invalidado

No deben usarse como conclusiones de ArtStation:

- “los perfiles de ArtStation usan X estructura”;
- “los breakdowns de ArtStation priorizan X”;
- “ArtStation demuestra X patrón de contratación”;
- “Alexander debe adaptar X desde ArtStation”.

La muestra no sostiene esas afirmaciones.

34.0.4 4. Lo que sí puede conservarse

El intento anterior sí aporta patrones generales de portfolio técnico.

Fuente	Plataforma	Uso válido	Fuerza
Potion Shader	Blog	shader breakdown	alta
Harry Alisavakis	Blog	autoridad técnica	alta
Julhe	GitHub Pages	posts shader	alta
Francisco Múrias	GitHub Pages	grid profesional	media
Alson Entuna	GitHub Pages	rol especializado	media
Alla Eddine Rakik	Framer	portfolio técnico	media
Omar Rinaz	GitHub Pages	CV visible	baja

34.0.5 5. Patrones generales válidos

5.1. Resultado visual primero

Un caso técnico debe abrir con el resultado:

```
Bloque de código: text
final render / gif / video / interactive demo / before-after
```

Para Alexander, esto aplica a:

- TwinSight X500;
- cross-section shader;
- exploded view;
- Inspect mode;
- Analyze mode;
- thermal heuristic view;
- CAD-to-WebGL pipeline.

5.2. Rol explícito

El portfolio debe decir qué hizo Alexander.

Formato recomendado:

```
Bloque de código: text
Role: Solo developer / technical artist / 3D pipeline owner
```

No basta con mostrar el proyecto como tesis.

5.3. Breakdown por problema

Cada case study debe responder:

```
Bloque de código: text
Problem → Constraint → Process → Technical solution → Result → Limitation
```

Esto es más fuerte que una galería visual.

5.4. Herramientas visibles

El stack debe aparecer en el hero o primera sección:

```
Bloque de código: text
Unity · WebGL · Blender · C# · UI Toolkit · Shader Graph/HLSL · 3D Optimization
```

5.5. Limitaciones declaradas

Para perfiles técnicos, declarar límites aumenta credibilidad.

Ejemplos aplicables:

- thermal view no es FEA;
- visual twin no es digital twin operativo;
- validación formativa si la muestra es pequeña;
- WebGL implica restricciones de memoria y carga.

34.0.6 6. Estructura recomendada para TwinSight X500

```
Bloque de código: text
1. Hero visual
2. One-line summary
3. Role / scope / tools
4. Problem
5. Constraints
6. CAD-to-WebGL pipeline
7. Interaction systems
8. Shader / visualization systems
9. Optimization notes
10. Validation evidence
11. Limitations
12. Links
```

34.0.7 7. Case studies derivados

Case study	Función	Prioridad
TwinSight X500	flagship	1
CAD-to-WebGL pipeline	pipeline	2
Cross-section shader	shader	3
Exploded view system	interaction	4

Case study	Función	Prioridad
Bottom-sheet UI	UX/system	5
Fastener inspection	detail system	6
Thermal heuristic view	analysis view	7

34.0.8 8. Estructura de página recomendada

```

Bloque de código: markdown
# TwinSight X500

## Summary
Interactive WebGL technical visualization of a Holybro X500 V2 drone assembly.

## Role
Solo developer / technical artist / 3D pipeline owner.

## Tools
Unity, WebGL, Blender, C#, UI Toolkit, Shader Graph/HLSL.

## Problem
2D manuals require users to mentally reconstruct spatial relationships.

## Constraints
Browser deployment, mobile-first UI, real-time interaction, academic validation.

## Technical Breakdown
- CAD cleanup and hierarchy normalization.
- Runtime part selection.
- Exploded view.
- Cross-section clipping.
- Bottom-sheet technical data.
- Heuristic thermal visualization.

## Result
A public WebGL prototype for inspection and study of a drone assembly.

## Limitations
Visual product twin, not operational digital twin.
Thermal mode is heuristic, not calibrated FEA.

## Links
Demo, repository, final report, manuals.

```

34.0.9 9. Qué evitar

Error	Riesgo
Grid sin breakdown	parece galería
Proyecto como “tesis” solamente	reduce valor laboral
Reclamar digital twin operativo	sobreventa
Mostrar métricas no cerradas	riesgo técnico
Ocultar limitaciones	baja credibilidad
No incluir video/gif	baja evidencia visual

34.0.10 10. Decisión operativa

Antes de convertir este módulo en benchmark final, ejecutar:

```

Bloque de código: text
A4_retry_artstation_only_benchmark

```

Hasta entonces, 28D debe usarse como:

- guía de portfolio técnico general;
- auditoría de error;
- estructura provisional de case studies;
- no como benchmark ArtStation final.

34.0.11 11. Prompt para A4-Retry

Bloque de código: text
Execute A4_retry_artstation_only_benchmark.

Context:

The previous A4 attempt failed. It collected portfolio/blog examples but did not properly research ArtStation profiles or
↪ ArtStation project breakdowns.

Objective:

Collect 20-35 public ArtStation examples relevant to Alexander's technical art and real-time 3D positioning.

Prioritize:

- Technical Artist profiles;
- Unity Technical Artist profiles;
- shader breakdown projects;
- VFX breakdown projects;
- real-time 3D breakdowns;
- optimization breakdowns;
- Blender technical breakdowns;
- XR / interactive 3D projects;
- technical visualization projects;
- portfolios with tools, GitHub, demo videos or detailed process notes.

Output structured tables only.

Fields:

- Artist / profile label
- ArtStation URL
- Source type: profile / resume / project / blog post / channel result
- Role target
- Seniority if visible
- Project type
- Breakdown structure
- Images used
- Videos used
- Technical text depth
- Tags / skills
- Tools shown
- Portfolio / GitHub links if visible
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:

- ArtStation sources only.
- Use public pages only.
- Do not invent missing data.
- If unavailable, write "not found".
- If ArtStation blocks access, record "blocked" and do not infer details.
- Include source URLs.
- Tables only.
- No narrative strategy.

Chapter 35

28D — ArtStation / Portfolio Benchmark

Fuente: 28D_artstation_portfolio_benchmark.md

Seccion: root

Estado: duplicate

Rol: Version parcial / duplicada

Lineas originales: 439

Ultima modificacion: 2026-06-11 17:52

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

— tipo: modulo_benchmark modulo: 28D_artstation_portfolio_benchmark estado: generado fecha: 2026-06-11
fuente_base: A4_artstation_portfolio_benchmark perfil_objetivo: Alexander Woodcock — Real-Time 3D / Unity
WebGL / Technical Visualization —

Contenido completo depurado

tipo: modulo_benchmark

modulo: 28D_artstation_portfolio_benchmark

estado: generado

fecha: 2026-06-11

fuente_base: A4_artstation_portfolio_benchmark

perfil_objetivo: Alexander Woodcock — Real-Time 3D / Unity WebGL / Technical Visualization

35.0.1 1. Estado del módulo

Este módulo convierte el intento **A4_artstation_portfolio_benchmark** en criterios prácticos para el portafolio de Alexander.

La muestra recuperada es útil, pero **no es exhaustiva**. El objetivo original era 20–35 ejemplos; la extracción real dejó 12 entradas utilizables por bloqueos de ArtStation, páginas con JavaScript y portafolios que no exponían suficiente contenido técnico.

Por tanto, este módulo debe usarse como:

- benchmark estructural;
- guía de presentación;
- patrón de case studies;
- criterio para decidir qué proyectos mostrar;
- no como estadística representativa del mercado.

35.0.2 2. Auditoría de calidad de A4

Criterio	Resultado	Lectura
Objetivo inicial	20–35	no cumplido
Ejemplos recuperados	12	parcial
ArtStation directo	débil	bloqueos
Blogs técnicos	fuerte	útiles
GitHub Pages	medio	variable
Portfolios visuales	medio	falta técnica
Evidencia usable	sí	con cautela

La parte más valiosa no vino de ArtStation puro, sino de **blogs técnicos, GitHub Pages y portafolios con proyectos técnicos visibles**.

35.0.3 3. Inventario consolidado

Label	Plataforma	Rol cercano	Tipo	Fuerza
Rakik	Framer	TA / Game Dev	portfolio	media
Potion Shader	Jettelly	Shader TA	breakdown	alta
Alisavakis	blog	Technical Artist	blog técnico	alta
Entuna	GitHub Pages	Lead TA	CV/portfolio	media
Rinaz	GitHub Pages	TA/Game Design	portfolio	baja
Múrias	GitHub Pages	Technical Artist	portfolio	media
Múrias Nebulas	GitHub Pages	VFX TA	project card	media
Múrias OpenGL	GitHub Pages	tools/rendering	project card	media
julhe SDF	GitHub Pages	shader dev	blog post	alta
julhe Optimize	GitHub Pages	shader dev	blog post	alta
julhe X5	GitHub Pages	shader dev	blog post	media
Rakik VFX Pack	Framer	VFX TA	project card	media

35.0.4 4. Patrones fuertes observados

4.1. El rol debe aparecer antes que el arte

Los perfiles útiles no se presentan solo como “3D artist”. Usan etiquetas más específicas:

```
Bloque de código: text
Technical Artist
Shader Developer
Pipeline & Tools Technical Artist
Game Developer & Technical Artist
VFX Technical Artist
Real-Time Look Development
```

Para Alexander, el posicionamiento más defendible es:

```
Bloque de código: text
Real-Time 3D Developer / Unity WebGL Technical Artist
```

Una variante más orientada a tesis y visualización:

```
Bloque de código: text
Technical Visualization Developer – Unity WebGL / Blender / Real-Time 3D
```

4.2. El proyecto debe abrir con resultado visual

Los mejores ejemplos muestran primero el resultado:

- imagen final;
- video corto;
- gif;
- comparación antes/después;
- captura de interfaz;
- viewport técnico;
- breakdown visual.

Para Alexander, cada caso debería abrir con una captura o gif del resultado final antes de explicar metodología.

4.3. El texto técnico debe ser específico

Los ejemplos fuertes explican:

- problema técnico;
- restricción;
- pipeline;
- herramientas;
- decisiones;
- resultado;
- limitaciones.

Los ejemplos débiles solo muestran imágenes sin explicar qué hizo el autor.

4.4. Los portfolios visuales necesitan páginas internas

Un grid de proyectos ayuda, pero no basta. Cada card debe llevar a una página con breakdown.

Una card sin página interna no demuestra capacidad técnica. Solo demuestra estética.

4.5. Los blogs técnicos funcionan como autoridad

Los ejemplos tipo Harry Alisavakis o julhe muestran que un blog técnico puede posicionar mejor que una galería visual si explica problemas concretos.

Para Alexander, esto favorece crear páginas tipo artículo para:

- CAD-to-WebGL pipeline;
- shader de corte;
- UI técnica en Unity;
- jerarquía de selección;
- optimización del modelo;
- visualización térmica heurística.

35.0.5 5. Qué debe adaptar Alexander

5.1. Hero de portfolio

Debe comunicar rol, stack y dominio en una sola pantalla.

Propuesta:

```
Bloque de código: text
Alexander Woodcock
Real-Time 3D Developer / Unity WebGL Technical Artist
Technical visualization, interactive 3D systems and optimized WebGL prototypes.
```

Stack visible:

Bloque de código: text
Unity · WebGL · Blender · C# · UI Toolkit · Shader Graph/HLSL · 3D Optimization

5.2. Proyecto principal

El proyecto principal debe ser **TwinSight X500**.

No debe presentarse solo como tesis. Debe presentarse como case study técnico:

Bloque de código: text
TwinSight X500 – Interactive WebGL Technical Visualization of a Drone Assembly

Rol sugerido:

Bloque de código: text
Solo developer / technical artist / 3D pipeline owner

Áreas demostrables:

- CAD-to-WebGL pipeline;
- Blender cleanup;
- Unity WebGL deployment;
- part selection;
- exploded view;
- cross-section shader;
- UI bottom sheet;
- heuristic thermal view;
- validation package.

5.3. Proyectos derivados

La tesis puede dividirse en varios mini case studies sin inventar proyectos nuevos.

Case study	Propósito	Prioridad
TwinSight X500	flagship	1
CAD-to-WebGL pipeline	pipeline técnico	2
Cross-section shader	shader/visualization	3
Exploded view system	interaction/system	4
Bottom-sheet UI	UX/tooling	5
Fastener inspection	hierarchy/detail	6
Thermal heuristic view	visual analysis	7

35.0.6 6. Estructura recomendada para cada case study

Cada página debe seguir esta secuencia:

Bloque de código: text
1. Hero visual
2. One-line summary
3. Role / tools / scope
4. Problem
5. Constraints
6. Pipeline
7. Technical breakdown
8. Result
9. Limitations
10. Links

Plantilla breve

```
Bloque de código: markdown
# Project Title

## Summary
One sentence describing the technical result.

## Role
Solo developer / technical artist / 3D pipeline owner.

## Tools
Unity, Blender, C#, WebGL, Shader Graph/HLSL, etc.

## Problem
What technical or user problem this solved.

## Constraints
Browser, mobile-first, performance, imported CAD, academic scope.

## Breakdown
Step-by-step explanation with screenshots.

## Result
What became visible, usable or measurable.

## Limitations
What the project does not claim.

## Links
Demo, GitHub, report, video.
```

35.0.7 7. Media mínima por proyecto

Proyecto	Media mínima	Extra útil
TwinSight X500	hero video	interaction gif
CAD pipeline	before/after	wireframe
Cross-section	gif	shader diagram
Exploded view	gif	hierarchy map
UI bottom sheet	screenshots	flow diagram
Thermal view	image pair	disclaimer
Fasteners	closeups	modular diagram

35.0.8 8. Profundidad técnica esperada

Nivel insuficiente

```
Bloque de código: text
I made a drone viewer in Unity.
```

Nivel aceptable

```
Bloque de código: text
I built a Unity WebGL prototype to inspect a drone assembly with part selection, exploded view and contextual technical
↪ information.
```

Nivel fuerte

```
Bloque de código: text
I converted a CAD-derived Holybro X500 V2 assembly into an optimized Unity WebGL technical visualization prototype. The system
↪ combines normalized runtime hierarchy, part-level selection, exploded assembly reading, shader-based cross-section and
↪ bottom-sheet technical documentation for browser-based inspection.
```

35.0.9 9. Qué no debe copiar Alexander

No copiar:

- layouts de otros portfolios;
- capturas o assets de terceros;
- textos de artículos de shaders;
- etiquetas senior si no son defendibles;
- “Technical Artist” como única etiqueta si el trabajo principal es visualización interactiva;
- promesas no visibles en la build pública;
- métricas sin evidencia;
- claims de digital twin operacional si no hay telemetría real.

35.0.10 10. Riesgos detectados para el portfolio de Alexander

Riesgo 1 — parecer solo académico

Mitigación:

Presentar la tesis como producto técnico interactivo, no solo como entrega universitaria.

Riesgo 2 — parecer solo modelador 3D

Mitigación:

Separar claramente pipeline, Unity runtime, UI, shaders y WebGL.

Riesgo 3 — mostrar demasiado informe y poca demo

Mitigación:

El portfolio debe abrir con demo visual, no con texto académico.

Riesgo 4 — sobrerreclamar capacidades

Mitigación:

Usar disclaimers: visual product twin, heuristic thermal view, no FEA, no telemetry.

35.0.11 11. Cambios concretos para el portfolio

Home

Debe tener:

```
Bloque de codigo: text
- rol claro;
- hero visual del dron;
- demo link;
- GitHub link;
- 3-5 project cards;
- stack técnico;
- contacto.
```

Project grid

Orden recomendado:

```
Bloque de codigo: text
1. TwinSight X500
2. CAD-to-WebGL Optimization Pipeline
3. Unity WebGL Interaction System
4. Shader-Based Cross-Section Tool
5. Technical UI / Bottom Sheet System
6. Optional: Blender Portrait / Character Study
```

Project pages

Cada página debe tener:

```
Bloque de código: text
- 1 video/gif;
- 3-6 imágenes;
- 1 diagrama;
- 1 bloque técnico;
- 1 bloque de limitaciones;
- links a demo/repo.
```

35.0.12 12. Relación con GitHub

El portfolio y GitHub deben reforzarse mutuamente.

Elemento	Portfolio	GitHub
Demo	visual primero	enlace técnico
README	resumen	detalle reproducible
Case study	explicación visual	código/manuales
Media	videos/gifs	screenshots
Evidencia	resultados	archivos fuente

35.0.13 13. Decisión de posicionamiento

La etiqueta más estratégica es:

```
Bloque de código: text
Real-Time 3D Developer / Unity WebGL Technical Artist
```

La etiqueta académica/profesional secundaria:

```
Bloque de código: text
Technical Visualization Developer
```

La etiqueta que debe evitarse como principal:

```
Bloque de código: text
3D Artist
```

No es falsa, pero reduce demasiado el perfil y oculta el valor técnico.

35.0.14 14. Próxima tarea

```
Bloque de código: text
A5_case_study_breakdown_benchmark
```

Objetivo:

Recolectar y analizar ejemplos públicos de case studies técnicos con estructura fuerte para adaptar la presentación de TwinSight X500.

Prompt sugerido:

```
Bloque de código: text
Execute A5_case_study_breakdown_benchmark.

Objective:
Collect 15-25 public technical case studies relevant to real-time 3D, Unity WebGL, technical visualization, shader breakdowns,
↳ CAD-to-real-time pipelines, optimization, interaction systems and technical art tools.
```

Output structured tables only.

Fields:

- Source label
- URL
- Platform
- Project type
- Role positioning
- Opening media
- Problem statement
- Constraints shown
- Pipeline shown
- Technical breakdown depth
- Tools shown
- Metrics shown
- Limitations shown
- Links provided
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:

- Public sources only.
- Do not invent missing data.
- If unavailable, write "not found".
- Keep table cells short.
- Include source URLs.
- Tables only.
- No strategy.

Chapter 36

28D — Benchmark actualizado de portafolios técnicos y arte 3D en tiempo real

Fuente: 28D_artstation_portfolio_benchmark_complete.md

Seccion: root

Estado: canonical

Rol: Version consolidada

Lineas originales: 855

Ultima modificacion: 2026-06-11 18:19

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

— module: 28D_artstation_portfolio_benchmark status: updated_complete source_task: A4_retry_accessible_portfolio_benchmark scope: accessible_portfolios_and_technical_breakdowns artstation_direct_access: blocked_by_cloudflare date: 2026-06-11 owner_profile: Alexander Woodcock Salomon target_positioning: - Real-Time 3D Developer - Unity Technical Artist - Unity WebGL Developer - Technical Visualization Developer - XR Developer - Simulation Developer - Tools / Pipeline Develop

Contenido completo depurado

module: 28D_artstation_portfolio_benchmark

status: updated_complete

source_task: A4_retry_accessible_portfolio_benchmark

scope: accessible_portfolios_and_technical_breakdowns

artstation_direct_access: blocked_by_cloudflare

date: 2026-06-11

owner_profile: Alexander Woodcock Salomon

target_positioning:

- Real-Time 3D Developer
 - Unity Technical Artist
 - Unity WebGL Developer
 - Technical Visualization Developer
 - XR Developer
 - Simulation Developer
 - Tools / Pipeline Developer
-

36.0.1 1. Estado del módulo

Este módulo reemplaza la versión anterior de `28D_artstation_portfolio_benchmark.md`.

La tarea original buscaba un benchmark centrado en ArtStation. Sin embargo, el acceso directo a ArtStation quedó bloqueado por verificación de Cloudflare. Por tanto, este módulo no debe presentarse como un benchmark exhaustivo de ArtStation.

El módulo actualizado usa fuentes públicas accesibles equivalentes: portafolios personales, GitHub Pages, Framer, blogs técnicos y breakdowns públicos de arte técnico, shaders, VFX, herramientas y 3D en tiempo real.

36.0.2 2. Decisión metodológica

ArtStation sigue siendo una fuente valiosa para evaluar presentación visual. No obstante, para estudiar estructura técnica, storytelling de proyecto, documentación, código, métricas y breakdowns, las fuentes accesibles recopiladas son más útiles que perfiles visuales bloqueados.

La decisión correcta es separar:

- **ArtStation directo:** pendiente por bloqueo de acceso.
- **Benchmark visual/técnico accesible:** válido para patrones de portafolio.
- **Case studies técnicos:** fuente principal para construir los proyectos de Alexander.
- **GitHub/README:** fuente principal para validar implementación y código.

36.0.3 3. Calidad del benchmark A4_retry

Criterio	Resultado
Fuentes accesibles	Sí
ArtStation directo	No
Portafolios técnicos	Sí
Blogs técnicos	Sí
Shaders/VFX	Sí
Unity/Unreal	Sí
Métricas	Escasas
Código visible	Parcial
Utilidad para Alexander	Alta
Exhaustividad	Media

36.0.4 4. Inventario de fuentes válidas

Fuente	Plataforma	Foco	Calidad
Alla Eddine Rakik	Framer	Tech art general	Media
Rakik — Boom Mike VR	Framer	VR demo	Baja-media
Rakik — Shader VFX Pack	Framer	Shader/VFX	Baja-media
Jettelly — Potion Shader	Blog	Unity shader breakdown	Alta
Harry Alisavakis	Blog	Technical art resources	Media
Omar Rinaz Costa	GitHub Pages	CV + tech art	Media
Alson Entuna	Personal site	Tools/pipeline	Media
Wolf van Veen	Personal site	Shaders/VFX	Alta
Francisco Múrias	GitHub Pages	VFX/rendering	Alta
Julhe — SDF Textures	Blog	Shader technique	Alta
Julhe — Shader Optimization	Blog	Shader performance	Alta
Julhe — X5 Shader	Blog	Shader framework	Alta

36.0.5 5. Fuentes fuertes

5.1 Jettelly — Real-Time Potion Shader Breakdown

URL: <https://jettelly.com/blog/a-real-time-potion-shader-breakdown-in-unity>

Es uno de los ejemplos más útiles porque no se limita a mostrar una imagen final. Presenta el problema técnico, la solución visual, el uso de Unity URP, Shader Graph y C#, y el comportamiento del efecto en movimiento.

Patrones adaptables:

- Título específico.
- Resultado visual temprano.
- Explicación de comportamiento.
- Capturas intermedias.
- Video o GIF del resultado.
- Separación entre efecto visual y técnica usada.

Aplicación para Alexander:

Este formato sirve para un breakdown de:

- modo X-Ray;
- shader de clipping/cross-section;
- visualización térmica heurística;
- resaltado de piezas;
- sistema de selección;
- controles WebGL.

Lo que no debe copiar:

No basta con decir “hecho en Unity”. El valor está en explicar la relación entre intención visual, limitaciones del motor, implementación y resultado.

5.2 Julhe — Always-Sharp SDF Textures

URL: <https://julhe.github.io/blog/always-sharp-sdf-textures/>

Este ejemplo funciona como referencia de comunicación técnica avanzada. Explica una técnica concreta, desarrolla el problema y muestra razonamiento técnico.

Patrones adaptables:

- Problema técnico acotado.
- Explicación conceptual.
- Visuales comparativos.
- Código o pseudocódigo.
- Resultado final verificable.

Aplicación para Alexander:

Sirve como modelo para escribir piezas técnicas sobre:

- optimización de lectura visual en WebGL;
- diseño de hotspots;
- claridad de bordes;
- visualización de piezas pequeñas;
- control de transparencia;
- UI técnica sobre modelos complejos.

Lo que no debe copiar:

No conviene publicar artículos demasiado matemáticos si el objetivo principal es empleabilidad inicial. La profundidad técnica debe ser suficiente, pero legible para recruiters técnicos y leads.

5.3 Julhe — How (Not) To Optimize Shaders

URL: <https://julhe.github.io/blog/how-not-to-optimize-shaders/>

Este ejemplo es especialmente relevante porque enseña mediante anti-patrones. No solo muestra qué hacer, sino qué evitar.

Patrones adaptables:

- Planteamiento de error común.
- Comparación entre enfoques.
- Análisis de rendimiento.
- Recomendaciones prácticas.
- Conclusión aplicable.

Aplicación para Alexander:

Puede inspirar un artículo propio:

- “How not to optimize a Unity WebGL technical viewer”.
- “Lessons from building a browser-based drone assembly viewer”.
- “What I would rebuild in my thesis viewer”.
- “Mobile-first constraints in Unity WebGL visualization”.

Lo que no debe copiar:

No debe convertirse en una crítica abstracta. Debe estar conectado a una experiencia propia y a decisiones reales del proyecto.

5.4 Francisco Múrias — Technical Art Portfolio

URL: <https://franciscomurias.github.io/>

Este portafolio es útil por su grilla visual, foco en proyectos técnicos, y combinación de arte técnico con rendering, VFX y herramientas.

Patrones adaptables:

- Home visual fuerte.
- Proyectos en tarjetas.
- Separación entre trabajos personales y profesionales.
- Enlaces a GitHub cuando aplica.
- Estética técnica sin sobreexplicar en la portada.

Aplicación para Alexander:

Sirve para estructurar una landing de portafolio con:

- TwinSight X500 como proyecto principal;
- miniaturas de módulos técnicos;
- tarjetas de herramientas;
- sección de breakdowns;
- enlaces a demo y GitHub.

Lo que no debe copiar:

No debe depender solo de imágenes atractivas. En el caso de Alexander, el valor competitivo está en el proceso técnico y la interacción WebGL.

5.5 Wolf van Veen — Technical Artist

URL: <https://wolfvanveen.art/>

Este ejemplo destaca por la organización visual mediante categorías: post-processing, shaders, particles, models y textures. Aunque el propio sitio indica que el portafolio está desactualizado, su estructura sigue siendo útil.

Patrones adaptables:

- Categorías visibles.
- Navegación simple.
- Galerías por proyecto.
- Modal o página individual por pieza.

- Herramientas mostradas por proyecto.

Aplicación para Alexander:

El portafolio de Alexander puede usar categorías similares, adaptadas a su perfil:

- Technical Visualization.
- Unity WebGL.
- Shaders & View Modes.
- Tools & Pipeline.
- 3D Assets.
- XR / Interactive.
- Case Studies.

Lo que no debe copiar:

No debe dejar proyectos sin explicación técnica. Cada tarjeta debe responder qué problema resolvió y cómo se implementó.

36.0.6 6. Fuentes medias

6.1 Alla Eddine Rakik — Technical Art Portfolio

URL: <https://technical-art-portfolio.io.framer.ai/>

El sitio tiene una estructura clara: presentación, herramientas, lenguajes y grilla de proyectos. Es útil como referencia visual de organización.

Aporta:

- landing directa;
- lista de software;
- proyectos por miniaturas;
- variedad de áreas: VR, 3D, VFX, OpenGL.

Limitación:

Los proyectos tienen poca profundidad técnica. Sirven como muestra visual, pero no como case studies completos.

Aplicación para Alexander:

Puede servir como modelo inicial de landing, pero debe complementarse con páginas de proyecto mucho más documentadas.

6.2 Omar Rinaz Costa — Portfolio

URL: <https://omarrinaz.github.io/>

Este portafolio sirve para observar estructura curricular: about, experiencia, educación, skills y portfolio. Es útil para la transición entre CV y portafolio.

Aporta:

- línea de tiempo;
- roles y fechas;
- educación visible;
- lista de responsabilidades;
- botón de descarga de CV.

Limitación:

Faltan métricas y proyectos técnicos profundos.

Aplicación para Alexander:

Puede inspirar una página About / CV, pero no debe ser el modelo principal para proyectos técnicos.

6.3 Alson Entuna — Lead Technical Artist Pipeline & Tools

URL: <https://alsonentuna.github.io/>

El sitio es útil porque comunica una transición profesional: gameplay programming hacia technical art, tools y pipeline.

Aporta:

- posicionamiento claro;
- foco en tools/pipeline;
- relato profesional;
- seniority explícito.

Limitación:

El home muestra más perfil que evidencia técnica. Para Alexander, esto sería insuficiente sin proyectos concretos.

Aplicación para Alexander:

Puede inspirar el texto de posicionamiento profesional:

Multimedia engineer focused on real-time 3D visualization, Unity WebGL, technical art systems and interactive product visualization.

6.4 Harry Alisavakis — Technical Art Blog

URL: <https://halisavakis.com/>

El blog funciona como repositorio de recursos y artículos. Es útil como referencia de consistencia y marca personal.

Aporta:

- publicaciones recurrentes;
- recursos técnicos;
- enfoque educativo;
- mezcla Unity/Unreal;
- comunidad técnica.

Limitación:

Es más blog que portafolio. No todo está organizado para evaluación rápida de reclutar.

Aplicación para Alexander:

Sirve como modelo para una sección Technical Notes o Breakdowns, no como home principal.

36.0.7 7. Fuentes débiles o incompletas

7.1 Rakik — Boom Mike VR

URL: <https://technical-art-portfolio.io/framer.ai/boom-mike>

Aporta una demo VR con video, pero poca explicación técnica. Es útil como referencia mínima de presentación visual.

No debe ser modelo principal para Alexander.

7.2 Rakik — Shader VFX Pack

URL: <https://technical-art-portfolio.io/framer.ai/shader-vfx-pack>

Aporta una muestra visual de efectos, pero no explica implementación, problema, parámetros ni resultados.

Debe evitarse este patrón si el objetivo es aplicar a roles técnicos.

36.0.8 8. Patrones de estructura para el portafolio de Alexander

8.1 Landing page

La landing debe responder en menos de 10 segundos:

- quién es Alexander;
- qué rol busca;
- cuál es su proyecto principal;
- qué evidencia técnica puede revisar el evaluador;
- dónde están demo, GitHub, CV y contacto.

Estructura recomendada:

```
Bloque de código: text
Hero
├─ Role headline
├─ One-line positioning
├─ CTA: View TwinSight X500
├─ CTA: GitHub
├─ CTA: Download CV

Featured Case Study
├─ TwinSight X500
├─ Unity WebGL technical viewer
├─ Demo / GitHub / Breakdown

Project Categories
├─ Technical Visualization
├─ Unity WebGL
├─ Shaders & View Modes
├─ Tools & Pipeline
├─ 3D Assets
├─ Technical Notes
```

8.2 Project card

Cada tarjeta de proyecto debe mostrar:

- nombre;
- rol;
- tecnología;
- tipo de problema;
- resultado visible;
- enlace a demo;
- enlace a breakdown;
- enlace a código si aplica.

Formato sugerido:

```
Bloque de código: markdown
## TwinSight X500

**Role:** Unity WebGL Developer / Technical Artist
**Type:** Browser-based 3D technical visualization
**Tools:** Unity 6000, Blender, WebGL, URP, GitHub Pages
**Focus:** Spatial inspection, part selection, exploded view, cross-section, heuristic thermal reading
**Demo:** [link]
**Code:** [link]
**Case Study:** [link]
```

8.3 Case study técnico

La estructura debe tomar como referencia Jettelly y Julhe.

Plantilla:

```
Bloque de código: markdown
```

```
# Project name

## Context
What problem existed?

## Goal
What the system needed to solve.

## Constraints
Browser, mobile, WebGL, asset size, usability, academic scope.

## Technical approach
Architecture, data, shaders, UI, pipeline.

## Implementation
Key systems and decisions.

## Results
Screenshots, video, metrics, validation.

## What I learned
Trade-offs, mistakes, next iteration.

## Links
Demo, GitHub, report, video.
```

36.0.9 9. Qué debe adaptar Alexander

9.1 De Jettelly

Adaptar:

- breakdown visual paso a paso;
- explicación de comportamiento dinámico;
- uso de imágenes y video;
- foco en Unity.

Aplicar a:

- thermal heuristic view;
- X-Ray mode;
- cross-section shader;
- exploded view;
- selection highlighting.

9.2 De Julhe

Adaptar:

- profundidad técnica;
- explicación de problema;
- anti-patronos;
- código o pseudocódigo;
- análisis de rendimiento.

Aplicar a:

- WebGL optimization;
- shader modes;
- UI performance;
- model hierarchy normalization;
- asset import pipeline.

9.3 De Francisco Múrias

Adaptar:

- grilla de proyectos;
- estética técnica;
- tarjetas visuales;
- enlaces a GitHub;
- separación por tipo de trabajo.

Aplicar a:

- home del portafolio;
- featured projects;
- technical demos;
- visual archive.

9.4 De Wolf van Veen

Adaptar:

- categorías claras;
- filtros por tipo de trabajo;
- modal o galería de proyecto;
- herramientas visibles por pieza.

Aplicar a:

- portfolio navigation;
- project taxonomy;
- visual browsing.

9.5 De Omar Rinaz

Adaptar:

- timeline de experiencia;
- formación académica;
- lista de roles;
- sección CV.

Aplicar a:

- About;
- CV page;
- career context.

36.0.10 10. Qué no debe copiar Alexander

Evitar:

- portafolios con solo imágenes;
- proyectos sin contexto;
- ausencia de métricas;
- falta de enlaces a demo;
- listas largas de software sin evidencia;
- describirse como senior sin respaldo;
- claims no verificables;
- usar ArtStation como única superficie;
- esconder el GitHub;
- omitir decisiones técnicas.

36.0.11 11. Recomendaciones concretas para el portafolio

11.1 Proyecto principal

TwinSight X500 debe ser el proyecto hero.

Título recomendado:

```
Bloque de código: text
TwinSight X500 – Unity WebGL Technical Visualization Viewer
```

Subtítulo recomendado:

```
Bloque de código: text
Browser-based interactive 3D viewer for inspecting the Holybro X500 V2 drone assembly, focused on spatial understanding, part
↩ hierarchy, exploded view, cross-section and heuristic visual analysis.
```

11.2 Categorías

Usar estas categorías:

```
Bloque de código: text
Technical Visualization
Unity WebGL
Shaders & View Modes
Tools & Pipeline
3D Production
Academic Research
Technical Notes
```

11.3 Case studies mínimos

Publicar al menos 4 piezas:

```
Bloque de código: text
1. TwinSight X500 – full case study
2. Cross-section shader / clipping plane
3. Drone CAD-to-Web pipeline
4. WebGL optimization / mobile constraints
```

Deseables:

```
Bloque de código: text
5. Fastener modular system
6. Exploded view logic
7. UI bottom-sheet interaction system
8. Heuristic thermal visualization
```

11.4 Medios visuales

Incluir:

- hero video corto;
- GIF de selección de piezas;
- GIF de exploded view;
- GIF de cross-section;
- capturas before/after;
- diagrama de pipeline;
- diagrama de arquitectura;
- tabla de métricas;
- enlace a demo.

11.5 Métricas

Agregar cuando sean verificables:

```
Bloque de código: text
FPS
frame time
build size
load time
memory
draw calls
triangles
texture memory
SUS
NASA-TLX Raw
task completion
```

No inventar métricas. Si no están medidas, marcar como pendiente.

36.0.12 12. Cambios concretos para módulos previos

12.1 Actualizar 17_cv_base_and_role_variants.md

Agregar una sección de proyectos técnicos con enfoque medible:

```
Bloque de código: markdown
### Featured Technical Project

**TwinSight X500 – Unity WebGL Technical Visualization Viewer**
Built an interactive browser-based 3D viewer for inspecting the Holybro X500 V2 drone assembly. Implemented part hierarchy,
↪ selection, exploded view, cross-section and visual analysis modes for technical study and usability evaluation.
```

Cuando existan métricas finales, añadir:

```
Bloque de código: markdown
Measured WebGL performance across load time, frame time and memory usage.
```

12.2 Actualizar README del repositorio

El README ya está bien orientado, pero puede mejorar como portafolio técnico si añade:

- GIF hero;
- project architecture diagram;
- pipeline diagram;
- screenshots table;
- direct link to case study;
- measured performance section;
- “What I built” section.

12.3 Actualizar landing de GitHub Pages

La landing debe priorizar:

1. Demo inmediata.
2. Caso de estudio.
3. Capturas del flujo.
4. Métricas verificadas.
5. Enlaces a manuales.
6. CV/portafolio.

36.0.13 13. Estructura recomendada del sitio personal

```

Bloque de codigo: text
/
├── Home
├── Projects
│   ├── TwinSight X500
│   ├── Cross-section Shader
│   ├── CAD-to-Web Pipeline
│   └── WebGL Optimization Notes
├── Technical Notes
│   ├── Unity WebGL constraints
│   ├── Shader view modes
│   └── 3D model optimization
├── About
├── CV
└── Contact

```

36.0.14 14. Taxonomía de proyectos

Categoría	Uso
Technical Visualization	TwinSight, drone viewer
Unity WebGL	Browser deployment
Shaders	X-Ray, thermal, clipping
Tools	pipeline utilities
3D Production	Blender, bake, optimization
Research	thesis validation
Notes	technical articles

36.0.15 15. Plantilla para página de proyecto

```

Bloque de código: markdown
---
title: TwinSight X500
role: Unity WebGL Developer / Technical Artist
tools: Unity, Blender, WebGL, URP
status: public demo
year: 2026
---

# TwinSight X500

## Summary

TwinSight X500 is a browser-based interactive 3D viewer for technical inspection of the Holybro X500 V2 drone assembly.

## Problem

Traditional 2D manuals force users to mentally reconstruct spatial relationships between parts.

## Solution

The project provides an interactive 3D environment with selection, part hierarchy, exploded view, cross-section and visual
↪ analysis modes.

## My Role

I designed and implemented the Unity WebGL prototype, prepared the 3D asset pipeline and developed the interaction systems
↪ used in the final thesis demo.

## Technical Stack

- Unity 6000
- WebGL
- URP
- Blender
- GitHub Pages
- C#

## Key Systems

- App state machine
- Part selection
- Bottom-sheet UI
- Exploded view
- Cross-section
- View modes

```

```

- Thermal heuristic visualization

## Constraints

- Browser deployment
- Mobile-first interaction
- Academic validation scope
- Complex imported hierarchy
- Performance constraints

## Results

Add measured metrics here when available.

## Media

Add GIFs, screenshots and video.

## Links

- Demo
- GitHub
- Report
- Technical manual

```

36.0.16 16. Plantilla para breakdown técnico

```

Bloque de codigo: markdown
# Cross-section shader breakdown

## What I wanted to solve

Users needed to inspect internal spatial relationships without losing context of the full drone assembly.

## Constraint

The solution had to work in Unity WebGL and remain usable on mobile devices.

## Approach

I implemented a shader-based clipping plane controlled through the Analyze mode UI.

## Implementation

Explain the clipping logic, material setup, UI control and limitations.

## Result

Show GIF, screenshots and performance notes.

## Lessons

Explain what worked, what failed and what would be improved.

```

36.0.17 17. Riesgos

Riesgo	Mitigación
Portafolio solo visual	Añadir breakdowns
Claims sin evidencia	Medir antes de publicar
GitHub oculto	Enlace visible
Proyecto académico percibido como menor	Enfatizar alcance técnico
Falta de seniority	Posicionarse como junior/mid técnico
Demasiado texto	Usar jerarquía visual
Falta de demo	Priorizar build estable

36.0.18 18. Orden operativo recomendado

```

Bloque de codigo: text
1. Congelar demo pública.
2. Capturar GIFs clave.
3. Medir rendimiento verificable.
4. Crear case study principal.
5. Crear 2-3 breakdowns técnicos.
6. Actualizar README.
7. Construir landing personal.

```


8. Ajustar CV.
9. Publicar LinkedIn/GitHub profile.

36.0.19 19. Prompt para A5

```
Bloque de código: text
Execute A5_case_study_benchmark.

Objective:
Analyze 3-5 strong accessible technical case studies relevant to Alexander's positioning:
Unity Technical Artist, Real-Time 3D Developer, Unity WebGL Developer, Technical Visualization Developer, Shader/Tools
↔ Developer.

Prioritize:
- Jettelly Real-Time Potion Shader Breakdown;
- Julhe shader breakdowns;
- Francisco Múrias technical projects;
- other accessible technical art breakdowns with code, images, video or metrics.

Output:
- source inventory;
- case study structure;
- storytelling pattern;
- technical depth;
- visual evidence;
- metrics used;
- what Alexander should adapt;
- project page template;
- breakdown article template.

Rules:
- Use public accessible sources only.
- Do not use ArtStation if blocked.
- Do not invent missing data.
- Keep tables concise.
- Provide actionable recommendations.
```

36.0.20 20. Conclusión

El benchmark actualizado muestra que el portafolio de Alexander no debe depender de una galería tipo ArtStation. Su ventaja competitiva está en combinar:

- evidencia visual;
- explicación técnica;
- demo WebGL;
- código público;
- métricas verificables;
- claridad de rol;
- estudios de caso.

La prioridad debe ser construir un portafolio híbrido: visual como un artista técnico, documentado como un desarrollador, y medible como un proyecto de ingeniería.

Chapter 37

29_portfolio_case_study_benchmark

Fuente: 29_portfolio_case_study_benchmark.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1011

Ultima modificacion: 2026-06-11 18:08

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This module benchmarks public case studies relevant to **TwinSight X500** and Alexander Woodcock Salomón's positioning as:

Contenido completo depurado

37.0.1 Module status

```
Bloque de codigo: text
Module: 29_portfolio_case_study_benchmark.md
Input task: A5_technical_visualization_case_studies
Date generated: 2026-06-11
Status: Partial but usable
Source type: public web case studies
```

37.0.2 Scope

This module benchmarks public case studies relevant to **TwinSight X500** and Alexander Woodcock Salomón's positioning as:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist
focused on Unity WebGL, CAD-to-realtime optimization, interactive technical visualization, and technical UI.
```

The source collection focused on case studies involving:

- Unity WebGL;
- browser-based interactive 3D;
- technical visualization;
- digital twins;
- XR / VR training;
- product configurators;
- CAD / BIM / reality-capture workflows;

- industrial visualization;
- spatial onboarding;
- training simulations.

37.0.3 Important limitation

The A5 collection produced **7 usable detailed case studies**, not the original target of 20–30.

This module is still useful because the collected examples have enough structure to extract case-study patterns for TwinSight. However, it should be treated as:

```
Bloque de codigo: text
Case-study benchmark v1
```

A later expansion can add:

```
Bloque de codigo: text
29B_demo_reels_benchmark.md
29C_artstation_breakdown_benchmark.md
29D_case_study_benchmark_expansion.md
```

37.1 1. Source inventory

37.1.1 1.1 Case studies collected

#	Case study	Company	Source link
1	WebXR Product Configurator	HapzXR	[Open](https://hapzxr.com/case-studies/webxr-product-configurator/)
2	Digital Twin of Production Facility	HapzXR	[Open](https://hapzxr.com/case-studies/digital-twin-of-production-facility/)
3	BIM Site Overlay Visualization	HapzXR	[Open](https://hapzxr.com/case-studies/bim-site-overlay-visualization/)
4	Construction Confined Space Simulation	HapzXR	[Open](https://hapzxr.com/case-studies/construction-confined-space-simulation/)
5	Work Spatial — AI-Driven Onboarding Platform	Virtual Verse Studio	[Open](https://virtualverse.studio/projects/work-spatial/)
6	NBK Virtugate — National Bank of Kuwait	Virtual Verse Studio	[Open](https://virtualverse.studio/projects/nbk-virtugate/)
7	Empathy Lab — UK Rail Sector Training	Virtual Verse Studio	[Open](https://virtualverse.studio/projects/empathy-lab/)

37.1.2 1.2 Supporting source pages

Source	Use	Link
Unity Industry	Context for Unity in industry	[Open](https://unity.com/industry)
Unity Digital Twin Applications	Digital twin context	[Open](https://unity.com/topics/digital-twin-applications-and-use-cases)
Virtual Verse Unity WebGL page	WebGL service positioning	[Open](https://virtualverse.studio/unity-webgl/)
Prevu3D Case Studies	Digital twin expansion source	[Open](https://www.prevu3d.com/digital-twin-case-studies/)

37.2 2. Benchmark table — identity and relevance

Case	Sector	Project type	Relevance
WebXR Product Configurator	Industrial equipment	Web 3D configurator	Very high
Digital Twin Facility	Industrial operations	Real-time digital twin	Very high
BIM Site Overlay	Construction	BIM / VR overlay	High
Confined Space Simulation	Construction safety	VR procedure training	Medium-high
Work Spatial	Enterprise onboarding	WebGL XR platform	High
NBK Virtugate	Banking onboarding	WebGL + VR learning	High
Empathy Lab	Rail training	VR soft-skill simulation	Medium

37.2.1 Relevance definition

Level	Meaning
Very high	Directly maps to TwinSight: WebGL, technical visualization, digital twin, product inspection, or browser-based 3D
High	Strongly useful for case-study structure or interaction design
Medium-high	Useful for training/procedural logic
Medium	Useful for presentation pattern, but weaker domain match

37.3 3. Detailed case-study notes

37.3.1 3.1 WebXR Product Configurator — HapzXR

Source: <https://hapzxr.com/case-studies/webxr-product-configurator/>

Confirmed facts

Field	Data
Source type	Company case study
Company	HapzXR
Region	India / enterprise XR services
Industry	Industrial equipment
Technology	WebGL, Three.js, WebAR
Project type	Browser-based 3D product configurator
Access model	Browser + WebAR
Metrics	3× engagement; 45% more qualified leads; 2-week shorter pre-order cycle
Confidence	High

Case structure

Bloque de código: text
 Headline
 Short value proposition
 Challenge
 Solution
 Results
 Project overview
 Snapshot
 Delivery flow
 Related work
 Call to action

Problem pattern

The buyer could not evaluate industrial equipment physically. Static catalogs and 2D renderings failed to communicate scale, detail, and configuration.

Solution pattern

A browser-based 3D configurator allowed users to:

- rotate;
- zoom;
- customize finishes;
- inspect internal components;
- use WebAR placement;
- send configuration data to CRM.

Direct relevance to TwinSight

This is the closest collected analogue to TwinSight’s **browser-based inspection and technical product understanding**. It shows how a 3D viewer becomes stronger when framed as a decision-support or sales-support tool, not just a model viewer.

What Alexander can adapt

- Frame TwinSight as an **interactive technical understanding tool**.
- Use “2D support was insufficient” as the problem.
- Emphasize browser access.
- Show interaction modes clearly.
- Include measurable outcomes if defensible.
- Present features as user-facing workflows, not just implementation details.

What not to copy

- Do not claim CRM integration.
- Do not claim WebAR unless implemented.
- Do not claim business conversion metrics.
- Do not overstate industrial client impact.

37.3.2 3.2 Digital Twin of Production Facility — HapzXR

Source: [\[https://hapzxr.com/case-studies/digital-twin-of-production-facility/\]](https://hapzxr.com/case-studies/digital-twin-of-production-facility/)(<https://hapzxr.com/case-studies/digital-twin-of-production-facility/>)

Confirmed facts

Field	Data
Source type	Company case study
Company	HapzXR
Industry	Chemical processing / facility operations
Technology	Unity / Unreal listed; IoT; web; VR
Project type	Real-time digital twin
Access model	Web dashboard + VR
Metrics	30% downtime reduction; 50% planning-time reduction
Confidence	High for page content; medium for stack specificity

Case structure

```
Bloque de código: text
Title
Value proposition
Challenge
Solution
Results
Project overview
Snapshot
Outcome summary
Delivery flow
CTA
```

Problem pattern

Monitoring systems lacked spatial context. Operators could see alerts but not easily understand where those alerts belonged in the facility.

Solution pattern

A navigable 3D facility twin connected equipment to:

- status;
- maintenance history;
- predictive health indicators;
- IoT sensor feeds;
- web monitoring;
- VR planning.

Direct relevance to TwinSight

TwinSight is not a live digital twin, but it is **digital-twin-adjacent** because it maps a real physical product into an interactive 3D structure for understanding spatial relationships.

What Alexander can adapt

- Use the term **digital-twin-adjacent technical visualization** carefully.
- Show how each component maps to real-world parts.
- Add a “future extension” section for sensor data, maintenance status, or assembly state.
- Emphasize spatial context as the core value.
- Structure the case study around “2D documentation lacks spatial context.”

What not to copy

- Do not claim live IoT.
- Do not claim predictive maintenance.
- Do not claim production facility scale.
- Do not call TwinSight a full digital twin unless live state/data integration is added.

37.3.3 3.3 BIM Site Overlay Visualization — HapzXR

Source: <https://hapzxr.com/case-studies/bim-site-overlay-visualization/>

Confirmed facts

Field	Data
Source type	Company case study
Company	HapzXR
Industry	Construction / AEC
Technology	Unity, BIM data layering, VR review
Project type	BIM / site overlay visualization
Access model	VR review
Metrics	Faster decisions; earlier mismatch detection
Confidence	Medium-high

Problem pattern

Stakeholders lacked a unified way to compare site reality against BIM plans.

Solution pattern

Immersive BIM walkthroughs overlaid site imagery and video onto BIM models to compare planned and actual states.

Direct relevance to TwinSight

The equivalent idea for TwinSight would be overlaying or connecting:

- manufacturer diagrams;
- assembly instructions;
- labels;
- inspection zones;
- exploded-view states;
- cross-section states;
- real part references.

What Alexander can adapt

- Add a case-study section called **From 2D support to spatial comparison**.
- Use before/after visuals:
- original 2D/manual support;
- optimized 3D viewer;
- exploded view;
- clipping;
- labels.
- Present each technical mode as a “comparison aid.”

What not to copy

- Do not claim BIM.
- Do not claim site photo alignment.
- Do not claim construction domain experience unless applying broadly to visualization.

37.3.4 3.4 Construction Confined Space Simulation — HapzXR

Source: <https://hapzxr.com/case-studies/construction-confined-space-simulation/>

Confirmed facts

Field	Data
Source type	Company case study
Company	HapzXR
Industry	Construction safety
Technology	Unity, Quest/PCVR, instructor review
Project type	VR procedural training
Access model	VR
Metrics	Procedural consistency and confidence; no hard numeric metric
Confidence	Medium-high

Problem pattern

Classroom briefings did not create enough readiness for high-risk confined-space operations.

Solution pattern

A simulation-first training flow guided users through:

- permit verification;
- pre-entry checks;
- hazard controls;
- communication;
- emergency escalation;
- guided and assessment modes.

Direct relevance to TwinSight

TwinSight is not a safety-training system, but it has strong overlap with procedural understanding and guided technical comprehension.

What Alexander can adapt

- Add a **task-based walkthrough** to the case study:
- identify component;
- inspect assembly relation;
- activate exploded view;
- use clipping;
- compare with 2D support.
- Show how the system supports procedural reasoning.
- Connect SUS/NASA-TLX evaluation to training-system validation language.

What not to copy

- Do not claim VR safety training.
- Do not claim procedural certification.
- Do not use “safety outcomes” unless evaluated.

37.3.5 3.5 Work Spatial — AI-Driven Onboarding Platform — Virtual Verse Studio

Source: https://virtualverse.studio/projects/work-spatial/

Confirmed facts

Field	Data
Source type	Portfolio case study
Company	Virtual Verse Studio
Region	Egypt / USA
Industry	Enterprise onboarding
Technology	WebGL, GenAI, Web3
Project type	Browser-based XR onboarding MVP
Access model	WebGL
Metrics	On-time MVP; investor/client demos
Confidence	High

Case structure

Bloque de código: text
Breadcrumb
Tags
Title
One-sentence value proposition
Image
Overview
Challenge
Solution
Key features
Results & impact
Client quote
Links
More portfolio items
CTA

Problem pattern

A startup needed a fast MVP combining multiple emerging technologies.

Solution pattern

The studio delivered a browser-based 3D workspace with:

- avatar navigation;
- generative AI;
- Web3 identity;
- modular architecture.

Direct relevance to TwinSight

This case is relevant for **portfolio structure** more than domain. It shows a concise, recruiter-readable format.

What Alexander can adapt

- Use short tags at the top:
- Unity WebGL
- Technical Visualization
- CAD-to-Realtime

- Drone Assembly
- Use a one-sentence project value proposition.
- Add a **Key Features** section before deep technical details.
- Add a **Results & Validation** section using SUS/NASA-TLX.
- Add direct links to demo, GitHub, video, and thesis if public.

What not to copy

- Do not include AI/Web3 as primary unless relevant.
- Do not turn TwinSight into a hype-stack case study.
- Do not overfocus on MVP speed if the project was academic and thesis-driven.

37.3.6 3.6 NBK Virtugate — National Bank of Kuwait — Virtual Verse Studio

Source: <https://virtualverse.studio/projects/nbk-virtugate/>

Confirmed facts

Field	Data
Source type	Portfolio case study
Company	Virtual Verse Studio
Region	Egypt / USA
Industry	Banking / corporate onboarding
Technology	Unity WebGL + VR
Project type	Gamified onboarding
Access model	Browser + VR
Metrics	Improved comprehension and engagement; no numeric data
Confidence	High

Problem pattern

Traditional employee orientation was passive, resource-heavy, and inconsistent across locations.

Solution pattern

A custom WebGL environment allowed employees to explore departments, history, tools, and branded spaces through avatars.

Direct relevance to TwinSight

This example is relevant because TwinSight also converts passive/2D information into an active interactive learning environment.

What Alexander can adapt

- Use the contrast:
 - passive documentation;
 - active spatial exploration.
- Emphasize browser-first deployment.
- Present technical information as an interactive environment.
- Use “no install required” if true for the deployed build.
- Include “desktop browser + future XR optional” only if framed as future.

What not to copy

- Do not over-gamify TwinSight unless game mechanics exist.
- Do not claim enterprise deployment.
- Do not claim engagement improvement unless measured.

37.3.7 3.7 Empathy Lab — UK Rail Sector Training — Virtual Verse Studio

Source: [\[https://virtualverse.studio/projects/empathy-lab/\]](https://virtualverse.studio/projects/empathy-lab/)(<https://virtualverse.studio/projects/empathy-lab/>)

Confirmed facts

Field	Data
Source type	Portfolio case study
Company	Virtual Verse Studio
Region	Egypt / USA
Industry	Rail / training
Technology	VR training
Project type	Scenario-based training
Access model	VR
Metrics	Measurable behavior improvement; no numeric data
Confidence	Medium

Problem pattern

Traditional role-play and lectures lacked realism for high-stress scenarios.

Solution pattern

The platform used first-person VR, perspective switching, behavioral design, and structured debriefs.

Direct relevance to TwinSight

The domain is weaker, but the **training logic** is useful. TwinSight can borrow the idea of scenario-driven learning and debrief-like explanation.

What Alexander can adapt

- Add a “User task flow” section.
- Explain how the viewer supports comprehension, not only viewing.
- Include user evaluation results.
- Use Think-Aloud findings as qualitative validation.

What not to copy

- Do not claim behavioral training.
- Do not claim soft-skill outcomes.
- Do not make TwinSight sound like a rail-training product.

37.4 4. Cross-case patterns

37.4.1 4.1 Common case-study structure

Across the strongest examples, the structure is:

Bloque de codigo: text

1. Title

2. One-line value proposition

3. Hero image or video

4. Problem / challenge

5. Solution

6. Key features

7. Results / impact

8. Technical snapshot

9. Delivery / process

10. Related work or CTA

37.4.2 4.2 Recommended TwinSight case-study structure

Bloque de codigo: text

1. Title

2. One-sentence value proposition

3. Demo video / hero GIF

4. Problem: 2D documentation fails spatial reasoning

5. Solution: Unity WebGL technical visualization

6. Key features

7. Technical implementation

8. CAD-to-realtime pipeline

9. Optimization results

10. Evaluation results

11. Role and contribution

12. Constraints and limitations

13. Next steps

14. Links

37.4.3 4.3 Feature framing pattern

Do not list features as isolated technical modules.
Frame them as user-facing workflows.

Technical feature	Better case-study framing
Component selection	Identify parts and relationships
Exploded view	Understand assembly hierarchy
Cross-section / clipping	Inspect hidden internal structures
Visual modes	Compare technical interpretations
Technical UI	Access part data while inspecting
CAD optimization	Make complex CAD browser-ready
WebGL deployment	Remove install friction
SUS / NASA-TLX	Validate usability and cognitive load

37.5 5. Metrics pattern

37.5.1 5.1 Metrics seen in collected examples

Metric type	Example
Engagement	3× engagement
Lead quality	45% more qualified leads
Time reduction	2-week shorter cycle
Downtime	30% reduction

Metric type	Example
Planning time	50% reduction
Comprehension	Improved engagement/comprehension
Behavior	Measurable behavior change
Decision speed	Faster decision cycles

37.5.2 5.2 Metrics available for TwinSight

TwinSight has unusually strong validation data for a portfolio project:

Metric	TwinSight value
Optimized triangles	95,617
Original CAD routes	6.5M+ triangles
Participants	12
SUS average	91.88
NASA-TLX Raw, 3D viewer	8.69
NASA-TLX Raw, 2D support	19.89
Task-condition records	96

37.5.3 5.3 How to present metrics

Use direct, defensible phrasing:

Bloque de código: text
Optimized CAD-derived drone geometry from multi-million-triangle source routes into a 95,617-triangle WebGL-ready model.

Bloque de código: text
Evaluated the viewer with 12 participants using SUS, NASA-TLX Raw, and Think-Aloud protocols.

Bloque de código: text
Observed lower perceived workload in the 3D viewer condition compared with 2D support.

Do not overstate causality.

Avoid:

Bloque de código: text
Proved that 3D is better than 2D.

Use:

Bloque de código: text
The formative evaluation suggested lower perceived workload and high usability for the interactive 3D viewer.

37.6 6. Visual pattern benchmark

37.6.1 6.1 Media types used

Media type	Seen in examples	Recommended for TwinSight
Hero image	Yes	Yes
Demo video	Yes	Yes

Media type	Seen in examples	Recommended for TwinSight
Screenshot sequence	Yes	Yes
Feature cards	Yes	Yes
Process diagram	Sometimes	Yes
Before/after	Often implied	Yes
Metrics cards	Yes	Yes
Client quote	Sometimes	Replace with thesis/evaluation quote
CTA	Yes	Yes

37.6.2 6.2 Required TwinSight visuals

Visual	Purpose
Hero drone viewer shot	Immediate comprehension
Exploded view shot	Assembly understanding
Clipping/cross-section shot	Technical inspection
Component selection shot	Interaction proof
Visual modes grid	Technical art proof
CAD/high-poly/optimized comparison	Pipeline proof
Triangle reduction card	Optimization proof
SUS/NASA-TLX card	Validation proof
Mobile/web view	WebGL deployment proof
Architecture/pipeline diagram	Developer credibility

37.7 7. Language patterns to adopt

37.7.1 Strong phrases from the benchmark

Use adapted versions of these patterns:

```
Bloque de codigo: text
browser-based 3D
real-time customization
interactive technical visualization
spatial context
navigable 3D replica
technical inspection
structured user flow
web-accessible experience
no-install deployment
measurable validation
faster understanding
component-level exploration
```

37.7.2 Phrases to avoid

Avoid unless directly true:

```
Bloque de codigo: text
live IoT integration
predictive maintenance
enterprise deployment
production-ready digital twin
WebAR placement
CRM integration
safety outcome
behavior change
client revenue impact
industrial metaverse
```

37.8 8. Recommended TwinSight case-study copy skeleton

37.8.1 Title

Bloque de código: text
TwinSight X500 – Unity WebGL Technical Visualization for Drone Assembly Inspection

37.8.2 Subtitle

Bloque de código: text
An interactive browser-based 3D viewer that helps users understand the spatial relationships between drone components through
↪ selection, exploded view, clipping, visual modes, and technical part panels.

37.8.3 Problem

Bloque de código: text
Traditional 2D assembly documentation forces users to mentally reconstruct spatial relationships between parts. For complex
↪ drone assemblies, this increases cognitive effort and makes hidden structures, orientation, and component hierarchy harder
↪ to understand.

37.8.4 Solution

Bloque de código: text
TwinSight X500 transforms CAD-derived drone geometry into an optimized Unity WebGL experience. Users can inspect components,
↪ isolate assemblies, activate exploded views, cut through structures, switch visual modes, and access technical information
↪ directly in the browser.

37.8.5 Key features

Bloque de código: text
- Component selection and highlighting
- Exploded assembly view
- Cross-section / clipping inspection
- Technical part information panels
- Realistic, X-Ray, blueprint, ghosted, wireframe, solid, and thermal-style modes
- CAD-to-realtime geometry optimization
- Mobile-first WebGL interaction design
- Usability and workload evaluation

37.8.6 Results

Bloque de código: text
- Reduced CAD-derived geometry from routes exceeding 6.5M triangles to 95,617 optimized triangles
- Evaluated with 12 participants
- SUS average: 91.88
- NASA-TLX Raw: 8.69 for 3D viewer vs 19.89 for 2D support
- 96 task-condition records

37.8.7 Role

Bloque de código: text
I developed the Unity WebGL integration, runtime interaction systems, technical UI, visualization modes, asset optimization
↪ workflow, Blender-based preparation, documentation, and evaluation pipeline, using AI assistance as an implementation
↪ support tool while retaining responsibility for architecture, debugging, integration, validation, and final technical
↪ decisions.

37.9 9. TwinSight case-study section order

37.9.1 Recommended final order

```
Bloque de codigo: text
01. Hero
02. One-line value proposition
03. Problem
04. Solution
05. Demo video
06. Key features
07. Technical pipeline
08. CAD-to-realtime optimization
09. Interaction systems
10. Visual modes
11. Evaluation results
12. Contribution and AI-assisted workflow
13. Constraints and limitations
14. Next steps
15. Links
```

37.9.2 Why this order

The benchmark case studies show that readers need value first, then details.

Do not start with academic methodology.

Start with:

```
Bloque de codigo: text
What problem did this solve?
What can the user do?
What technical proof exists?
What evidence validates it?
```

Then include academic details deeper.

37.10 10. Gaps in current benchmark

37.10.1 Missing categories

The current dataset lacks enough examples from:

Category	Status
Drone visualization	Missing
Robotics visualization	Missing
CAD-to-Unity case studies	Weak
Unity WebGL technical inspection tools	Weak
Medical 3D visualization	Missing
Aerospace assembly visualization	Weak
Three.js technical viewers	Missing
WebGPU industrial viewers	Missing
Academic HCI visualization demos	Missing

37.10.2 Suggested expansion query set

Use these later:

```
Bloque de codigo: text
"CAD to Unity case study"
"Unity technical visualization case study"
"WebGL 3D inspection tool case study"
"drone 3D visualization case study"
```



```
"robotics digital twin Unity case study"
"Three.js product configurator case study"
"interactive assembly instructions 3D case study"
"WebGL technical viewer CAD"
"Unity digital twin manufacturing case study"
"3D maintenance training Unity case study"
```

37.11 11. How this changes the roadmap

37.11.1 No major strategy change

The case-study benchmark confirms the existing direction:

```
Bloque de codigo: text
Unity WebGL + technical visualization + CAD-to-realtime + browser-based interaction
```

remains the strongest route.

37.11.2 Stronger emphasis needed

TwinSight should be presented less like:

```
Bloque de codigo: text
academic thesis viewer
```

and more like:

```
Bloque de codigo: text
technical visualization case study with measured usability and workload evaluation
```

37.11.3 Best market language

Use:

```
Bloque de codigo: text
technical visualization
interactive 3D
browser-based 3D
CAD-to-realtime
component inspection
spatial understanding
digital-twin-adjacent
training and assembly support
```

Use cautiously:

```
Bloque de codigo: text
digital twin
simulation
XR
technical training
```

Avoid as primary:

```
Bloque de codigo: text
game project
3D art project
academic demo
prototype only
AI-built app
```

37.12 12. Required updates to existing files

37.12.1 Update 08_twinsight_x500_case_study.md

Add:

- problem/solution/results structure;
- short value proposition;
- feature cards;
- metrics cards;
- source-inspired case-study layout;
- explicit “not a full digital twin” limitation;
- digital-twin-adjacent language.

37.12.2 Update 20_portfolio_copy_and_site_structure.md

Add:

- case-study page structure;
- portfolio card copy;
- feature labels;
- CTA labels;
- metrics section.

37.12.3 Update 21_twinsight_demo_video_script_and_shotlist.md

Add:

- first 5 seconds: value proposition;
- feature sequence based on problem-solution-result;
- metrics card near end;
- final links.

37.12.4 Update 17_cv_base_and_role_variants.md

Add one case-study bullet pattern:

```
Bloque de codigo: text
Built a Unity WebGL technical visualization prototype for drone assembly inspection, transforming CAD-derived assets into an
↪ optimized browser-based viewer with component selection, exploded view, clipping, visual modes, and usability/workload
↪ evaluation.
```

37.13 13. Next task recommendation

The next task should be:

```
Bloque de codigo: text
Fl_demo_reels_benchmark
```

37.13.1 Tool recommendation

Tool	Use
Agent mode	Yes
Deep Research	No

Reason:

The current file defines what the case study should say. The next missing piece is how the **demo video** should be edited and sequenced.

37.13.2 Prompt for next task

Bloque de codigo: text

Execute F1_demo_reels_benchmark.

Objective:
Collect public demo reel and project demo examples relevant to TwinSight X500:
technical artist demo reels, Unity technical reels, real-time 3D developer reels, technical visualization demos, XR/simulation
↔ demos, CAD-to-realtime/product visualization demos, WebGL interactive 3D demos.

Find 15-25 examples.

Output structured tables only.

Fields:
- Demo label
- URL
- Platform
- Role target
- Duration
- Opening structure
- Shot sequence
- Captions/text usage
- Voiceover yes/no
- Technical breakdown yes/no
- Tools shown
- Metrics shown
- Call-to-action
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:
- Use only public videos/pages.
- Do not invent duration if not visible.
- If unavailable, write "not found".
- Keep cells short.
- Tables only.
- Include URLs.
- No strategy.

Chapter 38

29B_demo_reels_benchmark

Fuente: 29B_demo_reels_benchmark.md

Seccion: root

Estado: canonical

Rol: Modulo final complementario

Lineas originales: 882

Ultima modificacion: 2026-06-11 18:20

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

El archivo anterior **no estaba completo** para el propósito real de 29B.

Contenido completo depurado

38.0.1 0. Estado del módulo

```
Bloque de codigo: text
Module: 29B_demo_reels_benchmark.md
Input task: F1_demo_reels_benchmark
Date generated: 2026-06-11
Status: Complete v1
Source type: public videos, public portfolio pages, public company demo/case pages
Use: TwinSight X500 demo video planning and portfolio presentation
```

38.0.2 1. Veredicto de completitud

El archivo anterior **no estaba completo** para el propósito real de 29B.

Funcionaba como un complemento general de casos de estudio, pero no como un módulo autónomo de **demo reels / video benchmark**. Le faltaban:

- separación clara entre demo reels personales, reels corporativos y casos de estudio con demo;
- análisis de duración;
- análisis de apertura;
- secuencia de shots;
- uso de captions;
- uso de voz en off;
- uso de breakdown técnico;
- extracción de patrones aplicables al video de TwinSight;
- shot list recomendado;
- estructura de edición;
- guion base;

- criterios de calidad;
- errores a evitar.

Este archivo reemplaza al 29B anterior.

38.0.3 2. Objetivo del benchmark

Este módulo analiza cómo se presentan públicamente proyectos comparables o parcialmente comparables a **TwinSight X500**, especialmente en video.

El objetivo no es copiar contenido visual, sino extraer patrones de presentación para diseñar un demo video profesional de 60–120 segundos para un perfil de:

```
Bloque de código: text
Real-Time 3D Developer / Unity Technical Artist
Unity WebGL · CAD-to-Realtime · Technical Visualization · Interactive 3D
```

38.0.4 3. Perfil de referencia: TwinSight X500

TwinSight X500 debe presentarse como:

```
Bloque de código: text
A Unity WebGL technical visualization prototype for drone assembly inspection.
```

La demo debe comunicar:

- problema de documentación 2D;
 - solución en 3D interactivo;
 - selección de componentes;
 - exploded view;
 - clipping / cross-section;
 - modos visuales;
 - UI técnica;
 - pipeline CAD-to-realtime;
 - optimización;
 - validación con usuarios;
 - propiedad técnica honesta, incluyendo AI-assisted implementation si se menciona.
-

38.0.5 4. Limitaciones de la recolección

La tarea F1 produjo una base útil, pero no perfecta.

Limitación	Impacto
YouTube oculta datos sin sesión	URLs exactas o descripciones pueden estar incompletas
Vimeo bloqueó acceso	algunos reels no fueron verificables
ArtStation bloqueó por seguridad	algunos reels no fueron accesibles
Muchos reels son visuales, no técnicos	falta detalle de stack
Empresas publican casos, no reels puros	se usan como referencia de video/case structure
No todos muestran duración	se marca como not found

38.0.6 Regla de uso

Este benchmark debe usarse como guía de edición y estructura, no como evidencia salarial ni de demanda laboral.

38.1 5. Source inventory

38.1.1 5.1 Fuentes principales

#	Source label	Type	URL
1	Robin Cai Visual Effects & Technical Art Demoreel 2025	Personal reel	[YouTube search](https://www.youtube.com/results?search_query=Robin+Cai+2025+Visual+Effects+%26+Technical+Art+Demoreel)
2	Unity Developer Demo Reel 2025	Personal reel	[YouTube search](https://www.youtube.com/results?search_query=Unity+Developer+Demo+Reel+2025+Multiplayer+WebGL+Technical+Art+Environment+Art)
3	Technical Visualization Demo Reel — Kinetic Vision	Corporate reel	[YouTube search](https://www.youtube.com/results?search_query=Kinetic+Vision+Technical+Visualization+Demo+Reel)
4	Kinetic Vision Technical Visualization service	Corporate service page	[Website](https://kinetic-vision.com/technical-visualization/)
5	Industrial3D recent animations and demo reels	Corporate reel page	[Website](https://industrial3d.com/3d-animation-in-new-york-ny/)
6	Work Spatial — AI-Driven Onboarding Platform	Portfolio case demo	[Website](https://virtualverse.studio/projects/work-spatial/)
7	NBK Virtugate — WebGL + VR Banking	Portfolio case demo	[Website](https://virtualverse.studio/projects/nbk-virtugate/)
8	Empathy Lab — UK Rail VR Training	Portfolio case demo	[Website](https://virtualverse.studio/projects/empathy-lab/)
9	WebXR Product Configurator	Case study demo	[Website](https://hapzxr.com/case-studies/webxr-product-configurator/)
10	Digital Twin of Production Facility	Case study demo	[Website](https://hapzxr.com/case-studies/digital-twin-of-production-facility/)
11	BIM Site Overlay Visualization	Case study demo	[Website](https://hapzxr.com/case-studies/bim-site-overlay-visualization/)
12	Construction Confined Space Simulation	Case study demo	[Website](https://hapzxr.com/case-studies/construction-confined-space-simulation/)

38.1.2 5.2 Source quality

Source type	Usefulness	Risk
Personal technical art reel	High	may be visual-heavy
Unity/WebGL personal reel	High	may be game-heavy
Corporate visualization reel	Medium-high	may be prerendered
Case-study demo page	High	not always video-first
VR training case	Medium	weaker WebGL fit
Digital twin case	High	may imply larger team/product

38.2 6. Benchmark: demo examples

38.2.1 6.1 Personal technical reels

Demo	Role target	Duration	Relevance
Robin Cai 2025 Visual Effects & Technical Art Demoreel	Technical Artist	1:42	Medium
Unity Developer Demo Reel 2025	Unity / WebGL Developer	1:17	High

Notes

Personal reels tend to:

- start fast;
- show work immediately;
- use short clips;
- rely on captions or title cards;
- avoid long explanations;
- show breadth of skill;
- end with name/contact/portfolio.

Application to TwinSight

TwinSight should not behave like a general art reel. It should be closer to a **technical product demo** with reel pacing.

Use:

```
Bloque de código: text
fast visual proof + captions + metrics + short technical breakdown
```

Do not use:

```
Bloque de código: text
music-only montage with unclear functionality
```

38.2.2 6.2 Corporate technical visualization reels

Demo	Company	Domain	Relevance
Technical Visualization Demo Reel	Kinetic Vision	Product/process/system visualization	High
Interactive 3D Media Reel	Industrial3D	Industrial 3D / simulation / product	Medium-high
2022 Demo Reel	Industrial3D	Industrial animation	Medium
Bechtel Digital Twin video	Industrial3D	Digital twin / engineering	Medium-high

Notes

Corporate reels tend to:

- show many projects quickly;
- emphasize industry credibility;

- use polished visuals;
- show machines, cutaways, assemblies, flows, or environments;
- focus on technical communication;
- rarely show implementation details.

Application to TwinSight

TwinSight should borrow:

- clarity;
- object close-ups;
- section/cutaway shots;
- before/after visuals;
- captions that explain value.

TwinSight should not copy:

- purely cinematic pacing;
- photorealism-first framing;
- lack of technical ownership;
- generic marketing language.

38.2.3 6.3 WebGL / browser-based case demos

Case	Platform	Core value	Relevance
Work Spatial	WebGL	browser-based XR onboarding	High
NBK Virtugate	WebGL + VR	browser onboarding + optional VR	High
WebXR Product Configurator	WebGL / WebAR	product inspection/configuration	Very high

Notes

Browser-based demos are the closest reference for TwinSight because they prove:

- no-install access;
- interactive 3D in browser;
- real-time interaction;
- fast comprehension;
- user-facing workflows.

Application to TwinSight

The video should explicitly show:

Bloque de codigo: text
Runs in browser
Interactive component selection
Real-time camera movement
Technical UI
No-install viewer

38.2.4 6.4 Digital twin / simulation case demos

Case	Domain	Core feature	Relevance
Digital Twin of Production Facility	industrial operations	live IoT + 3D facility	High
BIM Site Overlay Visualization	construction	BIM + overlay	Medium-high
Construction Confined Space Simulation	safety training	guided procedural training	Medium

Notes

These cases are valuable for:

- explaining spatial context;
- showing how 3D replaces fragmented documentation;
- presenting workflows;
- connecting interaction to user decisions;
- using results/metrics.

Application to TwinSight

TwinSight should use similar language:

```
Bloque de codigo: text
spatial context
guided inspection
component-level understanding
interactive technical support
reduced cognitive load
```

But avoid claiming:

```
Bloque de codigo: text
live IoT
predictive maintenance
production digital twin
VR training certification
```

38.3 7. Video pattern extraction

38.3.1 7.1 Ideal duration

Video type	Common duration	Recommended for TwinSight
Personal technical reel	60–120 s	90 s
Project demo	90–180 s	90–120 s
Deep breakdown	3–8 min	separate video
Social teaser	15–30 s	optional
Recruiter clip	45–75 s	optional

38.3.2 Recommendation

Create three video assets:

Asset	Duration	Use
Main demo	90–120 s	portfolio, LinkedIn Featured
Short teaser	30–45 s	LinkedIn posts
Technical breakdown	4–7 min	interviews, GitHub README

38.3.3 7.2 Opening structure

Weak opening:

```
Bloque de codigo: text
Logo fade-in → slow orbit → title after 20 seconds
```

Strong opening:

```
Bloque de codigo: text
0–3 s: final polished viewer shot
3–6 s: value proposition
6–10 s: problem contrast
```

Recommended opening caption:

```
Bloque de codigo: text
TwinSight X500
Unity WebGL technical visualization for drone assembly inspection
```

Second caption:

```
Bloque de codigo: text
From static 2D documentation to interactive spatial understanding
```

38.3.4 7.3 Shot pacing

Shot type	Recommended length
Hero reveal	3–5 s
UI interaction	4–6 s
Exploded view	5–7 s
Clipping plane	5–7 s
Visual modes grid	5–8 s
CAD optimization comparison	5–7 s
Metrics card	4–6 s
End card	3–5 s

Avoid shots longer than 8 seconds unless the narration requires it.

38.4 8. TwinSight demo video architecture

38.4.1 8.1 Main 90-second version

Bloque de	codigo: text
00-05 s	Hook / hero shot
05-12 s	Problem: 2D documentation lacks spatial context
12-22 s	Solution: browser-based Unity WebGL viewer
22-34 s	Component selection and technical panels
34-46 s	Exploded view
46-58 s	Cross-section / clipping
58-68 s	Visual modes
68-78 s	CAD-to-realtime optimization
78-86 s	Evaluation metrics
86-90 s	Role, stack, links

38.4.2 8.2 120-second version

Bloque de	codigo: text
00-05 s	Hero shot
05-12 s	Project title and positioning
12-22 s	Problem context
22-32 s	WebGL browser interaction
32-44 s	Component selection
44-56 s	Exploded view
56-68 s	Cross-section
68-80 s	Visual modes
80-92 s	CAD optimization pipeline
92-102 s	UI Toolkit / runtime systems
102-112 s	Evaluation metrics
112-120 s	Links / stack / role

38.4.3 8.3 30-second teaser

Bloque de	codigo: text
00-03 s	Hero shot
03-07 s	Select component
07-12 s	Exploded view
12-17 s	Clipping
17-22 s	Visual modes
22-27 s	Metrics card
27-30 s	Portfolio link / title

38.5 9. Required shot list for TwinSight

38.5.1 9.1 Core shots

#	Shot	Purpose
1	Full drone hero view	first impression
2	Browser frame / URL bar	WebGL proof
3	Component hover/select	interaction proof
4	Info panel opens	technical UI proof
5	Exploded view activation	assembly understanding
6	Exploded view close-up	hierarchy proof
7	Clipping plane movement	inspection proof
8	Internal structure revealed	technical visualization
9	Visual modes grid	technical art proof
10	X-ray / ghosted mode	inspection style
11	Wireframe / blueprint mode	engineering aesthetic
12	Thermal-style mode	visual experimentation
13	CAD high-poly vs optimized	pipeline proof
14	Triangle metric card	optimization proof
15	SUS / NASA-TLX card	validation proof
16	End card with stack	recruiter clarity

38.5.2 9.2 Optional shots

Shot	Use
Blender viewport	asset pipeline
Unity editor scene	implementation proof
UI Toolkit hierarchy	developer proof
Code snippet	C# proof
mobile viewport	web/mobile proof
GitHub README	public proof
thesis evaluation sheet	academic validation

38.6 10. Caption system

38.6.1 10.1 Caption style

Use short captions, not paragraphs.

Good:

```
Bloque de codigo: text
Component selection + technical panel
```

Bad:

```
Bloque de codigo: text
This system allows the user to select different drone components and view the associated information while navigating the
↪ interface.
```

38.6.2 10.2 Caption bank

```
Bloque de codigo: text
Unity WebGL technical visualization
Browser-based drone assembly inspection
Component-level selection
Technical part information
Exploded assembly view
Cross-section inspection
Multiple visual modes
CAD-to-realtime optimization
95,617 optimized triangles
Evaluated with SUS + NASA-TLX
Built with Unity, C#, URP, UI Toolkit, Blender
```

38.6.3 10.3 Metrics captions

```
Bloque de codigo: text
6.5M+ CAD triangle routes → 95,617 optimized triangles
12 validation participants
SUS average: 91.88
NASA-TLX Raw: 8.69 viewer vs 19.89 2D support
96 task-condition records
```

Use “viewer vs 2D support” only if the methodology supports that comparison.

38.7 11. Voiceover recommendation

38.7.1 Main recommendation

Use **captions + music only** for the public reel.

Add voiceover only for the long technical breakdown.

38.7.2 Reason

Recruiters often skim videos silently. Captions are safer.

38.7.3 If using voiceover

Keep it technical and restrained.

Example:

Bloque de codigo: text
TwinSight X500 is a Unity WebGL technical visualization prototype for drone assembly inspection. It converts CAD-derived
↪ assets into an optimized browser-based viewer with component selection, exploded view, cross-section tools, technical
↪ panels, and validation through SUS and NASA-TLX.

38.8 12. Technical breakdown video

Create a separate video, not the main reel.

38.8.1 Recommended length

Bloque de codigo: text
4-7 minutes

38.8.2 Structure

Bloque de codigo: text
1. Problem and thesis context
2. CAD source and optimization constraints
3. Blender cleanup / retopology / baking
4. Unity WebGL architecture
5. Runtime interaction systems
6. UI Toolkit interface
7. Visual modes
8. Evaluation methodology
9. Limitations
10. Next steps

38.8.3 What to show

- screen capture from Unity editor;
- scene hierarchy;
- C# scripts, briefly;
- shader graph, briefly;
- profiler if available;
- before/after asset counts;
- GitHub README;
- final browser build.

38.9 13. Editing rules

38.9.1 13.1 Must do

- show final output in first 5 seconds;
- use captions on every feature;
- keep shots short;
- show interaction, not just orbit shots;
- include metrics;
- include stack;
- include one clear end card;
- export cleanly at 1080p minimum;
- use consistent typography;
- avoid excessive transitions.

38.9.2 13.2 Must avoid

- long intro logos;
- slow cinematic-only orbit;
- unclear UI text;
- music louder than explanation;
- too many effects;
- pretending it is a shipped commercial product;
- hiding that it is an academic/portfolio prototype;
- overusing “AI” in the main reel;
- adding unimplemented XR/IoT/WebAR features.

38.10 14. Export specifications

38.10.1 Main reel

Parameter	Recommendation
Resolution	1920×1080
Frame rate	30 fps or 60 fps
Duration	90–120 s
Format	MP4 H.264
Audio	optional music
Captions	burned-in
Aspect	16:9
Primary use	portfolio, LinkedIn, README

38.10.2 LinkedIn teaser

Parameter	Recommendation
Resolution	1080×1080 or 1920×1080
Duration	30–45 s
Captions	mandatory
Hook	first 2 seconds
CTA	portfolio link / project name

38.10.3 Vertical teaser

Parameter	Recommendation
Resolution	1080×1920
Duration	20–30 s
Use	optional
Risk	UI may be hard to read

38.11 15. Recommended demo reel script

38.11.1 15.1 90-second caption script

```

Bloque de código: text
[00-05]
TwinSight X500
Unity WebGL technical visualization for drone assembly inspection

[05-12]
Problem:
2D documentation forces users to mentally reconstruct spatial relationships.

[12-22]
Solution:
An interactive browser-based 3D viewer for technical inspection.

[22-34]
Select components.
Highlight relationships.
Open technical part information.

[34-46]
Exploded view:
understand assembly hierarchy and spatial placement.

[46-58]
Cross-section tools:
inspect hidden structures and internal relationships.

[58-68]
Visual modes:
realistic, X-ray, ghosted, blueprint, wireframe, thermal-style.

[68-78]
CAD-to-realtime pipeline:
optimized complex CAD-derived assets for WebGL.

[78-86]
Evaluation:
SUS 91.88
NASA-TLX Raw: 8.69 viewer vs 19.89 2D support
12 participants

[86-90]
Unity · C# · WebGL · URP · UI Toolkit · Blender
Built by Alexander Woodcock Salomón

```

38.11.2 15.2 Shorter LinkedIn teaser script

```

Bloque de código: text
Static 2D assembly docs are hard to understand.

TwinSight X500 turns drone assembly into an interactive Unity WebGL viewer.

Select components.
Explode the assembly.
Cut through structures.
Switch visual modes.
Inspect technical data.

CAD-derived assets optimized for browser-based real-time 3D.

```

38.12 16. Portfolio embedding strategy

38.12.1 Where to place the video

Location	Video
Portfolio homepage	30–45 s teaser
TwinSight case study top	90–120 s main demo
GitHub README	compressed GIF + YouTube/Vimeo link
LinkedIn Featured	main video or case-study link
ArtStation	visual breakdown and short video
Interview follow-up	technical breakdown

38.12.2 Recommended hosting

Platform	Use
YouTube unlisted/public	main video
Vimeo	polished portfolio embed
GitHub README	GIF/video thumbnail
LinkedIn native upload	short teaser
Portfolio site	embedded main reel

38.13 17. GitHub README video integration

Use this structure:

```
Bloque de codigo: markdown
## Demo

[![TwinSight X500 Demo](./media/twinsight-demo-thumbnail.jpg)](https://youtube.com/...)

## Highlights
- Unity WebGL technical visualization
- CAD-to-realtime optimization
- Component selection and technical UI
- Exploded view and clipping tools
- Visual modes for inspection
- SUS + NASA-TLX evaluation
```

Add a lightweight GIF only if file size is controlled.

38.14 18. ArtStation video integration

ArtStation should not be only a final render gallery.

Use:

```
Bloque de codigo: text
1. 30–60 s video
2. final beauty screenshots
3. visual modes grid
4. topology / optimization screenshots
5. UI screenshots
6. breakdown captions
7. tools used
8. role and contribution
```


ArtStation tags:

```
Bloque de codigo: text
Unity
Technical Art
Realtime
WebGL
Blender
CAD Optimization
Visualization
Drone
Interactive 3D
```

38.15 19. What this changes in the roadmap

38.15.1 Keep

```
Bloque de codigo: text
TwinSight remains the flagship.
```

38.15.2 Add

```
Bloque de codigo: text
TwinSight needs a polished video asset before heavy applications.
```

38.15.3 Priority

The video is not optional. It is one of the highest-leverage artifacts because:

- recruiters skim;
- technical managers need proof fast;
- Unity/WebGL projects are hard to understand from text only;
- the candidate lacks formal industry experience;
- the video can compensate by showing proof.

38.16 20. Minimum viable demo video checklist

Before applications, produce at least:

```
Bloque de codigo: text
[ ] 30-45 s teaser
[ ] 90-120 s main demo
[ ] 10-15 strong screenshots
[ ] GitHub README embed
[ ] portfolio case-study embed
[ ] LinkedIn Featured entry
```

38.16.1 Acceptance criteria

The main demo is acceptable if a viewer understands within 20 seconds:

```
Bloque de codigo: text
what it is,
what problem it solves,
what the user can do,
why it is technically relevant.
```

38.17 21. Next recommended task

After this module, proceed with:

Bloque de codigo: text
F3_artstation_breakdown_benchmark

38.17.1 Tool recommendation

Tool	Use
Agent mode	Yes
Deep Research	No

38.17.2 Reason

The demo reel benchmark defines video structure. The next missing piece is the **visual breakdown style** for ArtStation / portfolio pages, especially:

- topology;
- materials;
- optimization;
- Blender portrait;
- technical art breakdown;
- visual modes;
- real-time asset presentation.

38.17.3 Prompt for next task

Bloque de codigo: text
Execute F3_artstation_breakdown_benchmark.

Objective:
Collect ArtStation or portfolio breakdowns useful for Alexander’s Blender portrait, TwinSight X500 and technical art
↔ presentation.

Find 15-25 examples involving:
- topology breakdown;
- material breakdown;
- grooming breakdown;
- lighting breakdown;
- shader breakdown;
- real-time asset breakdown;
- before/after optimization;
- character study presentation;
- technical art process documentation;
- Unity technical art breakdown;
- real-time visualization breakdown.

Output structured tables only.

Fields:
- Project label
- URL
- Platform
- Breakdown type
- Tools shown
- Images used
- Video/GIF used
- Technical text depth
- Tags
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:
- Public pages only.
- No invented data.

- If unavailable, write "not found".
- Keep cells short.
- Tables only.
- Include URLs.
- No strategy.

Chapter 39

29C_artstation_breakdown_benchmark

Fuente: 29C_artstation_breakdown_benchmark.md

Seccion: root

Estado: reference

Rol: Benchmark o complemento

Lineas originales: 838

Ultima modificacion: 2026-06-11 18:27

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

The previous F3 output was **not complete enough** as a final 29C module.

Contenido completo depurado

39.0.1 0. Module status

```
Bloque de codigo: text
Module: 29C_artstation_breakdown_benchmark.md
Input task: F3_artstation_breakdown_benchmark
Date generated: 2026-06-11
Status: Complete v1
Source type: public portfolio-pattern benchmark + accessible snippet-derived examples
Primary use: ArtStation / portfolio breakdown structure for TwinSight X500 and Blender portrait
```

39.0.2 1. Completion audit

The previous F3 output was **not complete enough** as a final 29C module.

It had useful general guidance, but it lacked:

- a clear module structure;
- source-quality classification;
- separate treatment of TwinSight and the Blender portrait;
- breakdown taxonomy;
- ArtStation post architecture;
- image requirements;
- technical captions;
- publication checklist;
- mistakes to avoid;
- direct update instructions for portfolio, GitHub and LinkedIn;
- a retry prompt for future deeper ArtStation collection.

This version completes the module as an actionable benchmark and production guide.

39.0.3 2. Scope

This module defines how Alexander should present technical-art and real-time 3D breakdowns on:

- ArtStation;
- personal portfolio;
- GitHub README;
- LinkedIn Featured;
- interview follow-up materials.

It focuses on two assets:

1. TwinSight X500

Unity WebGL technical visualization for drone assembly inspection.

1. Blender hyperrealistic portrait

Supporting 3D art / character-art signal, useful only if presented with technical breakdown depth.

39.0.4 3. Source limitations

ArtStation often blocks automated access through security checks. Several pages and snippets were visible through search, but full profile/project inspection was not reliably available.

Therefore, this module should be treated as:

Bloque de codigo: text
Portfolio breakdown structure benchmark v1

not as a complete database of verified ArtStation examples.

39.0.5 4. Source quality table

Source type	Reliability	Use
Fully accessible portfolio page	High	direct structure extraction
Search snippet from ArtStation	Medium	title/tag signal only
YouTube/Vimeo reel page	Medium	video structure
Studio case study	High	project narrative
General industry pattern	Medium	layout and content planning
Unopened ArtStation page	Low	do not cite as factual detail

39.0.6 5. Accessible benchmark signals

39.0.7 5.1 Search-snippet signals from ArtStation-like pages

Signal	What it indicates	Use for Alexander
Technical Artist Demo Reel 2025	reels are common for TA profiles	publish TwinSight video
Made with Unity	engine tag matters	tag Unity clearly
Unity foliage shader	shader breakdowns are portfolio assets	visual modes deserve breakdown
Unity URP water shader	URP is valid TA signal	state URP usage
Unity procedural fur shader	procedural/shader studies are valued	avoid claiming shader depth without proof
3D Rigger / Technical Artist Demo Reel	hybrid roles use reels	video must show systems
topology breakdown	expected for character/asset work	include wireframes
material breakdown	expected for realistic work	include texture/material maps
shader breakdown	expected for TA	include Shader Graph/code if present
game-ready breakdown	expected for real-time assets	include optimization metrics

39.0.8 5.2 Studio/case-study signals already collected

Source	Relevant signal
HapzXR WebXR Product Configurator	browser-based 3D + product interaction
HapzXR Digital Twin Facility	spatial context + operational visualization
HapzXR BIM Overlay	model comparison + overlay logic
Virtual Verse Work Spatial	concise project page structure
Virtual Verse NBK Virtugate	WebGL + VR onboarding presentation
Industrial3D reels	industrial technical animation language
Kinetic Vision reel	technical visualization as product/process/system communication

39.0.9 6. Breakdown taxonomy

39.0.10 6.1 Technical art breakdown types

Type	Typical content	Fit for TwinSight
Shader breakdown	graphs, nodes, render modes	High
Optimization breakdown	before/after geometry, texture budgets	Very high
Pipeline breakdown	CAD → Blender → Unity → WebGL	Very high
Tool breakdown	custom editor/runtime systems	High
Interaction breakdown	selection, UI, camera, states	Very high
Visual-mode breakdown	X-ray, ghosted, blueprint, thermal	Very high
Profiling breakdown	FPS, memory, load, draw calls	High
UX evaluation breakdown	SUS, NASA-TLX, Think-Aloud	Very high
Character art breakdown	topology, UVs, textures, grooming	Medium
Environment breakdown	scene assembly, lighting, props	Low-medium

39.0.11 6.2 Character / portrait breakdown types

Type	Typical content	Fit for Blender portrait
Sculpt breakdown	blockout to final sculpt	High
Topology breakdown	face loops, eyelids, mouth, deformation	High
UV breakdown	UDIMs/UVs, texture density	High
Skin material breakdown	albedo, roughness, SSS, displacement	High
Groom breakdown	hair guides, eyebrows, beard	High
Lighting breakdown	HDRI, key/fill/rim, camera	Medium-high
Render breakdown	passes, compositing	Medium
Real-time conversion	optimization for Unity/Unreal	High if done

39.0.12 7. ArtStation post architecture

39.0.13 7.1 Ideal structure for TwinSight ArtStation post

```

Bloque de código: text
01. Cover image
02. 30-60 second embedded video
03. One-sentence project summary
04. Role / contribution
05. Final screenshots
06. Feature breakdown
07. CAD-to-realtime pipeline
08. Optimization metrics
09. Visual modes breakdown
10. Unity interaction systems
11. UI / technical panels
12. Evaluation results
13. Tools used
14. Limitations
15. Links

```

39.0.14 7.2 Ideal structure for Blender portrait ArtStation post

Bloque de codigo: text

01. Cover render
02. Turntable video
03. Short project summary
04. Reference board
05. Sculpt stages
06. Topology
07. UVs
08. Texture maps
09. Skin shader / material setup
10. Grooming
11. Lighting
12. Final renders
13. Real-time/game-ready note if applicable
14. Tools used
15. Lessons learned

39.0.15 8. TwinSight breakdown specification

39.0.16 8.1 Cover

The cover should not be a generic drone render.
Use one of:

Cover option	Strength
Full drone with technical UI visible	best overall
Exploded view with labels	strongest assembly signal
Cross-section with UI panel	strongest inspection signal
Grid of visual modes	strongest technical-art signal

Recommended cover title:

Bloque de codigo: text

TwinSight X500
Unity WebGL Technical Visualization

Recommended subtitle:

Bloque de codigo: text

Drone assembly inspection · CAD-to-realtime · interactive 3D

39.0.17 8.2 Opening copy

Use short copy:

Bloque de codigo: text

TwinSight X500 is a Unity WebGL technical visualization prototype for inspecting the assembly of a Holybro X500 V2 drone. It
↔ converts CAD-derived assets into an optimized browser-based 3D viewer with component selection, exploded view, clipping
↔ tools, visual modes and usability/workload evaluation.

Do not open with:

Bloque de codigo: text

This was my thesis project about Design Science Research...

Academic methodology should appear later.

39.0.18 8.3 Role statement

Use:

Bloque de codigo: text
Role:
Real-time 3D development, Unity WebGL integration, C# runtime systems, technical UI, visual modes, Blender optimization
↔ workflow, evaluation design and documentation.

If AI assistance is mentioned:

Bloque de codigo: text
AI assistance was used for implementation support and iteration. Architecture, debugging, integration, validation, final
↔ decisions and project ownership remained my responsibility.

39.0.19 8.4 Key feature cards

Use concise cards:

Feature	Caption
Component selection	Inspect individual drone parts and relationships
Exploded view	Reveal assembly hierarchy and spatial placement
Cross-section	Cut through structures to expose hidden geometry
Visual modes	Switch between realistic, X-ray, ghosted, blueprint and wireframe views
Technical UI	Display part information while inspecting the model
WebGL deployment	Run the viewer directly in a browser
Evaluation	Measure usability and workload using SUS and NASA-TLX

39.0.20 8.5 CAD-to-realtime section

Recommended structure:

Bloque de codigo: text
CAD source
↓
conversion / tessellation
↓
Blender cleanup
↓
low-poly optimization
↓
UV / bake / material preparation
↓
Unity import
↓
WebGL deployment

Include:

- before/after triangle counts;
- wireframe comparison;
- modular fastener system;
- texture/material strategy;
- constraints for mobile/web;
- limitations.

39.0.21 8.6 Optimization metrics section

Use a metrics card layout.

Metric	Value
Optimized geometry	95,617 triangles
Source CAD routes	6.5M+ triangles
Participants	12
SUS average	91.88
NASA-TLX Raw, 3D viewer	8.69
NASA-TLX Raw, 2D support	19.89
Task-condition records	96

Add context:

Bloque de codigo: text

Evaluation was formative and academic, not a production deployment benchmark.

39.0.22 8.7 Visual modes breakdown

Create a grid:

Mode	What to show
Realistic	final material view
X-ray	internal structure readability
Ghosted	partial transparency
Blueprint	technical drawing aesthetic
Wireframe	topology / engineering signal
Solid	neutral inspection view
Thermal-style	qualitative visual analysis, not real FEA

Important wording:

Bloque de codigo: text

Thermal-style is a qualitative visual mode, not a physical thermal simulation.

39.0.23 8.8 Unity systems breakdown

Include one compact section:

Bloque de codigo: text

Runtime systems:

- selection/highlighting
- camera control
- exploded-view state
- clipping plane control
- visual mode switching
- bottom-sheet UI
- part metadata panel

Optional screenshot types:

- Unity hierarchy;
- script inspector;
- UI Toolkit panel;
- Shader Graph;
- C# snippet;
- profiler screenshot.

Do not include too much code on ArtStation. Put code detail in GitHub.

39.0.24 9. Blender portrait breakdown specification

39.0.25 9.1 Purpose

The Blender portrait should not be positioned as the main employability proof for technical visualization roles. It should function as:

Bloque de código: text
supporting evidence of high-end 3D fundamentals, anatomy, material work, grooming and visual discipline.

39.0.26 9.2 Required sections

Section	Required content
Final render	2–4 strongest images
Reference	board or brief reference explanation
Sculpt	blockout / mid / final
Topology	face loops and clean deformation
UVs	layout and density
Skin material	texture maps and shader setup
Eyes	iris/cornea material
Hair/groom	eyebrows, beard, eyelashes, hair guides
Lighting	setup diagram
Render	camera, focal length, compositing
Real-time adaptation	optional but valuable

39.0.27 9.3 What makes it relevant to technical art

Add a section:

Bloque de código: text
Technical focus:
- topology decisions
- material layering
- groom setup
- render optimization
- texture organization
- possible real-time conversion path

39.0.28 9.4 What to avoid

Do not frame it as:

Bloque de código: text
main portfolio proof for Unity WebGL roles.

Do frame it as:

Bloque de código: text
supporting 3D craft and technical presentation.

39.0.29 10. Image requirements

39.0.30 10.1 TwinSight images

Image	Priority
Hero final viewer	Required
Exploded view	Required

Image	Priority
Component selection	Required
Cross-section	Required
Visual modes grid	Required
CAD before/after	Required
Wireframe comparison	Required
UI panel close-up	Required
Evaluation metrics card	Required
Unity editor screenshot	Recommended
Blender cleanup screenshot	Recommended
WebGL browser screenshot	Required
GitHub / README preview	Optional

39.0.31 10.2 Blender portrait images

Image	Priority
Final portrait render	Required
Close-up face	Required
Side/profile view	Required
Wireframe overlay	Required
UV layout	Required
Texture maps	Required
Groom guides	Recommended
Lighting setup	Recommended
Clay render	Recommended
Turntable video	Recommended

39.0.32 11. Caption rules

39.0.33 11.1 Good caption style

Use:

```
Bloque de codigo: text
CAD-derived mesh optimized for browser-based real-time inspection.
```

Avoid:

```
Bloque de codigo: text
Here you can see the model that I worked on after importing it and making changes because it was too heavy and needed to work
↪ on the web.
```

39.0.34 11.2 TwinSight caption bank

```
Bloque de codigo: text
Unity WebGL technical visualization prototype
Component-level drone assembly inspection
CAD-to-realtime optimization pipeline
Exploded view for spatial hierarchy
Cross-section tool for hidden structures
Technical UI with part information
Multiple visual modes for inspection
Browser-based deployment, no install required
SUS and NASA-TLX formative evaluation
```

39.0.35 11.3 Portrait caption bank

```
Bloque de codigo: text
High-resolution sculpt
Clean facial topology
```

```

Skin material and texture layering
Eyebrow and beard grooming
Eye shader setup
Controlled studio lighting
Realistic render pass
Game-ready adaptation path

```

39.0.36 12. Tools section

39.0.37 12.1 TwinSight tools

Use a clean tools list:

```

Bloque de codigo: text
Unity
C#
Unity WebGL
URP
UI Toolkit
Shader Graph
Blender
Marmoset Toolbag
RizomUV
GitHub

```

Only include Marmoset/RizomUV if actually used in the final asset pipeline.

39.0.38 12.2 Portrait tools

```

Bloque de codigo: text
Blender
Cycles / Eevee Next
Substance 3D Painter
RizomUV / Blender UV tools
Marmoset Toolbag
Photoshop / Krita

```

Only include actual tools.

39.0.39 13. Tags

39.0.40 13.1 TwinSight ArtStation tags

```

Bloque de codigo: text
Unity
Unity WebGL
Technical Art
Real-Time 3D
Technical Visualization
CAD Optimization
Drone
Interactive 3D
CSharp
URP
Blender
Digital Twin
Simulation

```

Use Digital Twin cautiously.

39.0.41 13.2 Blender portrait tags

```

Bloque de codigo: text
Blender
Character Art
Portrait
Realistic Character
Topology
Grooming

```

Skin Shader
Lookdev
3D Art
Technical Art

39.0.42 14. Recommended layouts

39.0.43 14.1 TwinSight layout

Bloque de codigo: text
Hero image
Video
Problem/Solution
Feature grid
Pipeline diagram
Optimization before/after
Interaction screenshots
Visual modes grid
Evaluation metrics
Tools
Links

39.0.44 14.2 Portrait layout

Bloque de codigo: text
Hero render
Turntable
Reference
Sculpt process
Topology
UVs
Textures
Materials
Groom
Lighting
Final renders
Tools

39.0.45 15. ArtStation publication checklist

39.0.46 15.1 TwinSight checklist

Bloque de codigo: text
[] Cover image clear at thumbnail size
[] Video under 90 seconds embedded near top
[] Project explained in first 2 sentences
[] Role clearly stated
[] Unity WebGL visible
[] Component selection shown
[] Exploded view shown
[] Cross-section shown
[] Visual modes shown
[] CAD optimization shown
[] Metrics shown
[] Limitations stated
[] GitHub link added
[] Portfolio link added
[] Demo link added
[] Tags added

39.0.47 15.2 Portrait checklist

Bloque de codigo: text
[] Final render strong enough for cover
[] Turntable or image sequence included
[] References shown or described
[] Sculpt stages shown
[] Topology shown
[] UVs shown
[] Texture maps shown
[] Groom shown if relevant
[] Lighting shown
[] Tools listed
[] Technical contribution clear

39.0.48 16. Common mistakes

Mistake	Why it hurts
Only showing final renders	hides technical skill
No wireframes	weakens real-time credibility
No metrics	weakens technical proof
Long academic intro	recruiters disengage
Too much text	ArtStation is visual-first
No role statement	ownership unclear
Unclear AI usage	trust risk
Broken demo links	credibility loss
Claiming digital twin too strongly	overclaim risk
Hiding WebGL constraints	weak technical honesty

39.0.49 17. Recommended TwinSight ArtStation copy

39.0.50 17.1 Short intro

Bloque de código: text
TwinSight X500 is a Unity WebGL technical visualization prototype for drone assembly inspection. It transforms CAD-derived
↪ assets into an optimized browser-based 3D viewer with component selection, exploded view, cross-section tools, visual
↪ modes and technical UI panels.

39.0.51 17.2 Role

Bloque de código: text
Role: Real-time 3D development, Unity WebGL integration, C# runtime systems, technical UI, visual modes, Blender optimization
↪ workflow, documentation and formative evaluation.

39.0.52 17.3 Limitations

Bloque de código: text
This is an academic/portfolio prototype, not a deployed industrial digital twin. It does not include live IoT data, predictive
↪ maintenance, WebAR, or production monitoring. The thermal-style view is a qualitative visual mode, not a physical
↪ simulation.

39.0.53 17.4 Links section

Bloque de código: text
Links:
- Live demo: [add link]
- GitHub: [add link]
- Case study: [add link]
- Demo video: [add link]

39.0.54 18. Recommended Blender portrait ArtStation copy

39.0.55 18.1 Short intro

Bloque de código: text
A realistic Blender portrait study focused on facial anatomy, skin lookdev, grooming, lighting and technical presentation. The
↪ project is included as supporting evidence of 3D craft, material discipline and character-art fundamentals.

39.0.56 18.2 Role

Bloque de código: text
Role: sculpting, modeling, material setup, grooming, lighting, rendering and breakdown documentation.

39.0.57 18.3 Technical focus

Bloque de código: text
 Technical focus:

- clean facial topology
- skin material layering
- eye material setup
- eyebrow/beard grooming
- lighting and render control
- possible real-time adaptation path

39.0.58 19. Integration with existing roadmap

39.0.59 19.1 Impact on role strategy

This module does not change the main job-search strategy.

It reinforces:

Bloque de código: text
 TwinSight must be the flagship.
 The Blender portrait must be secondary.
 ArtStation must show process, not just beauty shots.

39.0.60 19.2 Impact on portfolio order

Recommended portfolio order:

Bloque de código: text

1. TwinSight X500
2. ARA Framework
3. Blender portrait
4. Smaller supporting projects

ArtStation-specific order:

Bloque de código: text

1. TwinSight X500 technical breakdown
2. Blender portrait breakdown
3. Shader/visual-mode studies if separated

39.0.61 20. Updates required to other files

39.0.62 20.1 Update 20_portfolio_copy_and_site_structure.md

Add:

Bloque de código: text
 ArtStation publication flow:

- TwinSight technical breakdown
- portrait technical breakdown
- visual-mode study posts

39.0.63 20.2 Update 21_twinsight_demo_video_script_and_shotlist.md

Add:

Bloque de código: text
 ArtStation-specific teaser export:

- 30-60 seconds
- no long intro
- burned-in captions
- feature-focused

39.0.64 20.3 Update 19_github_readme_templates.md

Add:

```
Bloque de codigo: text
README should link to ArtStation breakdown for visual process.
ArtStation should link back to GitHub for code/process credibility.
```

39.0.65 21. Next recommended task

Proceed with:

```
Bloque de codigo: text
30_live_job_postings_market_snapshot.md
```

Recommended subtask:

```
Bloque de codigo: text
B1_unity_realtime3d_live_jobs
```

39.0.66 Tool recommendation

Tool	Use
Agent mode	Yes
Deep Research	No

Reason:

The benchmark sequence for presentation artifacts now has:

```
Bloque de codigo: text
29 case-study benchmark
29B demo reels benchmark
29C ArtStation breakdown benchmark
```

The next highest value is to collect live job postings and compare actual requirements against the prepared materials.

39.0.67 22. Prompt for next task

```
Bloque de codigo: text
Execute B1_unity_realtime3d_live_jobs.

Objective:
Collect live job postings for roles aligned with Alexander's primary target:
Unity Developer, Real-Time 3D Developer, Unity Technical Artist, Unity WebGL Developer, Interactive 3D Developer, Technical
↔ Visualization Developer.

Candidate context:
Alexander is based in Colombia and targets remote international work. His strongest project is TwinSight X500, a Unity WebGL
↔ technical visualization prototype for drone assembly inspection with CAD-to-realtime optimization, C#, URP, UI Toolkit,
↔ Blender, exploded view, clipping, visual modes and usability/workload evaluation.

Output structured tables only.

Fields:
- Company
- Role title
- Job URL
- Country / region
- Remote scope
- Employment model
- Salary if visible
- Seniority
- Required skills
```


- Preferred skills
- Portfolio expectations
- GitHub expectations
- Work authorization notes
- Colombia remote eligibility
- Relevance to TwinSight
- Risk / mismatch
- Confidence
- Date checked

Rules:

- Use only live or recently listed postings.
- Do not assume remote means worldwide.
- Mark US-only, EU-only, LATAM-only, hybrid, onsite or unknown.
- Do not invent salary.
- If unavailable, write “not found”.
- Keep cells short.
- Include URLs.
- Tables only.
- No strategy.

Chapter 40

30 — Live job postings market snapshot

Fuente: 30_live_job_postings_market_snapshot.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 221

Ultima modificacion: 2026-06-11 19:53

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

— module: 30_live_job_postings_market_snapshot status: complete_from_B1_B2_B5_inputs created: 2026-06-12 source_modules: - agente_B1.pdf - agente_B2_digital_twin_simulation_live_jobs_partial_capture.md - agente_B5.pdf target_profile: Alexander Woodcock Salomon focus: - Unity Developer - Real-Time 3D Developer - Unity Technical Artist - Unity WebGL Developer - Interactive 3D Developer - Digital Twin / Simulation Developer - LATAM nearshore Unity roles —

Contenido completo depurado

module: 30_live_job_postings_market_snapshot

status: complete_from_B1_B2_B5_inputs

created: 2026-06-12

source_modules:

- agente_B1.pdf
- agente_B2_digital_twin_simulation_live_jobs_partial_capture.md
- agente_B5.pdf

target_profile: Alexander Woodcock Salomon

focus:

- Unity Developer
 - Real-Time 3D Developer
 - Unity Technical Artist
 - Unity WebGL Developer
 - Interactive 3D Developer
 - Digital Twin / Simulation Developer
 - LATAM nearshore Unity roles
-

40.0.1 1. Source status

Source	Scope	Status	Usability
B1	Unity / Real-Time 3D	Verified capture	High
B2	Digital Twin / Simulation	Partial snippet capture	Low-medium
B5	LATAM nearshore	Verified capture	High

B1 and B5 are usable as job-posting evidence. B2 is useful only as market signal because several job boards blocked access or required verification.

40.0.2 2. Consolidated market count

Category	Count	Evidence quality
Verified / captured postings	9	Medium-high
Snippet-only market signals	6	Low
LATAM-specific roles	3	Medium-high
Contractor / freelance signals	3	Medium
Digital twin / simulation signals	1	Low-medium

40.0.3 3. Verified / captured job postings

Company	Role	Region	Remote scope	Salary	Seniority	Fit	Confidence
RJ Games	Unity Developer	Cyprus	Remote anywhere	Negotiable	Middle	Medium	0.60
FLEXUS	Graphic Programmer / Render Developer	Ukraine	Remote	not found	Middle	High	0.70
HyperVR Games	Senior VR Unity Developer	Global / Europe	Remote flexible	Negotiable	Senior	High	0.75
Flat2VR Studios	Unity Technical Artist	EU & NA	Remote	not found	Mid-senior	High	0.65
Mudtek	VFX Artist, Unity / Shader Graph	Puerto Rico	Global remote	Salary proposed	Senior	Medium-high	0.70
DreamForge	Senior Unity Developer	United States	Fully remote	Competitive + equity	Senior	Medium-high	0.68
Azumo	Unity Developer	Latin America	LATAM remote	not found	Senior	High	medium
Insight Global	Sr. Unity Developer	LATAM offshore	LATAM only	\$29–\$43/hr	Senior	High	medium
Creamos	Senior Unity Developer	Latin America	LATAM remote	not found	Senior	Medium-high	medium

Bibia completa

40.0.4 4. Snippet-only / partial B2 records

Company / platform	Role	Region	Remote scope	Salary	Seniority	Fit	Confidence
Upwork	Digital Twin Development using Unity / Godot / Unreal	Unknown	Likely freelance remote	not found	not found	Very high	Medium-low
Indeed	Remote Unity Developer jobs	Remote / unknown	Unknown	not found	mixed	Medium	Low
Glassdoor	Unity Developer jobs in Remote	Remote / unknown	Unknown	not found	mixed	Medium	Low
Joble	Unity developer remote jobs	Unknown	Unknown	not found	mixed	Low-medium	Low
Remote Rocketship	Remote Unity Jobs	Remote / unknown	Unknown	not found	mixed	Medium	Low
Remotive	Unity search page	Remote / unknown	Unknown	not found	not found	Not usable	Low

Bibia completa

40.0.5 5. Role-family clustering

Role family	Postings	Signal strength	Alexander fit
Unity gameplay / app developer	5	High	Medium-high
Technical art / shaders / rendering	4	High	High
XR / VR / immersive apps	2	Medium	High
Web / HTML5 / WebGL-adjacent	2	Medium	High
Digital twin / simulation	1	Low-medium	Very high
LATAM nearshore	3	Medium-high	High

40.0.6 6. Requirement frequency snapshot

Requirement	Observed in	Market signal
Unity	9+	Core requirement
C#	8+	Core requirement
Performance optimization	6+	Strong differentiator
Shader / rendering	5+	Strong differentiator
Mobile iOS / Android	4+	Common requirement
VR / XR	3+	Specialized requirement
English B2/C1	3+	Nearshore requirement
Portfolio / reel	2+	Explicit in art/VFX roles
Web / HTML5	2	Relevant niche
Blender / 3D modeling	2	Useful supporting skill
Testing / validation	2	Engineering signal
Documentation	2	Engineering signal

40.0.7 7. Salary visibility

Salary type	Roles	Notes
Explicit hourly	1	Insight Global: \$29–\$43/hr
Negotiable	2	RJ Games, HyperVR Games
Applicant proposes salary	1	Mudtek
Competitive / equity	1	DreamForge
Not found	4	Flat2VR, FLEXUS, Azumo, Creamos
B2 snippet missing	6	Do not infer

40.0.8 8. Remote scope constraints

Scope	Roles	Constraint
LATAM only	3	Strong match for Colombia
EU & NA	1	Colombia likely excluded unless flexible
Global remote	2	Verify work authorization
Remote anywhere	1	Verify payment/legal terms
Unknown remote	6	Snippet-only; do not assume
US / Canada preference	0 in verified table	Not central in current sample

40.0.9 9. Contractor / freelance signals

Company / platform	Contractor signal	Evidence quality
Insight Global	Contract / permanent possible	Medium-high
Creamos	Freelance talent pool	Medium-high
Upwork	Freelance platform	Low-medium
Mudtek	Temporary contract possible	Medium

40.0.10 10. Best-fit postings for Alexander

Rank	Company	Role	Reason
1	Insight Global	Sr. Unity Developer	Unity + VR + Blender + LATAM
2	Creamos	Senior Unity Developer	Unity + web + portfolio + LATAM
3	HyperVR Games	Senior VR Unity Developer	Unity + XR + performance
4	Flat2VR Studios	Unity Technical Artist	Unity graphics pipeline + VR
5	FLEXUS	Graphic Programmer / Render Developer	shaders + rendering + math
6	Azumo	Unity Developer	LATAM + interactive 3D apps
7	Mudtek	VFX Artist, Unity / Shader Graph	Shader Graph + VFX
8	DreamForge	Senior Unity Developer	Unity + AI + gameplay
9	Upwork	Digital Twin Development	closest to thesis domain, but unverified

40.0.11 11. Gaps against current market

Gap	Evidence	Severity
Seniority pressure	Many roles senior	High
Shipped titles expectation	HyperVR, Mudtek	Medium-high
Mobile game emphasis	RJ, Azumo, DreamForge	Medium
Shader specialization	FLEXUS, Mudtek, Flat2VR	Medium-high
Portfolio proof	Creamos, Mudtek	High
English proof	Azumo, Creamos, HyperVR	High
WebGL niche scarcity	Few explicit roles	Medium
Digital twin scarcity	B2 weak capture	Medium

40.0.12 12. Positioning implication

The strongest positioning remains hybrid rather than purely game-dev:

```
Bloque de código: text
Unity WebGL / Real-Time 3D Developer
Technical Visualization + Technical Art
XR / Simulation-adjacent portfolio
C# + shaders + optimization + 3D pipeline
```

This market sample does not support relying only on “Unity Developer” as a generic title. The market is saturated with mobile/gameplay senior roles. Alexander’s differentiation should be technical visualization, WebGL, shader/view modes, optimization, CAD-to-web pipeline, and interactive inspection systems.

40.0.13 13. CV and portfolio implications

Asset	Required adjustment
CV headline	Mention Unity WebGL + technical visualization
Project section	Put TwinSight X500 first
Skills	Split Unity/C#, shaders, WebGL, Blender, optimization

Asset	Required adjustment
Portfolio	Add performance metrics and GIFs
GitHub	Expose README with architecture and demo
Case studies	Add cross-section, thermal view, CAD-to-web pipeline
English profile	State professional English only if defensible

40.0.14 14. Recommended role titles to target

Priority	Title
1	Unity Technical Artist
2	Unity WebGL Developer
3	Real-Time 3D Developer
4	Interactive 3D Developer
5	Technical Visualization Developer
6	XR Unity Developer
7	Simulation Developer
8	Unity Developer

40.0.15 15. Recommended search terms for next live-job pass

Search term	Purpose
Unity WebGL Developer	Direct fit
Interactive 3D Developer Unity	Broader fit
Technical Visualization Unity	Thesis-adjacent
Unity Technical Artist remote LATAM	Nearshore fit
Unity Shader Graph remote	Technical art fit
Unity XR Developer LATAM	XR fit
Digital Twin Unity freelance	Simulation fit
Real-Time 3D Developer remote	Non-game fit
Unity C# contractor LATAM	Contractor route

40.0.16 16. Interpretation rules for future searches

Rule	Reason
Do not assume “remote” means worldwide	Region restrictions are frequent
Separate LATAM-only from global remote	Colombia eligibility differs
Mark snippets separately	Snippets are not full postings
Do not infer salary	Most postings hide compensation
Check portfolio requirement manually	Often implied, not explicit
Record seniority exactly	Senior roles dominate this sample
Flag expired / talent-pool roles	Not all postings are immediate openings

40.0.17 17. Final market read

The current sample shows a viable but narrow path. Generic Unity roles skew toward senior game development, mobile, and shipped-title experience. The best fit for Alexander is not a generic game programmer lane; it is a technical visualization / technical art lane where Unity, C#, shaders, WebGL, optimization, Blender, and interactive 3D systems combine into one portfolio narrative.

The LATAM nearshore market is relevant, but small in this capture. The highest-value next step is not broad application volume. It is building a portfolio package that can satisfy the explicit expectations visible in Creamos, Mudtek, Flat2VR, Insight Global, and HyperVR: Unity proof, C# proof, optimization proof, visual proof, and project documentation.

Chapter 41

31_application_targeting_matrix_and_search_

Fuente: 31_application_targeting_matrix_and_search_protocol.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1073

Ultima modificacion: 2026-06-11 19:58

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This module translates the live-job snapshot into a practical protocol for deciding:

Contenido completo depurado

41.0.1 0. Module status

```
Bloque de codigo: text
Module: 31_application_targeting_matrix_and_search_protocol.md
Date generated: 2026-06-12
Status: Complete v1
Purpose: Convert job-market evidence into an application targeting system
Inputs: 30_live_job_postings_market_snapshot.md, B1, B2 partial, B5, 17, 18, 19B, 20B, 21B, 24, 25, 26, 27
Use: Application selection, search protocol, tracker scoring, CV/portfolio variant selection
```

41.1 1. Purpose

This module translates the live-job snapshot into a practical protocol for deciding:

```
Bloque de codigo: text
what roles to search,
what roles to apply to,
what roles to reject,
which CV/portfolio variant to use,
which gaps to close before applying,
and how to score opportunities consistently.
```

The goal is to prevent random applications and keep Alexander's search aligned with the strongest market-positioning evidence:

```
Bloque de codigo: text
Unity WebGL / Real-Time 3D / Technical Visualization / Technical Art / XR-adjacent
```

41.2 2. Market evidence summary from module 30

The live-job snapshot found:

Signal	Value
Verified / captured postings	9
Snippet-only market signals	6
LATAM-specific roles	3
Contractor / freelance signals	3
Digital twin / simulation signals	1

41.2.1 Key interpretation

The sample supports a narrow but viable route.

The strongest path is not:

```
Bloque de codigo: text
generic Unity Developer
```

The stronger path is:

```
Bloque de codigo: text
Unity WebGL / Real-Time 3D Developer
Technical Visualization + Technical Art
XR / Simulation-adjacent portfolio
C# + shaders + optimization + 3D pipeline
```

41.2.2 Market constraints

Constraint	Evidence
Many postings are senior	seniority pressure is high
Portfolio proof matters	Creamos and Mudtek explicitly signal portfolio/reel relevance
Technical art is relevant	Flat2VR, FLEXUS, Mudtek
XR/VR is relevant	HyperVR, Insight Global
LATAM remote exists	Azumo, Insight Global, Creamos
WebGL is niche but useful	Creamos/web platform, TwinSight differentiation
Digital twin is very relevant but undercaptured	B2 only partial

41.3 3. Targeting principle

Alexander should not compete primarily as:

```
Bloque de codigo: text
Junior Unity Gameplay Programmer
```

He should compete as:

```
Bloque de codigo: text
Unity WebGL / Real-Time 3D Developer with technical visualization, technical art and CAD-to-realtime proof.
```

This means the search strategy should prioritize roles where TwinSight is not just “a project,” but direct evidence.

41.4 4. Role priority matrix

41.4.1 4.1 Tier A — apply first

Role family	Why Tier A	Required proof
Unity Technical Artist Unity WebGL Developer Real-Time 3D Developer Technical Visualization Developer Interactive 3D Developer	strong technical art/rendering signal direct TwinSight evidence broad fit for interactive systems strongest thesis-domain fit broad non-game framing	visual modes, optimization, shaders, demo reel WebGL build, browser demo, README demo video, runtime interaction, C# case study, CAD pipeline, metrics browser demo, UI, interaction

41.4.2 4.2 Tier B — apply selectively

Role family	Why selective	Required proof
XR Unity Developer Simulation Developer	adjacent, but needs XR proof domain-relevant, but may require physics/engineering depth	frame as XR-adjacent unless VR exists clarify visualization vs simulation
Unity Developer Tools / Pipeline Developer Shader / Render Technical Artist	viable if not purely gameplay/mobile relevant but needs tooling proof attractive but risky	use Unity WebGL/technical visualization variant ARA + Unity pipeline documentation only if visual modes and shader proof are strong

41.4.3 4.3 Tier C — fallback only

Role family	Reason
Full-stack Developer Python Automation Developer LLM Application Developer 3D Artist / Blender Generalist Gameplay-only Unity Developer Mobile game Unity Developer	dilutes positioning unless role is 3D/web-adjacent secondary via ARA, not main route secondary unless applying to tools/automation weak compensation and weaker fit high competition; TwinSight less relevant apply only if WebGL/3D/optimization valued

41.4.4 4.4 Reject / avoid

Role type	Reject reason
Senior AAA Technical Artist Senior Shader Engineer Production Digital Twin Engineer Robotics Simulation Engineer Pure VFX Artist Pure 3D Character Artist Full-stack SaaS Developer Mobile casual gameplay roles	overclaim / shipped-title gap insufficient specialized shader proof may require IoT/cloud/industrial deployment may require ROS/physics/control systems TwinSight not enough; style mismatch secondary portfolio only unrelated proof weak alignment unless Unity/C# and remote LATAM are strong

41.5 5. Recommended public role titles

Use these titles in searches and materials.

41.5.1 5.1 Primary title

Bloque de código: text
Real-Time 3D Developer / Unity Technical Artist

41.5.2 5.2 Application-specific variants

Target role	Use this title
Unity WebGL	Unity WebGL / Interactive 3D Developer
Technical visualization	Technical Visualization Developer
Technical art	Unity Technical Artist
Real-time systems	Real-Time 3D Developer
XR	Unity XR / Real-Time 3D Developer
LATAM nearshore	Unity Developer — WebGL / Technical Visualization
Freelance	Unity WebGL Technical Visualization Developer

41.5.3 5.3 Titles to avoid for now

Bloque de código: text
Senior Unity Developer
Senior Technical Artist
Digital Twin Engineer
XR Specialist
Shader Engineer
Full-Stack Developer
AI Engineer
3D Generalist

Unless the role context specifically requires a simplified label.

41.6 6. Search term protocol

41.6.1 6.1 Primary search terms

Use these daily or 2–3 times per week.

Search term	Intent
Unity WebGL Developer remote	direct TwinSight fit
Unity Technical Artist remote	technical art fit
Real-Time 3D Developer remote	broad real-time fit
Interactive 3D Developer remote	non-game route
Technical Visualization Unity remote	thesis-domain fit
Unity C# contractor LATAM	nearshore contractor
Unity Developer LATAM remote	LATAM route
Unity Shader Graph remote	technical art route
Unity XR Developer remote	XR-adjacent route
Digital Twin Unity freelance	simulation/digital twin route

41.6.2 6.2 Secondary search terms

Search term	Use
WebGL 3D Developer remote	may find Three.js/WebGL roles
Unity visualization developer	technical visualization
3D product configurator developer	product visualization

Search term	Use
Simulation developer Unity	simulation-adjacent
XR technical artist Unity	technical art + XR
Unity tools developer remote	tooling route
CAD visualization developer	CAD-to-realtime route
Unity render developer remote	rendering/shader roles
Unity optimization remote	performance roles
Web 3D Unity remote	WebGL route

41.6.3 6.3 Experimental search terms

Use weekly.

Search term	Risk
industrial visualization Unity	low volume
assembly visualization 3D	niche
product visualization Unity WebGL	may be agency/freelance
digital twin visualization developer	may require IoT/cloud
real-time visualization engineer	may be senior
technical artist LATAM remote	low volume
Unity WebGPU developer	emerging/niche
simulation visualization engineer	may be engineering-heavy

41.7 7. Source protocol

41.7.1 7.1 Best sources by purpose

Source type	Use
company careers pages	best verification
Workable / Greenhouse / Lever	accessible direct postings
Remote Game Jobs	good for game/Unity remote
Work With Indies	indie technical art roles
InGameJob	game/Unity roles
LinkedIn Jobs	useful but must verify remote scope
Wellfound	startup / interactive / tools
Upwork	freelance digital twin / Unity gigs
Jobgether	LATAM remote / reposted roles
Insight Global / recruiters	contract nearshore
niche company pages	best for technical visualization

41.7.2 7.2 Lower-quality sources

Source	Use with caution
Indeed	broad, often noisy
Glassdoor	useful snippets, less detail
Jooble / Joble	aggregator noise
Remote Rocketship	useful signal, needs verification
generic SEO job pages	low trust

41.8 8. Remote eligibility protocol

Do not treat “remote” as enough.
Every posting must be labeled as one of:

Bloque de codigo: text

Worldwide
LATAM only
US only
Canada only
EU only
EU + NA
UK only
Hybrid
On-site
Unknown

41.8.1 8.1 Colombia suitability labels

Label	Meaning
Colombia-eligible	explicitly LATAM/global/contractor
Likely eligible	no restriction visible, but remote/global wording
Unclear	remote but region not defined
Likely not eligible	EU/NA/US-only
Not eligible	work authorization excludes Colombia

41.8.2 8.2 Immediate filter

Reject or postpone if:

Bloque de codigo: text

US work authorization required
EU work authorization required now
hybrid in city/country not accessible
security clearance required
onsite only

Exception:

Bloque de codigo: text

Save for future EU/Portugal/Germany path if strategically relevant.

41.9 9. Application scoring model

Score each opportunity from 1 to 5.

41.9.1 9.1 Fit score

Score	Definition
5	Direct TwinSight fit: Unity/WebGL/technical visualization/CAD/interactive 3D
4	Strong Unity/technical art/XR fit with one gap
3	General Unity role with partial overlap
2	Weak role with mostly unrelated requirements
1	Distraction

41.9.2 9.2 Probability score

Score	Definition
5	remote LATAM/contractor + portfolio-based + not senior-heavy
4	plausible with strong project proof
3	competitive but possible
2	mostly senior/shipped-title/local-only
1	unrealistic now

41.9.3 9.3 Compensation score

Score	Definition
5	likely USD 6k+/month path
4	likely USD 3k–6k/month
3	likely USD 1.5k–3k/month
2	low/unclear
1	below target or unpaid

41.9.4 9.4 Portfolio-fit score

Score	Definition
5	TwinSight directly answers role requirements
4	TwinSight + ARA/portrait answer most requirements
3	needs portfolio adaptation
2	current proof weak
1	unrelated portfolio

41.9.5 9.5 Region score

Score	Definition
5	LATAM remote / Colombia eligible
4	global remote
3	unclear but possible
2	EU/NA-only but maybe future
1	not eligible

41.9.6 9.6 Total score

Bloque de código: text
 Total = Fit + Probability + Compensation + Portfolio-fit + Region
 Max = 25

41.9.7 9.7 Action threshold

Total	Action
21–25	Apply now with tailored materials
17–20	Apply if materials are close
13–16	Save / apply only if low effort
9–12	Watchlist
≤8	Reject

41.10 10. Application priority matrix

41.10.1 10.1 Priority A

Apply within 24–48 hours.

Criteria:

```
Bloque de codigo: text
Fit ≥ 4
Probability ≥ 3
Portfolio-fit ≥ 4
Region ≥ 4
```

Typical roles:

```
Bloque de codigo: text
Unity WebGL Developer
Unity Technical Artist
Technical Visualization Developer
Interactive 3D Developer
LATAM Unity Developer with Web/VR/Blender
```

41.10.2 10.2 Priority B

Apply after minor tailoring.

Criteria:

```
Bloque de codigo: text
Fit ≥ 3
Probability ≥ 3
Portfolio-fit ≥ 3
Region ≥ 3
```

Typical roles:

```
Bloque de codigo: text
Unity Developer
XR Unity Developer
Unity Tools Developer
Shader Graph / VFX role
Simulation visualization role
```

41.10.3 10.3 Priority C

Save but do not prioritize.

Criteria:

```
Bloque de codigo: text
Fit 2–3
Probability 2–3
Portfolio-fit 2–3
```

Typical roles:

```
Bloque de codigo: text
mobile gameplay Unity
senior general Unity
broad game programmer
AI-assisted Unity role
```

41.10.4 10.4 Reject

Reject if:

```
Bloque de codigo: text
Region ≤ 2
Fit ≤ 2
Portfolio-fit ≤ 2
requires shipped AAA titles
requires 5-10 years production experience
requires relocation now
requires clearance
requires advanced shader/math/GPU beyond current proof
```

41.11 11. Material variant selection

41.11.1 11.1 CV variant map

Posting emphasis	CV variant
Unity WebGL	Unity WebGL / Interactive 3D CV
Technical Artist	Unity Technical Artist CV
Technical Visualization	Technical Visualization CV
XR / VR	XR-adjacent Unity CV
LATAM Unity	Unity Developer LATAM CV
Tools / Pipeline	Tools / Automation CV
AI + Unity	Unity + AI Tools CV
General Unity	Real-Time 3D Developer CV

41.11.2 11.2 Portfolio link priority

Role	First portfolio link
Unity WebGL	TwinSight case study
Technical Artist	TwinSight visual modes + breakdown
XR	TwinSight + future XR note
Simulation	TwinSight technical visualization
LATAM Unity	TwinSight + GitHub
Tools	ARA + TwinSight pipeline
Shader/VFX	visual modes + shader notes
3D art	Blender portrait + TwinSight secondary

41.11.3 11.3 GitHub link priority

Role	GitHub repo order
Unity / WebGL	TwinSight first
AI tools	ARA first, TwinSight second
Full-stack fallback	FitApp/AI News + TwinSight

Role	GitHub repo order
Technical art	TwinSight README + ArtStation
Tools/pipeline	ARA + TwinSight pipeline

41.12 12. Search workflow

41.12.1 12.1 Daily search, 30–45 minutes

Bloque de código: text

1. Search 3 primary terms.
2. Open only roles with remote/LATAM/global signal.
3. Record in tracker.
4. Score quickly.
5. Apply only to Priority A or strong B.

Primary daily rotation:

Bloque de código: text

Unity WebGL Developer remote
 Unity Technical Artist remote
 Unity Developer LATAM remote

41.12.2 12.2 Twice-weekly search, 60–90 minutes

Bloque de código: text

1. Search niche terms.
2. Check company career pages.
3. Check Workable/Greenhouse/Lever postings.
4. Search remote game boards.
5. Search Upwork/freelance only for technical visualization/digital twin.

Terms:

Bloque de código: text

Technical Visualization Unity remote
 Interactive 3D Developer remote
 Unity XR Developer LATAM
 Digital Twin Unity freelance

41.12.3 12.3 Weekly review

Bloque de código: text

1. Review tracker.
2. Identify recurring requirements.
3. Update CV keywords.
4. Update portfolio proof gaps.
5. Pick 5–10 applications for the week.

41.13 13. Tracker fields

Add these fields to the application tracker.

Bloque de código: text

Role family
 Priority A/B/C/Reject
 Fit score
 Probability score

```

Compensation score
Portfolio-fit score
Region score
Total score
Remote scope
Colombia eligibility
Seniority gap
Portfolio proof needed
CV variant
Portfolio link used
GitHub link used
Application status
Follow-up date
Interview risk notes

```

41.13.1 13.1 Tracker status labels

```

Bloque de codigo: text
Found
Scored
Saved
Applied
Follow-up sent
Interview
Rejected
Closed
Watchlist
Do not apply

```

41.14 14. Red flags

41.14.1 14.1 Hard red flags

Reject unless there is a strong reason.

```

Bloque de codigo: text
US work authorization required
EU work authorization required now
onsite only
security clearance
AAA shipped titles mandatory
7+ years required
lead/principal title
advanced GPU programming required
low/no pay
crypto/metaverse vague role with no product

```

41.14.2 14.2 Soft red flags

Proceed only if fit is high.

```

Bloque de codigo: text
mobile-only Unity
hypercasual games
pure gameplay
pure VFX
heavy multiplayer backend
React/full-stack primary
ARKit/ARCore production required
Unreal-only
native mobile required

```

41.15 15. Positive signals

41.15.1 15.1 Strong apply signals

```

Bloque de codigo: text

```

```

Unity WebGL
browser-based 3D
technical visualization
interactive 3D
CAD
digital twin visualization
XR training
simulation visualization
Unity Technical Artist
Shader Graph
URP/HDRP
optimization
Blender
LATAM remote
contractor
portfolio required

```

41.15.2 15.2 Especially strong combinations

Combination	Why strong
Unity + WebGL + remote	direct TwinSight proof
Unity + technical visualization	strongest domain match
Unity + Blender + VR/AR	matches B5/B1 signals
Unity + Shader Graph + optimization	technical art route
Unity + LATAM + C#	practical nearshore route
Digital twin + Unity + freelance	thesis-domain contractor route
Interactive 3D + product visualization	non-game commercial route

41.16 16. Portfolio gap-to-role map

41.16.1 16.1 Gaps observed in module 30

Gap	Severity	Closure artifact
Seniority pressure	High	strong project case study
Shipped-title expectation	Medium-high	demo video + GitHub + case study
Shader specialization	Medium-high	visual modes breakdown
Portfolio proof	High	TwinSight video + ArtStation + README
English proof	High	English CV + LinkedIn + video captions
WebGL niche scarcity	Medium	explicit WebGL positioning
Digital twin scarcity	Medium	digital-twin-adjacent wording
Mobile requirement	Medium	mention WebGL/mobile-first UI only if true

41.16.2 16.2 Closure order

```

Bloque de código: text
1. TwinSight demo video
2. TwinSight portfolio case study
3. GitHub README
4. ArtStation breakdown
5. LinkedIn Featured
6. English CV variant
7. Short technical breakdown video

```

41.17 17. Apply-now readiness

41.17.1 17.1 Minimum before serious applications

```
Bloque de codigo: text
[ ] English CV ready
[ ] LinkedIn updated
[ ] TwinSight case-study page public
[ ] TwinSight 90-second demo public
[ ] GitHub README updated
[ ] ArtStation breakdown started or published
[ ] application tracker active
```

41.17.2 17.2 Can apply before everything is perfect?

Yes, but only to:

```
Bloque de codigo: text
Priority A roles
LATAM remote roles
freelance/talent-pool roles
roles where TwinSight directly matches
```

Do not apply early to:

```
Bloque de codigo: text
senior technical art roles
shader-heavy roles
VR-heavy roles
AAA roles
```

until the proof package is stronger.

41.18 18. Weekly application targets

41.18.1 18.1 Conservative

```
Bloque de codigo: text
5-8 applications/week
only Priority A/B
high tailoring
```

41.18.2 18.2 Balanced

```
Bloque de codigo: text
10-15 applications/week
Priority A/B plus selected C
moderate tailoring
```

41.18.3 18.3 Aggressive

```
Bloque de codigo: text
20-30 applications/week
higher volume
more rejection risk
less tailoring
```

41.18.4 18.4 Recommended current mode

Bloque de código: text
Conservative → Balanced

Reason:

The portfolio package is still being finalized. Volume before proof will likely waste leads.

41.19 19. Application package by priority

41.19.1 19.1 Priority A package

Bloque de código: text
Tailored CV
Portfolio case-study link
GitHub README link
90-second demo video
Short cover note
Role-specific headline

41.19.2 19.2 Priority B package

Bloque de código: text
CV variant
Portfolio link
GitHub link
Short note

41.19.3 19.3 Priority C package

Bloque de código: text
Generic aligned CV
Portfolio link
No heavy tailoring

41.20 20. Cover note templates

41.20.1 20.1 Unity WebGL / technical visualization

Bloque de código: text
I'm applying because this role aligns closely with my Unity WebGL technical visualization work. My flagship project, TwinSight
↪ X500, is a browser-based 3D inspection prototype for drone assembly, built with Unity, C#, WebGL, URP, UI Toolkit and
↪ Blender. It includes component selection, exploded view, clipping tools, visual modes, CAD-to-realtime optimization and
↪ usability/workload evaluation.

41.20.2 20.2 Unity Technical Artist

Bloque de código: text
I'm interested in this role because it combines Unity, technical art, optimization and visual problem-solving. My main
↪ project, TwinSight X500, focuses on CAD-to-realtime optimization, visual inspection modes, WebGL constraints and runtime
↪ interaction systems in Unity.

41.20.3 20.3 LATAM remote Unity

Bloque de código: text
I'm based in Colombia and available for remote LATAM or international contractor work. My strongest Unity project is TwinSight
↔ X500, a WebGL technical visualization prototype with C#, URP, UI Toolkit, Blender optimization, component interaction and
↔ usability evaluation.

41.21 21. Search calendar

41.21.1 21.1 Monday

Bloque de código: text
Unity WebGL
Unity Technical Artist
LATAM Unity remote

41.21.2 21.2 Tuesday

Bloque de código: text
portfolio production
no broad searching

41.21.3 21.3 Wednesday

Bloque de código: text
Interactive 3D
Technical Visualization
Digital Twin Unity

41.21.4 21.4 Thursday

Bloque de código: text
apply to scored roles
follow-ups
tracker update

41.21.5 21.5 Friday

Bloque de código: text
XR Unity
Shader Graph
Remote game boards

41.21.6 21.6 Weekend

Bloque de código: text
portfolio/GitHub/LinkedIn improvements
weekly review

41.22 22. Company targeting protocol

41.22.1 22.1 Company types to prioritize

```
Bloque de codigo: text
technical visualization studios
XR training studios
industrial visualization companies
3D product configurator companies
Unity WebGL agencies
simulation visualization teams
digital twin visualization companies
LATAM nearshore software shops
remote game/interactive studios
```

41.22.2 22.2 Company types to deprioritize

```
Bloque de codigo: text
AAA studios requiring shipped titles
mobile hypercasual factories
pure outsourcing art studios
full-stack SaaS companies
defense/clearance-heavy companies
crypto/metaverse vaporware
local-only EU/US roles
```

41.23 23. Interview-readiness implications

41.23.1 23.1 Prepare answers for

```
Bloque de codigo: text
Why Unity WebGL?
How did you optimize CAD assets?
What was your exact role?
How much code did you write?
How did you use AI?
Is this a real digital twin?
How did you evaluate usability?
What would you improve next?
How would you adapt this to XR?
How would you work as a contractor from Colombia?
```

41.23.2 23.2 Highest-risk questions

```
Bloque de codigo: text
Do you have professional Unity experience?
Have you shipped commercial games?
How deep is your shader experience?
Can you work independently without AI?
Can you handle production deadlines?
```

41.23.3 23.3 Defense strategy

Do not pretend.

Use:

```
Bloque de codigo: text
I have limited formal industry experience, but my portfolio is built to demonstrate the relevant production-adjacent skills
↪ directly: Unity/C#, WebGL deployment, CAD optimization, runtime interaction, technical UI and structured evaluation.
```

41.24 24. Decision tree

41.24.1 24.1 Should Alexander apply?

```
Bloque de codigo: text
Is it remote or Colombia/LATAM eligible?
  No → reject or future watchlist.
  Yes → continue.

Does it require senior shipped-title experience?
  Yes → only apply if fit ≥ 5 and portfolio match ≥ 4.
  No → continue.

Does TwinSight directly support the role?
  Yes → score high.
  No → continue.

Can CV + portfolio be tailored in under 30 minutes?
  Yes → apply if total ≥ 17.
  No → save unless priority A.

Does the role strengthen the target path?
  Yes → apply.
  No → reject.
```

41.24.2 24.2 Fast reject test

Reject if three or more are true:

```
Bloque de codigo: text
senior only
mobile-only
onsite/hybrid outside Colombia
requires shipped commercial titles
requires Unreal only
requires advanced GPU programming
no portfolio relevance
unclear company/product
low/no pay
```

41.25 25. Final recommendation

41.25.1 Immediate target families

```
Bloque de codigo: text
1. Unity Technical Artist
2. Unity WebGL Developer
3. Real-Time 3D Developer
4. Technical Visualization Developer
5. Interactive 3D Developer
```

41.25.2 Secondary target families

```
Bloque de codigo: text
6. XR Unity Developer
7. Simulation Visualization Developer
8. Unity Tools / Pipeline Developer
9. Unity Developer LATAM remote
```

41.25.3 Fallback only

```
Bloque de codigo: text
10. Full-stack Developer
11. Python Automation Developer
12. 3D Artist / Blender Generalist
```

41.25.4 Current operating rule

Bloque de codigo: text
Do not maximize application volume yet.
Maximize proof quality, then apply selectively to high-fit roles.

41.26 26. Next module

Proceed with:

Bloque de codigo: text
32_application_materials_gap_closure_plan.md

41.26.1 Tool recommendation

Tool	Use
Agent mode	No
Deep Research	No

41.26.2 Purpose

Create a closure plan for the specific gaps identified in module 30 and operationalized in module 31:

Bloque de codigo: text
portfolio proof
GitHub proof
demo video
ArtStation breakdown
English CV
LinkedIn Featured
shader/visual modes proof
contractor readiness
interview defense

Chapter 42

32_application_materials_gap_closure_plan

Fuente: 32_application_materials_gap_closure_plan.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1007

Ultima modificacion: 2026-06-11 20:03

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This module converts the gaps identified in the live job snapshot and application targeting protocol into a concrete closure plan.

Contenido completo depurado

42.0.1 0. Module status

```
Bloque de codigo: text
Module: 32_application_materials_gap_closure_plan.md
Date generated: 2026-06-12
Status: Complete v1
Purpose: Close application-material gaps before serious job applications
Inputs: 17, 18, 19B, 20B, 21B, 24, 25, 26, 27, 30, 31
Use: execution plan for portfolio, CV, LinkedIn, GitHub, demo video, ArtStation and interview proof
```

42.1 1. Purpose

This module converts the gaps identified in the live job snapshot and application targeting protocol into a concrete closure plan.

The current job-market evidence shows that Alexander should not apply as a generic Unity candidate. The stronger path is:

```
Bloque de codigo: text
Unity WebGL / Real-Time 3D Developer
Technical Visualization + Technical Art
XR / Simulation-adjacent portfolio
C# + shaders + optimization + 3D pipeline
```

This means the application package must prove the following before serious application volume:

```

Bloque de codigo: text
1. Unity proof
2. C# proof
3. WebGL proof
4. technical visualization proof
5. CAD-to-realtime proof
6. technical-art / visual-mode proof
7. optimization proof
8. portfolio proof
9. GitHub proof
10. English professional communication proof
11. contractor/remote readiness proof
12. interview-defense readiness

```

42.2 2. Core market gaps to close

42.2.1 2.1 Gap table

Gap	Severity	Evidence source	Closure artifact
Seniority pressure	High	many postings are senior	strong proof package
Portfolio proof	High	portfolio/reel expected	TwinSight case study + demo
Shipped-title expectation	Medium-high	game/VR roles	show production-like process
Shader/rendering signal	Medium-high	technical art/render roles	visual modes breakdown
WebGL niche clarity	Medium	few explicit WebGL roles	make WebGL highly visible
Digital twin scarcity	Medium	weak B2 capture	use digital-twin-adjacent language
English proof	High	LATAM/remote roles	English CV + LinkedIn + video captions
Contractor readiness	Medium-high	LATAM/freelance signals	contractor packet
GitHub proof	High	technical credibility	README + media + repo hygiene
Interview risk	High	limited formal industry experience	answer bank + demo defense

42.3 3. Material stack required before serious applications

42.3.1 3.1 Required public assets

Asset	Status target	Priority
English CV	final and role-variant ready	1
LinkedIn profile	aligned with target role	1
TwinSight case-study page	public and polished	1
TwinSight 90-second demo	public	1
TwinSight GitHub README	updated	1
ArtStation TwinSight breakdown	published or drafted	2
30-second teaser	public	2
Technical breakdown video	optional but strong	3
Blender portrait breakdown	secondary support	3
ARA Framework README	secondary proof	3

42.3.2 3.2 Minimum application-ready package

Alexander can start applying to Priority A roles when these are done:

```

Bloque de codigo: text
[ ] English CV v1
[ ] LinkedIn headline/about updated
[ ] TwinSight case-study page public
[ ] TwinSight 90-second demo public
[ ] GitHub README updated
[ ] application tracker active

```

42.3.3 3.3 Strong package

For higher-quality roles, complete:

```
Bloque de codigo: text
[ ] ArtStation breakdown
[ ] short teaser video
[ ] technical breakdown video
[ ] ARA README
[ ] interview answer bank rehearsed
[ ] contractor readiness note
```

42.4 4. Closure roadmap

42.4.1 4.1 Phase 1 — Proof core

Goal:

```
Bloque de codigo: text
Make TwinSight understandable and credible in under 2 minutes.
```

Required outputs:

```
Bloque de codigo: text
1. 90-second TwinSight demo
2. TwinSight portfolio case-study page
3. GitHub README final
4. LinkedIn Featured entry
5. English CV with TwinSight first
```

Priority:

```
Bloque de codigo: text
Critical
```

42.4.2 4.2 Phase 2 — Technical depth

Goal:

```
Bloque de codigo: text
Make TwinSight defensible to technical leads.
```

Required outputs:

```
Bloque de codigo: text
1. technical breakdown video
2. ArtStation technical breakdown
3. architecture diagram
4. pipeline diagram
5. visual modes grid
6. optimization before/after images
```

Priority:

```
Bloque de codigo: text
High
```

42.4.3 4.3 Phase 3 — Market adaptation

Goal:

Bloque de codigo: text
Adapt materials to different target roles.

Required outputs:

Bloque de codigo: text
1. Unity WebGL CV variant
2. Unity Technical Artist CV variant
3. Technical Visualization CV variant
4. XR-adjacent CV variant
5. LATAM contractor CV variant
6. cover-note snippets by role

Priority:

Bloque de codigo: text
High

42.4.4 4.4 Phase 4 — Interview and contractor readiness

Goal:

Bloque de codigo: text
Reduce interview risk from limited formal industry experience.

Required outputs:

Bloque de codigo: text
1. project defense answers
2. AI-assisted workflow answer
3. contractor availability statement
4. remote work setup statement
5. rate/compensation decision bands
6. offer risk scorecard usage

Priority:

Bloque de codigo: text
Medium-high

42.5 5. TwinSight proof package

42.5.1 5.1 Required files

Bloque de codigo: text
twinsight_x500_demo_90s.mp4
twinsight_x500_teaser_30s.mp4
twinsight_x500_thumbnail.jpg
twinsight_x500_visual_modes_grid.png
twinsight_x500_metrics_card.png
twinsight_x500_pipeline_diagram.png
twinsight_x500_architecture_diagram.png
twinsight_x500_cad_optimization_before_after.png

42.5.2 5.2 Required portfolio sections

Bloque de codigo: text

Hero
Demo video
Problem
Solution
Key features
Technical pipeline
CAD-to-realtime optimization
Unity WebGL architecture
Visual modes
Evaluation
Role and contribution
AI-assisted workflow disclosure
Limitations
Links

42.5.3 5.3 Required GitHub sections

Bloque de codigo: text

Overview
Demo
Highlights
Screenshots
Technical Stack
Problem
Solution
Technical Pipeline
CAD-to-Realtime Optimization
Unity WebGL Architecture
Evaluation
My Role
AI-Assisted Workflow
Limitations
Roadmap
Setup
Links
Credits

42.5.4 5.4 Required ArtStation sections

Bloque de codigo: text

Cover image
Short video
Final screenshots
Feature breakdown
CAD optimization
Visual modes grid
UI / interaction breakdown
Metrics
Tools
Links

42.6 6. CV gap closure

42.6.1 6.1 Current CV risk

The market sample shows seniority pressure and portfolio expectations. A generic CV will underperform.
The CV must not say only:

Bloque de codigo: text

Unity Developer

It must say:

Bloque de codigo: text

Real-Time 3D Developer / Unity Technical Artist

or a role-specific variant.

42.6.2 6.2 CV headline variants

Variant	Use
Real-Time 3D Developer / Unity Technical Artist	default
Unity WebGL / Interactive 3D Developer	WebGL roles
Technical Visualization Developer	industrial/digital twin roles
Unity Technical Artist	shader/render/TA roles
Unity Developer — WebGL / Technical Visualization	LATAM nearshore roles

42.6.3 6.3 CV summary core

Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, C#, CAD-to-realtime optimization and interactive
↪ technical visualization. Built TwinSight X500, a browser-based drone assembly inspection prototype with component
↪ selection, exploded view, cross-section tools, visual modes, technical UI and SUS/NASA-TLX evaluation.

42.6.4 6.4 TwinSight CV bullet set

Use 3–5 bullets depending on role.

Default bullets

Bloque de codigo: text
- Built TwinSight X500, a Unity WebGL technical visualization prototype for drone assembly inspection using Unity, C#, URP, UI
↪ Toolkit and Blender.
- Implemented component selection, exploded view, cross-section/clipping tools, visual modes and technical information panels
↪ for interactive 3D inspection.
- Optimized CAD-derived source routes exceeding 6.5M triangles into a 95,617-triangle WebGL-ready model while preserving
↪ component-level readability.
- Evaluated the prototype with 12 participants using SUS, NASA-TLX Raw and Think-Aloud protocols, collecting 96 task-condition
↪ records.
- Documented technical constraints, usability findings and future improvements for digital-twin-adjacent technical
↪ visualization workflows.

Technical Artist variant

Bloque de codigo: text
- Developed visual inspection modes including realistic, X-ray, ghosted, blueprint, wireframe, solid and thermal-style
↪ qualitative views.
- Focused on rendering constraints, WebGL performance, visual readability and technical-art presentation for interactive
↪ inspection.

Unity WebGL variant

Bloque de codigo: text
- Deployed the prototype as a browser-based Unity WebGL viewer, emphasizing no-install access, runtime interaction and
↪ optimized 3D assets.

Technical Visualization variant

Bloque de codigo: text
- Designed the viewer to reduce spatial reasoning friction in drone assembly understanding compared with static 2D support.

42.6.5 6.5 CV proof checklist

Bloque de codigo: text
[] headline uses target role
[] TwinSight appears first

```
[ ] Unity/C#/WebGL visible in first screen
[ ] metrics included
[ ] portfolio link included
[ ] GitHub link included
[ ] demo video link included
[ ] AI use not overemphasized
[ ] no seniority overclaim
[ ] no unsupported digital twin claim
```

42.7 7. LinkedIn gap closure

42.7.1 7.1 LinkedIn headline

Recommended:

```
Bloque de código: text
Real-Time 3D Developer / Unity Technical Artist | Unity WebGL, C#, CAD-to-Realtime, Technical Visualization
```

Alternative:

```
Bloque de código: text
Unity WebGL / Interactive 3D Developer | Technical Visualization, CAD-to-Realtime, Blender
```

42.7.2 7.2 LinkedIn About opening

```
Bloque de código: text
I build interactive real-time 3D systems for technical visualization using Unity, C#, WebGL and Blender.

My flagship project, TwinSight X500, is a Unity WebGL drone assembly inspection prototype focused on CAD-to-realtime
↪ optimization, component selection, exploded view, cross-section tools, visual modes, technical UI and usability/workload
↪ evaluation.
```

42.7.3 7.3 Featured section order

```
Bloque de código: text
1. TwinSight X500 case study
2. TwinSight 90-second demo
3. GitHub repository
4. ArtStation breakdown
5. ARA Framework
```

42.7.4 7.4 LinkedIn proof checklist

```
Bloque de código: text
[ ] headline updated
[ ] About updated
[ ] TwinSight featured
[ ] demo video featured
[ ] GitHub featured
[ ] portfolio featured
[ ] skills reordered
[ ] experience/projects aligned
[ ] no broad unfocused positioning
```

42.7.5 7.5 Skills to prioritize

```
Bloque de código: text
Unity
C#
Unity WebGL
Technical Art
Real-Time 3D
Technical Visualization
Blender
```

```

Shader Graph
URP
UI Toolkit
CAD Optimization
Python

```

42.8 8. GitHub gap closure

42.8.1 8.1 Current risk

GitHub can either strengthen or weaken the application.

Weak GitHub signals:

```

Bloque de código: text
empty README
no screenshots
no demo
no setup
unclear ownership
broken links
large unexplained repo

```

Strong GitHub signals:

```

Bloque de código: text
clear README
demo thumbnail
screenshots
architecture explanation
metrics
limitations
setup instructions
clean repository description
topics
license/credits

```

42.8.2 8.2 Required updates

```

Bloque de código: text
[ ] repository description updated
[ ] topics added
[ ] README replaced with 19B content
[ ] media folder created
[ ] demo thumbnail added
[ ] screenshots added
[ ] video linked
[ ] portfolio linked
[ ] live demo linked
[ ] metrics included
[ ] limitations included
[ ] setup clarified
[ ] license/credits clarified

```

42.8.3 8.3 Repository description

```

Bloque de código: text
Unity WebGL technical visualization prototype for drone assembly inspection, featuring CAD-to-realtime optimization, component
↪ selection, exploded view, clipping tools and visual modes.

```

42.8.4 8.4 Topics

```

Bloque de código: text
unity
unity-webgl
csharp
webgl
technical-visualization

```

```
real-time-3d
technical-art
cad-optimization
interactive-3d
blender
drone
ui-toolkit
urp
```

42.9 9. Portfolio gap closure

42.9.1 9.1 Current risk

A portfolio without a complete TwinSight case study will not support the desired role family.

The case study must prove:

```
Bloque de codigo: text
not only that the app exists,
but that Alexander understands the technical decisions behind it.
```

42.9.2 9.2 Required page components

```
Bloque de codigo: text
[ ] hero image
[ ] 90-second video
[ ] problem section
[ ] solution section
[ ] feature grid
[ ] pipeline diagram
[ ] optimization metrics
[ ] architecture diagram
[ ] visual modes grid
[ ] evaluation metrics
[ ] role statement
[ ] limitations
[ ] links
```

42.9.3 9.3 Homepage changes

```
Bloque de codigo: text
[ ] headline aligned
[ ] skills strip visible
[ ] TwinSight first project
[ ] ARA second or third
[ ] Blender portrait secondary
[ ] contact CTA for remote roles
```

42.9.4 9.4 Portfolio acceptance test

A recruiter must understand within 20 seconds:

```
Bloque de codigo: text
what TwinSight is,
what technologies were used,
what the user can do,
why it is relevant to Unity/technical visualization roles.
```

42.10 10. ArtStation gap closure

42.10.1 10.1 Purpose

ArtStation should support:

```
Bloque de codigo: text
technical art credibility
visual presentation
breakdown discipline
3D craft
```

It is not the primary application platform for Unity WebGL roles, but it strengthens the proof package.

42.10.2 10.2 TwinSight ArtStation post

Required:

```
Bloque de codigo: text
[ ] cover image
[ ] short video
[ ] final screenshots
[ ] visual modes grid
[ ] CAD optimization before/after
[ ] wireframe comparison
[ ] UI screenshots
[ ] metrics card
[ ] tools list
[ ] links
```

42.10.3 10.3 Blender portrait ArtStation post

Required:

```
Bloque de codigo: text
[ ] final render
[ ] turntable or image sequence
[ ] references
[ ] sculpt stages
[ ] topology
[ ] UVs
[ ] material maps
[ ] grooming
[ ] lighting
[ ] tools
```

42.10.4 10.4 ArtStation warning

Do not let the Blender portrait dominate the professional positioning.

Use it as:

```
Bloque de codigo: text
supporting 3D craft proof
```

not as the flagship.

42.11 11. Demo video gap closure

42.11.1 11.1 Required videos

```
Bloque de codigo: text
[ ] 30-second teaser
[ ] 90-second main demo
[ ] 4-7 minute technical breakdown
```

42.11.2 11.2 Production priority

Bloque de código: text
 1. 90-second demo
 2. 30-second teaser
 3. technical breakdown

42.11.3 11.3 Must-show shots

Bloque de código: text
 [] browser/WebGL proof
 [] full drone hero
 [] component selection
 [] technical panel
 [] exploded view
 [] clipping
 [] visual modes
 [] CAD optimization
 [] metrics
 [] stack

42.11.4 11.4 Acceptance test

The video is acceptable if a viewer understands:

Bloque de código: text
 what it is,
 what problem it solves,
 what the user can do,
 why it is technically relevant

within 20 seconds.

42.12 12. English proof gap closure

42.12.1 12.1 Why this matters

LATAM remote and international roles repeatedly signal English communication requirements. Alexander's materials must demonstrate professional English before interviews.

42.12.2 12.2 Required English materials

Bloque de código: text
 [] English CV
 [] English LinkedIn
 [] English portfolio
 [] English README
 [] English video captions
 [] English project explanation script
 [] English interview answers

42.12.3 12.3 English demo script

Memorize:

Bloque de código: text
 TwinSight X500 is a Unity WebGL technical visualization prototype for drone assembly inspection. It transforms CAD-derived
 ↪ assets into an optimized browser-based 3D viewer with component selection, exploded view, cross-section tools, visual
 ↪ modes and technical information panels.

42.12.4 12.4 English risk

Avoid overcomplicated academic English.

Use clear professional English.

42.13 13. Shader / visual-mode proof closure

42.13.1 13.1 Why this matters

Several target roles signal shaders, rendering, optimization, Shader Graph, GPU/performance or technical art. TwinSight must show visual-mode proof.

42.13.2 13.2 Required proof

```
Bloque de codigo: text
[ ] visual modes grid
[ ] explanation of each mode
[ ] Shader Graph screenshot if used
[ ] material/renderer switching explanation
[ ] disclaimer for thermal-style mode
[ ] performance constraints note
```

42.13.3 13.3 Optional mini-project

If time allows, create a small standalone post:

```
Bloque de codigo: text
TwinSight Visual Modes Breakdown
```

Sections:

```
Bloque de codigo: text
realistic
X-ray
ghosted
blueprint
wireframe
thermal-style
implementation notes
limitations
```

42.14 14. Contractor readiness gap closure

42.14.1 14.1 Required contractor statement

```
Bloque de codigo: text
Based in Colombia and available for remote contractor/B2B work with international teams in Unity WebGL, interactive 3D and
↳ technical visualization.
```

42.14.2 14.2 Payment/admin items to prepare

```
Bloque de codigo: text
[ ] RUT status checked
[ ] invoicing process understood
[ ] payment methods identified
[ ] rate bands defined
[ ] timezone overlap statement ready
[ ] availability statement ready
[ ] contract red flags reviewed
```

42.14.3 14.3 Rate bands

Use internally:

```
Bloque de codigo: text
Entry acceptable: USD 1.5k-3k/month
Good target: USD 3k-6k/month
Ambitious target: USD 6k+/month
Hourly reference: derive from monthly target and workload
```

Do not publish rates on portfolio unless freelancing page requires it.

42.14.4 14.4 Remote work statement

```
Bloque de codigo: text
I am based in Colombia and can work with LATAM, North American or European teams depending on schedule overlap and contract
↔ structure.
```

42.15 15. Interview defense gap closure

42.15.1 15.1 Highest-risk topics

```
Bloque de codigo: text
limited formal industry experience
AI-assisted implementation
no shipped commercial Unity titles
shader depth
senior role expectations
digital twin terminology
WebGL performance
independent problem solving
```

42.15.2 15.2 Required answers

Prepare concise answers for:

```
Bloque de codigo: text
Tell me about TwinSight.
Why Unity WebGL?
How did you optimize CAD assets?
What was your exact role?
How much C# did you write?
How did you use AI?
Is this a real digital twin?
How did you evaluate it?
What would you improve?
Can you work without direct supervision?
```

42.15.3 15.3 Default positioning answer

```
Bloque de codigo: text
I have limited formal industry experience, but I built TwinSight X500 specifically to demonstrate the skills required for
↔ Unity WebGL and technical visualization work: C# runtime interaction, CAD-to-realtime optimization, technical UI, visual
↔ modes, WebGL deployment and structured evaluation.
```

42.16 16. Apply-readiness scoring

Score readiness from 0 to 100.

Area	Points
CV ready	15
LinkedIn ready	10
Portfolio case study	20
Demo video	15
GitHub README	15
ArtStation/breakdown	10
English materials	5
Interview defense	5
Contractor readiness	5
Total	100

42.16.1 16.1 Readiness interpretation

Score	Meaning
80–100	ready for serious applications
65–79	ready for selective Priority A roles
50–64	apply only to low-friction opportunities
<50	focus on materials first

42.17 17. Execution sprint

42.17.1 17.1 Sprint length

Bloque de código: text
10 working days

42.17.2 17.2 Sprint objective

Bloque de código: text
Reach 80/100 application readiness.

42.17.3 17.3 Day-by-day plan

Day	Output
1	capture TwinSight footage
2	edit 90-second demo
3	create 30-second teaser and thumbnail
4	finalize portfolio case-study page
5	update GitHub README and media
6	update LinkedIn profile and Featured
7	prepare ArtStation TwinSight breakdown
8	finalize English CV variants
9	rehearse interview answers
10	score first 20 roles and apply to 5–8

42.17.4 17.4 If time is limited

Prioritize:

```
Bloque de codigo: text
1. 90-second demo
2. portfolio page
3. GitHub README
4. CV
5. LinkedIn
```

ArtStation can follow.

42.18 18. Deliverable checklist

42.18.1 18.1 Must finish before serious applications

```
Bloque de codigo: text
[ ] 90-second TwinSight demo
[ ] TwinSight portfolio page
[ ] TwinSight GitHub README
[ ] English CV
[ ] LinkedIn profile update
[ ] application tracker scoring
```

42.18.2 18.2 Should finish soon after

```
Bloque de codigo: text
[ ] 30-second teaser
[ ] ArtStation breakdown
[ ] technical breakdown video
[ ] ARA README
[ ] Blender portrait breakdown
```

42.18.3 18.3 Optional later

```
Bloque de codigo: text
[ ] XR mini-demo
[ ] shader/visual-mode mini-post
[ ] WebGPU experiment
[ ] digital-twin sensor mockup
[ ] guided assembly mode
```

42.19 19. Final operating recommendation

Do not return to broad research until the proof package is public.

Current priority:

```
Bloque de codigo: text
Proof package > application volume > more research
```

The market has already shown the required proof signals:

```
Bloque de codigo: text
Unity
C#
WebGL
technical art
optimization
portfolio/reel
LATAM remote
English
Blender/3D support
```

The next bottleneck is not information. The bottleneck is public evidence.

42.20 20. Next module

Proceed with:

Bloque de codigo: text
33_portfolio_asset_production_sprint.md

42.20.1 Tool recommendation

Tool	Use
Agent mode	No
Deep Research	No

42.20.2 Purpose

Create an execution sprint focused on producing the actual assets:

Bloque de codigo: text
screenshots
video footage
thumbnail
metrics cards
pipeline diagram
visual modes grid
GitHub media folder
portfolio media package
ArtStation breakdown images

Chapter 43

33_portfolio_asset_production_sprint

Fuente: 33_portfolio_asset_production_sprint.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1025

Ultima modificacion: 2026-06-11 20:08

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This module converts the strategy and copy files into a concrete asset-production sprint.

Contenido completo depurado

43.0.1 0. Module status

```
Bloque de codigo: text
Module: 33_portfolio_asset_production_sprint.md
Date generated: 2026-06-12
Status: Complete v1
Purpose: Execute the production of portfolio assets needed before serious applications
Inputs: 08B, 19B, 20B, 21B, 29, 29B, 29C, 30, 31, 32
Use: production sprint for TwinSight media, GitHub media, portfolio visuals, ArtStation breakdown, LinkedIn proof
```

43.1 1. Purpose

This module converts the strategy and copy files into a concrete asset-production sprint.

The objective is to produce the public proof package required for applications:

```
Bloque de codigo: text
TwinSight video
portfolio media
GitHub media
ArtStation breakdown images
LinkedIn Featured assets
CV/portfolio visual proof
```

The bottleneck is no longer research.

The bottleneck is public, visible evidence.

43.2 2. Sprint goal

43.2.1 2.1 Final sprint outcome

By the end of the sprint, Alexander should have:

```
Bloque de codigo: text
1. a 90-second TwinSight demo video;
2. a 30-second teaser;
3. a strong thumbnail;
4. a screenshot package;
5. a visual modes grid;
6. a metrics card;
7. a CAD-to-realtime pipeline diagram;
8. an architecture diagram;
9. a GitHub /media folder;
10. a portfolio-ready media folder;
11. an ArtStation breakdown image set;
12. a LinkedIn Featured media package.
```

43.2.2 2.2 Minimum viable output

If time is limited, produce only:

```
Bloque de codigo: text
[ ] 90-second demo video
[ ] thumbnail
[ ] 8 screenshots
[ ] visual modes grid
[ ] metrics card
[ ] GitHub media folder
[ ] portfolio media folder
```

43.2.3 2.3 Strong output

```
Bloque de codigo: text
[ ] 90-second demo video
[ ] 30-second teaser
[ ] 4-7 minute technical breakdown
[ ] 12-20 screenshots
[ ] visual modes grid
[ ] optimization before/after
[ ] pipeline diagram
[ ] architecture diagram
[ ] metrics card
[ ] ArtStation breakdown package
[ ] LinkedIn teaser post package
```

43.3 3. Asset inventory

43.3.1 3.1 Required video assets

Asset	File name	Duration	Use	Priority
Main demo	twinsight_x500_demo_90s.mp4	90–120 s	portfolio, LinkedIn Featured, GitHub	1
Short teaser	twinsight_x500_teaser_30s.mp4	30–45 s	LinkedIn post, homepage	2
Technical breakdown	twinsight_x500_technical_breakdown_5min.mp4	4–5 min	interviews, GitHub, technical review	3
GIF preview	twinsight_x500_preview.gif	5–8 s	GitHub README	3

43.3.2 3.2 Required image assets

Asset	File name	Use	Priority
Thumbnail	twinsight_x500_thumbnail.jpg	YouTube/GitHub/portfolio	1
Hero viewer	01_hero_view.jpg	portfolio hero	1
Browser proof	02_browser_webgl_view.jpg	WebGL proof	1
Component selection	03_component_selection.jpg	feature card	1
Technical panel	04_technical_panel.jpg	UI proof	1
Exploded view	05_exploded_view.jpg	feature card	1
Cross-section	06_cross_section.jpg	feature card	1
Visual modes grid	07_visual_modes_grid.jpg	technical art proof	1
CAD optimization	08_cad_optimization_before_after.jpg	pipeline proof	1
Metrics card	09_metrics_card.jpg	validation proof	1
Pipeline diagram	10_pipeline_diagram.jpg	technical pipeline	2
Architecture diagram	11_architecture_diagram.jpg	Unity systems	2
Blender cleanup	12_blender_cleanup.jpg	process proof	2
Unity editor	13_unity_editor.jpg	implementation proof	2
GitHub preview	14_github_readme_preview.jpg	proof loop	3
ArtStation cover	15_artstation_cover.jpg	ArtStation	2

43.4 4. Folder structure

43.4.1 4.1 Master production folder

```
Bloque de codigo: text
twinsight_media_production/
├── 00_raw_capture/
│   ├── browser/
│   ├── unity_editor/
│   ├── blender/
│   └── metrics/
├── 01_selects/
├── 02_video_project/
├── 03_exports/
│   ├── video/
│   ├── images/
│   └── gifs/
├── 04_portfolio/
├── 05_github_media/
├── 06_artstation/
├── 07_linkedin/
└── 08_archive/
```

43.4.2 4.2 GitHub media folder

```
Bloque de codigo: text
media/
├── twinsight-demo-thumbnail.jpg
├── 01_viewer.jpg
├── 02_exploded_view.jpg
├── 03_cross_section.jpg
├── 04_visual_modes_grid.jpg
├── 05_technical_ui.jpg
├── 06_cad_optimization.jpg
├── 07_pipeline_diagram.jpg
├── 08_metrics_card.jpg
└── twinsight-preview.gif
```

43.4.3 4.3 Portfolio media folder

```
Bloque de codigo: text
portfolio/projects/twinsight-x500/
├── hero.jpg
├── demo_90s.mp4
└── teaser_30s.mp4
```

```
| thumbnail.jpg
| feature_component_selection.jpg
| feature_exploded_view.jpg
| feature_cross_section.jpg
| visual_modes_grid.jpg
| cad_optimization.jpg
| pipeline_diagram.jpg
| architecture_diagram.jpg
| metrics_card.jpg
```

43.5 5. Capture preparation

43.5.1 5.1 Before recording

```
Bloque de codigo: text
[ ] final WebGL build opens reliably
[ ] local build or hosted build tested
[ ] browser zoom set to readable scale
[ ] UI text is readable at 1080p
[ ] no debug UI visible unless intentional
[ ] final materials assigned
[ ] visual modes working
[ ] exploded view working
[ ] clipping tool working
[ ] component selection working
[ ] technical panels populated
[ ] browser performance acceptable
[ ] recording software tested
[ ] screen resolution fixed at 1920x1080 or higher
```

43.5.2 5.2 Build cleanup

```
Bloque de codigo: text
[ ] remove placeholder labels
[ ] remove console errors where possible
[ ] remove incomplete features from visible UI
[ ] hide unfinished buttons
[ ] ensure selected objects highlight cleanly
[ ] verify clipping plane does not break composition
[ ] reset default camera position
[ ] set default view to best hero angle
```

43.5.3 5.3 Visual cleanup

```
Bloque de codigo: text
[ ] consistent lighting
[ ] neutral background
[ ] no distracting browser tabs
[ ] no desktop notifications
[ ] clean cursor movement
[ ] readable color contrast
[ ] stable camera speed
```

43.6 6. Capture order

43.6.1 6.1 Capture sequence

Record in this order:

```
Bloque de codigo: text
1. hero orbit
2. browser/WebGL proof
3. navigation
4. component hover/select
5. technical panel
```

```

6. exploded view
7. exploded view close-up
8. clipping plane
9. cross-section internal view
10. visual modes one by one
11. visual modes grid source shots
12. CAD/high-poly source
13. optimized model / wireframe
14. Blender cleanup
15. Unity editor scene
16. metrics cards
17. GitHub README preview

```

43.6.2 6.2 Why this order

This matches the final video structure and reduces editing complexity.

Do not capture randomly.

43.7 7. Required shots

43.7.1 7.1 Hero shot

Requirement	Specification
Composition	drone centered, UI visible but not cluttered
Motion	slow orbit or camera pan
Duration	5–8 s raw
Output	hero image + opening video
Acceptance	understandable at thumbnail size

43.7.2 7.2 Browser proof shot

Requirement	Specification
Composition	browser frame visible
Motion	simple interaction
Duration	5–8 s
Purpose	prove WebGL/no-install
Caption	Browser-based Unity WebGL viewer

43.7.3 7.3 Component selection shot

Requirement	Specification
Action	hover/select component
UI	highlight + panel appears
Duration	6–10 s
Purpose	interaction proof
Caption	Component-level selection

43.7.4 7.4 Exploded view shot

Requirement	Specification
Action	activate exploded view
Motion	smooth separation
Duration	8–12 s
Purpose	assembly hierarchy proof
Caption	Exploded assembly view

43.7.5 7.5 Cross-section shot

Requirement	Specification
Action	move clipping plane
View	internal structures visible
Duration	8–12 s
Purpose	inspection proof
Caption	Cross-section inspection

43.7.6 7.6 Visual modes shot

Requirement	Specification
Modes	realistic, X-ray, ghosted, blueprint, wireframe, solid, thermal-style
Motion	switch modes cleanly
Duration	2–4 s per mode
Purpose	technical art proof
Caption	Multiple inspection modes

43.7.7 7.7 Optimization shot

Requirement	Specification
Content	CAD/high-poly vs optimized
View	wireframe or triangle comparison
Duration	5–8 s
Purpose	CAD-to-realtime proof
Caption	CAD-derived assets optimized for WebGL

43.7.8 7.8 Metrics shot

Requirement	Specification
Content	triangle count, SUS, NASA-TLX
Format	clean metric cards
Duration	5–8 s
Purpose	validation proof
Caption	Formative usability/workload evaluation

43.8 8. Screenshot package

43.8.1 8.1 Minimum screenshots

Bloque de codigo: text

01_hero_view.jpg
02_browser_webgl_view.jpg
03_component_selection.jpg
04_technical_panel.jpg
05_exploded_view.jpg
06_cross_section.jpg
07_visual_modes_grid.jpg
08_cad_optimization_before_after.jpg
09_metrics_card.jpg
10_pipeline_diagram.jpg

43.8.2 8.2 Screenshot requirements

Requirement	Standard
Resolution	1920×1080 minimum
UI	readable
Compression	high quality JPG or PNG
Color	consistent
Naming	numbered and descriptive
Cropping	consistent
Text	minimal and readable
Artifacts	none visible

43.8.3 8.3 Screenshot rejection criteria

Reject screenshot if:

Bloque de codigo: text

UI is unreadable
cursor blocks important detail
model is cropped poorly
lighting hides geometry
debug artifacts visible
text is too small
camera angle is ambiguous
feature is not understandable

43.9 9. Visual modes grid

43.9.1 9.1 Required grid

Create a 3×3 grid:

Bloque de codigo: text

Realistic | X-ray | Ghosted
Blueprint | Wireframe | Solid
Thermal-style | Clipping | Exploded

43.9.2 9.2 Grid requirements

Requirement	Standard
Same camera angle	yes

Requirement	Standard
Same crop	yes
Labels	short
Resolution	1920×1080 or 2400×1600
Background	consistent
Text	readable
Use	portfolio, ArtStation, GitHub

43.9.3 9.3 Thermal-style disclaimer

Include near the grid:

Bloque de código: text
Thermal-style is a qualitative visual mode, not a physical simulation.

43.10 10. Metrics card

43.10.1 10.1 Required metrics card

Use these values:

Bloque de código: text
95,617 optimized triangles
6.5M+ CAD source triangle routes
12 participants
SUS average: 91.88
NASA-TLX Raw: 8.69 viewer vs 19.89 2D support
96 task-condition records

43.10.2 10.2 Layout

Recommended layout:

Bloque de código: text
Top: TwinSight X500 – Evaluation & Optimization
Left: CAD optimization metrics
Right: usability/workload metrics
Bottom: academic/portfolio prototype note

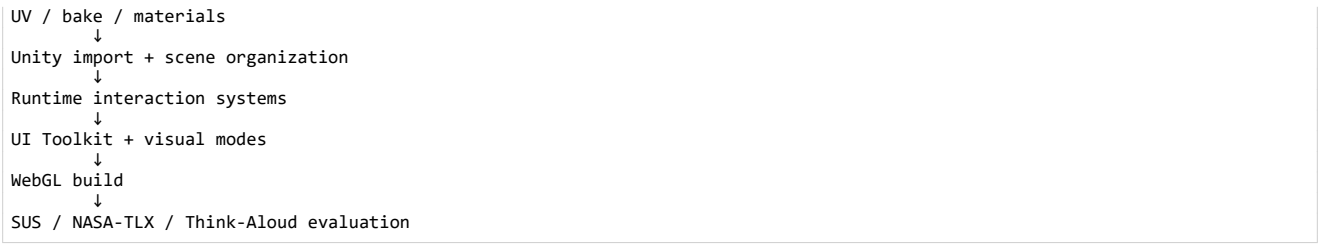
43.10.3 10.3 Safe footer

Bloque de código: text
Formative academic evaluation; results apply to tested tasks and participant sample.

43.11 11. Pipeline diagram

43.11.1 11.1 Required diagram

Bloque de código: text
CAD-derived source
↓
Tessellation / conversion
↓
Blender cleanup + optimization
↓



43.11.2 11.2 Diagram requirements

Requirement	Standard
Orientation	horizontal or vertical
Labels	concise
Tools	visible but not cluttered
Export	PNG/SVG
Use	portfolio + GitHub
Style	match portfolio theme

43.11.3 11.3 Tool accuracy rule

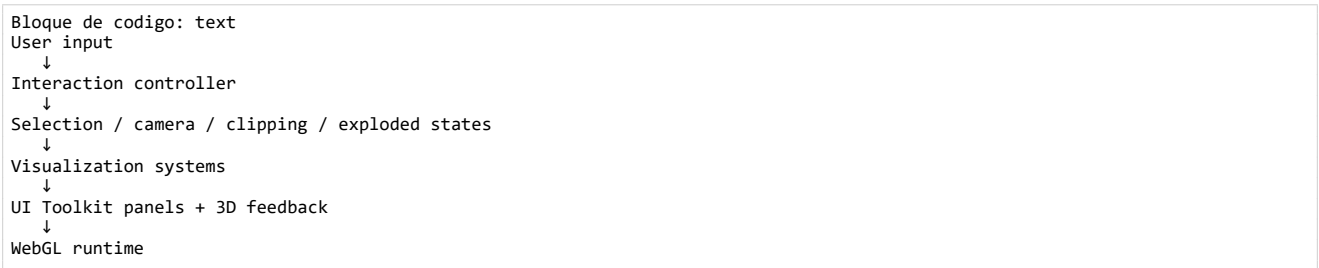
Only include:

```
Bloque de codigo: text
Marmoset
RizomUV
Substance
```

if actually used.
If planned but not used, omit or mark separately.

43.12 12. Architecture diagram

43.12.1 12.1 Required diagram



43.12.2 12.2 Optional expanded version

```
Bloque de codigo: text
Input Layer
Camera Controller
Selection System
State Managers
Visualization Mode Manager
Clipping Controller
Exploded View Controller
Part Metadata System
UI Toolkit
WebGL Build
```

43.12.3 12.3 Use

Use	Diagram version
portfolio	simple
GitHub	medium
technical breakdown video	expanded
interview	expanded

43.13 13. Video production

43.13.1 13.1 90-second demo timeline

```
Bloque de código: text
00-05 s Hero / title
05-12 s Problem
12-22 s Solution / WebGL
22-34 s Component selection
34-46 s Exploded view
46-58 s Cross-section
58-68 s Visual modes
68-78 s CAD optimization
78-86 s Evaluation metrics
86-90 s Stack / links
```

43.13.2 13.2 30-second teaser timeline

```
Bloque de código: text
00-03 s Hero
03-07 s Select component
07-12 s Exploded view
12-17 s Clipping
17-22 s Visual modes
22-27 s Metrics
27-30 s CTA
```

43.13.3 13.3 5-minute technical breakdown

```
Bloque de código: text
00:00-00:30 overview
00:30-01:10 problem
01:10-02:00 CAD optimization
02:00-02:50 Blender workflow
02:50-03:40 Unity architecture
03:40-04:30 interaction systems
04:30-05:10 visual modes
05:10-05:50 UI + evaluation
05:50-06:30 limitations / next steps
```

43.14 14. Captions

43.14.1 14.1 Main captions

```
Bloque de código: text
TwinSight X500
Unity WebGL technical visualization
From static 2D docs to spatial understanding
Browser-based drone assembly inspection
Component-level selection
Technical part information
Exploded assembly view
```

Cross-section inspection
Multiple visual modes
CAD-derived assets optimized for WebGL
SUS 91.88 · NASA-TLX Raw 8.69 vs 19.89
Unity · C# · WebGL · URP · UI Toolkit · Blender

43.14.2 14.2 Caption rules

Bloque de codigo: text
[] max 6–10 words
[] readable at mobile size
[] consistent position
[] high contrast
[] no paragraphs
[] no academic jargon in main reel

43.15 15. Thumbnail production

43.15.1 15.1 Recommended thumbnail

Bloque de codigo: text
Exploded drone view + UI panel + title

Text:

Bloque de codigo: text
TwinSight X500
Unity WebGL Technical Visualization

43.15.2 15.2 Thumbnail requirements

Requirement	Standard
readable at small size	yes
not too much text	yes
strong contrast	yes
shows actual project	yes
no stock imagery	yes
no academic title slide	yes

43.15.3 15.3 Thumbnail variants

Bloque de codigo: text
thumbnail_hero.jpg
thumbnail_exploded.jpg
thumbnail_cross_section.jpg
thumbnail_visual_modes.jpg

Pick one primary.

43.16 16. ArtStation package

43.16.1 16.1 Required ArtStation images

Bloque de codigo: text

```
01_cover.jpg
02_final_view.jpg
03_video_thumbnail.jpg
04_component_selection.jpg
05_exploded_view.jpg
06_cross_section.jpg
07_visual_modes_grid.jpg
08_cad_optimization.jpg
09_wireframe_comparison.jpg
10_ui_breakdown.jpg
11_metrics_card.jpg
12_tools_pipeline.jpg
```

43.16.2 16.2 ArtStation order

```
Bloque de codigo: text
cover
short video
final screenshots
feature breakdown
CAD optimization
visual modes grid
UI / interaction
metrics
tools
links
```

43.16.3 16.3 ArtStation text blocks

Use copy from:

```
Bloque de codigo: text
29C_artstation_breakdown_benchmark.md
```

Do not rewrite from scratch.

43.17 17. LinkedIn package

43.17.1 17.1 Required LinkedIn assets

```
Bloque de codigo: text
twinsight_teaser_30s.mp4
twinsight_thumbnail.jpg
twinsight_metrics_card.jpg
twinsight_visual_modes_grid.jpg
portfolio link
GitHub link
```

43.17.2 17.2 LinkedIn post package

Use:

```
Bloque de codigo: text
video teaser
short post
project link
3-5 hashtags
```

Recommended hashtags:

```
Bloque de codigo: text
#Unity3D
#TechnicalArt
#WebGL
#Realtime3D
#TechnicalVisualization
```

Use 3–5, not 15.

43.17.3 17.3 Featured section

Add in this order:

```
Bloque de codigo: text
1. TwinSight case study
2. TwinSight 90-second demo
3. GitHub repository
4. ArtStation breakdown
```

43.18 18. GitHub package

43.18.1 18.1 Required files

```
Bloque de codigo: text
media/twinsight-demo-thumbnail.jpg
media/01_viewer.jpg
media/02_exploded_view.jpg
media/03_cross_section.jpg
media/04_visual_modes_grid.jpg
media/05_technical_ui.jpg
media/06_cad_optimization.jpg
media/07_pipeline_diagram.jpg
media/08_metrics_card.jpg
media/twinsight-preview.gif
```

43.18.2 18.2 README update

Use:

```
Bloque de codigo: text
19B_github_readme_twinsight_final.md
```

43.18.3 18.3 GitHub acceptance criteria

```
Bloque de codigo: text
[ ] README explains project in first screen
[ ] demo thumbnail visible
[ ] screenshots visible
[ ] stack visible
[ ] metrics visible
[ ] limitations visible
[ ] links working
[ ] repository topics added
```

43.19 19. Portfolio package

43.19.1 19.1 Required media

```
Bloque de codigo: text
hero.jpg
demo_90s.mp4
teaser_30s.mp4
visual_modes_grid.jpg
metrics_card.jpg
pipeline_diagram.jpg
architecture_diagram.jpg
cad_optimization.jpg
feature screenshots
```


43.19.2 19.2 Copy source

Use:

```
Bloque de codigo: text
20B_portfolio_page_implementation_copy.md
```

43.19.3 19.3 Case-study structure source

Use:

```
Bloque de codigo: text
08B_twinsight_case_study_final_structure.md
```

43.20 20. Quality gates

43.20.1 20.1 Gate 1 — visual clarity

Pass if:

```
Bloque de codigo: text
[ ] a non-technical viewer understands it is a drone viewer
[ ] a technical viewer understands it is Unity/WebGL
[ ] UI is readable
[ ] features are visually obvious
```

43.20.2 20.2 Gate 2 — technical proof

Pass if:

```
Bloque de codigo: text
[ ] CAD optimization shown
[ ] runtime interaction shown
[ ] visual modes shown
[ ] evaluation shown
[ ] stack shown
```

43.20.3 20.3 Gate 3 — claim safety

Pass if:

```
Bloque de codigo: text
[ ] no full digital twin claim
[ ] no IoT claim
[ ] no real thermal simulation claim
[ ] no production deployment claim
[ ] no seniority overclaim
```

43.20.4 20.4 Gate 4 — application readiness

Pass if:

```
Bloque de codigo: text
[ ] portfolio page public
[ ] GitHub README public
[ ] demo video public
[ ] LinkedIn Featured updated
[ ] CV links point to working pages
```

43.21 21. 5-day compressed sprint

43.21.1 Day 1 — capture

Bloque de código: text
Record:

- hero
- browser
- selection
- exploded
- clipping
- visual modes
- CAD/Blender/Unity proof

43.21.2 Day 2 — edit

Bloque de código: text
Export:

- 90-second demo
- thumbnail
- 30-second teaser rough cut

43.21.3 Day 3 — visual assets

Bloque de código: text
Create:

- screenshots
- visual modes grid
- metrics card
- pipeline diagram
- architecture diagram

43.21.4 Day 4 — integration

Bloque de código: text
Update:

- portfolio
- GitHub README
- ArtStation draft
- LinkedIn Featured

43.21.5 Day 5 — QA and publish

Bloque de código: text
Check:

- all links
- all captions
- mobile readability
- claim safety
- application tracker links

43.22 22. 10-day full sprint

Day	Deliverable
1	footage capture
2	90-second demo edit
3	30-second teaser + thumbnail
4	screenshot package
5	visual modes grid + metrics card
6	pipeline + architecture diagrams
7	GitHub media folder + README
8	portfolio page media integration

Day	Deliverable
9	ArtStation breakdown
10	LinkedIn Featured + QA

43.23 23. Done definition

This sprint is complete when:

```
Bloque de codigo: text
[ ] TwinSight can be understood from the demo video alone.
[ ] TwinSight can be understood from the portfolio page alone.
[ ] TwinSight can be understood from the GitHub README alone.
[ ] The same positioning appears across portfolio, GitHub, LinkedIn and ArtStation.
[ ] Claims are safe and defensible.
[ ] Links are working.
[ ] Application tracker can use the project links.
```

43.24 24. Next module

Proceed with:

```
Bloque de codigo: text
34_weekly_application_execution_system.md
```

43.24.1 Tool recommendation

Tool	Use
Agent mode	No
Deep Research	No

43.24.2 Purpose

Create the weekly operating system for:

```
Bloque de codigo: text
searching roles
scoring roles
applying
following up
updating the tracker
reviewing metrics
improving materials
```

This should connect modules:

```
Bloque de codigo: text
25 tracker
26 dashboard
30 market snapshot
31 targeting matrix
32 material gaps
33 asset sprint
```

Chapter 44

34_weekly_application_execution_system

Fuente: 34_weekly_application_execution_system.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1128

Ultima modificacion: 2026-06-11 20:09

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This module defines the weekly execution system for Alexander's job-search roadmap.

Contenido completo depurado

44.0.1 0. Module status

```
Bloque de codigo: text
Module: 34_weekly_application_execution_system.md
Date generated: 2026-06-12
Status: Complete v1
Purpose: Weekly operating system for applications, follow-ups, tracker updates and portfolio iteration
Inputs: 25, 26, 27, 30, 31, 32, 33
Use: weekly execution cadence for job search and portfolio-driven applications
```

44.1 1. Purpose

This module defines the weekly execution system for Alexander's job-search roadmap.

The goal is not to maximize raw application volume. The goal is to run a controlled system that balances:

```
Bloque de codigo: text
portfolio proof
targeted applications
market feedback
follow-ups
interview preparation
material iteration
```

Current strategic position:

```
Bloque de codigo: text
Unity WebGL / Real-Time 3D Developer
Technical Visualization + Technical Art
XR / Simulation-adjacent portfolio
C# + shaders + optimization + 3D pipeline
```

44.2 2. Operating principle

44.2.1 2.1 Main rule

Bloque de codigo: text
Proof quality first, application volume second.

Because the current market evidence shows:

Bloque de codigo: text
- many roles are senior;
- portfolio/reel proof matters;
- technical-art/rendering proof matters;
- LATAM remote exists but is narrow;
- WebGL/technical visualization is a differentiator;
- generic Unity roles are competitive and often game/mobile-heavy.

44.2.2 2.2 Weekly loop

Bloque de codigo: text
1. Search
2. Score
3. Tailor
4. Apply
5. Follow up
6. Track
7. Review
8. Improve materials

This loop repeats weekly.

44.3 3. Weekly objective

44.3.1 3.1 Default weekly objective

Bloque de codigo: text
Submit 5-12 high-fit applications while improving one portfolio/material asset.

44.3.2 3.2 Application volume by readiness

Readiness level	Weekly applications	Material work
<50/100	0-3	heavy material production
50-64/100	3-5	portfolio/GitHub first
65-79/100	5-8	selective applications
80-100/100	10-15	steady iteration

44.3.3 3.3 Current recommended mode

Bloque de codigo: text
Selective application mode

Until the TwinSight proof package is public:

```
Bloque de codigo: text
5-8 applications/week
Priority A/B only
high tailoring
```

44.4 4. Weekly schedule

44.4.1 4.1 Monday — search and score

```
Bloque de codigo: text
Goal:
Build the week's candidate opportunity list.
```

Tasks:

```
Bloque de codigo: text
[ ] Search primary role terms.
[ ] Capture 10-20 postings.
[ ] Reject obvious mismatches.
[ ] Score remaining roles.
[ ] Select 5-12 target applications.
[ ] Assign CV variant.
[ ] Assign portfolio/GitHub links.
```

Primary search terms:

```
Bloque de codigo: text
Unity WebGL Developer remote
Unity Technical Artist remote
Unity Developer LATAM remote
Real-Time 3D Developer remote
Interactive 3D Developer remote
Technical Visualization Unity remote
```

Output:

```
Bloque de codigo: text
weekly_target_roles.csv or tracker rows updated
```

44.4.2 4.2 Tuesday — portfolio/material production

```
Bloque de codigo: text
Goal:
Improve proof before sending applications.
```

Tasks:

```
Bloque de codigo: text
[ ] Improve TwinSight demo/media.
[ ] Update case-study section.
[ ] Add one screenshot or diagram.
[ ] Improve GitHub README.
[ ] Update LinkedIn Featured if needed.
[ ] Prepare role-specific CV variant.
```

Priority order:

```
Bloque de codigo: text
1. 90-second demo
2. TwinSight case study
3. GitHub README
4. LinkedIn Featured
5. ArtStation breakdown
```

Output:

```
Bloque de codigo: text
one public material improvement
```

44.4.3 4.3 Wednesday — tailored applications

```
Bloque de codigo: text
Goal:
Submit the strongest applications.
```

Tasks:

```
Bloque de codigo: text
[ ] Apply to Priority A roles.
[ ] Apply to strong Priority B roles.
[ ] Use tailored CV variant.
[ ] Include correct portfolio link.
[ ] Include GitHub link when technically relevant.
[ ] Add short cover note if field exists.
[ ] Update tracker immediately.
```

Application target:

```
Bloque de codigo: text
3-6 applications
```

Output:

```
Bloque de codigo: text
tracker status = Applied
```

44.4.4 4.4 Thursday — follow-ups and recruiter contact

```
Bloque de codigo: text
Goal:
Increase response probability.
```

Tasks:

```
Bloque de codigo: text
[ ] Send follow-ups for applications older than 5-7 business days.
[ ] Connect with recruiters/hiring managers only for strong-fit roles.
[ ] Message companies where portfolio relevance is high.
[ ] Update tracker with follow-up status.
[ ] Save responses and objections.
```

Follow-up target:

```
Bloque de codigo: text
3-8 follow-ups
```

Output:

```
Bloque de codigo: text
tracker status = Follow-up sent
```

44.4.5 4.5 Friday — niche search and review

Bloque de código: text
Goal:
Capture niche roles and extract market learning.

Search terms:

Bloque de código: text
Digital Twin Unity freelance
Unity XR Developer LATAM
Unity Shader Graph remote
CAD visualization developer
Unity tools developer remote
Simulation visualization Unity
3D product configurator developer

Tasks:

Bloque de código: text
[] Search niche terms.
[] Save promising companies.
[] Add new recurring requirements to gap log.
[] Review weekly metrics.
[] Decide next week's material improvement.

Output:

Bloque de código: text
weekly_review.md or dashboard update

44.4.6 4.6 Weekend — optional deep work

Bloque de código: text
Goal:
Advance portfolio assets without application pressure.

Recommended tasks:

Bloque de código: text
[] Edit demo video.
[] Create ArtStation breakdown.
[] Improve Blender portrait breakdown.
[] Create technical breakdown video.
[] Add one mini-project if needed.
[] Practice interview answers.

44.5 5. Daily micro-routine

44.5.1 5.1 30-minute minimum routine

Bloque de código: text
10 min – check tracker and next action
10 min – search or apply to one role
10 min – improve one material or answer

44.5.2 5.2 90-minute standard routine

Bloque de código: text
15 min – review tracker
25 min – search and score
30 min – apply / follow up
20 min – improve material

44.5.3 5.3 3-hour deep work block

```
Bloque de código: text
30 min – choose target
90 min – portfolio/video/GitHub production
45 min – applications/follow-ups
15 min – tracker update
```

44.6 6. Application scoring workflow

44.6.1 6.1 Score every role

Use the scoring model from module 31.

Score type	Range
Fit	1–5
Probability	1–5
Compensation	1–5
Portfolio-fit	1–5
Region	1–5
Total	5–25

44.6.2 6.2 Action threshold

Total score	Action
21–25	Apply now
17–20	Apply if materials are close
13–16	Save / low-effort apply
9–12	Watchlist
≤8	Reject

44.6.3 6.3 Quick scoring template

```
Bloque de código: text
Role:
Company:
Remote scope:
Fit:
Probability:
Compensation:
Portfolio-fit:
Region:
Total:
Action:
CV variant:
Portfolio link:
GitHub link:
Risk:
```

44.7 7. Tracker workflow

44.7.1 7.1 Required tracker fields

```
Bloque de código: text
```

```

Company
Role title
URL
Date found
Date applied
Remote scope
Colombia eligibility
Seniority
Salary
Employment model
Role family
Priority
Fit score
Probability score
Compensation score
Portfolio-fit score
Region score
Total score
CV variant
Portfolio link used
GitHub link used
Cover note used
Status
Follow-up date
Response
Interview date
Risk notes
Next action

```

44.7.2 7.2 Status labels

```

Bloque de codigo: text
Found
Scored
Saved
Applied
Follow-up due
Follow-up sent
Interview
Technical test
Offer
Rejected
Closed
Watchlist
Do not apply

```

44.7.3 7.3 Tracker update rule

Update the tracker immediately after:

```

Bloque de codigo: text
finding a role
scoring a role
applying
following up
receiving a response
interview scheduling
rejection
offer

```

Do not rely on memory.

44.8 8. Application package selection

44.8.1 8.1 Package by role family

Role family	CV variant	First link	Second link
Unity WebGL	Unity WebGL CV	TwinSight case study	GitHub
Unity Technical Artist	Technical Artist CV	TwinSight breakdown	ArtStation
Real-Time 3D	Real-Time 3D CV	TwinSight demo	GitHub
Technical Visualization	Technical Visualization CV	TwinSight case study	demo
XR Unity	XR-adjacent CV	TwinSight + future XR note	demo

Role family	CV variant	First link	Second link
LATAM Unity	LATAM Unity CV	TwinSight	GitHub
Tools/Pipeline	Tools CV	ARA / pipeline	TwinSight
Shader/VFX	TA/shader variant	visual modes grid	ArtStation

44.8.2 8.2 Package by priority

Priority A

```
Bloque de codigo: text
[ ] tailored CV
[ ] portfolio case-study link
[ ] GitHub link
[ ] demo video link
[ ] short cover note
[ ] follow-up plan
```

Priority B

```
Bloque de codigo: text
[ ] CV variant
[ ] portfolio link
[ ] GitHub or demo link
[ ] brief note if available
```

Priority C

```
Bloque de codigo: text
[ ] generic aligned CV
[ ] portfolio link
[ ] no heavy tailoring
```

44.9 9. Cover note workflow

44.9.1 9.1 When to write a note

Write a note only if:

```
Bloque de codigo: text
the platform has a field,
the fit is Priority A/B,
and the note can mention specific overlap.
```

44.9.2 9.2 Note length

```
Bloque de codigo: text
50-120 words
```

44.9.3 9.3 Note formula

```
Bloque de codigo: text
1. Role fit
2. TwinSight evidence
3. Stack match
4. Remote/contractor availability if relevant
```

44.9.4 9.4 Example — Unity WebGL

Bloque de codigo: text

I'm applying because this role aligns closely with my Unity WebGL technical visualization work. My flagship project, TwinSight X500, is a browser-based 3D inspection prototype for drone assembly, built with Unity, C#, WebGL, URP, UI Toolkit and Blender. It includes component selection, exploded view, clipping tools, visual modes, CAD-to-realtime optimization and usability/workload evaluation.

44.9.5 9.5 Example — LATAM remote Unity

Bloque de codigo: text

I'm based in Colombia and available for remote LATAM or international contractor work. My strongest Unity project is TwinSight X500, a WebGL technical visualization prototype with C#, URP, UI Toolkit, Blender optimization, component interaction and usability evaluation.

44.10 10. Follow-up system

44.10.1 10.1 Follow-up timing

Situation	Follow-up
application submitted	after 5–7 business days
recruiter viewed profile	within 24–48 hours
interview completed	within 24 hours
technical test submitted	within 24 hours
no response after follow-up	one final follow-up after 7 days

44.10.2 10.2 Follow-up template

Bloque de codigo: text

Hi [Name],

I wanted to follow up on my application for [Role]. I'm especially interested because the role aligns with my Unity WebGL and technical visualization work.

My flagship project, TwinSight X500, is a browser-based Unity prototype for drone assembly inspection, with component selection, exploded view, clipping tools, visual modes and CAD-to-realtime optimization.

Portfolio: [link]
GitHub: [link]

Best,
Alexander

44.10.3 10.3 Follow-up rule

Do not follow up more than twice unless there is active conversation.

44.11 11. Weekly metrics

Track these every week.

44.11.1 11.1 Activity metrics

Metric	Target
roles found	20–40
roles scored	10–20
applications sent	5–12
follow-ups sent	3–8
portfolio updates	1–2
recruiter contacts	2–5

44.11.2 11.2 Quality metrics

Metric	Target
average fit score	≥4
average portfolio-fit score	≥4
Priority A applications	≥40%
rejected before applying	high
generic applications	low

44.11.3 11.3 Outcome metrics

Metric	Interpretation
response rate	material/targeting quality
recruiter screens	profile relevance
interviews	market fit
technical tests	proof credibility
rejections	expected, classify reasons
no responses	targeting/material issue

44.12 12. Weekly review template

Create a short review every Friday.

```
Bloque de código: markdown
# Weekly Application Review – [date]

## Numbers
- Roles found:
- Roles scored:
- Applications sent:
- Follow-ups sent:
- Responses:
- Interviews:
- Rejections:
- Portfolio updates:

## Best roles found
1.
2.
3.

## Rejection/no-response patterns
-

## New recurring requirements
-

## Portfolio/material gaps
-

## Next week focus
-
```

```
## Decision
- Continue same targeting:
- Adjust search terms:
- Improve materials:
```

44.13 13. Feedback loop

44.13.1 13.1 If no responses after 20 applications

Check:

```
Bloque de codigo: text
[ ] portfolio link visible?
[ ] demo video strong?
[ ] CV headline too broad?
[ ] applying to too-senior roles?
[ ] remote eligibility unclear?
[ ] cover note weak?
[ ] GitHub README incomplete?
```

44.13.2 13.2 If responses but no interviews

Check:

```
Bloque de codigo: text
[ ] CV experience gap too visible?
[ ] project description not business-relevant?
[ ] salary expectation mismatch?
[ ] English profile weak?
[ ] not enough commercial framing?
```

44.13.3 13.3 If interviews but no offers

Check:

```
Bloque de codigo: text
[ ] interview defense weak?
[ ] AI-use answer unclear?
[ ] technical depth not defended?
[ ] coding test gap?
[ ] shader/rendering gap?
[ ] remote contractor concerns?
```

44.14 14. Material improvement triggers

44.14.1 14.1 Improve CV if

```
Bloque de codigo: text
response rate <5% after 20 applications
recruiters misunderstand target role
roles keep rejecting for experience mismatch
```

44.14.2 14.2 Improve portfolio if

```
Bloque de codigo: text
people click but do not respond
recruiters ask what the project is
technical leads ask for demo proof
```

44.14.3 14.3 Improve GitHub if

Bloque de código: text
technical reviewers ask for code
README does not explain architecture
repo looks incomplete

44.14.4 14.4 Improve demo video if

Bloque de código: text
viewers do not understand project in 20 seconds
video is too slow
features are unclear
UI unreadable

44.14.5 14.5 Improve interview answers if

Bloque de código: text
interviews lead to no next step
AI-use question creates doubt
seniority gap becomes central

44.15 15. Application cadence modes

44.15.1 15.1 Conservative mode

Use when materials are incomplete.

Bloque de código: text
5-8 applications/week
Priority A only
high tailoring
portfolio improvement every week

44.15.2 15.2 Balanced mode

Use when proof package is public.

Bloque de código: text
10-15 applications/week
Priority A/B
moderate tailoring
weekly follow-ups

44.15.3 15.3 Aggressive mode

Use only after materials are stable and response data exists.

Bloque de código: text
20-30 applications/week
A/B/C selected
less tailoring
high tracker discipline

44.15.4 15.4 Recommended now

Bloque de código: text
Conservative until TwinSight proof package is public.
Balanced after portfolio + GitHub + demo are published.

44.16 16. Weekly time budget

44.16.1 16.1 Minimum viable

Bloque de codigo: text
5 hours/week

Allocation:

Bloque de codigo: text
2 h search/scoring
1.5 h applications
1 h follow-ups
0.5 h tracker/review

44.16.2 16.2 Recommended

Bloque de codigo: text
10-12 hours/week

Allocation:

Bloque de codigo: text
3 h search/scoring
3 h applications
2 h follow-ups/networking
3 h material improvement
1 h weekly review

44.16.3 16.3 Sprint mode

Bloque de codigo: text
20+ hours/week

Use when producing:

Bloque de codigo: text
demo video
portfolio page
GitHub media
ArtStation breakdown

44.17 17. Interview preparation loop

44.17.1 17.1 Weekly rehearsal

Once per week, rehearse:

Bloque de codigo: text
Tell me about TwinSight.
Why Unity WebGL?
How did you optimize CAD?
What was your role?
How did you use AI?
Is it a digital twin?
What would you improve?
Why should we hire you despite limited formal experience?

44.17.2 17.2 Answer quality criteria

Good answers are:

```
Bloque de codigo: text
specific
technical
honest
short
evidence-backed
not defensive
```

Bad answers are:

```
Bloque de codigo: text
vague
overexplaining
self-deprecating
overclaiming
too academic
```

44.18 18. Remote contractor workflow

44.18.1 18.1 Contractor readiness weekly check

```
Bloque de codigo: text
[ ] timezone overlap clear
[ ] monthly/hourly target defined
[ ] payment route known
[ ] invoicing/admin path understood
[ ] contract red flags reviewed
[ ] English intro ready
```

44.18.2 18.2 Contractor intro line

```
Bloque de codigo: text
I'm based in Colombia and available for remote contractor work with international teams, especially in Unity WebGL,
↩ interactive 3D and technical visualization.
```

44.18.3 18.3 Avoid

```
Bloque de codigo: text
I can work for any rate.
I have no professional experience but I can learn.
I can do everything.
```

Use:

```
Bloque de codigo: text
My strongest proof is TwinSight X500, a Unity WebGL technical visualization prototype that demonstrates the core skills
↩ required for this type of work.
```

44.19 19. Role-specific weekly targeting

44.19.1 19.1 Week A — Unity WebGL / Technical Visualization

Search:

Bloque de codigo: text
Unity WebGL Developer remote
Technical Visualization Unity remote
Interactive 3D Developer remote
CAD visualization developer
3D product configurator developer

Materials focus:

Bloque de codigo: text
TwinSight case study
GitHub README
demo video

44.19.2 19.2 Week B — Unity Technical Artist / Rendering

Search:

Bloque de codigo: text
Unity Technical Artist remote
Unity Shader Graph remote
Unity render developer remote
Technical Artist Unity remote LATAM

Materials focus:

Bloque de codigo: text
visual modes grid
shader/visual mode breakdown
ArtStation

44.19.3 19.3 Week C — LATAM Nearshore Unity

Search:

Bloque de codigo: text
Unity Developer LATAM remote
Unity C# contractor LATAM
Unity Developer Colombia remote
Unity Developer Latin America remote

Materials focus:

Bloque de codigo: text
English CV
contractor statement
portfolio links

44.19.4 19.4 Week D — XR / Simulation / Digital Twin

Search:

Bloque de codigo: text
Unity XR Developer remote
Simulation Developer Unity remote
Digital Twin Unity freelance
XR Technical Artist Unity

Materials focus:

Bloque de codigo: text
XR-adjacent framing
digital-twin-adjacent wording
limitations clarity

Repeat monthly.

44.20 20. No-go rules

Do not spend weekly application time on:

```
Bloque de codigo: text
roles that require relocation now
roles that require US/EU work authorization now
pure full-stack roles
pure 3D artist roles
senior AAA roles
unpaid internships unless strategically exceptional
unclear crypto/metaverse roles
roles with no relation to Unity/3D/technical visualization
```

Do not keep researching indefinitely.

If the proof package is incomplete, work on proof.

44.21 21. Weekly done definition

A week is successful if:

```
Bloque de codigo: text
[ ] at least 5 good roles were scored
[ ] at least 3-8 applications were sent, depending on readiness
[ ] tracker is updated
[ ] follow-ups were sent where due
[ ] one material asset improved
[ ] one weekly review was written
[ ] next week focus is defined
```

A week is not successful if:

```
Bloque de codigo: text
many roles were browsed but none were tracked
applications were sent without scoring
materials were not improved
follow-ups were forgotten
no learning was extracted
```

44.22 22. Monthly review

At the end of each month, review:

```
Bloque de codigo: text
total applications
response rate
interview rate
best-performing role titles
worst-performing role titles
recurring requirements
portfolio gaps
salary/rate signals
remote eligibility patterns
countries/regions with traction
```

44.22.1 22.1 Monthly decision

Choose one:

```
Bloque de codigo: text
continue same targeting
narrow to best role family
expand to adjacent role family
pause applications and improve materials
increase application volume
adjust salary expectations
```

44.23 23. Current recommended next action

Since the application system depends on public proof, the next practical action is:

```
Bloque de codigo: text
Execute the 33_portfolio_asset_production_sprint.
```

In operational terms:

```
Bloque de codigo: text
1. produce TwinSight video;
2. publish portfolio case study;
3. update GitHub README;
4. update LinkedIn Featured;
5. then apply weekly using this system.
```

44.24 24. Next module

Proceed with:

```
Bloque de codigo: text
35_interview_defense_and_technical_screen_system.md
```

44.24.1 Tool recommendation

Tool	Use
Agent mode	No
Deep Research	No

44.24.2 Purpose

Create a focused interview defense system for:

```
Bloque de codigo: text
TwinSight explanation
AI-assisted workflow
limited formal experience
Unity/C# technical questions
WebGL constraints
CAD optimization
technical art / visual modes
remote contractor readiness
salary/offer risk
```

Chapter 45

35_interview_defense_and_technical_screen_3

Fuente: 35_interview_defense_and_technical_screen_system.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 952

Ultima modificacion: 2026-06-11 20:13

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This module prepares Alexander to defend his profile in interviews for:

Contenido completo depurado

45.0.1 0. Module status

```
Bloque de codigo: text
Module: 35_interview_defense_and_technical_screen_system.md
Date generated: 2026-06-12
Status: Complete v1
Purpose: Interview defense system for Unity WebGL / Real-Time 3D / Technical Visualization roles
Inputs: 08B, 17, 18, 19B, 20B, 21B, 23, 24, 30, 31, 32, 33, 34
Use: interview preparation, recruiter screens, technical screens, project defense, salary/contractor conversations
```

45.1 1. Purpose

This module prepares Alexander to defend his profile in interviews for:

```
Bloque de codigo: text
Unity WebGL Developer
Real-Time 3D Developer
Unity Technical Artist
Technical Visualization Developer
Interactive 3D Developer
XR-adjacent Unity Developer
Simulation Visualization Developer
LATAM remote Unity contractor
```

The main risk is not lack of project evidence. The main risk is failing to explain the evidence clearly under interview pressure.

This system gives:

- concise answers;

- technical defense points;
 - risk-control language;
 - technical screen preparation;
 - behavioral answer patterns;
 - AI-assisted workflow disclosure;
 - contractor/remote readiness answers;
 - red-flag handling.
-

45.2 2. Interview positioning baseline

45.2.1 2.1 Default professional identity

Use this as the baseline identity:

```
Bloque de código: text
I'm a Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, C#, CAD-to-realtime optimization and interactive
↳ technical visualization.
```

45.2.2 2.2 Default project anchor

Use TwinSight as the anchor project:

```
Bloque de código: text
My flagship project is TwinSight X500, a Unity WebGL technical visualization prototype for drone assembly inspection. It
↳ converts CAD-derived drone assets into an optimized browser-based 3D viewer with component selection, exploded view,
↳ cross-section tools, visual modes, technical UI panels and usability/workload evaluation.
```

45.2.3 2.3 Default credibility frame

```
Bloque de código: text
I have limited formal industry experience in Unity, but I built TwinSight to demonstrate the relevant skills directly:
↳ Unity/C# implementation, WebGL deployment, CAD-to-realtime optimization, runtime interaction systems, technical UI, visual
↳ modes and structured evaluation.
```

45.2.4 2.4 What not to lead with

Do not open with:

```
Bloque de código: text
I am a student with no experience.
I used AI heavily.
This was only a thesis.
I am trying to get into any tech role.
I can learn anything.
```

Lead with evidence, then clarify limitations.

45.3 3. 30-second pitch

45.3.1 3.1 General version

```
Bloque de código: text
I'm a Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL and technical visualization. My flagship project
↳ is TwinSight X500, a browser-based Unity prototype for drone assembly inspection. It includes component selection,
↳ exploded view, clipping tools, visual modes, technical UI, CAD-to-realtime optimization and evaluation with SUS and
↳ NASA-TLX. I'm targeting remote roles where interactive 3D, optimization and technical visualization are useful.
```

45.3.2 3.2 Unity WebGL version

Bloque de código: text

I build browser-based interactive 3D experiences with Unity WebGL. My main project, TwinSight X500, turns CAD-derived drone geometry into an optimized WebGL viewer with selection, exploded view, clipping, visual modes and technical information panels. The project demonstrates Unity, C#, WebGL deployment, UI Toolkit and performance-aware asset preparation.

45.3.3 3.3 Technical Artist version

Bloque de código: text

My strongest technical-art proof is TwinSight X500: a Unity WebGL technical visualization prototype with CAD-to-realtime optimization, visual inspection modes, clipping, exploded view and technical UI. It shows my ability to bridge 3D asset preparation, runtime systems, visual readability and performance constraints.

45.3.4 3.4 Technical Visualization version

Bloque de código: text

I focus on interactive 3D tools for technical visualization. TwinSight X500 was built to address a specific problem: static 2D drone assembly documentation makes spatial relationships harder to understand. The solution is a browser-based Unity viewer where users can inspect parts, reveal assembly structure and compare visual modes interactively.

45.4 4. 2-minute project explanation

Use this when asked:

Bloque de código: text

Tell me about your main project.

45.4.1 Answer

Bloque de código: text

TwinSight X500 is a Unity WebGL technical visualization prototype for inspecting the assembly of a Holybro X500 V2 drone.

The problem was that static 2D assembly documentation forces users to mentally reconstruct spatial relationships between components, orientations and hidden structures. I wanted to test whether an interactive 3D viewer could support understanding more directly.

Technically, the project involved transforming CAD-derived drone geometry into an optimized real-time WebGL asset, organizing components for selection, implementing runtime interaction systems in Unity/C#, building technical UI panels with UI Toolkit, and adding features like exploded view, clipping/cross-section inspection and multiple visual modes.

The project was also evaluated as an academic prototype using SUS, NASA-TLX Raw and Think-Aloud methods. The formative results suggested high perceived usability and lower perceived workload for the 3D viewer condition compared with 2D support in the tested tasks.

For me, the value of the project is that it demonstrates the full pipeline: CAD-to-realtime optimization, Unity WebGL, runtime interaction, technical UI, technical visualization and evaluation.

45.5 5. Project defense: TwinSight questions

45.5.1 5.1 What problem were you solving?

Bloque de código: text

The core problem was spatial understanding. Drone assembly documentation is often shown through 2D diagrams, part lists and written instructions. Users must mentally reconstruct how components relate in 3D space. TwinSight addresses that by turning the drone assembly into an interactive 3D viewer where users can select parts, see hierarchy through exploded view, cut through geometry and inspect technical information directly in context.

45.5.2 5.2 Why Unity WebGL?

Bloque de código: text

Unity gave me a mature real-time 3D environment for interaction systems, visual modes, UI and asset organization. WebGL was
↳ important because the prototype needed to run in the browser without requiring a local install. That made it more
↳ accessible for evaluation and more aligned with technical visualization tools that need to be shared easily.

45.5.3 5.3 Why not Three.js?

Bloque de código: text

Three.js would also be valid for browser-based 3D. I chose Unity WebGL because my project needed a real-time engine workflow
↳ with scene hierarchy, interaction systems, UI Toolkit, rendering modes and a pipeline closer to Unity technical-art and
↳ real-time visualization roles. For this portfolio direction, Unity was the stronger strategic fit.

45.5.4 5.4 Is it a digital twin?

Bloque de código: text

I would describe it as digital-twin-adjacent, not a full production digital twin. It represents a real physical drone assembly
↳ in interactive 3D, but it does not currently include live sensor data, operational state synchronization, predictive
↳ maintenance or real-time telemetry. The current scope is technical visualization and assembly inspection.

45.5.5 5.5 Is it a simulation?

Bloque de código: text

It is not a physics simulation. It is an interactive technical visualization prototype. It simulates inspection workflows
↳ visually, but it does not simulate drone physics, structural loads, heat transfer or flight behavior.

45.5.6 5.6 What was the hardest technical part?

Bloque de código: text

The hardest part was balancing technical readability with WebGL constraints. CAD-derived geometry can be too heavy for browser
↳ deployment, so the challenge was preserving enough visual detail for inspection while reducing geometry and organizing the
↳ model into selectable components. The second challenge was coordinating multiple runtime states: selection, exploded view,
↳ clipping, visual modes and UI panels.

45.5.7 5.7 What would you improve next?

Bloque de código: text

I would improve three areas. First, I would deepen the metadata system so each component has richer technical information.
↳ Second, I would add guided assembly steps and search/filter tools. Third, I would explore WebGPU, XR inspection or
↳ sensor-state integration as future extensions, while keeping the current version honest as a WebGL technical visualization
↳ prototype.

45.6 6. CAD-to-realtime optimization defense

45.6.1 6.1 Explain the CAD pipeline

Bloque de código: text

The project started from CAD-derived geometry that was not suitable for direct real-time use. I inspected different
↳ conversion/tessellation routes, cleaned and optimized the assets in Blender, organized the model into logical components
↳ and prepared the geometry for Unity WebGL. The goal was not to preserve every CAD detail, but to preserve the visual
↳ information needed for assembly inspection while making the model usable in a browser.

45.6.2 6.2 Explain the triangle reduction

Bloque de código: text

The source routes exceeded 6.5M triangles in some forms, while the optimized runtime model was reduced to 95,617 triangles. I
→ treat that as a portfolio and academic-prototype optimization metric, not as a production benchmark. The main point is
→ that I could transform heavy CAD-derived data into a WebGL-ready inspection model while preserving component readability.

45.6.3 6.3 What did you decide to keep or remove?

Bloque de código: text

I prioritized silhouette, component identity, assembly relationships and features needed for inspection. I removed or
→ simplified geometry that did not contribute to understanding the assembly, especially excessive tessellation or internal
→ details that would not be visible or useful in the final viewer.

45.6.4 6.4 How did you handle fasteners?

Bloque de código: text

Fasteners were treated as modular repeated elements where possible. The goal was to avoid treating every screw as a unique
→ asset when repeated geometry could be handled more efficiently. For a future production version, I would further
→ standardize fastener families, metadata and instancing.

45.6.5 6.5 What would you do differently in a production pipeline?

Bloque de código: text

In production I would formalize the optimization pipeline more strictly: asset naming rules, automated validation, LOD
→ strategy, texture budgets, profiler targets, versioned exports, and clearer handoff between CAD, DCC tools and Unity. I
→ would also define target device constraints earlier.

45.7 7. Unity / C# technical screen defense

45.7.1 7.1 Runtime systems to explain

Be ready to explain:

Bloque de código: text

selection system
highlighting
camera/navigation
exploded-view state
clipping controller
visual-mode manager
UI Toolkit panels
part metadata
WebGL build configuration

45.7.2 7.2 Explain selection system

Bloque de código: text

The selection system connects user input to selectable drone components. When a part is selected, the viewer updates
→ highlighting and displays contextual technical information in the UI. The important design point is separating the
→ selectable component identity from the visual feedback and metadata presentation.

45.7.3 7.3 Explain exploded view

Bloque de código: text

The exploded view is a state-based transformation of assembly components. Instead of random separation, each component or
→ group needs a defined offset direction and target position so the assembly remains understandable. The value is preserving
→ spatial logic while making component relationships visible.

45.7.4 7.4 Explain clipping

Bloque de código: text

The clipping tool allows the user to cut through the model visually to inspect hidden structures. In Unity this can be handled
 ↳ through shader/material logic or clipping plane approaches depending on the implementation. The goal is not destructive
 ↳ geometry editing, but runtime visual inspection.

45.7.5 7.5 Explain visual modes

Bloque de código: text

The visual modes are inspection states that change how the model is rendered or interpreted visually. Some modes support
 ↳ readability, like realistic and solid modes. Others expose structure, like X-ray, ghosted, blueprint or wireframe modes.
 ↳ The thermal-style mode is qualitative only; it is not a physical simulation.

45.7.6 7.6 Explain UI Toolkit usage

Bloque de código: text

UI Toolkit was used to structure technical panels and interaction controls in a way that could support a more app-like
 ↳ interface. The interface needed to display component data while keeping the 3D inspection workflow visible.

45.7.7 7.7 Possible technical questions

Question	Answer direction
How do you avoid coupling UI and model logic?	separate component metadata, selection state and UI rendering
How do you optimize for WebGL?	geometry reduction, texture budget, draw calls, loading, UI simplicity
How do you manage visual modes?	central mode manager, materials/shaders/states
How do you debug WebGL builds?	browser console, Unity logs, simplified repro scenes
How do you handle large models?	split hierarchy, optimize mesh, instancing, LOD, compression
How do you scale to multiple products?	metadata schema, import pipeline, prefab conventions

45.8 8. WebGL constraints defense

45.8.1 8.1 What are WebGL constraints?

Bloque de código: text

WebGL deployment imposes stricter constraints than desktop builds: memory, loading time, browser compatibility, build size,
 ↳ shader support, compression, UI readability and runtime performance. For technical visualization, the challenge is making
 ↳ the experience accessible without sacrificing the core inspection value.

45.8.2 8.2 How did you address WebGL constraints?

Bloque de código: text

I focused on geometry optimization, reducing unnecessary CAD detail, organizing the scene hierarchy, keeping the UI readable
 ↳ and designing the interaction around core inspection tasks rather than adding unnecessary features. The project is still a
 ↳ prototype, so further profiling and build optimization would be part of the next step.

45.8.3 8.3 If asked about FPS/performance

Use only measured values if available.

If not final:

Bloque de código: text

The project was optimized around browser feasibility and reduced geometry, but I would avoid claiming final production
 ↳ performance until I run a complete profiling pass across target devices and browsers.

45.8.4 8.4 If asked about mobile

Bloque de código: text
The UI logic was designed with mobile-first constraints in mind, especially through bottom-sheet interaction patterns.
↪ However, final mobile performance and usability would require dedicated device testing before claiming production readiness.

45.9 9. Technical art / shader defense

45.9.1 9.1 How strong is your shader experience?

Bloque de código: text
My current shader/technical-art proof is strongest in visual inspection modes and material/render-state logic inside TwinSight. I would not claim to be a senior shader engineer. My strength is applying rendering and visual-mode techniques to improve technical readability in real-time 3D.

45.9.2 9.2 Explain visual modes as technical art

Bloque de código: text
The visual modes are not just aesthetic. They support different inspection tasks. For example, ghosted or X-ray-like views ↪ help reveal internal relationships, wireframe helps communicate structure, and blueprint-style rendering supports a technical drawing aesthetic. The design goal is visual readability for technical understanding.

45.9.3 9.3 If asked about URP/HDRP

Bloque de código: text
TwinSight is aligned with URP/WebGL constraints rather than high-end HDRP rendering. For browser deployment, URP is a more ↪ practical fit because it supports lightweight rendering and broad platform compatibility.

45.9.4 9.4 If asked about Shader Graph

Bloque de código: text
Shader Graph is useful for building and iterating visual modes without writing every shader manually. For deeper production ↪ shader work, I would still need to validate performance, compatibility and fallback behavior, especially for WebGL.

45.10 10. Evaluation defense

45.10.1 10.1 Why SUS and NASA-TLX?

Bloque de código: text
SUS provided a standardized usability measure, while NASA-TLX Raw helped capture perceived workload. Since the project ↪ compares interactive 3D support with 2D support for assembly-understanding tasks, both usability and workload were relevant.

45.10.2 10.2 What were the results?

Bloque de código: text
The formative evaluation involved 12 participants and 96 task-condition records. The SUS average was 91.88. NASA-TLX Raw was ↪ 8.69 for the 3D viewer condition and 19.89 for the 2D support condition.

45.10.3 10.3 What can you claim from the evaluation?

Bloque de código: text

I can claim that, in the tested tasks and sample, the prototype showed high perceived usability and lower perceived workload
↳ for the 3D viewer condition compared with 2D support. I would not claim universal superiority of 3D documentation based on
↳ this sample.

45.10.4 10.4 What are the limitations?

Bloque de código: text

The sample size was limited and academic. The evaluation was formative, not a large-scale industrial validation. It supports
↳ the portfolio and thesis argument, but it should not be overgeneralized as proof of production impact.

45.11 11. AI-assisted workflow defense

45.11.1 11.1 Short answer

Bloque de código: text

I used AI assistance as an implementation and iteration support tool. Final architecture decisions, debugging, Unity
↳ integration, asset preparation, evaluation design, documentation and project ownership remained my responsibility.

45.11.2 11.2 More detailed answer

Bloque de código: text

AI helped me accelerate parts of implementation and problem-solving, especially when exploring code structure or debugging
↳ options. But I still had to decide the architecture, integrate the systems in Unity, test behavior, adapt code to the
↳ actual project, prepare assets, validate the build and document the work. I see AI as a productivity tool, not a
↳ replacement for technical ownership.

45.11.3 11.3 If challenged: can you work without AI?

Bloque de código: text

Yes. AI can accelerate iteration, but it does not remove the need to understand the system. In a production environment I
↳ would follow the team's policy on AI usage. My priority would be maintainable code, clear documentation and reproducible
↳ decisions, whether AI is allowed or not.

45.11.4 11.4 If asked what AI generated

Bloque de código: text

I would separate it by category: code assistance, debugging suggestions, documentation support and planning. I would not
↳ describe AI as the author of the project. The integration, testing, adaptation and final decisions were mine.

45.11.5 11.5 Avoid

Do not say:

Bloque de código: text

AI did most of the code.
I just prompted.
I do not fully understand the code.
I can only work with ChatGPT.

45.12 12. Limited formal experience defense

45.12.1 12.1 Main answer

Bloque de codigo: text

I have limited formal industry experience in Unity, but I built my portfolio around production-adjacent proof rather than only coursework. TwinSight demonstrates Unity/C#, WebGL deployment, CAD-to-realtime optimization, technical UI, visual modes, runtime interaction systems and structured evaluation. I'm applying to roles where that evidence is directly relevant and where I can grow inside a real production team.

45.12.2 12.2 If asked why they should hire you

Bloque de codigo: text

Because I can bring a strong technical-visualization foundation, real Unity/WebGL project evidence and the ability to learn production workflows quickly. I'm not positioning myself as a senior developer. I'm positioning myself as a strong junior-to-mid candidate for interactive 3D and technical visualization work, with a serious flagship project and clear documentation.

45.12.3 12.3 If asked about shipped titles

Bloque de codigo: text

I do not have shipped commercial Unity titles yet. My strongest proof is TwinSight X500, which is a portfolio/academic prototype with real implementation, optimization, interaction systems and evaluation. For roles that require shipped game titles specifically, I understand that is a gap. For technical visualization or interactive 3D roles, TwinSight is more directly relevant.

45.13 13. Behavioral interview system

45.13.1 13.1 STAR format

Use:

Bloque de codigo: text

Situation
Task
Action
Result

Keep answers short.

45.13.2 13.2 Core stories to prepare

Story	Use
CAD optimization challenge	technical problem-solving
WebGL constraint management	engineering judgment
AI-assisted workflow	transparency
thesis evaluation	research/evidence
project procrastination recovery	resilience, if framed carefully
learning Unity/technical systems	growth
handling feedback from advisor/users	collaboration
prioritizing scope	product judgment

45.13.3 13.3 Story: CAD optimization

Bloque de código: text

Situation: The drone source geometry was too heavy for browser-based real-time use.

Task: Preserve assembly readability while reducing complexity.

Action: I explored conversion routes, cleaned and optimized the geometry, organized components and prepared the model for
↪ Unity WebGL.

Result: The runtime model reached 95,617 triangles while preserving component-level inspection value.

45.13.4 13.4 Story: evaluation

Bloque de código: text

Situation: I needed to test whether the 3D viewer was usable and whether it could reduce workload compared with 2D support.

Task: Design a formative evaluation.

Action: I used SUS, NASA-TLX Raw and Think-Aloud with 12 participants and structured task-condition records.

Result: The viewer condition showed high perceived usability and lower perceived workload in the tested tasks.

45.14 14. Technical screen preparation

45.14.1 14.1 Unity topics to review

Bloque de código: text

C# basics and OOP

MonoBehaviour lifecycle

ScriptableObjects

events/delegates

raycasting

materials/renderers

camera controls

UI Toolkit basics

prefab organization

build settings

WebGL constraints

profiling basics

45.14.2 14.2 Technical art topics to review

Bloque de código: text

URP basics

Shader Graph basics

material instances

render queues

transparency issues

wireframe/outline approaches

X-ray/ghosted visualization logic

optimization budgets

draw calls

triangles vs vertices

texture sizes

LOD and instancing

45.14.3 14.3 WebGL topics to review

Bloque de código: text

memory limits

build size

compression

browser compatibility

WebGL debugging

loading times

texture compression

single-thread limitations

mobile browser constraints

45.14.4 14.4 CAD-to-realtime topics to review

Bloque de código: text

```
tessellation
mesh cleanup
decimation
retopology
normal baking
UVs
scale/origin consistency
hierarchy organization
metadata mapping
performance budgeting
```

45.14.5 14.5 Evaluation topics to review

```
Bloque de codigo: text
SUS
NASA-TLX Raw
Think-Aloud
formative evaluation
sample limitations
task design
qualitative observations
```

45.15 15. Whiteboard / live screen risks

45.15.1 15.1 Likely technical exercises

Exercise	Preparation
explain project architecture	use diagram from 08B
debug selection logic	review raycasting / event flow
optimize scene	discuss mesh, materials, draw calls
implement simple interaction	practice Unity/C#
explain shader mode	prepare visual-mode breakdown
review code	keep repo readable
portfolio walkthrough	rehearse 5-minute demo

45.15.2 15.2 If you do not know

Use:

```
Bloque de codigo: text
I do not know the exact answer off the top of my head, but I would approach it by isolating the system, reproducing the issue,
↔ checking Unity logs/profiler data, simplifying the scene and validating one variable at a time.
```

Do not bluff.

45.16 16. Remote contractor interview answers

45.16.1 16.1 Availability

```
Bloque de codigo: text
I am based in Colombia and available for remote contractor work with international teams. I can overlap with LATAM, North
↔ American and some European schedules depending on the project cadence.
```

45.16.2 16.2 Work style

```
Bloque de codigo: text
I work best with clear milestones, documented requirements, short feedback loops and visible progress through demos or pull
↔ requests. For 3D/Unity work, I prefer sharing playable builds or short screen captures regularly.
```

45.16.3 16.3 Communication

Bloque de código: text
I can communicate in English and Spanish. For remote work, I rely on written summaries, task tracking, async updates and
↔ periodic calls to avoid ambiguity.

45.16.4 16.4 Contractor setup

Bloque de código: text
I am open to contractor/B2B arrangements. For legal and tax details, I would follow the required Colombian invoicing and
↔ social security obligations and align with the client's payment process.

Do not give legal/tax advice in interviews.

45.17 17. Salary / compensation defense

45.17.1 17.1 If asked expected compensation early

Bloque de código: text
I would like to understand the role scope, seniority expectations, contract structure and time commitment before giving a
↔ final number. For remote contractor roles, I'm targeting compensation aligned with LATAM international Unity work and the
↔ technical scope of the project.

45.17.2 17.2 If pressed for a range

Use a researched internal range, not random numbers.

Bloque de código: text
For full-time remote contractor work, depending on scope and seniority, I would generally evaluate opportunities in the USD
↔ 1.5k-3k/month entry range, USD 3k-6k/month strong range, and higher for more specialized or senior-level responsibilities.

Adapt to the actual role.

45.17.3 17.3 Do not say

Bloque de código: text
I will accept anything.
I do not know.
I just need experience.

45.18 18. Role-specific defense

45.18.1 18.1 Unity Technical Artist

Emphasize:

Bloque de código: text
visual modes
CAD optimization
URP/WebGL constraints
technical UI
performance-aware asset preparation

Avoid overclaiming:

Bloque de código: text
senior shader engineering
AAA production pipelines
advanced VFX

45.18.2 18.2 Unity WebGL Developer

Emphasize:

Bloque de código: text
browser deployment
WebGL constraints
runtime interaction
UI readability
optimized assets

Avoid:

Bloque de código: text
full-stack web claims
large-scale backend

45.18.3 18.3 Real-Time 3D Developer

Emphasize:

Bloque de código: text
interactive 3D systems
Unity/C#
camera/selection/clipping
runtime state management
technical visualization

Avoid:

Bloque de código: text
pure gameplay claims

45.18.4 18.4 Technical Visualization Developer

Emphasize:

Bloque de código: text
spatial understanding
assembly inspection
CAD-to-realtime
technical UI
evaluation

Avoid:

Bloque de código: text
production digital twin claims

45.18.5 18.5 XR Developer

Emphasize:

Bloque de código: text
3D interaction
technical training potential
spatial UI
future XR extension

Avoid:

Bloque de código: text
claiming shipped VR production

45.19 19. Questions to ask interviewers

45.19.1 19.1 Technical questions

```
Bloque de codigo: text
What is the target platform for your Unity/3D applications?
How do you currently handle asset optimization for real-time delivery?
Do you have existing technical-art or pipeline standards?
What are the main performance constraints?
How are builds reviewed or tested?
```

45.19.2 19.2 Product questions

```
Bloque de codigo: text
What user problem does the 3D experience need to solve?
Is the goal training, inspection, simulation, sales, configurator, or operations?
What metrics define success for the project?
```

45.19.3 19.3 Remote work questions

```
Bloque de codigo: text
How does the team handle async communication?
What timezone overlap is expected?
How are milestones and reviews structured?
```

45.19.4 19.4 Avoid early questions

Avoid leading with:

```
Bloque de codigo: text
vacation
benefits
how soon can I get promoted
can I work very little overlap
```

Ask compensation and contract questions after scope is clear.

45.20 20. Interview preparation checklist

45.20.1 20.1 Before interview

```
Bloque de codigo: text
[ ] review job posting
[ ] identify role family
[ ] choose 3 relevant TwinSight points
[ ] open portfolio link
[ ] open GitHub README
[ ] prepare demo video link
[ ] review AI-assisted answer
[ ] review digital twin limitation answer
[ ] prepare compensation range
[ ] prepare 3 questions
```

45.20.2 20.2 During interview

```
Bloque de codigo: text
[ ] lead with evidence
[ ] answer directly
[ ] do not overexplain
[ ] use specific metrics
[ ] clarify limitations
[ ] ask role-specific questions
[ ] take notes
```

45.20.3 20.3 After interview

```
Bloque de codigo: text
[ ] send thank-you/follow-up within 24 hours
[ ] update tracker
[ ] record objections/questions
[ ] update answer bank
[ ] prepare next-step material
```

45.21 21. Post-interview follow-up

45.21.1 21.1 Template

```
Bloque de codigo: text
Hi [Name],

Thank you for the conversation today. I appreciated learning more about [role/company/project].

As discussed, here are the main links for TwinSight X500:
- Portfolio case study: [link]
- GitHub: [link]
- Demo video: [link]

The project is most relevant to the role through Unity WebGL, C# runtime interaction, CAD-to-realtime optimization, technical
↪ UI, visual modes and usability/workload evaluation.

Best,
Alexander
```

45.21.2 21.2 If technical topic was unclear

```
Bloque de codigo: text
I also wanted to clarify one point from the interview: [brief clarification]. My current implementation handles [specific],
↪ and if I were extending it for production I would approach [specific] by [method].
```

45.22 22. Red flag handling

45.22.1 22.1 If they want senior experience

```
Bloque de codigo: text
I understand that the role may require more production experience than I currently have. My strongest value is in Unity WebGL,
↪ technical visualization and CAD-to-realtime workflows. If the team has room for a junior-to-mid profile with strong
↪ project evidence, I believe TwinSight is directly relevant.
```

45.22.2 22.2 If they want shipped games

```
Bloque de codigo: text
I do not have shipped commercial game titles yet. My portfolio is stronger in technical visualization than gameplay
↪ production. If the role is strictly shipped-game focused, I may not be the strongest fit. If it involves real-time 3D
↪ tools, visualization, optimization or technical UI, TwinSight is much more aligned.
```

45.22.3 22.3 If they question Colombia remote

```
Bloque de codigo: text
I am based in Colombia and can work remotely with international teams. I can align with agreed timezone overlap and contractor
↪ processes. I prefer clear milestones, async updates and frequent build/demo reviews.
```

45.22.4 22.4 If they question formal degree status

Bloque de codigo: text

My thesis project is complete and pending defense. The project itself is already at the portfolio stage, and I can share the
↔ case study, demo and technical documentation.

45.23 23. Self-evaluation after each interview

Score from 1 to 5.

Area	Score
Explained TwinSight clearly	1–5
Defended technical choices	1–5
Handled AI question	1–5
Handled experience gap	1–5
Asked good questions	1–5
Avoided overclaiming	1–5
Maintained concise answers	1–5
Follow-up sent	yes/no

45.23.1 Improvement note

After each interview, write:

Bloque de codigo: text

What question surprised me?
Where did I overexplain?
Where did I sound weak?
What proof should I add to portfolio?
What answer should I rehearse?

45.24 24. Immediate preparation tasks

Before the first serious interview:

Bloque de codigo: text

[] memorize 30-second pitch
[] memorize 2-minute TwinSight explanation
[] prepare AI-assisted answer
[] prepare digital-twin-adjacent answer
[] prepare limited-experience answer
[] review CAD optimization explanation
[] review Unity architecture explanation
[] prepare portfolio walkthrough
[] prepare 3 questions for employer
[] prepare follow-up template

45.25 25. Next module

Proceed with:

Bloque de codigo: text

36_public_profile_launch_sequence.md

45.25.1 Tool recommendation

Tool	Use
Agent mode	No
Deep Research	No

45.25.2 Purpose

Create the exact launch sequence for making the public proof package visible:

```
Bloque de codigo: text
portfolio publish
GitHub update
LinkedIn update
ArtStation update
demo video upload
README links
Featured section
first LinkedIn post
tracker link update
```

Chapter 46

36_public_profile_launch_sequence

Fuente: 36_public_profile_launch_sequence.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 1167

Ultima modificacion: 2026-06-11 20:19

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

This module defines the exact sequence for making Alexander's public professional profile coherent and application-ready.

Contenido completo depurado

46.0.1 0. Module status

```
Bloque de codigo: text
Module: 36_public_profile_launch_sequence.md
Date generated: 2026-06-12
Status: Complete v1
Purpose: Public launch sequence for Alexander's professional proof package
Inputs: 08B, 18, 19B, 20B, 21B, 29B, 29C, 31, 32, 33, 34, 35
Use: publish and synchronize portfolio, GitHub, LinkedIn, ArtStation, demo video and application tracker
```

46.1 1. Purpose

This module defines the exact sequence for making Alexander's public professional profile coherent and application-ready.

The launch must connect:

```
Bloque de codigo: text
Portfolio
GitHub
LinkedIn
ArtStation
Demo video
CV
Application tracker
```

The objective is to create a consistent proof loop:

Bloque de codigo: text
 LinkedIn → Portfolio → Demo video → GitHub → ArtStation → Contact / Apply

and:

Bloque de codigo: text
 CV → Portfolio → GitHub → Demo video → Interview defense

46.2 2. Launch principle

46.2.1 2.1 Main rule

Bloque de codigo: text
 Do not publish disconnected assets.

Each public profile must point to the same core positioning:

Bloque de codigo: text
 Real-Time 3D Developer / Unity Technical Artist
 Unity WebGL · C# · CAD-to-Realtime · Technical Visualization

Each public asset must reinforce TwinSight X500 as the flagship proof.

46.2.2 2.2 What must be consistent everywhere

Element	Required wording
Primary title	Real-Time 3D Developer / Unity Technical Artist
Flagship project	TwinSight X500
Core stack	Unity, C#, WebGL, URP, UI Toolkit, Blender
Core domain	technical visualization, CAD-to-realtime
Prototype status	academic/portfolio prototype
Digital twin wording	digital-twin-adjacent, not full digital twin
Thermal mode wording	qualitative visual mode, not simulation
AI wording	AI-assisted implementation support, final ownership retained

46.3 3. Launch assets checklist

46.3.1 3.1 Required before public launch

Bloque de codigo: text
 [] portfolio homepage updated
 [] TwinSight case-study page public
 [] TwinSight 90-second demo uploaded
 [] TwinSight thumbnail exported
 [] GitHub README updated
 [] GitHub media folder added
 [] LinkedIn headline/About updated
 [] LinkedIn Featured section updated
 [] English CV links updated
 [] application tracker links updated

46.3.2 3.2 Strong launch additions

```
Bloque de codigo: text
[ ] 30-second teaser uploaded natively to LinkedIn
[ ] ArtStation TwinSight breakdown published
[ ] technical breakdown video uploaded
[ ] ARA Framework README cleaned
[ ] Blender portrait breakdown published
```

46.3.3 3.3 Launch should not wait for

```
Bloque de codigo: text
perfect long technical breakdown
every secondary project polished
new job-market research
more LinkedIn benchmarking
all possible ArtStation posts
```

46.4 4. Launch order

46.4.1 4.1 Correct order

```
Bloque de codigo: text
1. Publish demo video
2. Update GitHub README
3. Publish portfolio case study
4. Update homepage
5. Update LinkedIn profile
6. Add LinkedIn Featured items
7. Publish ArtStation breakdown
8. Update CV links
9. Update application tracker templates
10. Publish first LinkedIn post
```

46.4.2 4.2 Why this order

Step	Reason
demo first	all pages need video link
GitHub second	provides technical proof
portfolio third	central public hub
homepage fourth	routes viewers to case study
LinkedIn fifth	sends traffic to finished hub
ArtStation later	supports visual/technical proof
CV after links	avoids broken URLs
tracker last	ensures applications use final links

46.5 5. Asset naming and URL standards

46.5.1 5.1 Public URLs needed

```
Bloque de codigo: text
Portfolio homepage: [add]
TwinSight case study: [add]
Live WebGL demo: [add]
GitHub repo: [add]
Demo video: [add]
ArtStation breakdown: [add]
LinkedIn profile: [add]
```


46.5.2 5.2 File naming

Use consistent naming:

```
Bloque de codigo: text
twinsight_x500_demo_90s.mp4
twinsight_x500_teaser_30s.mp4
twinsight_x500_thumbnail.jpg
twinsight_x500_visual_modes_grid.jpg
twinsight_x500_metrics_card.jpg
twinsight_x500_pipeline_diagram.jpg
twinsight_x500_architecture_diagram.jpg
```

46.5.3 5.3 Slug naming

Recommended portfolio slug:

```
Bloque de codigo: text
/projects/twinsight-x500
```

Recommended GitHub repo description:

```
Bloque de codigo: text
Unity WebGL technical visualization prototype for drone assembly inspection.
```

46.6 6. Portfolio launch

46.6.1 6.1 Homepage hero

Use:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist
```

Subheadline:

```
Bloque de codigo: text
Unity WebGL, C# and Blender workflows for interactive technical visualization, CAD-to-realtime optimization and browser-based
↔ 3D inspection.
```

CTA buttons:

```
Bloque de codigo: text
View TwinSight X500
Watch demo
View GitHub
Contact
```

46.6.2 6.2 Homepage project order

```
Bloque de codigo: text
1. TwinSight X500
2. ARA Framework
3. Blender Portrait Study
4. AI News Aggregator
5. FitApp-Free
```

46.6.3 6.3 TwinSight project card

Bloque de codigo: text

TwinSight X500

Unity WebGL technical visualization for drone assembly inspection.

Interactive browser-based 3D viewer with component selection, exploded view, clipping tools, visual modes, CAD-to-realtime ↩ optimization and SUS/NASA-TLX evaluation.

Tags:

Bloque de codigo: text

Unity · C# · WebGL · Technical Visualization · CAD-to-Realtime · Blender · UI Toolkit

46.6.4 6.4 TwinSight case-study page must include

Bloque de codigo: text

[] hero image

[] 90-second demo video

[] problem

[] solution

[] key features

[] technical pipeline

[] optimization metrics

[] Unity WebGL architecture

[] visual modes

[] evaluation

[] role and contribution

[] AI-assisted workflow disclosure

[] limitations

[] links

46.6.5 6.5 Portfolio QA

Bloque de codigo: text

[] mobile layout readable

[] video loads

[] all buttons work

[] project title visible above fold

[] stack visible above fold

[] metrics readable

[] limitations included

[] no broken images

[] no placeholder text

[] no unsupported claims

46.7 7. GitHub launch

46.7.1 7.1 Update repository description

Use:

Bloque de codigo: text

Unity WebGL technical visualization prototype for drone assembly inspection.

or longer:

Bloque de codigo: text

Unity WebGL technical visualization prototype for drone assembly inspection, featuring CAD-to-realtime optimization, component ↩ selection, exploded view, clipping tools and visual modes.

46.7.2 7.2 Add topics

```
Bloque de codigo: text
unity
unity-webgl
csharp
webgl
technical-visualization
real-time-3d
technical-art
cad-optimization
interactive-3d
blender
drone
ui-toolkit
urp
```

Optional:

```
Bloque de codigo: text
digital-twin
simulation
xr
usability-testing
nasa-tlx
sus
```

Use optional topics only if wording is controlled.

46.7.3 7.3 Add /media folder

Required:

```
Bloque de codigo: text
media/twinsight-demo-thumbnail.jpg
media/01_viewer.jpg
media/02_exploded_view.jpg
media/03_cross_section.jpg
media/04_visual_modes_grid.jpg
media/05_technical_ui.jpg
media/06_cad_optimization.jpg
media/07_pipeline_diagram.jpg
media/08_metrics_card.jpg
media/twinsight-preview.gif
```

46.7.4 7.4 README update

Use:

```
Bloque de codigo: text
19B_github_readme_twinsight_final.md
```

Minimum README requirements:

```
Bloque de codigo: text
[ ] title
[ ] demo thumbnail
[ ] live demo link
[ ] portfolio case study link
[ ] highlights
[ ] screenshots
[ ] stack
[ ] pipeline
[ ] optimization metrics
[ ] evaluation metrics
[ ] role
[ ] limitations
[ ] setup
[ ] credits/license
```

46.7.5 7.5 GitHub QA

Bloque de codigo: text

```
[ ] README renders correctly
[ ] images load
[ ] markdown tables fit
[ ] links work
[ ] repo description updated
[ ] topics added
[ ] no private/secret data
[ ] no broken relative paths
[ ] license/credits clear
[ ] setup instructions honest
```

46.8 8. Demo video launch

46.8.1 8.1 Upload targets

Platform	Asset
YouTube	90-second demo
Vimeo	optional polished embed
LinkedIn native	30-second teaser
GitHub README	thumbnail + link
Portfolio	embedded 90-second demo
ArtStation	30–60 second video

46.8.2 8.2 Main video title

Bloque de codigo: text

```
TwinSight X500 – Unity WebGL Technical Visualization for Drone Assembly Inspection
```

46.8.3 8.3 Main video description

Bloque de codigo: text

```
TwinSight X500 is a Unity WebGL technical visualization prototype for inspecting the assembly of a Holybro X500 V2 drone.

It includes component selection, exploded assembly view, cross-section/clipping tools, multiple visual modes, technical
↪ information panels, CAD-to-realtime optimization and formative usability/workload evaluation using SUS and NASA-TLX Raw.

Stack: Unity, C#, WebGL, URP, UI Toolkit, Blender.

Portfolio: [link]
GitHub: [link]
Live demo: [link]
```

46.8.4 8.4 Video tags

Bloque de codigo: text

```
Unity
Unity WebGL
Technical Visualization
Technical Art
Real-Time 3D
CAD Optimization
Drone Assembly
C#
Blender
```

46.8.5 8.5 Video QA

```
Bloque de codigo: text
[ ] first 5 seconds show final output
[ ] captions readable
[ ] no unsupported claims
[ ] video description has links
[ ] thumbnail readable
[ ] audio level acceptable
[ ] comments/settings intentional
```

46.9 9. LinkedIn launch

46.9.1 9.1 Update headline

Recommended:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist | Unity WebGL, C#, CAD-to-Realtime, Technical Visualization
```

Alternative:

```
Bloque de codigo: text
Unity WebGL / Interactive 3D Developer | Technical Visualization, CAD-to-Realtime, Blender
```

46.9.2 9.2 Update About section

Opening:

```
Bloque de codigo: text
I build interactive real-time 3D systems for technical visualization using Unity, C#, WebGL and Blender.

My flagship project, TwinSight X500, is a Unity WebGL drone assembly inspection prototype focused on CAD-to-realtime
↪ optimization, component selection, exploded view, cross-section tools, visual modes, technical UI and usability/workload
↪ evaluation.
```

Add:

```
Bloque de codigo: text
I am based in Colombia and available for remote opportunities in Unity WebGL, real-time 3D, technical visualization and
↪ technical art.
```

46.9.3 9.3 Featured section order

```
Bloque de codigo: text
1. TwinSight X500 case study
2. TwinSight 90-second demo
3. GitHub repository
4. ArtStation breakdown
5. ARA Framework
```

46.9.4 9.4 LinkedIn project entry

Title:

```
Bloque de codigo: text
TwinSight X500 – Unity WebGL Technical Visualization
```

Description:

```
Bloque de codigo: text
Browser-based 3D viewer for drone assembly inspection, built with Unity, C#, WebGL, URP, UI Toolkit and Blender. Includes
↪ CAD-to-realtime optimization, component selection, exploded view, cross-section tools, visual modes and SUS/NASA-TLX
↪ evaluation.
```

46.9.5 9.5 LinkedIn skills order

Prioritize:

```
Bloque de codigo: text
Unity
C#
Unity WebGL
Real-Time 3D
Technical Art
Technical Visualization
Blender
Shader Graph
URP
UI Toolkit
CAD Optimization
Python
```

46.10 10. First LinkedIn launch post

46.10.1 10.1 Technical version

```
Bloque de codigo: text
I built TwinSight X500, a Unity WebGL technical visualization prototype for drone assembly inspection.

The goal was to move beyond static 2D assembly documentation and support spatial understanding through interactive 3D
↔ inspection.

Core features:
- component selection
- exploded assembly view
- cross-section inspection
- technical part panels
- multiple visual modes
- CAD-to-realtime optimization
- SUS / NASA-TLX evaluation

Stack:
Unity, C#, WebGL, URP, UI Toolkit, Blender.

Demo / case study:
[link]
```

46.10.2 10.2 Short version

```
Bloque de codigo: text
TwinSight X500 – Unity WebGL technical visualization for drone assembly inspection.

CAD-derived assets optimized into an interactive browser-based viewer with component selection, exploded view, clipping tools,
↔ visual modes and usability/workload evaluation.

Stack:
Unity, C#, WebGL, URP, UI Toolkit, Blender.

[link]
```

46.10.3 10.3 Contractor-oriented version

```
Bloque de codigo: text
I'm sharing TwinSight X500, my flagship Unity WebGL technical visualization project.

It is a browser-based 3D viewer for drone assembly inspection, built around CAD-to-realtime optimization, component selection,
↔ exploded view, clipping tools, visual modes and technical UI.

I'm currently targeting remote roles or contract work in Unity WebGL, real-time 3D, technical visualization and technical art.

Case study:
[link]
```

46.10.4 10.4 Hashtags

Use 3–5 only:

```
Bloque de codigo: text
#Unity3D
#WebGL
#TechnicalArt
#Realtime3D
#TechnicalVisualization
```

Avoid too many hashtags.

46.11 11. ArtStation launch

46.11.1 11.1 Title

```
Bloque de codigo: text
TwinSight X500 – Unity WebGL Technical Visualization
```

46.11.2 11.2 Description

```
Bloque de codigo: text
A browser-based technical visualization prototype for inspecting the assembly of a Holybro X500 V2 drone. Built in Unity WebGL
↪ with C#, URP, UI Toolkit and Blender, the project focuses on CAD-to-realtime optimization, component selection, exploded
↪ view, cross-section tools and visual inspection modes.
```

46.11.3 11.3 Post order

```
Bloque de codigo: text
1. Cover image
2. Short video
3. Final screenshots
4. Feature breakdown
5. CAD optimization
6. Visual modes grid
7. UI and interaction
8. Evaluation metrics
9. Tools
10. Links
```

46.11.4 11.4 Tags

```
Bloque de codigo: text
Unity
Unity WebGL
Technical Art
Real-Time 3D
Technical Visualization
CAD Optimization
Drone
Interactive 3D
CSharp
URP
Blender
```

46.11.5 11.5 ArtStation QA

```
Bloque de codigo: text
[ ] cover image strong
[ ] title role-aligned
[ ] video near top
[ ] breakdown images included
[ ] not just beauty shots
[ ] tools listed
```

```
[ ] portfolio/GitHub links included
[ ] no overclaiming
```

46.12 12. CV launch update

46.12.1 12.1 Update links

Every CV variant must include:

```
Bloque de codigo: text
Portfolio: [link]
GitHub: [link]
LinkedIn: [link]
TwinSight case study: [link]
```

Optional:

```
Bloque de codigo: text
Demo video: [link]
ArtStation: [link]
```

46.12.2 12.2 CV headline

Default:

```
Bloque de codigo: text
Real-Time 3D Developer / Unity Technical Artist
```

Role-specific variants:

```
Bloque de codigo: text
Unity WebGL / Interactive 3D Developer
Technical Visualization Developer
Unity Technical Artist
Unity Developer – WebGL / Technical Visualization
```

46.12.3 12.3 CV project first bullet

```
Bloque de codigo: text
Built TwinSight X500, a Unity WebGL technical visualization prototype for drone assembly inspection using Unity, C#, URP, UI
↳ Toolkit and Blender.
```

46.12.4 12.4 CV QA

```
Bloque de codigo: text
[ ] links work
[ ] headline matches role
[ ] TwinSight first
[ ] metrics included
[ ] no unsupported claims
[ ] one-page version available
[ ] ATS-friendly formatting
```

46.13 13. Application tracker update

46.13.1 13.1 Add final links to tracker defaults

```
Bloque de codigo: text
Default portfolio link:
Default TwinSight case-study link:
Default GitHub link:
Default demo video link:
Default LinkedIn link:
Default ArtStation link:
```

46.13.2 13.2 Add asset-readiness columns

```
Bloque de codigo: text
Portfolio link ready
GitHub link ready
Demo link ready
ArtStation link ready
CV variant ready
LinkedIn updated
```

46.13.3 13.3 Add public-proof status

```
Bloque de codigo: text
Incomplete
MVP ready
Published
QA passed
Needs update
```

46.14 14. Cross-linking map

46.14.1 14.1 Required proof loop

```
Bloque de codigo: text
Portfolio → Demo video
Portfolio → GitHub
Portfolio → ArtStation
Portfolio → LinkedIn / Contact

GitHub → Portfolio
GitHub → Demo video
GitHub → Live WebGL demo

LinkedIn → Portfolio
LinkedIn → Demo video
LinkedIn → GitHub

ArtStation → Portfolio
ArtStation → GitHub
ArtStation → Demo video

CV → Portfolio
CV → GitHub
CV → LinkedIn
```

46.14.2 14.2 Broken loop risk

Do not publish assets that point to empty pages.

Bad:

```
Bloque de codigo: text
LinkedIn → portfolio page under construction
GitHub → missing images
CV → dead demo link
ArtStation → no GitHub
```

46.15 15. Launch QA checklist

46.15.1 15.1 Technical QA

```
Bloque de codigo: text
[ ] portfolio loads
[ ] GitHub README images load
[ ] video plays
[ ] mobile view readable
[ ] links work
[ ] images optimized
[ ] no large broken media
[ ] no console-breaking portfolio errors
```

46.15.2 15.2 Content QA

```
Bloque de codigo: text
[ ] role title consistent
[ ] TwinSight description consistent
[ ] stack consistent
[ ] metrics consistent
[ ] limitations included
[ ] AI disclosure controlled
[ ] digital twin wording safe
[ ] thermal mode disclaimer present
```

46.15.3 15.3 Recruiter QA

A recruiter must see within 20 seconds:

```
Bloque de codigo: text
[ ] what Alexander does
[ ] what TwinSight is
[ ] what stack was used
[ ] where to watch the demo
[ ] where to view GitHub
```

46.15.4 15.4 Technical lead QA

A technical reviewer must find within 60 seconds:

```
Bloque de codigo: text
[ ] architecture
[ ] pipeline
[ ] optimization
[ ] runtime systems
[ ] evaluation
[ ] limitations
```

46.16 16. Launch sequence timeline

46.16.1 Day 1 — video and GitHub

```
Bloque de codigo: text
[ ] upload 90-second demo
[ ] export thumbnail
[ ] update GitHub README
[ ] add media folder
[ ] test all GitHub images/links
```

46.16.2 Day 2 — portfolio

```
Bloque de codigo: text
[ ] publish TwinSight case study
[ ] update homepage
[ ] add demo embed
[ ] add GitHub link
[ ] add CTA
[ ] mobile QA
```

46.16.3 Day 3 — LinkedIn

```
Bloque de codigo: text
[ ] update headline
[ ] update About
[ ] update Featured section
[ ] update skills
[ ] update project entry
[ ] add portfolio/GitHub/demo links
```

46.16.4 Day 4 — ArtStation

```
Bloque de codigo: text
[ ] publish TwinSight breakdown
[ ] add video
[ ] add visual modes grid
[ ] add CAD optimization
[ ] add metrics card
[ ] add links
```

46.16.5 Day 5 — CV and tracker

```
Bloque de codigo: text
[ ] update CV links
[ ] update CV variants
[ ] update tracker default links
[ ] test link path from CV to portfolio/GitHub
[ ] prepare first 5 applications
```

46.16.6 Day 6 — first launch post

```
Bloque de codigo: text
[ ] publish LinkedIn post
[ ] respond to comments
[ ] save engagement notes
[ ] update tracker if leads appear
```

46.17 17. Soft launch vs hard launch

46.17.1 17.1 Soft launch

Use if assets are good but not perfect.

```
Bloque de codigo: text
portfolio published
GitHub ready
LinkedIn Featured updated
no public announcement yet
apply selectively
```

46.17.2 17.2 Hard launch

Use when public post is ready.

```
Bloque de codigo: text
LinkedIn post
ArtStation post
demo video public
portfolio fully linked
```

46.17.3 17.3 Recommended path

```
Bloque de codigo: text
Soft launch first.
Hard launch after QA.
```

Reason:

This allows applications to begin without forcing a public announcement before QA is complete.

46.18 18. Post-launch monitoring

46.18.1 18.1 First 48 hours

Check:

```
Bloque de codigo: text
[ ] links
[ ] video playback
[ ] LinkedIn profile views
[ ] GitHub repo views/stars if available
[ ] portfolio analytics if available
[ ] comments/messages
[ ] recruiter reactions
```

46.18.2 18.2 First week

Track:

```
Bloque de codigo: text
profile views
connection requests
messages
portfolio clicks
GitHub visits
application responses
questions received
```

46.18.3 18.3 Use questions as feedback

If people ask:

```
Bloque de codigo: text
What is this?
```

Then the hero/thumbnail is unclear.

If they ask:

```
Bloque de codigo: text
Did you build this?
```

Then the role/contribution is unclear.

If they ask:

Bloque de codigo: text
Is it a real digital twin?

Then limitation wording must be more visible.

If they ask:

Bloque de codigo: text
Can I try it?

Then live demo link must be more prominent.

46.19 19. Public claim safety

46.19.1 19.1 Safe public claims

Bloque de codigo: text
Unity WebGL technical visualization prototype
browser-based 3D viewer
CAD-derived geometry optimized for WebGL
component selection
exploded view
cross-section tools
visual modes
formative usability/workload evaluation
AI-assisted implementation support
digital-twin-adjacent

46.19.2 19.2 Unsafe claims unless implemented

Bloque de codigo: text
production digital twin
industrial deployment
IoT integration
predictive maintenance
real thermal simulation
FEA
enterprise-ready
senior-level shipped product

46.19.3 19.3 Required disclaimers

Bloque de codigo: text
academic/portfolio prototype
not a full production digital twin
thermal-style mode is qualitative, not physical simulation
formative evaluation, not large-scale industrial validation

46.20 20. Launch failure modes

46.20.1 20.1 Failure mode: too academic

Symptom:

Bloque de codigo: text
people do not understand the value quickly

Fix:

```
Bloque de codigo: text
move thesis/methodology lower
lead with demo, problem, solution, features
```

46.20.2 20.2 Failure mode: too visual, not technical

Symptom:

```
Bloque de codigo: text
looks like a 3D render, not a developer/TA project
```

Fix:

```
Bloque de codigo: text
add pipeline, architecture, metrics, GitHub, runtime systems
```

46.20.3 20.3 Failure mode: overclaiming

Symptom:

```
Bloque de codigo: text
interviewers challenge digital twin/AI/thermal claims
```

Fix:

```
Bloque de codigo: text
tighten limitations and safe wording
```

46.20.4 20.4 Failure mode: broken proof loop

Symptom:

```
Bloque de codigo: text
viewer cannot move from LinkedIn to portfolio/GitHub/demo
```

Fix:

```
Bloque de codigo: text
cross-link all assets
```

46.21 21. Final pre-application gate

Do not begin serious application volume until:

```
Bloque de codigo: text
[ ] Portfolio case study is public.
[ ] GitHub README is updated.
[ ] Demo video is public.
[ ] LinkedIn Featured has TwinSight.
[ ] CV links work.
[ ] Tracker has final links.
```

Selective Priority A applications are allowed before full hard launch if:

```
Bloque de codigo: text
portfolio + GitHub + demo are functional
```

46.22 22. Next module

Proceed with:

Bloque de código: text

37_application_tracker_and_metrics_upgrade.md

46.22.1 Tool recommendation

Tool	Use
Spreadsheet tooling	Yes, if modifying workbook
Agent mode	No
Deep Research	No

46.22.2 Purpose

Upgrade the tracker system to include:

Bloque de código: text

targeting scores
proof links
readiness status
follow-up dates
role-family segmentation
remote eligibility
material variant used
response metrics
weekly review fields

If the existing spreadsheet is available, modify it. If not, generate a new tracker template.

Chapter 47

Alexander Woodcock Salomon Portfolio

Fuente: README.md

Seccion: root

Estado: canonical

Rol: Modulo principal

Lineas originales: 22

Ultima modificacion: 2026-06-11 21:07

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Public portfolio website for international remote employability, centered on TwinSight X500 and the Human / CG Cookie Blender character pipeline study.

Contenido completo depurado

Public portfolio website for international remote employability, centered on TwinSight X500 and the Human / CG Cookie Blender character pipeline study.

47.0.1 Development

```
Bloque de codigo: bash
npm install
npm run dev
```

47.0.2 Build

```
Bloque de codigo: bash
npm run build
```

47.0.3 Content Model

Project copy, routes, placeholders, links, and focus variants are stored in `src/data`.

Unset public links intentionally remain as bracketed placeholders such as `[GITHUB_URL]`, `[DEMO_VIDEO_URL]`, `[CV_PDF_URL]`, and `[EMAIL]` until final assets are ready.

Part II

Research Markdown completo

Appendix A

A4 — ArtStation / Portfolio Benchmark

Fuente: Research/agent_A4_artstation_portfolio_benchmark (1).md

Seccion: research

Estado: research_duplicate

Rol: Research / estudio de mercado

Lineas originales: 139

Ultima modificacion: 2026-06-11 18:06

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

— tipo: benchmark_fuente modulo: A4_artstation_portfolio_benchmark estado: corregido_invalidado fecha: 2026-06-11 reemplaza: A4_artstation_portfolio_benchmark_recuperado motivo: el intento anterior no investigó perfiles/proyectos ArtStation de forma suficiente —

Contenido completo depurado

tipo: benchmark_fuente

modulo: A4_artstation_portfolio_benchmark

estado: corregido_invalidado

fecha: 2026-06-11

reemplaza: A4_artstation_portfolio_benchmark_recuperado

motivo: el intento anterior no investigó perfiles/proyectos ArtStation de forma suficiente

A.0.1 1. Corrección de alcance

El intento anterior de A4 **no cumple el objetivo original**.

La solicitud era recolectar ejemplos públicos de **ArtStation y portfolios** relevantes para technical art, real-time 3D, shaders, VFX, optimización, Blender, XR y visualización técnica.

La salida recuperada incluyó principalmente:

- Framer portfolios;
- GitHub Pages portfolios;
- blogs técnicos;
- artículos externos;
- project cards sin desglose interno.

No incluyó un benchmark real de perfiles/proyectos de ArtStation. Por tanto, este archivo queda como **registro de intento fallido**, no como benchmark válido.

A.0.2 2. Auditoría del intento anterior

Criterio	Resultado	Estado
Objetivo	20–35 ejemplos	no cumplido
Entradas recuperadas	12	parcial
Perfiles ArtStation	0	inválido
Proyectos ArtStation	0–1	insuficiente
Portfolios externos	9+	usable
Blogs técnicos	3+	usable
Benchmark ArtStation	no	inválido
Benchmark portfolio general	sí	parcial

A.0.3 3. Entradas externas recuperadas

Estas entradas pueden conservarse como referencia de **estructura de portfolio**, pero no deben contarse como ArtStation benchmark.

Label	URL	Plataforma	Rol cercano	Tipo	Fuerza
Alla Eddine Rakik	https://technical-art-portfolio.io/framer.ai	Framer	TA / Game Dev	portfolio	media
Potion Shader	https://jettelly.com/blog/a-real-time-potion-shader-breakdown-in-unity	Blog	Shader TA	breakdown	alta
Harry Alisavakis	https://halisavakis.com	Blog	Technical Artist	blog técnico	alta
Alson Entuna	https://alsonentuna.github.io	GitHub Pages	Lead TA	portfolio / CV	media
Omar Rinaz Costa	https://omarrinaz.github.io	GitHub Pages	TA / Game Design	portfolio	baja
Francisco Múrias	https://franciscomurias.github.io	GitHub Pages	Technical Artist	portfolio	media
julhe	https://julhe.github.io	GitHub Pages	Shader developer	blog técnico	alta

A.0.4 4. ArtStation pendiente de verificación

Estas fuentes aparecieron como candidatos durante el proceso o en contexto previo, pero **no fueron auditadas correctamente**. Deben verificarse en A4-Retry antes de usarse como evidencia.

Candidato	URL	Tipo esperado	Estado	Uso permitido ahora
Mike Spadaro	https://mjspadaro.artstation.com/resume	ArtStation resume	pendiente	no usar como benchmark
Eureka Thurston	https://www.artstation.com/eurekathurston	ArtStation profile	pendiente	no usar como benchmark
Genshin Shader Breakdown	https://www.artstation.com/artwork/wJZ4Gg	ArtStation project	pendiente	no usar como benchmark
ArtStation Technical Art channel	https://www.artstation.com/channels/technical_art	ArtStation channel	pendiente	solo exploración
ArtStation VR/AR channel	https://www.artstation.com/channels/virtual_and_augmented_reality	ArtStation channel	pendiente	solo exploración

A.0.5 5. Decisión

A4 debe repetirse como una tarea separada:

```
Bloque de código: text
A4_retry_artstation_only_benchmark
```

No se deben derivar decisiones finales de portfolio desde ArtStation hasta ejecutar ese retry.

Los ejemplos externos sí pueden alimentar criterios generales de presentación:

- abrir con resultado visual;
- incluir breakdown técnico;
- mostrar herramientas;
- separar rol, problema, restricciones y resultado;
- usar imágenes, GIFs o video corto;
- evitar grids sin explicación.

A.0.6 6. Prompt operativo para A4-Retry

```
Bloque de código: text
Execute A4_retry_artstation_only_benchmark.

Context:
The previous A4 attempt failed. It collected portfolio/blog examples but did not properly research ArtStation profiles or
↳ ArtStation project breakdowns.

Objective:
Collect 20-35 public ArtStation examples relevant to Alexander's technical art and real-time 3D positioning.

Prioritize:
- Technical Artist profiles;
- Unity Technical Artist profiles;
- shader breakdown projects;
- VFX breakdown projects;
- real-time 3D breakdowns;
- optimization breakdowns;
- Blender technical breakdowns;
- XR / interactive 3D projects;
- technical visualization projects;
- portfolios with tools, GitHub, demo videos or detailed process notes.

Output structured tables only.

Fields:
- Artist / profile label
- ArtStation URL
- Source type: profile / resume / project / blog post / channel result
- Role target
- Seniority if visible
- Project type
- Breakdown structure
- Images used
- Videos used
- Technical text depth
- Tags / skills
- Tools shown
- Portfolio / GitHub links if visible
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:
- ArtStation sources only.
- Use public pages only.
- Do not invent missing data.
- If unavailable, write "not found".
- If ArtStation blocks access, record "blocked" and do not infer details.
- Include source URLs.
- Tables only.
- No narrative strategy.
```

Appendix B

A4 — ArtStation / Portfolio Benchmark

Fuente: Research/agent_A4_artstation_portfolio_benchmark.md

Seccion: research

Estado: research_duplicate

Rol: Research / estudio de mercado

Lineas originales: 48

Ultima modificacion: 2026-06-11 17:52

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

— tipo: benchmark_fuente modulo: A4_artstation_portfolio_benchmark estado: recuperado fecha: 2026-06-11
base: resultados parciales de A4 —

Contenido completo depurado

tipo: benchmark_fuente

modulo: A4_artstation_portfolio_benchmark

estado: recuperado

fecha: 2026-06-11

base: resultados parciales de A4

B.0.1 Nota de alcance

La extracción A4 quedó **parcial**. El objetivo original era reunir 20–35 ejemplos, pero varias páginas de ArtStation activaron controles de seguridad o dependían de JavaScript. Este archivo recupera la información utilizable del intento A4 y la conserva como evidencia de trabajo, no como benchmark exhaustivo.

B.0.2 Tabla 1 — Inventario recuperado

Perfil / proyecto	URL	Plataforma	Rol objetivo	Seniority	Tipo	Breakdown	Media	Texto técnico	Tools visibles	Links visibles	Confianza	
Alla Eddine Rakik	https://technical-art-portfolio.io/framer.ai	Framer	Game Developer / Technical Artist	no encontrado	Portfolio	Home + proyectos	imágenes, gifs	medio	Unity, Unreal, Blender, Substance	GitHub	media	
Potion Shader	https://jettelly.com/blog/a-real-time-potion-shader-breakdown-in-unity	Blog	Shader / TA	no encontrado	Shader breakdown	artículo + demo	imágenes, video	alto	Unity URP, Shader Graph, C#	no encontrado	alta	
Harry Alisavakis	https://halisavakis.com	Blog	Technical Artist	no encontrado	Blog técnico	posts seriadados	imágenes	alto	Unity, shaders	redes	alta	
Alson Entuna	https://alsonentuna.github.io	GitHub Pages	Lead Technical Artist	Lead	Portfolio / CV	hero + CV	imágenes	medio	CryEngine, pipeline tools	resume	media	
Omar Rinaz Costa	https://omarrinaz.github.io	GitHub Pages	Game Designer / TA	no encontrado	Portfolio	avatar + CV	imagen	bajo	no encontrado	CV	media	
Francisco Múrias	https://franciscomurias.github.io	GitHub Pages	Technical Artist	no encontrado	Portfolio	grid profesional	imágenes	medio	Unreal, tools	no encontrado	media	
Múrias — Nebulas	https://franciscomurias.github.io/#portfolio	GitHub Pages	VFX / TA	no encontrado	VFX	project card	imagen	medio	volumetrics	no encontrado	media	
Múrias — OpenGL	https://franciscomurias.github.io/#portfolio	GitHub Pages	Tools / rendering	no encontrado	Tool / rendering	project card	imagen	medio	OpenGL	no encontrado	media	
julhe — SDF	https://julhe.github.io	GitHub Pages	Shader developer	no encontrado	Blog post	post técnico	imagen	alto	SDF, shaders	no encontrado	alta	
julhe — Optimize Shaders	https://julhe.github.io	GitHub Pages	Shader developer	no encontrado	Blog post	post técnico	imagen	alto	GPU, sampling	no encontrado	alta	
julhe — X5 Shader	https://julhe.github.io	GitHub Pages	Shader developer	no encontrado	Blog post	post técnico	imagen	medio	shader framework	no encontrado	media	
Rakik — Shader VFX Pack	https://technical-art-portfolio.io/framer.ai/projects	Framer	VFX Technical Artist	no encontrado	VFX pack	project card	imagen	bajo	Unity shaders	no encontrado	media	

Biblia completa

APPENDIX B. A4 — ARTSTATION / PORTFOLIO BENCHMARK

B.0.3 Tabla 2 — Qué adaptar y qué evitar

Perfil / proyecto	Adaptable para Alexander	No copiar
Alla Eddine Rakik	Categorías por proyecto; stack visible	Proyectos sin detalle técnico
Potion Shader	Final primero; luego breakdown	Texto o capturas exactas
Harry Alisavakis	Blog técnico como autoridad	Frecuencia artificial de posts
Alson Entuna	Título de rol muy específico	Hero visual poco profesional
Omar Rinaz Costa	Botón directo de CV	Falta de casos técnicos
Francisco Múrias	Grid por tipo de trabajo	Cards sin explicación
Múrias — Nebulas	Proyecto VFX como pieza visual fuerte	No explicar contribución
Múrias — OpenGL	Tool/rendering como prueba técnica	Solo screenshot sin contexto
Julhe — SDF	Artículos profundos por problema	Copiar ejemplos de shader
Julhe — Optimize Shaders	Optimización como contenido central	Abusar de jerga sin demo
Julhe — X5 Shader	Framework shader documentado	Presentar frameworks incompletos
Rakik — Shader VFX Pack	Pack visual agrupado	No publicar demos o links

Appendix C

B2_digital_twin_simulation_live_jobs_partial_c

Fuente: Research/agente_B2_digital_twin_simulation_live_jobs_partial_capture.md
Seccion: research
Estado: research_snapshot
Rol: Research / estudio de mercado
Lineas originales: 104
Ultima modificacion: 2026-06-11 19:30
Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

No se logró construir una tabla B2 completa de ofertas verificadas. Sí se capturaron varios resultados públicos desde snippets de buscador antes de que los portales bloquearan el acceso.

Contenido completo depurado

C.0.1 Estado

Bloque de codigo: text
Task: B2 – Digital Twin / Simulation live jobs
Date checked: 2026-06-12
Status: Incomplete / partial capture
Reason: multiple job boards blocked automated browsing or required verification.

C.0.2 Qué se logró recolectar

No se logró construir una tabla B2 completa de ofertas verificadas. Sí se capturaron varios resultados públicos desde snippets de buscador antes de que los portales bloquearan el acceso.
Estos registros deben tratarse como:

Bloque de codigo: text
snippet-level evidence, not fully verified postings

No usar como base para aplicar sin reabrir manualmente cada URL.

C.0.3 Tabla parcial — resultados capturados

Company / platform	Role	URL	Date checked	Region	Remote scope	Salary	Seniority	Requirements captured	Portfolio expectations	Work authorization	Contractor possibility	Relevance Alexander	Notes
Upwork	Digital Twin Development using Unity / Godot / Unreal Engine	https://www.upwork.com/freelance-jobs/apply/Digital-Twin-Development-using-Unity-Godot-Unreal-Engine_~021033003486070320/	2026-06-12	Unknown	Likely remote freelance	not found	not found	Digital twin, Unity/Godot/Unreal, 3D modeling, simulation, real-time data integration	not found	Unknown	High; Upwork freelance format	Very high; digital twin + Unity + simulation + contractor	B2 Digital Twin Simulation Live Jobs Partial Capture
Indeed	Remote Unity Developer jobs	https://www.indeed.com/q-unity-developer-1-remote-jobs.html	2026-06-12	Remote / unknown	Unknown; not assumed worldwide	not found	mixed	Search snippet showed 507 remote Unity developer openings	not found	Unknown	Unknown	Medium; Unity remote market signal, not specific to digital twin	APENDIX C. B2 DIGITAL_TWIN_SIMULATION_LIVE_JOBS_PARTIAL_CAPTURE
Glassdoor	Unity Developer jobs in Remote	https://www.glassdoor.com/Job/remote-unity-developer-jobs-SRCH_IL.0,6_IS11047_K07,22.htm	2026-06-12	Remote / unknown	Unknown; not assumed worldwide	not found	mixed	Search snippet showed 75 Unity developer jobs in Remote, June 2026	not found	Unknown	Unknown	Medium; Unity remote market signal	
Joble	Unity developer remote jobs	https://joble.org/jobs-unity-developer-remote	2026-06-12	Unknown	Unknown	not found	mixed	Search snippet showed many Unity developer remote vacancies	not found	Unknown	Unknown	Low-medium; weak source quality	
Remote Rocketship	Remote Unity Jobs	https://www.remoterocketship.com/jobs/unity	2026-06-12	Remote / unknown	Unknown	not found	mixed	Search snippet showed 103 remote Unity engineer positions	not found	Unknown	Unknown	Medium; useful if manually reopened later	
Remotive	Unity search page	https://remotive.com/remote-jobs/search?search=unity	2026-06-12	Remote / unknown	Unknown	not found	not found	Page returned 404 / unavailable during browsing	not found	Unknown	Unknown	Not usable from current attempt	

C.0.4 Access failures captured

Platform	Result
Upwork	Cloudflare verification blocked full job page
Remote Rocketship	blocked by Cloudflare/security page
Indeed	additional verification required
Remotive	search URL returned 404 / page not found
DuckDuckGo	usable for snippets only
Google	affected by prior blocked/session state

C.0.5 Interpretation constraints

Constraint	Meaning
Snippet-only rows	Not enough to apply or cite as final evidence
Remote scope unknown	Do not assume Colombia eligibility
Salary missing	Do not infer compensation
Work authorization missing	Must verify manually
Contractor possibility inferred only for Upwork	Upwork format suggests freelance, but full terms were inaccessible
B2 incomplete	Requires agent/manual browsing or another search path

C.0.6 What this partial capture suggests

Signal	Strength
Unity remote postings exist broadly	Medium
Digital twin + Unity freelance work appears on Upwork	Medium-high
Full live-job verification failed	High
B2 needs rerun through agent or manual browser	High

C.0.7 Recommended rerun prompt

```

Bloque de código: text
Retry B2_digital_twin_simulation_live_jobs with broader sources and lower target volume.

Objective:
Collect 8-15 live or recently listed postings related to Digital Twin, Simulation, Unity, Real-Time 3D, XR, Technical
↔ Visualization, Industrial Visualization.

Prioritize accessible sources:
- company career pages;
- LinkedIn job snippets if visible;
- Wellfound;
- Workable;
- Greenhouse;
- Lever;
- Otta / Welcome to the Jungle if accessible;
- Upwork only if full posting visible;
- job board snippets only if clearly marked snippet-level.

Fields:
company, role, URL, date checked, region, remote scope, salary if visible, seniority, requirements, portfolio expectations,
↔ work authorization, contractor possibility, relevance to Alexander, confidence.

```

Rules:

- Do not assume remote means worldwide.
- If full posting is blocked, mark as snippet-only.
- Do not invent salary.
- If unavailable, write not found.
- Tables only.

Appendix D

Estado actual

Fuente: Research/cerrando_sistema.md

Seccion: research

Estado: research_snapshot

Rol: Research / estudio de mercado

Lineas originales: 340

Ultima modificacion: 2026-06-11 15:11

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Sí. Falta **cerrar el sistema**, no necesariamente investigar mucho más. Ya tienes una base modular suficientemente amplia, pero todavía hay brechas entre **investigación** y **ejecución real**.

Contenido completo depurado

Sí. Falta **cerrar el sistema**, no necesariamente investigar mucho más. Ya tienes una base modular suficientemente amplia, pero todavía hay brechas entre **investigación** y **ejecución real**.

D.0.1 Estado actual

Ya está cubierto el bloque estratégico principal:

```
Bloque de codigo: text
00_master_audit_blueprint.md
01_source_of_truth_profile.md
02_market_role_fit_and_positioning.md
03_salary_benchmark_and_remote_colombia.md
04_eu_portugal_germany_mobility_strategy.md
05_language_strategy_and_roi.md
06_education_courses_rebelway_masters_phd.md
07_portfolio_strategy_and_project_architecture.md
08_twinsight_x500_case_study.md
09_ara_ai_tools_and_secondary_projects.md
10_github_linkedin_cv_strategy.md
11_company_targets_job_boards_recruiters.md
12_application_system_and_tracker.md
13_interview_negotiation_and_offer_risk.md
14_30_60_90_execution_plan.md
15_final_integrated_strategy.md
```

Eso ya funciona como **investigación base + plan estratégico**.

Pero todavía no es un sistema completamente ejecutable porque faltan artefactos operativos.

D.1 Qué falta

D.1.1 1. Auditoría final de consistencia

Antes de producir CV, LinkedIn, portfolio y trackers, conviene hacer una pasada final:

```
Bloque de código: text
16_final_consistency_audit_and_gap_register.md
```

Debe revisar:

- contradicciones entre secciones;
- claims peligrosos;
- datos que quedaron débiles;
- secciones que dependen de verificación futura;
- qué archivos son estrategia y cuáles son ejecución;
- qué acciones deben hacerse antes de aplicar;
- qué cosas no se deben afirmar públicamente.

Herramientas: no Deep Research, no agente.

Esto sería el cierre lógico antes de pasar a producción de materiales.

D.1.2 2. Archivos ejecutables que todavía faltan

La investigación dice qué hacer, pero faltan documentos de uso directo:

```
Bloque de código: text
17_cv_base_and_role_variants.md
18_linkedin_final_rewrite.md
19_github_readme_templates.md
20_portfolio_copy_and_site_structure.md
21_twinsight_demo_video_script_and_shotlist.md
22_outreach_templates_english_spanish.md
23_interview_answer_bank.md
24_offer_evaluation_scorecard.md
25_application_tracker_template.csv / .xlsx
26_weekly_execution_dashboard.md
```

Estos son los que realmente convierten la estrategia en acción.

D.1.3 3. Validación externa pendiente

La sección con más incertidumbre sigue siendo:

```
Bloque de código: text
11_company_targets_job_boards_recruiters.md
```

Ahí sí hay riesgo de que algunos datos estén incompletos o no actualizados, porque empresas, careers pages y remote policies cambian constantemente.

No intentaría perfeccionar las 120 empresas ahora. Haría esto:

Pasada práctica

Elegir solo:

```
Bloque de código: text
Top 30 empresas A-priority
Top 20 empresas B-priority
Top 20 job boards / recruiters
```

Y verificar:

- careers URL;
- si hay roles vivos;
- si aceptan remoto;
- si remoto significa worldwide, LATAM, US-only o EU-only;
- si hay contractor/B2B;
- idioma;
- seniority;
- si TwinSight encaja.

Herramienta recomendada: modo agente, pero solo para esa verificación concreta.

No usaría agente para todo el documento.

D.2 Qué haría ahora

D.2.1 Paso 1 — Generar 16_final_consistency_audit_and_gap_register.md

Este archivo debe responder:

```
Bloque de código: text
¿Qué está cerrado?
¿Qué falta?
¿Qué tiene riesgo?
¿Qué requiere verificación humana?
¿Qué requiere agente?
¿Qué debe producirse como material final?
¿Qué se puede ejecutar ya?
¿Qué no se debe hacer todavía?
```

Esto crea un puente entre la investigación y los materiales finales.

Herramientas: ninguna.

D.2.2 Paso 2 — Crear el paquete de aplicación

Después de aprobar el gap register, generaría en este orden:

1. CV

```
Bloque de código: text
17_cv_base_and_role_variants.md
```

Debe incluir:

- CV base;
- CV para Real-Time 3D / Unity;
- CV para Technical Visualization / Digital Twin;
- CV para Unity WebGL / Interactive 3D;
- CV para Tools / Python Automation;
- bullet bank;
- reglas de edición.

2. LinkedIn

```
Bloque de código: text
18_linkedin_final_rewrite.md
```

Debe incluir:

- headline final;
- About;
- Experience;
- Education;
- Projects;
- Featured;
- Skills;
- recruiter keywords;
- availability statement.

3. GitHub

```
Bloque de código: text  
19_github_readme_templates.md
```

Debe incluir:

- README TwinSight;
- README ARA;
- repo descriptions;
- pinned repo order;
- disclaimer de IA;
- disclaimer de assets/CAD;
- GitHub profile README.

4. Portfolio

```
Bloque de código: text  
20_portfolio_copy_and_site_structure.md
```

Debe incluir:

- copy listo para web;
- home page;
- about;
- project cards;
- case study TwinSight;
- ARA;
- Blender portrait;
- contact;
- estructura MVP.

5. Demo video

```
Bloque de código: text  
21_twinsight_demo_video_script_and_shotlist.md
```

Debe incluir:

- guion de 60 s;
- guion de 120 s;
- lista de tomas;
- orden visual;
- captions;

- versión sin voz;
 - versión para LinkedIn.
-

D.3 Lo que falta como información tuya

Para cerrar con máxima precisión, faltan datos concretos:

D.3.1 TwinSight

- URL final del repo público.
- URL final del demo WebGL si existe.
- Capturas finales.
- Video o grabación.
- Métricas finales verificadas.
- Texto exacto aprobado de la tesis.
- Qué partes del CAD pueden mostrarse públicamente.
- Qué assets deben tener disclaimer.

D.3.2 CV / LinkedIn

- Historial laboral exacto con fechas.
- Nombre de clientes freelance si pueden mencionarse.
- Si hay proyectos pagados que puedan mostrarse.
- Nivel real de inglés: certificado o solo autodeclarado.
- Fecha estimada de sustentación.
- Disponibilidad semanal exacta después de sustentación.

D.3.3 Legal / contractor

- Si ya tienes RUT.
- Si ya facturas electrónicamente.
- Si tienes contador.
- Método preferido de pago.
- Si aceptarías contrato por hora, mensual retainer o fixed-price.
- Mínimo neto real aceptable.

D.3.4 Portafolio

- Dominio o plataforma elegida.
 - Si quieres GitHub Pages, Framer, Webflow, Notion, HTML propio o WordPress.
 - Estilo visual preferido.
 - Si quieres portfolio en inglés solamente o bilingüe.
-

D.4 Roadmap seguro

D.4.1 Fase A — Cierre documental

Bloque de código: text
16_final_consistency_audit_and_gap_register.md

Objetivo: asegurar que nada se contradice.

D.4.2 Fase B — Materiales públicos

```
Bloque de código: text
17_cv_base_and_role_variants.md
18_linkedin_final_rewrite.md
19_github_readme_templates.md
20_portfolio_copy_and_site_structure.md
21_twinsight_demo_video_script_and_shotlist.md
```

Objetivo: quedar listo para aplicar.

D.4.3 Fase C — Sistema de aplicación

```
Bloque de código: text
22_outreach_templates_english_spanish.md
23_interview_answer_bank.md
24_offer_evaluation_scorecard.md
25_application_tracker_template.xlsx
26_weekly_execution_dashboard.md
```

Objetivo: operar la búsqueda laboral.

D.4.4 Fase D — Verificación externa puntual

Usar modo agente para:

```
Bloque de código: text
Top 30 companies verification
Top 20 job boards verification
Top recruiter list verification
```

Objetivo: no aplicar a listas muertas o mal filtradas.

D.4.5 Fase E — Ejecución 30 días

Aplicar, medir, corregir.

D.5 Recomendación inmediata

El siguiente archivo debería ser:

```
Bloque de código: text
16_final_consistency_audit_and_gap_register.md
```

Herramientas:

Herramienta	Usar
Deep Research	No
Modo agente	No

Después de ese archivo, pasamos a generar materiales finales de aplicación. Ahí ya no estaremos investigando: estaremos produciendo el kit real para buscar trabajo.

Appendix E

D2_conferences_events_demo_venues

Fuente: Research/D2_conferences_events_demo_venues.md

Seccion: research

Estado: research_snapshot

Rol: Research / estudio de mercado

Lineas originales: 272

Ultima modificacion: 2026-06-11 21:07

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Opportunity name	URL	Organizer	Region	Opportunity type	Domain	Status	Deadline	Date checked							
Confidence										SIGGRAPH 2026 — Emerging Technologies	<https://s2026.siggraph.org/>	ACM SIGGRAPH	USA + virtual	Demo/exhibit track	Computer graphics / interactive tech
2026 likely closed; event upcoming	not found	2026-06-11	Medium		SIGGRAPH 2026 — Real-Time Live!	<https://s2026.siggraph.org/>	ACM SIGGRAPH								

Contenido completo depurado

E.0.1 0. Module status

Bloque de codigo: text
Module: D2_conferences_events_demo_venues.md
Date checked: 2026-06-11
Status: Complete v1 – traditional web research
Purpose: Map conferences, demo venues, poster/showcase tracks and public event channels where TwinSight X500 could plausibly
↔ be submitted, demoed, exhibited, or used as portfolio evidence.
Candidate: Alexander Woodcock Salomón
Primary project: TwinSight X500
Positioning: Real-Time 3D Developer / Unity Technical Artist

E.0.2 1. Scope notes

Bloque de codigo: text
Current-cycle information changes quickly.
Fields not visible in accessible public pages are marked as "not found".
Remote participation is not assumed unless clearly visible.
Travel-heavy venues are scored lower on probability.
General paper venues are minimized because D3 covers publication/poster venues in more depth.

E.0.3 2. Opportunity identity table

Opportunity name	URL	Organizer	Region	Opportunity type	Domain	Status	Deadline	Date checked	Confidence
SIGGRAPH 2026 — Emerging Technologies	https://s2026.siggraph.org/	ACM SIGGRAPH	USA + virtual	Demo/exhibit track	Computer graphics / interactive tech	2026 likely closed; event upcoming	not found	2026-06-11	Medium
SIGGRAPH 2026 — Real-Time Live!	https://s2026.siggraph.org/	ACM SIGGRAPH	USA + virtual	Live demo track	Real-time graphics	2026 likely closed; event upcoming	not found	2026-06-11	Medium
SIGGRAPH 2026 — Immersive Pavilion	https://s2026.siggraph.org/	ACM SIGGRAPH	USA + virtual	Immersive exhibit	XR / spatial computing	2026 likely closed; event upcoming	not found	2026-06-11	Medium
SIGGRAPH 2026 — Posters	https://s2026.siggraph.org/	ACM SIGGRAPH	USA + virtual	Poster venue	CG / HCI / visualization	2026 likely closed; event upcoming	not found	2026-06-11	Medium
SIGGRAPH Asia 2026 — Emerging Technologies	https://asia.siggraph.org/2026/	ACM SIGGRAPH Asia	Malaysia	Demo/exhibit track	CG / interactive tech	Upcoming / annual	not found	2026-06-11	Medium
SIGGRAPH Asia 2026 — Real-Time Live!	https://asia.siggraph.org/2026/	ACM SIGGRAPH Asia	Malaysia	Live demo track	Real-time graphics	Upcoming / annual	not found	2026-06-11	Medium
IEEE VIS 2026 — Posters	https://ieeevis.org/year/2026/welcome	IEEE VGTC	USA	Poster venue	Visualization / visual analytics	Upcoming / annual	not found	2026-06-11	Medium
IEEE VIS 2026 — Arts / Experience / Demos	https://ieeevis.org/year/2026/welcome	IEEE VGTC	USA	Showcase / demo	Visualization systems	Upcoming / annual	not found	2026-06-11	Low
EuroVis 2026 — Posters	https://www.eurovis.org/	Eurographics + IEEE VGTC	UK / Europe	Poster venue	Visualization	Upcoming / annual	not found	2026-06-11	Medium
EuroVis 2026 — Short Papers / Demos	https://www.eurovis.org/	Eurographics + IEEE VGTC	UK / Europe	Short/demo venue	Visualization	Upcoming / annual	not found	2026-06-11	Low
ACM Web3D 2026 — Demos	https://web3d.siggraph.org/	ACM / Web3D Consortium	Unknown / rotating	Demo venue	Web 3D / WebGL / X3D	Upcoming / annual	not found	2026-06-11	Low
ACM Web3D 2026 — Posters	https://web3d.siggraph.org/	ACM / Web3D Consortium	Unknown / rotating	Poster venue	Web 3D	Upcoming / annual	not found	2026-06-11	Low
ACM CHI 2027 — Interactivity	https://chi2027.acm.org/	ACM SIGCHI	Unknown	Interactive demo venue	HCI / interaction	Annual / likely recurring	not found	2026-06-11	Low
ACM CHI 2027 — Late-Breaking Work	https://chi2027.acm.org/	ACM SIGCHI	Unknown	LBW/poster venue	HCI / usability	Annual / likely recurring	not found	2026-06-11	Low
ACM CHI 2027 — Student Research Competition	https://chi2027.acm.org/	ACM SIGCHI	Unknown	Student research venue	HCI / usability	Annual / likely recurring	not found	2026-06-11	Low
ACM DIS 2026 — Student Design Competition	https://dis.acm.org/2026/	ACM DIS	Europe	Student design venue	Interaction design	Closed / annual	27 Feb 2026 snippet	2026-06-11	Medium
ACM UIST 2026 — Student Innovation Contest	https://www.emotibit.com/uist-student-innovation-contest-2026/	ACM UIST + EmotiBit	USA	Student innovation demo	HCI / sensors / XR	Open / upcoming	not found	2026-06-11	High
ACM UIST 2026 — Demos / Posters	https://uist.acm.org/2026/	ACM UIST	USA	Demo/poster venue	UI software / HCI	Upcoming / annual	not found	2026-06-11	Low
IEEE VR 2027 — Research Demos	https://ieeevr.org/	IEEE VGTC	Rotating	Demo venue	VR / 3D UI	Annual / likely recurring	not found	2026-06-11	Low
IEEE VR 2027 — Posters	https://ieeevr.org/	IEEE VGTC	Rotating	Poster venue	VR / 3D UI	Annual / likely recurring	not found	2026-06-11	Low
IEEE ISMAR 2026 — Demos	https://www.ismar.net/	IEEE / ACM / VGTC	Rotating	Demo venue	AR / MR	Upcoming / annual	not found	2026-06-11	Low
IEEE ISMAR 2026 — Posters	https://www.ismar.net/	IEEE / ACM / VGTC	Rotating	Poster venue	AR / MR	Upcoming / annual	not found	2026-06-11	Low
ACM VRST 2026 — Demos / Posters	https://vrst.acm.org/	ACM SIGCHI / SIGGRAPH	Rotating	Demo/poster venue	VR systems	Annual / likely recurring	not found	2026-06-11	Low
Laval Virtual — ReVolution Research	https://www.laval-virtual.com/	Laval Virtual	France	XR demo/exhibit	Immersive tech research	Annual / likely recurring	not found	2026-06-11	Medium
Laval Virtual — ReVolution Students	https://www.laval-virtual.com/	Laval Virtual	France	Student XR exhibit	Immersive tech	Annual / likely recurring	not found	2026-06-11	Medium
Laval Virtual Awards	https://www.laval-virtual.com/	Laval Virtual	France	Awards/showcase	XR / immersive tech	Annual / likely recurring	not found	2026-06-11	Medium
AWE USA — Auggie Awards	https://www.awexr.com/	AWE XR	USA	Awards / expo	AR / VR / XR	Annual / likely recurring	not found	2026-06-11	Medium
AWE USA — Startup Pitch / Expo	https://www.awexr.com/	AWE XR	USA	Pitch / demo expo	XR / spatial computing	Annual / likely recurring	not found	2026-06-11	Low
Games for Change Festival — XR for Change	https://festival.gamesforchange.org/	Games for Change	USA + hybrid history	Festival / awards	Games / XR for impact	Annual / likely recurring	not found	2026-06-11	Medium
Serious Play Conference — Serious Play Awards	https://www.seriousplayconf.com/awards/	Serious Play Conference	USA / online history	Awards / conference	Serious games / training	Annual / likely recurring	not found	2026-06-11	Low
Unity Awards — Best Student Project / Industry Innovation	https://unity.com/awards	Unity	Global	Awards / showcase	Unity projects	2025 closed; annual recurring	not found	2026-06-11	High
The Rookies Awards — Game Dev / Real-Time / 3D	https://www.therookies.co/contests	The Rookies	Global online	Student portfolio awards	3D / game / real-time art	Annual / likely recurring	not found	2026-06-11	Medium
SXSW Innovation Awards	https://www.sxsw.com/awards/	SXSW	USA	Awards / showcase	Innovation / XR / AI	Annual / likely recurring	not found	2026-06-11	Medium
Prix Ars Electronica — Interactive Art+	https://ars.electronica.art/prix/en/	Ars Electronica	Austria / global	Awards / festival	Interactive media art	Annual / likely recurring	not found	2026-06-11	Medium
A MAZE. Awards & Festival	https://amaze-berlin.de/	A MAZE.	Germany	Festival / awards	Games / playful media	Annual / likely recurring	not found	2026-06-11	Low
HCII 2026 — Student Design Competition	https://2026.hci.international/	HCI International	Canada	Student design competition	HCI / UX	Upcoming / annual	not found	2026-06-11	Medium
HCII 2026 — World Usability Day Design Challenge	https://2026.hci.international/	HCI International	Canada	Design challenge / awards	Usability / UX	Upcoming / annual	not found	2026-06-11	Medium
Falling Walls Lab	https://falling-walls.com/lab/	Falling Walls Foundation	Global / local labs	Science pitch event	Research / innovation	Annual / likely recurring	varies by local lab	2026-06-11	Medium

E.0.4 3. Eligibility and logistics table

Opportunity name	Eligibility	Colombia eligibility	Student eligibility	Individual/team	Online/remote participation	Travel required?	Cost / fee	Submission format	Required materials
SIGGRAPH 2026 — Emerging Technologies	Open to contributors; track rules vary	Unknown; international accepted generally	Student not required	Individual/team	Virtual elements possible; demo may require in-person	Likely yes for exhibits	Registration/travel cost likely	Demo/poster submission	Abstract, media, demo video
SIGGRAPH 2026 — Real-Time Live!	Open to contributors; track rules vary	Unknown; international accepted generally	Student not required	Individual/team	Virtual elements possible; demo may require in-person	Likely yes for exhibits	Registration/travel cost likely	Demo/poster submission	Abstract, media, demo video
SIGGRAPH 2026 — Immersive Pavilion	Open to contributors; track rules vary	Unknown; international accepted generally	Student not required	Individual/team	Virtual elements possible; demo may require in-person	Likely yes for exhibits	Registration/travel cost likely	Demo/poster submission	Abstract, media, demo video
SIGGRAPH 2026 — Posters	Open to contributors; track rules vary	Unknown; international accepted generally	Student not required	Individual/team	Virtual elements possible; demo may require in-person	Likely yes for exhibits	Registration/travel cost likely	Demo/poster submission	Abstract, media, demo video
SIGGRAPH Asia 2026 — Emerging Technologies	Open to contributors; track rules vary	Unknown; international accepted generally	Student not required	Individual/team	Virtual elements possible; demo may require in-person	Likely yes for exhibits	Registration/travel cost likely	Demo/poster submission	Abstract, media, demo video
SIGGRAPH Asia 2026 — Real-Time Live!	Open to contributors; track rules vary	Unknown; international accepted generally	Student not required	Individual/team	Virtual elements possible; demo may require in-person	Likely yes for exhibits	Registration/travel cost likely	Demo/poster submission	Abstract, media, demo video
IEEE VIS 2026 — Posters	Researchers/practitioners	Unknown	Student possible	Individual/team	Usually in-person; remote not assumed	Likely yes	Registration/travel cost	Poster/demo submission	Abstract, poster, video/demo
IEEE VIS 2026 — Arts / Experience / Demos	Researchers/practitioners	Unknown	Student possible	Individual/team	Usually in-person; remote not assumed	Likely yes	Registration/travel cost	Poster/demo submission	Abstract, poster, video/demo
EuroVis 2026 — Posters	Researchers/practitioners	Unknown	Student possible	Individual/team	Usually in-person; remote not assumed	Likely yes	Registration/travel cost	Poster/demo submission	Abstract, poster, video/demo
EuroVis 2026 — Short Papers / Demos	Researchers/practitioners	Unknown	Student possible	Individual/team	Usually in-person; remote not assumed	Likely yes	Registration/travel cost	Poster/demo submission	Abstract, poster, video/demo
ACM Web3D 2026 — Demos	Researchers/developers	Unknown	Student possible	Individual/team	Hybrid not assumed	Likely	Registration/travel cost	Demo/poster	Abstract, demo, video, web link
ACM Web3D 2026 — Posters	Researchers/developers	Unknown	Student possible	Individual/team	Hybrid not assumed	Likely	Registration/travel cost	Demo/poster	Abstract, demo, video, web link
ACM CHI 2027 — Interactivity	HCI contributors; track-specific	Unknown	Student possible; SRC requires student	Individual/team	Remote not assumed	Likely	Registration/travel cost	Interactivity/LBW/SRC	Extended abstract, video, demo
ACM CHI 2027 — Late-Breaking Work	HCI contributors; track-specific	Unknown	Student possible; SRC requires student	Individual/team	Remote not assumed	Likely	Registration/travel cost	Interactivity/LBW/SRC	Extended abstract, video, demo
ACM CHI 2027 — Student Research Competition	HCI contributors; track-specific	Unknown	Student possible; SRC requires student	Individual/team	Remote not assumed	Likely	Registration/travel cost	Interactivity/LBW/SRC	Extended abstract, video, demo
ACM DIS 2026 — Student Design Competition	Student teams	Unknown	Yes	Team	Remote not assumed	Likely	Registration/travel cost	Pictorial/design submission	Design brief, visuals, prototype
ACM UIST 2026 — Student Innovation Contest	Student innovators; proposals selected	Likely; travel required	Yes	Team/individual	Proposal online; final in-person	Yes	Travel + registration likely	Proposal + demo	Proposal, EmotiBit prototype
ACM UIST 2026 — Demos / Posters	HCI/UI contributors	Unknown	Student possible	Individual/team	Remote not assumed	Likely	Registration/travel cost	Demo/poster	Abstract, video, demo
IEEE VR 2027 — Research Demos	XR/VR/MR contributors	Unknown	Student possible	Individual/team	Remote not assumed	Likely	Registration/travel cost	Demo/poster	Abstract, poster, demo, video
IEEE VR 2027 — Posters	XR/VR/MR contributors	Unknown	Student possible	Individual/team	Remote not assumed	Likely	Registration/travel cost	Demo/poster	Abstract, poster, demo, video
IEEE ISMAR 2026 — Demos	XR/VR/MR contributors	Unknown	Student possible	Individual/team	Remote not assumed	Likely	Registration/travel cost	Demo/poster	Abstract, poster, demo, video
IEEE ISMAR 2026 — Posters	XR/VR/MR contributors	Unknown	Student possible	Individual/team	Remote not assumed	Likely	Registration/travel cost	Demo/poster	Abstract, poster, demo, video
ACM VRST 2026 — Demos / Posters	XR/VR/MR contributors	Unknown	Student possible	Individual/team	Remote not assumed	Likely	Registration/travel cost	Demo/poster	Abstract, poster, demo, video
Laval Virtual — ReVolution Research	XR projects; categories vary	Unknown	Students accepted in student track	Individual/team	Expo likely in-person	Yes	Travel/booth cost possible	Project application + exhibit	Prototype, demo video, description
Laval Virtual — ReVolution Students	XR projects; categories vary	Unknown	Students accepted in student track	Individual/team	Expo likely in-person	Yes	Travel/booth cost possible	Project application + exhibit	Prototype, demo video, description
Laval Virtual Awards	XR projects; categories vary	Unknown	Students accepted in student track	Individual/team	Expo likely in-person	Yes	Travel/booth cost possible	Project application + exhibit	Prototype, demo video, description
AWE USA — Auggie Awards	XR companies/creators; award rules vary	Unknown	Not student-specific	Individual/team/company	Mostly in-person expo	Likely	Submission/attendance cost unknown	Award/pitch/demo	Product/project description, video
AWE USA — Startup Pitch / Expo	XR companies/creators; award rules vary	Unknown	Not student-specific	Individual/team/company	Mostly in-person expo	Likely	Submission/attendance cost unknown	Award/pitch/demo	Product/project description, video
Games for Change Festival — XR for Change	Games/XR for social impact	Unknown	Student categories possible	Individual/team	Hybrid history; not assumed	Possible	Entry/attendance cost unknown	Award/festival submission	Playable/demo, impact statement
Serious Play Conference — Serious Play Awards	Serious games/training projects	Unknown	Student possible	Individual/team	Online history varies	Possible	Entry cost unknown	Awards submission	Game/training demo, description
Unity Awards — Best Student Project / Industry Innovation	Unity-made projects	Likely global	Student category exists in past	Individual/team/studio	Online submission	No travel required unless invited	No fee found	Awards nomination	Project page, video, build
The Rookies Awards — Game Dev / Real-Time / 3D	Students/recent graduates	Likely global	Yes	Individual	Online	No	not found	Portfolio contest	Breakdown, images, video
SXSW Innovation Awards	Innovation projects; categories vary	Unknown	Not student-specific	Individual/team/company	Submission online; event in-person	Likely	Submission/attendance cost likely	Awards submission	Project brief, video, demo link
Prix Ars Electronica — Interactive Art+	Artists/creators/researchers	Likely global	Student not required	Individual/team	Online submission; festival in-person	Possible	not found	Awards submission	Documentation, media, project text
A MAZE. Awards & Festival	Games/playful media creators	Likely global	Not required	Individual/team	Submission online; festival in-person	Possible	Submission fee unknown	Game/art submission	Playable/video, description
HCII 2026 — Student Design Competition	College/university students or design teams	Unknown	Yes for SDC	Team/individual	Remote not assumed	Likely	Registration/travel cost	Design submission/award	Poster, design report, prototype
HCII 2026 — World Usability Day Design Challenge	College/university students or design teams	Unknown	Yes for SDC	Team/individual	Remote not assumed	Likely	Registration/travel cost	Design submission/award	Poster, design report, prototype
Falling Walls Lab	Researchers/startups/students pitch innovation	Local lab rules vary	Student possible	Individual	Local events; finals in Berlin	Yes if selected	Travel cost possible	3-min pitch	Pitch deck, product demo, visuals

E.0.5 4. Fit and value table

Opportunity name	TwinSight fit	Why it fits	Risks / mismatch	Workload estimate	Prestige value	Portfolio value	Funding value	Networking value
SIGGRAPH 2026 — Emerging Technologies	High	Interactive technical demo / XR venue	Requires polished exhibit; travel	High	High	High	Low	High
SIGGRAPH 2026 — Real-Time Live!	High	Direct real-time demo venue	Very high bar; deadlines likely passed	High	High	High	Low	High
SIGGRAPH 2026 — Immersive Pavilion	High	Interactive technical demo / XR venue	Requires polished exhibit; travel	High	High	High	Low	High
SIGGRAPH 2026 — Posters	High	CG/interactive visualization poster	Competitive; academic framing	Medium	High	High	Low	High
SIGGRAPH Asia 2026 — Emerging Technologies	High	Interactive technical demo / XR venue	Requires polished exhibit; travel	High	High	High	Low	High
SIGGRAPH Asia 2026 — Real-Time Live!	High	Direct real-time demo venue	Very high bar; deadlines likely passed	High	High	High	Low	High
IEEE VIS 2026 — Posters	High	Visualization + evaluation fit	Needs stronger data/visual analytics framing	Medium	High	High	Low	High
IEEE VIS 2026 — Arts / Experience / Demos	High	Visualization + evaluation fit	Needs stronger data/visual analytics framing	Medium	High	High	Low	High
EuroVis 2026 — Posters	High	Visualization venue in Europe	Travel + academic framing	Medium	High	High	Low	Medium
EuroVis 2026 — Short Papers / Demos	High	Visualization venue in Europe	Travel + academic framing	Medium	High	High	Low	Medium
ACM Web3D 2026 — Demos	Very high	Direct browser 3D/WebGL fit	May prefer web standards over Unity WebGL	Medium	Medium	High	Low	Medium
ACM Web3D 2026 — Posters	Very high	Direct browser 3D/WebGL fit	May prefer web standards over Unity WebGL	Medium	Medium	High	Low	Medium
ACM CHI 2027 — Interactivity	Very high	HCI/usability/evaluation fit	Needs rigorous HCI framing	Medium	High	High	Low	High
ACM CHI 2027 — Late-Breaking Work	Very high	HCI/usability/evaluation fit	Needs rigorous HCI framing	Medium	High	High	Low	High
ACM CHI 2027 — Student Research Competition	Very high	HCI/usability/evaluation fit	Needs rigorous HCI framing	Medium	High	High	Low	High
ACM DIS 2026 — Student Design Competition	Medium	Design/process angle	Student team and design framing	Medium	Medium	Medium	Low	Medium
ACM UIST 2026 — Student Innovation Contest	Low	HCI venue but sensor-specific	Requires new EmotiBit project	High	High	Low	Low	High
ACM UIST 2026 — Demos / Posters	Medium	UI systems venue	Needs novel UI/software contribution	High	High	Medium	Low	High
IEEE VR 2027 — Research Demos	Medium	3D UI / XR-adjacent	No VR implementation yet	Medium	High	Medium	Low	High
IEEE VR 2027 — Posters	Medium	3D UI / XR-adjacent	No VR implementation yet	Medium	High	Medium	Low	High
IEEE ISMAR 2026 — Demos	Medium	MR/AR industrial visualization adjacent	No AR implementation yet	High	High	Medium	Low	High
IEEE ISMAR 2026 — Posters	Medium	MR/AR industrial visualization adjacent	No AR implementation yet	High	High	Medium	Low	High
ACM VRST 2026 — Demos / Posters	Medium	VR systems / technical systems	No VR build yet	Medium	Medium	Medium	Low	Medium
Laval Virtual — ReVolution Research	Medium	XR/immersive demo ecosystem	Needs XR reframing or exhibit budget	Medium	Medium	High	Low	High
Laval Virtual — ReVolution Students	Medium	XR/immersive demo ecosystem	Needs XR reframing or exhibit budget	Medium	Medium	High	Low	High
Laval Virtual Awards	Medium	XR/immersive demo ecosystem	Needs XR reframing or exhibit budget	Medium	Medium	High	Low	High
AWE USA — Auggie Awards	Medium	XR/product demo ecosystem	May require company/product maturity	Medium	Medium	Medium	Low	High
AWE USA — Startup Pitch / Expo	Medium	XR/product demo ecosystem	May require company/product maturity	Medium	Medium	Medium	Low	High
Games for Change Festival — XR for Change	Low-medium	Impact/training angle possible	Needs social/learning impact narrative	Medium	Medium	Medium	Low	Medium
Serious Play Conference — Serious Play Awards	Medium	Training/inspection learning angle	Needs serious-game/training framing	Medium	Medium	Medium	Low	Medium
Unity Awards — Best Student Project / Industry Innovation	Very high	Direct Unity project showcase	Annual timing; competition broad	Low	Medium	High	Low	Medium
The Rookies Awards — Game Dev / Real-Time / 3D	Medium	Student portfolio visibility	May value art over technical visualization	Low	Medium	High	Low	Medium
SXSW Innovation Awards	Medium	Innovation/showcase venue	High cost and broad competition	High	High	Medium	Low	High
Prix Ars Electronica — Interactive Art+	Medium	Interactive media art angle	Needs artistic/public framing	Medium	High	Medium	Low	High
A MAZE. Awards & Festival	Low	Interactive media/games showcase	Weak technical visualization fit	Low	Medium	Medium	Low	Medium
HCII 2026 — Student Design Competition	High	HCI/design/usability fit	Travel and academic framing	Medium	Medium	High	Low	Medium
HCII 2026 — World Usability Day Design Challenge	High	HCI/design/usability fit	Travel and academic framing	Medium	Medium	High	Low	Medium
Falling Walls Lab	Medium	Science/innovation pitch	Requires 3-min public pitch; broad competition	Low	Medium	Medium	Low	Medium

E.0.6 5. Scoring table

Opportunity name	Fit score	Probability score	Value score	Priority
SIGGRAPH 2026 — Emerging Technologies	5	2	5	Watchlist
SIGGRAPH 2026 — Real-Time Live!	5	2	5	Watchlist
SIGGRAPH 2026 — Immersive Pavilion	4	2	4	Watchlist
SIGGRAPH 2026 — Posters	4	2	5	B
SIGGRAPH Asia 2026 — Emerging Technologies	5	2	5	Watchlist
SIGGRAPH Asia 2026 — Real-Time Live!	5	2	5	Watchlist
IEEE VIS 2026 — Posters	5	3	5	A
IEEE VIS 2026 — Arts / Experience / Demos	5	3	5	A
EuroVis 2026 — Posters	4	2	4	B
EuroVis 2026 — Short Papers / Demos	4	2	4	B
ACM Web3D 2026 — Demos	5	3	4	A
ACM Web3D 2026 — Posters	5	3	4	A
ACM CHI 2027 — Interactivity	5	3	5	A
ACM CHI 2027 — Late-Breaking Work	5	3	5	A
ACM CHI 2027 — Student Research Competition	5	3	5	A
ACM DIS 2026 — Student Design Competition	4	2	4	B
ACM UIST 2026 — Student Innovation Contest	2	2	3	Watchlist
ACM UIST 2026 — Demos / Posters	4	2	4	B
IEEE VR 2027 — Research Demos	4	2	5	B
IEEE VR 2027 — Posters	4	2	5	B
IEEE ISMAR 2026 — Demos	4	2	5	B
IEEE ISMAR 2026 — Posters	4	2	5	B
ACM VRST 2026 — Demos / Posters	4	3	4	B
Laval Virtual — ReVolution Research	4	3	4	A
Laval Virtual — ReVolution Students	4	3	4	A
Laval Virtual Awards	4	3	4	A
AWE USA — Auggie Awards	3	2	4	B
AWE USA — Startup Pitch / Expo	3	2	4	B
Games for Change Festival — XR for Change	3	3	3	C
Serious Play Conference — Serious Play Awards	3	3	3	C
Unity Awards — Best Student Project / Industry Innovation	5	4	4	A
The Rookies Awards — Game Dev / Real-Time / 3D	3	4	4	B
SXSW Innovation Awards	3	1	4	Watchlist
Prix Ars Electronica — Interactive Art+	3	2	4	Watchlist
A MAZE. Awards & Festival	2	3	3	C
HCII 2026 — Student Design Competition	4	2	4	B
HCII 2026 — World Usability Day Design Challenge	4	2	4	B
Falling Walls Lab	3	3	3	C

E.0.7 6. Source table

Opportunity name	Source URLs	Notes
SIGGRAPH 2026 — Emerging Technologies	https://s2026.siggraph.org/	Next cycle: Annual; status: 2026 likely closed; event upcoming; deadline: not found; confidence: Medium
SIGGRAPH 2026 — Real-Time Live!	https://s2026.siggraph.org/	Next cycle: Annual; status: 2026 likely closed; event upcoming; deadline: not found; confidence: Medium
SIGGRAPH 2026 — Immersive Pavilion	https://s2026.siggraph.org/	Next cycle: Annual; status: 2026 likely closed; event upcoming; deadline: not found; confidence: Medium
SIGGRAPH 2026 — Posters	https://s2026.siggraph.org/	Next cycle: Annual; status: 2026 likely closed; event upcoming; deadline: not found; confidence: Medium
SIGGRAPH Asia 2026 — Emerging Technologies	https://asia.siggraph.org/2026/	Next cycle: 2026 Kuala Lumpur; status: Upcoming / annual; deadline: not found; confidence: Medium
SIGGRAPH Asia 2026 — Real-Time Live!	https://asia.siggraph.org/2026/	Next cycle: Annual; status: Upcoming / annual; deadline: not found; confidence: Medium
IEEE VIS 2026 — Posters	https://ieeevis.org/year/2026/welcome	Next cycle: 2026 Boston; status: Upcoming / annual; deadline: not found; confidence: Medium
IEEE VIS 2026 — Arts / Experience / Demos	https://ieeevis.org/year/2026/welcome	Next cycle: Annual; status: Upcoming / annual; deadline: not found; confidence: Low
EuroVis 2026 — Posters	https://www.eurovis.org/	Next cycle: 2026 Nottingham; status: Upcoming / annual; deadline: not found; confidence: Medium
EuroVis 2026 — Short Papers / Demos	https://www.eurovis.org/	Next cycle: Annual; status: Upcoming / annual; deadline: not found; confidence: Low
ACM Web3D 2026 — Demos	https://web3d.siggraph.org/	Next cycle: Annual; status: Upcoming / annual; deadline: not found; confidence: Low
ACM Web3D 2026 — Posters	https://web3d.siggraph.org/	Next cycle: Annual; status: Upcoming / annual; deadline: not found; confidence: Low
ACM CHI 2027 — Interactivity	https://chi2027.acm.org/	Next cycle: 2027; status: Annual / likely recurring; deadline: not found; confidence: Low
ACM CHI 2027 — Late-Breaking Work	https://chi2027.acm.org/	Next cycle: 2027; status: Annual / likely recurring; deadline: not found; confidence: Low
ACM CHI 2027 — Student Research Competition	https://chi2027.acm.org/	Next cycle: 2027; status: Annual / likely recurring; deadline: not found; confidence: Low
ACM DIS 2026 — Student Design Competition	https://dis.acm.org/2026/	Next cycle: Annual; status: Closed / annual; deadline: 27 Feb 2026 snippet; confidence: Medium
ACM UIST 2026 — Student Innovation Contest	https://www.emotibit.com/uist-student-innovation-contest-2026/	Next cycle: Annual; status: Open / upcoming; deadline: not found; confidence: High
ACM UIST 2026 — Demos / Posters	https://uist.acm.org/2026/	Next cycle: Annual; status: Upcoming / annual; deadline: not found; confidence: Low
IEEE VR 2027 — Research Demos	https://ieeevr.org/	Next cycle: 2027; status: Annual / likely recurring; deadline: not found; confidence: Low
IEEE VR 2027 — Posters	https://ieeevr.org/	Next cycle: 2027; status: Annual / likely recurring; deadline: not found; confidence: Low
IEEE ISMAR 2026 — Demos	https://www.ismar.net/	Next cycle: Annual; status: Upcoming / annual; deadline: not found; confidence: Low
IEEE ISMAR 2026 — Posters	https://www.ismar.net/	Next cycle: Annual; status: Upcoming / annual; deadline: not found; confidence: Low
ACM VRST 2026 — Demos / Posters	https://vrst.acm.org/	Next cycle: Annual; status: Annual / likely recurring; deadline: not found; confidence: Low
Laval Virtual — ReVolution Research	https://www.laval-virtual.com/	Next cycle: 2027 likely; status: Annual / likely recurring; deadline: not found; confidence: Medium
Laval Virtual — ReVolution Students	https://www.laval-virtual.com/	Next cycle: 2027 likely; status: Annual / likely recurring; deadline: not found; confidence: Medium
Laval Virtual Awards	https://www.laval-virtual.com/	Next cycle: Annual; status: Annual / likely recurring; deadline: not found; confidence: Medium
AWE USA — Auggie Awards	https://www.awexr.com/	Next cycle: 2027 likely; status: Annual / likely recurring; deadline: not found; confidence: Medium
AWE USA — Startup Pitch / Expo	https://www.awexr.com/	Next cycle: Annual; status: Annual / likely recurring; deadline: not found; confidence: Low
Games for Change Festival — XR for Change	https://festival.gamesforchange.org/	Next cycle: Annual; status: Annual / likely recurring; deadline: not found; confidence: Medium
Serious Play Conference — Serious Play Awards	https://www.seriousplayconf.com/awards/	Next cycle: Annual; status: Annual / likely recurring; deadline: not found; confidence: Low
Unity Awards — Best Student Project / Industry Innovation	https://unity.com/awards	Next cycle: Annual; status: 2025 closed; annual recurring; deadline: not found; confidence: High
The Rookies Awards — Game Dev / Real-Time / 3D	https://www.therookies.co/contests	Next cycle: Annual; status: Annual / likely recurring; deadline: not found; confidence: Medium
SXSW Innovation Awards	https://www.sxsw.com/awards/	Next cycle: Annual; status: Annual / likely recurring; deadline: not found; confidence: Medium
Prix Ars Electronica — Interactive Art+	https://ars.electronica.art/prix/en/	Next cycle: Annual; status: Annual / likely recurring; deadline: not found; confidence: Medium
A MAZE. Awards & Festival	https://amaze-berlin.de/	Next cycle: Annual; status: Annual / likely recurring; deadline: not found; confidence: Low

Opportunity name	Source URLs	Notes
HCII 2026 — Student Design Competition	https://2026.hci.international/	Next cycle: Annual; status: Upcoming / annual; deadline: not found; confidence: Medium
HCII 2026 — World Usability Day Design Challenge	https://2026.hci.international/	Next cycle: Annual; status: Upcoming / annual; deadline: not found; confidence: Medium
Falling Walls Lab	https://falling-walls.com/lab/	Next cycle: Annual; status: Annual / likely recurring; deadline: varies by local lab; confidence: Medium

E.0.8 7. High-priority extraction table

Priority	Opportunity cluster	Practical use
A	Web3D, CHI, IEEE VIS, Unity Awards, Laval Virtual	Best alignment with TwinSight proof
B	SIGGRAPH posters, SIGGRAPH Asia, EuroVis, DIS, UIST demos, IEEE VR, ISMAR, VRST, AWE, The Rookies, HCII	Strong but higher eligibility/travel/adaptation burden
C	Games for Change, Serious Play, A MAZE, Falling Walls	Useful only if reframed as training, learning, or public-impact demo
Watchlist	SIGGRAPH Emerging Tech / Real-Time Live, SXSW, Ars Electronica, UIST SIC	High prestige or visibility but high mismatch/cost/barrier

E.0.9 8. D2-to-D3 boundary

Belongs in D2	Belongs in D3
Demo tracks	Full paper tracks
Interactivity tracks	Journal-first publication routes
Student showcases	Academic publication venues
Expo/showcase awards	Poster-only publication plans
Public demo venues	Paper/poster submission strategy
Portfolio visibility events	Research dissemination strategy

E.0.10 9. Immediate evidence requirements for future submissions

Asset	Needed for
90-second TwinSight demo	Unity Awards, Web3D, Laval Virtual, AWE
30-second teaser	public showcase / festival submissions
Technical breakdown video	SIGGRAPH, Web3D, IEEE VIS, CHI
Case-study page	all venues
GitHub README	Web3D, CHI, technical reviewers
Metrics card	IEEE VIS, CHI, HCII
Visual modes grid	SIGGRAPH, Unity Awards, The Rookies
Pipeline diagram	Web3D, IEEE VIS, CHI
Live WebGL build	Unity Awards, Web3D, demos

Appendix F

Resumen Ejecutivo

Fuente: Research/deep-research-report_00.md

Seccion: research

Estado: research_snapshot

Rol: Research / estudio de mercado

Lineas originales: 461

Ultima modificacion: 2026-06-11 12:18

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Alexander Woodcock Salomón es un **Technical Artist y Desarrollador 3D en Tiempo Real** (Unity, WebGL, C#, Blender, optimización CAD-realidad) con fuerte formación en ingeniería multimedia y electrónica. Su meta es asegurar **empleos mejor remunerados** (idealmente remotos desde Colombia) en los próximos 3–6 meses, con un plan de 12–24 meses para duplicar o triplicar su salario y lograr movilidad internacional. Dada su experiencia académica (tesis sobre visualización técnica

Contenido completo depurado

Alexander Woodcock Salomón es un **Technical Artist y Desarrollador 3D en Tiempo Real** (Unity, WebGL, C#, Blender, optimización CAD-realidad) con fuerte formación en ingeniería multimedia y electrónica. Su meta es asegurar **empleos mejor remunerados** (idealmente remotos desde Colombia) en los próximos 3–6 meses, con un plan de 12–24 meses para duplicar o triplicar su salario y lograr movilidad internacional. Dada su experiencia académica (tesis sobre visualización técnica de dron en Unity WebGL) y sus proyectos con IA asistida, el enfoque de búsqueda se orientará a roles de *Technical Artist*, *3D interactivo*, *XR/simulación/digital twin* y *desarrollo de herramientas Python/IA*, en industrias de simulación industrial, visualización técnica, videojuegos y tecnologías creativas.

Los objetivos salariales de USD 1.5k, 3k y 6k mensuales corresponden a rangos desde niveles locales bajos hasta medianos-globales. Por ejemplo, el salario medio remoto de un *Technical Artist* es de ~\$90k anuales (~\$7.5k/mes). Así, \$1.5k/mes (\$18k/año) queda por debajo de ese promedio global, \$3k/mes (\$36k/año) se acerca a roles junior intermedios internacionales, y \$6k/mes (\$72k/año) es realista para posiciones mid-level en mercados maduros. Estos datos indican que la búsqueda debe priorizar ofertas globales mejor remuneradas (EE.UU./UE) mientras se aprovechan los canales de contratación remota desde Colombia.

En Colombia, el salario típico de un ingeniero de software (~COP 123 M anuales, ~USD 31k) es inferior a los objetivos internacionales. Por tanto, el plan se centra en **trabajo remoto/contrato B2B internacional**. Se preparará el candidato para contratación remota: obtener RUT, afiliarse a EPS/ARL, usar facturación electrónica DIAN, y manejar pagos internacionales (Wise/Payoneer).

Las barreras legales actuales impiden asumir empleo en la UE sin permiso: ni residencia alemana ni ciudadanía portuguesa están vigentes. Aunque el candidato espera futuro pasaporte portugués por linaje (requiere 3 años de residencia según ley reciente) y casarse en Alemania (facilitaría visa familiar sin idioma), estos son logros a mediano plazo (>2 años). De momento, la prioridad es el trabajo remoto desde Colombia. Se explorarán puestos freelance/contratista mediante EOR o plataformas globales, garantizando cumplimiento fiscal colombiano.

En resumen, la estrategia combina:

- **Investigación de mercado y priorización de roles** (Technical Artist, 3D interactivo, XR/simulación, Python/AI) con ajuste al perfil técnico real.
- **Estrategia de CV/LinkedIn/GitHub/portafolio** basada en proyectos clave (p.ej. TwinSight X500) y palabras clave ATS (Unity, C#, WebGL, optimización 3D, Blender, Python).
- **Plan 30/60/90 días** concreto: limpiar y publicar proyectos relevantes (GitHub, ArtStation), optimizar perfiles profesionales, iniciar contactos y aplicaciones.
- **Plan de acciones de networking y entrevistas**: secuencias de outreach en inglés y español, preparación de respuestas a temas sensibles (baja experiencia formal, uso de IA en proyectos) y estrategia de negociación basada en datos de salario global.

Cada recomendación se fundamenta en datos recientes (2024–2026) y normas locales. Por ejemplo, para empleo remoto desde Colombia se considerarán modelos de contratación B2B/contratista documentando IVA exportación (0%) y usando pagos internacionales formales. En la ruta internacional, se evitará sobrevalorar las oportunidades europeas inmediatas: la residencia portuguesa exige 3 años en Portugal y la ruta alemana requiere matrimonio y solicitud de visa (sin dominio del alemán). La búsqueda inicial se enfocará en empresas globales remotas de EE.UU., Canadá, Reino Unido, Países Bajos y Australia con alta contratación remota en tech art, XR y simulación.

F.1 Diagnóstico de la posición de mercado actual

El perfil de Alexander combina **ingeniería multimedia, herramientas 3D y Python/IA**. A nivel laboral:

- **Experiencia**: proyectos de portfolio/académicos (p.ej. prototipo WebGL de dron, herramientas de automatización AI). Aún no hay experiencia formal de industria en technical art/Unity, sólo freelance y proyectos personales. Esto es común en candidatos junior de campos emergentes.
- **Habilidades destacadas**: Unity/C#, WebGL/URP, Blender (retopología y optimización CAD), UI Toolkit, Shader Graph, Python, LLM y pipeline automation. Estos responden a demandas de industrias de *visualización técnica y simulación*. Sin embargo, debe validarse experiencia con frameworks web (FastAPI, React) u otras tecnologías secundarias si se postula a full-stack o PyDev.
- **Idiomas**: español nativo, inglés C1, alemán nulo. Esto califica para roles internacionales en inglés; alemán ofrece poca ventaja en corto plazo.
- **Ubicación**: Colombia – mercado local limitado en tech art (salarios bajos, pocas grandes empresas en videojuegos). El candidato apunta a mercado global remoto: USA, Canadá, Europa, LATAM nearshore.
- **Legal**: No tiene permiso UE inmediato. Portugal por linaje podría llegar, pero requiere residir 3 años tras el cambio de ley. La ruta alemana por cónyuge (futura) permite estadía/sitio de trabajo en Alemania sin alemán, pero esto es incierto y a mediano plazo.

Entorno salarial: En EEUU/UE, salarios para *Technical Artists* y *Desarrolladores 3D/AR/VR* rondan USD 80k–130k anuales. Es decir, unos USD 6.7k–10.8k mensuales. **OBJETIVOS SALARIALES**:

- **USD 1.5k/mes (\$18k/año)**: muy bajo comparado a niveles mundiales; situaría a Alexander por debajo de la mediana local (≈USD 21k–31k/año para devs en Colombia). Quizá factible solo en proyectos freelance locales o pasantías.
- **USD 3k/mes (\$36k/año)**: aún bajo para EE.UU./Europa, equivale a nivel muy junior. Sin embargo, está por encima de salarios colombianos medios para devs (≈USD 27k–41k). Probable en el corto plazo como contratista B2B remota en startups o empresas nearshore.
- **USD 6k/mes (\$72k/año)**: rango medio-alto global. Muy plausible para mid-level Technical Artist o XR/Sim Developer en mercados clave. Meta final ambiciosa pero alcanzable.

Tendencias de mercado (2024-2026): La demanda de perfiles Unity/AR/VR y visualización 3D sigue activa por la industria de videojuegos, metaverso, simulación militar/industrial y sectores de visualización (salud, ingeniería). Empresas remotas están creciendo, y portales de empleo remoto listan numerosas vacantes para *Unity Technical Artist* y *3D Developer*. Por ejemplo, Indeed y ZipRecruiter muestran centenares de ofertas remotas para "Unity Technical Artist" con salarios medios de USD 70k–140k. Sin embargo, la competencia es alta (Glassdoor indica ~47 candidatos por oferta de Unity dev en 2025). Diferenciarse requerirá portafolio sólido y contactos de la industria.

Resumen del diagnóstico: Alexander tiene un perfil técnico fuerte pero junior, con proyectos relevantes (Twin-Sight X500) pero sin experiencia corporativa significativa. Su costo local es bajo, su potencial salarial global es alto. La estrategia debe resaltar sus logros académicos/prácticos, posicionarlo como candidato capacitado en Technical Art/3D interactivo, y conectar con el mercado remoto antes de depender de vías migratorias. Se recomienda una búsqueda priorizando empresas globales con mentalidad remota y proyectos de visualización técnica, ajustando expectativas de salario inicial (c. \$18–36k/año), mientras se avanza hacia posiciones mejor pagadas (\$72k/año) en 12-24 meses.

F.2 Declaración de posicionamiento recomendada

Basado en habilidades y proyectos clave, se propone la siguiente declaración de posicionamiento:

“Technical Artist & Desarrollador 3D en Tiempo Real especializado en Unity, WebGL y Blender, con experiencia en optimización CAD-2-realidad y creación de visualizaciones técnicas interactivas. Uso Python e IA para automatizar pipelines de producción.”

Este enunciado resalta su especialización Unity/WebGL, optimización 3D, visualización técnica e integración de AI/Python. Se puede ajustar según el rol:

- **Unity Technical Artist / 3D Developer:** “Technical Artist Unity con enfoque en render en tiempo real y UI interactivas.”
- **Interactive 3D Developer / WebGL Developer:** “Desarrollador 3D interactivo (Unity WebGL) para visualización técnica.”
- **XR/Simulación/Digital Twin:** “Desarrollador XR/Simulación especializado en gemelos digitales para manufactura y drones.”
- **Python/AI Tooling:** “Desarrollador Python & AI construyendo herramientas de automatización de investigación (e.g. ARA Framework).”

Se recomienda priorizar Unity/WebGL técnicos en industrias de visualización industrial, drones, energía o defensa, antes que juegos puros (donde suele requerirse portafolio artístico más extenso). El candidato puede mencionar videojuegos/tecnología creativa como pasión, pero por estrategia resaltar **productividad / visualización técnica** (dónde hay más demanda remunerativa). Ejemplo: “Real-Time 3D Developer (Unity) con foco en simulación industrial y visualización técnica, complementado con herramientas de IA para mejorar pipelines.”

La combinación “Technical Artist & Real-Time 3D Developer” cubre múltiples roles listados (1,2,3,4,5,6,8), mientras que el componente “Python AI Tooling” (punto 11) se presentaría como secundario en CV/LinkedIn por si las oportunidades de visualización no fructifican. La ruta full-stack (12) puede omitirse a favor de especialización técnica, aunque conocimientos web/Python pueden destacarse en sección de habilidades o proyectos.

F.3 Matriz de priorización de roles

Rol candidato	Demanda Relativa	Ajuste al Perfil	Salario Potencial (@EE.UU./UE)	Comentarios
Unity Technical Artist / 3D Dev	Alta (videojuegos, tech art)	Muy alta	\\$80k–\\$130k+/año	Enfocar portafolio en optimización Blender, shaders, UI interactiva.
Real-Time 3D / WebGL Developer	Alta (industrial, simulación, web)	Alta	\\$70k–\\$120k/año (XR, simulación)	Prototipos interactivos como TwinSight X500 atraen en industria técnica.
XR / Simulación / Digital Twin Developer	Media-alta (AR/VR crece)	Media-alta	\\$90k–\\$140k/año	Requiere enfatizar hardware/no obstante; mostrar adaptabilidad de skills.
Technical Visualization (industrial)	Media (ingeniería, salud)	Media	\\$80k–\\$120k/año	Enfatizar proyectos CAD-3D, visualización de procesos técnicos.
Game Technical Artist	Media (videojuegos AAA o indie)	Media-baja	\\$70k–\\$130k/año	Alto costo de entrar sin experiencia significativa; mejor pivot a sectores civiles.
Tools Developer / Pipeline (C#/Python)	Baja-media	Media	\\$60k–\\$100k/año	Podría sumar valor, pero mercado saturado de dev tools; se usa como skill adicional.

Rol candidato	Demanda Relativa	Ajuste al Perfil	Salario Potencial (@EE.UU./UE)	Comentarios
Python/AI Tooling Developer	Alta (startups, IA/LLM)	Media	\\$80k–\\$140k/año	Mantener proyecto ARA en stack, pero no abandonar el enfoque 3D principal.
Full-stack Developer (Secundario)	Media (back-end/front-end)	Media-baja	\\$70k–\\$120k/año	Como ruta secundaria si otras fallan; mejorar React/FastAPI podría abrir puertas B2B.

Conclusión rol-prioridad: Priorizar *Unity Technical Artist / Real-Time 3D* enfocado en industrias de simulación/visualización (puntos 1,2,3,5,6,8). Incluir *XR/Simulación* (7,8) como roles afines. Posicionamiento secundario en *AI/Python* (11) para roles de herramienta si conviene, pero sin diluir la narrativa principal en 3D. Se descartaría dedicación a full-stack puro (12), más allá de lo básico web, pues aleja del objetivo 3D.

F.4 Tabla de priorización de mercados

Región/País	Mercado remoto/contratista	Demanda 3D/Unity/Tech Art	Salario USD ≈ (junior–mid)	Comentarios estratégicos
EE.UU. / Canadá	Muy alta (tech hubs, estudios juegos, startups)	Alta (videojuegos, AR/VR)	\\$60k–\\$130k+/año (jun-senior)	Prioridad alta. Contratos B2B o EOR posibles. Buen pago pero alta competencia.
Europa Occidental (DE, UK, NL, Nordics)	Media-Alta (empresas tecnológicas, simulación)	Alta (industria, automotriz, VR)	€40k–€80k/año junior, mid €60k–€100k (≈\\$45k–\\$110k)	De alta calidad, aunque posible barrera idiomática (salvo inglés en tech). Evaluar Alemania tras reubicación. UK/Países Bajos muy remotos con inglés.
Escandinavia (Suecia, Noruega, Finlandia)	Media (tech y juegos)	Alta (juegos, visualización médica)	Similar Europa (~\\$50k–100k)	Alto costo de vida, buen inglés, sectores VR/medical. Menos remotos.
Sudeste Asiático (por evid)	Baja (pocos roles remotos locales, pocos estudios)	Baja	\\$30k–\\$50k	Dejar como opción final; salaje bajo relativo.
LatAm (remoto): México, Brasil, Argentina	Alta (nearshore, outsourcing)	Media (outsourcing dev/games)	USD 25k–50k (contratista)	Buen ajuste idiomático (español/inglés). Competencia local media. Salarios menores comparados a EE.UU., pero posibles para Ingresos iniciales.
España / Portugal	Media (creciente tech, gaming)	Media (videojuegos, digital twin)	€20k–€40k junior, mid €30k–€60k (\\$22k–\\$65k)	Cuidado con burocracia UE. Inglés a veces suficiente; portugués útil a largo plazo.
Alemania / Europa centro	Media-Baja (sector manufactura y VR)	Media (ingeniería, simulación)	€30k–€60k (\\$33k–\\$66k)	Idioma puede ser barrera; visa familiar al casarse en Alemania abre mercado.
India / Middle East Remoto global (empresas fully-remote)	Baja (focos distintos) Muy Alta	Baja Alta (todas)	\\$15k–30k variable (\\$40k–\\$120k)	No estratégico. Plataformas remotas (GitLab, Automattic) buscan perfiles tecnológicos, aunque no necesariamente 3D. Buena oportunidad freelance / contract.

Recomendaciones de mercado: Dar prioridad a **EE.UU./Canadá** y **Europa occidental** por remuneración y demanda técnica. También explorar **LatAm para puestos remotos bilingües** (empresas nearshore y startups) que ofrezcan USD/competitividad. Países de habla inglesa facilitan entrevistas sin dominio alemán. Preparar CV y portafolio adaptado a cada mercado (ej. traducir al inglés, formatos internacionales). Evitar mercados locales de salario bajo (Colombia) y Asia semieconómica.

F.5 Tabla de benchmarking salarial

Se presenta un cuadro resumen de rangos salariales estimados por mercado y rol, en USD mensuales (equivalente bruto aproximado):

Rol / Mercado	EE.UU. remoto (USD)	UE remoto (USD)	Contratista B2B Col (USD)	Fuente/Comentarios
Unity Technical Artist / Dev	\\$6,000–10,000 (mid)	\\$4,500–8,000 (mid)	\\$2,000–3,500	Glassdoor: Tech Artist remoto ~\$90k/~\$7.5k
XR/AR/VR Developer	\\$8,000–12,000	\\$5,000–9,000	\\$2,500–4,000	XR dev Glassdoor med ~\$113k
Technical Visualization	\\$6,000–10,000	\\$4,000–8,000	\\$2,000–3,500	Similar a above, roles industriales altos.
3D WebGL/Interactive Dev	\\$5,000–9,000	\\$4,000–7,000	\\$2,000–3,000	Unity WebGL enfocado a web/móviles, quizá menor que AR/VR.
Game Technical Artist	\\$6,000–12,000	\\$5,000–9,000	\\$1,800–3,000	Alta mediana EE.UU., pero baja medio local; campo competitivo.
Python/AI Tools Dev	\\$7,000–12,000	\\$5,000–9,000	\\$2,500–4,000	STEM/AI roles IT mediano-alto, en AI startups puede subir.
Junior Generalist (LatAm)	\\$2,500–4,000	\\$2,000–3,500	\\$1,500–2,500	Salarios latinos en USD, a veces pagados en dólares.

Fuentes: Datos de Glassdoor para roles relacionados, reportes ZipRecruiter/Levels.fyi. Los rangos UE son aproximados (Euro a USD). Los valores locales B2B/Col se estiman en base a tarifas freelance en plataformas tech (~\$15–30/hora según experiencia) y salidas típicas de proyecto (~\$2000–3500/mes inicial).

Comentarios: Los valores en EE.UU./UE representan salarios brutos anuales convertidos mensualmente. Los salarios como contratista B2B desde Colombia suelen pactarse en USD (pagos internacionales) y son competitivos para salarios LATAM, pero inferiores al nivel EE.UU. Por ejemplo, un contrato mid-level Unity puede rondar ~\$25–50/hora (USD) en Outsourcing, unos ~\$4k–8k/mes, mientras que un contrato similar en EE.UU. fijo ofrece más.

F.6 Análisis de factibilidad de remoto desde Colombia

Trabajar remoto desde Colombia como contratista (B2B) es viable, pero requiere cumplir normativas:

- **Formato legal:** Se utiliza un *Contrato de Prestación de Servicios* a nombre de un trabajador independiente. No hay retención de impuestos por el cliente extranjero, pero el consultor/contratista (Alexander) debe facturar legalmente.
- **Impuestos:** El contratista presenta *Formulario 210* anual de renta y paga IVA 19% sobre servicios locales. Sin embargo, los servicios exportados (facturables a empresas extranjeras) están gravados al 0%. Debe registrarse en RUT (Registro Único Tributario) y emitir **factura electrónica** mediante DIAN (con IVA 0%). Sin RUT válido, aplican retenciones altas.
- **Seguridad social:** Como independiente, Alexander debe cotizar autónomamente a salud/pensión (EPS y Colpensiones/otra AFP) y ARL (riesgos laborales) por su cuenta. Los clientes pueden exigir comprobante de ARL vigente. Estos aportes garantizan cobertura.
- **Pago:** Lo ideal es transferencia bancaria internacional o plataformas como Payoneer/TransferWise (Wise). Se debe declarar correctamente el movimiento de divisas al Banco de la República. Papaya aconseja pagos por wire USD o PayPal (respetando reglas AML).
- **Otros:** Registrar empresa unipersonal/independiente (“persona natural comerciante”) o al menos RUT. Llevar contabilidad básica (facturas, comprobantes EPS/ARL). Eventualmente considerar EOR (no necesario si se gestiona B2B directo).

Conclusiones: Conformar una formalidad básica (RUT, facturación electrónica, ARL/seguridad social) es esencial para acceder a clientes internacionales y evitar penalidades. Se recomienda revisar guías legales: Umbrex y Mismo coinciden en procedimientos de contratación y evita’er “misclasificación”. Se incluirá un checklist: “Registrar RUT, contratar ARL, abrir cuenta COP/USD, habilitar Factura Electrónica DIAN, definir condiciones de pago internacional”.

F.7 Estrategia futura UE (Alemania/Portugal)

- **Alemania:** Si la pareja alemana de Alexander logra empleo allí, Alexander puede solicitar visa de reagrupación familiar. Como cónyuge de ciudadano alemán (UE), no se requiere alemán para la visa. Entonces obtendría permiso de residencia/trabajo sin mayores trabas y podrá buscar empleo on-site. Sin embargo, esto depende de los planes reales de la pareja (no garantizado). A medio plazo, si la pareja se traslada, Alexander debe maximizar su inglés, quizá empezar alemán básico para integración. No asumir privilegios alemanes antes de casarse y obtener visa.
- **Portugal (ciudadanía Sephardí):** Actualmente la ley exige *residencia física 3 años en Portugal* para obtener la ciudadanía por linaje. Si Alexander inicia este proceso, deberá residir en Portugal como mínimo 3 años (puede ser no consecutivos) para poder naturalizar. En el corto plazo (3–6 meses) esto no está disponible. Es un proyecto a largo plazo (≥3 años). No debe mencionarse como “inminente” para reclutadores, solo como meta personal eventual. Conviene mantenerlo en privado en los primeros meses.
- **Otras opciones UE:** Existen visas de residencia para trabajadores digitales en países como España, Portugal (sin citizenship), pero requieren ingresos mínimos sostenibles (~€2000–3000/mes) y a veces prueba de fondos. Podría explorarse en 12–24 meses si tiene experiencia laboral internacional. Por ahora, es secundario.

F.8 Estrategia de CV

Se preparará un CV orientado a ATS (un página) en inglés, con claro perfil y bullets cuantificados. Estructura:

- **Encabezado:** Nombre, título (p.ej. “*Technical Artist & Unity Developer*”), localización (Colombia), correo/LinkedIn/GitHub.
- **Resumen profesional (2–3 líneas):** Destacar perfil “*Multimedia Engineer / Technical Artist especializado en Unity WebGL y optimización 3D; apasionado por visualización técnica y producción asistida por IA.*”
- **Habilidades clave:** Lista de Skills en bloque (Unity, C#, Blender, WebGL, Shader Graph, Python, IA, etc.) con términos ATS.
- **Experiencia (proyectos relevantes):** Dado que no hay empleo formal relevante, se listarán proyectos clave como si fueran “freelance/independiente”:
- *TwinSight X500 (Proyecto de Tesis – Interactive 3D Drone Visualization):* bullets con métricas (“Optimización de CAD 6.5M→95k triángulos; desarrollo Unity WebGL, UI técnico; evaluaciones SUS 91.88, NASA-TLX 8.69 vs 19.89”). Mostrar liderazgo técnico y alcance.
- *ARA Framework (Automatización de Investigación Académica, Python):* bullets (“Desarrollo de sistema de agentes LLM; integraciones LangGraph, scraping de papers; validado parcialmente con prototipos. Presentar como proof-of-concept de AI tooling.”).
- *Otros proyectos destacados:* (Adaptador: Por completarse en CV bullets: ai-news-aggregator, FitApp-Free, retrato Blender). Cada uno con logros medibles (p.ej. FitApp: “Back-end Flask FastAPI con +100 usuarios activos”; retrato Blender: “Pipeline completo característico (modelado, texturizado, iluminación, render)”).
- **Educación:**
- UNAD – Ingeniero Multimedia (Agosto 2021 – Jul 2026). Indicar mención de *tesis TwinSight X500 / WebGL* (no “ARA Framework”). Resaltar materias de gráficos, software.
- UNAL – Ingeniería Electrónica (Marzo 2013 – Nov 2018, 8 semestres completados). Frase honesta: “Studied Electronic Engineering (80% of curriculum) at Universidad Nacional de Colombia” o “Completed advanced coursework in Electronic Engineering”.
- **Idiomas y herramientas:** Listado de idiomas (Español nativo, Inglés C1). Herramientas (Unity, Blender, Git, etc.).
- **Cursos:** Mencionar sin certificar como “Training en pipeline de personajes” o “3D animation” (p.ej. “CG Cookie: High-Fidelity Character Pipeline & Topology (self-study)”), si es de alguna relevancia.

Palabras clave ATS: Unity, Real-Time 3D, WebGL, C#, Python, Shader Graph, Performance Optimization, XR, Visualization, Technical Artist, Blender, GameDev, Automation, LangGraph, LLM, AI Tools, GitHub, LinkedIn, Portfolio.

Se desarrollarán **4 variantes de CV** focalizadas:

1. *Unity Technical Artist / Real-Time 3D:* Enfatizar proyectos TwinSight, Shaders, Blender optimization. Habil-

idades Unity centradas.

2. *Unity WebGL / Interactive 3D*: Poner TwinSight y habilidades web (URP, WASM, UI Toolkit). Mostrar portafolio web.
3. *XR / Simulación / Digital Twin*: Añadir palabras clave XR/VR, sim. Más énfasis en prototipos industriales, CAD.
4. *Python AI Tools*: Destacar ARA Framework, ai-news, habilidades Python/AI. Dejar TwinSight como secundario, enfocándose en LLM/multiagent (evaluar si combinar o mantener separado del main profile).

F.9 Banco de bullets CV

Ejemplos de viñetas cuantificables para CV (en inglés):

- **TwinSight X500 (Technical Visualization):**
 - “Developed Unity WebGL prototype for interactive 3D visualization of a Holybro X500 drone; optimized CAD assemblies from 6.5M to 95K triangles for real-time performance.”
 - “Implemented exploded view, cross-section, selection UI and multiple render modes (realistic, x-ray, blueprint, thermal) using Unity URP and Shader Graph.”
 - “Orchestrated academic evaluation with 12 participants: achieved SUS=91.9 and NASA-TLX raw=8.69 (3D) vs 19.89 (2D), demonstrating UX improvement.”
 - “Led asset optimization in Blender (retopology, baking) and pipeline from STEP CAD to low-poly game-ready models.”
- **ARA Framework (AI Research Automation):**
 - “Built a multi-agent Python system (LangGraph, LangChain) for automating literature search and report generation; integrated Semantic Scholar API and browser automation (Playwright/MarkItDown).”
 - “Designed agents for niche analysis, literature review, architecture drafting and content synthesis; validated with initial mock outputs.”
 - “Prepared the architecture and prototype pipeline (Redis, Groq LLaMA model) to demonstrate LLM orchestration in research workflows.”
- **AI News Aggregator:**
 - “Developed full-stack AI-driven news platform (FastAPI, React, PostgreSQL, Redis) to crawl and analyze tech news via LLM pipelines.”
 - “Implemented multi-agent pipeline for news scraping and sentiment analysis; tested MVP with [n] feed integrations.”
- **FitApp-Free (Fitness App):**
 - “Prototyped mobile fitness app backend (Flask/FastAPI) for calisthenics training; features user progress tracking, injury alerts, and adaptive routines.”
 - “Configured PostgreSQL data models for workouts and progress; integrated React frontend scaffolding.”
- **Blender Retrato Hiperrealista:**
 - “Modelled and rendered a high-fidelity human portrait in Blender following CG Cookie pipeline, demonstrating advanced topology, sculpting, texturing and lighting skills.”
 - “Produced wireframe topology charts, UV layout and shading breakdowns to show technical process; published case study on ArtStation.”
- **Generales:**
 - “Published and maintain GitHub repos for portfolio; cleaned/organized code (private→public) and curated README for recruiters.”
 - “Launched personal portfolio site (HTML/CSS/JS) with project demos, CV, and case studies to showcase work to international recruiters.”

Estos bullets deben adaptarse a cada versión del CV, usando verbos de acción y datos numéricos (por ejemplo, porcentaje de reducción de polígonos o métricas de usabilidad) cuando sea posible. Evitar el término “AI” sin contexto: en entrevistas se enfatizará que fue *asistencia* y su rol fue diseño/programación.

F.10 Estrategia de LinkedIn

Foto de perfil: Profesional, fondo neutral, rostro claramente visible. Si es posible, una imagen "técnica" (p.ej. frente a modelado 3D) pero preferiblemente primer plano amistoso/profesional. Evitar selfies informales.

Titular ("Headline"): Enfocado en palabras clave técnicas. Ejemplos:

- *Unity Technical Artist & Tools Dev – Unity, WebGL, C#.*
- *Real-Time 3D Developer (Unity/WebGL) – Technical Visualization.*
- *Technical Artist | Unity/WebGL | 3D Optimization.*
- *Multimedia Engineer / Unity Developer – IA & Python Tooling.*
- *Technical Artist – Blender, CAD-to-RealTime, AI-Driven Pipelines.*

Se darán ~15 opciones similares y se elegirá la más concisa/potente. Ejemplo final: **"Technical Artist & Real-Time 3D Developer (Unity, C#, WebGL) – 3D Visualization & AI Tooling"**.

Extracto/"About" (EN): 2–3 párrafos cortos. Debe ser equilibrado entre profesional y personal, "premium". Mencionar punto fuerte y metas: ej. "Multimedia Engineer with Unity & Blender expertise, passionate about transforming CAD data into interactive 3D experiences. Completed thesis "TwinSight X500": Unity WebGL drone visualization with 6.5M→95K triangle optimization. Seeking remote roles in Technical Art/Visualization to leverage my background in realtime graphics and AI-assisted tools. Spanish native, English fluent, open to global opportunities." En español: síntesis breve de misión ("ayudar a ingenieros a entender sus diseños 3D, construyendo interfaces web inmersivas..."), pero preferiblemente en inglés para coherencia internacional. No repetir CV/portafolio, sino contar una historia concisa del perfil, indicando disponibilidad remota ("based in Colombia, open to remote or EU relocation") y aspiraciones (ej. "dynamics roles requiring 3D dev & creative tech"). Evitar tono modesto, ir seguro de logros.

Experiencia / Proyectos en LinkedIn: Reescribir "Independent Technical Artist":

- Título: "Freelance Technical Artist / Unity Developer" (Mar 2024–Presente).
- Descripción: "Desarrollo herramientas automatizadas y optimización de activos 3D; integración de assets URP; creación de shaders (dissolve, hologramas, etc.). Gestión de pipeline Blender→Unity para proyectos de visualización. Destacar "TwinSight X500 (Tesis): Unity WebGL para visualización dron, optimización 99% de malla y UI técnica", y "ARA Framework: pipeline IA para investigación académica usando LangGraph". Incluir métricas breves.

Educación:

- UNAD Multimedia (2021–2026). Descripción: "Engineering multidisciplinary (3D, software, matemática) con énfasis en gráficos y realidad virtual; tesis final: *TwinSight X500, prototipo web 3D interactivo*."
- UNAL Electrónica (2013–2018, 80% curr.). Descripción: "Advanced coursework in Electronic Engineering (electronics, systems modeling, optimization). Systems simulation, física, lógica digital, etc."

Voluntariado ARCADE – "Student Researcher (Feb 2025–Presente): investigación en visualización 3D y XR en entornos educativos."

Sección "Proyectos": Incluir "TwinSight X500 – WebGL Drone Assembly" (descripción breve, e.g. "Unity WebGL app for interactive drone assembly visualization (optimización CAD, UI, cross-sections) – *GitHub/Proyecto*"). "ARA Framework – LLM Research Tool", "Blender Portrait – Character pipeline" (ligar a ArtStation/GitHub).

Destacados ("Featured"): Ordenar: Portafolio/Media principal (video demo o imágenes TwinSight), GitHub, ArtStation, publicaciones relevantes. TwinSight debe ser lo primero, luego ARA (si público), luego otros proyectos.

Skills (Aptitudes): Priorizar 7–10 más relevantes: *Unity, 3D Graphics, C#, WebGL, Technical Art, Python, Git, 3D Optimization, VR/AR, Shader Graph, Blender*. Reordenar para que aparezcan primeros. Añadir "B2B Contracts" o "Remote Work" en idiomas o intereses si aplica. El apartado de "Languages" ya está bien (Spanish, English).

Perfil/Apariencia: Banner de LinkedIn sugerido: Una composición profesional de renderizado 3D de un dron o escenas técnicas (ej. captura del TwinSight X500) con texto superpuesto como "Unity | Technical Artist | 3D Developer". Foto profesional (no cara al sol, etc.).

Actividad: Publicar contenido relevante periódicamente: estudios propios (por ejemplo, avances de TwinSight X500, infografías), artículos cortos (ej. "Optimizing CAD models for real-time: mi experiencia con dron X500"), re-publicar noticias de XR/técnica. Comentar en posts de Unity/Blender, participar en grupos de "Tech Art" o "Unity Developer" en LinkedIn, ArtStation. Esto mejora visibilidad.

Estrategia de palabras clave (recruiters): Incluir en resumen/títulos variantes bilingües: *Technical Artist, Desar-*

rollador 3D, Unity Developer, WebGL Engineer, XR Developer, Visualization Engineer, Herramientas de Producción IA, etc. El algoritmo de LinkedIn: asegurarse de mencionar “remote” y “Colombia” (o LatAm) en la sección “Información” para aparecer en búsquedas como “Unity developer remote Latin America”. Ejemplo booleano para reclutadores:

```
Bloque de código
("Unity Technical Artist" OR "3D Developer" OR "Interactive 3D Developer" OR "WebGL Developer" OR "XR Developer" OR "Digital
↳ Twin Developer") AND (remote OR Colombia OR LATAM OR remoto)
```

Esto ayuda a que reclutadores que buscan perfiles 3D/Unity/remote nos encuentren.

Disponibilidad geográfica en perfil: Mencionar “Based in Colombia (Remote only) / Willing to relocate to EU (esp. Germany/Portugal) if visa provided”, en la sección de “Intereses” o “Resumen” sin sonar pretencioso: por ejemplo, “Open to remote roles globally; Long-term EU mobility (via Portuguese citizenship in process)”. Esto semáforo: nada de afirmar “Ciudadano UE inminente” sin cumplir requisitos.

Otras: Evitar inconsistencia: corregir en LinkedIn actual que pone “Tesis: ARA Framework” si en realidad fue TwinSight. Usar sincronización: el título “Technical Artist & Tools Developer” está bien, acentuar “Optimization, WebGL” en la sección. Asegurarse de quitar proyectos irrelevantes o sacar el portafolio duplicado en Educación.

F.11 Estrategia de GitHub

- **Repositorios clave:** Definir 3–5 repos “estelares”. Poner en **pinned** (Bio de GitHub):
 1. *WebGL-Thesis-Proposal* (TwinSight X500) – principal. Limpieza: añadir README explicativo conciso, capturas, licencia. Incluir *Demo deploy* link (si posible host en GitHub Pages/WebGL).
 2. *ARA-Framework* – si se decide público. README claro: “Python LLM Orchestration for research”. Señalar que es proto/incompleto.
 3. *ai-news-aggregator* – si finaliza mínimamente, poner con demo/README si lo publica.
 4. *3D Character (Blender Portrait)* – usar ArtStation como medio y GitHub solo para wireframes etc.
 5. *GitHub-intro / Portfolio repos* – eliminar, archivar o combinar.
- **Público vs privado:** Convertir en público los proyectos más fuertes (TwinSight, proyectos de AI/pipeline). Mantener privado o archivar códigos sin valor reclutable (ej. proyectos obsoletos, antiguos), o señalarlos como **portfolio antiguo**. Poner notas README aclarando estado (beta, no mantenimiento).
- **README:** En cada repo, inicializar con un “recruiter-friendly” README: breve descripción, tecnología, rol propio (ej. “Lead developer and integrator”), demos/video embed, métricas clave (TiPs). Evitar jergas de nicho y destacar resultados/uso.
- **GitHub Pages:** Crear página portfolio vinculado a delarge95.github.io (por ejemplo con los proyectos destacados).
- **Etiquetado:** Crear etiquetas y temas relevantes (Unity, C#, WebGL, Python).
- **Actividad:** Contribuir a repos públicos/iniciativas Unity o CG (por visibilidad). Documentar algunos issues/respuestas en comunidades (StackOverflow).
- **Código limpio:** Borrar demos privados con información sensible, comprimir assets, etiquetar commits.
- **IA Asistida:** Si se ha usado ChatGPT en código, no ocultarlo pero dejar claro que el trabajo final fue suyo. En README, se puede mencionar “Auxiliary use of AI tools for code scaffolding; all design and logic by author.”

F.12 Estrategia de portafolio

- **Sitio portfolio (MVP):** Construir un sitio web sencillo (hosteable en GitHub Pages) que enlace a proyectos: *TwinSight X500* como caso estrella, con video demo breve (60–90s), capturas, descripción. Incluir *ARA Framework*, *AI-news* solo si se considera relevante, pero indicarlo como “herramientas internas” (secundario). *Blender Portrait* en sección Art/Graphics. Cada proyecto con resumen, rol, tecnologías, logros cuantificables (tomados de CV bullets).
- **Case studies:** Para cada proyecto principal (TwinSight, ARA, Blender), preparar página con:
- **Problema/desafío:** contexto (p.ej. necesidad de entender dron industrial en 3D).
- **Solución:** enfoque técnico (UX, tecnologías usadas).

- **Rol personal:** qué contribuciones propias se hicieron (enfatar partes críticas).
- **Resultado:** métricas o estado (objetivos logrados, validaciones).
- **Lecciones / Futuro:** mejoras pendientes o aprendizajes (muestra reflexión profesional).

Incluir screenshots (wireframe, editor Unity, UI, explosion, heatmap de performance, etc.), gráficos (reducción polígonos, resultados de SUS/TLX).

- **ArtStation:** Complementario. Cargar retrato Blender. Estructura técnica: imágenes del modelo wireframe/texture/bake, explicaciones breves en cada. Conectar a **Technical Art:** mostrar fluidez con Blender (no solo arte final). Possibly mostrar antes/después wireframe→render.
- **PDF/Deck:** Tener un PDF unificado de portfolio para enviar por email. Que incluya resumen de habilidades, proyectos clave y enlaces. A menudo reclutadores aprecian un “one-pager” visual. Mantener en inglés.
- **Demo reel/video:** Un video de 60–120s para TwinSight X500:
- **Guión:** abrir con contexto (“App to explore drone assembly in 3D”), mostrar interfaz UI seleccionable (click en partes), demonstration of explosion mode, cross-section, x-ray etc. Resaltar performance en móvil/web. Concluir con “Ver proyecto completo en GitHub/portafolio.”
- **Shots:** listamos escenas cortas: vista general del dron (rotando), click part (muestra panel info), explosion (animación separación de partes), modos visuales (cambiar a x-ray, thermal, blueprint), dispositivo móvil (mostrar que corre en web mobile).
- Añadir voz o texto superpuesto breve: “Interactive 3D visualization; CAD→real-time optimization; implemented in Unity WebGL.”

F.13 Company target list (ejemplos destacados)

Generar lista de ~100 posibles empleadores/agencias, priorizados por fit. Ejemplos por categoría (**resume**):

- **Videojuegos AAA/indie:** Ubisoft (FR/Global) – Technical Artist (remoto USA/FR), Microsoft Xbox Game Studios (US), Insomniac Games (US), Bungie (US), Cloud Imperium Games (Star Citizen), Unity Technologies (US), Epic Games (US), Tencent/Nexon (Asia).
- **Realidad Virtual / Simulación:** Meta (US, XR Research), Microsoft (Hololens), NVIDIA (Omniverse), Varjo (FI), Niantic (US), Magic Leap (US), Unity (iluminación).
- **Defensa/Espacio:** Boeing (US), Lockheed Martin (US), Northrop Grumman (US), Raytheon (US), Airbus Defence (EU), DLR (Alemania).
- **Industria / CAD / Automoción:** Siemens Digital Industries (DE), Dassault Systèmes (FR), ANSYS (US), Bentley Systems (US/UK), AutoDesk (US). Roles en visualización o digital twin.
- **Aeroespacial/Drones:** NASA (JPL/GSFC), Cessna/Textron, SpaceX (Motion Graphics, simulación), Holybro (EE.UU.).
- **Visualización médica/educativa:** Siemens Healthineers (DE), GE Healthcare (US), Philips (NL).
- **Agrícola / Energía:** John Deere (US), Schlumberger (US, AR/VR oilfield), ABB (Suiza/SE), Schneider Electric (FR).
- **Plataformas / Consultoría tecnológica:** Globant (ARG, oficinas en COL), Endava, Wipro, Accenture (gamedev/tech art divisions).
- **Startups creativas / AI:** MagicLeap, FormLabs, Cranial (AI demo), VR education startups.
- **Outsourcing LATAM:** UruIT (Col), PSL Corp (Col), Arkano (UY), Demium (ARG), Levvel (adapt tech in Colombia).
- **Gobierno/Educación:** Labs de VR en universidades (p.ej. ETH Zurich simulation lab, Fraunhofer IGD), proyectos financiamiento europeo.

Para cada empresa, idealmente se anotarían *remote likelihood* (p.ej. Unity/Epic/KLaber son fully remote friendly), salarios estimados, roles a aplicar (“Technical Artist”, “3D Dev”), idioma principal, etc. Esto requiere investigación puntual por empresa (carreras, Glassdoor reviews). En esta respuesta no es posible listar 100 con detalle citado, pero se recomienda al candidato crear una hoja de cálculo con criterios: cercanía industrial, historial de contratación remota, culturas de startup, etc.

F.14 Bolsas de empleo y comunidades

- **General / remotas:** LinkedIn (filtros remotos Unity), Indeed, Glassdoor, ZipRecruiter. Plataformas de empleo remoto: We Work Remotely, RemoteOK, Working Nomads, FlexJobs.
- **Tech art / Unity:** Unity Connect, Polycount Jobs (foros), ArtStation Jobs, Gamedev.net, CGSociety.
- **XR / AR/VR:** XR Bootcamp job board, Augmented Reality developer Slack, arvr.slack.com, XR Dive.
- **Digital twin / Industrial:** LinkedIn Grupos (Digital Twin Forum), Big Data & IoT Slack, Industry forums (Product Visualization Discords).
- **Freelance:** Upwork, Freelancer, Toptal (temporales solo), Gigster para devs, Cord, Malt (Europa).
- **LATAM específicas:** elempleo.com, compuTrabajo, Workana; comunidades BogotáTech, Medellín GDC; Tech recruitment firms nearshore (MuleSoft LATAM, etc).
- **Comunidades/Discord/Slack:** GameDev Underground (Discord), Unreal Engine Slack, Unity Developer Groups, Blender Artists Forum. LinkedIn grupos: "Technical Artists World", "Unity Developers", "XR/AR Professionals".

F.15 Sistema de aplicación y métricas

- **Rutina semanal:** Dedicar 2–3 horas diarias (20–25 semanales) a aplicaciones/alcance. Objetivo ~5-10 aplicaciones semanales (calidad > cantidad).
- **A/B/C categorías:** Categorizar vacantes:
 - *A (alta probabilidad):* Experiencia ideal, fit total.
 - *B (medio):* Falta experiencia formal, pero habilidades clave están.
 - *C (difícil):* Roles severos (senior AAA, necesitan portafolio top).

Priorizar A/B, aspirar a 50% customización personal.

- **Customización:** Adaptar CV+cover letter breve (3-4 líneas) a descripción del puesto: mencionar tecnologías clave del anuncio, ej. "Su oferta pide Unity + WebGL + C#: he liderado X proyecto con esas stack."
- **Herramientas:** Llevar un tracker en Notion/Google Sheets con columnas: Empresa, Rol, Fuente (link), Contacto (link perfil reclutador), Fecha aplica, Estado, Notas (adaptaciones, fecha follow-up). Analizar métricas mensuales (# apps, # entrevistas, % respuestas).
- **Outreach y seguimiento:** Conectar en LinkedIn con reclutadores y leads (ver scripts abajo). Seguir con 1–2 mensajes de seguimiento cada 10 días después de aplicar si no contestan.
- **Rechazos:** Establecer criterio: si oferta < USD 1.5k/mo neto o condiciones contractuales inestables (sin contrato claro), declinar con cortesía.
- **Balance:** Semana activa de aplicaciones + networking + mejora portafolio. No saturarse, mantener aprendizaje (cursos/idioma) paralelamente.

F.16 Plan 30/60/90 días (tentativo)

Días 1–30:

- Finalizar defensa de tesis.
- **CV y LinkedIn:** Crear base CV en inglés y variante específico Unity; reescribir LinkedIn según arriba; optimizar GitHub (limpieza repos, public repos finalizados).
- **Portafolio:** Lanzar sitio web simple (GitHub Pages) con TwinSight X500 y Avatar. Crear demo video breve TwinSight.
- **Redes:** Aumentar contactos LinkedIn de industrias target; enviar solicitudes con mensaje personalizado (ver scripts). Unirme a foros/Discord relevantes.
- **Setup administrativo:** Obtener RUT, Afiliaciones EPS/ARL, abrir cuenta COP+USD y vincular Payoneer/Wise. Configurar facturación (Factura electrónica DIAN).
- **Busqueda:** Preparar lista de empresas (+20 prioritarias), reclutadores de Unity/art tech, grupos de Slack. Postular a ofertas iniciales fáciles (revisión de portales/remotos).
- **Tracker:** Establecer en Notion/Sheets y revisar avances semanalmente.

- **Aprendizaje:** Comenzar curso o proyecto corto para cubrir gap (p.ej. React básico o optimización Unity).

Días 31–60:

- **Aplicaciones:** Aumentar número (10–15 por semana), aplicar a roles fuera de portafolio inicial. Personalizar cada solicitud.
- **Portafolio/Proyectos:** Subir mejoras en proyectos (TwinSight, ARA). Decidir si publicar ARA (limpieza final). Producir un caso de estudio corto para blog/LinkedIn sobre TwinSight o AI en CV.
- **Networking:** Realizar contactos directos con empresas (LinkedIn InMails, email), asistir a meetups virtuales/bootcamps de tech art. Pedir recomendaciones de profesores/mentores (script profesor).
- **Preparación entrevista:** Practicar preguntas comunes (Unity, gráficos, optimización). Preparar “pitch” de proyectos (TwinSight, ARA). Ensayar respuesta “¿Cuánto código fue tuyo vs IA?” (p.ej. “lideré la arquitectura e integración; la IA ayudó en esqueleto de código, pero implementé y probé cada módulo”).
- **Ajustes:** Recoger feedback de primeras entrevistas (si las hay) y mejorar CV/gabs. Ajustar estrategia de rol si es necesario (ej. mayor foco Python si Unity se resiste).
- **Documentación:** Crear plantillas de cover letters, CV, extracts LinkedIn por sector. Empezar a enviar saldos para Alemania/Portugal (idioma o doc, si aplica a largo plazo).

Días 61–90:

- **Escalar outreach:** Aplicar a vacantes más ambiciosas (niveles B/C), incluyendo empresas top (puede usar referidos). Hacer seguimiento intensivo a contactos creados.
- **Entrevistas:** Idealmente tener entrevistas técnicas y de cultura. Prepararse con portafolio digital (video, slides).
- **Negociación:** Investigar salarios específicos de ofertas obtenidas. Practicar respuestas a “salario esperado” (basado en research de salario medio, rango deseado, ajustes por remoto).
- **Ofertas:** Evaluar propuestas, priorizar aquellas con mejor remuneración y potencial. Negociar en base a rangos del mercado. Rechazar ofertas muy lejos de objetivo sin comprometer esperanza.
- **Capacitación continua:** Dedicarse a gap skills (React, Three.js, Shader Graph avanzado) según feedback de entrevistas.
- **Preparación EU:** Si en aparece opción alemana/portuguesa, planificar documentos. Mantener opciones abiertas.

F.17 Scripts de acercamiento (Inglés y Español)

(Plantillas básicas; personalizar con nombres/motivos específicos.)

- **Reclutador / Conexión LinkedIn (EN):**

“Hi [Nombre], I’m Alexander Woodcock, a Unity Technical Artist based in Colombia. I found your profile when searching for [role] opportunities at [Compañía]. With expertise in Unity/WebGL and CAD-to-3D visualization (see my TwinSight X500 thesis), I’m very interested in [empresa/puesto]. Could we connect? I’d love to discuss possible fits. Thank you!”

- **Líder técnico de Tech Art (EN):**

“Hello [Nombre], I’m a Technical Artist (Unity/WebGL) working on an interactive drone assembly prototype (see GitHub WebGL-Thesis-Proposal). I admire [empresa]’s work in [campo XR/simulación/etc.]. I’d appreciate 15 minutes to share how my Unity/C# & optimization skills could support your team’s projects.”

- **Fundador estudio creativo (ES):**

“Hola [Nombre], soy Alexander Woodcock, Ingeniero Multimedia especializado en Unity y visualización 3D (ej. prototipo web de dron en mi tesis). He visto el trabajo de [empresa] en [campo relevante] y me interesa sumarme como Technical Artist/Desarrollador 3D. ¿Podríamos conversar sobre posibles colaboraciones remotas? Adjunto mi portafolio.”

- **Director técnico Unity (EN):**

“Hi [Nombre], as a Senior Unity Dev at [empresa], you might be seeking strong Technical Artists. I’m a passionate Unity/WebGL developer (8+ semesters engineering) with a portfolio in 3D optimization (6.5M→95K polys). Could we discuss any opportunities at [empresa]? I’m targeting remote or contractor roles.”

- **Empresa XR/Simulación (EN):**

“Good day, [Nombre]. I’m contacting you about potential roles in XR/simulation at [empresa]. My background includes Unity WebGL, C# tools dev, and AI-assisted pipelines. At [proyecto], I built a 3D drone visualization with real-time interactivity. Would love to contribute to [empresa]’s simulation projects.”

- **Reclutador videojuego (ES):**

“Hola [Nombre], actualmente freelance en Unity/C# con experiencia en herramientas de optimización 3D. Vi que [empresa] busca Technical Artist para juegos en Unity. Aunque mi pasión es también la simulación, tengo portafolio con shaders y UI Unity. Me encantaría explorar oportunidades de contrato remoto.”

- **Follow-up Postulación (EN):**

“Hello [Nombre], I wanted to follow up on my application for [puesto] submitted on [fecha]. I remain very interested in this role. My Unity and 3D expertise (TwinSight prototype) align well with the job requirements. Please let me know if you need any further info. Thanks again for considering me.”

- **Aplicación especulativa (EN):**

“Hi [Nombre], I’m Alexander, a Unity Technical Artist in Colombia. I’m reaching out speculatively as I admire [empresa]’s work. I have experience in real-time 3D visualization and AI tool development. If there are upcoming Unity/3D positions (even contractor), I’d be keen to apply. Thank you for your time.”

- **Solicitud de referencia (EN):**

“Hi [Nombre], I’m looking to expand my network in the [XR/Technical Art] field. As someone respected in the community, could you kindly refer me to any roles or contacts? I’d be glad to support or informally discuss how my skills (Unity, WebGL, Python) could fit.”

- **Profesor / Asesor (ES):**

“Estimado [Profesor], le escribo para solicitarle una recomendación/introducción a contactos en la industria 3D. Como exalumno de [UNAD/UNAL], apreciaría su orientación sobre cómo aprovechar mis proyectos (p.ej. tesis TwinSight X500) para oportunidades internacionales.”

- **Cliente freelance anterior (ES):**

“Hola [Nombre], espero esté bien. Quería pedirle un testimonio breve sobre el proyecto que hicimos juntos (optimización web). Estoy enfocando mi portafolio en trabajos 3D y un comentario suyo sería valioso.”

- **Expectativas salariales (EN):**

“Based on my research and the responsibilities of this position, I expect a salary in the range of [ej. \ \$70k–\ \$80k] annually. Of course, I’m open to discussion depending on the overall package.” (Luego alinear con objetivos: \ \$1,500->\ \$18k mínimo, según rol; \ \$3k->\ \$36k; \ \$6k->\ \$72k)

- **Disponibilidad remota desde Colombia (ES):**

“Disponibilidad completa remota desde Colombia; constituiría mi condición habitual de trabajo. Cuento con RUT y facturación oficial. Estoy abierto a contratos directos o B2B.”

- **Configuración contractor (ES):**

“Como contratista independiente, puedo facturar legalmente servicios exportados (IVA 0%) y cuento con RUT. Prefiero pagos vía transferencia internacional (Wise/PayPal). ¿La empresa utiliza EOR/Deel o prefiere un contrato directo? Estoy preparado para acordar términos transparentes.”

- **Movilidad UE futura (EN):**

“I plan to pursue Portuguese citizenship (ancestry) in the next 2–3 years, which may facilitate EU relocation. For now, I seek fully remote or international contractor roles that don’t require current EU work authorization.”

- **Preguntas beneficios/contrato (EN):**

“Could you clarify if the position is contractor or W2? And what benefits or payment frequency are offered? Also, is there an expected full-time schedule (hours) for this contractor role?”

- **Rechazo oferta inestable (EN):**

“Thank you for the offer. Unfortunately, the compensation (USD X/month) is below my minimum requirement (USD 1500). Additionally, the role’s scope and contract terms are not yet clear. I prefer stable, well-defined positions. I appreciate your time and wish you the best.”

- **Ausencia de experiencia formal (EN):**

“While I lack formal studio experience, my portfolio projects (ex: TwinSight X500) demonstrate practical Unity and 3D pipeline skills. I am a quick learner with strong fundamentals (ING. Electrónica 80% cursado) ready to apply in a professional context.”

- **Uso de IA en proyectos (EN):**

“I used AI tools as coding assistants (to draft boilerplate) but personally architected, integrated, and debugged all code. My contributions include the core algorithms and systems design. For example, I guided the AI generation and refined every script.”

- **“¿Por qué contratarlo?” (ES):**

“Poseo un perfil único que combina habilidades técnicas (Unity, C#, optimización 3D, Python) con experiencia en proyectos end-to-end. Mi tesis TwinSight validó con métricas la mejora en usabilidad. Aunque sin trayectoria empresarial clásica, ya demostré liderazgo técnico y creatividad aplicada. Estoy motivado y preparado para agregar valor real desde el día uno.”

F.18 Preparación para entrevistas y negociación

Entrevistas técnicas: Esperar preguntas sobre Unity/WebGL (cómo optimizar performance, cómo funcionan shaders, ejemplos de pipelines). Reforzar bases de gráficos, estructuras de datos 3D, experiencia con UI Toolkit. Practicar resolución de problemas en vivo (p.ej. pseudocódigo). En cuanto a IA/Scripted: decir que integra AI, pero no se espera que copie conocimiento exacto.

Preguntas comunes: “Háblame de un proyecto en que lideraste [TwinSight]. ¿Cómo resolviste X reto de optimización? ¿Cómo calibraste el sistema UX?”; “¿Cómo explicarías tu uso de IA en el desarrollo?”; “Expectativas salariales/visado.”; “¿Por qué Unity vs Unreal?”; “¿Cómo te mantienes actualizado en técnicas de render?”. Preparar respuestas basadas en hechos (mencionar métricas de TwinSight, frameworks).

Negociación salarial: Llevar rangos de referencia (ej. “Unity Tech Artist senior en EEUU ~ \\$100k”), comunicar ambición pero flexibilidad. Empezar por arriba del mínimo aspirado (~\\$2k en Colombia convertidos) y ajustar. Tomar en cuenta beneficios adicionales (equity, remote stipend). Discutir modalidad (contratista vs empleado) y cobertura de impuestos (explicar cómo tributará en Colombia, esperar no deducir impuestos del pago).

Checklist de oferta: Además de salario, verificar:

- Tipo de contrato (W2, 1099, EOR).
- Días vacacionales, flexibilidad horario.
- Cobertura de herramientas/licencias.
- Responsabilidades vs título (nivel real del puesto).
- Opciones de crecimiento (ojos a 12–24 meses).

F.19 Estrategias de búsqueda final

- **Conservadora:** Enfocarse en ofertas seguras remotas en empresas medianas (ej. outsourcing game studios, consultoras tecnológicas) con salarios modestos pero estables. Prepararse con CV pulido para entrevistas típicas.
- **Equilibrada:** Mezclar solicitudes a startups internacionales remotas de nicho (visualización, IA) y empresas establecidas (Unity, Autodesk). Usar portafolio activo, EOR/Deel si es necesario para asegurar pago. Mantener networking activo y preparación de habilidades (tal vez Inglés perfeccionado).
- **Agresiva:** Target empresas grandes y elevadas (Google AR, Meta, Boeing VR, etc), networkear directamente con altos cargos, presionarse a aplicar a ~20 empresas top. Mejorar más habilidades (Shader Graph avanzado, Three.js). Uso intensivo de LinkedIn y referidos.

Recomendación: Combinación equilibrada: aplicar tanto en puestos “probables” (nearshore, mediana tech, \36k/año nivel) como “aspiracionales” (grandes, \72k/año). Mientras se gana confianza y experiencia de entrevistas, ir escalando metas. Balancear búsquedas según prioridades salariales y viabilidad legal.

F.20 Brechas de habilidades (ROI)

Priorizadas por impacto en empleabilidad:

- **Unity Performance & Profiling:** Conocer Profiler, optimizar GPU/CPU (milestones visuales).
- **Unity WebGL específica:** Build/WebGL size limits, compression, Addressables para RAM.
- **Shaders (HLSL avanzado):** Más allá de Shader Graph, escribir shaders manualmente.
- **Three.js/WebGPU:** Diversificar a Web 3D (por si aplican a roles WebGL puro).
- **React y full-stack básico:** Mejorar React en ARA/ai-news, entender REST/DB.
- **XR básico:** Fundamentos AR/VR (tracking, SteamVR/ARKit).
- **Estructura C#:** Arquitectura limpia (MVVM, patterns) para soporte a escalas grandes.
- **FastAPI/Backend:** Pulir FastAPI para la ruta Python.
- **LangGraph/LLM Aplicaciones:** Entender mejor orquestación multi-agent (LangChain nuevo).
- **Blender polishing:** Esculpir/retopología/nody vs linear workflow.
- **Videography:** Edición básica (Premiere), storyboarding para demo.
- **Escritura técnica:** Redactar documentación de proyectos y CV fuerte.
- **Idiomas:** inglés (presentations), alemán/portugués básico (entrevistas introductorias).
- **Gestión de proyectos Agile:** Familiarizarse con JIRA, Scrum (para roles corporativos).

Cada skill a abordar según necesidad detectada por entrevistas y ofertas: cursos online, tutoriales, prácticas rápidas.

F.21 Conclusión y pasos siguientes

Alexander debe ejecutar este plan de manera disciplinada. En resumen:

- **Posicionamiento:** Presentarse como Technical Artist Unity/WebGL, poniendo énfasis en proyectos concretos (TwinSight X500).
- **Outreach:** Utilizar LinkedIn proactivamente, impulsar su visibilidad en comunidades especializadas, y construir una red de contactos.
- **Herramientas:** Establecer formalidades contractuales (RUT, facturación), y familiarizarse con EOR/plataformas (Deel, Payoneer) para agilizar la contratación.
- **Metas cortas:** Dentro de 3 meses debe tener entrevistas con empresas internacionales, preparando bases para una oferta remota.
- **Metas largas:** Consolidar un empleo que le pague >USD 3k/mes al año, y planificar residencia europea (matrimonio) como respaldo. Seguir mejorando portfolio con nuevos demos y aprendizajes técnicos.

Finalmente, revisar periódicamente el progreso: ajustar roles objetivo según respuestas del mercado, optimizar CV/LinkedIn tras feedback, y mantener actitud analítica (“ingenieril”) en cada paso. Con una estrategia fundamentada en datos actuales y centrada en sus fortalezas técnicas, Alexander puede maximizar sus oportunidades de un empleo internacional bien remunerado en 3–6 meses.

Appendix G

Auditoría por Secciones del Informe de Estrategia de Empleo

Fuente: Research/deep-research-report_01_structure.md

Seccion: research

Estado: research_snapshot

Rol: Research / estudio de mercado

Lineas originales: 469

Ultima modificacion: 2026-06-11 12:19

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Modulo incluido completo para revision documental.

Contenido completo depurado

G.0.1 1. Resumen Ejecutivo

- **Contenido actual:** Describe al candidato y metas salariales (USD 1.5k/3k/6k), menciona brevemente perfil (Technical Artist, multimedia, simulación) y resume las estrategias clave (investigación de mercado, posicionamiento, red de contactos, plan 30/60/90, negociación).
- **Faltantes / Correcciones:**
- **Cifras con fuente:** Incluir referencias actuales (2024–2026) para salarios y demanda (por ejemplo, Glassdoor o reportes de industria) en lugar de afirmaciones sin cita.
- **Prioridades claras:** Resumir al final las acciones inmediatas con datos (ej. “Optimizar CV/LinkedIn, iniciar contactos en EEUU/EU, metas de entrevistas”).
- **Detalle de objetivos:** Explicitar plazos y métricas (e.g. “en 3 meses >X entrevistas”).
- **Vincular proyectos:** Mencionar brevemente proyectos clave (TwinSight X500, ARA) como evidencia del perfil.
- **Lenguaje objetivo:** Evitar frases vagas o aspiracionales sin soporte (p.ej. “estrategia combinada” necesita detalle concreto).

G.0.2 2. Diagnóstico de la posición de mercado actual

- **Contenido actual:** Enumera experiencia (proyectos académicos/personal), habilidades técnicas (Unity, Blender, Python, etc.), idiomas, ubicación, situación legal y metas salariales con rangos y comentarios. Incluye resumen final del perfil y metas.
- **Faltantes / Correcciones:**

- **Validar datos salariales:** Verificar con fuentes actuales salarios en EEUU/UE para Technical Artist y roles similares. Corregir rangos de USD 6.7k–10.8k mensuales según datos (Glassdoor, PayScale) para diferentes niveles.
- **Oferta vs demanda:** Añadir referencias de demanda de cada rol (ej. informes de empleo Unity/XR) para justificar etiquetas “Alta/Media”.
- **Proyectos relevantes:** Incluir que el candidato completó tesis sobre visualización técnica (TwinSight) y proyectos de IA (ARA) como experiencia práctica. Añadir su experiencia freelance (sitios web) como experiencia relevante.
- **Estudios incompletos:** Explicar claramente educación: “Ingeniería Multimedia (UNAD) – tesis finalizada, en espera de defensa” y “Carrera de Ingeniería Electrónica (UNAL) 80% cursada”. Recomendar terminología (e.g. “Advanced coursework in Electronic Engineering”).
- **Idiomas:** Aclarar nivel exacto: Inglés C1 (certificado posible con exámenes, pruebas realizadas?), evaluar si incluir habilidades de lectura de manuales técnicos en alemán o portugués básico.
- **Aspectos legales y logísticos:** Precisar que como no EU, necesita contrato internacional; resumir procesos (sumario de Fe de Vida, DIAN, RUT) con referencia a fuentes oficiales.
- **Datos de mercado:** Añadir tendencias (p.ej. aumento de roles VR/visualización técnica en 2025) con enlaces a estudios o reportes recientes.
- **Competencia:** Mencionar brevemente perfil de competidores (otros junior/medios Unity tech artists) para calibrar ambición salarial.
- **Conclusión diagnóstica:** Reforzar que el perfil es junior pero técnicamente sólido, con gap en experiencia formal; recalcar plan de subir escalones salariales mediante proyectos y certificación.

G.0.3 3. Declaración de posicionamiento recomendada

- **Contenido actual:** Propuesta de enunciado principal en inglés y variantes por rol (Technical Artist Unity, 3D interactivo, XR/Digital Twin, Python/AI).
- **Faltantes / Correcciones:**
- **Enunciado clave:** Refinar el tagline único. Asegurar que incluya habilidades distintivas: “*Unity Technical Artist & Real-Time 3D Developer (Blender/URP, Shader Graph), especializado en optimización CAD-to-RT y visualización técnica interactiva con IA.*” (por ejemplo).
- **Coherencia de idioma:** Decidir si usar español o inglés según público objetivo. LinkedIn y CV profesional irían en inglés.
- **Enfatizar Blender/CAD:** Añadir Blender y optimización CAD explícitamente en el posicionamiento, dado su uso en TwinSight.
- **Comparación de roles:** Fundamentar la elección de roles prioritarios en evidencia: e.g. número de vacantes Unity Tech Artist vs Game TA. Incluir roles adicionales (p.ej. CAD-to-Realtime Technical Artist explícito, Tools Developer) según mercado.
- **Ajuste del enunciado:** Alinear con habilidades claves (C#, Python), pero no confundir al ATS; quizá mencionar IA/ML como secundario (apoyando desarrollo).
- **Evaluar singulares vs combinación:** Decidir si separar “Technical Artist” y “3D Developer” en dos perfiles o unirlos. Analizar qué títulos atraen más reclutadores (usar Google Jobs o LinkedIn para frecuencia de títulos).
- **Coherencia con CV/LinkedIn:** Garantizar que el enunciado coincida con la declaración en resumen profesional del CV y titular de LinkedIn.
- **Eliminar ambigüedades:** Eliminar dobles sentidos (“destrezas en Python para producción” podría explicarse mejor, p.ej. “Automatización de pipelines con Python/IA”).

G.0.4 4. Matriz de priorización de roles

- **Contenido actual:** Tabla con 7 roles (Unity TA, Real-Time/WebGL Dev, XR/Simulación, Vis Tec, Game TA, Tools Dev, Python/AI Tools, Full-stack secundario) con demanda, ajuste perfil, salario, comentarios. Se concluye priorizar roles Unity/real-time, luego XR/industrial, evitar sólo games o full-stack.
- **Faltantes / Correcciones:**
- **Datos cuantitativos:** Sustentar “Demanda Alta/Media” con datos reales (p.ej. cantidad de ofertas en LinkedIn/Indeed para cada rol). Añadir referencias o al menos estimaciones de portales de empleo.
- **Nuevos roles:** Incluir “CAD-to-Realtime Technical Artist” como rol separado si es un nicho, y “Interactive 3D/WebGL Developer” como rol específico (actual lista mezcla con Unity TA y XR).

- **Remoto vs on-site:** Añadir columna sobre viabilidad remota del rol (ej. Unity TA remote es común, Game TA menos).
- **Freelance/contract:** Incluir columna de viabilidad contrac_tor (p.ej. Python/AI Tools alta, dado proyectos cortos).
- **Localidad:** Notar qué roles demandan relocalización (XR en fabricantes, Visualización mediana).
- **Competencia:** Mencionar nivel de competencia: ej. Game TA alta competencia, lo que baja prioridad.
- **Fuente de ajuste:** Explicar cómo se calificó “ajuste al perfil” (ej. puntaje según skills).
- **Reordenar prioridades:** Analizar si “Digital Twin” podría ser rol separado o incluido en “Simulación Industrial”.
- **Detalles de salario:** Agregar rango salarial esperado por rol comparando EEUU/EU vs Colombia, con cita (como en tabla salarial).
- **Feedback de candidatos similares:** Ver si otros técnicos en roles similares publican experiencias o usan título específico (foros de Unity o StackOverflow Careers).

G.0.5 5. Tabla de priorización de mercados

- **Contenido actual:** Tabla con regiones (EEUU/Canadá, Europa Occidental, Escandinavia, Sudeste Asiático, LATAM remota, España/Portugal, Alemania/Europa Centro, India/Middle East, Remoto global) con demanda, salario estimado y comentarios.
- **Faltantes / Correcciones:**
- **Completar filas incompletas:** La fila “Sudeste Asiático” aparece cortada (“por evid”) sin datos claros. Incluir países relevantes (India, Singapur) y rangos reales.
- **Agregar Australia/Nueva Zelanda:** Omitidos en la tabla; agregarlos con demanda medio-alta y salarios (Sydney/ANZ tech).
- **Claridad en LATAM remoto:** Diferenciar mercado local (Colombia, etc.) de empresas LATAM que contratan remotos (p.ej. Globant). Indicar salarios típicos (USD 1-3k para junior LATAM remoto).
- **Salario y fuente:** Para cada región, incluir rangos salariales actualizados con referencias (reportes salary, e.g. PayScale por país). Ejemplo: USA/Can: junior Unity TA ~\$70k/año (Glassdoor).
- **Competencia lingüística:** Notar requerimientos: e.g. en NL/Nordics idioma inglés suele bastar, en DE requeriría alemán.
- **Remoto global:** Específico sobre fully-remote companies (GitLab, etc. pagan por nivel global; incluir si ofrecen contratos B2B).
- **Formato:** Asegurar consistencia en formato (unas filas tienen “USD ≈ (junior–mid)”, otras no completan).
- **EOR/Contratación:** Mencionar para cada región la facilidad de contratación B2B o necesidad de EOR.
- **Movilidad futura:** Indicar para España/Portugal y Alemania cómo influye la potencial ciudadanía en 2 años. P.ej., España/Portugal permite residencia con EU pass.
- **Fuentes:** Citar rankings globales de oportunidades remotas (p.ej. reporte Stack Overflow 2025 sobre trabajos remotos).
- **Comentarios precisos:** Completar comentarios truncados (p.ej. añada período “mercado remoto: We Work Remotely...” a comentarios breves).

G.0.6 6. Tabla de benchmarking salarial

- **Contenido actual:** Tabla por rol y mercado (EE.UU., UE, Contratista Col) con rangos en USD. Se mencionan datos de Glassdoor para Tech Artist y XR.
- **Faltantes / Correcciones:**
- **Fuentes exactas:** Para cada cifra, incluir fuente específica (por ejemplo, Glassdoor o Levels.fyi). Indicar si rango es junior/mid-level.
- **Distinción empleado vs contratista:** La columna “Contratista B2B Col” necesita fundamentarse en pagos freelance. Agregar comparación con salario local de ingeniería.
- **Salario neto vs bruto:** Especificar si USD son brutos anuales/mesuales. Aclarar que USD mensuales mencionados suponen ~USD 36–144k/año según nivel.
- **Rangos más realistas:** Verificar si US/UE dado junior-mid son esos rangos (parecen altos). Ajustar con datos reales (Glassdoor muestra ~\$80–120k/año = ~\$6.7–10k mensuales para Technical Artist senior).
- **Otros roles:** Incluir “Interactive 3D Developer” con WebGL (podría equipararse a “Frontend 3D Dev” en unas cifras).

- **Regionalización:** Podría incluir “Remoto LatAm” como columna separada.
- **Formato de fila:** La fila “3D WebGL/Interactive Dev” está truncada; terminar la explicación.
- **Contexto local:** Mencionar rango de la meta inicial (USD 1.5k) respecto a salarios locales en Colombia (p.ej. COP \$5–6M).
- **Actualización 2026:** Buscar publicaciones recientes sobre salarios remotos tech (ej. reportes de Deel/Remote).
- **Comentarios adicionales:** Explicar por qué roles de “AI/Tools” pueden superar a tech art en startups.
- **Presentación clara:** Tal vez dividir en dos tablas (US vs UE) para facilitar lectura.

G.0.7 7. Análisis de factibilidad de remoto desde Colombia

- **Contenido actual:** Descripción del contrato B2B (facturación, impuestos, seguridad social) y opciones de pago internacional (Wise, PayPal), afiliaciones (EPS, ARL), EOR vs directo.
- **Faltantes / Correcciones:**
- **Actualización normativa:** Verificar cambios recientes en DIAN sobre facturación electrónica, retenciones y cotizaciones para independientes.
- **Fuentes oficiales:** Agregar enlaces a DIAN o libros tributarios para sustentar los procesos (por ejemplo, instructivo DIAN sobre exportación de servicios).
- **Alternativas de pago:** Incluir soluciones modernas: plataformas EOR (Deel, Remofirst), bancos internacionales (Mercantil, Bancolombia corresponsal), servicios como Payoneer, Bitwage.
- **Trámites clave:** Especificar pasos concretos con plazos (e.g. obtener RUT en 1 semana, afiliación ARL en X días).
- **Impuestos:** Detallar cómo se calcula el impuesto de renta de un contratista (tarifa máxima ~39%) y declarar facturas.
- **Seguridad social:** Confirmar porcentajes (EPS ~12.5%, ARL según riesgo, etc.) y plazo de pago mensual.
- **Afiliaciones E.U.:** Mencionar no necesaria, pero plan para seguro de salud si viaja o se muda (e.g. EHIC temporal).
- **Visados digitales:** Analizar si programas de visa digital nomad existen para colombianos (por ejemplo, Georgia o Portugal citizen).
- **Checklist:** Incluir lista de verificación: RUT, Cámara de comercio (si aplica), cuenta USD, firma electrónica para facturación.
- **Ejemplos / casos:** Referir estudios o guías de freelancers colombianos exportando servicios de TI.
- **Riesgos:** Advertir sobre fluctuación COP/USD e implicaciones en ingreso.

G.0.8 8. Estrategia futura UE (Alemania/Portugal)

- **Contenido actual:** Enumera opciones: casarse con pareja alemana (permiso de residencia), obtener pasaporte portugués (ley sefardí), otras visas UE. Se menciona no asumir privilegios antes de cumplir requisitos.
- **Faltantes / Correcciones:**
- **Leyes precisas:** Citar normativa: proceso de residencia para cónyuge en Alemania (siempre y cuando casados, idioma mínimo B1); leyes portugués sefardí (tiempo de espera ~2 años completando trámite) con fuente oficial.
- **Tiempos y requisitos:** Aclarar plazos aproximados: e.g. “en 24 meses se puede solicitar ciudadanía portuguesa si reúne documentos (certificado sefardí, antecedentes penales)”.
- **Visas de trabajo alternas:** Investigar visas tech de Portugal/España (p.ej. startup visa), Alemania (EU Blue Card: requiere título y X salario).
- **Idioma:** Incluir análisis de rentabilidad de aprender alemán vs portugués (basado en demanda de trabajo; por ej. Portugal tiene mercado IT en inglés, Alemania menos para no UE).
- **Mercado local DE/PT:** Mencionar oportunidades específicas (empresas que valoran portugués/EU passport).
- **Estrategia paso a paso:** Sugerir ruta: (1) conseguir empleo remoto estable, (2) tramitar ciudadanía portuguesa antes de emigrar, (3) casarse para visa/residencia DE si surge oportunidad.
- **Referencias oficiales:** Enlazar información de gobiernos sobre reunificación familiar DE y ciudadanía PT.
- **Riesgos:** Señalar que cambios políticos pueden afectar (ej. ley sefardí está sujeta a cambios), no confiar solo en promesas futuras.

- **Impacto en búsqueda:** Explicar cómo esto influye en enfocarse en roles remotos ahora sin asumir cambio inmediato de estatus legal.

G.0.9 9. Estrategia de CV

- **Contenido actual:** Instrucciones para CV en inglés ATS-friendly. Se sugiere estructura (Encabezado, Resumen, Habilidades, Experiencia por proyectos, Educación, Idiomas, Cursos). Menciona brevemente proyectos TwinSight y ARA en experiencia.
- **Faltantes / Correcciones:**
- **Formato preciso:** Confirmar 1 página, con diseño claro (tipografía legible, uso moderado de negrita). Referir guías ATS para alineación de texto.
- **Términos en inglés:** Traducir secciones a inglés (actuales están mezcladas). Ej: “Summary” en lugar de “Resumen profesional”.
- **Detalles de encabezado:** Asegurar inclusión de LinkedIn/GitHub/Portfolio links activos.
- **Resumen profesional:** Ejemplo de párrafo concreto en inglés que resuma logros (“Completed Master’s thesis on 3D drone visualization...” etc.) en 2-3 líneas.
- **Experiencia por proyectos:** Listar proyectos como posiciones (ej. “Independent Technical Artist”). Incluir fechas tentativas (Mar 2024–Present) para proyectos clave.
- **Educación:** Adaptar nomenclatura internacional: “UNAD – B.S. in Multimedia Engineering (expected 2026)”. Para UNAL: “Electronic Engineering coursework (80% completed)”.
- **Cursos relevantes:** Mencionar en inglés los cursos relevantes (sin certificado) bajo sección “Professional Development” o “Courses”. E.g. “Completed CGCookie High-Fidelity Character Pipeline (self-study)”.
- **Palabras clave ATS:** Incluir más variantes: “Interactive 3D, Visual Computing, GPU Optimization” para cubrir más búsquedas.
- **Logros cuantificables:** Asegurar que cada bullet muestre impacto o métrica (porcentaje, tiempo ahorrado) como en TwinSight bullets.
- **Orden:** Organizar experiencias desde TwinSight (tesis) luego ARA, luego proyectos menores.
- **Sin jergas:** Evitar términos coloquiales; usar “3D asset optimization”, “interactive UI”, etc.
- **Sinónimos:** Garantizar se mencionan roles objetivo (“Unity Developer”, “Technical Artist”) repetidamente.
- **Error de tesis:** Asegurar que CV no describa erróneamente la tesis (debe ser TwinSight, no ARA).
- **Revisión bilingüe:** Aunque CV será en inglés, comprobar que el inglés sea preciso (por ejemplo, no “obtained possibility”).

G.0.10 10. Estructura completa del CV

- **Análisis:** Falta detallar la estructura final con secciones y orden.
- **Pendientes:**
- Diseñar esquema definitivo de CV en inglés de 1 página con secciones: Encabezado, Perfil, Habilidades técnicas, Experiencia relevante (proyectos), Educación, Idiomas, Cursos.
- Definir título profesional preciso (ej. “Technical Artist & 3D Developer”).
- Establecer secciones específicas: “Projects” en lugar de Experiencia laboral tradicional (pues es junior).
- Incluir íconos o logos (si lo permite ATS) para LinkedIn/GitHub/portafolio.
- Verificar formato: márgenes, uso de viñetas, consistencia.
- Asegurar que la línea de disponibilidad (“Full-time, willing to relocate”) esté clara.
- Crear ejemplo de contenido breve para cada sección (en inglés) para guiar la redacción final.

G.0.11 11. Variantes de CV (4 perfiles)

- **Contenido actual:** Se menciona la necesidad de 4 variantes y su enfoque por rol (Unity TA, WebGL, XR/Digital Twin, Python/AI).
- **Faltantes / Correcciones:**
- **Definir diferencias clave:** Para cada variante, especificar qué mover hacia arriba/abajo y qué enfatizar. Ej.:

1. *Unity TA/3D:* poner experiencia TwinSight arriba, resaltar Unity, Shaders, Blender.

2. *WebGL/Interactive 3D*: destacar TwinSight con énfasis en WebGL, URP, UI Toolkit; incluir cualquier experiencia web (FastAPI quizás).
3. *XR/Simulación/Digital Twin*: mencionar ARA menos; resaltar prototipos industriales, XR, Bluetooth frameworks (aunque no listados aún); integrar el retrato Blender como prueba de habilidad 3D.
4. *Python/AI Tools*: Priorizar ARA Framework y AI News; posicionar proyectos de IA; mencionar brevemente habilidades web como secundaria.

- **Alineación de keywords**: Adaptar los títulos (“Technical Artist” vs “Developer”) y habilidades según el rol.
- **Personalización del summary**: Cambiar el resumen de perfil para cada rol destacando la meta (p.ej. “Experienced in interactive visualization...” vs “Skilled in Python ML pipelines...”).
- **Orden de proyectos**: Variar orden de los proyectos en Experiencia para resaltar el más relevante al rol.
- **Uso de plantilla**: Crear plantilla base y ajustar manualmente cada variante para mantener coherencia.
- **Validar ATS**: Probar cada variante con filtros simples (p.ej. pasándolo por un sitio que detecte keywords).

G.0.12 12. Banco de viñetas para CV

- **Contenido actual**: Varias viñetas cuantificadas para TwinSight, ARA, FitApp y Retrato Blender.
- **Faltantes / Correcciones**:
- **Completar vacíos**: Incluir viñetas para el proyecto *AI News Aggregator*. Ejemplo: “Developed full-stack AI news platform (FastAPI, React, PostgreSQL) for automated content aggregation and analysis.”
- **Uniformidad de idioma**: Convertir las viñetas en inglés o español consistente (preferiblemente inglés para CV internacional). Actualmente hay mezcla. Ej. “Launched personal portfolio site...” debe traducirse a inglés.
- **Proyectos extra**: Añadir viñetas para experiencias laborales previas (campamento YMCA, call center) si relevantes (p.ej. “Improved English proficiency by teaching activities...”).
- **Errores**: Corregir typos (“fitapp” vs “FitApp”, “AI News-aggregator”), asegurar consistencia (p. ej. unidades “12 participants” vs “12 participants; SUS=91.88”).
- **Asignación específica**: Etiquetar viñetas para cada CV variante (al pie o con notas) para fácil uso.
- **Lecciones aprendidas**: Incluir al menos una viñeta general que destaque la autodirección/soporte (por ej. “Self-directed learner with X hours of online training” si se desea).
- **Validar métricas**: Revisar que las cifras sean correctas (Triángulos, SUS scores) y citar fuente de estudio (tesis).

G.0.13 13. Estrategia completa de LinkedIn

- **Contenido actual**: Sugerencias de foto, headline (varias opciones), extracto “About”, experiencia (gaps señalados), banner, actividad, keywords, boolean string ejemplo, menciones geográficas, errores actuales (tesis).
- **Faltantes / Correcciones**:
- **Titulares**: Generar las 15 opciones requeridas (todas en inglés, sin caracteres especiales) e identificar la mejor con base en claridad y keywords. Ejemplos a refinar: “Technical Artist – Unity/WebGL & 3D Visualization” etc. Usar estudios de LinkedIn (Capgemini estudia headliners con más clics).
- **Headline final**: Elegir la opción que combine enfoque técnico y palabras clave. Justificar elección.
- **Extracto/About (en inglés)**: Redactar borrador profesional de 2-3 párrafos, que equilibre perfil técnico y metas. Debe incluir: su background multimedia, pasión por visualizar datos 3D, proyectos clave y búsqueda de roles remotos. (Evitar jerga, en tono positivo/confidente.)
- **Experiencia actualizada**: Reescribir “Independent Technical Artist” en LinkedIn con viñetas sintéticas (en inglés) de TwinSight y ARA, utilizando bullets preparadas. Borrar errores (tesis ARA vs TwinSight). Añadir voluntariado ARCADE e idiomas.
- **Proyectos/Featured**: Crear secciones de “Featured” para TwinSight (con enlace GitHub/demo), ARA, Retrato. Ordenarlos según relevancia. Indicar qué añadir (videos, capturas).
- **Educación**: Arreglar descripción: “Thesis: TwinSight X500 interactive 3D visualization (pending defense)” en UNAD; “Electronic Eng – 80% coursework” en UNAL. No mencionar ARA si no es tesis formal.
- **Orden de habilidades**: Reordenar skills destacando Unity, C#, Python, Blender, real-time 3D al inicio.
- **Banner conceptual**: Diseñar idea (e.g. collage de drone wireframe + Unity interface) con texto sutil “Technical Artist – Unity & 3D”.
- **Foto**: Recordar usar fondo neutro, buen encuadre, vestimenta profesional-casual (evitar selfies).

- **Actividad:** Planificar publicar ~1/semana: reflexiones de proyecto (TwinSight), avances de cursos, artículos relevancia (Tech Art). Unirse/participar en grupos LinkedIn de Unity y XR.
- **Palabras clave (recruiters):** Expandir lista: “Interactive 3D Developer, CAD Visualization, Digital Twin, XR Simulation, Technical Artist Unity, Python Automation”. Incluir sinónimos en inglés.
- **Boolean searches:** Incluir 2–3 strings adicionales (e.g. (“Unity” OR “Unreal”) AND “Remote” AND (colombia OR latam)). Verificar sintaxis.
- **Disponibilidad:** En “About” mencionar claramente “Open to 100% remote roles based in Colombia” y “long-term EU mobility plan”. Evitar frases ambiguas (“soon EU citizen”).
- **Consistencia:** Corregir o eliminar información contradictoria (p.ej. portada de LinkedIn actual, proyectos en educación).
- **Idioma:** Decidir el idioma principal (en inglés para audiencia internacional). Si el perfil está en inglés, toda la info (actividades, comentarios) en inglés.

G.0.14 14. Auditoría y plan de acción de GitHub

- **Contenido actual:** Recomendaciones para repos clave (TwinSight, ARA, ai-news, blender portrait), públicos vs privados, README recruiter-friendly, GitHub Pages, etiquetado, actividad, limpieza de código, manejo de AI.
- **Faltantes / Correcciones:**
- **Repositorios faltantes:** Revisar repos mencionados en plan pero no listados: *deus-ex-machina*, *skills-introduction-to-github*, otros en GitHub (buscar privados pendientes). Decidir status (publicar/archivar).
- **Priorizar destacados:** Confirmar 3–5 repos con mayor impacto: p.ej. TwinSight (portfolio principal), ARA Framework, Blender portrait (si contiene procesos), *ai-news-aggregator* (si es mínimo viable), *3Dweb*.
- **Readmes:** Escribir README detallado para TwinSight y ARA con enfoque en uso (no solo código). Incluir capturas/gifs si es posible y enlace a demo WebGL. Para repos con código incompleto, aclarar “proyecto en desarrollo”.
- **GitHub Pages:** Generar un sitio estático (delarge95.github.io) que sirva de portfolio mínimo, enlazando proyectos.
- **Estandarizar etiquetas:** Usar tags (archivos .github/release.toml o en descripción) que incluyan TechArt, Unity, WebGL, Blender, Python.
- **Organización:** Mover repos irrelevantes (ej. repositorios personales básicos) a una organización diferente o marcarlos como “archived”.
- **Actividad social:** Contribuir a otros repos (como issues/respuestas en Unity o StackOverflow) y subir al menos un issue o PR público (demostrar colaboración).
- **Herramientas de análisis:** Usar GitHub Copilot o Advanced Security insights para mejorar la calidad del código y documentar cambios.
- **Cumplimiento legal:** Confirmar que no hay violación de licencias de terceros en los repos. Limpiar cualquier asset con copyright (usar propias texturas).
- **Metadata:** En profile de GitHub, incluir bio breve con rol deseado y ubicación.

G.0.15 15. Estrategia de portafolio

- **Contenido actual:** Sugerencias: sitio web MVP, case studies (TwinSight, ARA, Retrato), uso de ArtStation, PDF one-pager, demo reel/video de TwinSight con guión y shots.
- **Faltantes / Correcciones:**
- **Sitio portfolio:** Definir tecnología (HTML/CSS/JS estático, React, o Hugo) y hosting (GitHub Pages, Netlify). Prototipo fácil de actualizar.
- **Estructura web:** Incluir páginas separadas: Home (resumen breve + enlace a LinkedIn/GitHub), Portafolio (videos/demos), Blog/Artículos (1-2 posts técnicos), Contacto.
- **Case studies completas:** Para cada proyecto principal, detallar: título, rol, tecnologías, desafíos, soluciones, resultados con métricas (ej. “Reducción a 95k polígonos – validado con 12 usuarios”). Incluir referencias a contexto académico (TwinSight es tesis UNAD).
- **AI News Aggregator:** Si se publica (ver punto 18), prever sección propia o mencionar en “Otros proyectos relevantes”.
- **FitApp-Free:** Evaluar su lugar: si se hace público, incluir como proyecto web completo; si no, mencionarlo brevemente en CV pero evitar en portafolio principal.

- **ArtStation:** Completar perfil: subir retrato Blender con breakdown (wireframe, texturas). Posiblemente crear “Technical breakdown” para cada artwork (topología, shaders, grooming). Incluir al menos 3-5 piezas (pueden ser variaciones del retrato).
- **PDF Portfolio:** Crear un PDF/slide deck breve (4-6 slides) que contenga el resumen de TwinSight y ARA, listo para enviar por email. Diseñarlo profesional (InDesign, Canva).
- **Demo reel/video:** Planificar grabación de la demo: escribir guión en inglés, preparar gráficos. Definir formato (Full HD, narración en voz o texto).
- **Organización de imágenes:** Crear repositorios de capturas/screenshots (alta calidad) para usar en CV, LinkedIn, portafolio. Etiquetarlos (TwinSight-assembly, ARA-architecture, etc).
- **Accesibilidad:** Considerar subtítulos o descripciones para el video/demo.
- **Mantenimiento:** Plan de actualizar el portafolio 2 veces al año con proyectos nuevos.

G.0.16 16. Plantilla de caso de estudio – TwinSight X500

- **Faltantes / Requerido:** Crear un outline completo del caso de estudio:
- **Antecedentes:** Describir el problema (dificultad de entender manuales CAD de un dron). Contexto: “Proyecto de tesis UNAD 2026”.
- **Objetivos:** Qué se quería lograr (mejorar comprensión, interactividad, pruebas de usabilidad).
- **Tecnologías:** Unity WebGL, C#, URP, Blender (retopología), UI Toolkit, mediciones SUS/NASA-TLX.
- **Rol del candidato:** Detallar tareas propias (programación, UI, optimización, modelado de hélices, coordinación del estudio).
- **Features clave:** Lista de funcionalidades (exhibición ensamblaje, vistas explosivas, cross-section, modos visuales X-ray, blueprint, térmico).
- **Proceso:** Resumen de pipeline (de CAD/STEP a modelos optimizados, uso de Blender y Unity).
- **Resultados:** Métricas concretas (reducción de 6.5M a 95k tris, SUS 91.88 vs 59 de base, NASA-TLX 8.69 vs 19.89).
- **Impacto:** Qué demostró este proyecto (validación de usabilidad, potencial comercial).
- **Lecciones aprendidas:** Qué se mejoraría (p.ej. mayor optimización WebGL, integrar feedback más, thermal dinámico).
- **Medios:** Incluir diagramas o imágenes (bloque de Unity, esquemas de uso, foto del dron).
- **Palabras clave ATS:** Unity, WebGL, CAD, 3D Visualization, Technical Artist.
- **Evidencias:** Enlaces a demo WebGL si existe, screenshots, link repo GitHub.

G.0.17 17. Plantilla de caso de estudio – ARA Framework

- **Faltantes / Requerido:** Desarrollar similar estructura:
- **Problema:** Investigación académica consume mucho tiempo (búsqueda bibliográfica, organización).
- **Solución:** Sistema multi-agente en Python (LangGraph, LangChain, Redis) para automatizar generación de reportes.
- **Rol propio:** Diseñó la arquitectura de agentes (nicho, investigación, síntesis), desarrolló la integración con APIs (Semantic Scholar, MarkItDown), coordinó prototipos y pruebas.
- **Tecnologías:** Python 3.12, LangGraph, Groq LLaMA 3.3, LangChain, Redis, Playwright, Semantic Scholar API.
- **Estado:** Mencionar que es prototipo (“proof-of-concept”), nivel de completitud actual (p.ej. agentes 1–3 funcionales, se planea mejorar).
- **Resultados iniciales:** Ejemplo de output (resúmenes de papers, nicho propuesto) para demostrar funcionalidad.
- **Desafíos:** Velocidad, precisión del LLM, volumen de datos.
- **Lecciones:** Necesidad de curación humana, escalabilidad de agentes, preguntas futuras (fine-tuning LLMs).
- **Imágenes:** Diagrama de arquitectura multi-agente, capturas del flujo de trabajo.
- **Visibilidad:** Link a código y breve demo (si se crea un ejemplo público).

G.0.18 18. Recomendación – AI News Aggregator

- **Faltantes / Requerido:** Decidir si incluir o descartar:
- **Estado actual:** El proyecto está privado y requiere ajustes. Determinar si vale la pena completarlo (posible demostrar habilidad full-stack con AI).
- **Viabilidad:** Evaluar tiempo necesario para dejarlo publicable (quizá un MVP con fuente de noticias pública y análisis básico).
- **Posicionamiento:** Si se publica, presentarlo como “AI-driven content platform” en CV y portafolio secundario. Resaltar tecnología Python/React/AI.
- **Completar readme/demo:** Si se hace público, escribir README con casos de uso (e.g. “agrega noticias de IA de múltiples fuentes y las analiza”).
- **Incluir en portafolio:** Solo como “otro proyecto”, no primer plano. Si no, archivar privado y no mencionarlo.
- **CV:** Incluir 1-2 bullets destacando habilidades tecnológicas (FastAPI, React, extracción automática).

G.0.19 19. Recomendación – FitApp-Free

- **Faltantes / Requerido:** Evaluar relevancia:
- **Objetivo:** Es proyecto personal de backend/fitness. Determinar si respalda el enfoque técnico-art o distrae.
- **Si continuar:** Podría mostrar habilidades de app móvil y lógica de programación, útil para full-stack secundario. Subirlo parcialmente (screens/demo de UI).
- **Si no:** Mantener privado y no incluir en los principales materiales.
- **Menciones:** Si se completa, agregar bullet CV: “Desarrollé backend de app de fitness con seguimiento de progreso de entrenamiento en Python/Flask y PostgreSQL.”
- **Portafolio:** Solo incluir en sección “Otros proyectos de software” con breve descripción.

G.0.20 20. Recomendación – Retrato hiperrealista en Blender

- **Contenido actual:** Se incluyen viñetas.
- **Faltantes / Correcciones:**
- **Mostrar en ArtStation:** Subir la imagen final y desglosar técnica: etapas de modelado, sculpt, texturizado, grooming, shading y luces.
- **Case study:** En portafolio, incluir sección de “Character Pipeline”: describir flujo de trabajo de topología, UVs, NODES usados en shading, post-procesado.
- **Conectar con Technical Art:** Explicar que demuestra “pipeline de personaje de alta calidad” (competencia en retopología, materiales complejos).
- **Capturas:** Incluir wireframes y mapas de normales/materiales en portafolio/CV (para destacar habilidad técnica).
- **CV:** Usar bullets ya escritos, asegurarse de traducir a inglés consistente.

G.0.21 21. Plan de demo reel/video (60–120s)

- **Faltantes / Requerido:**
- **Guión detallado:** Escribir guión en inglés: introducción breve (texto/audio): objetivo del proyecto; destacar 2-3 demos clave.
- **Secuencia de escenas:** Listar shots con tiempos aproximados:
 1. *Contexto:* Texto “Interactive 3D visualization for drone assembly” + vista general drone (5s).
 2. *UI/Interactividad:* Mostrar selección de pieza y panel de info (5s).
 3. *Explotar:* Exploded view del dron (5s).
 4. *Clipping:* Section view (5s).
 5. *Modos Visuales:* Animación cambiando a X-Ray/Blueprint/Termal (15s total).
 6. *Mobile/WebGL:* Interfaz en navegador móvil (5s).
 7. *Estadísticas/UI final:* Mostrar datos SUS/TLX o gráfico rápido (5s).
- **Narración/subtítulos:** Preparar frases clave: “Optimized drone CAD data for real-time Unity visualization...” (inglés claro).

- **Herramientas:** Plan de grabación (Unity Recorder para video, edición en Premiere o DaVinci).
- **Formato:** Exportar en 1080p, considerar versión de 60s vertical (mobile).

G.0.22 22. Lista de capturas/screenshots técnicos

- **Faltantes / Requerido:**
- **TwinSight:** Capturas de: dron completo, dron wireframe, Unity editor mostrando jerarquía, ventana de inspector de pieza, modos (X-Ray, Blueprint), gráficos de datos (tabla reducciones).
- **Blender:** Imágenes de: modelo high-poly vs retopology, UV layout, wireframe sobre render, materiales.
- **ARA Framework:** Diagrama de agentes, captura de un output de texto sintetizado (si disponible).
- **Generales:** Logo del proyecto (si se hace) o un diagrama 3D representativo.
- **Formatos:** Guardar al menos 1920×1080, nombrar archivos descriptivos (e.g. TwinSight_exploded.png).
- **Uso:** Incluir estas imágenes en el portfolio web y en documentos PDF.

G.0.23 23. Banco de palabras clave

- **Faltantes / Requerido:**
- **Keywords inglés:** Unity, C#, Real-Time 3D, WebGL, URP, Interactive Visualization, Technical Artist, GPU Optimization, Shader Graph, Blender, CAD, Digital Twin, XR/AR/VR, LLM, AI Tools, Python, FastAPI, .NET, IGRA.
- **Keywords español (para adaptaciones locales):** Artista Técnico, 3D Interactivo, Visualización Técnica, Optimización 3D, Programador Unity, Inteligencia Artificial, Automación Python.
- **Sinónimos y variantes:** E.g. en inglés: “3D Visualization Developer”, “VR Developer”, “Simulation Engineer”.
- **Palabras de impacto:** “Optimization”, “Real-Time Rendering”, “Low-Poly Workflow”, “Interactive UI”, “Data Visualization”, “Digital Twin”.
- **Aspectos blandos:** “Problem-solving”, “Agile development” (para sistemas corporativos).
- **Tecnologías secundarias:** “Git/GitHub”, “React”, “FastAPI”, “SQL”, “Blender Modeling”, “Shader Programming”.
- **Instrucciones:** Usar este banco para CV/LinkedIn/GitHub descripciones y títulos de repositorios para ser encontrado por reclutadores.

G.0.24 24. Lista de empresas/objetivos (≥100)

- **Contenido actual:** Ejemplos puntuales por categoría (juegos, VR, defensa, etc.) en párrafos.
- **Faltantes / Correcciones:**
- **Ampliar lista:** Elaborar tabla con al menos 100 empresas, incluyendo:
- **Videojuegos (AAA/indie):** Ubisoft, Electronic Arts, Nintendo, CD Projekt, Insomniac, Valve, Wargaming, Larian Studios, Naughty Dog, etc.
- **XR/Simulación:** Unity Technologies, Epic Games, Oculus (Meta), Magic Leap, Varjo, Nvidia Omniverse, Mitsubishi Electric (simulación VR), Boeing XR, Airbus A&D.
- **Digital Twins/Industria:** Siemens Digital Industries, Bosch, PTC (Creo/ThingWorx), ANSYS, Siemens Energy (gems), Dassault Systèmes, AutoDesk.
- **Drons/Aeroespacial:** DJI (China), Lockheed Martin (división drones), Boeing, SpaceX (grafismo VR intern), Holybro (asistente), General Dynamics.
- **MedTech/Educación:** Siemens Healthineers, Philips, GE Healthcare (ARX labs), Boston Scientific (VR training), univers. Harvard VR lab, Unity Reflect para arquitectura.
- **Energía/Defensa:** Raytheon, Northrop Grumman (Simulators), DARPA XR programs, Boeing, Leonardo (Italia defense XR).
- **Consultorías TI:** Accenture (immersive tech), Globant, ThoughtWorks, Endava, Softtek, PSL (Medellín), UruIT, TCS, Wipro (digital twin projects).
- **Outsourcing LATAM:** Grupo RutaN, Enola Labs (CO), Levvel, Belatrix (LATAM offices).
- **Tech Startups:** Magic Leap, Formlabs, Hugging Face (AI), Neuralink (neurovisualización), startups de digital twin o medtech en Europa.
- **Empresas remotas:** GitLab, Automattic, InVision (tech artists), empresas fully-remote de XR.

- **Columnas requeridas:** País/región, tipo (sector), roles buscados, % remoto (basado en ofertas actuales), idioma requerido, sueldo estimado, link carrera, Contactos de LinkedIn, estrategia (aplicación directa, referidos).
- **Priorización:** Marcar empresas top-20 high-fit (p.ej. Unity, Siemens) con mayor prioridad.
- **Fuentes:** Usar LinkedIn/Glassdoor/jsereros para confirmar presencia de roles Unity/3D en esas compañías.
- **Notas legales:** Indicar para cada si permiten contratos B2B (p.ej. consultorías sí, product companies menos).
- **Diversificar:** Incluir al menos 10 empresas por cada región objetivo (EEUU, UE, LATAM, Asia remota).
- **Reclutadores/contactos:** Anotar head hunters especializados (p.ej. Arteria en España, Robert Half Tech).

G.0.25 25. Bolsas de empleo y comunidades

- **Contenido actual:** Listado de portales generales (LinkedIn, Indeed, WWR), sitios específicos (Unity Connect, Polycount, ArtStation, Gamedev, CGSociety), XR (XR Bootcamp jobs, Slack), digital twin (grupos LinkedIn, Slack), freelance (Upwork, Toptal, Malt), LATAM locales, comunidades Discord y LinkedIn.
- **Faltantes / Correcciones:**
- **Fuentes adicionales:** Añadir portales como Glassdoor, StackOverflow Jobs (para remote), Angellist (startups remotas).
- **Comunidades:** Listar Discords/Slack específicos: “Tech Art Slack”, “VR/AR Association”, “Digital Twin Forum”. Grupos de Facebook o Meetups (e.g. XR development en Berlín).
- **LATAM:** Incluir comunidades hispanas: “Game Developers Colombia”, “Desarrolladores Unity LatAm”, foros de Spanish Game Dev.
- **Canales de reclutamiento:** Referir newsletters y accounts de Twitter/LinkedIn para ofertas (p.ej. @RemoteOk, @XRJobs).
- **Freelance específico:** Añadir sitios enfocados a tech (Gun.io, Hired, Freelancer.com, Turing.com).
- **Palabra clave en portales:** Ejemplo de filtros recomendados (ej. “Unity AND remote AND Latin America” en Remotive).
- **Eventos y conferencias:** Mencionar participación en conferencias virtuales (GDC, SIGGRAPH virtual) y cómo obtener redes.
- **Frecuencia de seguimiento:** Sugerir revisión periódica de estas fuentes (diariamente, semanalmente).

G.0.26 26. Sistema de aplicación y métricas

- **Contenido actual:** Propuesta de rutina semanal (20–25h), apps por semana (5–10), categorías A/B/C de vacantes, personalización ~50%, uso de tracker, seguimiento a reclutadores después de 10 días, criterios de rechazo (salario < USD1.5k, condiciones inestables).
- **Faltantes / Correcciones:**
- **Criterios A/B/C detallados:** Definir ejemplos concretos para cada categoría, basados en los requisitos vs perfil (por ej. A: Unity roles remotos con 3D; C: juegos AAA si no hay experiencia).
- **Herramientas de automatización:** Mencionar uso de herramientas (e.g. LinkedIn Easy Apply, mail merge en e-mails) sin perder personalización.
- **Plantillas:** Preparar plantillas básicas de correo de aplicación/carta de presentación adaptables (en inglés).
- **KPIs:** Especificar métricas de seguimiento: #postulaciones, #entrevistas, ratio de respuesta, #contactos nuevos. Revisión semanal y ajuste de estrategia según tasa de éxito.
- **Diversificación:** Asegurar incluir aplicaciones internas (LinkedIn) y externas (portales especializados).
- **Freelance como alternativa:** Mantener presencia en plataformas freelance como Plan B si las aplicaciones no rinden los frutos esperados.
- **Calendario:** Sugerir días dedicados a cada tarea: e.g. L-Mi aplicaciones, J-V networking/cursos.
- **Criterios de seguimiento:** Ej. si no hay respuesta en 2 semanas, enviar mensaje de seguimiento.
- **Revisión de ofertas:** Incluir chequeo de elementos críticos (contratista vs empleado, timings) antes de aceptar entrevista.

G.0.27 27. Plantilla de seguimiento (Notion/Google Sheets)

- **Faltantes / Requerido:**

- **Columnas recomendadas:** Empresa, Puesto, Link oferta, Contacto reclutador, Fecha de aplicación, Estado (Aplicada, Entrevista, Oferta, Rechazada), Fecha de última acción, Acción siguiente programada, Fuente (LinkedIn/Portal), Tipo A/B/C, Observaciones.
- **Fórmulas/KPIs:** Incluir cálculos automáticos: total apps, porcentaje de respuesta, % entrevistas.
- **Etiquetas/Colores:** Por prioridad A/B/C (semáforo) para visualizar foco.
- **Ejemplo inicial:** Proveer un par de filas ejemplo ya completas (p.ej. aplicación a Unity y a empresa de simulación).
- **Instrucciones de uso:** Breve guía en la misma hoja sobre cómo actualizar tras cada avance.

G.0.28 28. Rutina semanal de búsqueda

- **Contenido actual:** Mencionado implícitamente en el sistema de aplicación.
- **Faltantes / Correcciones:**
- **Horario recomendado:** Proponer bloques diarios: mañanas (1-2h enviar apps), tardes (1h networking/LinkedIn), fines de semana (revisar resultados, cursos).
- **Bloques de aprendizaje:** Incluir tiempo fijo semanal para cursos o práctica (p.ej. 5-7h/semana).
- **Networking periódico:** Programa semanal para contactarse con X personas nuevas (p.ej. martes y jueves dedicar 30 min a LinkedIn).
- **Revisión y ajuste:** Una sesión semanal (p.ej. domingo) para analizar métricas, actualizar tracker y plan del siguiente bloque de 7 días.
- **Balance personal:** Incluir descansos y ejercicio (recomendado 30 min/día) para evitar burnout en periodo de búsqueda.
- **Flexibilidad:** Ajustar rutina según respuestas; estar listo para cambiar enfoque (ej. más tiempo a aplicaciones si falta respuesta).

G.0.29 29. Plan tentativo 30/60/90 días

- **Contenido actual:** Detallado en el informe.
- **Faltantes / Correcciones:**
- **Más métricas:** Definir objetivos específicos (e.g. “30 días: CV listo + 10 contactos LinkedIn; 60 días: 15 apps enviadas, 3 entrevistas técnicas”).
- **Responsabilidades diarias:** Asignar tareas diarias concretas (p.ej. lunes revisar portales, miércoles editar video, viernes networking).
- **Revisiones periódicas:** Programar reuniones de revisión propia cada 2 semanas para evaluar progresos.
- **Plan de contingencia:** Qué hacer si no se logran metas intermedias (p.ej. reevaluar CV o roles si tras 30 días no hay entrevistas).
- **Preparación de recursos:** Asegurar terminar tesis y obtener títulos/constancias necesarios antes de aplicar.
- **Actualización continua:** Incluir “mantenerse informado de noticias del sector” (leer blogs, papers) para entrevistas.

G.0.30 30. Scripts de acercamiento (Inglés/Español)

- **Contenido actual:** Plantillas para varios escenarios (reclutador, lead técnico, fundador, director Unity, rechazo oferta, explicar falta experiencia, AI, etc.) tanto en EN como ES.
- **Faltantes / Correcciones:**
- **Cobertura total:** Incluir plantillas para los casos faltantes solicitados:
- *Game studio recruiter (EN/ES)* (similar a tech art, pero indicando pasión por juegos).
- *Aplicación especulativa (EN)* a empresa sin oferta específica (hablar de interés en posibles posiciones).
- *Petición de recomendación/Referral (EN)* a contacto interno (“I see you work at X, I'd love if you could refer me for Unity roles...”).
- *Profesor/mentor (ES/EN)* solicitando carta de recomendación (“Spanish: Profesor X, agradezco su orientación...”).
- *Seguimiento después de la entrevista (EN)* (“Thank you for your time, reaffirm interés, contact details”).

- **Disponibilidad remota desde Colombia (ES/EN)** (“Disponible full-time remoto desde Colombia, con equipo propio”).
- **Personalización:** Asegurar que cada script tenga variables [Nombre, Empresa] claras.
- **Claridad en salario y contrato:** Mantener plantillas sobre negociación (ej. “Would you clarify if this is full-time or contract?”).
- **Traducciones exactas:** Revisar que los mensajes en español sean formales y correctos gramaticalmente.
- **Tono profesional:** Ajustar scripts para que no suenen genéricos ni presumidos; ejemplos: agradecer oportunidad, mostrar empatía al responder una negativa.
- **Guía de uso:** Acompañar cada plantilla con nota breve de cuándo aplicarla.

G.0.31 31. Preparación para entrevistas

- **Contenido actual:** Breve mención de posibles preguntas técnicas Unity/WebGL y comunes. Tácticas de uso de IA discutidas. Estrategia de negociación salarial esbozada. Checklist de oferta.
- **Faltantes / Correcciones:**
- **Preguntas técnicas específicas:** Listar preguntas probables (e.g. “Explicar ciclo de evento en Unity”, “Cómo optimizar un Shader Graph”, “¿Qué es LOD en modelos 3D?”). Preparar respuestas concisas.
- **Preguntas de comportamiento:** Practicar el famoso “Cuéntame sobre ti”, “Fortalezas y debilidades”, “¿Por qué esta empresa?”. Tener ejemplos con el perfil.
- **Presentación del portfolio:** Preparar versión demo/tour de TwinSight para mostrar pantalla. Ensayar narración breve del proyecto.
- **Exposición de proyectos:** Preparar “storytelling” de cada proyecto clave usando método STAR (Situación, Tarea, Acción, Resultado).
- **Idiomas:** Practicar entrevista en inglés, posiblemente con un amigo o tutor para evaluar fluidez.
- **Técnicas de entrevista virtual:** Verificar setup (cámara, fondo, micrófono) y conexión antes de cada entrevista online.
- **Preguntas al reclutador:** Preparar de antemano qué preguntar (“team composition?”, “roadmap del producto?”, “oportunidad de capacitación?”).
- **Conocimiento de la empresa:** Investigar 1-2 noticias recientes o productos de la empresa entrevistante para demostrar interés.
- **Código/ejercicio:** Anticipar si puede haber test de código (pocas chances en TechArt, pero tal vez tomar un desafío Unity simple).

G.0.32 32. Estrategia de negociación

- **Contenido actual:** Menciona llevar rangos de referencia, considerar modalidad (W2/1099/EOR), beneficios adicionales (vacaciones, flexibilidad).
- **Faltantes / Correcciones:**
- **Rango objetivo:** Establecer meta salarial mínima (USD 3k) y justificativo (mercado remoto Senior en LATAM rondan X).
- **Costos adicionales:** Incluir cálculo de deducciones sociales en Colombia para ofertar un neto aceptable.
- **Beneficios no monetarios:** Mencionar negociar días de vacaciones, patrocinios de formación, equipo de trabajo provisto, cobertura de salud privada (muy relevante para contratistas).
- **Contratación internacional:** Tener en cuenta propuestas de EOR: negociar pagos mensuales en USD, uso de moneda estable.
- **Estrategias:** Ej. “price anchoring”: dar primero un rango en que 3k+ sea aceptable. Practicar respuestas tipo “Mi expectativa está alrededor de [rango calculado] en este mercado.”
- **Documentación:** Llevar comparativas salariales impresas (Glassdoor/PayScale) si se entrevista presencial, para sustentar cifras.
- **Títulos de cargos:** Asegurar que al nivel de oferta corresponde su rol (p.ej. que “Engineer” suene a Senior).
- **Plan B:** Si no alcanza el salario deseado, negociar otros beneficios (bono por objetivo, stock, entrenamiento).

G.0.33 33. Lista de señales de alerta (Red Flags)

- **Faltantes / Requerido:**

- **Condiciones contractuales:** No aceptar cláusulas confusas, pago exclusivamente en retenciones (alejarse de “no hay contrato claro”).
- **Firma de NDA excesivo:** Cuidado con requisitos de confidencialidad que bloqueen publicar trabajos profesionales.
- **Plazos de pago dudosos:** Empresas que tardan en pagar o pagan en moneda local devaluada.
- **Alta rotación:** Compañías con reclamos de despidos masivos recientes.
- **Entrevistas vagas:** Entrevistadores sin claridad en el rol o cronograma (puede indicar improvisación).
- **Oferta demasiado baja:** Cualquier oferta < USD 1.5k (meta inicial) por rol técnico completo.
- **Comunicaciones extrañamente informales:** Reclutadores pidiendo entrevistas solo por WhatsApp o sin confirmar detalles.
- **Falta de recorrido legal:** Ofertas en efectivo sin facturación posible.
- **Expertise desconocida:** Empresas no verificables o con mala reputación online.
- **Exceso de tests:** Pedir tarea enorme no pagada antes de negociar salario.
- **Contratos poco claros:** Indicar incongruencias entre lo prometido (beneficios, horario) y lo escrito.

G.0.34 34. Estrategias de búsqueda (conservadora, equilibrada, agresiva)

- **Contenido actual:** Breve descripción de tres enfoques y recomendación de combinar estrategias equilibradas.
- **Faltantes / Correcciones:**
- **Estrategia conservadora:** Aplicar principalmente a empresas “seguras” y ofertas estables (consultoras, empresas consolidadas) que ofrecen trabajo remoto estándar. Enfocar en roles con fuerte correspondencia al perfil. Prepararse para aceptar salario inicial modesto mientras gana experiencia.
- **Estrategia equilibrada:** Mezclar candidaturas a medianas/pequeñas startups internacionales (posibles sueldos más altos, pero riesgo medio), y consultoras/estudios de tecnología en LATAM (salario menor, estabilidad, facilidad legal). Mantener portfolio en inglés y atención al reclutamiento global.
- **Estrategia agresiva:** Postular a roles en empresas top-tier (Google AR/Unity, NVIDIA, Meta VR) aun sin experiencia directa; networking intensivo con headhunters; contratación como consultor independiente con tarifas más altas. Requiere high volume de outreach y preparación para rechazos.
- **Comparativa:** Tabla breve de pros/contras para cada (salario vs riesgo vs tiempo de contratación).
- **Recomendación final:** Explicar por qué se adopta un mix equilibrado: balance entre oportunidades asequibles y aspiraciones altas, asegurando avance en cada etapa.
- **Asignación de recursos:** Recomendaciones numéricas (ej. 50% aplicaciones enfoque conservador, 30% equilibrado, 20% agresivo) con revisión trimestral.

G.0.35 35. Brechas de habilidades (ROI) – roadmap

- **Contenido actual:** Lista priorizada de habilidades a desarrollar (Unity profiling, WebGL optimización, Shader/HLSL, Three.js, React, XR, C# patterns, Python FastAPI, LangGraph, Blender, video/editing, escritura técnica, idiomas, Agile).
- **Faltantes / Correcciones:**
- **Categorizar por prioridad:** Asignar corto (<3 meses), mediano (3–12 meses) y largo plazo (>12m) a cada habilidad.
- **Integración con cursos:** Relacionar cada ítem con cursos concretos (e.g. “Unity WebGL – curso oficial Unity o tutorial en línea”, “Shader/HLSL – Coursera”).
- **Medición de progreso:** Sugerir pequeños proyectos o pruebas para validar cada habilidad (p.ej. mejorar perfil de un demo, subir un shader propio a GitHub).
- **Idiomas específico:** Para alemán/portugués, fijar meta mínima (hablar con fluidez básica en 1 año, con apps semanales).
- **Herramientas complementarias:** Git avanzado (branching, CI/CD) y metodologías ágiles (uso básico de JIRA/Trello).
- **Soft skills:** Añadir comunicación en inglés fluida y trabajo en equipo virtual.
- **Fuentes:** Citar cursos recomendados para cada área (platzi, Coursera, Rebelway).
- **Planificación:** Crear plan de estudio semanal con fechas objetivo (ej. “Terminar Shader Graph avanzado en 4 semanas”).

G.0.36 36. Conclusiones y próximos pasos

- **Contenido actual:** Resume brevemente el plan: posicionamiento como Unity TA, enfoque en Networking, formalidades contractuales, metas 3–6 meses.
- **Faltantes / Correcciones:**
- **Resumen ejecutivo:** Destacar 3 hallazgos clave: (1) “Perfil técnico sólido pero junior, necesita visibilidad y experiencia práctica”, (2) “Objetivo salarial de 3k–6k requiere inicio en roles remotos bien remunerados con proyección”, (3) “Estrategia clara en CV/LinkedIn + networking es vital”.
- **Metas específicas:** Reiterar objetivos concretos (e.g. “Conseguir al menos 3 ofertas de entrevista en 3 meses, alcanzar USD 3k en 6-12 meses”).
- **Iteración continua:** Enfatizar la necesidad de revisar el plan cada mes usando las métricas del tracker, y ajustar variantes de CV/objetivos en base a feedback.
- **Mentalidad:** Recordar persistencia y aprendizaje constante (en cursos, idioma, proyectos) mientras se busca empleo.
- **Agradecimientos/documentación:** Notar que todo el material (CV, scripts, plan) estará disponible como recursos reutilizables.
- **Plan a largo plazo:** Vincular con objetivo 12-24 meses (p. ej. “avanzar hacia roles senior o migración EU tras mejorar perfil”).

G.0.37 37. Formación continua (Cursos y especializaciones)

- **Faltantes / Requerido:** Incluir una sección con:
- **Cursos completados:** Resumir brevemente los cursos relevantes hechos (sin certificado) listados en el Excel (CG Cookie, Animation, Modeling, Compositing, Videoviva, Web interactiva, Autodesk, etc.) para mostrar habilidades autodidactas.
- **Cursos recomendados a corto plazo (0–6 meses):** Ej. cursos en línea de Unity avanzado (URP optimización, GPU instancing), Blender avanzado (Hard Ops, iluminación técnica), React/Three.js (para complementar WebGL), Git y prácticas ágiles. Incluir los cursos de Rebelway accesibles: Houdini Pyro, Nuke avanzado, Katana.
- **Mediano plazo (6–12 meses):** Especializaciones o bootcamps: p.ej. certificación Unity, curso de XR Foundation (ARKit/ARCore), diplomado en Visualización 3D (si existe). Aprendizaje de idiomas técnicos (alemán B1/A2).
- **Largo plazo (>12 meses):** Opciones académicas: maestrías o posgrados en campos afines (p.ej. Maestría en Realidad Virtual, Ingeniería de medios interactivos). Mencionar universidades UE o becas posibles.
- **Idiomas:** Detallar meta: Inglés fluido (nivel C2) continuo, empezar portugués básico (Grupo Beta Archivos sefardí), alemán como segunda prioridad con aplicabilidad mediana.
- **Metodología:** Integrar cursos en la rutina semanal, usar proyectos reales para práctica (p.ej. implementar reel de efectos con Houdini).
- **Fuentes:** Referenciar plataformas: Udemy, Coursera, Domestika, Rebelway, CGCookie; mencionar uso de webinars y tutoriales gratuitos.

Appendix H

Auditoría de roles y posicionamiento

Fuente: Research/deep-research-report_02.md
Seccion: research
Estado: research_snapshot
Rol: Research / estudio de mercado
Lineas originales: 108
Ultima modificacion: 2026-06-11 12:16
Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Se evaluaron las 23 familias de roles propuestas en función de la demanda real (2025–2026), encaje con el perfil de Alexander y su objetivo de trabajo remoto desde Colombia. A continuación se resumen las conclusiones clave en tablas y listas.

Contenido completo depurado

Se evaluaron las 23 familias de roles propuestas en función de la demanda real (2025–2026), encaje con el perfil de Alexander y su objetivo de trabajo remoto desde Colombia. A continuación se resumen las conclusiones clave en tablas y listas.

H.0.1 1. Tabla de priorización de roles

Rol / Posición	Prioridad (Alta/Media/Baja)	Comentarios clave (demanda, requisitos, salario)
Unity Technical Artist (Artista técnico Unity)	Alta	Rol frecuente en empresas de videojuegos y simulación; portafolio exigente (Shaders, optimización en Unity/URP). Salarios senior ~ \$130–150k. Requiere 3–5+ años de experiencia y dominio de C#, Python, herramientas 3D. Encaja con TwinSight (Unity+optimización). Remoto posible con EOR/contratación.
Real-Time 3D Developer / Interactive 3D Developer	Alta	Roles amplios que incluyen desarrollo interactivo (WebGL, Three.js) y visualización 3D en tiempo real. Demanda creciente en VR/AR y visualización técnica. TwinSight es ejemplo directo. Salarios muy variables; startups WebGL pueden pagar hasta ~\$125k. Requiere Unity/WebGL y optimización. Buen encaje con perfil (C#, WebGL, UX Toolkit).
Technical Artist (general) (Artista técnico)	Media-Alta	Demanda sólida en estudios de juegos y simulación (pipeline entre arte y dev). Similares requisitos a Unity TA pero más genérico (puede incluir Unreal, Houdini, etc.). Salarios promedio ~ \$138k EE. UU. (higher than entry targets). Requiere experiencia en herramientas 3D (Photoshop, Maya, Houdini) y scripting. Buen encaje por know-how gráfico, pero muy competitivo.

Rol / Posición	Prioridad (Alta/Media/Baja)	Comentarios clave (demanda, requisitos, salario)
XR/AR/VR Developer (Desarrollador XR/AR/VR)	Media	Roles emergentes en empresas de realidad extendida (Meta, Google, startups). Requiere experiencia en Unity XR, C#, diseño de experiencias 3D. Ofertas Sr. XR dev ~ \$70–100k; puestos remotos (~\$90k) existen. Perfil puede pivotar hacia AR/VR usando TwinSight si amplía skills. Idiomas hablados pueden requerirse (inglés). Demanda creciente en simulación/educación/industrial.
Technical Visualization / Digital Twin Developer	Media	Orientado a industrias (manufactura, energía, defensa) que necesitan crear gemelos digitales o visualizaciones técnicas. Roles pocos en cantidad, pero suelen requerir Unity o motores 3D (como en Cincinnati ejemplo). Salarios en EE. UU. ~\$100–160k. Gran encaje para TwinSight (drone CAD→Unity). Remoto posible (muchas consultoras), pero competido.
Unity / WebGL Developer (general)	Media	Amplia demanda global en videojuegos y aplicaciones 3D web. Pueden incluir desde front-end 3D hasta AR/VR. Ejemplo: Unity developer industrial (HMI) con Unity. Salario promedio WebGL ~ \$125k en startups, pero en promedio USA ~\$21/h. Remoto LATAM viable; ejemplos pagaron \$5–6k mensuales (Somewhere). Perfil coincide en parte con TwinSight/WebGL; requiere marketing de portafolio web.
Tools/Pipeline Developer (Desarrollador de herramientas)	Media	Enfocado en scripting y automatización (blender, pipeline, ARA). Los roles puros son raros; a veces aparece en estudios grandes. Salario medio un Python dev (~\$80k–120k). TwinSight/ARA evidencia interés, pero necesitan traducir a aplicación de producto (p.ej. editor de Unity o herramienta CAD). Competencia moderada, requiere demostrar proyectos útiles (ARA framework). Remoto factible.
3D Product Visualization Developer (Gráficos 3D industria)	Media	Empresas de e-commerce o manufactura crean vistas 3D de productos. Requiere Blender/Unity para presentaciones interactivas. Demanda menor pero creciente; salarios variables (p.ej. industria automotriz/healthcare 80–120k). Encaje alto con Blender Portrait, pero hay mucha competencia de artistas 3D y demanda de UI/UX.
Simulation / Engineering Developer	Media	Incluye roles de simulación industrial o entrenamiento (e.g. EEUU, Defensa). A menudo Unity/Unreal con C#. Ofertas muestran desarrolladores 3D XR para el ejército (~\$70–90k). Requiere background técnico (física/matemática) que Alexander tiene. Permite aprovechar skills físicos. Remoto limitado (muchos on-site en Defensa), aunque consultoras contratan.
Shader Technical Artist	Media	Rol especializado en shaders, efectos gráficos (Unity/Unreal). Demanda específica en producción AAA o VFX. Salarios senior altos (\$100k+). Requiere portafolio de shaders (Shader Graph, HLSL) y conocimiento de URP. Perfil Alexander tiene bases (Shader Graph, HLSL básico). Puede usarse en portafolios.
Full-Stack Developer	Baja	Demanda general en software (ya sea web, apps). Pero se aleja del eje 3D/Unity. Aunque hay muchas vacantes globales y remotas (React/Python/DB), sería desviar la carrera. Salarios ~ \$60–120k. Se considera ruta de contingencia si no se consigue otro rol, pero no alineado con portafolio técnico-artístico.
3D Artist / Generalist (Blender)	Baja	Rol creativo puro, muy competitivo. Perfil de Alexander es técnico (no enfocarse en arte estático). Además, portafolios de arte 3D saturados (mostrados en ArtStation). Salario medio menor (\$40–60k). Se puede incluir un Blender Portrait de respaldo, pero no debe ser el foco principal.
AI Tools / Python Developer	Baja	Roles de inteligencia artificial e integración de LLM (ej. <i>AI Tools Developer</i> , <i>LLM Developer</i>). Creciente demanda (data science, ML), pero requiere experiencia sólida en ML. ARA/LLM aquí es un prototipo interno; no es un producto terminado. Ofertas de Python genérico son comunes (50–120k), pero el foco de Alexander es 3D más que IA. Puede usarse de forma secundaria (herramientas de pipeline), pero no es primaria en corto plazo.

Rol / Posición	Prioridad (Alta/Media/Baja)	Comentarios clave (demanda, requisitos, salario)
CAD-to-Realtime Tech Artist	Baja	Extremadamente nicho. No aparecen ofertas específicas de “CAD-to-Realtime” como título. Implica industria CAD (autocad, NX) + optimización 3D. Podría encajar en <i>Technical Visualization</i> o <i>Digital Twin</i> , pero no parece un rol autónomo. Más bien, es una habilidad dentro de dev de visualización técnica.

Fuentes: Se consultaron portales de empleo y comparadores salariales. Por ejemplo, ZipRecruiter indica un salario medio ~\$138k/año para *Technical Artist* en EE.UU.. LinkedIn/Indeed muestran *Unity Developer* LATAM ofreciendo \$5–6k/mes. El sector Web3 startup reporta \$125k/yr para *WebGL Developer*. Para XR dev remotos se ven rangos \$70–100k. Los datos reflejan el mercado a mediados de 2026 y los ajustamos a la realidad de Colombia (contratos B2B o EOR).

H.0.2 2. Roles secundarios y rutas de contingencia

- **Technical Artist general (no Unity):** Similar al rol principal pero ampliando a otras tecnologías (Unreal, Maya, Houdini). Como ruta secundaria, útil si no hay vacantes Unity específicas. Exige experiencia similar.
- **XR/Simulation Developer:** Desarrollo de aplicaciones VR/AR o simulaciones técnicas. Puede apalancar el mismo skillset (3D, C#) y la formación en física, pero suele requerir manejo de dispositivos específicos (oculus, hololens) y aún es competitivo.
- **Interactive/WebGL Developer:** Especializado en experiencias 3D Web (Three.js, WebAssembly). Aprovecha la tesis TwinSight (WebGL) y la habilidad Python/C#. No tan bien pagado como otros, pero buena puerta de entrada.
- **Tools/Pipeline Developer:** Para roles internos de automatización (Unity plugins, herramientas de animación). Utiliza ARA/AI skills. Mercado limitado, a veces parte del equipo de tech art. Salario medio.
- **Game Technical Artist:** Centrado en videojuegos AAA. Muy competitivo para un perfil *junior*. Buena fuente de motivación y apta si se mejora portfolio con demos de juegos. Pero en 3–6 meses quizá poco realista llegar a ofrecer.

Rutas de contingencia (fallback): full-stack (desarrollador web/C#) como último recurso; proyectos freelance (Upwork, Freelancer) en Unity/WebGL; apoyo en diseño 3D de producto (bajos pagos pero refuerza portfolio); continuar freelance local mientras se busca remoto.

H.0.3 3. Posicionamiento principal recomendado

Basado en la tabla anterior, se sugiere un **posicionamiento único centrado en 3D interactivo y visualización técnica real-time**. Por ejemplo:

“Desarrollador 3D en tiempo real y Artista Técnico especializado en Unity, C# y WebGL, enfocado en visualización técnica interactiva (simulaciones, gemelos digitales, visualización CAD a tiempo real).”

Este enunciado destaca Unity/C#, WebGL y la idea de visualización técnica (TwinSight) sin limitarse a videojuegos. Enfatiza habilidades fuertes (optimización, herramientas 3D) y el beneficio clave (“visualización técnica interactiva”).

H.0.4 4. Rutas secundarias de posicionamiento

- **Interactive 3D/WebGL Developer:** Enfocarse en portales, sitios o experiencias web 3D. LinkedIn y GitHub deben reflejar proyectos con Three.js/WebGL.
- **XR/AR/VR Developer:** Resaltar interés y base física/simulación; incluir cualquier demostración experimental en AR/VR.
- **Digital Twin/Simulación:** Mencionar perspectiva industrial; destacar reducción de polígonos CAD→3D (TwinSight).
- **Python/IA Tooling (ARA):** Como ruta auxiliar, puede presentar el framework ARA como muestra de habilidad en Python y LLM, pero aclarando que es prototipo.

H.0.5 5. Rutas de contingencia

En caso de dificultades iniciales, considerar:

- Proyectos freelance cortos en Unity/WebGL (contratación por horas o app móvil simple).
- Roles de desarrollador general (C#, .NET) en empresas de software (salario más bajo, usa portfolio como backend).
- Mejorar portafolio con cursos prácticos (Blender, shader).
- Preparación de certificaciones (Unity, etc.) para ganar credibilidad sin gastar años.

H.0.6 6. Keywords ATS sugeridas por rol

Para optimizar CV/LinkedIn/GitHub ante sistemas de búsqueda:

- **Unity Technical Artist:** Unity, C#, Shader Graph, URP, optimization, pipeline, content creation tools (Photoshop/Maya/Houdini), technical art, real-time rendering, VR, WebGL.
- **Technical Artist (general):** Technical Artist, game pipeline, VFX, Houdini, Substance, tool development, Python, C++, 3D optimization.
- **Real-Time 3D/Interactive Dev:** Unity, WebGL, Three.js, JavaScript, graphics performance, C#, interactive UI, UI Toolkit, runtime systems, mobile optimization.
- **Unity/WebGL Developer:** Unity3D, WebGL, HTML5, JavaScript, CI/CD, Addressables, profiling, Async/Await, performance.
- **Technical Visualization/Digital Twin:** Digital Twin, CAD, PiXYZ, Blender, tessellation, simulation, BIM, MAYK (Manufacturing, Aerospace, Militar, etc.), STL.
- **XR/AR/VR Developer:** XR, AR, VR, HoloLens, ARCore/ARKit, 3D interaction, spatial mapping, Unity XR, Oculus, Meta.
- **Tools/Pipeline Dev:** Python, LLM, LangChain, FastAPI, Blender Add-on, CI/CD, automation, editor scripting, software tooling.
- **Shader Technical Artist:** HLSL, ShaderLab, URP, High Definition Render Pipeline (HDRP), Shader Graph, GPU Compute, post-processing.
- **Python/AI Tools Developer:** Python, LLM, AI, Machine Learning, LangGraph, agent, automation, TensorFlow, PyTorch, API.
- **Full-Stack (fallback):** React, Node.js, FastAPI, PostgreSQL, REST, Docker, AWS, Git.
- **3D Artist / Generalist:** Blender, 3D modeling, texturing, rigging, UV unwrapping, PBR materials, lighting.

H.0.7 7. Gaps de evidencia por rol

- **Roles CAD to Realtime y CAD Centric:** No hay muchos anuncios directos; suelen encajar en *Technical Visualization*. Investigar empresas de ingeniería/defensa (p.ej. Unity for Industry, Dassault, Siemens NX).
- **XR/AR:** Falta dato concreto de demanda para entrada JR; solo posiciones senior visibles (Google, Meta). Investigación de nicho VR é importante.
- **Herramientas/AI (ARA, AI-Tools):** Roles específicos casi no aparecen en portales. ARA es un prototipo interno; debería definirse si se hará open-source. Riesgo de canibalizar el branding si se muestra como “proyecto principal” sin mercado directo.
- **Full-Stack:** Evidencia abundante, pero no alineada con meta principal; uso como evidencia de flexibilidad si se convierte en opción secundaria.
- **Cursos/Rebelway:** No hemos evaluado aún si presentar cursos avanzados (Blender, CG) en CV/LinkedIn. Debe validarse cuáles aportan por sí mismos sin certificación.
- **Idioma Alemán:** No se encontró evidencia de rol inmediato que exija alemán; generalmente Inglés es suficiente en remote global, alemán solo en roles locales presenciales.

H.0.8 8. Contradicciones estratégicas a evitar

- **Documentación de proyectos:** No confundir el contenido de tesis. La tesis final es *TwinSight X500* (dron) – no el ARA Framework. En CV/LinkedIn, se debe corregir para evitar malentendidos.

- **Citizenship/Movilidad:** No afirmar derechos de trabajo en Europa sin haber obtenido ciudadanía o visa. Enfatizar “remoto desde Colombia” y mencionar la ciudadanía portuguesa como plan futuro, no actual.
- **Senioridad exagerada:** Evitar términos como “Senior” o “Expert” en CV si no se cuenta experiencia formal (ACTA mostró rol técnico sin empleo fijo). Mejor usar “Junior-Mid” y enfocar en proyectos concretos.
- **Enfoque de rol mixto:** No mezclar en un mismo perfil demasiadas áreas sin conexión (p.ej. “Full-Stack” + “Artista técnico” + “Data”). Esto diluye la estrategia. Enfocar en “3D en tiempo real” prioritariamente.
- **Exagerar aporte de IA:** En entrevistas/portafolio, aclarar que ARA fue un proyecto personal prototipo, donde se usó IA como herramienta, pero **no** autopresentarlo como producto comercial. La responsabilidad técnica final fue del candidato.
- **Expectativas salariales altas prematuramente:** Evitar anunciar objetivos de \$6k/mes en 6 meses sin experiencia comprobable. Mejor posicionarse para ofertas de \$3k–\$4k/mes inicialmente (viajar hasta \$6k con mejoras en skills).
- **Uso de certificaciones no oficiales:** No dar demasiada importancia a cursos sin certificación (CG Cookie, Alive!), mencionarlos como “formación práctica” si es necesario, pero no como credenciales.

H.0.9 9. Recomendación final: principales familias de rol (3–5) a atacar

1. **Artista Técnico/Desarrollador 3D en tiempo real (Unity):** Este es el rol principal. Incluye posiciones como *Unity Technical Artist*, *Technical Artist*, *Real-Time 3D Developer*. Enfatizar su experiencia en proyectos Unity (TwinSight), optimización de activos CAD→Unity, y habilidades en C#/URP/Shader Graph. Apuntar a empresas de simulación, digital twins e incluso videojuegos. Salario objetivo: \$2–3k mensuales para empezar, escalando rápido con experiencia (target a \$6k en 1–2 años acorde al mercado).
1. **Desarrollador XR/Simulación/Digital Twin:** Roles en realidad extendida y simulación industrial (posiciones de *XR Developer*, *Simulation Developer*, *Digital Twin Developer*). Utiliza la formación en electrónica/física y Unity para crear simulaciones inmersivas. Enfocar en empresas que trabajen con entrenamiento militar, salud, manufactura o ciudades inteligentes. Ejemplos: desarrollos AR en Unity, gemelos digitales en Unity/Houdini, apps VR técnicas. Requieren aprendizaje de SDKs XR (ARCore, etc.). Su potencial salarial puede llegar a \$70–90k (EE. UU.).
1. **Desarrollador WebGL/Interactivo 3D:** Familias como *Interactive 3D Developer*, *Unity WebGL Developer* y *Frontend 3D Visualization Developer*. Aquí aprovecha la tesis TwinSight (WebGL) y habilidades de frontend. Empresas de tecnologías Web3, marketing inmersivo y educación ofrecen roles remotos. Aunque el salario inicial es más bajo (\$40–60k), se sube rápido con portafolio y experiencia. Destacar proyectos personales en WebGL (TwinSight) en GitHub/LinkedIn.
1. **Herramientas de Pipeline / Automatización (AI):** Como ruta secundaria ambiciosa, presentar ARA Framework como ejemplo de destrezas en Python y LLM. Roles en empresas que integran LLM en flujos de trabajo (p.ej. GenAI, data-mining) pueden interesar. Sin embargo, **no priorizar** esta ruta ahora; usarla más como diferenciador suplementario que como foco.
1. **Fallback – Full-Stack / Generalista:** Solo en caso de necesidad inmediata. Posee conocimientos de Python/FastAPI/React pero el mercado es altamente competido. Se sugiere como opción secundaria baja prioridad, manteniendo siempre el foco en roles 3D/Unity arriba.

Resumen: Priorizar propuestas alineadas a *Unity/C# en 3D interactivo y visualización técnica*, evitando distraerse en roles horizontales sin conexión al eje. Seguir estas familias de rol, con la estrategia de portafolio y networking correspondiente, maximizará las probabilidades de conseguir empleo remoto bien remunerado en 6–12 meses.

Fuentes: Datos obtenidos de ofertas reales y reportes de mercado (ZipRecruiter, Indeed, Wellfound). Ejemplos: Oferta *Technical Artist* en Activision (Carlsbad) pedía 5+ años y pagaba ~\$130–156k; posiciones Unity en LATAM ofrecen \$5–6k mensuales; mercado WebGL Startups muestra salarios promedio \$125k. Estos respaldan las estimaciones anteriores.

Appendix I

Executive Summary

Fuente: Research/deep-research-report_03.md

Seccion: research

Estado: research_snapshot

Rol: Research / estudio de mercado

Lineas originales: 106

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Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Salary ranges: In the US, mid-level Unity/3D roles (~Unity Developer, Technical Artist) typically pay **\$60–110K/yr**, with senior positions reaching **\$150K–\$200K**. Technical visualization/XR/simulation roles (e.g. “Digital Twin Engineer”) command similar or higher pay (US median ~\$156K). In Canada and Western Europe mid-level Unity devs earn roughly **\$60–90K/yr**; senior positions ~\$90–110K (USD equivalents). In Spain and Portugal salaries are much lower (mid-level ga

Contenido completo depurado

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Contract vs Employee: Tech contractors often earn ~10–15% more gross than employees. For example, a US engineer might make ~\$172K/yr as a contractor vs \$156K/yr salaried. However, contractors forgo employer-paid benefits (health, social security, retirement) and must handle taxes and compliance themselves. Employees have steadier income and benefits (EPS, pension contributions, paid leave) but lower gross pay.

Remote viability: Most of these roles can be done remotely for international companies. Latin American markets (US/Canada/EU firms hiring LATAM talent) often pay above local rates. Many remote-first companies (via EOR services) hire Unity/3D and dev talent worldwide. Payment platforms like Wise, Payoneer or EOR providers (Deel, Remote, Oyster) are commonly used.

Key findings:

- Entry (0–2 yrs): expect ~\$20–40K/yr in Colombia or ~\$30–50K via nearshore remote.
- 3–6 mo plan (grow to ~\$1.5K/mo): focus on securing a contractor or junior remote role (e.g. online agencies or B2B contracts via Upwork, Toptal).
- 6–12 mo plan (grow to ~\$3K/mo): leverage proven project (TwinSight) to negotiate higher rates or move to mid-level roles in EU/US markets.
- 12–24 mo plan (~\$6K/mo): target senior/architect roles or continuous contracting with US/EU companies (likely via EOR or establishing a LLC).

Each recommended role (see next section) has ample demand, but proof-of-skill requirements vary. TwinSight X500 directly supports Unity/WebGL and technical visualization roles. The ARA Framework and other AI projects underpin tooling and Python developer claims (helpful for AI/tooling roles, less so for Unity).

Resumen para Colombia (breve):

- **Rangos salariales:** En EE. UU., Unity/3D e ingenieros en 3D real pago medio ~\$60–110K/año; en Canadá/Europa Occidental ~\$60–90K/año; en España ~€48–70K. En Colombia (contracting remoto) un senior gana ~\$34/h (~\$68K/año); salarios locales mucho menores (~\$20–30K/año).
- **Empleado vs Contratista:** Contratistas tienden a ganar ~10-15% más bruto, pero sin beneficios (salud, pensión pagados por empleador). Empleados ganan menos pero con seguridad social y prestaciones.
- **Viabilidad remota:** Roles de Unity, 3D y Python suelen permitir trabajo remoto. Se usan plataformas como Wise/Payoneer y EORs (Deel/Oyster) para pagos internacionales. En Colombia se debe registrar en el RUT como exportador de servicios (exento de IVA) y declarar renta normal; se aportan ~12.5% a salud y 16% (de 40% base) a pensión.

I.1 Salary Benchmarking by Role & Region

Role / Experience	USA (annual)	Europe (West) / UK	Colombia (remote)	Notes / Sources
Unity Technical Artist (mid-senior)	~\$80K–\$126K (mid); Senior base ~\$115K	UK: ~\$45K–93K; Ger: ~\$50K–95K; Spain: ~\$35K (junior)	~\$20K–30K (empleado)	Mid-level Unity Technical Artist pays approx. \$80–126K in US. Senior leads exceed \$115K+.
Unity / Real-Time 3D Dev (mid)	~\$60K–\$80K (median \$80K)	Canada: ~\$62K–80K; UK: ~\$48K–65K; Ger: ~\$50K–70K	~\$20K–30K (empleado)	US median Unity Developer ~\$80K. Senior/architect roles pay more (~\$100K+).
Tech Visualization / Digital Twin (mid-senior)	~\$117K–\$218K (median \$156K) (Digital Twin Eng.)	EU: roughly \$70K–120K for experienced roles (game/VR industry)	~\$20K–30K (empleado)	E.g. "Digital Twin Engineer" Glassdoor median \$156K. Specialized XR/Sim roles follow game dev ranges.
Tools/Pipeline / Python Dev (mid)	~\$100K (mid); ~\$130K+ (senior)	Europe: ~\$70K–100K (mid)	~\$34/hr senior (~\$68K/yr)	US Python dev median ~\$129K; senior ~\$142K. Mid-level ~80–100K.
Game Tech/Graphical Roles	~ (see above)	Europe: lower generally (\$60K–90K)	~ (as above)	Game Developer salaries: USA ~\$114K, Latin America ~\$37K. Spain mid ~€48K–70K.

- All figures are approximate and vary by source: Glassdoor/job surveys and industry reports.
- Colombia contractor rates (via Lemon.io) suggest **\$34/hr** for senior devs. Local *contratación* salaries are much lower (often ~\$20–30K/yr for devs).
- *Junior/Entry*: expect 50–60% of mid figures; *Senior*: add ~20–30%.

Spanish summary: En EE. UU. una posición Unity/3D técnica paga típicamente **USD 60–110K/año** (median ~\$80K), subiendo a +\$120K en niveles altos. Roles especializados en visualización 3D o simulación reportan medianas similares o mayores (p.ej. ~\$156K para "Digital Twin"). En Canadá/Reino Unido/Alemania, los sueldos son ~30–40% menores (ej. Unity dev medio ~USD 60–80K). En España/Portugal rondan ~€48K–70K. En Colombia (trabajo remoto) los contratos sénior son ~USD 68K/año; empleos locales ganan ~\$20–30K.

I.2 Contractor vs Employee Comparison

- **Contractor (independent):** Typically **10–15% higher pay** than salaried roles. E.g. BridgeView: full-time \$156K vs contract \$172K. Contractors handle *all* own benefits (health insurance, retirement, etc.). They invoice gross rates and must pay taxes/contributions themselves. No paid leave or job stability guaranteed. Short-term projects and flexibility are perks, but you cover (or omit) benefits.
- **Employee (salaried):** Lower gross pay but includes employer-paid benefits (health/EPS, pension contributions, paid vacation/holidays). In Colombia, an employee has ~9% pension (and 4% parafiscales) withheld and employer adds another 12.5% to EPS and 16% of base to pension. Contractors in Colombia must voluntarily affiliate to EPS and pension (about 12.5% + 16% of 40% of income).

Aspect	Contractor	Employee
Pay	Higher hourly/daily rate	Lower base salary
Benefits	<i>None provided.</i> Must arrange own health/pension.	Health & pension contributions split with employer; paid leave.
Taxes	Responsible for own filings (Renta, IVA exempt if export).	Employer withholds/declares for you.
Flexibility	High (choose projects, hours); no long-term commitment.	Fixed schedule, stability.
Employer requirements	Need RUT registration, invoicing, self-managed seguridad social.	Only need CV/apply; employer handles legalities.

Spanish checklist (Empleado vs Contratista):

- **Contratista:** Emite facturas al exterior. Gana ~10-15% más bruto. Sin beneficios: debe cotizar EPS/pensión (12.5% + 16% de 40%), seguridad (ARL). No hay vacaciones pagas ni cesantías.
- **Empleado:** Cobra salario neto menor, pero recibe cobertura de salud/pensión del empleador, vacaciones, cesantías. La empresa hace retenciones de nómina (10% ARL, 12.5% EPS, 8.5% pensión por empleado).

1.3 Payment Platforms and Contracts

Popular platforms: Wise (Cuenta Multidivisa), Payoneer, PayPal y tarjetas Prepagas son habituales. Wise/Payoneer ofrecen cuentas en USD/Euros para recibir pagos con bajas comisiones. PayPal es universal pero cobra ~2.9%+\$0.30 por transacción (alto costo). Para contratos formales, **EOR/Payroll** como Deel, Remote, Oyster o Papaya son muy usados: ellos actúan como empleador local y facturan al cliente, simplificando cumplimiento y pagos (a cambio de tarifa).

Método	Tipo	Comisiones	Pros	Contras
Wise	Cuenta bancaria FI	~0.4%–1% + fijo	Baja conversión, rápido, multi-USD.	Tarjeta Wise <i>no disponible en COL</i> .
Payoneer	Cuenta bancaria	Gratis recibir con tarjeta	Amplia aceptación (Upwork, freelances). Tarjeta física/virtual posible.	Tarjeta USD \$35/año. Dólar a pesos: ~3%.
PayPal	E-wallet	~2.9%+\$0.30 + FX cost	Ampliamente aceptado (clientes globales).	Altas comisiones; política incierta.
Deel/Remote	EOR/Payroll (USD)	10–15%+ (servicio EOR)	Manejan impuestos/seguridad; pago garantizado.	Costoso; menos control directo.
Oyster/Papaya	EOR/Pago global	~3–5% gestión + tasas	Similar a Deel; mayor presencia en EU/Asia.	Spread bancario; soporte variable.
Transferencia Bancaria (SWIFT)	Banco local	~0.5% + tarifa fija	Directo, seguro.	Altas tarifas fijas (>USD 30), lenta.

Consejo: Para proyectos pequeños o freelancers, Wise/Payoneer + factura electrónica es ágil. Para rol de largo plazo/alto ingreso, usar una EOR (Deel, Remote) evita trámites.

1.4 Colombian Contractor Administrative Checklist

- **Registro Único Tributario (RUT):** Actualizar el RUT con actividad “*Exportador de servicios*”. Obligatorio facturar exportaciones legales.
- **Factura Electrónica:** Si prevés >3,500 UVT (~COP 183M) en un año, debes facturar electrónicamente. Se recomienda habilitarlo antes. Recuerda: los servicios exportados van exentos de IVA (Art. 481 literal c ET), por lo que en tus facturas no cobrarás IVA.
- **Impuesto de Renta:** Debes declarar ingresos globales. Las exportaciones no retienen IVA ni impuestos por parte del cliente extranjero. Sin embargo, tu renta la pagarás sobre la utilidad neta de tus servicios al exterior (tarifa progresiva ~19%-39%). Usa régimen *régimen común* o *régimen simple de tributación* según convenga.

- **Seguridad Social:** Afiliación como independiente: cotiza **EPS (salud) ~12.5%** del 40% de tu ingreso base y **Pensión ~16%** del 40%. También ARL (riesgos, ~0.5% base) y cajas de compensación si aplican. Considéralo en tu tarifario.
- **Pagos Internacionales:** Abre cuentas en USD/EUR (Wise, Payoneer). Conserva comprobantes de pagos al exterior. Usa contratos escritos (p. ej. emitidos vía Upwork/Deel/contrato de prestación de servicios) que describan servicio y monto neto en moneda extranjera. Emite factura electrónica en pesos (convertir monto USD a COP) para cumplir con DIAN.
- **Contabilidad:** Lleva registros claros de facturas, contratos y recibos de pago. Considera asesoría contable para régimen simple o común.

Resumen (lista de chequeo): _“Soy exportador de servicios”: a) Inscribir RUT con actividad exportadora; b) Facturar electrónicamente si supero 183M COP; c) No cobrar IVA (exento como exportación); d) Cotizar salud (12.5%), pensión (16% de 40%) y ARL; e) Usar Wise/Payoneer o EORs para cobros extranjeros; f) Registrar contablemente todos los ingresos/gastos._

1.5 Timeline to Income Targets

```
Bloque de código: mermaid
gantt
    title 30/60/90 & 12-Month Income Growth
    dateFormat YYYY-MM-DD
    section Income Targets
    Reach \$1.5K/m (USD) :t1, 2026-07-01, 3m
    Reach \$3K/m :t2, after t1, 6m
    Reach \$6K/m :t3, after t2, 9m
    section Key Actions
    Optimize portfolio & CV :a1, 2026-07-01, 2m
    Active networking & apply :a2, after a1, 6m
    Upskill (courses, demos) :a3, 2026-09-01, 9m
    Interview prep & negotiations :a4, after a3, 3m
```

- **0–3 meses:** Finalizar portafolio MVP (TwinSight case study, demo video), pulir CV/LinkedIn, comenzar aplicaciones remotas (LATAM/US freelancers). *Objetivo inicial: ganar contratos o empleo de ~\$1.5K/mes.*
- **3–6 meses:** Revisar y elevar tarifas tras primeras ofertas; expandir red (LinkedIn, comunidades game/tech). Publicar/oportunidades (Upwork, Tecla, remote job boards). *Meta: consolidar ~\$3K/mes y casos de éxito (TwinSight demo, ARA preview).*
- **6–12 meses:** Apuntar a roles o contratos mejor pagados (senior mid/full-stack Unity, vis. técnica, AI tooling en EU/US). Negociar mejores condiciones, posiblemente con EOR. Considerar cursos avanzados (e.g. XR/Shader especialización) para alcanzar ~\$6K/mes.

Nota: El camino rápido implica ganar experiencia visible y cartas de recomendación pronto; ser conservador/balanceado en volumen de apps vs personalización; buscar propuestas de valor únicas (digital twins, AI).

Resumen final: El reporte evidencia que los roles con portafolio fuerte pueden alcanzar desde \$1.5K/mes remotos hasta >\$6K/mes con 12-24 meses de progresión (cita ejemplos y metas). En Colombia, la vía más viable inicial es freelance/contratista internacional (US/EU) aprovechando proyectos reales (TwinSight), con soporte en plataformas globales (Wise, Payoneer, EOR) y cumplimiento tributario local (RUT, facturación exportación, seguridad social). Cada objetivo salarial incremental se sustenta en alcanzar un nivel de experiencia/portafolio requerido para el mercado objetivo.

Fuentes destacadas: Glassdoor (salarios Unity dev/tech art), informes de mercados de contratación (Alcor, Qubit Labs), datos Colombia (Lemon.io) y guías tributarias (DIAN/actualicese). En todos los casos se diferencian cifras **verificadas** (encuestas oficiales) de **estimadas** (informes o muestras de oferta), señalando donde hay incertidumbre.

Appendix J

Benchmark salarial y trabajo remoto (Colombia)

Fuente: Research/deep-research-report_04.md

Seccion: research

Estado: research_snapshot

Rol: Research / estudio de mercado

Lineas originales: 89

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Resumen de orientacion

Este documento integra datos recientes (2025–2026) de salarios y contratación remota para perfiles 3D/Unity y roles afines. Incluye: rangos salariales por rol y región, tarifas de contratistas, costos contratistas/EOR, y detalles fiscales colombianos. Las cifras provienen de fuentes verificadas (Howdy, Glassdoor, PayScale, ERI SalaryExpert, Tecla, etc.) y reflejan situaciones reales de mercado.

Contenido completo depurado

Este documento integra datos recientes (2025–2026) de salarios y contratación remota para perfiles 3D/Unity y roles afines. Incluye: rangos salariales por rol y región, tarifas de contratistas, costos contratistas/EOR, y detalles fiscales colombianos. Las cifras provienen de fuentes verificadas (Howdy, Glassdoor, PayScale, ERI SalaryExpert, Tecla, etc.) y reflejan situaciones reales de mercado.

J.0.1 Resumen de hallazgos clave

- **Brecha US vs LATAM:** En EE.UU. un **Desarrollador Unity/3D en tiempo real** senior suele ganar en torno a \$100–140 mil/año (por ejemplo, Glassdoor reporta sueldo medio ~\$113 K), mientras que en América Latina el promedio es mucho menor (~\$55–67 K/año según Howdy). Contratando en LATAM se ahorra ~60% comparado con EE.UU..
- **Europa Occidental:** En Alemania el salario promedio para **Unity/3D** es ~€74 K/año, en España ~€50 K. Técnicos con especialidad en Unity reciben similares (€70–75 K/año en Alemania). Países como Países Bajos o Nordics pagarían algo más (~€80–100 K), pero no hallamos datos precisos.
- **Roles técnicos y especializados:** Un **Artista Técnico Unity (Technical Artist)** en EE.UU. ronda \$75–80 K, y en Alemania ~€70 K. Un **desarrollador WebGL/3D** en startups US promedia ~\$113 K (rango muy amplio), similar a un Unity Dev. Roles XR/AR/VR en EE.UU. promedian ~\$113 K (Glassdoor) – son nicho pero cotizan alto (hasta ~\$150 K). Roles de **herramientas/pipeline o Python/IA** se ubican en rangos similares a desarrolladores back-end (~\$60–90 K medio en EE.UU.), aunque no hay datos específicos detallados.

- **Full-stack y 3D Artist (fallbacks):** Full-stack en Colombia se estima ~\$30–78 K (según nivel), muy por debajo de Europa/EE.UU. Un **3D Artist/Blender** generalista ganaría menos que desarrollador (tal vez \$30–50 K en EE.UU, mucho menos en LATAM). Estos roles no son prioridad principal: sirven de respaldo en caso necesario.
- **Contratación remota y EOR:** Cada vez más empresas usan **EOR** (Employer of Record) por temas de cumplimiento. Trabajar como contratista independiente desde Colombia conlleva retener impuestos (aprox. 28–30% para salud/pensión) y posiblemente IVA 19% (aunque se puede tramitar exención por “exportación de servicios” con RUT exportador). Plataformas globales (Wise, PayPal) facilitan pagos internacionales.
- **Metas de ingreso:** Un ingreso remoto bruto de **USD 1.5K/mes** (18K/año) es factible en 0–3 meses contratando desde LATAM (sueldo junior local o contrato medio); **USD 3K/mes** (36K/año) es meta 3–6 meses con buen portfolio (mid-junior internacional, freelance fuerte); **USD 6K/mes** (72K/año) es difícil a corto plazo (necesita rol US/EU senior, o proyecto digital twin/IA con fuerte tracción). Tras obtener ciudadanía portuguesa (o visa EU), el rango salarial objetivo subiría, pues permitiría roles mejor pagados en Europa.

J.0.2 Salarios por rol y región

A continuación, se resumen los rangos salariales anuales (brutos) por rol y región. Los valores son aproximados y varían según experiencia, industria y empresa.

- **Desarrollador 3D/Real-Time (Unity/C#, WebGL):**
 - *EE.UU.:* base típico de **\$80K–100K** anual para perfil mid-level; seniors pueden llegar a **\$115K–150K+**. Glassdoor reporta rango \$60K–\$106K (mediana ~\$80K).
 - *Alemania:* promedio ~€74K/año (rango ~€50K–90K).
 - *España/Portugal:* promedio ~€50K/año (junior €35K, senior €57K). Portugal es similar a España o algo menos.
 - *LatAm (Colombia):* Unity Dev promedio ~COP 108M/año (~USD \$30K); entry ~COP 76M (\$21K), senior ~COP 124M (\$34K). Howdy estima ingeniero LATAM ~USD 55K–67K en general.
 - *Remoto global (contractor):* mercados freelance pagan típicamente +30–60% que LATAM interno. Plataformas como Lemon indican para un Unity Senior remotos ~US\$57–74/h (~\$120K–150K año full-time).
 - *Notas:* Roles Unity/WebGL suelen requerir portafolio con demos (ej. WebGL build), repositorios; TwinSight X500 es prueba fuerte para este perfil.
- **Artista Técnico (Technical Artist Unity):**
 - *EE.UU.:* promedio ~\$75K/año (rango típicamente \$60–119K según experiencia).
 - *Alemania:* ~€70K/año (rango €49K–86K).
 - *España:* no hay datos específicos, posiblemente €40K–50K rango medio.
 - *LatAm:* menores, quizá USD 20K–40K; Howdy destaca que roles TI/Ingeniería lideran salarios, pero TA suele verse más como rol 3D/artístico. ARA (framework Python) no es factor directo en salario TA.
 - *Notas:* El portafolio de TwinSight X500 demuestra capacidad TA (optimización, UI técnica); se espera evidencia de trabajo en shaders, pipelines o herramientas para aplicar a este rol.
- **Desarrollador WebGL/3D (navegador interactivo):**
 - *EE.UU.:* Wellfound reporta salario medio ~**\$113 K/año** para WebGL en startups (rango amplio \$38K–\$245K). Estos roles son híbridos web/3D.
 - *Europa:* se ubican en similar que Unity Dev (tal vez €60K–€100K para senior). No hay fuente específica.
 - *LatAm:* similar a Unity Dev LATAM (~USD 20–30K).
 - *Notas:* Experiencia concreta con WebGL (ej. builds en Unity o Three.js) aumenta competitividad. La vacante suele requerir demo en navegador (TwinSight X500 es ejemplo relevante).
- **Desarrollador XR/AR/VR:**
 - *EE.UU.:* Glassdoor estima promedio **\$112.7 K/año** (rango típico \$86K–\$149K). VR/AR es especialidad en auge.
 - *Alemania/EU:* posiblemente €60K–€100K para senior, pero datos limitados.
 - *LatAm:* faltan datos claros; probablemente mas bajos (USD 20–40K).
 - *Notas:* Cualquiera experiencia en AR/VR (por ejemplo, Unity para Meta Quest, etc.) es un plus salarial (ver Lemon: AR/VR +12–20% premium). Sin embargo, rol XR puro es secundario al perfil principal.
- **Desarrollador de Herramientas/Pipeline (Python, Data, QA):**

- **EE.UU.:** similar a desarrollador back-end: ~\$60–100K dependiendo de especialidad. Python/LLM dev puede superar, pero no hay cifra concreta.
- **Alemania:** ~\$50–80K.
- **LatAm:** ~\$20–40K. Por ejemplo, desarrolladores Python/jr en Latinoamérica suelen cobrar 15–25 mil USD para juniors, y los seniors de software ~60K (incluyendo mercado general).
- **Notas:** El proyecto ARA (LangGraph) es muestra de habilidades Python/IA, útil para roles de automatización de investigación o prototipos, pero no para un rol centralizado sin portafolio de producto.
- **Full-Stack / Backend (fallback):**
 - **EE.UU.:** ~\$80K base (rango \$60–120K+ según experiencia).
 - **LatAm:** Tecla sitúa full-stack Colombia en ~USD 30–105K según nivel (junior ~\$30–42K, senior ~\$78–105K).
 - **Notas:** Aunque Alexander tiene conocimientos de JS/Python (ej. FitApp, AI News), este no debe ser enfoque principal. Sin embargo, fortalecer backend (p.ej. API robusta) puede abrir roles en empresas no de 3D.
- **3D Artist / Modelador (fallback):**
 - **EE.UU.:** un 3D generalista suele ganar \$40–60K en empresas pequeñas; en industrias como publicidad podría llegar a \$70K. No hay fuente fiable.
 - **LatAm:** mucho menor, tal vez USD 10–20K (salaries de artistas 3D freelancers en LATAM suelen ser bajos).
 - **Notas:** El retrato hiperrealista en Blender es demo de habilidades 3D, pero no es prueba de workflow de juego/pipeline. Es más bien un activo secundario de portafolio y no una ruta de carrera central.

Las **metas de ingreso** propuestas (USD 1.5K, 3K, 6K mensuales) dependen de modalidad y experiencia:

- **USD 1.5K/mes (~18K/año):** viable con roles junior en LATAM (por ejemplo full-stack JR) o subcontratos locales. Se requiere CV+portafolio básico y buena red de aplicaciones freelance.
- **USD 3K/mes (~36K/año):** factible en 3–6 meses apuntando a contratos remotos internacionales junior fuerte o mid-level latino. Ejemplo: contratista remoto medio en Latinoamérica según Tarifa Howdy ~USD 60–75K/año (~\$5–6K/mes) (full cost), lo cual deja al trabajador algo menos neto.
- **USD 6K/mes (~72K/año):** difícil a corto plazo. Requeriría rol norteamericano/EU de nivel medio-alto o senior (por ejemplo, **Digital Twin/Simulación**, o **Unity Senior en empresa europea**). En 12–24 meses, con portafolio sólido (TwinSight, validación académica) y tal vez ciudadanía portuguesa, podría alcanzarse en Alemania o Portugal.

J.0.3 Contratación remota y modalidades

- **Contratos locales vs internacionales:** Trabajar con *empresa colombiana* implica salario local (bajo) y aportes de nómina obligatorios (aprox. 28.5% del sueldo). Contratar *directamente para empresa extranjera* como contratista B2B aumenta ingresos brutos (sin deducir impuestos en Colombia), pero exige manejar retenciones, IVA y seguridad social.
- **Employer of Record (EOR):** Plataformas (e.g. Deel, Oyster) permiten al empleador extranjero contratar legalmente en Colombia. Según Howdy, la tendencia es fuerte hacia EOR por cumplimiento. Como ventaja, el contratista recibe beneficios similares a local (EPS, pensión retenidos) pero suele sacrificar algo del ingreso neto (ya que el empleador paga ~20.5% adicional de SS). Sin embargo, EOR ofrece estabilidad y suele ser mejor opción para roles full-time remotos formales.
- **Tarifas de contratistas:** A nivel mundial, los seniors Unity/3D cobran ~\$50–75/h (~\$100K–150K año). En LATAM, según una guía de reclutamiento, *junior* puede rondar \$25K/año y *senior* \$60–75K. Por ejemplo, un desarrollador Python senior en Colombia podría negociar ~\$3,000–4,000 USD al mes como contratista (antes de gastos locales). Estas tarifas deben ajustarse por la competencia y especialización.
- **Efecto neto (bruto a neto):** Hay que reservar ~30–40% del ingreso bruto para impuestos y SS. Por ejemplo, USD 3,000 brutos mensuales (~COP 10.5M) implicarían cotizaciones y provisiones (salud 12.5%, pensión 16%, ARL 0.5–6% según riesgo) por unos COP 3–4M, dejando neto ~USD 1,800–2,100. Además conviene usar medios de pago internacionales (Wise, Payoneer) para minimizar costos de cambio.

J.0.4 Recomendaciones estratégicas

- **Prioridad 0–3 meses:** Enfocarse en roles remotos de **Unity/WebGL Interactivo** o **Desarrollo 3D en tiempo real** nivel junior-medio. Utilizar portafolio existente (TwinSight X500, demos en WebGL) para aplicar.

Ofertas en agencias LATAM (ej. cerca de \$3–5K brutos al mes) o empresas pequeñas startup. Consolidar perfil GitHub con ejemplos de optimización, shaders, interactividad 3D.

- **3–6 meses:** Apuntar a posiciones que valoren su especialidad “CAD-to-realtime” y UI técnico (p.ej. simulación, visualización industrial). Podría incluir Ingeniería Digital/Digital Twin (automatización, GIS, IoT). El pasaporte portugués futuro abre roles en UE; en este periodo todavía sería remoto. Expandir red en job boards especializados (p.ej. Unity Connect, StackOverflow Jobs).
- **12–24 meses:** Con pasaporte PT (~2027) o gracias a reagrupación familiar (Alemania), moverse a roles internos en la UE: e.g. Unity Developer sénior, Technical Artist en industria 3D (AR/VR industrial, automoción). Esto justificaría aspirar a USD 6K+/mes y beneficios completos UE (salud, pensión, protección laboral). Mientras tanto, prepararse para entrevistas internacionales (idiomas, certificados de Blender/Unity).

En resumen, la ruta más coherente es enfocarse en **Desarrollador 3D/Unity/WebGL** (perfil híbrido dev+TA) como objetivo principal, usando TwinSight como carta de presentación. Full-stack y 3D generalista quedan como respaldo secundario. Las oportunidades en IA/automatización (proyecto ARA) pueden servir para roles de tooling interno, pero no deben desviar el foco principal 3D.

J.0.5 Próxima sección

```
Bloque de código: markdown
04_eu_portugal_germany_mobility_strategy.md
```

Para la estrategia de movilidad (ciudadanía portuguesa, matrimonios, visados) recomiendo activar **Deep Research**. Es necesario consultar fuentes oficiales (SEF Portugal, Gobierno Alemán, web de visados UE) para obtener requisitos, tiempos y costos actualizados (2024–2026). No usaremos modo agente aún; el perfil de estudios requiere más análisis documental que recolección masiva.

Appendix K

Estrategia de idiomas y ROI

Fuente: Research/deep-research-report_05.md

Seccion: research

Estado: research_snapshot

Rol: Research / estudio de mercado

Lineas originales: 108

Ultima modificacion: 2026-06-11 13:02

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Resumen ejecutivo: El **inglés** es imperativo: prácticamente todos los roles técnicos internacionales exigen inglés fluido (mínimo B2/C1). Por eso, se recomienda certificarlo (IELTS/TOEFL) lo antes posible. El **portugués** es clave para operar en Portugal (una creciente hub tecnológica en Europa) y para acceder a mercados lusófonos; se sugiere alcanzar al menos nivel B1–B2 y luego optar por el examen oficial CELPE-Bras (costo ~US\$120). El **alemán** es relevante por la

Contenido completo depurado

Resumen ejecutivo: El **inglés** es imperativo: prácticamente todos los roles técnicos internacionales exigen inglés fluido (mínimo B2/C1). Por eso, se recomienda certificarlo (IELTS/TOEFL) lo antes posible. El **portugués** es clave para operar en Portugal (una creciente hub tecnológica en Europa) y para acceder a mercados lusófonos; se sugiere alcanzar al menos nivel B1–B2 y luego optar por el examen oficial CELPE-Bras (costo ~US\$120). El **alemán** es relevante por la posible mudanza a Alemania con su pareja alemana: al casarse y registrarse, obtiene derecho de libre movimiento en la UE (puede entrar con su pasaporte colombiano y luego obtener residencia, pudiendo trabajar sin restricciones). Sin embargo, conviene nivel mínimo A2–B1 para la vida diaria; a largo plazo, subir a B2/C1 mejoraría sus oportunidades locales. En resumen, el orden de prioridad es: **1) Inglés** (mantener e intensificar hasta C1), **2) Portugués** (objetivo B2 en 6–12 meses), **3) Alemán** (objetivo B1–B2 en 12–24 meses). A continuación se desarrolla esta estrategia con datos, tablas de ROI, cronograma y plan de acción.

K.0.1 Demanda de idiomas en tecnología

En el sector tecnológico global, el **inglés** es la lengua franca. Estudios recientes muestran que alrededor del 10–15% de las ofertas laborales en Europa (por ejemplo, en Francia) incluyen “Inglés” como requisito, y cuando se pide, exige niveles altos (generalmente B2 o C1). En tech roles (desarrollo Unity, 3D, simulación, etc.) es norma “fluidez” (C1 según CEFR). En consecuencia, un certificado de inglés avalado (TOEFL/IELTS/Cambridge) **mejorará significativamente el perfil** y la competitividad salarial.

El **portugués** ha ganado relevancia pues Portugal impulsa su sector tech (Tech Visa, startups). Muchos empleos tecnológicos en Portugal se ofrecen en inglés, pero para vivir allí es “*condición no oficial*” saber portugués. Candidatos con portugués pueden destacar en empresas portuguesas e incluso en posiciones globales vinculadas a Brasil. Datos de mercado indican que el portugués es uno de los idiomas de mayor demanda en Europa (sobre todo en tech y servicios).

El **alemán** es muy útil en Alemania (la economía más grande de Europa). Aun cuando ciudades tech como Berlín reclutan en inglés (en 2026 Berlín es un “hub Inglés-primero” en startups), muchas empresas alemanas valoran al menos un nivel conversacional. **Clave:** al casarse con ciudadana alemana, Alexander podrá trabajar en Alemania sin trámite de visado especial, gracias al libre movimiento UE. No obstante, aprender alemán le ayudará en la integración social y ampliará sus opciones fuera del nicho anglófono.

K.0.2 Niveles CEFR recomendados por rol

Para todos los roles objetivo, **inglés B2/C1** es el mínimo realista. Algunas orientaciones por familia de rol (asumiendo perfil junior/mid con TwinSight X500):

- **Unity/3D Developer & Technical Artist:** B2 mínimo, C1 recomendado. Interactuar con docs, equipos internacionales y test de usabilidad requiere comprensión avanzada.
- **XR/AR/VR y Simulación/Visualización Técnica:** B2 en inglés. Alemania y Portugal ofrecen proyectos en inglés. El portugués podría ser útil en simulación o gemelos digitales europeos.
- **Full-stack / Python (herramientas IA):** B2 en inglés. El alemán es útil si trabaja en research/IA en Alemania.
- **Ruta Portugal:** inglés B2 + portugués B1 rápido, luego B2.
- **Ruta Alemania (con cónyuge):** inglés B2, alemán B1 para cotidiano (idealmente B2 para roles profesionales).

En síntesis, se recomienda:

- **Inglés:** apuntar a **C1** (certificado IELTS/TOEFL). Nivel mínimo **B2** para el primer trabajo internacional.
- **Portugués:** empezar objetivo **B1** (mantenido) y escalar a **B2**. Examen CELPE-Bras si necesitara certificación.
- **Alemán:** objetivo inicial **A2–B1** (en 12 meses) y luego **B2** si persigue empleo directo en Alemania. Su pareja alemana exime de ciertas exigencias de idioma en visados.

K.0.3 ROI por idioma y rol

La siguiente tabla resume el **impacto estimado** de cada idioma en oportunidades y salario, por familia de rol. Se califican de forma cualitativa la **mejora de ofertas** y **potencial salarial**, más un rango de **tiempo de estudio** (de B1 a meta) y **coste aproximado** de cursos+certificados:

Idioma	Mejora de ofertas (probabilidad)	Potencial aumento salarial	Tiempo a nivel objetivo	Coste aprox. (cursos+exámenes)
Inglés	Muy alta (necesario en TODOS los roles globales)	+10–20% (ofertas internacionales)	12–18 meses (de B1 a C1)	~\$2.000 (cursos intensivos + IELTS/TOEFL (~\$250))
Portugués	Alta (clave en roles Portugal/Brasil)	+5–10% (roles lusófonos)	6–12 meses (de B1 a B2)	~\$1.000 (curso intensivo + CELPE-Bras ~\$120)
Alemán	Media (en Alemania, beneficios legales)	+5–10% (roles Alemania)	12–24 meses (de A1 a B2)	~\$1.200 (curso intensivo + Goethe B1 (€259))

- **Notas:** La mejora de ofertas refleja principalmente la *admisibilidad* a mercados específicos. Inglés eleva todas las oportunidades globales. Portugués abre puertas en Portugal/Brasil; alemán, en Alemania. El aumento salarial (%) es estimado. Por ejemplo, dominando el idioma local puede negociarse mejor salario, o tener prioridad en entrevistas. El coste incluye cursos en línea o presenciales intensivos y tasas de examen (citadas arriba).

K.0.4 Plan de estudio y cronograma (3/6/12/24 meses)

A continuación se esquematiza un posible cronograma de estudio para alcanzar los niveles meta en 24 meses. Cada bloque corresponde a clases, práctica intensiva y certificaciones:

```
Bloque de codigo: mermaid
gantt
    title Plan de idiomas: 24 meses (2026-2027)
```



```

dateFormat YYYY-MM
section Inglés
A1\u2013B2 (Consolidar B2)\t :done, a1, 2026-06, 4m
Preparaci\u00f3n IELTS TOEFL :active, a2, 2026-10, 8m
section Portugu\u00e9s
Curso intensivo portugu\u00e9s B1 :por1, 2026-06, 3m
Conversaci\u00f3n / Inmersi\u00f3n :por2, 2026-09, 6m
section Alem\u00e1n
Curso intensivo alem\u00e1n A1\u2013B1:ger1, 2026-06, 6m
Conversaci\u00f3n \u2014 Goethe B1 :ger2, 2026-12, 12m
Goethe B2 / Aprendizaje continuo :ger3, 2027-12, 12m

```

En este plan hipotético se asume nivel base **B1 de inglés** (coherente con el perfil y estudio previo), y **A1 en alemán/portugués**. Por ejemplo:

- De 0 a 6 meses: consolidar **B2 en inglés** (clases avanzadas, práctica diaria), completar **B1 en portugués** (curso intensivo) y **A1–B1 en alemán** (curso intensivo).
- De 6 a 12 meses: **C1 en inglés** (preparación de IELTS/TOEFL y conversación avanzada), **B2 en portugués** (más práctica, cursos de cultura) y **B1 en alemán** (práctica oral, Goethe B1).
- A 12–24 meses: mantener inglés con uso profesional, afianzar portugués en contexto real (viajar a Portugal), y elevar alemán a nivel B2 (Goethe B2).

Los tiempos son aproximados y dependen de intensidad. Estudios estiman que pasar de B1 a B2 requiere unas 150–200 horas de instrucción, y de B2 a C1 otras 200–250 horas. Dedicando 10 horas semanales, se podría lograr B2 en ~6 meses e IELTS C1 en ~12 meses.

K.0.5 Diagrama de decisión de idiomas

Este diagrama ayuda a priorizar según objetivo: trabajo remoto inmediato o migración UE a medio plazo.

```

Bloque de código: mermaid
flowchart LR
    Start((¿Objetivo?)) --> Remoto[Trabajo remoto internacional 3\u20136m]
    Start --> Movilidad[Movilidad UE 12\u201324m]
    Remoto --> Ingl[Ingl\u00e9s B2\u2013C1 (certificaci\u00f3n IELTS/TOEFL)]
    Ingl --> Fin1[Fin1[Foco: Ofertas Globales (silenciamiento de requisitos locales)]]
    Movilidad --> Destino[Destino{Destino EU}]
    Destino --> Portugal[Portugal (Tech Visa)\nPortug\u00e9s B1\u2013B2 + ingl\u00e9s]
    Destino --> Alemania[Alemania (conyugue UE)\nIngl\u00e9s + Alem\u00e1n A2\u2013B1]
    Portugal --> Fin2[Fin2[Residencia legal \u2014 y oportunidades tech en Lisboa/Oporto]]
    Alemania --> Fin3[Fin3[Residencia con s\u00edmbolo y mercado tech de Berl\u00edn/Frankfurt]]

```

- **Remoto 3–6m:** En el corto plazo, priorizar **inglés**. Con nivel alto de inglés (certificado), puede aplicar a cualquier oferta global sin barreras lingüísticas, enfocándose en skills técnicos.
- **Movilidad UE 12–24m:** Si el objetivo es reubicar a Europa, elegir el destino (por pasaporte portugués o cónyuge alemán). Para **Portugal**, añadir portugués (meta B2) facilitará la vida diaria y requisitos (aunque no es obligatorio para Tech Visa). Para **Alemania**, el inglés es suficiente para la mayoría de roles tech (especialmente en Berlín), pero aprender alemán (A2–B1) acelerará la integración, aunque no es requisito para el permiso de residencia de cónyuge. En ambos casos, mantener el inglés alto es crítico.

K.0.6 Certificaciones y pruebas

- **Inglés:** Se recomienda IELTS Academic o TOEFL iBT. Objetivo ~7.0 IELTS o 95 TOEFL (equivalente a C1). Costo ~\$250. Certificados oficiales añaden credibilidad ante reclutadores y procesos migratorios. Alternativa: Cambridge C1 Advanced (CPE).
- **Portugués:** El certificado oficial es **CELPE-Bras** (expedido por Brasil). Si bien no se exige para visas portuguesas, sirve como prueba formal de nivel (cubre B1–C2). La tarifa actual es \$120. Se sugiere certificar tras alcanzar B2.
- **Alemán:** Exámenes del Goethe-Institut (A1–C2). Se recomendarían **Goethe-Zertifikat B1** (259 €) y luego **B2** (289 €). Incluso un certificado B1 demuestra compromiso ante empleadores. El TestDaF (200 €) es otra opción, pero orientada a estudios.
- **Plan de pruebas:** Registrar fechas de exámenes internacionales con anticipación. Ejemplo: dentro de 6–12 meses presentar IELTS o TOEFL; en 12–18 meses, CELPE-Bras o Goethe B1; a 24 meses, Goethe B2.

K.0.7 Plan de acción

1. **Inglés (3–6–12m):** Inscribirse en curso intensivo avanzado (online o presencial) desde ahora. Practicar conversación con nativos (italki/Tandem) y simulacros IELTS/TOEFL. Meta: *B2 en 6m, C1 en 12m*. Presentar examen oficial (p.ej. IELTS) antes de fin de 2026.
2. **Portugués (3–6–12m):** Iniciar curso de portugués básico (aprovechando similitudes con español). Objetivo *B1 en 3–6m*. Después, curso intermedio-práctico (videos, medios) para alcanzar *B2 en 12m*. Planear examen CELPE-Bras hacia 2027 una vez nivel B2.
3. **Alemán (6–12–24m):** Tomar clases de alemán desde nivel A0. Meta *A2–B1 en 12m* con curso intensivo o convivencia (ej. visita a Alemania/charlar con su pareja). Continuar estudio hasta B2 en 24m. Prepararse para Goethe B1 a un año.
4. **Uso diario y herramientas:** Incorporar apps (Duolingo, Babbel) y podcasts para práctica diaria. Aprovechar reuniones de intercambio idiomático. Leer documentación técnica en inglés (mantener vocabulario).
5. **Certificaciones:** Reservar fechas de examen (como IELTS, Goethe) con antelación para cumplir objetivos temporales.
6. **Acciones administrativas:** Una vez en Portugal o Alemania, realizar cursos locales (por ejemplo, Goethe-Institut Alemania) para velocidad de integración. Mantener perfil de LinkedIn actualizado con certificaciones en idiomas alcanzados.

Conclusión: Avanzar de forma consistente en cada idioma maximizará sus opciones: en 3–6 meses el enfoque total en inglés le abrirá ofertas globales; en 12 meses la combinación con portugués y alemán básico preparará su transición a Europa. Logrando los niveles propuestos antes de 24 meses, fortalecerá notablemente su empleabilidad y justificación salarial en cada mercado objetivo.

Fuentes: Se ha revisado literatura actualizada (2025–2026) y ofertas reales para roles técnicos. Las recomendaciones se basan en:

- Estudios de mercado laboral (e.g. ETS 2024 sobre inglés en ofertas; tendencias de idiomas en Europa).
- Páginas oficiales de certificaciones (Goethe, Celpe-Bras, IELTS).
- Políticas de movilidad de la UE (Make-it-in-Germany sobre reagrupación familiar, certificaciones lingüísticas).
- Análisis estratégico (patrones de ofertas en 2025–2026). Las cifras de horas de aprendizaje provienen de EF. Las estrategias propuestas combinan hechos verificados con estimaciones realistas para el perfil de Alexander.

Appendix L

Estrategia formativa en Unity/Real-Time 3D y visualización técnica

Fuente: Research/deep-research-report_06.md

Seccion: research

Estado: research_snapshot

Rol: Research / estudio de mercado

Lineas originales: 114

Ultima modificacion: 2026-06-11 13:20

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Resumen Ejecutivo: Se analizan programas formativos (Rebelway, MOOCs, certificaciones Unity, másters) para un perfil colombiano orientado a Unity/3D en tiempo real. Rebelway ofrece cursos especializados en VFX y Unreal, con proyectos prácticos y certificación (p.ej. *Realtime FX for Games*, 25h/8 semanas, ~\$698). Los cursos online (Coursera, Udemy, YouTube) son baratos o gratuitos pero carecen de acreditación oficial. La especialización de Michigan State (Coursera) permit

Contenido completo depurado

Resumen Ejecutivo: Se analizan programas formativos (Rebelway, MOOCs, certificaciones Unity, másters) para un perfil colombiano orientado a Unity/3D en tiempo real. Rebelway ofrece cursos especializados en VFX y Unreal, con proyectos prácticos y certificación (p.ej. *Realtime FX for Games*, 25h/8 semanas, ~\$698). Los cursos online (Coursera, Udemy, YouTube) son baratos o gratuitos pero carecen de acreditación oficial. La especialización de Michigan State (Coursera) permite crear 4 juegos en Unity en ~2–6 meses. Unity ofrece certificaciones pagas (examen remoto, sin datos de precio público) para validar competencias. Másters relevantes incluyen el **MBVRIA** (Especialización en Visualización 3D en Tiempo Real, 12 meses online, Universidad butic/Unreal Engine, beca ~€8.000) y el **M.Sc. en Computación Gráfica, VR y Simulación** de U-tad (1 año presencial, 60 ECTS). Las rutas formativas combinan cursos básicos (0–3m), especializaciones prácticas (3–9m) y formación avanzada (9–24m) con proyectos finales (reels, demos WebGL) para maximizar empleabilidad. Se estima que con esta ruta el candidato podría alcanzar ingresos mensuales ~USD 1.5k al inicio (nivel entry colombiano), ~USD 3k en 1–2 años (junior/mid en remoto) y ~USD 6k en 2–3 años (senior o puestos internacionales). Los riesgos incluyen alta inversión de tiempo/coste en másters versus beneficio incierto frente a portafolios sólidos; por ello se prioriza una combinación de autoaprendizaje intensivo y cursos focalizados. Se recomienda presentar los cursos no certificados enfatizando proyectos y habilidades demostradas (incluir enlaces a demos y repos) en el CV/LinkedIn.

L.0.1 Metodología de evaluación

Se revisaron fuentes oficiales (páginas de proveedores, universidades, plataformas de aprendizaje) y testimonios verificados. Para cada programa se tabulan características clave (nombre, proveedor, modalidad, duración,

coste, reconocimiento profesional, requisitos, habilidades, evidencia de empleabilidad, tiempo estimado a retorno, recomendación para cada rol). Se consideraron las 9 familias de roles objetivo (desarrollador 3D, técnico de Unity, WebGL, visualización técnica, XR/AR/VR, herramientas/pipeline, AI, full-stack y 3D artist). Al no hallar datos abiertos (precios de másters, algunos costes de certificación), se marcó “no encontrado”. El ROI se estima contrastando costes de estudio con rangos salariales locales e internacionales. Además, se analizan recursos gratuitos vs formales para evidenciar habilidades ante reclutadores.

L.0.2 Comparativa de programas

Cursos Rebelway

Nombre	Proveedor	Modalidad	Duración	Coste (USD)	Reconocimiento	Requisitos	Habilidades clave	Empleabilidad (alumni)	Tiempo a ROI	Recomendación rol
Realtime FX for Games & Cinematics (Houdini + Unreal)	Rebelway	Online (8 sem.)	25+ h (8 sem.)	698	Certificado SideFX (curso oficial)	Nivel intermedio (Houdini)	VFX en tiempo real, partículas, simulación (Houdini+Unreal)	Egresados: trabajos en MPC, Blizzard	~6–12 meses (aprendizaje + portafolio)	Alta: 10,4,5; Baja: 3,6,5
Intro to Unreal Engine Beginner Game Design	Rebelway	Online (9 sem.)	25+ h (9 sem.)	698	Certificado SideFX (curso oficial)	Ninguno (principiantes)	Fundamentos Unreal Engine 4/5, entornos 3D, FX básicos	Egresados: MPC, Blizzard	~6 meses (aprendizaje básico)	Alta: 10,4,5; Baja: 3,6,5

Notas: Ambos cursos son 100% online y “SideFX Certified”. Rebelway destaca 5,000+ alumnos globales y #1 en encuestas de industria. Se enfocan en roles técnicos de 3D/VFX (desarrollador 3D, técnico-artista Unity, XR artístico) y menos en Python/AI o full-stack. Su ROI histórico es alto: un alumno logró un empleo en 6 meses.

Cursos en línea (Udemy/Coursera/YouTube)

Nombre / Ejemplo	Plataforma	Modalidad	Duración (ej.)	Coste	Reconocimiento	Requisitos	Habilidades clave	Empleabilidad	Tiempo a ROI	Recomendación rol
Unity Game Dev Specialization (Michigan StateU, Coursera)	Coursera	Online (flexible)	~2–4 meses (10h/sem)	~\$50/mes (suscripción)	Certificado Coursera (MSU)	Ninguno inicial	Diseño de juegos 2D/3D en Unity, C#, UI, prototipos	MSU, reconocido en EE.UU.	~3–6 meses (curso + proyectos)	Alta: 1,2,3,4,5,7,8,9; Media:4,5,7,8,9
C# Programming for Unity (Univ. Colorado, Coursera)	Coursera	Online (flexible)	~3–6 meses	~\$50/mes	Certificado Coursera (Colorado)	Básicos de programación	C# para juegos, patrones OOP, debugging Unity	Universidad acreditada	~4 meses (curso intensivo)	Alta: 1,2,3,4,5,7,8,9; Media:6,7,8,9
Cursos autodidactas Unity 3D (Udemy, YouTube)	Udemy/YouTube	Online (asíncrono)	Varios cursos (~20–50h c/u)	Gratis/\$10–100	Sin certificación oficial	Ninguno	Fundamentos Unity, scripting C#, prototipos 2D/3D	Depende de portafolio; no acreditan directamente	~0–3 meses (con curriculum)	Media:2; Baja (como sustento)

Notas: Los MOOCs de Coursera ofrecen contenido académico (MSU, Colorado) con certificado opcional (por suscripción). Udemy/YT son de bajo costo/gratis, *no* otorgan títulos, pero sirven para aprender habilidades iniciales. En el CV, lo ideal es mencionar proyectos concretos realizados y solo listar estos cursos bajo “Formación complementaria”. Su utilidad ante reclutadores es indirecta: más valen los resultados prácticos y repos en GitHub que el certificado.

Certificaciones Unity

Nombre	Proveedor	Modalidad	Duración	Coste USD	Reconocimiento	Requisitos	Habilidades clave	Empleabilidad	Tiempo a ROI	Recomendación rol
Unity Certified Associate: Game Dev	Unity/Pearson VUE	Examen online	90 min + prep	~USD 149 (aprox.)	Certificación oficial Unity	Haber construido proyectos Unity	Conocimientos Unity global (animación, física, UI, scripting)	Reconocido por empleadores de juegos	~3–6 meses (estudio + examen)	Alta: 1, 3, 6, 7, 8, 9, 10; Baja: 4, 5

Bloque completa

APPENDIX L. ESTRATEGIA FORMATIVA EN UNITY/REAL-TIME 3D Y VISUALIZACIÓN TÉCNICA

Notas: La certificación oficial Unity (Associate) valida habilidades generales de desarrollo de juegos en Unity. Se obtiene al aprobar un examen (90 min) en línea. Aunque no es un título académico, demuestra dominio técnico ante reclutadores. Sin embargo, es opcional: un portafolio sólido es más valorado. El costo exacto varía (no se publica; estimado ~150 USD) y se necesita preparación (Unity Learn, cursos propios o oficiales).

Másteres y posgrados

Nombre	Institución	Modalidad	Duración	Coste (USD)	Reconocimiento	Requisitos admisión	Temas / Skills	Empleabilidad (egresados)	ROI estimado	Recomendación rol
MBVRIA – Visualización 3D en Tiempo Real (UE)	butic The New School (ESP)	Online (12 meses)	12 meses, 430h	~no encontrado (becas €8.000)	Título propio (Unreal Engine)	Grado o experiencia en 3D/VFX	Unreal Engine, 3ds Max, Substance, VFX, IA aplicada	Convenios con empresas; formación oficial	1–2 años (Alta inversión)	Alta: 1,4,5; Media: 3,6; Baja: 7–9
M.Sc. en Comput. Gráf., VR y Simulación	U-tad (España) / UCJC (ESP)	Presencial (MAD)	1 año (60 ECTS)	~no encontrado	Máster Universitario oficial	Titulación universitaria; idioma inglés B2	Simulación, gráficos por computador, VR/AR, programación C++/Python	Colaboración en I+D, proyectos con empresas	1–2 años (Alta inversión)	Alta: 1,4,5; Media: 2,5,6; Baja: 3,6–9
(Opcional) M.Sc. en Visualización Arquitectónica	butic / Autodesk (ESP)	Online/híbrido	~1 año	no encontrado	Título propio (Autodesk/Unreal)	Grado/licenciatura	Archviz, VR, Unreal Engine	Orientado a arquitectura, equiparable a MBVRIA	1–2 años	Alta: 4,5; Media: 1,2; otros
PhD en Computación Gráfica / VR	Universidades (varias)	Presencial	~4 años	(becas/disertación)	Doctorado	Máster, méritos investigación	Investigación avanzada en VR/AR, gráficos 3D	Carrera académica/investigativa	4+ años	Media: 4- Baja:

Notas: Los másteres españoles citados son 1 año (60 ECTS). El MBVRIA de butic ofrece “título propio” certificado por Unreal Engine y Autodesk, con amplia cobertura de herramientas 3D e IA, y está orientado a roles de Visualización Técnica y producción en tiempo real. U-tad (centro oficial de Universidad Camilo José Cela) forma expertos en simulación y VR aplicada a ingeniería y VFX. Los programas presenciales suelen tener tasas elevadas (el MCGRVS de U-tad ronda los €9.000/año). El retorno de inversión de un máster es a largo plazo (1–2 años al menos), pero mejora credibilidad y red profesional. En cambio, un PhD (no prioritario aquí) lleva años y está enfocado a investigación más que desarrollo práctico inmediato.

L.0.3 Cursos sin certificado y recursos gratuitos

Los cursos no certificados (Udemy, YouTube, documentación oficial de Unity/Unreal) son útiles para aprender y mejorar el portafolio, pero ofrecen **poco reconocimiento formal**. Los reclutadores prefieren **proyectos demostrables** antes que sellos de cursos gratuitos. Es aconsejable:

- En el CV/LinkedIn incluirlos en “Formación complementaria” y remarcar habilidades adquiridas.
- Enfatizar **links a demos/repositorios** creados durante esos cursos (videojuegos pequeños, demos WebGL, entornos VR, etc.).
- Obtener insignias digitales (p.ej. Coursera) o microcredentials, para añadir valor.

L.0.4 Ruta formativa priorizada

Se propone un plan escalonado con hitos y entregables:

Bloque de código: mermaid
timeline

```

title Ruta formativa (Junio 2026 - 2028)
2026-06-11 : *Inicio del plan:* Autoaprendizaje básico (C#, Unity Engine, modelado 3D básico)
2026-09-01 : Finalización de cursos introductorios (Unity/Unreal en Udemy/Coursera)
2026-12-01 : Certificado Unity Associate (prep intensiva + examen), primer demo WebGL interactiva
2027-03-01 : Curso avanzado (Rebelway o similar) + portafolio (Proyecto 3D técnico)
2027-09-01 : Especialización adicional (XR, Python/AI para real-time), segundo demo/presentación
2028-03-01 : (Opcional) Inicio de máster o bootcamp intensivo
2028-12-01 : Proyecto final amplio (juego/visualización compleja), preparación para roles senior

```

Bloque de código: mermaid
flowchart LR

```

A[Inicio: Fundamentos 3D/Unity] --> B[Cursos online (Udemy, YouTube)]
B --> C[Certificación Unity Associate]
C --> D[Proyecto 1: Demo WebGL básica]
D --> E[Cursos avanzados (Rebelway/Coursera especializaciones)]
E --> F[Proyecto 2: Entorno 3D complejo o minijuego]
F --> G[Opcional: Máster / Bootcamp presencial]
G --> H[Proyecto 3: Portafolio completo (demo VR/AR o simulación)]
H --> I[Empleo Junior (<$1.5k/mes)]
I --> J[Mid-level ~2-3 años (~$3k/mes)]
J --> K[Senior+ (>$6k/mes, roles remotos/internacionales)]

```

Entregables clave: al final de cada fase, tener al menos un proyecto funcional (juego Unity, demo WebGL, visualización VR) con repositorio público. En 0–3m: manejo de Unity básico y C#, un demo sencillo; 3–9m: varios prototipos (2D/3D) y reel VFX pequeño; 9–24m: proyecto mayor (por ejemplo, simulador VR, juego completo o gemelo digital). Estos trabajos serán la evidencia principal ante empleadores/clientes.

L.0.5 Costes estimados y ROI

- **Coste total formativo:** incluidas certificaciones (~\$150) y cursos (Rebelway ~\$700), más posible máster (~€8k–€9k). Un bootcamp/Rebelway + algunos MOOC podría sumar ~\$1k–2k. Un máster eleva el coste a >\$10k.
- **Ingresos esperados:** en Colombia un desarrollador junior gana ~\$24k–42k/año (~USD 2k–3.5k/mes), un mid \$41k–66k (~3.5k–5.5k/mes), senior \$61k–92k (~5–7.5k/mes). De forma remota en 2026, el promedio es ~\$59k/año (~\$4.9k/mes).
- **Escenarios de ROI:**
 - 0–6m: con formación autodidacta puede obtener un contrato junior (~\$1.5–2k/mes locales) si ya tiene portafolio básico.
 - 6–18m: tras certificaciones y cursos avanzados, puede aspirar a puestos mid-level ~\$3k/mes (~USD 36k/año), amortizando los gastos iniciales.

- 18–36m: con experiencia y proyectos consolidados (o un máster), aspiraciones senior/internacionales ~\$6k/mes (USD 72k/año) son viables en el mercado global.

En general, el retorno dependerá de la demanda del rol específico y la capacidad para demostrar habilidades (por eso los proyectos y certificaciones importan).

L.0.6 Riesgos y alternativas

- **Riesgo Máster:** alta inversión de tiempo y dinero con retorno incierto en el corto plazo (especialmente fuera de España). Alternativa: cursos cortos o bootcamps (p.ej. Rebelway) que son más rápidos y enfocados en práctica.
- **Riesgo Bootcamp:** menor prestigio académico; cubren áreas específicas (VFX o juego). Necesitan ser complementados con auto-estudio amplio.
- **Autodidacta:** es la opción menos costosa, pero exige disciplina. Es posible quedar rezagado sin dirección. Se recomienda combinar con cursos estructurados (MOOCs) y comunidades (GitHub, foros).
- **Financiación:** buscar becas (MBVRIA ofrece hasta €8000), pagos fraccionados (Rebelway 3 cuotas) y recursos gratuitos (Unity Learn, libros, vídeos).

L.0.7 Recomendaciones para el CV/LinkedIn

- Destacar **proyectos propios:** videos de juegos o demos, enlaces a repos (GitHub, Itch.io) con descripciones claras.
- Incluir **certificaciones oficiales** (Unity Certified) destacadas.
- Listar **cursos relevantes** en formación, pero enfatizar el **resultado** (“Curso de Unity – proyecto X realizado”). No detallar cursos gratuitos insignificantes.
- Subrayar tecnologías: “Unity 3D, C#, Unreal Engine, Houdini, Blender, VR/AR, Git, Python”, etc.
- Mencionar **idiomas** si aplica (p.ej. inglés avanzado).

Fuentes: Sitios oficiales de cursos y universidades, testimonios de exalumnos, y datos salariales regionales. († Los precios se basan en información pública; marcamos “no encontrado” donde no es divulgada).

Appendix M

Resumen Ejecutivo

Fuente: Research/deep-research-report_11.md

Seccion: research

Estado: research_snapshot

Rol: Research / estudio de mercado

Lineas originales: 148

Ultima modificacion: 2026-06-11 13:54

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

- **Listas de empresas objetivo:** Se compilan compañías que contratan perfiles Unity Developer, Technical Artist, Technical Visualization Engineer, Digital Twin Engineer, XR Engineer y AI Tooling Engineer, distinguiendo entre posiciones **remotas, por contrato (freelance/B2B)** y **locales (presenciales)** en Europa (especialmente Portugal y Alemania) y Colombia. Cada empresa se prioriza según: adecuación técnica del rol, idioma requerido y factibilidad de contratación (

Contenido completo depurado

Se presenta una estrategia de búsqueda activa orientada a un perfil Senior en Unity/Technical Artist/Visualization/XR/AI-tooling, enfocado en posiciones remotas (globales), por contrato (freelance) y empleos locales en Europa (Portugal, Alemania) y Colombia. Se identifican empresas objetivo, portales de empleo y agencias especializadas, así como comunidades profesionales para networking. Se proponen cronogramas de contactos (30/60/90 días), métricas clave de éxito (aplicaciones, entrevistas, ofertas) y plantillas de outreach. Además, se analizan riesgos jurídicos (clasificación laboral, fiscalidad, EOR) con recomendaciones concretas.

M.1 Objetivos

- **Listas de empresas objetivo:** Se compilan compañías que contratan perfiles Unity Developer, Technical Artist, Technical Visualization Engineer, Digital Twin Engineer, XR Engineer y AI Tooling Engineer, distinguiendo entre posiciones **remotas, por contrato (freelance/B2B)** y **locales (presenciales)** en Europa (especialmente Portugal y Alemania) y Colombia. Cada empresa se prioriza según: adecuación técnica del rol, idioma requerido y factibilidad de contratación (e.g. ciudadanía portuguesa futura, ruta de visado alemana).
- **Metodología de búsqueda:** Se usan portales oficiales y reconocidos (LinkedIn, Glassdoor, Indeed, RemoteOK, WeWorkRemotely, AngelList, Qubit Labs, etc.), además de sitios de la industria (ArtStation, Stack Overflow Jobs, Job Boards especializados en juegos y tecnología). Se consultan fuentes en español, inglés, portugués y alemán cuando estén disponibles. Las ofertas se extraen directamente de páginas corporativas o via CV recruiters (para citas de requisitos y localización).

M.2 Metodología

1. **Fuentes principales:** Prioridad a fuentes oficiales y bolsas de empleo: sitios de carreras de empresas, LinkedIn, Glassdoor, Indeed, RemoteOK, WeWorkRemotely, AngelList, InGameJob, etc. Se utilizan también informes de salario (Howdy, Howmuch.net, Lemon.io) para estimar rangos salariales por ubicación.
2. **Criterios de selección:** Se filtran roles que mencionen Unity/C#/VR/AR/XR/Digital Twin/AI Tools en descripciones. Se prefieren empresas que sean compatibles con trabajo remoto o que tengan presencia en las regiones objetivo. Se registra idioma en la oferta (e.g. inglés, español, portugués, alemán). Se estima encaje ("probabilidad alta/med/baja") basándose en requisitos técnicos y soft skills del candidato.
3. **Extracción de datos:** Para cada oferta relevante se extraen: nombre de la empresa, ubicación (país/sede), modalidad (remote/contractor/local), link a la oferta o página de carreras, requisitos clave (tecnologías, experiencia, idiomas), rango salarial (cuando es indicado o estimado), idioma del puesto, nota sobre movilidad/visa. Si no hay cifras salariales, se dan rangos promedio con fuente (p.ej. Howdy, Lemon.io).
4. **Reclutadores y comunidades:** Se identifican agencias de selección especializadas (p.ej. Qubit Labs, Remote Staff, GetonBoard) y redes profesionales (Discord y Slack de Unity/XR, meetups locales, comunidades de exalumnos de ingeniería). Se preparan plantillas de contacto adaptadas (contractor vs empleo fijo) para LinkedIn y email, enfocados en destacar adaptabilidad remota y visión internacional.
5. **Roadmap 30/60/90 días:** Se detalla un plan de actividades (ver diagrama), con metas cuantificables (número de aplicaciones enviadas, entrevistas agendadas, ofertas recibidas). Incluye seguimiento (follow-ups) y ajuste de estrategia tras cada fase.
6. **Riesgos y mitigación:** Se analizan riesgos de *misclassification* (ser contratado como empleado vs contratista), obligaciones fiscales (IVA, seguridad social) y estructuras tipo *EOR* (Employer of Record) para empleo remoto internacional. Se dan recomendaciones (usar contratos por proyecto claros, plataformas de pago de nómina locales, etc.).

M.3 Empresas objetivo por rol y modalidad

A continuación, para cada rol clave se presenta una tabla con empresas seleccionadas, país/sede, modalidad, link, requisitos clave, salario estimado, idioma requerido, probabilidad de encaje y notas de movilidad/visa:

M.3.1 Unity Developer

Empresa	País (sede)	Modalidad	Link (oferta/careers)	Requisitos clave	Salario / Tarifa	Idioma	Encaje	Movilidad / Visa
Avataris GmbH	Austria (Augsburg)	Remoto (Freelance, EMEA)	[Oferta Unity Dev (remota)](https://remotegamejobs.com/jobs/avataris-gmbh-unity-developer-remote-job-13e6422f-22af-4181-988a-ca2fb72a15a5)	Unity, C#, experiencia con IA/avatares, colaboración con artists, inglés avanzado	No especificado (startup AI)	Inglés	Alta	Remoto (no se requiere visa; posición EMEA)
MarcoPolo Learning	EUA (Global)	Remoto (Full-time)	[Oferta Unity Dev (remota)](https://remoteok.com/remote-jobs/86329-remote-unity-developer-marcopolo-learning)	Unity, C#, experiencia móvil iOS/Android, CS background	~\$70k–120k/año (estimado)	Inglés	Alta	Remoto (EEUU; requiere estatus nómada)
HyperVR Games	Letonia (Riga)	Remoto (Full-time)	[LinkedIn: Senior VR Unity Dev](https://gamejobhunter.com/offer/hypervr-senior-vr-developer)	5+ años Unity/C# en VR, pipeline de juego, frameworks VR (OpenXR, etc.), inglés	No especificado (lea negociación)	Inglés	Alta	Remoto (acepta candidatos en EU)
Arkadium (Gryphon)	Portugal (Lisboa)	Local (Híbrido)	[Empleos Arkadium Lisboa](https://pt.indeed.com/ofertas?q=Unity+Developer&l=Lisboa)	Desarrollo de juegos, C#/.NET, UI/UX, colaborativo	~€40k–60k/año (senior en PT)	Inglés/Portugués	Media	Establecimiento local; visa UE ideal
Ubisoft Berlin	Alemania (Berlin)	Local (Onsite)	[Careers Ubisoft Germany](https://www.ubisoft.com/en-us/careers)	Proyectos AAA, Unity/C++, experiencia multijugador	~€60k–80k/año (senior)	Inglés/Alemán	Media	Onsite (requiere CD c/alemán)

M.3.2 Technical Artist

Empresa	País (sede)	Modalidad	Link (oferta/careers)	Requisitos clave	Salario / Tarifa	Idioma	Encaje	Movilidad / Visa
SKC Games Studio	Reino Unido / UE	Remoto (Full-time)	[Oferta Technical Artist (remota)](https://www.workwithindies.com/careers/skc-games-studio-technical-artist)	3–5 años como Technical Artist en Unity, shaders, pipelines, optimización	No indicado (s/rango UK indie)	Inglés	Alta	Remoto (solo UK/EU); si no, visado si UE/apto con pasaporte (UE)
People Can Fly	Polonia (Varsovia)	Local (Onsite)	[Careers PCF – Senior TA (Warsaw)](https://careers.smartrecruiters.com/peoplecanfly)	Artista técnico sénior, Unity/C#, Houdini/Procedural, AAA gaming	~€50k–70k/año (senior en Polonia)	Inglés	Media	Onsite (permiso EU, inglés)
Yggdrasil Gaming	Polonia (Cracovia)	Local (B2B contractor)	[Builtin Tech Artist (Cracovia)](https://builtin.com/job/technical-artist-b2b-agreement/3636641)	Bridge arte/desarrollo, Photoshop, Spine, optimización 2D, 2+ años	No disponible (contrato B2B)	Inglés	Media	Contrato freelance (posible EOR); necesidade contrato local
Artifex Mundi	Polonia (Katowice)	Local	[Artifex Careers](https://www.artifexmundi.com/careers/)	Pipeline de arte, Unity tools, UI, experiencia 3D/2D	~50k zł–70k zł (senior PL)	Inglés/Polaco	Media	Onsite (permiso de trabajo, PLN)

M.3.3 Technical Visualization Engineer

Empresa	País (sede)	Modalidad	Link (oferta/careers)	Requisitos clave	Salario estimado	Idioma	Encaje	Movilidad / Visa
EP Group (EngineeringPeople)	Alemania (Dornstadt)	Remoto/Contrato	[Oferta Visualization Engineer](https://www.ep-group.de/karriere)	Experiencia en CAD/3D, visualización técnica, Python, Teamcenter	€55k–70k/año	Alemán/Inglés	Media	Puede requerir reubicación local; visa UE
Jacobs Engineering	Alemania	Local (Onsite)	[Jacobs Careers](https://jobs.jacobs.com)	Ingeniería de simulación, 3D CAD (CATIA/Creo), Visualización	€60k–80k/año	Inglés/Alemán	Media	Onsite (permiso de trabajo UE)

Nota: Los roles de *Technical Visualization* suelen requerir formación en ingeniería/cad, manejo de herramientas CAD/PLM y colaboración en proyectos de visualización industrial. Las cifras salariales son aproximadas (p.ej. ingreso promedio en Alemania) ya que las ofertas concretas raramente especifican salarios.

M.3.4 Digital Twin Engineer

Empresa	País (sede)	Modalidad	Link (oferta/careers)	Requisitos clave	Salario estimado	Idioma	Encaje	Movilidad / Visa
Next Ventures	Alemania (Walldorf)	Contrato	[Oferta Digital Twin (contract)](https://next-ventures.us/jobs/manufacturing-simulation-digital-twin-production/)	Creación y mantenimiento de Gemelo Digital, SAP/PLM, simulación de planta, análisis de flujo de producción	€70k–100k/año (consultoría DE)	Inglés/Alemán*	Media	Contrato (Germania); necesita permiso UE*
Luxoft (DXC)	Alemania (Remoto)	Remoto	[Luxoft Careers (Digital Twin)](https://career.luxoft.com/jobs)	Simulación, Python/Java, integración MES/MOM, IoT, PLM	~€60k–80k/año (senior)	Inglés	Media	Remoto (posible IR35; visa UK/EU)
Siemens Digital Industries	Alemania	Híbrido/Remoto	[Siemens Jobs – Digital Twin](https://jobs.sw.siemens.com)	Experiencia en simulación 3D, modelado de sistemas, C/C++/Python, PTC/Teamcenter	~€65k–90k/año	Inglés/Alemán	Alta	Híbrido (permiso UE o EOR; visa según contrato)

Nota: **Luxoft y Siemens** frecuentemente abren roles de simulación y gemelos digitales. Se requiere experiencia en herramientas de PLM/PLT (Teamcenter, VRED, PTC, etc.), lenguajes de scripting y entornos industriales. En portales de empleo suelen aparecer como “Simulation Engineer” o “Digital Twin Consultant”. La oferta de Next Ventures muestra el tipo de responsabilidades (gestión modelo planta, cálculos de producción).

M.3.5 XR Engineer / AR/VR Engineer

Empresa	País (sede)	Modalidad	Link (oferta/careers)	Requisitos clave	Salario estimado	Idioma	Encaje	Movilidad / Visa
Bertrandt (Ford)	Reino Unido (Laindon)	Contrato (IR35)	[Oferta XR Engineer (grad 2026)](https://www.brightnetwork.co.uk/graduate-jobs/bertrandt-uk-ltd/xr-engineer-extended-reality-laindon-2026)	Experiencia con Unreal/VRED/CAD, visualización 3D, datos (PowerBI), Python/C++, sistemas XR (HMDs)	~£30k–40k/año (junior)	Inglés	Media	Contrato UK (IR35, se exige elegibilidad UK)
Varjo	Finlandia (Espoo)	Remoto híbrido	[Varjo Careers (VR jobs)](https://varjo.com/jobs)	Hardware VR, Unity/Unreal, C#/C++, renderizado y tracking	~€50k–70k/año	Inglés	Media	Remoto (permiso Finlandia UE)
Avantis Education	Reino Unido (Gloucester)	Híbrido	[LinkedIn: Senior XR Engineer](https://www.linkedin.com/jobs/search/?keywords=XR%20Engineer)	Unity, XR (VR/AR), plugins VR/handtracking, C#/NET	~£35k–45k (senior UK)	Inglés	Media	Onsite (permiso UE en futuro; Brexit)
Niantic (Proximal Labs)	EE.UU. (Remoto EU)	Remoto	[Niantic Careers – XR Roles](https://nianticlabs.com/jobs)	ARKit, Unity/Unreal, backend servicios geo, Python/Go	~\$90k–130k/año	Inglés	Baja	Remoto (visa EE.UU., usar EOR/contrato)

Nota: Los roles XR engloban desde desarrollo de hardware (campos HMD) hasta software de realidad extendida. Bertrandt ofrece un puesto de *XR Engineer* en el sector automotriz. Compañías como **Varjo** (hardware VR finlandés) o integraciones de **Unreal/Unity con aplicaciones industriales** suelen buscar estos perfiles. Niantic se incluye como ejemplo líder AR; sin embargo, el candidato debería gestionar la visa (EEUU) o apuntar a EOR.

M.3.6 AI Tooling Engineer

Empresa	País (sede)	Modalidad	Link (oferta/careers)	Requisitos clave	Salario estimado	Idioma	Encaje	Movilidad / Visa
team.blue Global	Bélgica/EU (Remoto)	Remoto (Full-time)	[Oferta AI Tooling Engineer](https://careers.team.blue/jobs/7837774-ai-tooling-engineer)	5+ años dev plataformas internas, experiencia con agentes IA (Claude, GPT), devtools, API, scripting, docs	~€80k–100k/año (senior EU)	Inglés	Alta	Remoto EU (sin visado, elegibilidad laboral)
Supabase	Remoto Global (EMEA)	Remoto (Full-time)	[JobGether: AI Tooling Engineer](https://jobgether.com) (agregador)	TypeScript/JavaScript, IA/ML, desarrollo de APIs, devops para IA	~\$70k–90k/año (EMEA)	Inglés	Media	Remoto (contrato global)
Canonical	Reino Unido	Remoto (UK/EU)	[Canonical Careers – AI Roles](https://canonical.com/careers)	Montaje de plataformas AI, Python/C++, Kubernetes, herramientas MLOps	~£50k–70k/año UK	Inglés	Baja	Remoto (sin paquete visa; UE pronto)

Nota: **team.blue** es un ejemplo concreto de startup europea buscando IA interna. En general, el perfil requiere fuerte expertise en frameworks de agentes IA (p.ej. Claude, LangChain), lenguajes de scripting (Python, TypeScript) y desarrollo de herramientas internas. Las empresas suelen pedir nivel senior y trabajo remoto. Se debe comprobar si requieren ciudadanía local (team.blue no patrocina visas).

M.4 Recruiters, Agencias y Comunidades

- **Recruiters/Agencias especializadas:** Se recomienda contactar agencias focalizadas en tecnología e ingeniería (p.ej. **Qubit Labs**, **Hays IT**, **Michael Page** especializado TI) y plataformas de freelance para Unity/AI (como **Lemon.io**, **Toptal**, **Upwork**). Para contratistas remotos en Europa: **Malt** (contratos B2B), **Toptal** (talento freelance). Para EE.UU. remota: considerar **RemoteOK**, **WeWorkRemotely**, y sitios nicho en gaming/VR.
- **Comunidades y redes profesionales:** Únase a servidores de Discord y Slack como *Unity Developer Community*, *GameDev Slack*, *XR Bootcamp Slack*, *XR/AR/MR Meetup Groups* (p.ej. XR Brussels/Barcelona, Unity Lisbon Meetup) y congresos virtuales (GDC, AWE). También grupos de LinkedIn (ej. *VR/AR Association*, *AI developers*). Participar en meetups online relacionados con Realidad Virtual, Digital Twins, AI.
- **Plantillas de outreach:** Adaptar mensajes según modalidad:
- **Contratista:** “Hola [Nombre], soy desarrollador Unity con X años en VR/AI. Me interesa colaborar vía proyecto contract, apporto [skill clave] para [tipo de proyectos]. ¿Podemos coordinar llamada para explorar necesidades?”
- **Empleo local:** “Estimado [Nombre], como profesional con experiencia en [tecnología] busco oportunidad de empleo en [ubicación]. Poseo pasaporte PT (en 2 años) y alto dominio de [idioma]. Adjunto CV/portfolio. ¿Hay openings en [empresa] para mi perfil?”.
- Para LinkedIn InMail: breve y personalizado, mencionando mutuas conexiones o interés específico (e.g. un proyecto de la empresa).

M.5 Estrategia 30/60/90 días y métricas

Bloque de código: mermaid
gantt

```

title Plan de 90 días - Búsqueda Activa
dateFormat YYYY-MM-DD
section Preparación inicial
  Actualizar CV/Portfolio :done, 2026-06-12, 7d
  Investigar empresas objetivo :done, 2026-06-19, 7d
section Primer mes (30d)
  Envío de aplicaciones (fase I) :active, 2026-06-26, 14d
  Outreach LinkedIn/recruiters :2026-06-26, 14d
  Métricas meta: 30-50 apps, 10 entrevistas :2026-06-26, 14d
section Segundo mes (60d)
  Seguimiento / Networking :2026-07-10, 14d
  Segunda ronda de aplicaciones: 2026-07-24, 14d
  Métricas meta: +50 apps, 20 entrevistas, 2 ofertas :2026-07-10, 28d
section Tercer mes (90d)
  Refinar estrategia y CV :2026-08-01, 7d
  Tercera ronda/apply final :2026-08-08, 21d
  Preparación entrevistas finales :2026-08-29, 2d
  Métricas meta finales: 100 apps, 30 entrevistas, 5 ofertas :2026-08-01, 30d

```

- **Scripts y seguimiento:** Tras aplicar, enviar email/LinkedIn de seguimiento a los 7–10 días si no hay respuesta. Mantener registro (empresa, fecha, contacto). Para cada entrevista, enviar agradecimiento y reiterar interés.
- **Métricas clave:** Número de aplicaciones (ideal >30/mensual), entrevistas obtenidas (1 de cada ~10 apps), ofertas generadas (>3 en 90 días). Ajustar foco según resultados (p.ej. enfocar más en remote si local no responde).

M.6 Riesgos y mitigación

- ****Malaprendizaje laboral (*misclassification*):**** Asegurarse de la modalidad contractuales. Para contratos freelance/B2B en Colombia o Portugal, clarificar que no hay relación laboral subordinada (usar facturación por servicios, evitar depender de un solo empleador).

- **Fiscalidad e impuestos:** Si se trabaja como contratador remoto, revisar diferencias fiscales: ej. si factura desde Portugal con pasaporte PT, tributa según ley portuguesa; en Colombia si residencia fiscal local, considerar declarar renta/IVA. Se recomienda conversar con contador especializado en expatriados.
- **Plataformas EOR (Employer of Record):** Para empresas que requieren presencia legal (p.ej. EE.UU. u UE), usar agencias EOR (Deel, Remote.com, OysterHR) que permitan contratación internacional legal. Esto evita necesidad de crear entidad propia. Ventaja: simplifica nómina y compliance, pero con comisión.
- **Licencias y NDA:** En proyectos de simulación o Defensa/industria, puede haber NDA/contratos especiales. Leer con cuidado.

M.7 Activar “Modo Agente” y entregables esperados

Cuando sea necesario recolectar masivamente ofertas o contactos, se usará un agente (herramienta de scraping) para extraer datos estructurados. Los entregables típicos serían archivos CSV con columnas: Empresa, Título del Puesto, Enlace, País, Modalidad, Requisitos principales, Fecha de publicación, Contactos (RR.HH./recruiters), etiquetas (p.ej. Unity/XR). Estos se pueden filtrar y actualizar periódicamente (p.ej. cada semana).

M.8 Diagramas de flujo

Bloque de código: mermaid
 flowchart LR
 A[Iniciar: Perfil preparado] --> B[Identificar empresas objetivos]
 B --> C[Enviar solicitudes (CV + carta)]
 C --> D[Enviar correo a reclutadores/agencias]
 C --> E[Networking (LinkedIn, Slack, Meetups)]
 D --> F[Seguimiento (Follow-up \u00e1gil)]
 E --> F
 F --> G[Entrevistas iniciales]
 G --> H[Entrevistas técnicas finales]
 H --> I[Recepción ofertas / negociación]

M.9 Conclusiones

La estrategia combina búsqueda proactiva (mercado abierto) con networking focalizado. Se priorizan empresas que ya contratan perfiles similares (por ejemplo, startups de IA/VR en EU y grandes industriales en DE) y se adaptan los mensajes según modalidad (remoto vs local). El candidato debe usar su futura ciudadanía portuguesa y la ruta de visado alemana como ventajas competitivas (evitar barreras de inmigración). Las tablas anteriores sirven de guía inicial de compañías y requisitos; se deben actualizar periódicamente con nuevas ofertas. Finalmente, se recomienda documentar todo el proceso (cronograma, respuestas, entrevistas) para evaluar y ajustar la estrategia en cada fase.

Fuentes: Se citan sitios de empleo y perfiles de trabajo relevantes, p.ej. ofertas de Unity/VR a través de RemoteOK, anuncios de Technical Artist en WorkWithIndies, portales de empresas (team.blue, Next Ventures, Bertrandt), y datos salariales publicados (Howdy, Lemon.io). Estas referencias ilustran requisitos y condiciones actuales del mercado.

Appendix N

13. Entrevistas técnicas, negociación y riesgos de oferta

Fuente: Research/deep-research-report_13.md

Seccion: research

Estado: research_snapshot

Rol: Research / estudio de mercado

Lineas originales: 172

Ultima modificacion: 2026-06-11 14:33

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Resumen ejecutivo: Este informe brinda pautas prácticas para preparar entrevistas técnicas y negociaciones de oferta adaptadas al perfil de un candidato colombiano especializado en Unity/3D interactivo con portafolio TwinSight X500. Cubre prácticas actuales de entrevistas (preguntas y pruebas típicas), rangos salariales por región (LATAM, EE.UU., UE, Alemania, Portugal), estrategias de negociación como contratista frente a empleado (tarifas, impuestos, beneficios, NDAs, I

Contenido completo depurado

Resumen ejecutivo: Este informe brinda pautas prácticas para preparar entrevistas técnicas y negociaciones de oferta adaptadas al perfil de un candidato colombiano especializado en Unity/3D interactivo con portafolio TwinSight X500. Cubre prácticas actuales de entrevistas (preguntas y pruebas típicas), rangos salariales por región (LATAM, EE.UU., UE, Alemania, Portugal), estrategias de negociación como contratista frente a empleado (tarifas, impuestos, beneficios, NDAs, IP, no competencia), y un checklist legal/riesgo (facturación internacional, IVA/IRPF, permisos de trabajo, export control, firmas electrónicas, cláusulas contractuales). Se identifican señales de alerta en ofertas de trabajo, se proponen ejemplos de preguntas para entrevistas técnicas y gerenciales, y se ofrece orientación sobre cómo presentar pruebas técnicas y portafolio (incluyendo métricas clave y video demo). También se sugieren estrategias de negociación salarial según la prioridad de las compañías (A/B/C), ejemplos de cláusulas contractuales con texto modelo breve, y recomendaciones operativas finales (scripts de respuesta, timing, criterios de aceptación/rechazo).

N.0.1 1. Entrevistas técnicas: preparación y formato

En el perfil buscado, las entrevistas técnicas combinan preguntas teóricas, ejercicios prácticos (código en vivo o take-home projects), diseño de sistemas y revisión de código. Por ejemplo, Tokio School señala que los ejercicios habituales incluyen **preguntas teóricas**, resolución de algoritmos en vivo, diseño arquitectónico de sistemas y revisión de código existente. En roles Unity/3D interactivo se suman preguntas específicas del motor (objetos, físicas, animaciones) y tareas prácticas sobre optimización o automatización. Es común la siguiente secuencia:

- **Contacto inicial (screening):** repaso de CV/portafolio.

- **Entrevista técnica:** preguntas de C# y algoritmos (p. ej. OOP, estructuras de datos) o sobre el motor Unity (por ejemplo, ciclo de vida de un *GameObject*, funcionalidad de `Update()` vs `FixedUpdate()`, manejo de *Prefabs*, tipos de *Colliders*, etc.). Adaface documenta estas preguntas típicas: “¿Qué es un *GameObject* en Unity y cómo se relaciona con un Componente?”, “¿Qué propósito tienen las funciones `Update()` y `FixedUpdate()`?”, “¿Cómo se crea un *Prefab* y por qué son útiles?”, “¿Cuáles son los tipos principales de *Collider* en Unity y para qué sirven?”, entre otras.
- **Prueba práctica:** puede ser un *take-home* (pequeño prototipo o reto de optimización) o un ejercicio de codificación en vivo (por ejemplo, escribir un script en C# para resolver un problema concreto, diseñar un pipeline sencillo en Unity o escribir un shader básico). En algunos casos se hace *pair programming* en línea o revisión de código preexistente.
- **Entrevista con líderes:** discusión de arquitectura, retroalimentación sobre el portafolio, preguntas de diseño de sistemas o pipeline (por ejemplo: “¿Cómo estructurarías un pipeline de importación de assets o de simulación?”). También se pueden hacer preguntas situacionales o de experiencia (“Describe un reto de rendimiento que hayas resuelto”).
- **Entrevista HR / cultural:** preguntas generales (motivación, disponibilidad, expectativas salariales).

En resumen, el candidato debe prepararse para resolver en directo algoritmos y problemas de programación (como se señala en) y al mismo tiempo demostrar conocimiento profundo de Unity y la cadena CAD→gráficos en tiempo real. Se recomienda practicar con ejemplos de preguntas Unity/TA (ver ejemplos arriba) y repasar ejercicios de optimización (por ejemplo, manejo eficiente de miles de *GameObjects*, uso de *coroutines* vs *threads*, perfilado de escena, etc.). Además, es útil repasar conceptos de gráficos (Shader Graph, iluminación, VR-specific optimizations) y buenas prácticas de pipeline (automatización en Unity Editor, uso de Git y resolución de conflictos).

N.0.2 2. Rangos salariales por región (USD/mes aproximados)

A continuación se resumen rangos salariales brutos mensuales aproximados para un perfil Unity/Real-Time 3D, según región de contratación:

Región/ubicación	Rango ~\$ USD/mes	Fuente
Latinoamérica (remoto)	~3.000 – 5.000	Datos contractor
EE.UU. (remoto)	~6.000 – 12.000	Datos contractor
Europa (remoto, W. Europa)	~5.000 – 9.000	Estimación (EE.UU. -10/20%)
Alemania (on-site)	~4.500 – 6.500 (~50k–75k €)	Freelancermap
Portugal/España (on-site)	~2.500 – 4.000 (~20k–35k €)	Freelancermap

- **Latinoamérica (remoto):** Un perfil senior en Latinoamérica cobra típicamente **\$35/hr** (~\$5.6k/mes), rango medio \$27–47/hr.
- **EE.UU. (remoto):** Según datos, un Senior Unity dev EE.UU. cobra mediana **\$57/hr** (~\$9.1k/mes); en top markets llega a \$72–89/hr (~\$13–15k/mes).
- **Europa (remoto):** Los salarios remotos en Europa Occidental suelen ser algo menores que en EE.UU. (por ejemplo, 5–10% menos). Para puestos equivalentes estimamos ~\$5–9k/mes.
- **Alemania:** Según Freelancermap, un Unity developer gana **50–75k € al año** (aprox \$4.4k–\$6.6k/mes).
- **Portugal/España:** En España/Portugal los rangos suelen ser 20–35k € (~\$2.3k–\$3.9k/mes).

Estos valores son brutos y dependen de experiencia y tipo de empresa. En general, a un desarrollador Unity remoto en EE.UU. se le puede pagar el doble que a uno en LATAM, mientras que la contratación en Europa (especialmente Europa del Este/LatAm) ofrece ahorros para la empresa sin pérdida de calidad. Las cifras anteriores ayudan a fijar expectativas salariales: por ejemplo, una oferta de **\$5k/mes** para puesto remote EE.UU. sería baja; **>\$7k/mes** está en rango competitivo para un perfil senior, mientras que en LATAM se podría considerar aceptable desde \$3–4k.

N.0.3 3. Negociación: contratista vs empleado

- **Estatus y tarifa:** Como contratista independiente (freelance/consultor), el candidato debe negociar una tarifa **por día u hora** que cubra su salario neto esperado más el costo de impuestos y beneficios. Según el IRS de EE.UU., al empleador *no se retiene ni paga impuestos* por servicios de un contratista, a diferencia

de un empleado. Por tanto, la tarifa bruta que acuerde el contratista debe incluir la parte de salud/seguridad social y beneficios que una empresa normalmente asumiría (en EE.UU. ~15.3% de Seguro Social/Medicare, en Colombia aportes independientes, etc.). Como regla, una tarifa diaria de freelance suele equivaler a ~2–3 veces el prorrateo diario de un salario corporativo, para compensar la falta de prestaciones.

- **Impuestos y facturación:** El contratista debe encargarse de sus impuestos. En Colombia, las prestaciones de servicios exportadas usualmente llevan **retención en la fuente** (~10% sobre el pago) pero no IVA (0% por exportación de servicios). En España/UE, la factura a cliente empresarial UE se emite sin IVA (inversión del sujeto pasivo); a cliente fuera de la UE (exportación) suele ser exenta de IVA. Se debe verificar el régimen fiscal local: en Colombia puede requerirse inscripción como exportador y declarar renta por esos ingresos (retención del 10%). El contratista debe informar al cliente (p. ej. con factura internacional) su estatus fiscal.
- **Beneficios y seguridad:** Como empleado se recibirían seguro médico, vacaciones pagadas, cotizaciones sociales, etc. Como contratista no; por eso la negociación de la tarifa debe compensar (o bien solicitar una cláusula de pago adicional para cubrir esos beneficios). En EE.UU. por ejemplo, un contratista 1099 no recibe 401k ni seguro, por lo que la tarifa suele ser mayor. En Europa se consideran aportes de autónomo vs empleado.
- **Duración y exclusividad:** El contrato independiente suele ser de corto plazo (proyecto o por horas) y sin prestaciones de preaviso amplio. Un empleado tiene, según país, un periodo de aviso (ej. 1 mes). Es importante negociar la cláusula de terminación anticipada (por ejemplo, 15 días vs 1 mes). También revisar acuerdos de exclusividad: de empleado a veces se espera dedicación exclusiva, mientras que como contratista se puede negociar libertad para otros proyectos no competidores.
- **Cláusulas de confidencialidad/IP:** Tanto empleado como contratista suelen firmar NDA. Conviene clarificar la **propiedad intelectual**: típicamente el empleador se quedará con el código/proyecto, pero se puede negociar derechos de uso (por ejemplo, licencias perpetuas sobre herramientas genéricas que uno lleve). Verificar cláusulas de no competencia: si es muy amplias, tratarlas (por ejemplo, limitar geografía o duración).
- **Seguridad laboral:** Un contrato como empleado incluye seguro de desempleo y derechos laborales. Como contratista no. Esto se compensa con mayor remuneración. Revisar también si el contrato es “local” (requiere estar registrado como autónomo o LLC local) o un contrato de servicios internacional (facturación en dólares/euros). Cada país tiene distintas obligaciones (por ejemplo, IVA español, IRPF español del 15% retención al autónomo, etc.), así que confirmar mecanismos de pago (transferencia internacional, factura con VAT, etc.) durante la negociación.

Fuente: El IRS explica que los contratistas no conllevan retenciones de Seguro Social/Medicare del pagador, lo que implica que el trabajador independiente asume todos los impuestos por su cuenta.

N.0.4 4. Riesgos legales y checklist contractual

Antes de aceptar una oferta (especialmente remota/internacional) conviene revisar estos puntos clave:

- **Contrato por escrito:** Confirmar alcance de trabajo, deliverables, duración, forma de pago, moneda, jurisdicción legal. Evitar contratos ambiguos. Pedir versión final y plazos razonables para leerlo.
- **Facturación internacional:** Si cobra como freelance a empresa extranjera, la factura debe cumplir requisitos (con NIF/VAT europeo o sin IVA si es exportación, etc.). Por ejemplo, facturar servicios a cliente UE B2B sin IVA (inversión de sujeto pasivo). Si es fuera de la UE, emita con IVA 0% (exportación). Conservar documentación que justifique la exportación.
- **Impuestos locales:** Confirmar retenciones exigibles. En Colombia, las empresas retienen ~10% (sin IVA). En España, por ejemplo, un autónomo en facturas a empresa debe retener 15% (IRPF) si no aplica el régimen de 0%. Evaluar darse de alta en el régimen correspondiente (autónomo, freelance con NIF, etc.).
- **Permiso de trabajo / reubicación:** Si el puesto es remoto desde Colombia, normalmente no se requiere visa. Pero si hay planes de reubicación (Alemania o Portugal), revisar apoyo de la empresa para visa/permiso. Verificar que el empleo formal sea compatible con el futuro status: por ejemplo, Alemania puede requerir visa de empleo (y el candidato, al casarse con ciudadano alemán, podrá gestionar residencia). Obtener claro si ofrecen sponsor de visa, reembolso de gastos de traslado, etc.
- **Seguridad de datos y export controls:** Si el trabajo toca tecnologías sensibles (VR/AR en sectores regulados, simulaciones militares, criptografía, etc.), podría requerirse comprobación de antecedentes o licencia de exportación. Confirmar que el proyecto no infringe leyes de exportación (por ejemplo, trabajo con CAD en sector defensa).
- **Cumplimiento corporativo:** Verificar reputación de la empresa (sitio web, proyectos previos, reviews). Cuidado con empresas desconocidas sin presencia pública.
- **Cláusulas de NDA y propiedad intelectual:** Revisar que el NDA sea equilibrado (proteja datos confiden-

ciales de ambas partes). Las cláusulas de propiedad intelectual deben indicar claramente que la empresa adquiere derechos sobre el trabajo realizado, y qué (si algo) retiene el candidato (por ejemplo, uso de conocimiento general o código reutilizable).

- **No competencia / exclusividad:** Negociar límites razonables de duración (por ejemplo, 6 meses) y alcance (solo proyectos directamente competidores o mismos clientes). Para roles contratador es habitual evitar cláusulas muy amplias que impidan trabajar en proyectos distintos a largo plazo.
- **Período de prueba / terminación:** Aclarar si existe “período de prueba” o cláusula de prueba y de cuánto tiempo. En contratos de consultoría, buscar cláusula de salida anticipada con preaviso (p.ej. 15–30 días), para no quedar indefinido. Como referencia, RRHH en contratos suele exigir 1 mes de aviso para terminar.
- **Beneficios y pagos:** En ofertas de trabajo, detallan si el salario es bruto o neto, y qué cubre (e.g. tickets restaurante, bonos, stock options). Revisar también periodicidad de pago (mensual/bimestral) y penalizaciones por retrasos.
- **Seguridad social y salud:** Si va como “empleado local” en Alemania/España, preguntar tasas de contribución y beneficios (seguro salud, pensión). Si es empleado remoto en Colombia (caso menos común, pero posible con empresas colombianas), verificar prestaciones locales obligatorias.
- **Otros riesgos:** Evitar ofertas que pidan inversión inicial (pago por tool u oportunidad), que no den acceso al GitLab/repositorio real o que exijan períodos de prueba sin pago. Cualquier petición inusual (pagar por software o equipo propio sin compensación) es señal de alerta.

N.0.5 5. Señales de alerta en ofertas

Durante el proceso se deben detectar posibles fraudes o malas oportunidades. Señales de alerta típicas:

- **Salario “demasiado bueno para ser cierto”:** ofertas con sueldos muy por encima del mercado sin entrevistas técnicas justificadas suelen ser estafas o confusiones.
- **Falta de entrevista técnica o de equipo:** si avanzan solo con HR y no hay pruebas técnicas (o las entrevistas son solo de ventas) podría indicar falta de profesionalismo en la empresa.
- **Petición de dinero o equipo propio:** una empresa legítima nunca pedirá al candidato pagos adelantados por software, cursos, equipos, etc.
- **Dirección o referencia poco claras:** dificultad para verificar la existencia de la empresa, perfiles inexistentes en LinkedIn o mala reputación en buscadores.
- **Contratos verbales o vagos:** ausencia de oferta escrita detallada (salario, horas, responsabilidades) es un riesgo; siempre exigir contrato o al menos un offer letter por escrito.
- **Presión excesiva:** si insisten en “aceptar ya” sin tiempo para consultar el contrato o negociar, es sospechoso.
- **Incompatibilidad con la experiencia:** ofertas que piden habilidades no relacionadas con Unity/3D (por ejemplo, sólo ventas, MLM, o multilevel marketing) no corresponden al rol.
- **Pagos en esquemas dudosos:** recibir pago únicamente por criptomonedas u otros métodos no convencionales sin explicación, es un riesgo.

N.0.6 6. Preguntas de entrevista (técnicas y gerenciales)

Preparar ejemplos de preguntas que se podrían hacer:

- **Preguntas técnicas Unity/3D** (basadas en Adaface):
 - “¿Qué es un *GameObject* en Unity y cómo se relaciona con un Componente?”.
 - “Explique `Update()` vs `FixedUpdate()` y cuándo usar cada uno”.
 - “¿Cómo crear un *Prefab* y por qué son útiles?”.
 - “¿Cuáles son los principales tipos de *Collider* en Unity y para qué se utilizan?”.
 - Preguntas de C#: “¿Cuáles son los pilares de la POO? (herencia, polimorfismo, etc.)”.
 - “¿Cómo optimizarías el rendimiento de un juego móvil con muchos objetos físicos?” (p. ej. culling, LOD, batching).
 - “Explique cómo implementar un sistema de guardado de datos en Unity (serialización, `PlayerPrefs`, archivos)”.
 - “Describa un sistema de UI o animación que haya implementado, o cómo integraría un SDK externo (análíticas, AR) en Unity.”
- **Preguntas para Technical Artist / pipeline:**

- “¿Cómo diagnosticaría y mejoraría el rendimiento de un shader o escena compleja?”
- “Dado un conjunto de assets CAD, ¿cómo los importaría a Unity optimizando poliedros y texturas?”
- “Diseñe brevemente un pipeline de build automatizado para varios entornos (dev/test/prod) en Unity.”
- “¿Cómo mantendría datos de simulación (digital twin) sincronizados entre el modelo CAD y el juego en tiempo real?”
- **Preguntas gerenciales / situacionales:**
 - “¿Qué proyecto de su portafolio destaca más, y qué métrica (FPS, memoria, calidad visual) mejoró gracias a su aporte?”
 - “Dé un ejemplo de un bug difícil que resolvió en Unity. ¿Cómo lo diagnosticó?”
 - “¿Cómo priorizaría tareas si le asignan mejorar la velocidad de carga vs. la calidad visual en un juego?”
 - “¿Qué herramientas internas o librerías ha creado o mejorado, y cómo benefician al equipo?”

Se recomienda preparar también preguntas para el entrevistador, por ejemplo: “¿Qué retos técnicos enfrenta el equipo actualmente? ¿Qué expectativas tienen respecto al workflow o a la creación de nuevas herramientas?”, “¿Cuál es la estructura del equipo de producción?”, etc.

N.0.7 7. Preparación de pruebas técnicas y portafolio

Para maximizar el impacto del portafolio (incluyendo TwinSight X500 y proyectos secundarios):

- **Contexto claro:** Cada proyecto debe llevar una breve descripción de su propósito y tu rol (p.ej. “Desarrollo del pipeline CAD→Unity para simulación industrial; yo fui responsable del importador de modelos, optimización de mallas y creación de dashboards de visualización”).
- **Métricas específicas:** Mostrar datos cuantitativos (p.ej. reducción de polígono en %, aumento de FPS, tamaño de build). Por ejemplo, “Optimicé la escena X de 30 FPS a 60 FPS en un dispositivo móvil, reduciendo un 40% de draw calls” o “Disminuí el tiempo de carga inicial de 10s a 3s implementando loading asíncrono”.
- **Artefactos reproducibles:** Incluir enlaces a repositorios (GitHub/GitLab) con código fuente o ejemplos de escenas (puede ser solo fragmentos clave, no necesariamente todo el proyecto). Acompañar de capturas de pantalla o diagramas de flujo del proceso.
- **Video demo:** Un video corto (60-120s) mostrando la aplicación o demo final en acción, resaltando las características principales y la interacción (por ejemplo, cómo reaccionan los objetos CAD al comando del usuario, o cómo el entorno 3D visualiza datos en tiempo real). Este video puede ser alojado en ArtStation o YouTube con enlace en el portafolio.
- **Herramientas y pipeline:** Si desarrollaste herramientas internas (scripts de Unity Editor, nodos de shader, plugins), mostrar ejemplos (captura del Editor o del código) con descripción de su funcionalidad y cómo automatizaron tareas.
- **Documentación:** Un README claro que explique cómo ejecutar o visualizar el proyecto, qué tecnologías usa (Unity version, librerías, etc.) y qué parte aporta tu esfuerzo personal.
- **Proyectos secundarios relevantes:** Incluir brevemente proyectos de realidad aumentada (ARA) o Blender (retrato) si muestran habilidades técnicas (por ejemplo, un shader procedural en Blender o un prototipo WebGL AR). Sin embargo, enfatizar TwinSight como proyecto principal.
- **Al preparar pruebas técnicas:** practicar resolver pequeños retos de Unity bajo tiempo (por ejemplo, crear una lógica sencilla en C# en 20–30 minutos) y repasar algoritmos comunes. Tener ejemplos listos de código limpio. Al enfrentar un *take-home*, seguir buenas prácticas: escribir tests o al menos comentarios que expliquen el código, incluir instrucciones de uso y resultados esperados.

N.0.8 8. Estrategia de negociación salarial por prioridad

Según la clasificación previa de compañías (A/B/C), sugerimos diferentes objetivos salariales y argumentos al negociar:

- **Empresas prioritarias A (grandes estudios, tech líderes, UE/EE.UU.):** Apuntar a la cúspide del rango salarial. Preparar un **rango objetivo** alto (p.ej. EE.UU. remoto: \ \$8k–\ \$10k/mes; UE: \ \$6k–\ \$8k/mes) basado en datos de mercado. Argumentar con el valor único de TwinSight y habilidades especializadas (RT3D, CAD). Si es contrato freelance en EE.UU., proponer tarifa diaria proporcional (p.ej. \ \$600–\ \$800/día). Enfatizar disposición a hacer contrapartida (p.ej. asumir tareas extra, mayor disponibilidad horaria) a cambio

de mejor paquete. No descartar ofrecer cierta flexibilidad (p.ej. stock options o bonus por cumplimientos) si no bajan mucho el efectivo.

- **Empresas prioridad B (mid-size UE o LATAM remoto):** Fijar rango medio (por ejemplo \\$5k–\\$7k/mes en EE.UU., \\$4k–\\$6k/mes en UE/LATAM). Negociar beneficios extra en vez de salario bruto (p.ej. pago de herramientas, cursos, días extra de PTO). Mostrar interés y cómo se puede asumir mayor responsabilidad gradualmente.
- **Empresas prioridad C (startups pequeñas, locales o roles secundario):** Ser más flexible. Un contrato en LATAM local podría rondar \\$2k–\\$3k/mes (o tarifa de \\$20–\\$30/h). Aquí se puede negociar *package* (tal vez un modesto salario + participación), o buscar compensaciones no monetarias (trabajo remoto, horas reducidas con horario fijo). Sin embargo, tener un límite mínimo claro (ej. no aceptar menos de \\$2k/mes netos). En roles de aprendizaje (junior), negociar entrenamiento concreto a cambio de tarifa más baja inicialmente.

En toda negociación, es útil presentar ejemplos de salarios del mercado (usar datos de [23] y [35]) y recordar la diferencia contractor/empleado. Siempre dejar algún espacio para la “contraoferta” del reclutador, sin mostrar el mínimo deseado en la primera cifra.

N.0.9 9. Ejemplos de cláusulas contractuales

Algunas cláusulas comunes y cómo ajustarlas:

- **No competencia:** “Durante X tiempo tras la terminación, no prestará servicios para competidores en Y industria.” Propuesta: Limitar a 6–12 meses y sector específico, o eliminar geografía amplia. Por ejemplo: **“No competir en proyectos basados en el mismo modelo de simulación en industrias diferentes (defensa, medicina, etc.) fuera de [países acordados].”**
- **Propiedad intelectual:** “Toda IP desarrollada será propiedad exclusiva de la empresa.” Se puede negociar: **“La propiedad del código fuente será de la empresa, otorgándole al contratista licencia de uso personal para su portafolio.”** Además, especificar que ciertas herramientas genéricas preexistentes no se transfieren.
- **Confidencialidad (NDA):** “No divulgar información sensible X (cliente, código, algoritmos).” Se acepta, pero pedir límite de tiempo (p.ej. 2 años) y definición clara de “confidencial”.
- **Limitación de responsabilidad:** Incluir que el contratista no será responsable de daños indirectos o fallos por defecto del software sin malicia.
- **Condiciones de pago:** “Pago a 30 días tras facturar.” Asegurar que sea calendarizado o quincenal para flujo de caja.
- **Cláusula de terminación:** Definir período de preaviso razonable (p.ej. 15 días) para ambas partes.
- **Bonificaciones y revisiones:** Si aplica, establecer criterios claros de revisión salarial anual o condiciones de bonus por desempeño.

En negociación, se sugiere proponer ejemplos de redacción breve: por ejemplo, para no competencia se puede introducir “salvo que sea requerido por ley local” o reducir “alcance global” a “proyectos específicos”. Para NDAs, pedir que incluya excepción de información ya pública. Estos ajustes muestran profesionalismo y conocimiento legal.

N.0.10 10. Recomendaciones operativas finales

- **Scripts de comunicación:** Preparar frases clave. Por ejemplo, para confirmar interés en la oferta: “Estoy muy entusiasmado con la oportunidad. Mi experiencia en [TwinSight X500, optimización] me permite aportar en [punto fuerte del rol]. Me gustaría saber más sobre [cultura/stack/tareas].” Al rechazar amablemente: “Agradezco la oferta, pero tras evaluar los términos he decidido que no es adecuado para mis objetivos en este momento.”
- **Calendario de aplicación:** Responder a cada contacto en <3 días, preparar material específico (CV, carta de presentación orientada, link a portfolio) para cada aplicación. Mantener rastreador (ver sección 12) con fechas de entrevistas y próximos pasos.
- **Entrevista de hiring manager:** Además de preguntas técnicas, prepare preguntas para entender prioridades del puesto (“¿Cómo definen el éxito en este rol?”), estructura del equipo y crecimiento futuro.
- **Preparación personal:** Dormir bien la noche anterior, revisar proyectos relevantes justo antes, tener ejemplos listos con cifras. Simular explicar TwinSight y otros demos en 1-2 minutos para pitch elevator.

- **Negociación final:** Tras recibir oferta, analizar con calma. Hacer contraoferta basada en cifras del mercado y beneficios comparables. Si la empresa pone condiciones (por ejemplo, exclusividad total), valorar costo/beneficio. Es recomendable solicitar 2–3 días para revisar el contrato final.
- **Aceptar vs rechazar:** Aceptar cuando la oferta cumpla expectativas salariales mínimas (según prioridad) y condiciones laborales claras (sin cláusulas abusivas). Rechazar si el salario está muy por debajo del mercado, o si el contrato tiene riesgos legales inaceptables (p.ej. cláusula de exclusividad indefinida). En caso de dudas, es válido expresar reservas y negociar ajustes antes de firmar.

Diagramas (flujos de proceso):

Bloque de código: mermaid

flowchart LR

```
A[Envío de CV/Portafolio] --> B[Screening inicial (RRHH)]
B --> C[Entrevista técnica / Coding Challenge]
C --> D[Prueba práctica o demo]
D --> E[Entrevista con Líder técnico]
E --> F[Entrevista final (recursos humanos)]
F --> G[Oferta de empleo]
```

Bloque de código: mermaid

flowchart LR

```
Offer[Oferta Recibida] --> Review[Revisar Condiciones (salario, beneficios)]
Review --> Negotiation[Negoaciación de Términos (salario, cláusulas)]
Negotiation --> Counter[Contraoferta (ajustes documentados)]
Counter --> Agreement[Acuerdo Final (firma de contrato)]
```

Fuentes: Datos salariales y de mercado según informes actualizados; normativa IRS sobre contratistas; guía fiscal para facturación internacional; preguntas Unity de Adaface; recomendaciones de preparación técnica. Cualquier supuesto no especificado (por ejemplo, conversión EUR/USD) se ha estimado con tasas aproximadas recientes.

Appendix O

Resumen ejecutivo

Fuente: Research/deep-research-report_C3.md

Seccion: research

Estado: research_snapshot

Rol: Research / estudio de mercado

Lineas originales: 85

Ultima modificacion: 2026-06-11 20:19

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

En todos los casos se recomiendan candidaturas tempranas y **portafolio o proyecto previo** fuerte (especialmente Technical Art y Game Tech). Muchas becas/EIT aplican solo a nacionales UE, por lo que Alexander debería considerar opciones con tasas reducidas (ej. Estados europeos o becas internas). Los másteres listados son oficializados por universidades reconocidas (no simples cursos), con estructuras curriculares transparentes y conexiones industriales (prácticas y empresas)

Contenido completo depurado

Para **Alexander** (Tech Artist con Unity, Blender, WebGL y 3D interactivo), los másteres de **Technical Art/Game Tech** y **Simulación/Digital Twins** ofrecen el mayor retorno de inversión (ROI), seguidos de cerca por **Computer Graphics**; los programas **XR/Immersive** quedan en tercero. Se destacan en cada categoría:

- **Technical Art/Game Tech (ROI ALTO):** programas como el *MSc Technical Art & VFX* de Abertay (Escocia), *MSc Technical Art for Games & VFX* de Escape/Coventry (Londres) y *MSc Computer Animation & Visual Effects* de Bournemouth (Reino Unido) están diseñados específicamente para **bridgers art-código**. Capacitan en Houdini, Unreal/Unity, shaders y optimización, con alta demanda en juegos, VFX e inmersivo.
- **Simulación/Digital Twins (ROI ALTO):** el *EIT Master Autonomous Systems & Intelligent Robots* (doble grado EU) y el *Master Game Technology* de BUas (Países Bajos) abordan IA, robótica, IoT y simulación. Ambos incluyen prácticas/tesis ligadas a la industria (USD, defensa, automoción). Preparan para roles en automatización y digitalización industrial.
- **Computer Graphics (ROI MEDIO):** másteres intensivos en fundamentos teóricos. Ejemplos: *MSc Computer Graphics, Vision & Imaging* de UCL (Londres) y *Master en Ciencias de la Computación (Visual & Interactive Computing)* de ETH Zurich (Suiza). Imparten gráficos por computador, visión por computador y GPU (C++, OpenGL), pero son muy exigentes y costosos (p. ej. UCL £42,700/año). Adecuados para R&D y PhD, aunque menos orientados a producción inmediata.
- **XR/Immersivo (ROI MEDIO):** programas como *MSc Virtual Reality & AR* de Lund (Suecia) y *EIT Master HCI & XR* (EU) combinan diseño UX, interacción espacial y multimedia. Enseñan VR/AR, seguimiento de movimiento, desarrollo en Unity/Unreal. Son innovadores, pero los costos para no-EU (Lund 370.000 SEK) o la exigencia de co-financiamiento (becas EIT sólo para EU) moderan el ROI.

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son oficializados por universidades reconocidas (no simples cursos), con estructuras curriculares transparentes y conexiones industriales (prácticas y empresas asociadas).

O.0.1 Principales programas comparados

Programa (Campo)	Univ./Institución	País	Modalidad	Duración	Costes (aprox.)	Idioma	Entrada / Requisitos	Becas/ayudas	Plazo aplic.	Temas clave / Tecnologías (Unity, Unreal, etc.)	Salidas (puestos clave)	Compatibilidad
MSc Technical Art & VFX (Technical Art)	Abertay University, Dundee	Reino Unido	Presencial (campus) o parcial	1 año (full-time)	Int. £19,950 (2026)	Inglés	Grado 2:2 en arte 3D, CS o similar; portafolio	Becas internac. hasta £3,000/año	Aplicación abierta (int.)	Houdini, Unreal, USD/MtIX, VFX, Python/C# (Unity)	Technical Artist, VFX Artist, TD, R&D para juegos/VFX	Alta (directa)
MSc Technical Art for Games & VFX	Escape Studios/Coventry Univ.	Reino Unido	Presencial (Londres) / Online	1 año (full-time)	Reino Unido: ~£12,850 (campus UK students), Int. £19,995	Inglés	Grado en CS, Programación, Games o experiencia equivalente	Descuentos/becas por fechas (p. ej. aplicar antes 22 Jun 2026 para intake Sept)	22 Jun 2026 (búsq. Becas)	Unreal Engine, Houdini, FX en tiempo real, Python/HLSL, optimización	Technical Artist, Programador de VFX, Ingeniero en UE5, TD	Muy alta (match exacto)
MSc Computer Anim. & VFX	Bournemouth Univ. (NCCA)	Reino Unido	Presencial	1 año (full-time)	UK/ROI: £11,550; Int. £19,688	Inglés	Grado 2:2 en compu/arte; portafolio técnico opcional	Becas académicas (£2k), BAFTA (Reino Unido)	Aplicación: antes 2026 (sep.)	C++, Python, Shader Dev, efectos en 3D, pipeline de animación, Houdini, Maya	Technical Director, TD de Animación, Artista Técnico, Programador Gráficos	Alta (uso Unity/C++)
MSc Computer Graphics, Vision & Imaging	UCL – Computer Science	Reino Unido	Presencial	1 año (full-time)	UK: £21,500; Int: £42,700 (2026)	Inglés	Grado en CS/IT (o afin); GRE opcional; portafolio no requerido específico	Becas UCL (como Chevening, etc.)	Enero 2026 (indef.)	CG avanzada, Visión por Computador, Machine Learning, VR opcional	Graphics Engineer, Investigador en Visión/VR, PhD en Computación Gráfica	Media (alto nivel)
MSc Virtual Reality & AR	Lund University	Suecia	Presencial	2 años (full-time)	No EU: SEK 370.000 total (≈€36k)	Inglés	Grado en Ing. Eléctrica/Informática/Media; portafolio no especificado	Becas MBA Lund; voluntariados Riksbank (sólo UE/EEA)	Enero 2026	VR/AR UX, Interacción 3D, CG, procesamiento de imagen, interacción humano-ordenador	XR Developer, Especialista UI/UX 3D, Investigador XR, Role en empresas VR (Meta, etc.)	Media (XR es volátil)
Master HCI & XR (doble)	EIT Digital Master School	Europa (varias)	Presencial (EU)	2 años (dual degree)	Matricula variable; EIT ofrece becas: exención total/tarifa para EU	Inglés	Grado CS, InfoSystems, Diseño, CogSci, Ing; habilidades mínimas en programación y diseño	Becas EIT (excelencia plena/scholarships medio) para EU	1 Jun 2026 (app.)	Diseño UX para XR, Computación gráfica, IA en XR, emprendimiento tecnológico	XR Interaction Developer, XR Game Developer, Investigador HCI, Emprendedor tecnológico XR	Media-Alta (internac.)
Master Autonomous Systems	EIT Digital Master School	Europa (varias)	Presencial (EU)	2 años (dual degree)	Matricula variable; becas EIT: exención/scholarships para EU	Inglés	Grado Ing. Eléctrica/Robótica/CS/Electrónica; C++ sólidas, matemáticas; experiencia en sistemas, IoT	Becas EIT de excelencia (EU/EEE)	1 Jun 2026 (PP3)	Robótica, IA/ML, IoT, sistemas embebidos, control, visión artificial	Ingeniero de Automatización, R&D en robótica, Consultor IoT, CTO startups tecnológicas	Alta (industria crece)
MSCS – Game Development	USC Viterbi (USC)	EEUU	Presencial (Los Ángeles)	~1 año (full-time)	~USD 70,000 total (est.)	Inglés	Grado en CS (o similar); buen GPA; GRE no requerido (2027)	Becas internas USC, aid financiera (muy competitiva)	Dic 2025 (becas); Ene 2026 (final)	Infraestructura juegos, simulación masiva, Immersion tech, cognición en juegos	Desarrollador de videojuegos, Ingeniero de simulación, R&D en juegos/serious games	Baja (costo/visado)
Master CS (Visual & Interactive)	ETH Zurich	Suiza	Presencial	1.5 años (full-time)	CHF 730/sem (≈€700/sem) – casi gratis	Inglés	Grado en CS/Ingeniería, expediente sobresaliente; se acepta Background matemático	Becas ETH (CSCS, etc.); sin matrícula por ser Suiza	15 Mar 2026 (invierno suizo)	Computación gráfica, CV avanzada, GPUs, realidad virtual, algoritmos geométricos	Researcher en gráficos/visión, PhD, Ingeniero en visualización de datos	Media (alto nivel)
Master Game Technology	BUas – Breda University	P. Bajos	Presencial	1 año (full-time)	~€15,000 total (estim.)	Inglés	Experiencia/Grado relacionado con juegos; portfolio creativo + propuesta de investigación	Becas NL (Orange Tulip), financ. nacional/internac.	1 Jul 2026 (recomendado)	Investigación en juegos (procedural, ML, gráficos, interfaces), pipelines de producción	Investigador/Especialista en juegos, R&D en arte/programación de juegos	Alta (investigación)

Fuentes: páginas oficiales de cada programa (véase referencias).

Ranking recomendado (Top 10)

1. **Abertay – MSc Technical Art & VFX (UK)**: Máster pionero en Technical Art. Combina Houdini+Unreal, con acceso a estudio de producción real. Alta empleabilidad específica. ROI *muy alto*.
2. **Coventry/Escape – MSc Technical Art for Games & VFX (UK)**: Diseñado con la industria (CrazyLabs, Improbable), tecnologías UE/Houdini. Enfoque práctico con mentores. ROI *muy alto*.
3. **Bournemouth (NCCA) – MSc Computer Anim. & VFX (UK)**: Énfasis en programación (C++/Python, shaders) y creación de efectos. Facilitado por el ranking #2 en Europa de animación. ROI *alto*.
4. **EIT Digital – Master Autonomous Systems (EU)**: Doble grado EU con prácticas obligatorias. Fuerte en robótica, IA/ML e IoT. Orientado a industria 4.0/digital twins. ROI *alto*.
5. **Breda (BUas) – Master Game Technology (NL)**: Máster de investigación aplicada en juegos. Enfocado en innovación y pipelines (procedural content, ML, graficos). Desarrolla destrezas investigativas y prácticas. ROI *alto*.
6. **UCL – MSc Computer Graphics, Vision & Imaging (UK)**: Programa académico líder en CG y visión con módulos avanzados. Caro (42.7k GBP) y exigente, pero prestigioso. ROI *medio-alto* (ideal PhD tracks).
7. **EIT Digital – Master HCI & XR (EU)**: Doble grado EU con startups y minor en emprendimiento. Especialista en diseño inmersivo y sistemas interactivos. Caro pero ofrece becas EIT. ROI *medio-alto*.
8. **ETH Zurich – Master CS (Visual & Interactive) (CH)**: Énfasis matemático/científico en gráficos 3D y visión. Matrícula baja (730 CHF/sem). Alta demanda en R&D, aunque perfil teórico. ROI *medio*.
9. **Lund University – MSc VR & AR (SE)**: Enfocado en aplicaciones VR/AR (UX, imagen, interacción). No cobrará matrícula a EU, pero altos costos vida. ROI *medio*.
10. **USC Viterbi – MS Computer Science (Game Dev) (EEUU)**: Programa intensivo de desarrollo de juegos e ingeniería de simulación. Excelente reputación, pero muy caro y competitivo. ROI *bajo-medio* (grandes costos, OPT STEM).

O.0.2 Línea de tiempo de plazos clave

Bloque de código: mermaid
 gantt
 dateFormat YYYY-MM-DD
 section 2026
 USC (USC Game Dev) :crit, done, 2026-01-15, 0d
 Abertay (UK) :crit, 2026-06-22, 0d
 Escape (UK) :crit, 2026-06-22, 0d
 EIT Digital (AP3 cerrada) :crit, 2026-06-01, 0d
 EIT Digital (AP4) :crit, 2026-11-30, 0d
 Lund (Suecia) :crit, 2026-01-15, 0d
 BUas Breda (Game Tech) :crit, 2026-07-01, 0d

O.0.3 Criterios de decisión vs. programas

Bloque de código: mermaid
 flowchart TD
 Perfil[Alexander: Tech Artist (Unity/Blender/WebGL)]
 Perfil --> Interes[Áreas de interés]
 Interes --> TechArt[**Technical Art / Game Tech**]
 Interes --> Graphics[**Computer Graphics / VFX**]
 Interes --> XR[**VR/AR / Interacción**]
 Interes --> Simulacion[**Simulación / Robótica**]
 TechArt --> Abertay_A[MSc Technical Art & VFX (Abertay)]
 TechArt --> Escape_B[MSc Technical Art for Games & VFX (Coventry/Escape)]
 TechArt --> Bournemouth_C[MSc Comp. Anim. & VFX (Bournemouth)]
 Graphics --> UCL_D[MSc CG, Vision & Imaging (UCL)]
 Graphics --> ETH_E[Master CS (Visual Computing) (ETH Zurich)]
 XR --> Lund_F[MSc VR & AR (Lund Univ.)]
 XR --> EITXR_G[Master HCI & XR (EIT Digital)]
 Simulacion --> EITSim_H[Master Autonomous Systems & Robots (EIT Digital)]
 Simulacion --> GameTech_I[Master Game Technology (BUas Breda)]
 classDef foco fill:#f2f2f2;
 TechArt,Graphics,XR,Simulacion,Perfil,Interes class:foco;

En la tabla comparativa y en la recomendación anterior se ponderó **compatibilidad con el perfil actual de Alexander (software y pipeline de juegos/visualización)** y la empleabilidad nearshore. Los programas del Reino Unido y Europa facilitan visados (pos-Brexit, pasaporte UE no aplica, pero hay opciones de patrocinio), y muchos ofrecen prácticas/asesoramiento profesional. Se sugiere enfocarse en *Technical Art* como prioridad, seguido por *Game Technology* aplicada a simulación/industria, considerando siempre requisitos de idioma e inversión financiera.

Archivos adjuntos: Detalle completo de cada programa en formato Markdown (incluye tablas por programa, requisitos detallados, fuentes).

Appendix P

Becas nacionales (Portugal, Alemania)

Fuente: Research/deep-research-report_C4.md
Seccion: research
Estado: research_snapshot
Rol: Research / estudio de mercado
Lineas originales: 22
Ultima modificacion: 2026-06-11 20:28
Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Programa/Beca	Organización	URL oficial	Fecha comprob.	País/Ámbito	Tipo	Cobertura (monto)	Eleg-
ibilidad clave	Plazos	Proceso de solicitud	Obligaciones	Work auth	Prob. éxito	Relevancia	Confianza
Fuentes							

Contenido completo depurado

P.0.1 Becas nacionales (Portugal, Alemania)

Programa/Beca	Organización	URL oficial	Fecha comprob.	País/Ámbito	Tipo	Cobertura (monto)	Elegibilidad clave	Plazos	Proceso de solicitud	Obligaciones	Work auth	Prob. éxito	Bibliografía completa	Resumen
DAAD – Study Scholarships (STEM)	DAAD (Servicio Alemán de Intercambio Acad.)	https://www.daad.de/en/\{-\}	Jun 2026	Alemania	Becas (estudios máster)	992 €/mes de mantenimiento; 460 €/año adicionales; seguro de salud; contribución viaje (aeropuerto más cercano). No cubre matrícula.	Tit. universitario relevante (ing., CS, STEM). Destinadas a países en desarrollo (incluye Latinoamérica). Dominio inglés/Alemán (nivel depende del máster); a veces IELTS/TOEFL requ.	Convocatoria anual (solicitudes en octubre → inicios en otoño siguiente).	Aplicación online en portal DAAD, con CV, cartas recomendación, comprobantes académicos, propuesta de estudios.	Exclusivamente académicas; posibilidad trabajo parcial (20 h/sem; visado estudiante).	Permiso de residencia de estudiante en Alemania; empleo parcial (~20 h/sem) permitido.	Media (competitiva; muchos aplicantes)		APPENDIX P. BECAS NACIONALES (PORTUGAL, ALEMANIA)

P.0.2 Becas UE / Erasmus Mundus

Programa/Beca	Organización	URL oficial	Fecha comprob.	País/Ámbito	Tipo	Cobertura (monto)	Elegibilidad clave	Plazos	Proceso de solicitud	Obligaciones	Work auth	Prob. éxito	Becado	Requisitos
IMLEX – MSc en Imaging & Light in XR (EMJMD)	Consortio Erasmus Mundus (FI, BE, JP, FR)	https://www.imlex.org	Jun 2026	UE-Japón (Movilidad)	Beca Erasmus Mundus	Beca completa: exención de matrícula (9.000 €/año para no-EU, exentos si becado); estipendio mensual (bolsa EMJM); cubre visado y viajes.	Grado/licenciatura en Informática, Electrónica, Física, Matemáticas u otros STEM. Conocimientos en Computación Gráfica o XR. Inglés alto (IELTS ≈6.5). Sociedad con motiv. investigadora.	Convocatoria anual (postulaciones Dic–Ene).	Solicitud vía portal Studyinfo (submit CV, cartas, portafolio proyectos, motivación).	Estudios a tiempo completo; dedicación exclusiva (no trabajo externo); posible participación en proyectos de investigación/laboratorio.	Visa de estudiante UE/Japón; permite trabajo parcial (<20 h/sem) según visado.	Baja (muy competitivo; 10 becas/año).	10	Formación en el extranjero, experiencia profesional, idiomas, etc.
IPCVai – MSc en Image Proc., Computer Vision & AI (EMJMD)	Consortio Erasmus Mundus (Hungria, España, Francia)	https://ipcv.eu	Jun 2026	Unión Europea	Beca Erasmus Mundus	Beca completa: exención matrícula + €1.400/mes durante 24 meses + €2.000–3.000/año para desplazamiento + 1000 € instalación + seguro.	Grado en Informática, Ingeniería Eléctrica, Matemáticas/Física o similares con interés en visión por computador. Inglés TOEFL/IELTS (min. 6.5). Preselección de expediente académico.	Convocatoria anual (plazo: oct–ene para inicio en septiembre 2026).	Aplicación online: currículo, cartas recomendación, transcripts, project portfolio, entrevista posible.	Estudios full time; dedicación exclusiva.	Visa UE (Hungria) con permiso de residente; trabajo parcial permitido (≈20 h/sem) bajo visado estudiante.	Baja (solo ~15 becas anuales).	15	Al menos un idioma, experiencia profesional, etc.

P.0.3 Becas Investigación / PhD

Programa/Beca	Organización	URL oficial	Fecha comprob.	País/Ámbito	Tipo	Cobertura (monto)	Elegibilidad clave	Plazos	Proceso de solicitud	Obligaciones	Work auth	Prob. éxito	Bibliografía completa	Referencia
FCT – Bolsa de Doutoramento (Línea General)	FCT (Fund. Ciência e Tecnologia, Portugal)	https://www.fct.pt/financiamento/bolsas-de-investigacao-para-doutoramento	Jun 2026	Portugal	Beca (investigación)	€1.901,21/mes (Bolsa BPD en Portugal) + seguro; cursos doctorales; matrícula gratuita.	Cualquier nacionalidad; inscrito o por inscribirse en doctorado portugués en STEM (CS/Ing. etc.); Master con alta distinción. Inglés u otro (doc. oficiales).	Convocatoria anual (p.ej. marzo 2026 abre).	Solicitud online en portal FCT; documentos: projeto de investigación, CV, diplomados, cartas recomendação.	Investigación doctoral full-time; dedicação exclusiva (no contrato de trabajo remunerado fuera de la beca).	Visa estudiante (residência por formação avançada); no se suele combinar con empleo comercial.	Media (relativa al estudiante competitivo, ~15% de aceptación)	Bibliografía completa	Fin Un po €1 m in Ro XI he ár Un in do Al co 3D ac ial gr
DAAD – Research Grants para Doctorandos (E.g., SP/DL)	DAAD (Alemania)	https://www.daad.de/ {ver convocatoria específica}	Jun 2026	Alemania	Becas (estudios e investigación)	Aprox. €1200/mes (estipendio doctoral) + seguro + viaje (valor depende de programa).	Cualquier nacionalidad; candidatos con doctorando planeado en Universidad alemana, con M.Sc. relevante. Normalmente programa de estancia de 6–12 meses como doctorando.	Convocatoria periódica (Depende de programa; p.ej., Spring 2026).	Formulario DAAD, CV, cartas recomendación, plan de trabajo doctoral.	Investigación (short-term research stays).	Visa académica; trabajo parcial (hasta 20 h/sem) permitido bajo visado.	Baja (becas limitadas; requiere proyecto claro)	Bibliografía completa	APPENDIX P. BECAS NACIONALES (PORTUGAL, ALEMANIA)

Notas: Las tablas anteriores listan programas de financiación para másteres y doctorados relevantes al perfil técnico deseado. La probabilidad de éxito es estimada para candidatos LATAM: baja cuando hay alta demanda (pocas becas), media para programas con muchos beneficiarios (DAAD, FCT). Se han priorizado fuentes oficiales (DAAD, FCT, consorcios Erasmus), y las cifras monetarias se citan de documentos oficiales. Para requisitos de idioma, normalmente se exige inglés (p.ej. TOEFL/IELTS) y a veces dominio del idioma local (alemán/portugués). En general, estas becas suponen residencia presencial (p.ej. a tiempo completo), pero en todos los casos la normativa de estancia de estudiante permite trabajo parcial (~20 h/sem) autorizado por visa. Las obligaciones de enseñanza o investigación suelen ser “dedicación exclusiva” al proyecto de estudios (salvo que se especifique función docente).

Appendix Q

E1 — Technical Volunteering Opportunities

Fuente: Research/E1_technical_volunteering_opportunities.md

Seccion: research

Estado: research_snapshot

Rol: Research / estudio de mercado

Lineas originales: 471

Ultima modificacion: 2026-06-11 21:07

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

— module: E1_technical_volunteering_opportunities status: complete date_checked: 2026-06-11 scope: technical volunteering, civic tech, humanitarian mapping, XR accessibility, games for social impact, digital twins, geospatial tools, nonprofit AI/data support target_profile: - Unity Technical Artist - Unity WebGL Developer - Real-Time 3D Developer - Technical Visualization Developer - XR Developer - Simulation / Digital Twin Developer - Tools / Pipeline Developer - Python au

Contenido completo depurado

module: E1_technical_volunteering_opportunities

status: complete

date_checked: 2026-06-11

scope: technical volunteering, civic tech, humanitarian mapping, XR accessibility, games for social impact, digital twins, geospatial tools, nonprofit AI/data support

target_profile:

- Unity Technical Artist
- Unity WebGL Developer
- Real-Time 3D Developer
- Technical Visualization Developer
- XR Developer
- Simulation / Digital Twin Developer
- Tools / Pipeline Developer
- Python automation / tooling developer

owner_profile: Alexander Woodcock Salomon

Q.0.1 1. Executive summary

This module maps technical volunteering routes that can produce credible experience for Alexander without requiring a formal job offer.

The strongest opportunities are not generic volunteering roles. They are technical contribution paths where Alexander can produce visible artifacts:

- public repositories;
- data visualizations;
- GIS or map outputs;
- XR accessibility resources;
- small Unity/WebGL demos;
- documentation;
- workflow automation;
- humanitarian mapping contributions;
- civic-tech prototypes.

The highest ROI paths for Alexander are:

1. Humanitarian mapping / geospatial visualization.
2. XR accessibility / inclusive immersive design.
3. Games and immersive media for social impact.
4. AI/data support for nonprofits.
5. Open engineering / CAD / digital-twin adjacent communities.
6. Civic-tech prototyping.

These can support future applications to roles in Unity WebGL, technical visualization, XR, simulation, digital twins and tools/pipeline.

Q.0.2 2. Selection criteria

Criterion	Meaning
Public access	Public website or community
Remote potential	Can contribute from Colombia
Technical relevance	3D, XR, GIS, data, tools, Unity, WebGL
Portfolio value	Can create proof of work
Social impact	Nonprofit, civic, humanitarian, education
Low barrier	No formal employment needed
Credibility	Recognized organization or active ecosystem
Risk	Avoid unpaid pseudo-jobs

Q.0.3 3. Top opportunities table

Rank	Opportunity	Organization	URL	Type	Remote	Main skills	Portfolio output	Fit	Biblia completa
1	Humanitarian mapping	HOT OSM	https://www.hotosm.org/	Humanitarian GIS	Yes	OSM, GIS, QA	Mapping record	Very high	
2	Missing Maps	Red Cross / MSF / HOT	https://www.missingmaps.org/	Remote mapping	Yes	OSM, validation	Mapathon proof	High	
3	GIS volunteer missions	GISCorps	https://www.giscorps.org/	GIS volunteering	Mixed	GIS, data, maps	Mission record	High	
4	XR accessibility projects	XR Access	https://xraccess.org/	XR accessibility	Yes	XR, Unity, WebXR	Accessible XR notes	Very high	
5	Social-impact games	Games for Change	https://www.gamesforchange.org/	Games for good	Mixed	Game design, Unity	Social-impact prototype	High	
6	Nonprofit AI/data support	DataKind	https://www.datakind.org/	Data for good	Mixed	Python, data, AI	Analysis/demo	Medium-high	
7	Tech partners for NGOs	Tech To The Rescue	https://www.techtotherescue.org/	Pro-bono tech	Team-based	AI, product, engineering	NGO tech project	Medium-high	
8	Civic tech network	Code for All	https://codeforall.org/	Civic tech	Mixed	Web, data, UX	Civic prototype	Medium-high	
9	Skilled nonprofit projects	Catchafire	https://www.catchafire.org/	Skills-based volunteering	Yes	Web, design, data	Scoped deliverable	Medium	
10	Civic project marketplace	DemocracyLab	https://www.democracylab.org/	Civic tech projects	Yes	Web, UX, data	Public project	Medium	
11	Open drone mapping	OpenDroneMap	https://www.opendronemap.org/	Open geospatial	Yes	Python, photogrammetry	3D maps/tooling	Very high	
12	Open engineering design	Open Source Ecology	https://www.opensourceecology.org/	Open hardware	Mixed	CAD, FreeCAD, docs	CAD/3D contribution	Medium-high	
13	Wikimedia 3D/media/data	Wikimedia Commons / Wikidata	https://commons.wikimedia.org/	Open knowledge	Yes	3D files, metadata	Public uploads/data	Medium	
14	Citizen science	Zooniverse	https://www.zooniverse.org/	Science volunteering	Yes	Visual classification	Low technical proof	Low-medium	
15	Nonprofit tech community	TechSoup / TechSoup Connect	https://www.techsoup.org/	Tech-for-good network	Mixed	nonprofit tech	Networking / events	Medium	
16	Volunteer matching	VolunteerMatch	https://www.volunteermatch.org/	Opportunity search	Yes	varies	Depends on project	Medium	
17	Open geospatial foundation	OSGeo	https://www.osgeo.org/	Open geospatial	Yes	GIS, docs, code	OSS/GIS contribution	High	
18	Open CAD ecosystem	FreeCAD	https://www.freecad.org/	CAD open source	Yes	Python, CAD, docs	Technical tooling	Medium-high	
									APPENDIX Q. E1 — TECHNICAL VOLUNTEERING OPPORTUNITIES

Q.0.4 4. Detailed opportunity inventory

4.1 Humanitarian mapping and geospatial volunteering

Opportunity	Organization	URL	Contribution mode	Technical fit	Portfolio value	Notes
HOT OSM	Humanitarian OpenStreetMap Team	https://www.hotosm.org/	Mapping, validation, community tasks	GIS, maps, disaster response	High	Strong fit for spatial data and technical visualization
HOT Tasking Manager	HOT OSM	https://tasks.hotosm.org/	Remote mapping tasks	OSM, QA, geodata	High	Concrete contribution history
Missing Maps	Red Cross / MSF / HOT	https://www.missingmaps.org/	Mapathons, remote mapping	GIS, humanitarian ops	Medium-high	Good entry point
GISCorps	URISA / GISCorps	https://www.giscorps.org/	Skilled GIS missions	GIS, geocoding, web maps	High	More selective
OSGeo	Open Source Geospatial Foundation	https://www.osgeo.org/	Code, docs, community	GIS tooling, WebGIS	High	Strong for technical visualization
OpenDroneMap	ODM community	https://www.opendronemap.org/	Code, docs, forum, testing	Photogrammetry, 3D models	Very high	Best 3D/geospatial bridge

Why this matters for Alexander

Humanitarian mapping creates direct evidence of spatial reasoning, data quality, GIS, validation, remote collaboration and public-interest technical work. OpenDroneMap adds the strongest connection to 3D reconstruction, aerial imagery, point clouds and digital twin workflows.

Best first action

Create a small public case study:

```
Bloque de codigo: text
Humanitarian Mapping to 3D Visualization:
from OSM / aerial data to a Unity or WebGL spatial viewer
```

4.2 XR accessibility and inclusive immersive technology

Opportunity	Organization	URL	Contribution mode	Technical fit	Portfolio value	Notes
XR Access	Cornell Tech initiative	https://xraccess.org/	Projects, research, Slack, resources	XR, accessibility, UX	High	Highly aligned
XR Accessibility GitHub	XR Access / XR Association	https://xraccessibility.github.io/	Docs, code snippets, resources	Unity, WebXR, ARCore	Very high	Strong technical artifact
XR Access Symposium	XR Access	https://xraccess.org/symposium/	Events, talks, networking	XR research	Medium	Good for visibility
Immersive Web community	W3C ecosystem	https://immersive-web.github.io/	Specs, issues, examples	WebXR, WebGL	High	More technical, less NGO-like

Why this matters for Alexander

XR accessibility can connect his Unity/WebGL skill set with a socially useful niche. It also allows him to build technical demos that are not just visual, but interaction-focused and user-centered.

Best first action

Create an accessibility breakdown:

```
Bloque de código: text
Accessible inspection mode for a Unity WebGL 3D technical viewer
```

Possible elements:

- large labels;
- contrast modes;
- reduced motion;
- keyboard navigation;
- simplified interaction states;
- screen-reader-friendly metadata outside canvas;
- color-blind-safe highlighting;
- touch target scaling.

4.3 Games, simulations and immersive media for social impact

Opportunity	Organization	URL	Contribution mode	Technical fit	Portfolio value	Notes
Games for Change G4C Festival	G4C Games for Change	https://www.gamesforchange.org/ https://festival.gamesforchange.org/	Volunteer, juror, community, festival Talks, networking, showcases	Game dev, XR, serious games Games + impact	High Medium	Relevant to social-impact games Not pure volunteering
G4C Learn Serious games communities	Games for Change various	https://learn.gamesforchange.org/ not found	Education programs Mentoring, jams	Teaching, game design Unity, design	Medium Medium	Useful if mentoring Must vet case by case

Why this matters for Alexander

Games for Change is useful because it frames Unity and real-time 3D as tools for education, training, empathy and social impact. This can support a profile beyond entertainment games.

Best first action

Create one small serious-game prototype:

```
Bloque de codigo: text
Interactive 3D training module for technical assembly or safety education
```

This can reuse his thesis technology.

4.4 AI, data and nonprofit technical support

Opportunity	Organization	URL	Contribution mode	Technical fit	Portfolio value	Notes
DataKind	DataKind	https://www.datakind.org/	Data projects, events, volunteering	Python, data, AI	Medium-high	Less 3D, strong tech-for-good
DataDive events	DataKind	https://www.datakind.org/datadive-events/	Short data events	Python, analysis	Medium	Good for quick evidence
Tech To The Rescue	TTTR	https://www.techtotherescue.org/	Pro-bono tech partnerships	AI, product, software	Medium-high	Often team/company oriented
Tech To The Rescue projects	TTTR	https://app.techtotherescue.org/	Available NGO projects	AI, engineering	Medium-high	Check project access
Catchafire	Catchafire	https://www.catchafire.org/	Scoped pro-bono projects	Web, design, data	Medium	Good for small deliverables
DemocracyLab	DemocracyLab	https://www.democracylab.org/	Civic tech projects	Web, UX, data	Medium	Good for public contributions
VolunteerMatch	VolunteerMatch	https://www.volunteermatch.org/	Search virtual tech roles	varies	Medium	Use filters carefully

Why this matters for Alexander

These platforms can generate short, bounded projects that look credible on a CV if the output is measurable. They are less aligned to Unity/3D, but useful for Python automation, dashboards, data visualization and nonprofit tech.

Best first action

Search for terms:

```
Bloque de código: text
data visualization
map
GIS
3D
Unity
web developer
dashboard
automation
AI prototype
```

4.5 Open engineering, CAD and digital-twin adjacent volunteering

Opportunity	Organization	URL	Contribution mode	Technical fit	Portfolio value	Notes
Open Source Ecology	OSE	https://www.opensourceecology.org/	CAD, docs, design sprints	FreeCAD, 3D design	Medium-high	Strong engineering narrative
FreeCAD	FreeCAD community	https://www.freecad.org/	Docs, Python workbenches, testing	CAD, Python, UI	High	Strong tools/pipeline fit
OpenSCAD	OpenSCAD community	https://openscad.org/	Code, docs, examples	parametric CAD	Medium-high	Good for procedural design
OpenDroneMap	ODM	https://www.opendronemap.org/	Photogrammetry tooling	3D reconstruction	Very high	Strongest digital-twin bridge
OSGeo projects	OSGeo	https://www.osgeo.org/projects/	GIS tools	maps, geodata	High	Good for visualization

Why this matters for Alexander

This category bridges his drone thesis, CAD-to-web pipeline and technical visualization focus. It is more defensible than generic volunteering.

Best first action

Make a public technical note:

Bloque de código: text
Using open geospatial / CAD data for browser-based 3D technical visualization

4.6 Open knowledge and citizen science

Opportunity	Organization	URL	Contribution mode	Technical fit	Portfolio value	Notes
Wikimedia Commons	Wikimedia	https://commons.wikimedia.org/	3D uploads, media metadata	3D files, documentation	Medium	Useful if uploading models
Wikidata	Wikimedia	https://www.wikidata.org/	Structured data	data modeling	Medium	Useful for semantic metadata
Zooniverse	Citizen Science Alliance	https://www.zooniverse.org/	Image/data classification	visual analysis	Low-medium	Low coding evidence
Galaxy Zoo / science projects	Zooniverse	https://www.zooniverse.org/projects	Visual classification	perception, data	Low	Use only if relevant

Why this matters for Alexander

This path has lower direct job ROI, but it can build evidence of data discipline, metadata, public documentation and visual classification.

Q.0.5 5. Ranking by relevance to Alexander

Priority	Path	ROI	Reason
1	OpenDroneMap + HOT OSM	Very high	3D, mapping, drone, digital twin
2	XR Access	Very high	XR, Unity/WebXR, accessibility
3	GISCorps	High	Skilled GIS missions
4	Games for Change	High	Unity + impact narrative
5	OSGeo	High	GIS software ecosystem
6	Open Source Ecology	Medium-high	CAD + open engineering
7	FreeCAD / OpenSCAD	Medium-high	Python + CAD tools
8	DataKind	Medium-high	Python/data credibility
9	Tech To The Rescue	Medium-high	NGO tech projects
10	Code for All	Medium	civic tech network
11	DemocracyLab	Medium	project marketplace
12	Catchafire	Medium	short pro-bono tasks
13	Wikimedia	Medium	open knowledge
14	Zooniverse	Low-medium	citizen science, low coding
15	VolunteerMatch / Idealist	Variable	depends on listing

Q.0.6 6. Recommended contribution strategy

6.1 Best 30-day path

Week	Action	Output
1	Join HOT OSM and map 2–3 tasks	OSM profile proof
1	Join XR Access Slack/newsletter	community entry
2	Contribute to OpenDroneMap docs/forum	first public contribution
2	Build tiny Unity/WebGL map viewer	technical demo
3	Add accessibility mode to demo	XR/a11y case
4	Publish case study	portfolio page

6.2 Best 60-day path

Track	Deliverable
Humanitarian GIS	OSM mapping + validation record
Technical visualization	WebGL viewer using map/drone/CAD data
Accessibility	reduced motion, contrast, keyboard navigation
Documentation	public README and article
Social impact	explain use case and stakeholders

6.3 Best 90-day path

Milestone	Evidence
1 public contribution	PR, issue, docs, or forum solution
1 volunteering profile	HOT / GISCorps / G4C / DataKind
1 demo	Unity WebGL or WebXR prototype

Milestone	Evidence
1 case study 1 CV bullet	problem, process, result measurable contribution

Q.0.7 7. Portfolio project ideas from E1

Project 1 — Humanitarian 3D map viewer

Field	Value
Source	HOT / OSM / OpenDroneMap
Stack	Unity WebGL or Three.js
Output	browser-based spatial viewer
Value	humanitarian mapping + 3D visualization
Difficulty	medium
ROI	very high

Project 2 — Accessible XR inspection mode

Field	Value
Source	XR Access resources
Stack	Unity, WebXR, UI Toolkit
Output	accessibility mode
Value	inclusive technical visualization
Difficulty	medium
ROI	high

Project 3 — NGO technical dashboard

Field	Value
Source	DataKind / Catchafire / DemocracyLab
Stack	Python, JS, maps
Output	data dashboard
Value	measurable nonprofit impact
Difficulty	low-medium
ROI	medium

Project 4 — Open engineering CAD-to-Web demo

Field	Value
Source	Open Source Ecology / FreeCAD
Stack	FreeCAD, Blender, Unity WebGL
Output	interactive assembly viewer
Value	direct match to thesis
Difficulty	medium-high
ROI	very high

Q.0.8 8. CV language generated from E1

Conservative version

Bloque de código: text
Contributed to humanitarian and open technical communities through geospatial mapping, documentation and prototype development, focusing on spatial data, accessibility and browser-based 3D visualization.

Stronger version after evidence exists

Bloque de código: text
Built a browser-based 3D visualization prototype using open geospatial and CAD workflows, integrating humanitarian mapping data and accessibility-oriented interaction modes for technical inspection use cases.

Bullet format

Bloque de código: text
- Contributed to open geospatial and humanitarian mapping workflows using OpenStreetMap-based tools, focusing on data quality, spatial interpretation and technical visualization.
- Developed a Unity/WebGL prototype translating open spatial data into an interactive 3D inspection workflow.
- Documented accessibility-oriented interaction patterns for XR/WebGL technical viewers, including contrast, reduced motion and keyboard-accessible navigation.

Q.0.9 9. Risk assessment

Risk	Severity	Mitigation
unpaid work with no output	high	choose public artifacts
generic volunteering	medium	pick technical tasks
no portfolio proof	high	document every contribution
too much time spent mapping	medium	cap weekly hours
irrelevant nonprofit tasks	medium	filter for GIS, 3D, XR, data
private NGO work cannot be shown	medium	ask for anonymized case study
weak technical signal	high	build demos, not just attend events
overclaiming impact	high	state exact contribution only

Q.0.10 10. Best search terms

Use these terms inside volunteer platforms:

Bloque de código: text
GIS
mapping
OpenStreetMap
data visualization
dashboard
web developer
Python
automation
AI
Unity
XR
accessibility
3D
digital twin
simulation
technical documentation

Avoid:

Bloque de código: text
 social media assistant
 graphic design only
 newsletter formatting
 generic website maintenance
 fundraising calls
 event staffing only

Q.0.11 11. Evidence notes

Source	Evidence used
UNV Online Volunteering	remote volunteer model; online assignments
HOT / OSM	humanitarian mapping and Tasking Manager ecosystem
Missing Maps	remote mapping for vulnerable areas
GISCorps	skilled GIS volunteering
Code for All	global civic tech network
Code for America	civic tech and public-service software
DataKind	data science and AI for social impact
Tech To The Rescue	pro-bono tech partnerships for social impact
XR Access	volunteer-led accessible XR projects
Games for Change	games and immersive media for social impact
OpenDroneMap	open drone mapping, point clouds, 3D models
Open Source Ecology	open engineering and remote design collaboration
Catchafire	skills-based volunteering
Wikimedia	open 3D/media/data contribution
Zooniverse	citizen science classification

Q.0.12 12. Final recommendation

Alexander should prioritize **one geospatial/digital-twin path** and **one XR/accessibility path**.

Recommended sequence:

Bloque de código: text
 1. HOT OSM / Missing Maps
 2. OpenDroneMap
 3. XR Access
 4. Games for Change
 5. DataKind or Tech To The Rescue

This sequence produces a coherent story:

Bloque de código: text
 I build browser-based 3D technical visualization systems for real-world spatial, educational and humanitarian use cases.

That story is stronger than generic volunteering and more aligned with Unity WebGL, technical art, simulation and digital twins.

Appendix R

paquete_continuidad_agent_research

Fuente: Research/paquete_continuidad_agent_research.md

Seccion: research

Estado: research_snapshot

Rol: Research / estudio de mercado

Lineas originales: 932

Ultima modificacion: 2026-06-11 16:50

Deduplicacion conservadora: 0 bloques repetidos omitidos; 0 caracteres repetidos no reimpresos.

Resumen de orientacion

Este documento permite continuar el proyecto en otro chat sin perder continuidad técnica.

Contenido completo depurado

R.0.1 0. Propósito del paquete

Este documento permite continuar el proyecto en otro chat sin perder continuidad técnica.

El foco actual no es rehacer la investigación base, sino **continuar la fase de investigaciones con agente** para robustecer el roadmap internacional de Alexander Woodcock Salomón.

El flujo activo es:

Bloque de codigo: text
1. Ejecutar una tarea concreta con modo agente.
2. Pegar el resultado en el chat.
3. Auditar calidad, corregir vacíos y convertirlo en un módulo .md.
4. Pasar a la siguiente tarea.
5. Al final, integrar todo en una actualización del roadmap.

R.0.2 1. Contexto base del proyecto

Perfil

Bloque de codigo: text
Nombre: Alexander Woodcock Salomón
Ubicación: Colombia
Idioma nativo: Español
Inglés: C1 autopercebido
Alemán: inicial / no funcional
Portugués: estratégicamente relevante por ciudadanía portuguesa esperada

Posicionamiento profesional recomendado

Bloque de código: text
Real-Time 3D Developer / Unity Technical Artist
focused on interactive technical visualization, CAD-to-realtime optimization, and Unity WebGL deployment.

Proyecto principal

Bloque de código: text
TwinSight X500
Unity WebGL technical visualization prototype for inspecting and understanding the assembly of a Holybro X500 V2 drone.

Stack principal

Bloque de código: text
Unity
C#
Unity WebGL
URP
UI Toolkit
Shader Graph
Blender
CAD-to-realtime optimization
Technical visualization
Interactive 3D
Python automation
Git/GitHub

Situación laboral

Bloque de código: text
Formal industry experience in Unity / Technical Art: limited
Strongest proof: thesis / portfolio project
Target: remote international work from Colombia
Preferred model: contractor / B2B
Minimum initial target: USD 1.5k/month
Realistic 6-month target: USD 3k/month
Ambitious 12-24 month target: USD 6k/month

Movilidad

Bloque de código: text
Current EU work authorization: No
Portuguese passport/citizenship: expected in approximately 2 years, considered highly likely by the user
Germany route: possible through legal marriage/residence route with German partner, but not current work authorization
Immediate strategy: remote from Colombia
Medium-term strategy: EU/Germany/Portugal once legally enabled

R.0.3 2. Documentos ya generados

El sistema principal ya llegó hasta:

Bloque de código: text
00-16: research, strategy, audit and planning
17-24: application materials and decision systems
25: spreadsheet tracker
26: weekly execution dashboard
27: conservative/balanced/aggressive scenarios
28: first agent benchmark module

Archivos importantes ya generados

Bloque de código: text
17_cv_base_and_role_variants.md
18_linkedin_final_rewrite.md

```
19_github_readme_templates.md
20_portfolio_copy_and_site_structure.md
21_twinsight_demo_video_script_and_shotlist.md
22_outreach_templates_english_spanish.md
23_interview_answer_bank.md
24_offer_evaluation_scorecard.md
25_application_tracker_template.xlsx
26_weekly_execution_dashboard.md
27_strategy_scenarios_conservative_balanced_aggressive.md
28_competitor_profile_benchmark.md
```

Estado del archivo 28

El archivo:

```
Bloque de código: text
28_competitor_profile_benchmark.md
```

fue generado a partir de la tarea:

```
Bloque de código: text
A1_linkedin_profiles_benchmark
```

Contiene:

- benchmark superficial de perfiles LinkedIn encontrados por snippets públicos;
- familias de rol comparables;
- headline patterns;
- señales que Alexander debería copiar;
- señales que no debería copiar;
- implicaciones para CV, LinkedIn y portafolio.

Limitación del archivo 28

La búsqueda de LinkedIn fue limitada porque muchas páginas requieren login. Por tanto, el archivo 28 debe considerarse:

```
Bloque de código: text
surface-level LinkedIn positioning benchmark
```

No debe considerarse un análisis completo de perfiles.

R.0.4 3. Estado actual de tareas con agente

Completado

Código	Tarea	Estado	Salida
A1	LinkedIn profiles benchmark	Completada	28_competitor_profile_benchmark.md

Intentada pero incompleta

Código	Tarea	Estado	Problema
A2	Public CV / resume benchmark	Incompleta	El agente encontró pocas fuentes y terminó diciendo que no podía recopilar más

Próxima acción recomendada

Reintentar A2 con un prompt más robusto y menos restrictivo.

Razón:

- la primera ejecución de A2 fue insuficiente;
- sí encontró fuentes útiles, pero no las explotó bien;
- se deben permitir fuentes como portfolios personales, PDFs, resume databases, university samples, Art-Station resume pages y páginas personales;
- se debe pedir mínimo 12–20 ejemplos, no 30–50;
- debe aceptar ejemplos anónimos o sample resumes siempre que sean concretos.

R.0.5 4. Regla de uso de agente

El agente se usa para:

```
Bloque de código: text
public web collection
structured source data
live verification
tables
URLs
date checked
```

No se usa para:

```
Bloque de código: text
final strategy
CV writing
LinkedIn writing
portfolio copy
negotiation judgment
legal interpretation
narrative synthesis
```

La síntesis final la hace el asistente después de recibir los datos.

R.0.6 5. Secuencia general de nuevos módulos

Estos son los módulos adicionales propuestos después del sistema 00–27.

```
Bloque de código: text
28_competitor_profile_benchmark.md
29_portfolio_case_study_benchmark.md
30_live_job_postings_market_snapshot.md
31_verified_company_targets_expanded.md
32_education_scholarships_masters_fellowships.md
33_competitions_awards_hackathons_thesis_opportunities.md
34_volunteering_open_source_experience_pipeline.md
35_recruiter_outreach_database.md
36_content_strategy_linkedin_artstation_github.md
37_salary_contracting_legal_update.md
38_agent_research_synthesis_and_roadmap_update.md
```

Ajuste necesario

Como 28_competitor_profile_benchmark.md ya fue generado solamente con A1, las próximas tareas de benchmark profesional pueden manejarse de dos formas:

Opción recomendada Crear archivos complementarios:

```
Bloque de código: text
28B_public_cv_resume_benchmark.md
28C_github_profiles_benchmark.md
28D_artstation_portfolio_benchmark.md
```

Luego crear:

```
Bloque de código: text
28E_competitor_benchmark_synthesis.md
```

Opción alternativa Regenerar el archivo 28 como versión completa:

```
Bloque de código: text
28_competitor_profile_benchmark_v2.md
```

La opción recomendada es **28B / 28C / 28D / 28E**, porque evita sobreescribir trabajo ya hecho.

R.0.7 6. Orden inmediato recomendado

Próximas tareas

Orden	Código	Tarea	Usar agente	Deep Research
1	A2-Retry	Public CV / resume benchmark	Sí	No
2	A3	GitHub profiles benchmark	Sí	No
3	A4	ArtStation / portfolio benchmark	Sí	No
4	A5	Technical visualization case studies	Sí	No
5	F1	Demo reels benchmark	Sí	No
6	F3	ArtStation breakdown benchmark	Sí	No
7	B1	Unity / Real-Time 3D live jobs	Sí	No
8	B2	Digital Twin / Simulation live jobs	Sí	No
9	B5	LATAM nearshore jobs	Sí	No
10	B6	Top 30 company verification	Sí	No

Motivo del orden

Primero se mejora el benchmark de presentación profesional:

```
Bloque de código: text
CV → GitHub → ArtStation/Portfolio → Case Studies → Demo reels
```

Luego se pasa al mercado vivo:

```
Bloque de código: text
jobs → companies → recruiters → education → competitions → volunteering
```

Esto evita gastar usos de agente en ofertas que pueden caducar antes de que los materiales estén listos.

R.0.8 7. Prompt recomendado para reintentar A2

Usar este prompt en el próximo chat o en el mismo chat, con modo agente activo.

```
Bloque de código: text
Execute A2_retry_public_cv_resume_benchmark.

Context:
We already attempted A2, but the result was incomplete. This retry should be more flexible and collect concrete public
↪ resume/CV examples from personal websites, PDF resumes, resume databases, university sample packets, GitHub resume pages,
↪ ArtStation resume pages, portfolio CV pages, and public blind CVs.

Objective:
Collect 12-20 public CV/resume examples relevant to Alexander's target positioning:
Real-Time 3D Developer / Unity Technical Artist / Unity WebGL Developer / Technical Visualization Developer / XR Developer /
↪ Simulation Developer / Tools-Pipeline Developer.

Acceptable source types:
- real candidate public PDF resumes;
- personal website resume pages;
- portfolio pages with resume section;
- GitHub resume READMEs;
- ArtStation resume pages;
- blind CVs from reputable recruiting databases;
- university sample resumes if they contain concrete examples relevant to game development, technical art, creative
↪ technology, computational design, Unity, XR, simulation, visualization, or technical tools.

Do not use generic advice articles unless they include a concrete resume sample.

Output structured tables only.

Fields:
- Source label
- Resume/CV URL
- Source type
- Role target
- Country/region if visible
- Seniority
- Resume structure
- Headline/title used
- Summary style
- Skills organization
- Project/experience structure
- Metrics used
- Education wording
- Portfolio/GitHub links if visible
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:
- Use only publicly available information.
- Do not invent missing data.
- If unavailable, write "not found".
- Keep cells short.
- Include source URLs.
- Tables only.
- Do not write strategy.
- Do not write narrative recommendations.
- Mark generated/sample resumes as "sample", not real candidate.
```

Salida esperada después de A2-Retry

Cuando el agente entregue resultados, generar:

```
Bloque de código: text
28B_public_cv_resume_benchmark.md
```

Ese archivo debe contener:

- fuentes recolectadas;
- patrones de estructura de CV;
- qué adaptar al CV de Alexander;
- qué evitar;
- actualización concreta para 17_cv_base_and_role_variants.md;
- prompt para la siguiente tarea A3.

R.0.9 8. Prompt para A3 — GitHub profiles benchmark

Usar después de cerrar A2-Retry.

```
Bloque de código: text
Execute A3_github_profiles_benchmark.

Objective:
Collect public GitHub profile examples relevant to Alexander's positioning:
Real-Time 3D Developer, Unity Technical Artist, Unity WebGL Developer, Technical Visualization Developer, XR Developer,
↔ Simulation Developer, Tools/Pipeline Developer, Python automation/tooling developer.

Find 20-35 GitHub profiles or repositories.

Prioritize:
- Unity projects with strong README;
- WebGL / Three.js / WebGPU projects;
- technical visualization projects;
- simulation or digital twin projects;
- Unity tools/editor tools;
- technical artist tools;
- Python automation / LLM tooling repos;
- portfolios using GitHub Pages.

Output structured tables only.

Fields:
- GitHub profile/repo label
- URL
- Source type: profile / repo / GitHub Pages / organization
- Role relevance
- Main technologies
- README structure
- Pinned repos if profile
- Project media used
- Badges/topics used
- Demo links
- Portfolio links
- License/disclaimer handling
- Documentation quality
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:
- Use only public GitHub data.
- Do not clone private repos.
- Do not invent unavailable data.
- If unavailable, write "not found".
- Tables only.
- No narrative strategy.
- Include source URLs.
```

Expected output file:

```
Bloque de código: text
28C_github_profiles_benchmark.md
```

R.0.10 9. Prompt para A4 — ArtStation / portfolio benchmark

```
Bloque de código: text
Execute A4_artstation_portfolio_benchmark.

Objective:
Collect public ArtStation and portfolio examples relevant to Alexander's technical art and real-time 3D positioning.

Find 20-35 examples across:
- Unity Technical Artist portfolios;
- Technical Artist portfolios;
- shader breakdowns;
- VFX breakdowns;
- real-time 3D portfolios;
- 3D optimization breakdowns;
- Blender technical breakdowns;
- character/portrait technical breakdowns;
- XR / interactive 3D portfolios;
- technical visualization portfolios.

Output structured tables only.

Fields:
- Artist / profile label
```

- URL
- Platform
- Role target
- Seniority if visible
- Project type
- Breakdown structure
- Images used
- Videos used
- Technical text depth
- Tags/skills
- Tools shown
- Portfolio/GitHub links if visible
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:

- Use only public pages.
- Do not invent missing data.
- If unavailable, write “not found”.
- Tables only.
- No narrative strategy.
- Include source URLs.

Expected output file:

Bloque de código: text
28D_artstation_portfolio_benchmark.md

R.0.11 10. Prompt para A5 — Technical visualization case studies

Bloque de código: text
Execute A5_technical_visualization_case_studies.

Objective:

Collect public case studies relevant to TwinSight X500 and Alexander’s positioning:
Unity WebGL technical visualization, CAD-to-realtime, digital twin visualization, product configurators, simulation
↔ visualization, XR training, industrial visualization, technical inspection tools.

Find 20-30 examples from:

- company case studies;
- portfolio case studies;
- Unity showcases;
- WebGL/Three.js showcases;
- digital twin demos;
- industrial/product visualization studios;
- XR simulation demos.

Output structured tables only.

Fields:

- Case study title
- URL
- Company/person
- Industry
- Technology stack if visible
- Project type
- Problem statement
- Solution summary
- Features shown
- Media used
- Metrics/results if visible
- Case study structure
- Relevance to TwinSight
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:

- Use public sources only.
- Prefer official case studies or direct portfolio pages.
- Do not invent stack or metrics.
- If unavailable, write “not found”.
- Tables only.
- No strategy.
- Include source URLs.

Expected output file:

Bloque de código: text
29_portfolio_case_study_benchmark.md

R.0.12 11. Prompt para F1 — Demo reels benchmark

```
Bloque de código: text
Execute F1_demo_reels_benchmark.

Objective:
Collect public demo reel examples relevant to TwinSight X500:
technical artist demo reels, Unity technical reels, real-time 3D developer reels, technical visualization demos, XR/simulation
↔ demos, CAD-to-realtime/product visualization demos.

Find 15-25 examples.

Output structured tables only.

Fields:
- Demo label
- URL
- Platform
- Role target
- Duration
- Opening structure
- Shot sequence
- Captions/text usage
- Voiceover yes/no
- Technical breakdown yes/no
- Tools shown
- Metrics shown
- Call-to-action
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:
- Use only public videos/pages.
- Do not invent duration if not visible.
- If unavailable, write "not found".
- Tables only.
- Include URLs.
```

Expected output file:

```
Bloque de código: text
29B_demo_reels_benchmark.md
```

R.0.13 12. Prompt para F3 — ArtStation breakdown benchmark

```
Bloque de código: text
Execute F3_artstation_breakdown_benchmark.

Objective:
Collect ArtStation or portfolio breakdowns useful for Alexander's Blender portrait and technical art presentation.

Find 15-25 examples involving:
- topology breakdown;
- material breakdown;
- grooming breakdown;
- lighting breakdown;
- shader breakdown;
- real-time asset breakdown;
- before/after optimization;
- character study presentation;
- technical art process documentation.

Output structured tables only.

Fields:
- Project label
- URL
- Platform
- Breakdown type
- Tools shown
- Images used
- Video/GIF used
- Technical text depth
- Tags
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:
- Public pages only.
- No invented data.
```

- If unavailable, write "not found".
- Tables only.
- Include URLs.

Expected output file:

Bloque de código: text
29C_artstation_breakdown_benchmark.md

R.0.14 13. Mercado vivo: tareas posteriores

B1 — Unity / Real-Time 3D live jobs

Expected output:

Bloque de código: text
30_live_job_postings_market_snapshot.md

Prompt resumido:

Bloque de código: text
Collect live job postings for Unity Developer, Real-Time 3D Developer, Unity Technical Artist, Unity WebGL Developer,
↔ Interactive 3D Developer.

Fields:
company, role, URL, date checked, region, remote scope, salary if visible, seniority, requirements, portfolio expectations,
↔ work authorization, contractor possibility, relevance to Alexander, confidence.

Tables only. No strategy. Do not assume remote means worldwide.

B2 — Digital Twin / Simulation live jobs

Output can append to:

Bloque de código: text
30_live_job_postings_market_snapshot.md

B5 — LATAM nearshore live jobs

Output can append to:

Bloque de código: text
30_live_job_postings_market_snapshot.md

B6 — Top 30 company verification

Expected output:

Bloque de código: text
31_verified_company_targets_expanded.md

R.0.15 14. Educación y oportunidades

Future tasks:

```
Bloque de código: text
C1_high_roi_short_courses
C2_rebelway_deep_audit
C3_masters_computer_graphics_xr_simulation
C4_scholarships_portugal_germany_eu
C5_phd_research_assistantships
D1_thesis_innovation_competitions
D2_conferences_events_demo_venues
D3_paper_poster_publication_venues
E1_technical_volunteering_opportunities
E2_open_source_unity_webgl_projects
E3_remote_internships_apprenticeships
E4_aligned_freelance_gig_opportunities
```

Expected modules:

```
Bloque de código: text
32_education_scholarships_masters_fellowships.md
33_competitions_awards_hackathons_thesis_opportunities.md
34_volunteering_open_source_experience_pipeline.md
```

R.0.16 15. Síntesis final después de todas las tareas

Cuando estén listos los módulos 28B–37, generar:

```
Bloque de código: text
38_agent_research_synthesis_and_roadmap_update.md
```

Ese módulo debe responder:

```
Bloque de código: text
1. Qué cambió frente al roadmap original.
2. Qué no cambió.
3. Qué empresas suben a prioridad A.
4. Qué roles bajan de prioridad.
5. Qué cambios aplicar a CV.
6. Qué cambios aplicar a LinkedIn.
7. Qué cambios aplicar a GitHub.
8. Qué cambios aplicar al portafolio.
9. Qué cursos o programas sí justifican tiempo.
10. Qué becas o maestrías entran al roadmap 12-24 meses.
11. Qué concursos o venues valen la pena.
12. Qué oportunidades de experiencia práctica son viables.
13. Qué ajustes hacer al tracker.
14. Qué cambia en el plan 30/60/90.
```

R.0.17 16. Criterios de calidad para resultados de agente

Un resultado de agente es útil si tiene:

```
Bloque de código: text
URLs verificables
fecha de revisión
fuente concreta
datos estructurados
campos "not found" cuando no hay información
distinción entre perfil real y sample
confianza por fila
celdas cortas
sin narrativa innecesaria
```

Un resultado de agente es insuficiente si:

```
Bloque de codigo: text
termina diciendo que no puede buscar
trae solo 2-3 ejemplos
trae artículos genéricos sin ejemplos
mezcla estrategia con datos
inventa información no visible
no incluye URLs
no incluye fecha
no separa muestras reales de samples
```

Cuando un resultado es insuficiente:

```
Bloque de codigo: text
1. No generar módulo final todavía, o generar módulo marcándolo como incompleto.
2. Reintentar con prompt más específico.
3. Reducir volumen pedido.
4. Ampliar fuentes aceptables.
5. Pedir salida parcial.
```

R.0.18 17. Estado exacto para continuar

Última acción completada correctamente

```
Bloque de codigo: text
A1_linkedin_profiles_benchmark completada.
28_competitor_profile_benchmark.md generado.
```

Última acción fallida o incompleta

```
Bloque de codigo: text
A2_public_cv_resume_benchmark intentada, pero insuficiente.
```

Próxima acción exacta

```
Bloque de codigo: text
Ejecutar A2_retry_public_cv_resume_benchmark con modo agente activado.
```

Después de recibir resultados A2-Retry

Generar:

```
Bloque de codigo: text
28B_public_cv_resume_benchmark.md
```

Luego indicar:

```
Bloque de codigo: text
Siguiendo tarea: A3_github_profiles_benchmark
Modo agente: Sí
Deep Research: No
```

R.0.19 18. Instrucción para el próximo chat

Copiar este bloque al iniciar un chat nuevo:

Bloque de código: text
Continuamos el proyecto Laboral de Alexander Woodcock Salomón.

No rehacer la estrategia base. Ya existen módulos 00-27, el tracker 25 y el módulo 28 de benchmark LinkedIn.

Estado actual:

- A1_linkedin_profiles_benchmark completado.
- 28_competitor_profile_benchmark.md generado.
- A2_public_cv_resume_benchmark fue intentado pero quedó incompleto.
- Próxima tarea: ejecutar A2_retry_public_cv_resume_benchmark con agente.
- Después de recibir los datos, generar 28B_public_cv_resume_benchmark.md.
- Mantener el flujo: agente recolecta datos → asistente audita/sintetiza → genera .md → propone siguiente tarea.

Usar la estrategia central:

Real-Time 3D Developer / Unity Technical Artist focused on Unity WebGL, CAD-to-realtime optimization and technical
↪ visualization.

No usar Deep Research salvo que se pida explícitamente para investigación profunda. Para las próximas tareas usar modo agente,
↪ no Deep Research.

R.0.20 19. Prompt compacto para continuar de inmediato

Usar este prompt como siguiente paso:

Bloque de código: text
Execute A2_retry_public_cv_resume_benchmark.

Context:

A2 was attempted but incomplete. Retry with broader source types and lower target volume.

Objective:

Collect 12-20 concrete public resume/CV examples relevant to Real-Time 3D Developer, Unity Technical Artist, Unity WebGL
↪ Developer, Technical Visualization Developer, XR Developer, Simulation Developer, Tools/Pipeline Developer.

Accept:

- real public PDF resumes;
- personal website resume pages;
- portfolio CV pages;
- GitHub resume READMEs;
- ArtStation resume pages;
- blind CVs from recruiting databases;
- university sample resumes if they include concrete examples relevant to Unity, game development, technical art, creative
↪ technology, computational design, XR, simulation, visualization, or technical tools.

Output structured tables only.

Fields:

- Source label
- Resume/CV URL
- Source type
- Role target
- Country/region if visible
- Seniority
- Resume structure
- Headline/title used
- Summary style
- Skills organization
- Project/experience structure
- Metrics used
- Education wording
- Portfolio/GitHub links if visible
- What Alexander can adapt
- What Alexander should not copy
- Confidence
- Date checked

Rules:

- Public sources only.
- Do not invent data.
- If unavailable, write "not found".
- Keep cells short.
- Include source URLs.
- Tables only.
- No strategy.
- Mark generated/sample resumes as sample, not real candidate.

R.0.21 20. Prioridad real

El proyecto ya tiene suficiente estrategia.

La prioridad ahora es:

Bloque de código: text
1. mejorar evidencia pública;
2. comparar contra perfiles reales;
3. aplicar cambios a CV / LinkedIn / GitHub / portfolio;
4. empezar aplicaciones con tracker;
5. usar agente para verificar oportunidades vivas.

No volver a abrir una investigación general sin razón.

Part III

Anexos

Inventario de PDFs de Research

PDF	Tamano	Ultima modificacion	Enlace
agent_03.pdf	106.4 KB	2026-06-11 12:23	file:///E:/Laboral/Research/agent_03.pdf
agent_11.pdf	1672.5 KB	2026-06-11 14:15	file:///E:/Laboral/Research/agent_11.pdf
agent_A2.pdf	89.9 KB	2026-06-11 17:33	file:///E:/Laboral/Research/agent_A2.pdf
agent_A4.pdf	58.0 KB	2026-06-11 18:15	file:///E:/Laboral/Research/agent_A4.pdf
agent_C1.pdf	60.2 KB	2026-06-11 19:54	file:///E:/Laboral/Research/agent_C1.pdf
agent_C2.pdf	70.5 KB	2026-06-11 20:04	file:///E:/Laboral/Research/agent_C2.pdf
agent_D1.pdf	53.8 KB	2026-06-11 20:40	file:///E:/Laboral/Research/agent_D1.pdf
agent_D1b.pdf	56.9 KB	2026-06-11 20:52	file:///E:/Laboral/Research/agent_D1b.pdf
agent_F1.pdf	56.6 KB	2026-06-11 18:14	file:///E:/Laboral/Research/agent_F1.pdf
agent_F3.pdf	41.5 KB	2026-06-11 18:24	file:///E:/Laboral/Research/agent_F3.pdf
agente_B1.pdf	39.0 KB	2026-06-11 19:20	file:///E:/Laboral/Research/agente_B1.pdf
agente_B5.pdf	27.1 KB	2026-06-11 19:29	file:///E:/Laboral/Research/agente_B5.pdf

Indice de Historic

Archivo	Ruta	Tamano	Nota
alexander_woodcock_context.md	Historic/alexander_woodcock_context.md	10 KB	Historico/desactualizado; no gobierna la estrategia actual.
contexto_maestro_alexander (descatualizado, aún sin terminar proyecto).md	Historic/contexto_maestro_alexander (descatualizado, aún sin terminar proyecto).md	3.8 KB	Historico/desactualizado; no gobierna la estrategia actual.
deep-research-report (8).md	Historic/deep-research-report (8).md	56.7 KB	Historico/desactualizado; no gobierna la estrategia actual.
estrategia_idiomas_mercado_laboral_tecnica_estrategia_idiomas_mercado_laboral_tecnica_artist.md	Historic/estrategia_idiomas_mercado_laboral_tecnica_estrategia_idiomas_mercado_laboral_tecnica_artist.md	26.7 KB	Historico/desactualizado; no gobierna la estrategia actual.
job_search_strategy.md	Historic/job_search_strategy.md	4.4 KB	Historico/desactualizado; no gobierna la estrategia actual.
master_schedule.md	Historic/master_schedule.md	5.9 KB	Historico/desactualizado; no gobierna la estrategia actual.
plan_audit.md	Historic/plan_audit.md	6.8 KB	Historico/desactualizado; no gobierna la estrategia actual.
portfolio_strategy.md	Historic/portfolio_strategy.md	3.6 KB	Historico/desactualizado; no gobierna la estrategia actual.
ticktick_schedule.md	Historic/ticktick_schedule.md	3.8 KB	Historico/desactualizado; no gobierna la estrategia actual.

Puente a Obsidian y portafolio web

La navegacion viva del sistema esta en `_obsidian/00_HOME.md`.

La carpeta de portafolio web esta en [portfolio_web](#).